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CRYPTOGRAPHY AND NETWORK SECURITY

INTRODUCTION TO MODERN CRYPTOGRAPHY

Second Edition

Jonathan Katz
Yehuda Lindell



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Jonathan Katz

University of Maryland
College Park, MD, USA

Yehuda Lindell

Bar-Ilan University
Ramat Gan, Israel



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Preface

The goal of our book remains the same as in the first edition: to present the basic paradigms and principles of modern cryptography to a general audience with a basic mathematics background. We have designed this book to serve as a textbook for undergraduate- or graduate-level courses in cryptography (in computer science, electrical engineering, or mathematics departments), as a general introduction suitable for self-study (especially for beginning graduate students), and as a reference for students, researchers, and practitioners.

There are numerous other cryptography textbooks available today, and the reader may rightly ask whether another book on the subject is needed. We would not have written this book—nor worked on revising it for the second edition—if the answer to that question were anything other than an unequivocal *yes*. What, in our opinion, distinguishes our book from other available books is that it provides a *rigorous* treatment of modern cryptography in an *accessible* manner appropriate for an introduction to the topic.

Our focus is on *modern* (post-1980s) cryptography, which is distinguished from classical cryptography by its emphasis on definitions, precise assumptions, and rigorous proofs of security. We briefly discuss each of these in turn (these principles are explored in greater detail in Chapter 1):

- **The central role of definitions:** A key intellectual contribution of modern cryptography has been the recognition that *formal definitions of security are an essential first step in the design of any cryptographic primitive or protocol*. The reason, in retrospect, is simple: if you don't know what it is you are trying to achieve, how can you hope to know when you have achieved it? As we will see in this book, cryptographic definitions of security are quite strong and—at first glance—may appear impossible to achieve. One of the most amazing aspects of cryptography is that efficient constructions satisfying such strong definitions can be proven to exist (under rather mild assumptions).
- **The importance of precise assumptions:** As will be explained in Chapters 2 and 3, many cryptographic constructions cannot currently be proven secure in an unconditional sense. Security often relies, instead, on some widely believed (though unproven) assumption(s). The modern cryptographic approach dictates that *any such assumption must be clearly stated and unambiguously defined*. This not only allows for objective evaluation of the assumption but, more importantly, enables rigorous proofs of security as described next.

- **The possibility of proofs of security:** The previous two principles serve as the basis for the idea that *cryptographic constructions can be proven secure* with respect to clearly stated definitions of security and relative to well-defined cryptographic assumptions. This concept is the essence of modern cryptography, and is what has transformed the field from an art to a science.

The importance of this idea cannot be overemphasized. Historically, cryptographic schemes were designed in a largely ad hoc fashion, and were deemed to be secure if the designers themselves could not find any attacks. In contrast, modern cryptography advocates the design of schemes with formal, mathematical proofs of security in well-defined models. Such schemes are *guaranteed* to be secure unless the underlying assumption is false (or the security definition did not appropriately model the real-world security concerns). By relying on long-standing assumptions (e.g., the assumption that “factoring is hard”), it is thus possible to obtain schemes that are extremely unlikely to be broken.

A unified approach. The above principles of modern cryptography are relevant not only to the “theory of cryptography” community. The importance of precise definitions is, by now, widely understood and appreciated by developers and security engineers who use cryptographic tools to build secure systems, and rigorous proofs of security have become one of the requirements for cryptographic schemes to be standardized.

Changes in the Second Edition

In preparing the second edition, we have made a conscious effort to integrate a more practical perspective (without sacrificing a rigorous approach). This is reflected in a number of changes and additions we have made:

- We have increased our coverage of *stream ciphers*, introducing them as a variant of pseudorandom generators in Section 3.3.1, discussing stream-cipher modes of operation in Section 3.6.1, and describing modern stream-cipher design principles and examples in Section 6.1.
- We have emphasized the importance of *authenticated encryption* (see Section 4.5) and have added a section on secure communication sessions.
- We have moved our treatment of hash functions into its own chapter (Chapter 5), have included some standard applications of cryptographic hash functions (Section 5.6), and have added a section on hash-function design principles and widely used constructions (Section 6.3). We have also improved our treatment of birthday attacks (covering small-space birthday attacks in Section 5.4.2) and have added a discussion of rainbow tables and time/space tradeoffs (Section 5.4.3).

- We have included several important attacks on implementations of cryptography that arise in practice, including chosen-plaintext attacks on chained-CBC encryption (Section 3.6.2), padding-oracle attacks on CBC-mode encryption (Section 3.7.2), and timing attacks on MAC verification (Section 4.2).
- After much deliberation, we have decided to introduce the random-oracle model much earlier in the book (Section 5.5). This allows us to give a proper, integrated treatment of standardized, widely used public-key encryption and signature schemes in later chapters, instead of relegating them to second-class status in a chapter at the end of the book.
- We have strengthened our coverage of elliptic-curve cryptography (Section 8.3.4) and have added a discussion of its impact on recommended key lengths (Section 9.3).
- In the chapter on public-key encryption, we introduce the KEM/DEM paradigm as a form of hybrid encryption (see Section 11.3). We also cover DHIES/ECIES in addition to the RSA PKCS #1 standards.
- In the chapter on digital signatures, we now describe the construction of signatures from identification schemes using the Fiat–Shamir transform, with the Schnorr signature scheme as a prototypical example. We have also improved our coverage of DSA/ECDSA. We include brief discussions of SSL/TLS and signcryption, both of which serve as culminations of everything covered up to that point.
- In the “advanced topics” chapter, we have amplified our treatment of homomorphic encryption, and have included sections on secret sharing and threshold encryption.

Beyond the above, we have also edited the entire book to make extensive corrections as well as smaller adjustments, including more worked examples, to improve the exposition. Several additional exercises have also been added.

Guide to Using This Book

This section is intended primarily for instructors seeking to adopt this book for their course, though the student picking up this book on his or her own may also find it a useful overview.

Required background. We have structured the book so that the only formal prerequisite is a course on discrete mathematics. Even here we rely on very little material: we assume familiarity with basic (discrete) probability and modular arithmetic. Students reading this book are also expected to have had some exposure to algorithms, mainly to be comfortable reading pseudocode and to be familiar with big- $\mathcal{O}$ notation. Many of these concepts are reviewed in Appendix A and/or when first used in the book.

Notwithstanding the above, the book does use definitions, proofs, and abstract mathematical concepts, and therefore requires some mathematical maturity. In particular, the reader is assumed to have had some exposure to proofs at the college level, whether in an upper-level mathematics course or a course on discrete mathematics, algorithms, or computability theory.

Suggestions for course organization. The core material of this book, which we recommend should be covered in any introductory course on cryptography, consists of the following (in all cases, starred sections are excluded; more on this below):

- *Introduction and Classical Cryptography:* Chapters 1 and 2 discuss classical cryptography and set the stage for modern cryptography.
- *Private-Key (Symmetric) Cryptography:* Chapter 3 on private-key encryption, Chapter 4 on message authentication, and Chapter 5 on hash functions provide a thorough treatment of these topics.

We also highly recommend covering Section 6.2, which deals with block-cipher design; in our experience students really enjoy this material, and it makes the abstract ideas they have learned in previous chapters more concrete. Although we do consider this core material, it is not used in the rest of the book and so can be safely skipped if desired.

- *Public-Key (Asymmetric) Cryptography:* Chapter 8 gives a self-contained introduction to all the number theory needed for the remainder of the book. The material in Chapter 9 is not used subsequently; however, we do recommend at least covering Section 9.3 on recommended key lengths. The public-key revolution is described in Chapter 10. Ideally, all of Chapters 11 and 12 should be covered; those pressed for time can pick and choose appropriately.

We are typically able to cover most of the above in a one-semester (35-hour) undergraduate course (omitting some proofs and skipping some topics, as needed) or, with some changes to add more material on theoretical foundations, in the first three-quarters of a one-semester graduate course. Instructors with more time available can proceed at a more leisurely pace or incorporate additional topics, as discussed below.

Those wishing to cover additional material, in either a longer course or a faster-paced graduate course, will find that the book is structured to allow flexible incorporation of other topics as time permits (and depending on the interests of the instructor). Specifically, the starred (*) sections and chapters may be covered in any order, or skipped entirely, without affecting the overall flow of the book. We have taken care to ensure that none of the core material depends on any of the starred material and, for the most part, the starred sections do not depend on each other. (When they do, this dependence is explicitly noted.)

We suggest the following from among the starred topics for those wishing to give their course a particular flavor:

- *Theory*: A more theoretically inclined course could include material from Section 3.2.2 (semantic security); Chapter 7 (one-way functions and hard-core predicates, and constructing pseudorandom generators, functions, and permutations from one-way permutations); Section 8.4 (one-way functions and collision-resistant hash functions from number-theoretic assumptions); Section 11.5.3 (RSA encryption without random oracles); and Section 12.6 (signatures without random oracles).
- *Mathematics*: A course directed at students with a strong mathematics background—or being taught by someone who enjoys this aspect of cryptography—could incorporate Section 4.6 (information-theoretic MACs in finite fields); some of the more advanced number theory from Chapter 8 (e.g., the Chinese remainder theorem and the Miller–Rabin primality test); and all of Chapter 9.

In either case, a selection of advanced topics from Chapter 13 could also be included.

Feedback and Errata

Our goal in writing this book was to make modern cryptography accessible to a wide audience beyond the “theoretical computer science” community. We hope you will let us know if we have succeeded. The many enthusiastic emails we have received in response to our first edition have made the whole process of writing this book worthwhile.

We are always happy to receive feedback. We hope there are no errors or typos in the book; if you do find any, however, we would greatly appreciate it if you let us know. (A list of known errata will be maintained at <http://www.cs.umd.edu/~jkatz/imc.html>.) You can email your comments and errata to jkatz@cs.umd.edu and lindell@biu.ac.il; please put “Introduction to Modern Cryptography” in the subject line.

Acknowledgments

For the second edition: We are grateful to the many readers of the first edition who have sent us comments, suggestions, and corrections that helped to greatly improve the book. Discussions with Claude Crépeau, Bill Gasarch, Gene Itkis, Leonid Reyzin, Tom Shrimpton, and Salil Vadhan regarding the content and overall “philosophy” of the book were especially fruitful. We also thank Bar Alon, Gilad Asharov, Giuseppe Ateniese, Amir Azodi, Omer Berkman, Sergio de Biasi, Aurora Bristor, Richard Chang, Qingfeng Cheng, Kwan Tae Cho, Kyliah Clarkson, Ran Cohen, Nikolas Coukouma, Dana Dachman-Soled, Michael Fang, Michael Farcasin, Pooya Farshim, Marc Fischlin, Lance

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Part I

Introduction and Classical Cryptography

Chapter 1

Introduction

1.1 Cryptography and Modern Cryptography

The *Concise Oxford English Dictionary* defines cryptography as “the art of writing or solving codes.” This is historically accurate, but does not capture the current breadth of the field or its present-day scientific foundations. The definition focuses solely on the *codes* that have been used for centuries to enable secret communication. But cryptography nowadays encompasses much more than this: it deals with mechanisms for ensuring integrity, techniques for exchanging secret keys, protocols for authenticating users, electronic auctions and elections, digital cash, and more. Without attempting to provide a complete characterization, we would say that modern cryptography involves *the study of mathematical techniques for securing digital information, systems, and distributed computations against adversarial attacks*.

The dictionary definition also refers to cryptography as an *art*. Until late in the 20th century cryptography was, indeed, largely an art. Constructing good codes, or breaking existing ones, relied on creativity and a developed sense of how codes work. There was little theory to rely on and, for a long time, no working definition of what constitutes a good code. Beginning in the 1970s and 1980s, this picture of cryptography radically changed. A rich theory began to emerge, enabling the rigorous study of cryptography as a *science* and a mathematical discipline. This perspective has, in turn, influenced how researchers think about the broader field of computer security.

Another very important difference between classical cryptography (say, before the 1980s) and modern cryptography relates to its adoption. Historically, the major consumers of cryptography were military organizations and governments. Today, cryptography is everywhere! If you have ever authenticated yourself by typing a password, purchased something by credit card over the Internet, or downloaded a verified update for your operating system, you have undoubtedly used cryptography. And, more and more, programmers with relatively little experience are being asked to “secure” the applications they write by incorporating cryptographic mechanisms.

In short, cryptography has gone from a heuristic set of tools concerned with ensuring secret communication for the military to a science that helps secure systems for ordinary people all across the globe. This also means that cryptography has become a more central topic within computer science.

Goals of this book. Our goal is to make the basic principles of modern cryptography accessible to students of computer science, electrical engineering, or mathematics; to professionals who want to incorporate cryptography in systems or software they are developing; and to anyone with a basic level of mathematical maturity who is interested in understanding this fascinating field. After completing this book, the reader should appreciate the security guarantees common cryptographic primitives are intended to provide; be aware of standard (secure) constructions of such primitives; and be able to perform a basic evaluation of new schemes based on their proofs of security (or lack thereof) and the mathematical assumptions underlying those proofs. It is not our intention for readers to become experts—or to be able to design new cryptosystems—after finishing this book, but we have attempted to provide the terminology and foundational material needed for the interested reader to subsequently study more advanced references in the area.

This chapter. The focus of this book is the formal study of modern cryptography, but we begin in this chapter with a more informal discussion of “classical” cryptography. Besides allowing us to ease into the material, our treatment in this chapter will also serve to motivate the more rigorous approach we will be taking in the rest of the book. Our intention here is not to be exhaustive and, as such, this chapter should not be taken as a representative historical account. The reader interested in the history of cryptography is invited to consult the references at the end of this chapter.

1.2 The Setting of Private-Key Encryption

Classical cryptography was concerned with designing and using *codes* (also called *ciphers*) that enable two parties to communicate secretly in the presence of an eavesdropper who can monitor all communication between them. In modern parlance, codes are called *encryption schemes* and that is the terminology we will use here. Security of all classical encryption schemes relied on a secret—a *key*—shared by the communicating parties in advance and unknown to the eavesdropper. This scenario is known as the *private-key* (or *shared-/secret-key*) setting, and private-key encryption is just one example of a cryptographic primitive used in this setting. Before describing some historical encryption schemes, we discuss private-key encryption more generally.

In the setting of private-key encryption, two parties share a key and use this key when they want to communicate secretly. One party can send a message, or *plaintext*, to the other by using the shared key to *encrypt* (or “scramble”) the message and thus obtain a *ciphertext* that is transmitted to the receiver. The receiver uses the same key to *decrypt* (or “unscramble”) the ciphertext and recover the original message. Note the same key is used to convert the

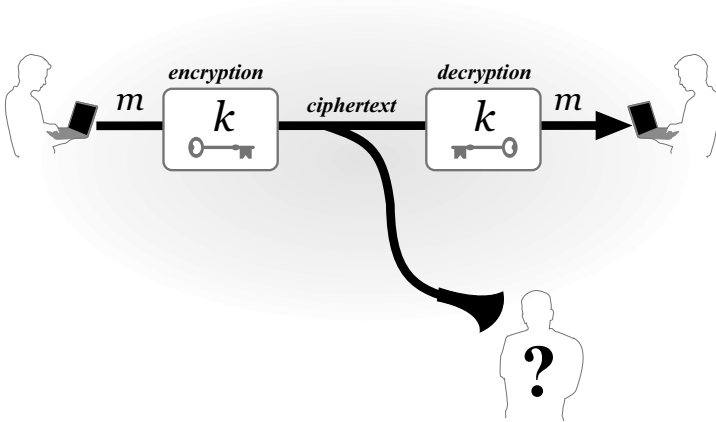


FIGURE 1.1: One common setting of private-key cryptography (here, encryption): two parties share a key that they use to communicate securely.

plaintext into a ciphertext and back; that is why this is also known as the *symmetric-key* setting, where the symmetry lies in the fact that both parties hold the same key that is used for encryption and decryption. This is in contrast to *asymmetric*, or *public-key*, encryption (introduced in Chapter 10), where encryption and decryption use different keys.

As already noted, the goal of encryption is to keep the plaintext hidden from an eavesdropper who can monitor the communication channel and observe the ciphertext. We discuss this in more detail later in this chapter, and spend a great deal of time in Chapters 2 and 3 formally defining this goal.

There are two canonical applications of private-key cryptography. In the first, there are two distinct parties separated in *space*, e.g., a worker in New York communicating with her colleague in California; see Figure 1.2. These two users are assumed to have been able to securely share a key in advance of their communication. (Note that if one party simply sends the key to the other over the public communication channel, then the eavesdropper obtains the key too!) Often this is easy to accomplish by having the parties physically meet in a secure location to share a key before they separate; in the example just given, the co-workers might arrange to share a key when they are both in the New York office. In other cases, sharing a key securely is more difficult. For the next several chapters we simply assume that sharing a key is possible; we will revisit this issue in Chapter 10.

The second widespread application of private-key cryptography involves the same party communicating with itself over *time*. (See Figure 1.2.) Consider, e.g., disk encryption, where a user encrypts some plaintext and stores the resulting ciphertext on their hard drive; the same user will return at a later

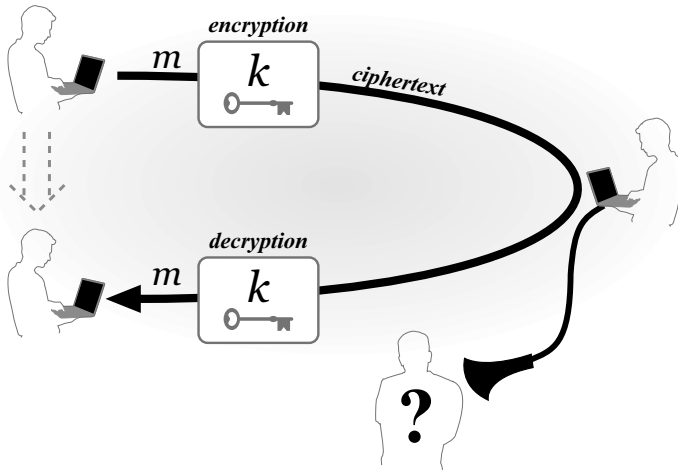


FIGURE 1.2: Another common setting of private-key cryptography (again, encryption): a single user stores data securely over time.

point in time to decrypt the ciphertext and recover the original data. The hard drive here serves as the communication channel on which an attacker might eavesdrop by gaining access to the hard drive and reading its contents. “Sharing” the key is now trivial, though the user still needs a secure and reliable way to remember/store the key for use at a later point in time.

The syntax of encryption. Formally, a private-key encryption scheme is defined by specifying a *message space* $\mathcal{M}$ along with three algorithms: a procedure for generating keys (**Gen**), a procedure for encrypting (**Enc**), and a procedure for decrypting (**Dec**). The message space $\mathcal{M}$ defines the set of “legal” messages, i.e., those supported by the scheme. The algorithms have the following functionality:

1. The *key-generation algorithm* **Gen** is a probabilistic algorithm that outputs a key k chosen according to some distribution.
2. The *encryption algorithm* **Enc** takes as input a key k and a message m and outputs a ciphertext c . We denote by $\text{Enc}_k(m)$ the encryption of the plaintext m using the key k .
3. The *decryption algorithm* **Dec** takes as input a key k and a ciphertext c and outputs a plaintext m . We denote the decryption of the ciphertext c using the key k by $\text{Dec}_k(c)$.

An encryption scheme must satisfy the following correctness requirement: for every key k output by **Gen** and every message $m \in \mathcal{M}$, it holds that

$$\text{Dec}_k(\text{Enc}_k(m)) = m.$$

In words: encrypting a message and then decrypting the resulting ciphertext (using the same key) yields the original message.

The set of all possible keys output by the key-generation algorithm is called the *key space* and is denoted by $\mathcal{K}$. Almost always, **Gen** simply chooses a uniform key from the key space; in fact, one can assume without loss of generality that this is the case (see Exercise 2.1).

Reviewing our earlier discussion, an encryption scheme can be used by two parties who wish to communicate as follows. First, **Gen** is run to obtain a key k that the parties share. Later, when one party wants to send a plaintext m to the other, she computes $c := \text{Enc}_k(m)$ and sends the resulting ciphertext c over the public channel to the other party.¹ Upon receiving c , the other party computes $m := \text{Dec}_k(c)$ to recover the original plaintext.

Keys and Kerckhoffs’ principle. As is clear from the above, if an eavesdropping adversary knows the algorithm **Dec** as well as the key k shared by the two communicating parties, then that adversary will be able to decrypt any ciphertexts transmitted by those parties. It is for this reason that the communicating parties must share the key k securely and keep k completely secret from everyone else. Perhaps they should keep the decryption algorithm **Dec** secret, too? For that matter, might it not be better for them to keep all the details of the encryption scheme secret?

In the late 19th century, Auguste Kerckhoffs argued the opposite in a paper he wrote elucidating several design principles for military ciphers. One of the most important of these, now known simply as *Kerckhoffs’ principle*, was:

The cipher method must not be required to be secret, and it must be able to fall into the hands of the enemy without inconvenience.

That is, an encryption scheme should be designed to be secure *even if* an eavesdropper knows all the details of the scheme, so long as the attacker doesn’t know the key being used. Stated differently, security should not rely on the encryption scheme being secret; instead, Kerckhoffs’ principle demands that *security rely solely on secrecy of the key*.

There are three primary arguments in favor of Kerckhoffs’ principle. The first is that it is significantly easier for the parties to maintain secrecy of a short key than to keep secret the (more complicated) algorithm they are using. This is especially true if we imagine using encryption to secure the communication between all pairs of employees in some organization. Unless each pair of parties uses their own, unique algorithm, some parties will know the algorithm used by others. Information about the encryption algorithm might be leaked by one of these employees (say, after being fired), or obtained by an attacker using reverse engineering. In short, it is simply unrealistic to assume that the encryption algorithm will remain secret.

¹We use “:=” to denote deterministic assignment, and assume for now that **Enc** is deterministic. A list of common notation can be found in the back of the book.

Second, in case the honest parties' shared, secret information *is* ever exposed, it will be much easier for them to change a key than to replace an encryption scheme. (Consider updating a file versus installing a new program.) Moreover, it is relatively trivial to generate a new random secret, whereas it would be a huge undertaking to design a new encryption scheme.

Finally, for large-scale deployment it is significantly easier for users to all rely on the same encryption algorithm/software (with different keys) than for everyone to use their own custom algorithm. (This is true even for a single user who is communicating with several different parties.) In fact, it is desirable for encryption schemes to be *standardized* so that (1) compatibility is ensured by default and (2) users will utilize an encryption scheme that has undergone public scrutiny and in which no weaknesses have been found.

Nowadays Kerckhoffs' principle is understood as advocating that cryptographic designs be made completely public, in stark contrast to the notion of "security by obscurity" which suggests that keeping algorithms secret improves security. It is very dangerous to use a proprietary, "home-brewed" algorithm (i.e., a non-standardized algorithm designed in secret by some company). In contrast, published designs undergo public review and are therefore likely to be stronger. Many years of experience have demonstrated that it is very difficult to construct good cryptographic schemes. Therefore, our confidence in the security of a scheme is much higher if it has been extensively studied (by experts other than the designers of the scheme) and no weaknesses have been found. As simple and obvious as it may sound, the principle of open cryptographic design (i.e., Kerckhoffs' principle) has been ignored over and over again with disastrous results. Fortunately, today there are enough secure, standardized, and widely available cryptosystems that there is no reason to use anything else.

1.3 Historical Ciphers and Their Cryptanalysis

In our study of "classical" cryptography we will examine some historical encryption schemes and show that they are insecure. Our main aims in presenting this material are (1) to highlight the weaknesses of an "ad hoc" approach to cryptography, and thus motivate the modern, rigorous approach that will be taken in the rest of the book, and (2) to demonstrate that simple approaches to achieving secure encryption are unlikely to succeed. Along the way, we will present some central principles of cryptography inspired by the weaknesses of these historical schemes.

In this section, plaintext characters are written in **lower case** and ciphertext characters are written in **UPPER CASE** for typographical clarity.

Caesar's cipher. One of the oldest recorded ciphers, known as *Caesar's*

cipher, is described in *De Vita Caesarum, Divus Iulius* (“The Lives of the Caesars, the Deified Julius”), written in approximately 110 CE:

There are also letters of his to Cicero, as well as to his intimates on private affairs, and in the latter, if he had anything confidential to say, he wrote it in cipher, that is, by so changing the order of the letters of the alphabet, that not a word could be made out...

Julius Caesar encrypted by shifting the letters of the alphabet 3 places forward: **a** was replaced with **D**, **b** with **E**, and so on. At the very end of the alphabet, the letters wrap around and so **z** was replaced with **C**, **y** with **B**, and **x** with **A**. For example, encryption of the message **begin the attack now**, with spaces removed, gives:

EHJLQWKHDWWDFNQRZ.

An immediate problem with this cipher is that the encryption method is *fixed*; there is no key. Thus, anyone learning how Caesar encrypted his messages would be able to decrypt effortlessly.

Interestingly, a variant of this cipher called ROT-13 (where the shift is 13 places instead of 3) is still used nowadays in various online forums. It is understood that this does not provide any cryptographic security; it is used merely to ensure that the text (say, a movie spoiler) is unintelligible unless the reader of a message consciously chooses to decrypt it.

The shift cipher and the sufficient key-space principle. The *shift cipher* can be viewed as a keyed variant of Caesar’s cipher.² Specifically, in the shift cipher the key k is a number between 0 and 25. To encrypt, letters are shifted as in Caesar’s cipher, but now by k places. Mapping this to the syntax of encryption described earlier, the message space consists of arbitrary length strings of English letters with punctuation, spaces, and numerals removed, and with no distinction between upper and lower case. Algorithm **Gen** outputs a uniform key $k \in \{0, \dots, 25\}$; algorithm **Enc** takes a key k and a plaintext and shifts each letter of the plaintext forward k positions (wrapping around at the end of the alphabet); and algorithm **Dec** takes a key k and a ciphertext and shifts every letter of the ciphertext *backward* k positions.

A more mathematical description is obtained by equating the English alphabet with the set $\{0, \dots, 25\}$ (so **a** = 0, **b** = 1, etc.). The message space $\mathcal{M}$ is then any finite sequence of integers from this set. Encryption of the message $m = m_1 \cdots m_\ell$ (where $m_i \in \{0, \dots, 25\}$) using key k is given by

$$\text{Enc}_k(m_1 \cdots m_\ell) = c_1 \cdots c_\ell, \quad \text{where } c_i = [(m_i + k) \bmod 26].$$

(The notation $[a \bmod N]$ denotes the remainder of a upon division by N , with $0 \leq [a \bmod N] < N$. We refer to the process mapping a to $[a \bmod N]$

²In some books, “Caesar’s cipher” and “shift cipher” are used interchangeably.

as *reduction modulo N* ; we will have more to say about this beginning in Chapter 8.) Decryption of a ciphertext $c = c_1 \cdots c_\ell$ using key k is given by

$$\text{Dec}_k(c_1 \cdots c_\ell) = m_1 \cdots m_\ell, \quad \text{where } m_i = [(c_i - k) \bmod 26].$$

Is the shift cipher secure? Before reading on, try to decrypt the following ciphertext that was generated using the shift cipher and a secret key k :

OVDTHUFWVZZPISLRLFZHYLAOLYL.

Is it possible to recover the message without knowing k ? Actually, it is trivial! The reason is that there are only 26 possible keys. So one can try to decrypt the ciphertext using every possible key and thereby obtain a list of 26 candidate plaintexts. The correct plaintext will certainly be on this list; moreover, if the ciphertext is “long enough” then the correct plaintext will likely be the only candidate on the list that “makes sense.” (The latter is not necessarily true, but will be true most of the time. Even when it is not, the attack narrows down the set of potential plaintexts to at most 26 possibilities.) By scanning the list of candidates it is easy to recover the original plaintext.

An attack that involves trying every possible key is called a *brute-force* or *exhaustive-search* attack. Clearly, for an encryption scheme to be secure it must not be vulnerable to such an attack.³ This observation is known as the *sufficient key-space principle*:

Any secure encryption scheme must have a key space that is sufficiently large to make an exhaustive-search attack infeasible.

One can debate what amount of effort makes a task “infeasible,” and an exact determination of feasibility depends on both the resources of a potential attacker and the length of time the sender and receiver want to ensure secrecy of their communication. Nowadays, attackers can use supercomputers, tens of thousands of personal computers, or graphics processing units (GPUs) to speed up brute-force attacks. To protect against such attacks the key space must therefore be very large—say, of size at least 2^{70} , and even larger if one is concerned about long-term security against a well-funded attacker.

The sufficient key-space principle gives a *necessary* condition for security, but not a *sufficient* one. The next example demonstrates this.

The mono-alphabetic substitution cipher. In the shift cipher, the key defines a map from each letter of the (plaintext) alphabet to some letter of the (ciphertext) alphabet, where the map is a fixed shift determined by the key. In the *mono-alphabetic substitution cipher*, the key also defines a map on the alphabet, but the map is now allowed to be *arbitrary* subject only to the constraint that it be one-to-one so that decryption is possible. The key

³Technically, this is only true if the message space is larger than the key space; we will return to this point in Chapter 2. Encryption schemes used in practice have this property.

space thus consists of all *bijections*, or *permutations*, of the alphabet. So, for example, the key that defines the following permutation

a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
X	E	U	A	D	N	B	K	V	M	R	O	C	Q	F	S	Y	H	W	G	L	Z	I	J	P	T

(in which a maps to X, etc.) would encrypt the message **tellhimaboutme** to **GDOOKVCXEFLGCD**. The name of this cipher comes from the fact that the key defines a (fixed) substitution for individual characters of the plaintext.

Assuming the English alphabet is being used, the key space is of size $26! = 26 \cdot 25 \cdot 24 \cdots 2 \cdot 1$, or approximately 2^{88} , and a brute-force attack is infeasible. This, however, does not mean the cipher is secure! In fact, as we will show next, it is easy to break this scheme even though it has a large key space.

Assume English-language text is being encrypted (i.e., the text is grammatically correct English writing, not just text written using characters of the English alphabet). The mono-alphabetic substitution cipher can then be attacked by utilizing statistical patterns of the English language. (Of course, the same attack works for any language.) The attack relies on the facts that:

1. For any key, the mapping of each letter is fixed, and so if **e** is mapped to **D**, then every appearance of **e** in the plaintext will result in the appearance of **D** in the ciphertext.
2. The frequency distribution of individual letters in the English language is known (see Figure 1.3). Of course, very short texts may deviate from this distribution, but even texts consisting of only a few sentences tend to have distributions that are very close to the average.

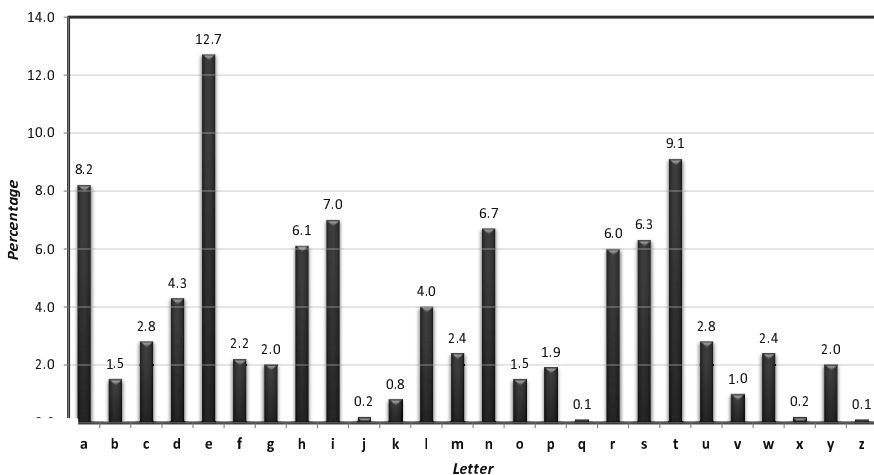


FIGURE 1.3: Average letter frequencies for English-language text.

The attack works by tabulating the frequency distribution of characters in the ciphertext, i.e., recording that **A** appeared 11 times, **B** appeared 4 times, and so on. These frequencies are then compared to the known letter frequencies of normal English text. One can then guess parts of the mapping defined by the key based on the observed frequencies. For example, since **e** is the most frequent letter in English, one can guess that the most frequent character in the ciphertext corresponds to the plaintext character **e**, and so on. Some of the guesses may be wrong, but enough of the guesses will be correct to enable relatively quick decryption (especially utilizing other knowledge of English, such as the fact that **u** generally follows **q**, and that **h** is likely to appear between **t** and **e**). We conclude that although the mono-alphabetic substitution cipher has a large key space, it is still insecure.

It should not be surprising that the mono-alphabetic substitution cipher can be quickly broken, since puzzles based on this cipher appear in newspapers (and are solved by some people before their morning coffee!). We recommend that you try to decipher the following ciphertext—this should convince you how easy the attack is to carry out. (Use Figure 1.3 to help you.)

JGRMQOYGHMVBJWRWQFPWHGFFDQGFPPFZRKBEEBJIZQQOCIBZKLFAGQVVFZWWE
 OGWOPFGFHWOLPHRLLOLDFMFGQWBLWBWQOLKFWBYLBLYLSFLJGRMQBOLWJVFP
 FWQVHQWFFPQOQVFPQOCFPOGFWFJIGFQVHLHLROQVFGWJVFPOLFQVQVFILE
 OGQILHQFQGIQVVOVSFAFGBWQVHQWIVJWJVFPFWHGFIWIHZZRQGBABHZQOCGFHX

An improved attack on the shift cipher. We can use letter-frequency tables to give an improved attack on the shift cipher. Our previous attack on the shift cipher required decrypting the ciphertext using each possible key, and then checking which key results in a plaintext that “makes sense.” A drawback of this approach is that it is somewhat difficult to automate, since it is difficult for a computer to check whether a given plaintext “makes sense.” (We do not claim that it would be impossible, as the attack could be automated using a dictionary of valid English words. We only claim that it would not be trivial to automate.) Moreover, there may be cases—we will see one later—where the plaintext characters are distributed just like English-language text even though the plaintext itself is not valid English, in which case checking for a plaintext that “makes sense” will not work.

We now describe an attack that does not suffer from these drawbacks. As before, associate the letters of the English alphabet with $0, \dots, 25$. Let p_i , with $0 \leq p_i \leq 1$, denote the frequency of the i th letter in normal English text (ignoring spaces, punctuation, etc.). Calculation using Figure 1.3 gives

$$\sum_{i=0}^{25} p_i^2 \approx 0.065. \quad (1.1)$$

Now, say we are given some ciphertext and let q_i denote the frequency of the i th letter of the alphabet in this ciphertext; i.e., q_i is simply the number

of occurrences of the i th letter of the alphabet in the ciphertext divided by the length of the ciphertext. If the key is k , then p_i should be roughly equal to q_{i+k} for all i , because the i th letter is mapped to the $(i+k)$ th letter. (We use $i+k$ instead of the more cumbersome $[i+k \bmod 26]$.) Thus, if we compute

$$I_j \stackrel{\text{def}}{=} \sum_{i=0}^{25} p_i \cdot q_{i+j}$$

for each value of $j \in \{0, \dots, 25\}$, then we expect to find that $I_k \approx 0.065$ (where k is the actual key), whereas I_j for $j \neq k$ will be different from 0.065. This leads to a key-recovery attack that is easy to automate: compute I_j for all j , and then output the value k for which I_k is closest to 0.065.

The Vigenère (poly-alphabetic shift) cipher. The statistical attack on the mono-alphabetic substitution cipher can be carried out because the key defines a fixed mapping that is applied letter-by-letter to the plaintext. Such an attack could be thwarted by using a *poly-alphabetic substitution cipher* where the key instead defines a mapping that is applied on *blocks* of plaintext characters. Here, for example, a key might map the 2-character block **ab** to **DZ** while mapping **ac** to **TV**; note that the plaintext character **a** does not get mapped to a fixed ciphertext character. Poly-alphabetic substitution ciphers “smooth out” the frequency distribution of characters in the ciphertext and make it harder to perform statistical analysis.

The *Vigenère cipher*, a special case of the above also called the poly-alphabetic *shift* cipher, works by applying several independent instances of the shift cipher in sequence. The key is now viewed as a *string* of letters; encryption is done by shifting each plaintext character by the amount indicated by the next character of the key, wrapping around in the key when necessary. (This degenerates to the shift cipher if the key has length 1.) For example, encryption of the message **tellhimaboutme** using the key **cafe** would work as follows:

Plaintext:	tellhimaboutme
Key (repeated):	cafecafecafeca
Ciphertext:	VEQPJIREDOZXOE

(The key need not be an English word.) This is exactly the same as encrypting the first, fifth, ninth, ... characters with the shift cipher and key **c**; the second, sixth, tenth, ... characters with key **a**; the third, seventh, ... characters with **f**; and the fourth, eighth, ... characters with **e**. Notice that in the above example **l** is mapped once to **Q** and once to **P**. Furthermore, the ciphertext character **E** is sometimes obtained from **e** and sometimes from **a**. Thus, the character frequencies of the ciphertext are “smoothed out,” as desired.

If the key is sufficiently long, cracking this cipher appears daunting. Indeed, it had been considered by many to be “unbreakable,” and although it was invented in the 16th century, a systematic attack on the scheme was only devised hundreds of years later.

Attacking the Vigenère cipher. A first observation in attacking the Vigenère cipher is that *if the length of the key is known* then attacking the cipher is relatively easy. Specifically, say the length of the key, also called the *period*, is t . Write the key k as $k = k_1 \cdots k_t$ where each k_i is a letter of the alphabet. An observed ciphertext $c = c_1 c_2 \cdots$ can be divided into t parts where each part can be viewed as having been encrypted using a shift cipher. Specifically, for all $j \in \{1, \dots, t\}$ the ciphertext characters

$$c_j, c_{j+t}, c_{j+2t}, \dots$$

all resulted by shifting the corresponding characters of the plaintext by k_j positions. We refer to the above sequence of characters as the j th *stream*. All that remains is to determine, for each of the t streams, which of the 26 possible shifts was used. This is not as trivial as in the case of the shift cipher, because it is no longer possible to simply try different shifts in an attempt to determine when decryption of a stream “makes sense.” (Recall that a stream does not correspond to consecutive letters of the plaintext.) Furthermore, trying to guess the entire key k at once would require a brute-force search through 26^t different possibilities, which is infeasible for large t . Nevertheless, we can still use letter-frequency analysis to analyze each stream independently. Namely, for each stream we tabulate the frequency of each ciphertext character and then check which of the 26 possible shifts yields the “right” probability distribution for that stream. Since this can be carried out independently for each stream (i.e., for each character of the key), this attack takes time $26 \cdot t$ rather than time 26^t .

A more principled, easier-to-automate approach is to use the improved method for attacking the shift cipher discussed earlier. That attack did not rely on checking for a plaintext that “made sense,” but only relied on the underlying frequency distribution of characters in the plaintext.

Either of the above approaches gives a successful attack when the key length is known. What if the key length is unknown?

Note first that as long as the maximum length T of the key is not too large, we can simply repeat the above attack T times (for each possible value $t \in \{1, \dots, T\}$). This leads to at most T different candidate plaintexts, among which the true plaintext will likely be easy to identify. So an unknown key length is not a serious obstacle.

There are also more efficient ways to determine the key length from an observed ciphertext. One is to use *Kasiski’s method*, published in the mid-19th century. The first step here is to identify repeated patterns of length 2 or 3 in the ciphertext. These are likely the result of certain bigrams or trigrams that appear frequently in the plaintext. For example, consider the common word “the.” This word will be mapped to different ciphertext characters, depending on its position in the plaintext. However, if it appears twice in the same relative position, then it will be mapped to the same ciphertext characters. For a sufficiently long plaintext, there is thus a good chance that “the” will be mapped repeatedly to the same ciphertext characters.

Consider the following concrete example with the key **beads** (spaces have been added for clarity):

Plaintext:	the man and the woman retrieved the letter from the post office
Key:	bea dsb ead sbe adsbe adsbeadsb ead sbears bead sbe adsb eadsbe
Ciphertext:	ULE PSO ENG LII WREBR RHLSMEYWE XHH DFXTHJ GVOP LII PRKU SFIADI

The word **the** is mapped sometimes to ULE, sometimes to LII, and sometimes to XHH. However, it is mapped *twice* to LII, and in a long enough text it is likely that it would be mapped multiple times to each of these possibilities. Kasiski's observation was that the distance between such repeated appearances (assuming they are not coincidental) must be a multiple of the period. (In the above example, the period is 5 and the distance between the two appearances of LII is 30, which is 6 times the period.) Therefore, the greatest common divisor of the distances between repeated sequences (assuming they are not coincidental) will yield the key length t or a multiple thereof.

An alternative approach, called the *index of coincidence method*, is more methodical and hence easier to automate. Recall that if the key length is t , then the ciphertext characters

$$c_1, c_{1+t}, c_{1+2t}, \dots$$

in the first stream all resulted from encryption using the same shift. This means that the frequencies of the characters in this sequence are expected to be identical to the character frequencies of standard English text *in some shifted order*. In more detail: let q_i denote the observed frequency of the i th English letter in this stream; this is simply the number of occurrences of the i th letter of the alphabet divided by the total number of letters in the stream. If the shift used here is j (i.e., if the first character k_1 of the key is equal to j), then for all i we expect $q_{i+j} \approx p_i$, where p_i is the frequency of the i th letter of the alphabet in standard English text. (Once again, we use q_{i+j} in place of $q_{[i+j \bmod 26]}$.) But this means that the sequence $q_0, \dots, q_{25}$ is just the sequence $p_0, \dots, p_{25}$ shifted j places. As a consequence (cf. Equation (1.1)):

$$\sum_{i=0}^{25} q_i^2 \approx \sum_{i=0}^{25} p_i^2 \approx 0.065.$$

This leads to a nice way to determine the key length t . For $\tau = 1, 2, \dots$, look at the sequence of ciphertext characters $c_1, c_{1+\tau}, c_{1+2\tau}, \dots$ and tabulate $q_0, \dots, q_{25}$ for this sequence. Then compute

$$S_\tau \stackrel{\text{def}}{=} \sum_{i=0}^{25} q_i^2.$$

When $\tau = t$ we expect $S_\tau \approx 0.065$, as discussed above. On the other hand, if τ is not a multiple of t we expect that all characters will occur with roughly equal

probability in the sequence $c_1, c_{1+\tau}, c_{1+2\tau}, \dots$, and so we expect $q_i \approx 1/26$ for all i . In this case we will obtain

$$S_\tau \approx \sum_{i=0}^{25} \left(\frac{1}{26} \right)^2 \approx 0.038.$$

The smallest value of τ for which $S_\tau \approx 0.065$ is thus likely the key length. One can further validate a guess τ by carrying out a similar calculation using the second stream $c_2, c_{2+\tau}, c_{2+2\tau}, \dots$, etc.

Ciphertext length and cryptanalytic attacks. The above attacks on the Vigenère cipher require a longer ciphertext than the attacks on previous schemes. For example, the index of coincidence method requires c_1, c_{1+t}, c_{1+2t} (where t is the actual key length) to be sufficiently long in order to ensure that the observed frequencies match what is expected; the ciphertext itself must then be roughly t times larger. Similarly, the attack we showed on the mono-alphabetic substitution cipher requires a longer ciphertext than the attack on the shift cipher (which can work for encryptions of even a single word). This illustrates that a longer key can, in general, require the cryptanalyst to obtain more ciphertext in order to carry out an attack. (Indeed, the Vigenère cipher can be shown to be secure if the key is as long as what is being encrypted. We will see a similar phenomenon in the next chapter.)

Conclusions. We have presented only a few historical ciphers. Beyond their historical interest, our aim in presenting them was to illustrate some important lessons. Perhaps the most important is that *designing secure ciphers is hard*. The Vigenère cipher remained unbroken for a long time. Far more complex schemes have also been used. But a complex scheme is not necessarily secure, and all historical schemes have been broken.

1.4 Principles of Modern Cryptography

As should be clear from the previous section, cryptography was historically more of an art than a science. Schemes were designed in an ad hoc manner and evaluated based on their perceived complexity or cleverness. A scheme would be analyzed to see if any attacks could be found; if so, the scheme would be “patched” to thwart that attack, and the process repeated. Although there may have been agreement that some schemes were *not* secure (as evidenced by an especially damaging attack), there was no agreed-upon notion of what requirements a “secure” scheme should satisfy, and no way to give evidence that any specific scheme was secure.

Over the past several decades, cryptography has developed into more of a science. Schemes are now developed and analyzed in a more systematic

manner, with the ultimate goal being to give a rigorous *proof* that a given construction is secure. In order to articulate such proofs, we first need *formal definitions* that pin down exactly what “secure” means; such definitions are useful and interesting in their own right. As it turns out, most cryptographic proofs rely on currently unproven *assumptions* about the algorithmic hardness of certain mathematical problems; any such assumptions must be made explicit and be stated precisely. An emphasis on definitions, assumptions, and proofs distinguishes *modern* cryptography from classical cryptography; we discuss these three principles in greater detail in the following sections.

1.4.1 Principle 1 – Formal Definitions

One of the key contributions of modern cryptography has been the recognition that formal definitions of security are *essential* for the proper design, study, evaluation, and usage of cryptographic primitives. Put bluntly:

If you don't understand what you want to achieve, how can you possibly know when (or if) you have achieved it?

Formal definitions provide such understanding by giving a clear description of what threats are in scope and what security guarantees are desired. As such, definitions can help guide the design of cryptographic schemes. Indeed, it is much better to formalize what is required *before* the design process begins, rather than to come up with a definition *post facto* once the design is complete. The latter approach risks having the design phase end when the designers' patience is exhausted (rather than when the goal has been met), or may result in a construction achieving *more* than is needed at the expense of efficiency.

Definitions also offer a way to evaluate and analyze what is constructed. With a definition in place, one can study a proposed scheme to see if it achieves the desired guarantees; in some cases, one can even *prove* a given construction secure (see Section 1.4.3) by showing that it meets the definition. On the flip side, definitions can be used to conclusively show that a given scheme is *not* secure, insofar as the scheme does not satisfy the definition. In particular, note that the attacks in the previous section do not automatically demonstrate that any of the schemes shown there is “insecure.” For example, the attack on the Vigenère cipher assumed that sufficiently long English text was being encrypted, but could the Vigenère cipher be “secure” if short English text, or compressed text (which will have roughly uniform letter frequencies), is encrypted? It is hard to say without a formal definition in place.

Definitions enable a meaningful comparison of schemes. As we will see, there can be multiple (valid) ways to define security; the “right” one depends on the context in which a scheme is used. A scheme satisfying a weaker definition may be more efficient than another scheme satisfying a stronger definition; with precise definitions we can properly evaluate the trade-offs between the two schemes. Along the same lines, definitions enable secure usage of schemes. Consider the question of deciding which encryption scheme

to use for some larger application. A sound way to approach the problem is to first understand what notion of security is required for that application, and then find an encryption scheme satisfying that notion. A side benefit of this approach is *modularity*: a designer can “swap out” one encryption scheme and replace it with another (that also satisfies the necessary definition of security) without having to worry about affecting security of the overall application.

Writing a formal definition forces one to think about what is essential to the problem at hand and what properties are extraneous. Going through the process often reveals subtleties of the problem that were not obvious at first glance. We illustrate this next for the case of encryption.

An example: secure encryption. A common mistake is to think that formal definitions are not needed, or are trivial to come up with, because “everyone has an intuitive idea of what security means.” This is not the case. As an example, we consider the case of encryption. (The reader may want to pause here to think about how they would formally define what it means for an encryption scheme to be secure.) Although we postpone a formal definition of secure encryption to the next two chapters, we describe here informally what such a definition should capture.

In general, a security definition has two components: a security guarantee (or, from the attacker’s point of view, what constitutes a successful attack on the scheme) and a threat model. The security guarantee defines what the scheme is intended to prevent the attacker from doing, while the threat model describes the power of the adversary, i.e., what actions the attacker is assumed able to carry out.

Let’s start with the first of these. What should a secure encryption scheme guarantee? Here are some thoughts:

- *It should be impossible for an attacker to recover the key.* We have previously observed that if an attacker can determine the key shared by two parties using some scheme, then that scheme cannot be secure. However, it is easy to come up with schemes for which key recovery is impossible, yet the scheme is blatantly insecure. Consider, e.g., the scheme where $\text{Enc}_k(m) = m$. The ciphertext leaks no information about the key (and so the key cannot be recovered if it is long enough) yet the message is sent in the clear! We thus see that inability to recover the key is not sufficient for security. This makes sense: the aim of encryption is to protect the *message*; the key is a means for achieving this but is, in itself, unimportant.
- *It should be impossible for an attacker to recover the entire plaintext from the ciphertext.* This definition is better, but is still far from satisfactory. In particular, this definition would consider an encryption scheme secure if its ciphertexts revealed 90% of the plaintext, as long as 10% of the plaintext remained hard to figure out. This is clearly unacceptable in most common applications of encryption; for example, when encrypting

a salary database, we would be justifiably upset if 90% of employees' salaries were revealed!

- *It should be impossible for an attacker to recover any character of the plaintext from the ciphertext.* This looks like a good definition, yet is still not sufficient. Going back to the example of encrypting a salary database, we would not consider an encryption scheme secure if it reveals whether an employee's salary is more than or less than \$100,000, even if it does not reveal any particular digit of that employee's salary. Similarly, we would not want an encryption scheme to reveal whether employee *A* makes more than employee *B*.

Another issue is how to formalize what it means for an adversary to “recover a character of the plaintext.” What if an attacker correctly guesses, through sheer luck or external information, that the least significant digit of someone's salary is 0? Clearly that should not render an encryption scheme insecure, and so any viable definition must somehow rule out such behavior as being a successful attack.

- *The “right” answer: regardless of any information an attacker already has, a ciphertext should leak no additional information about the underlying plaintext.* This informal definition captures all the concerns outlined above. Note in particular that it does not try to define what information about the plaintext is “meaningful”; it simply requires that *no* information be leaked. This is important, as it means that a secure encryption scheme is suitable for all potential applications in which secrecy is required.

What is missing here is a precise, mathematical formulation of the definition. How should we capture an attacker's prior knowledge about the plaintext? And what does it mean to (not) leak information? We will return to these questions in the next two chapters; see especially Definitions 2.3 and 3.12.

Now that we have fixed a security *goal*, it remains to specify a *threat model*. This specifies what “power” the attacker is assumed to have, but does not place any restrictions on the adversary's *strategy*. This is an important distinction: we specify what we assume about the adversary's abilities, but we do *not* assume anything about *how it uses* those abilities. It is impossible to foresee what strategies might be used in an attack, and history has proven that attempts to do so are doomed to failure.

There are several plausible options for the threat model in the context of encryption; standard ones, in order of increasing power of the attacker, are:

- **Ciphertext-only attack:** This is the most basic attack, and refers to a scenario where the adversary just observes a ciphertext (or multiple ciphertexts) and attempts to determine information about the underlying plaintext (or plaintexts). This is the threat model we have been

implicitly assuming when discussing classical encryption schemes in the previous section.

- **Known-plaintext attack:** Here, the adversary is able to learn one or more plaintext/ciphertext pairs generated using some key. The aim of the adversary is then to deduce information about the underlying plaintext of some *other* ciphertext produced using the same key.

All the classical encryption schemes we have seen are trivial to break using a known-plaintext attack; we leave a demonstration as an exercise.

- **Chosen-plaintext attack:** In this attack, the adversary can obtain plaintext/ciphertext pairs (as above) for plaintexts *of its choice*.
- **Chosen-ciphertext attack:** The final type of attack is one where the adversary is additionally able to obtain (some information about) the *decryption* of ciphertexts of its choice, e.g., whether the decryption of some ciphertext chosen by the attacker yields a valid English message. The adversary’s aim, once again, is to learn information about the underlying plaintext of some *other* ciphertext (whose decryption the adversary is unable to obtain directly).

None of these threat models is inherently better than any other; the right one to use depends on the environment in which an encryption scheme is deployed.

The first two types of attack are the easiest to carry out. In a ciphertext-only attack, the only thing the adversary needs to do is eavesdrop on the public communication channel over which encrypted messages are sent. In a known-plaintext attack it is assumed that the adversary somehow also obtains ciphertexts corresponding to known plaintexts. This is often easy to accomplish because not all encrypted messages are confidential, at least not indefinitely. As a trivial example, two parties may always encrypt a “hello” message whenever they begin communicating. As a more complex example, encryption may be used to keep quarterly-earnings reports secret until their release date; in this case, anyone eavesdropping on the ciphertext will later obtain the corresponding plaintext.

In the latter two attacks the adversary is assumed to be able to obtain encryptions and/or decryptions of plaintexts/ciphertexts of its choice. This may at first seem strange, and we defer a more detailed discussion of these attacks, and their practicality, to Section 3.4.2 (for chosen-plaintext attacks) and Section 3.7 (for chosen-ciphertext attacks).

1.4.2 Principle 2 – Precise Assumptions

Most modern cryptographic constructions cannot be proven secure unconditionally; such proofs would require resolving questions in the theory of computational complexity that seem far from being answered today. The result of

this unfortunate state of affairs is that proofs of security typically rely on *assumptions*. Modern cryptography requires any such assumptions to be made explicit and mathematically precise. At the most basic level, this is simply because mathematical proofs of security require this. But there are other reasons as well:

1. *Validation of assumptions:* By their very nature, assumptions are statements that are not proven but are instead conjectured to be true. In order to strengthen our belief in some assumption, it is necessary for the assumption to be studied. The more the assumption is examined and tested without being refuted, the more confident we are that the assumption is true. Furthermore, study of an assumption can provide evidence of its validity by showing that it is implied by some other assumption that is also widely believed.

If the assumption being relied upon is not precisely stated, it cannot be studied and (potentially) refuted. Thus, a pre-condition to increasing our confidence in an assumption is having a precise statement of what exactly is being assumed.

2. *Comparison of schemes:* Often in cryptography we are presented with two schemes that can both be proven to satisfy some definition, each based on a different assumption. Assuming all else is equal, which scheme should be preferred? If the assumption on which the first scheme is based is *weaker* than the assumption on which the second scheme is based (i.e., the second assumption implies the first), then the first scheme is preferable since it may turn out that the second assumption is false while the first assumption is true. If the assumptions used by the two schemes are not comparable, then the general rule is to prefer the scheme that is based on the better-studied assumption in which there is greater confidence.
3. *Understanding the necessary assumptions:* An encryption scheme may be based on some underlying building block. If some weaknesses are later found in the building block, how can we tell whether the encryption scheme is still secure? If the underlying assumptions regarding the building block are made clear as part of proving security of the scheme, then we need only check whether the required assumptions are affected by the new weaknesses that were found.

A question that sometimes arises is: rather than prove a scheme secure based on some other assumption, why not simply assume that the construction *itself* is secure? In some cases—e.g., when a scheme has successfully resisted attack for many years—this may be a reasonable approach. But this approach is never preferred, and is downright dangerous when a new scheme is being introduced. The reasons above help explain why. First, an assumption that has been tested for several years is preferable to a new, ad hoc assumption

that is introduced along with a new construction. Second, there is a general preference for assumptions that are simpler to state, since such assumptions are easier to study and to (potentially) refute. So, for example, an assumption that some mathematical problem is hard to solve is simpler to study and evaluate than the assumption that an encryption scheme satisfies a complex security definition. Another advantage of relying on “lower-level” assumptions (rather than just assuming a construction is secure) is that these low-level assumptions can typically be used in other constructions. Finally, low-level assumptions can provide *modularity*. Consider an encryption scheme whose security relies on some assumed property of one of its building blocks. If the underlying building block turns out *not* to satisfy the stated assumption, the encryption scheme can still be instantiated using a different component that is believed to satisfy the necessary requirements.

1.4.3 Principle 3 – Proofs of Security

The two principles described above allow us to achieve our goal of providing a rigorous *proof* that a construction satisfies a given definition under certain specified assumptions. Such proofs are especially important in the context of cryptography where there is an attacker who is actively trying to “break” some scheme. Proofs of security give an iron-clad guarantee—relative to the definition and assumptions—that no attacker will succeed; this is much better than taking an unprincipled or heuristic approach to the problem. Without a proof that no adversary with the specified resources can break some scheme, we are left only with our intuition that this is the case. Experience has shown that intuition in cryptography and computer security is disastrous. There are countless examples of unproven schemes that were broken, sometimes immediately and sometimes years after being developed.

Summary: Rigorous vs. Ad Hoc Approaches to Security

Reliance on definitions, assumptions, and proofs constitutes a rigorous approach to cryptography that is distinct from the informal approach of classical cryptography. Unfortunately, unprincipled, “off-the-cuff” solutions are still designed and deployed by those wishing to obtain a quick solution to a problem, or by those who are simply unknowledgable. We hope this book will contribute to an awareness of the rigorous approach and its importance in developing provably secure schemes.

1.4.4 Provable Security and Real-World Security

Much of modern cryptography now rests on sound mathematical foundations. But this does not mean that the field is no longer partly an *art* as well. The rigorous approach leaves room for creativity in developing definitions suited to contemporary applications and environments, in proposing new

mathematical assumptions or designing new primitives, and in constructing novel schemes and proving them secure. There will also, of course, always be the art of *attacking* deployed cryptosystems, even if they are proven secure. We expand on this point next.

The approach taken by modern cryptography has revolutionized the field, and helps provide confidence in the security of cryptographic schemes deployed in the real world. But it is important not to overstate what a proof of security implies. A proof of security is always relative to the *definition* being considered and the *assumption(s)* being used. If the security guarantee does not match what is needed, or the threat model does not capture the adversary's true abilities, then the proof may be irrelevant. Similarly, if the assumption that is relied upon turns out to be false, then the proof of security is meaningless.

The take-away point is that provable security of a scheme does not necessarily imply security of that scheme in the real world.⁴ While some have viewed this as a drawback of provable security, we view this optimistically as illustrating the *strength* of the approach. To attack a provably secure scheme in the real world, it suffices to focus attention on the definition (i.e., to explore how the idealized definition differs from the real-world environment in which the scheme is deployed) or the underlying assumptions (i.e., to see whether they hold). In turn, it is the job of cryptographers to continually refine their definitions to more closely match the real world, and to investigate their assumptions to test their validity. Provable security does not end the age-old battle between attacker and defender, but it does provide a framework that helps shift the odds in the defender's favor.

References and Additional Reading

In this chapter, we have studied just a few of the known historical ciphers. There are many others of both historical and mathematical interest, and we refer the reader to textbooks by Stinson [168] or Trappe and Washington [169] for further details. The important role cryptography has played throughout history is a fascinating subject covered in books by Kahn [97] and Singh [163].

Kerckhoffs' principles were elucidated in [103, 104]. Shannon [154] was the first to pursue a rigorous approach to cryptography based on precise definitions and mathematical proofs; we explore his work in the next chapter.

⁴Here we are not even considering the possibility of an incorrect *implementation* of the scheme. Poorly implemented cryptography is a serious problem in the real world, but this problem is somewhat outside the scope of cryptography *per se*.

Exercises

- 1.1 Decrypt the ciphertext provided at the end of the section on mono-alphabetic substitution ciphers.
- 1.2 Provide a formal definition of the **Gen**, **Enc**, and **Dec** algorithms for the mono-alphabetic substitution cipher.
- 1.3 Provide a formal definition of the **Gen**, **Enc**, and **Dec** algorithms for the Vigenère cipher. (Note: there are several plausible choices for **Gen**; choose one.)
- 1.4 Implement the attacks described in this chapter for the shift cipher and the Vigenère cipher.
- 1.5 Show that the shift, substitution, and Vigenère ciphers are all trivial to break using a chosen-plaintext attack. How much chosen plaintext is needed to recover the key for each of the ciphers?
- 1.6 Assume an attacker knows that a user's password is either **abcd** or **bedg**. Say the user encrypts his password using the shift cipher, and the attacker sees the resulting ciphertext. Show how the attacker can determine the user's password, or explain why this is not possible.
- 1.7 Repeat the previous exercise for the Vigenère cipher using period 2, using period 3, and using period 4.
- 1.8 The shift, substitution, and Vigenère ciphers can also be defined over the 128-character ASCII alphabet (rather than the 26-character English alphabet).
 - (a) Provide a formal definition of each of these schemes in this case.
 - (b) Discuss how the attacks we have shown in this chapter can be modified to break each of these modified schemes.

Chapter 2

Perfectly Secret Encryption

In the previous chapter we presented historical encryption schemes and showed how they can be broken with little computational effort. In this chapter, we look at the other extreme and study encryption schemes that are *provably* secure even against an adversary with unbounded computational power. Such schemes are called *perfectly secret*. Besides rigorously defining the notion, we will explore conditions under which perfect secrecy can be achieved. (Beginning in this chapter, we assume familiarity with basic probability theory. The relevant notions are reviewed in Appendix A.3.)

The material in this chapter belongs, in some sense, more to the world of “classical” cryptography than to the world of “modern” cryptography. Besides the fact that all the material introduced here was developed before the revolution in cryptography that took place in the mid-1970s and 1980s, the constructions we study in this chapter rely only on the first and third principles outlined in Section 1.4. That is, precise mathematical definitions are used and rigorous proofs are given, but it will not be necessary to rely on any unproven computational assumptions. It is clearly advantageous to avoid such assumptions; we will see, however, that doing so has inherent limitations. Thus, in addition to serving as a good basis for understanding the principles underlying modern cryptography, the results of this chapter also justify our later adoption of all three of the aforementioned principles.

Beginning with this chapter, we will define security and analyze schemes using probabilistic experiments involving algorithms making randomized choices; a basic example is given by communicating parties’ choosing a random key. Thus, before returning to the subject of cryptography *per se*, we briefly discuss the issue of generating randomness suitable for cryptographic applications.

Generating randomness. Throughout the book, we will simply assume that parties have access to an unlimited supply of independent, unbiased random bits. In practice, where do these random bits come from? In principle, one could generate a small number of random bits by hand, e.g., by flipping a fair coin. But such an approach is not very convenient, nor does it scale.

Modern *random-number generation* proceeds in two steps. First, a “pool” of high-entropy data is collected. (For our purposes a formal definition of entropy is not needed, and it suffices to think of entropy as a measure of unpredictability.) Next, this high-entropy data is processed to yield a sequence of nearly independent and unbiased bits. This second step is necessary since high-entropy data is not necessarily uniform.

For the first step, some source of unpredictable data is needed. There are several ways such data can be acquired. One technique is to rely on external inputs, for example, delays between network events, hard-disk access times, keystrokes or mouse movements made by the user, and so on. Such data is likely to be far from uniform, but if enough measurements are taken the resulting pool of data is expected to have sufficient entropy. More sophisticated approaches—which, by design, incorporate random-number generation more tightly into the system at the hardware level—have also been used. These rely on physical phenomena such as thermal/shot noise or radioactive decay. Intel has recently developed a processor that includes a digital random-number generator on the processor chip and provides a dedicated instruction for accessing the resulting random bits (after they have been suitably processed to yield independent, unbiased bits, as discussed next).

The processing needed to “smooth” the high-entropy data to obtain (nearly) uniform bits is a non-trivial one, and is discussed briefly in Section 5.6.4. Here, we just give a simple example to give an idea of what is done. Imagine that our high-entropy pool results from a sequence of *biased* coin flips, where “heads” occurs with probability p and “tails” with probability $1 - p$. (We do assume, however, that the result of any coin flip is *independent* of all other coin flips. In practice this assumption is typically not valid.) The result of 1,000 such coin flips certainly has high entropy, but is not close to uniform. We can obtain a uniform distribution by considering the coin flips in pairs: if we see a head followed by a tail then we output “0,” and if we see a tail followed by a head then we output “1.” (If we see two heads or two tails in a row, we output nothing, and simply move on to the next pair.) The probability that any pair results in a “0” is $p \cdot (1 - p)$, which is exactly equal to the probability that any pair results in a “1,” and we thus obtain a uniformly distributed output from our initial high-entropy pool.

Care must be taken in how random bits are produced, and using poor random-number generators can often leave a good cryptosystem vulnerable to attack. One should use a random-number generator that is *designed for cryptographic use*, rather than a “general-purpose” random-number generator, which is not suitable for cryptographic applications. In particular, the `rand()` function in the C `stdlib.h` library is *not* cryptographically secure, and using it in cryptographic settings can have disastrous consequences.

2.1 Definitions

We begin by recalling and expanding upon the syntax that was introduced in the previous chapter. An encryption scheme is defined by three algorithms `Gen`, `Enc`, and `Dec`, as well as a specification of a (finite) *message space* $\mathcal{M}$

with $|\mathcal{M}| > 1$.¹ The key-generation algorithm **Gen** is a probabilistic algorithm that outputs a key k chosen according to some distribution. We denote by $\mathcal{K}$ the (finite) *key space*, i.e., the set of all possible keys that can be output by **Gen**. The encryption algorithm **Enc** takes as input a key $k \in \mathcal{K}$ and a message $m \in \mathcal{M}$, and outputs a ciphertext c . We now allow the encryption algorithm to be probabilistic (so $\text{Enc}_k(m)$ might output a different ciphertext when run multiple times), and we write $c \leftarrow \text{Enc}_k(m)$ to denote the possibly probabilistic process by which message m is encrypted using key k to give ciphertext c . (In case **Enc** is deterministic, we may emphasize this by writing $c := \text{Enc}_k(m)$). Looking ahead, we also sometimes use the notation $x \leftarrow S$ to denote uniform selection of x from a set S .) We let $\mathcal{C}$ denote the set of all possible ciphertexts that can be output by $\text{Enc}_k(m)$, for all possible choices of $k \in \mathcal{K}$ and $m \in \mathcal{M}$ (and for all random choices of **Enc** in case it is randomized). The decryption algorithm **Dec** takes as input a key $k \in \mathcal{K}$ and a ciphertext $c \in \mathcal{C}$ and outputs a message $m \in \mathcal{M}$. We assume *perfect correctness*, meaning that for all $k \in \mathcal{K}$, $m \in \mathcal{M}$, and any ciphertext c output by $\text{Enc}_k(m)$, it holds that $\text{Dec}_k(c) = m$ with probability 1. Perfect correctness implies that we may assume **Dec** is deterministic without loss of generality, since $\text{Dec}_k(c)$ must give the same output every time it is run. We will thus write $m := \text{Dec}_k(c)$ to denote the process of decrypting ciphertext c using key k to yield the message m .

In the definitions and theorems below, we refer to probability distributions over $\mathcal{K}$, $\mathcal{M}$, and $\mathcal{C}$. The distribution over $\mathcal{K}$ is the one defined by running **Gen** and taking the output. (It is almost always the case that **Gen** chooses a key uniformly from $\mathcal{K}$ and, in fact, we may assume this without loss of generality; see Exercise 2.1.) We let K be a random variable denoting the value of the key output by **Gen**; thus, for any $k \in \mathcal{K}$, $\Pr[K = k]$ denotes the probability that the key output by **Gen** is equal to k . Similarly, we let M be a random variable denoting the message being encrypted, so $\Pr[M = m]$ denotes the probability that the message takes on the value $m \in \mathcal{M}$. The probability distribution of the message is not determined by the encryption scheme itself, but instead reflects the likelihood of different messages being sent by the parties using the scheme, as well as an adversary's uncertainty about what will be sent. As an example, an adversary may know that the message will either be **attack today** or **don't attack**. The adversary may even know (by other means) that with probability 0.7 the message will be a command to attack and with probability 0.3 the message will be a command not to attack. In this case, we have $\Pr[M = \text{attack today}] = 0.7$ and $\Pr[M = \text{don't attack}] = 0.3$.

K and M are assumed to be independent, i.e., what is being communicated by the parties is independent of the key they happen to share. This makes sense, among other reasons, because the distribution over $\mathcal{K}$ is determined by

¹If $|\mathcal{M}| = 1$ there is only one message and no point in communicating, let alone encrypting.

the encryption scheme itself (since it is defined by Gen), while the distribution over $\mathcal{M}$ depends on the context in which the encryption scheme is being used.

Fixing an encryption scheme and a distribution over $\mathcal{M}$ determines a distribution over the space of ciphertexts $\mathcal{C}$ given by choosing a key $k \in \mathcal{K}$ (according to Gen) and a message $m \in \mathcal{M}$ (according to the given distribution), and then computing the ciphertext $c \leftarrow \text{Enc}_k(m)$. We let C be the random variable denoting the resulting ciphertext and so, for $c \in \mathcal{C}$, write $\Pr[C = c]$ to denote the probability that the ciphertext is equal to the fixed value c .

Example 2.1

We work through a simple example for the shift cipher (cf. Section 1.3). Here, by definition, we have $\mathcal{K} = \{0, \dots, 25\}$ with $\Pr[K = k] = 1/26$ for each $k \in \mathcal{K}$.

Say we are given the following distribution over $\mathcal{M}$:

$$\Pr[M = \mathbf{a}] = 0.7 \quad \text{and} \quad \Pr[M = \mathbf{z}] = 0.3.$$

What is the probability that the ciphertext is B? There are only two ways this can occur: either $M = \mathbf{a}$ and $K = 1$, or $M = \mathbf{z}$ and $K = 2$. By independence of M and K , we have

$$\begin{aligned} \Pr[M = \mathbf{a} \wedge K = 1] &= \Pr[M = \mathbf{a}] \cdot \Pr[K = 1] \\ &= 0.7 \cdot \left(\frac{1}{26}\right). \end{aligned}$$

Similarly, $\Pr[M = \mathbf{z} \wedge K = 2] = 0.3 \cdot \left(\frac{1}{26}\right)$. Therefore,

$$\begin{aligned} \Pr[C = \mathbf{B}] &= \Pr[M = \mathbf{a} \wedge K = 1] + \Pr[M = \mathbf{z} \wedge K = 2] \\ &= 0.7 \cdot \left(\frac{1}{26}\right) + 0.3 \cdot \left(\frac{1}{26}\right) = 1/26. \end{aligned}$$

We can calculate conditional probabilities as well. For example, what is the probability that the message $\mathbf{a}$ was encrypted, given that we observe ciphertext B? Using Bayes' Theorem (Theorem A.8) we have

$$\begin{aligned} \Pr[M = \mathbf{a} \mid C = \mathbf{B}] &= \frac{\Pr[C = \mathbf{B} \mid M = \mathbf{a}] \cdot \Pr[M = \mathbf{a}]}{\Pr[C = \mathbf{B}]} \\ &= \frac{0.7 \cdot \Pr[C = \mathbf{B} \mid M = \mathbf{a}]}{1/26}. \end{aligned}$$

Note that $\Pr[C = \mathbf{B} \mid M = \mathbf{a}] = 1/26$, since if $M = \mathbf{a}$ then the only way $C = \mathbf{B}$ can occur is if $K = 1$ (which occurs with probability $1/26$). We conclude that $\Pr[M = \mathbf{a} \mid C = \mathbf{B}] = 0.7$. $\diamond$

Example 2.2

Consider the shift cipher again, but with the following distribution over $\mathcal{M}$:

$$\Pr[M = \mathbf{kim}] = 0.5, \Pr[M = \mathbf{ann}] = 0.2, \Pr[M = \mathbf{boo}] = 0.3.$$

What is the probability that $C = \text{DQQ}$? The only way this ciphertext can occur is if $M = \text{ann}$ and $K = 3$, or $M = \text{boo}$ and $K = 2$, which happens with probability $0.2 \cdot 1/26 + 0.3 \cdot 1/26 = 1/52$.

So what is the probability that ann was encrypted, conditioned on observing the ciphertext DQQ ? A calculation as above using Bayes' Theorem gives $\Pr[M = \text{ann} \mid C = \text{DQQ}] = 0.4$. $\diamond$

Perfect secrecy. We are now ready to define the notion of *perfect secrecy*. We imagine an adversary who knows the probability distribution over $\mathcal{M}$; that is, the adversary knows the likelihood that different messages will be sent. This adversary also knows the encryption scheme being used; the only thing unknown to the adversary is the key shared by the parties. A message is chosen by one of the honest parties and encrypted, and the resulting ciphertext transmitted to the other party. The adversary can *eavesdrop* on the parties' communication, and thus observe this ciphertext. (That is, this is a ciphertext-only attack, where the attacker gets only a single ciphertext.) For a scheme to be perfectly secret, observing this ciphertext should have *no effect* on the adversary's knowledge regarding the actual message that was sent; in other words, the *a posteriori* probability that some message $m \in \mathcal{M}$ was sent, conditioned on the ciphertext that was observed, should be no different from the *a priori* probability that m would be sent. This means that the ciphertext reveals nothing about the underlying plaintext, and the adversary learns absolutely nothing about the plaintext that was encrypted. Formally:

DEFINITION 2.3 *An encryption scheme $(\text{Gen}, \text{Enc}, \text{Dec})$ with message space $\mathcal{M}$ is perfectly secret if for every probability distribution over $\mathcal{M}$, every message $m \in \mathcal{M}$, and every ciphertext $c \in \mathcal{C}$ for which $\Pr[C = c] > 0$:*

$$\Pr[M = m \mid C = c] = \Pr[M = m].$$

(The requirement that $\Pr[C = c] > 0$ is a technical one needed to prevent conditioning on a zero-probability event.)

We now give an equivalent formulation of perfect secrecy. Informally, this formulation requires that the probability distribution of the ciphertext does not depend on the plaintext, i.e., for any two messages $m, m' \in \mathcal{M}$ the distribution of the ciphertext when m is encrypted should be identical to the distribution of the ciphertext when m' is encrypted. Formally, for every $m, m' \in \mathcal{M}$, and every $c \in \mathcal{C}$,

$$\Pr[\text{Enc}_K(m) = c] = \Pr[\text{Enc}_K(m') = c] \quad (2.1)$$

(where the probabilities are over choice of K and any randomness of Enc). This implies that the ciphertext contains no information about the plaintext, and that it is impossible to distinguish an encryption of m from an encryption of m' , since the distributions over the ciphertext are the same in each case.

LEMMA 2.4 *An encryption scheme $(\text{Gen}, \text{Enc}, \text{Dec})$ with message space $\mathcal{M}$ is perfectly secret if and only if Equation (2.1) holds for every $m, m' \in \mathcal{M}$ and every $c \in \mathcal{C}$.*

PROOF We show that if the stated condition holds, then the scheme is perfectly secret; the converse implication is left to Exercise 2.4. Fix a distribution over $\mathcal{M}$, a message m , and a ciphertext c for which $\Pr[C = c] > 0$. If $\Pr[M = m] = 0$ then we trivially have

$$\Pr[M = m \mid C = c] = 0 = \Pr[M = m].$$

So, assume $\Pr[M = m] > 0$. Notice first that

$$\Pr[C = c \mid M = m] = \Pr[\text{Enc}_K(M) = c \mid M = m] = \Pr[\text{Enc}_K(m) = c],$$

where the first equality is by definition of the random variable C , and the second is because we condition on the event that M is equal to m . Set $\delta_c \stackrel{\text{def}}{=} \Pr[\text{Enc}_K(m) = c] = \Pr[C = c \mid M = m]$. If the condition of the lemma holds, then for *every* $m' \in \mathcal{M}$ we have $\Pr[\text{Enc}_K(m') = c] = \Pr[C = c \mid M = m'] = \delta_c$. Using Bayes' Theorem (see Appendix A.3), we thus have

$$\begin{aligned} \Pr[M = m \mid C = c] &= \frac{\Pr[C = c \mid M = m] \cdot \Pr[M = m]}{\Pr[C = c]} \\ &= \frac{\Pr[C = c \mid M = m] \cdot \Pr[M = m]}{\sum_{m' \in \mathcal{M}} \Pr[C = c \mid M = m'] \cdot \Pr[M = m']} \\ &= \frac{\delta_c \cdot \Pr[M = m]}{\sum_{m' \in \mathcal{M}} \delta_c \cdot \Pr[M = m']} \\ &= \frac{\Pr[M = m]}{\sum_{m' \in \mathcal{M}} \Pr[M = m']} = \Pr[M = m], \end{aligned}$$

where the summation is over $m' \in \mathcal{M}$ with $\Pr[M = m'] \neq 0$. We conclude that for every $m \in \mathcal{M}$ and $c \in \mathcal{C}$ for which $\Pr[C = c] > 0$, it holds that $\Pr[M = m \mid C = c] = \Pr[M = m]$, and so the scheme is perfectly secret. ■

Perfect (adversarial) indistinguishability. We conclude this section by presenting another equivalent definition of perfect secrecy. This definition is based on an *experiment* involving an adversary passively observing a ciphertext and then trying to guess which of two possible messages was encrypted. We introduce this notion since it will serve as our starting point for defining computational security in the next chapter. Indeed, throughout the rest of the book we will often use experiments of this sort to define security.

In the present context, we consider the following experiment: an adversary $\mathcal{A}$ first specifies two arbitrary messages $m_0, m_1 \in \mathcal{M}$. One of these two

messages is chosen uniformly at random and encrypted using a random key; the resulting ciphertext is given to $\mathcal{A}$. Finally, $\mathcal{A}$ outputs a “guess” as to which of the two messages was encrypted; $\mathcal{A}$ *succeeds* if it guesses correctly. An encryption scheme is *perfectly indistinguishable* if *no* adversary $\mathcal{A}$ can succeed with probability better than $1/2$. (Note that, for any encryption scheme, $\mathcal{A}$ can succeed with probability $1/2$ by outputting a uniform guess; the requirement is simply that no attacker can do any better than this.) We stress that no limitations are placed on the computational power of $\mathcal{A}$.

Formally, let $\Pi = (\text{Gen}, \text{Enc}, \text{Dec})$ be an encryption scheme with message space $\mathcal{M}$. Let $\mathcal{A}$ be an adversary, which is formally just a (stateful) algorithm. We define an experiment $\text{PrivK}_{\mathcal{A}, \Pi}^{\text{eav}}$ as follows:

The adversarial indistinguishability experiment $\text{PrivK}_{\mathcal{A}, \Pi}^{\text{eav}}$:

1. The adversary $\mathcal{A}$ outputs a pair of messages $m_0, m_1 \in \mathcal{M}$.
2. A key k is generated using Gen , and a uniform bit $b \in \{0, 1\}$ is chosen. Ciphertext $c \leftarrow \text{Enc}_k(m_b)$ is computed and given to $\mathcal{A}$. We refer to c as the **challenge ciphertext**.
3. $\mathcal{A}$ outputs a bit b' .
4. The output of the experiment is defined to be 1 if $b' = b$, and 0 otherwise. We write $\text{PrivK}_{\mathcal{A}, \Pi}^{\text{eav}} = 1$ if the output of the experiment is 1 and in this case we say that $\mathcal{A}$ **succeeds**.

As noted earlier, it is trivial for $\mathcal{A}$ to succeed with probability $1/2$ by outputting a random guess. Perfect indistinguishability requires that it is impossible for any $\mathcal{A}$ to do better.

DEFINITION 2.5 *Encryption scheme $\Pi = (\text{Gen}, \text{Enc}, \text{Dec})$ with message space $\mathcal{M}$ is perfectly indistinguishable if for every $\mathcal{A}$ it holds that*

$$\Pr [\text{PrivK}_{\mathcal{A}, \Pi}^{\text{eav}} = 1] = \frac{1}{2}.$$

The following lemma states that Definition 2.5 is equivalent to Definition 2.3. We leave the proof of the lemma as Exercise 2.5.

LEMMA 2.6 *Encryption scheme Π is perfectly secret if and only if it is perfectly indistinguishable.*

Example 2.7

We show that the Vigenère cipher is *not* perfectly indistinguishable, at least for certain parameters. Concretely, let Π denote the Vigenère cipher for the message space of two-character strings, and where the period is chosen uniformly in $\{1, 2\}$. To show that Π is not perfectly indistinguishable, we exhibit an adversary $\mathcal{A}$ for which $\Pr [\text{PrivK}_{\mathcal{A}, \Pi}^{\text{eav}} = 1] > \frac{1}{2}$.

Adversary $\mathcal{A}$ does:

1. Output $m_0 = \mathbf{aa}$ and $m_1 = \mathbf{ab}$.
2. Upon receiving the challenge ciphertext $c = c_1c_2$, do the following: if $c_1 = c_2$ output 0; else output 1.

Computation of $\Pr [\text{PrivK}_{\mathcal{A},\Pi}^{\text{eav}} = 1]$ is tedious but straightforward.

$$\begin{aligned} \Pr [\text{PrivK}_{\mathcal{A},\Pi}^{\text{eav}} = 1] &= \frac{1}{2} \cdot \Pr [\text{PrivK}_{\mathcal{A},\Pi}^{\text{eav}} = 1 \mid b = 0] + \frac{1}{2} \cdot \Pr [\text{PrivK}_{\mathcal{A},\Pi}^{\text{eav}} = 1 \mid b = 1] \\ &= \frac{1}{2} \cdot \Pr[\mathcal{A} \text{ outputs } 0 \mid b = 0] + \frac{1}{2} \cdot \Pr[\mathcal{A} \text{ outputs } 1 \mid b = 1], \end{aligned} \quad (2.2)$$

where b is the uniform bit determining which message gets encrypted. $\mathcal{A}$ outputs 0 if and only if the two characters of the ciphertext $c = c_1c_2$ are equal. When $b = 0$ (so $m_0 = \mathbf{aa}$ is encrypted) then $c_1 = c_2$ if either (1) a key of period 1 is chosen, or (2) a key of period 2 is chosen, and both characters of the key are equal. The former occurs with probability $\frac{1}{2}$, and the latter occurs with probability $\frac{1}{2} \cdot \frac{1}{26}$. So

$$\Pr[\mathcal{A} \text{ outputs } 0 \mid b = 0] = \frac{1}{2} + \frac{1}{2} \cdot \frac{1}{26} \approx 0.52.$$

When $b = 1$ then $c_1 = c_2$ only if a key of period 2 is chosen and the first character of the key is one more than the second character of the key, which happens with probability $\frac{1}{2} \cdot \frac{1}{26}$. So

$$\Pr[\mathcal{A} \text{ outputs } 1 \mid b = 1] = 1 - \Pr[\mathcal{A} \text{ outputs } 0 \mid b = 1] = 1 - \frac{1}{2} \cdot \frac{1}{26} \approx 0.98.$$

Plugging into Equation (2.2) then gives

$$\Pr [\text{PrivK}_{\mathcal{A},\Pi}^{\text{eav}} = 1] = \frac{1}{2} \cdot \left(\frac{1}{2} + \frac{1}{2} \cdot \frac{1}{26} + 1 - \frac{1}{2} \cdot \frac{1}{26} \right) = 0.75 > \frac{1}{2},$$

and the scheme is not perfectly indistinguishable. $\diamond$

2.2 The One-Time Pad

In 1917, Vernam patented a perfectly secret encryption scheme now called the *one-time pad*. At the time Vernam proposed the scheme, there was no proof that it was perfectly secret; in fact, there was not yet a notion of what perfect secrecy was. Approximately 25 years later, however, Shannon introduced the definition of perfect secrecy and demonstrated that the one-time pad achieves that level of security.

CONSTRUCTION 2.8

Fix an integer $\ell > 0$. The message space $\mathcal{M}$, key space $\mathcal{K}$, and ciphertext space $\mathcal{C}$ are all equal to $\{0, 1\}^\ell$ (the set of all binary strings of length ℓ).

- **Gen:** the key-generation algorithm chooses a key from $\mathcal{K} = \{0, 1\}^\ell$ according to the uniform distribution (i.e., each of the 2^ℓ strings in the space is chosen as the key with probability exactly $2^{-\ell}$).
- **Enc:** given a key $k \in \{0, 1\}^\ell$ and a message $m \in \{0, 1\}^\ell$, the encryption algorithm outputs the ciphertext $c := k \oplus m$.
- **Dec:** given a key $k \in \{0, 1\}^\ell$ and a ciphertext $c \in \{0, 1\}^\ell$, the decryption algorithm outputs the message $m := k \oplus c$.

The one-time pad encryption scheme.

In describing the scheme we let $a \oplus b$ denote the *bitwise exclusive-or* (XOR) of two binary strings a and b (i.e., if $a = a_1 \cdots a_\ell$ and $b = b_1 \cdots b_\ell$ are ℓ -bit strings, then $a \oplus b$ is the ℓ -bit string given by $a_1 \oplus b_1 \cdots a_\ell \oplus b_\ell$). In the one-time pad encryption scheme the key is a uniform string of the same length as the message; the ciphertext is computed by simply XORing the key and the message. A formal definition is given as Construction 2.8. Before discussing security, we first verify correctness: for every key k and every message m it holds that $\text{Dec}_k(\text{Enc}_k(m)) = k \oplus k \oplus m = m$, and so the one-time pad constitutes a valid encryption scheme.

One can easily prove perfect secrecy of the one-time pad using Lemma 2.4 and the fact that the ciphertext is uniformly distributed regardless of what message is encrypted. We give a proof based directly on the original definition.

THEOREM 2.9 *The one-time pad encryption scheme is perfectly secret.*

PROOF We first compute $\Pr[C = c \mid M = m']$ for arbitrary $c \in \mathcal{C}$ and $m' \in \mathcal{M}$. For the one-time pad,

$$\begin{aligned} \Pr[C = c \mid M = m'] &= \Pr[\text{Enc}_K(m') = c] = \Pr[m' \oplus K = c] \\ &= \Pr[K = m' \oplus c] \\ &= 2^{-\ell}, \end{aligned}$$

where the final equality holds because the key K is a uniform ℓ -bit string. Fix any distribution over $\mathcal{M}$. For any $c \in \mathcal{C}$, we have

$$\begin{aligned} \Pr[C = c] &= \sum_{m' \in \mathcal{M}} \Pr[C = c \mid M = m'] \cdot \Pr[M = m'] \\ &= 2^{-\ell} \cdot \sum_{m' \in \mathcal{M}} \Pr[M = m'] \\ &= 2^{-\ell}, \end{aligned}$$

where the sum is over $m' \in \mathcal{M}$ with $\Pr[M = m'] \neq 0$. Bayes' Theorem gives:

$$\begin{aligned}\Pr[M = m \mid C = c] &= \frac{\Pr[C = c \mid M = m] \cdot \Pr[M = m]}{\Pr[C = c]} \\ &= \frac{2^{-\ell} \cdot \Pr[M = m]}{2^{-\ell}} \\ &= \Pr[M = m].\end{aligned}$$

We conclude that the one-time pad is perfectly secret. ■

The one-time pad was used by several national-intelligence agencies in the mid-20th century to encrypt sensitive traffic. Perhaps most famously, the “red phone” linking the White House and the Kremlin during the Cold War was protected using one-time pad encryption, where the governments of the US and USSR would exchange extremely long keys using trusted couriers carrying briefcases of paper on which random characters were written.

Notwithstanding the above, one-time pad encryption is rarely used any more due to a number of drawbacks it has. Most prominent is that *the key is as long as the message*.² This limits the usefulness of the scheme for sending very long messages (as it may be difficult to securely share and store a very long key), and is problematic when the parties cannot predict in advance (an upper bound on) how long the message will be.

Moreover, the one-time pad—as the name indicates—*is only secure if used once* (with the same key). Although we did not yet define a notion of secrecy when multiple messages are encrypted, it is easy to see that encrypting more than one message with the same key leaks a lot of information. In particular, say two messages m, m' are encrypted using the same key k . An adversary who obtains $c = m \oplus k$ and $c' = m' \oplus k$ can compute

$$c \oplus c' = (m \oplus k) \oplus (m' \oplus k) = m \oplus m'$$

and thus learn the exclusive-or of the two messages or, equivalently, exactly where the two messages differ. While this may not seem very significant, it is enough to rule out any claims of perfect secrecy for encrypting two messages using the same key. Moreover, if the messages correspond to natural-language text, then given the exclusive-or of two sufficiently long messages it is possible to perform frequency analysis (as in the previous chapter, though more complex) and recover the messages themselves. An interesting historical example of this is given by the *VENONA project*, as part of which the US and UK were able to decrypt ciphertexts sent by the Soviet Union that were mistakenly encrypted with repeated portions of a one-time pad over several decades.

²This does not make the one-time pad useless, since it may be easier for two parties to share a key at some point in time before the message to be communicated is known.

2.3 Limitations of Perfect Secrecy

We ended the previous section by noting some drawbacks of the one-time pad encryption scheme. Here, we show that these drawbacks are not specific to that scheme, but are instead *inherent* limitations of perfect secrecy. Specifically, we prove that *any* perfectly secret encryption scheme must have a key space that is at least as large as the message space. If all keys are the same length, and the message space consists of all strings of some fixed length, this implies that the key is at least as long as the message. In particular, the key length of the one-time pad is optimal. (The other limitation—namely, that the key can be used only once—is also inherent if perfect secrecy is required; see Exercise 2.13.)

THEOREM 2.10 *If $(\text{Gen}, \text{Enc}, \text{Dec})$ is a perfectly secret encryption scheme with message space $\mathcal{M}$ and key space $\mathcal{K}$, then $|\mathcal{K}| \geq |\mathcal{M}|$.*

PROOF We show that if $|\mathcal{K}| < |\mathcal{M}|$ then the scheme cannot be perfectly secret. Assume $|\mathcal{K}| < |\mathcal{M}|$. Consider the uniform distribution over $\mathcal{M}$ and let $c \in \mathcal{C}$ be a ciphertext that occurs with non-zero probability. Let $\mathcal{M}(c)$ be the set of all possible messages that are possible decryptions of c ; that is

$$\mathcal{M}(c) \stackrel{\text{def}}{=} \{m \mid m = \text{Dec}_k(c) \text{ for some } k \in \mathcal{K}\}.$$

Clearly $|\mathcal{M}(c)| \leq |\mathcal{K}|$. (Recall that we may assume Dec is deterministic.) If $|\mathcal{K}| < |\mathcal{M}|$, there is some $m' \in \mathcal{M}$ such that $m' \notin \mathcal{M}(c)$. But then

$$\Pr[M = m' \mid C = c] = 0 \neq \Pr[M = m'],$$

and so the scheme is not perfectly secret. ■

Perfect secrecy with shorter keys? The above theorem shows an inherent limitation of schemes that achieve perfect secrecy. Even so, individuals occasionally claim they have developed a radically new encryption scheme that is “unbreakable” and achieves the security of the one-time pad without using keys as long as what is being encrypted. The above proof demonstrates that such claims *cannot* be true; anyone making such claims either knows very little about cryptography or is blatantly lying.

2.4 *Shannon's Theorem

In his work on perfect secrecy, Shannon also provided a characterization of perfectly secret encryption schemes. This characterization says that, under certain conditions, the key-generation algorithm **Gen** must choose the key *uniformly* from the set of all possible keys (as in the one-time pad); moreover, for every message m and ciphertext c there is a *unique* key mapping m to c (again, as in the one-time pad). Beyond being interesting in its own right, this theorem is a useful tool for proving (or disproving) perfect secrecy of suggested schemes. We discuss this further after the proof.

The theorem as stated here assumes $|\mathcal{M}| = |\mathcal{K}| = |\mathcal{C}|$, meaning that the sets of plaintexts, keys, and ciphertexts all have the same size. We have already seen that for perfect secrecy we must have $|\mathcal{K}| \geq |\mathcal{M}|$. It is easy to see that correct decryption requires $|\mathcal{C}| \geq |\mathcal{M}|$. Therefore, in some sense, encryption schemes with $|\mathcal{M}| = |\mathcal{K}| = |\mathcal{C}|$ are “optimal.”

THEOREM 2.11 (Shannon's theorem) *Let $(\text{Gen}, \text{Enc}, \text{Dec})$ be an encryption scheme with message space $\mathcal{M}$, for which $|\mathcal{M}| = |\mathcal{K}| = |\mathcal{C}|$. The scheme is perfectly secret if and only if:*

1. *Every key $k \in \mathcal{K}$ is chosen with (equal) probability $1/|\mathcal{K}|$ by algorithm **Gen**.*
2. *For every $m \in \mathcal{M}$ and every $c \in \mathcal{C}$, there exists a unique key $k \in \mathcal{K}$ such that $\text{Enc}_k(m)$ outputs c .*

PROOF The intuition behind the proof is as follows. To see that the stated conditions imply perfect secrecy, note that condition 2 means that any ciphertext c could be the result of encrypting any possible plaintext m , because there is some key k mapping m to c . Since there is a *unique* such key, and each key is chosen with equal probability, perfect secrecy follows as for the one-time pad. For the other direction, perfect secrecy immediately implies that for every m and c there is at least one key mapping m to c . The fact that $|\mathcal{M}| = |\mathcal{K}| = |\mathcal{C}|$ means, moreover, that for every m and c there is exactly *one* such key. Given this, each key must be chosen with equal probability or else perfect secrecy would fail to hold. A formal proof follows.

We assume for simplicity that **Enc** is deterministic. (One can show that this is without loss of generality here.) We first prove that if the encryption scheme satisfies conditions 1 and 2, then it is perfectly secret. The proof is essentially the same as the proof of perfect secrecy for the one-time pad, so we will be relatively brief. Fix arbitrary $c \in \mathcal{C}$ and $m \in \mathcal{M}$. Let k be the unique key, guaranteed by condition 2, for which $\text{Enc}_k(m) = c$. Then,

$$\Pr[C = c \mid M = m] = \Pr[K = k] = 1/|\mathcal{K}|,$$

where the final equality holds by condition 1. So

$$\Pr[C = c] = \sum_{m \in \mathcal{M}} \Pr[\text{Enc}_K(m) = c] \cdot \Pr[M = m] = 1/|\mathcal{K}|.$$

This holds for any distribution over $\mathcal{M}$. Thus, for any distribution over $\mathcal{M}$, any $m \in \mathcal{M}$ with $\Pr[M = m] \neq 0$, and any $c \in \mathcal{C}$, we have:

$$\begin{aligned} \Pr[M = m \mid C = c] &= \frac{\Pr[C = c \mid M = m] \cdot \Pr[M = m]}{\Pr[C = c]} \\ &= \frac{\Pr[\text{Enc}_K(m) = c] \cdot \Pr[M = m]}{\Pr[C = c]} \\ &= \frac{|\mathcal{K}|^{-1} \cdot \Pr[M = m]}{|\mathcal{K}|^{-1}} = \Pr[M = m], \end{aligned}$$

and the scheme is perfectly secret.

For the second direction, assume the encryption scheme is perfectly secret; we show that conditions 1 and 2 hold. Fix arbitrary $c \in \mathcal{C}$. There must be some message m^* for which $\Pr[\text{Enc}_K(m^*) = c] \neq 0$. Lemma 2.4 then implies that $\Pr[\text{Enc}_K(m) = c] \neq 0$ for every $m \in \mathcal{M}$. In other words, if we let $\mathcal{M} = \{m_1, m_2, \dots\}$, then for each $m_i \in \mathcal{M}$ we have a nonempty set of keys $\mathcal{K}_i \subset \mathcal{K}$ such that $\text{Enc}_k(m_i) = c$ if and only if $k \in \mathcal{K}_i$. Moreover, when $i \neq j$ then $\mathcal{K}_i$ and $\mathcal{K}_j$ must be disjoint or else correctness fails to hold. Since $|\mathcal{K}| = |\mathcal{M}|$, we see that each $\mathcal{K}_i$ contains only a single key k_i , as required by condition 2. Now, Lemma 2.4 shows that for any $m_i, m_j \in \mathcal{M}$ we have

$$\Pr[K = k_i] = \Pr[\text{Enc}_K(m_i) = c] = \Pr[\text{Enc}_K(m_j) = c] = \Pr[K = k_j].$$

Since this holds for all $1 \leq i, j \leq |\mathcal{M}| = |\mathcal{K}|$, and $k_i \neq k_j$ for $i \neq j$, this means each key is chosen with probability $1/|\mathcal{K}|$, as required by condition 1. ■

Shannon's theorem is useful for deciding whether a given scheme is perfectly secret. Condition 1 is easy to check, and condition 2 can be demonstrated (or contradicted) without having to compute any probabilities (in contrast to working with Definition 2.3 directly). As an example, perfect secrecy of the one-time pad is trivial to prove using Shannon's theorem. We stress, however, that the theorem only applies when $|\mathcal{M}| = |\mathcal{K}| = |\mathcal{C}|$.

References and Additional Reading

The one-time pad is popularly credited to Vernam [172], who filed a patent on it, but recent historical research [25] shows that it was invented some

35 years earlier. Analysis of the one-time pad had to await the groundbreaking work of Shannon [154], who introduced the notion of perfect secrecy.

In this chapter we studied perfectly secret *encryption*. Some other cryptographic problems can also be solved with “perfect” security. A notable example is the problem of message authentication where the aim is to prevent an adversary from (undetectably) modifying a message sent from one party to another. We study this problem in depth in Chapter 4, discussing “perfectly secure” message authentication in Section 4.6.

Exercises

- 2.1 Prove that, by redefining the key space, we may assume that the key-generation algorithm **Gen** chooses a key uniformly at random from the key space, without changing $\Pr[C = c \mid M = m]$ for any m, c .

Hint: Define the key space to be the set of all possible random tapes for the randomized algorithm **Gen**.

- 2.2 Prove that, by redefining the key space, we may assume that **Enc** is deterministic without changing $\Pr[C = c \mid M = m]$ for any m, c .

- 2.3 Prove or refute: An encryption scheme with message space $\mathcal{M}$ is perfectly secret if and only if for every probability distribution over $\mathcal{M}$ and every $c_0, c_1 \in \mathcal{C}$ we have $\Pr[C = c_0] = \Pr[C = c_1]$.

- 2.4 Prove the second direction of Lemma 2.4.

- 2.5 Prove Lemma 2.6.

- 2.6 For each of the following encryption schemes, state whether the scheme is perfectly secret. Justify your answer in each case.

(a) The message space is $\mathcal{M} = \{0, \dots, 4\}$. Algorithm **Gen** chooses a uniform key from the key space $\{0, \dots, 5\}$. **Enc_k**(m) returns $[k + m \bmod 5]$, and **Dec_k**(c) returns $[c - k \bmod 5]$.

(b) The message space is $\mathcal{M} = \{m \in \{0, 1\}^\ell \mid \text{the last bit of } m \text{ is } 0\}$. **Gen** chooses a uniform key from $\{0, 1\}^{\ell-1}$. **Enc_k**(m) returns ciphertext $m \oplus (k\|0)$, and **Dec_k**(c) returns $c \oplus (k\|0)$.

- 2.7 When using the one-time pad with the key $k = 0^\ell$, we have **Enc_k**(m) = $k \oplus m = m$ and the message is sent in the clear! It has therefore been suggested to modify the one-time pad by only encrypting with $k \neq 0^\ell$ (i.e., to have **Gen** choose k uniformly from the set of *nonzero* keys of length ℓ). Is this modified scheme still perfectly secret? Explain.

2.8 Let Π denote the Vigenère cipher where the message space consists of all 3-character strings (over the English alphabet), and the key is generated by first choosing the period t uniformly from $\{1, 2, 3\}$ and then letting the key be a uniform string of length t .

- (a) Define $\mathcal{A}$ as follows: $\mathcal{A}$ outputs $m_0 = \text{aab}$ and $m_1 = \text{abb}$. When given a ciphertext c , it outputs 0 if the first character of c is the same as the second character of c , and outputs 1 otherwise. Compute $\Pr[\text{PrivK}_{\mathcal{A}, \Pi}^{\text{eav}} = 1]$.
- (b) Construct and analyze an adversary $\mathcal{A}'$ for which $\Pr[\text{PrivK}_{\mathcal{A}', \Pi}^{\text{eav}} = 1]$ is greater than your answer from part (a).

2.9 In this exercise, we look at different conditions under which the shift, mono-alphabetic substitution, and Vigenère ciphers are perfectly secret:

- (a) Prove that if only a single character is encrypted, then the shift cipher is perfectly secret.
- (b) What is the largest message space $\mathcal{M}$ for which the mono-alphabetic substitution cipher provides perfect secrecy?
- (c) Prove that the Vigenère cipher using (fixed) period t is perfectly secret when used to encrypt messages of length t .

Reconcile this with the attacks shown in the previous chapter.

2.10 Prove that a scheme satisfying Definition 2.5 must have $|\mathcal{K}| \geq |\mathcal{M}|$ without using Lemma 2.4. Specifically, let Π be an arbitrary encryption scheme with $|\mathcal{K}| < |\mathcal{M}|$. Show an $\mathcal{A}$ for which $\Pr[\text{PrivK}_{\mathcal{A}, \Pi}^{\text{eav}} = 1] > \frac{1}{2}$.

Hint: It may be easier to let $\mathcal{A}$ be randomized.

2.11 Assume we require only that an encryption scheme ($\text{Gen}, \text{Enc}, \text{Dec}$) with message space $\mathcal{M}$ satisfy the following: For all $m \in \mathcal{M}$, we have $\Pr[\text{Dec}_K(\text{Enc}_K(m)) = m] \geq 2^{-t}$. (This probability is taken over choice of the key as well as any randomness used during encryption.) Show that perfect secrecy can be achieved with $|\mathcal{K}| < |\mathcal{M}|$ when $t \geq 1$. Prove a lower bound on the size of $\mathcal{K}$ in terms of t .

2.12 Let $\varepsilon \geq 0$ be a constant. Say an encryption scheme is ε -perfectly secret if for every adversary $\mathcal{A}$ it holds that

$$\Pr[\text{PrivK}_{\mathcal{A}, \Pi}^{\text{eav}} = 1] \leq \frac{1}{2} + \varepsilon.$$

(Compare to Definition 2.5.) Show that ε -perfect secrecy can be achieved with $|\mathcal{K}| < |\mathcal{M}|$ when $\varepsilon > 0$. Prove a lower bound on the size of $\mathcal{K}$ in terms of ε .

2.13 In this problem we consider definitions of perfect secrecy for the encryption of *two* messages (using the same key). Here we consider distributions over *pairs* of messages from the message space $\mathcal{M}$; we let M_1, M_2 be random variables denoting the first and second message, respectively. (We stress that these random variables are not assumed to be independent.) We generate a (single) key k , sample a pair of messages (m_1, m_2) according to the given distribution, and then compute ciphertexts $c_1 \leftarrow \text{Enc}_k(m_1)$ and $c_2 \leftarrow \text{Enc}_k(m_2)$; this induces a distribution over pairs of ciphertexts and we let C_1, C_2 be the corresponding random variables.

- (a) Say encryption scheme $(\text{Gen}, \text{Enc}, \text{Dec})$ is *perfectly secret for two messages* if for all distributions over $\mathcal{M} \times \mathcal{M}$, all $m_1, m_2 \in \mathcal{M}$, and all ciphertexts $c_1, c_2 \in \mathcal{C}$ with $\Pr[C_1 = c_1 \wedge C_2 = c_2] > 0$:

$$\begin{aligned} \Pr[M_1 = m_1 \wedge M_2 = m_2 \mid C_1 = c_1 \wedge C_2 = c_2] \\ = \Pr[M_1 = m_1 \wedge M_2 = m_2]. \end{aligned}$$

Prove that *no* encryption scheme can satisfy this definition.

Hint: Take $c_1 = c_2$.

- (b) Say encryption scheme $(\text{Gen}, \text{Enc}, \text{Dec})$ is *perfectly secret for two distinct messages* if for all distributions over $\mathcal{M} \times \mathcal{M}$ where the first and second messages are guaranteed to be different (i.e., distributions over pairs of *distinct* messages), all $m_1, m_2 \in \mathcal{M}$, and all $c_1, c_2 \in \mathcal{C}$ with $\Pr[C_1 = c_1 \wedge C_2 = c_2] > 0$:

$$\begin{aligned} \Pr[M_1 = m_1 \wedge M_2 = m_2 \mid C_1 = c_1 \wedge C_2 = c_2] \\ = \Pr[M_1 = m_1 \wedge M_2 = m_2]. \end{aligned}$$

Show an encryption scheme that provably satisfies this definition.

Hint: The encryption scheme you propose need not be efficient, although an efficient solution is possible.

Part II

Private-Key (Symmetric) Cryptography

Chapter 3

Private-Key Encryption

In the previous chapter we saw some fundamental limitations of perfect secrecy. In this chapter we begin our study of modern cryptography by introducing the weaker (but sufficient) notion of *computational* secrecy. We will then show how this definition can be used to bypass the impossibility results shown previously and, in particular, how a short key (say, 128 bits long) can be used to encrypt many long messages (say, gigabytes in total).

Along the way we will study the fundamental notion of *pseudorandomness*, which captures the idea that something can “look” completely random even though it is not. This powerful concept underlies much of modern cryptography, and has applications and implications beyond the field as well.

3.1 Computational Security

In Chapter 2 we introduced the notion of perfect secrecy. While perfect secrecy is a worthwhile goal, it is also unnecessarily strong. Perfect secrecy requires that *absolutely no information* about an encrypted message is leaked, even to an eavesdropper *with unlimited computational power*. For all practical purposes, however, an encryption scheme would still be considered secure if it leaked only a *tiny* amount of information to eavesdroppers with *bounded computational power*. For example, a scheme that leaks information with probability at most 2^{-60} to eavesdroppers investing up to 200 years of computational effort on the fastest available supercomputer is adequate for any real-world application. Security definitions that take into account computational limits on the attacker, and allow for a small probability of failure, are called *computational*, to distinguish them from notions (like perfect secrecy) that are *information-theoretic* in nature. Computational security is now the *de facto* way in which security is defined for all cryptographic purposes.

We stress that although we give up on obtaining perfect security, this does not mean we do away with the rigorous mathematical approach. Definitions and proofs are still essential, and the only difference is that we now consider weaker (but still meaningful) definitions of security.

Computational security incorporates two relaxations relative to information-

theoretic notions of security (in the case of encryption, both these relaxations are necessary in order to go beyond the limitations of perfect secrecy discussed in the previous chapter):

1. *Security is only guaranteed against efficient adversaries that run for some feasible amount of time.* This means that given enough time (or sufficient computational resources) an attacker may be able to violate security. If we can make the resources required to break the scheme larger than those available to any realistic attacker, then for all practical purposes the scheme is unbreakable.
2. *Adversaries can potentially succeed (i.e., security can potentially fail) with some very small probability.* If we can make this probability sufficiently small, we need not worry about it.

To obtain a meaningful theory, we need to precisely define the above relaxations. There are two general approaches for doing so: the *concrete approach* and the *asymptotic approach*. These are described next.

3.1.1 The Concrete Approach

The concrete approach to computational security quantifies the security of a cryptographic scheme by explicitly bounding the maximum success probability of any (randomized) adversary running for some specified amount of time or, more precisely, investing some specific amount of computational effort. Thus, a concrete definition of security takes roughly the following form:

A scheme is (t, ε) -secure if any adversary running for time at most t succeeds in breaking the scheme with probability at most ε .

(Of course, the above serves only as a general template, and for the above statement to make sense we need to define exactly what it means to “break” the scheme in question.) As an example, one might have a scheme with the guarantee that no adversary running for at most 200 years using the fastest available supercomputer can succeed in breaking the scheme with probability better than 2^{-60} . Or, it may be more convenient to measure running time in terms of CPU cycles, and to construct a scheme such that no adversary using at most 2^{80} cycles can break the scheme with probability better than 2^{-60} .

It is instructive to get a feel for the large values of t and the small values of ε that are typical of modern cryptographic schemes.

Example 3.1

Modern private-key encryption schemes are generally assumed to give almost optimal security in the following sense: when the key has length n —and so the key space has size 2^n —an adversary running for time t (measured in, say, computer cycles) succeeds in breaking the scheme with probability at most

$ct/2^n$ for some fixed constant c . (This simply corresponds to a brute-force search of the key space, and assumes no preprocessing has been done.)

Assuming $c = 1$ for simplicity, a key of length $n = 60$ provides adequate security against an adversary using a desktop computer. Indeed, on a 4 GHz processor (that executes 4×10^9 cycles per second) 2^{60} CPU cycles require $2^{60}/(4 \times 10^9)$ seconds, or about 9 years. However, the fastest supercomputer at the time of this writing can execute roughly 2×10^{16} floating point operations per second, and 2^{60} such operations require only about 1 minute on such a machine. Taking $n = 80$ would be a more prudent choice; even the computer just mentioned would take about 2 years to carry out 2^{80} operations.

(The above numbers are for illustrative purposes only; in practice $c > 1$, and several other factors—such as the time required for memory access and the possibility of parallel computation on a network of computers—significantly affect the performance of brute-force attacks.)

Today, however, a recommended key length might be $n = 128$. The difference between 2^{80} and 2^{128} is a *multiplicative factor* of 2^{48} . To get a feeling for how big this is, note that according to physicists' estimates the number of seconds since the Big Bang is on the order of 2^{58} .

If the probability that an attacker can successfully recover an encrypted message in one year is at most 2^{-60} , then it is much more likely that the sender and receiver will both be hit by lightning in that same period of time. An event that occurs once every hundred years can be roughly estimated to occur with probability 2^{-30} in any given second. Something that occurs with probability 2^{-60} in any given second is 2^{30} times *less* likely, and might be expected to occur roughly once every 100 billion years. $\diamond$

The concrete approach is important in practice, since concrete guarantees are what users of a cryptographic scheme are ultimately interested in. However, precise concrete guarantees are difficult to provide. Furthermore, one must be careful in interpreting concrete security claims. For example, a claim that no adversary running for 5 years can break a given scheme with probability better than ε begs the questions: what type of computing power (e.g., desktop PC, supercomputer, network of hundreds of computers) does this assume? Does this take into account future advances in computing power (which, by Moore's Law, roughly doubles every 18 months)? Does the estimate assume the use of "off-the-shelf" algorithms, or dedicated software implementations optimized for the attack? Furthermore, such a guarantee says little about the success probability of an adversary running for 2 years (other than the fact that it can be at most ε) and says nothing about the success probability of an adversary running for 10 years.

3.1.2 The Asymptotic Approach

As partly noted above, there are some technical and theoretical difficulties in using the concrete-security approach. These issues must be dealt with in

practice, but when concrete security is not an immediate concern it is convenient instead to use an *asymptotic* approach to security; this is the approach taken in this book. This approach, rooted in complexity theory, introduces an integer-valued *security parameter* (denoted by n) that parameterizes both cryptographic schemes as well as all involved parties (namely, the honest parties as well as the attacker). When honest parties initialize a scheme (i.e., when they generate keys), they choose some value n for the security parameter; for the purposes of this discussion, one can think of the security parameter as corresponding to the length of the key. The security parameter is assumed to be known to any adversary attacking the scheme, and we now view the running time of the adversary, as well as its success probability, as functions of the security parameter rather than as concrete numbers. Then:

1. We equate “efficient adversaries” with randomized (i.e., probabilistic) algorithms running in time *polynomial in n* . This means there is some polynomial p such that the adversary runs for time at most $p(n)$ when the security parameter is n . We also require—for real-world efficiency—that honest parties run in polynomial time, although we stress that the adversary may be much more powerful (and run much longer than) the honest parties.
2. We equate the notion of “small probabilities of success” with success probabilities *smaller than any inverse polynomial in n* (see Definition 3.4). Such probabilities are called *negligible*.

Let PPT stand for “probabilistic polynomial-time.” A definition of asymptotic security then takes the following general form:

A scheme is secure if any PPT adversary succeeds in breaking the scheme with at most negligible probability.

This notion of security is *asymptotic* since security depends on the behavior of the scheme for sufficiently large values of n . The following example makes this clear.

Example 3.2

Say we have a scheme that is asymptotically secure. Then it may be the case that an adversary running for n^3 minutes can succeed in “breaking the scheme” with probability $2^{40} \cdot 2^{-n}$ (which is a negligible function of n). When $n \leq 40$ this means that an adversary running for 40^3 minutes (about 6 weeks) can break the scheme with probability 1, so such values of n are not very useful. Even for $n = 50$ an adversary running for 50^3 minutes (about 3 months) can break the scheme with probability roughly $1/1000$, which may not be acceptable. On the other hand, when $n = 500$ an adversary running for 200 years breaks the scheme only with probability roughly 2^{-500} . $\diamond$

As indicated by the previous example, we can view the security parameter as a mechanism that allows the honest parties to “tune” the security of a scheme to some desired level. (Increasing the security parameter also increases the time required to run the scheme, as well as the length of the key, so the honest parties will want to set the security parameter as small as possible subject to defending against the class of attacks they are concerned about.) Viewing the security parameter as the key length, this corresponds roughly to the fact that the time required for an exhaustive-search attack grows exponentially in the length of the key. The ability to “increase security” by increasing the security parameter has important practical ramifications, since it enables honest parties to defend against increases in computing power. The following example gives a sense of how this might play out in practice.

Example 3.3

Let us see the effect that the availability of faster computers might have on security in practice. Say we have a cryptographic scheme in which the honest parties run for $10^6 \cdot n^2$ cycles, and for which an adversary running for $10^8 \cdot n^4$ cycles can succeed in “breaking” the scheme with probability at most $2^{-n/2}$. (The numbers are intended to make calculations easier, and are not meant to correspond to any existing cryptographic scheme.)

Say all parties are using 2 GHz computers and the honest parties set $n = 80$. Then the honest parties run for $10^6 \cdot 6400$ cycles, or 3.2 seconds, and an adversary running for $10^8 \cdot (80)^4$ cycles, or roughly 3 weeks, can break the scheme with probability only 2^{-40} .

Say 8 GHz computers become available, and all parties upgrade. Honest parties can increase n to 160 (which requires generating a fresh key) and maintain a running time of 3.2 seconds (i.e., $10^6 \cdot 160^2$ cycles at $8 \cdot 10^9$ cycles/second). In contrast, the adversary now has to run for over 8 million seconds, or more than 13 weeks, to achieve a success probability of 2^{-80} . The effect of a faster computer has been to make the adversary’s job *harder*. $\diamond$

Even when using the asymptotic approach it is important to remember that, ultimately, when a cryptosystem is deployed in practice a concrete security guarantee will be needed. (After all, one must decide on some value of n .) As the above examples indicate, however, it is generally the case that an asymptotic security claim can be translated into a concrete security bound for any desired value of n .

The Asymptotic Approach in Detail

We now discuss more formally the notions of “polynomial-time algorithms” and “negligible success probabilities.”

Efficient algorithms. We have defined an algorithm to be efficient if it runs in polynomial time. An algorithm A runs in polynomial time if there exists a

polynomial p such that, for every input $x \in \{0, 1\}^*$, the computation of $A(x)$ terminates within at most $p(|x|)$ steps. (Here, $|x|$ denotes the length of the string x .) As mentioned earlier, we are only interested in adversaries whose running time is polynomial in the security parameter n . Since we measure the running time of an algorithm in terms of the length of its input, we sometimes provide algorithms with the security parameter written in unary (i.e., as 1^n , or a string of n ones) as input. Parties (or, more precisely, the algorithms they run) may take other inputs besides the security parameter—for example, a message to be encrypted—and we allow their running time to be polynomial in the (total) length of their inputs.

By default, we allow all algorithms to be probabilistic (or randomized). Any such algorithm may “toss a coin” at each step of its execution; this is a metaphorical way of saying that the algorithm can access an unbiased random bit at each step. Equivalently, we can view a randomized algorithm as one that, in addition to its input, is given a uniformly distributed *random tape* of sufficient length¹ whose bits it can use, as needed, throughout its execution.

We consider randomized algorithms by default for two reasons. First, randomness is essential to cryptography (e.g., in order to choose random keys and so on) and so honest parties must be probabilistic; given this, it is natural to allow adversaries to be probabilistic as well. Second, randomization is practical and—as far as we know—gives attackers additional power. Since our goal is to model *all* realistic attacks, we prefer a more liberal definition of efficient computation.

Negligible success probability. A negligible function is one that is asymptotically smaller than any inverse polynomial function. Formally:

DEFINITION 3.4 *A function f from the natural numbers to the non-negative real numbers is negligible if for every positive polynomial p there is an N such that for all integers $n > N$ it holds that $f(n) < \frac{1}{p(n)}$.*

For shorthand, the above is also stated as follows: for every polynomial p and *all sufficiently large values of n* it holds that $f(n) < \frac{1}{p(n)}$. An equivalent formulation of the above is to require that for all constants c there exists an N such that for all $n > N$ it holds that $f(n) < n^{-c}$. We typically denote an arbitrary negligible function by negl .

Example 3.5

The functions 2^{-n} , $2^{-\sqrt{n}}$, and $n^{-\log n}$ are all negligible. However, they approach zero at very different rates. For example, we can look at the minimum value of n for which each function is smaller than $1/n^5$:

¹If the algorithm in question runs for $p(n)$ steps on inputs of length n , then a random tape of length $p(n)$ is sufficient since the attacker can read at most one random bit per time step.

1. Solving $2^{-n} < n^{-5}$ we get $n > 5 \log n$. The smallest integer value of n for which this holds is $n = 23$.
2. Solving $2^{-\sqrt{n}} < n^{-5}$ we get $n > 25 \log^2 n$. The smallest integer value of n for which this holds is $n \approx 3500$.
3. Solving $n^{-\log n} < n^{-5}$ we get $\log n > 5$. The smallest integer value of n for which this holds is $n = 33$.

From the above you may have the impression that $n^{-\log n}$ approaches zero more quickly than $2^{-\sqrt{n}}$. However, this is incorrect; for all $n > 65536$ it holds that $2^{-\sqrt{n}} < n^{-\log n}$. Nevertheless, this does show that for values of n in the hundreds or thousands, an adversarial success probability of $n^{-\log n}$ is preferable to an adversarial success probability of $2^{-\sqrt{n}}$. $\diamond$

A technical advantage of working with negligible success probabilities is that they obey certain closure properties. The following is an easy exercise.

PROPOSITION 3.6 *Let negl_1 and negl_2 be negligible functions. Then,*

1. *The function negl_3 defined by $\text{negl}_3(n) = \text{negl}_1(n) + \text{negl}_2(n)$ is negligible.*
2. *For any positive polynomial p , the function negl_4 defined by $\text{negl}_4(n) = p(n) \cdot \text{negl}_1(n)$ is negligible.*

The second part of the above proposition implies that if a certain event occurs with only negligible probability in a certain experiment, then the event occurs with negligible probability even if the experiment is repeated polynomially many times. (This relies on the union bound; see Proposition A.7.) For example, the probability that n fair coin flips all come up “heads” is negligible. This means that even if we repeat the experiment of flipping n coins polynomially many times, the probability that *any* of those experiments result in n heads is still negligible.

A corollary of the second part of the above proposition is that if a function g is *not* negligible, then neither is the function $f(n) \stackrel{\text{def}}{=} g(n)/p(n)$ for any positive polynomial p .

Asymptotic Security: A Summary

Any security definition consists of two parts: a definition of what is considered a “break” of the scheme, and a specification of the power of the adversary. The power of the adversary can relate to many issues (e.g., in the case of encryption, whether we assume a ciphertext-only attack or a chosen-plaintext attack). However, when it comes to the *computational* power of the adversary, we will from now on model the adversary as efficient and thus only consider adversarial strategies that can be implemented in probabilistic polynomial

time. Definitions will also always be formulated so that a break that occurs with negligible probability is not considered significant. Thus, the general framework of any security definition will be as follows:

A scheme is *secure* if for every *probabilistic polynomial-time* adversary $\mathcal{A}$ carrying out an attack of some formally specified type, the probability that $\mathcal{A}$ succeeds in the attack (where success is also formally specified) is *negligible*.

Such a definition is *asymptotic* because it is possible that for small values of n an adversary can succeed with high probability. In order to see this in more detail, we expand the term “negligible” in the above statement:

A scheme is *secure* if for every PPT adversary $\mathcal{A}$ carrying out an attack of some formally specified type, and for every positive polynomial p , there exists an integer N such that when $n > N$ the probability that $\mathcal{A}$ succeeds in the attack is less than $\frac{1}{p(n)}$.

Note that nothing is guaranteed for values $n \leq N$.

On the Choices Made in Defining Asymptotic Security

In defining the general notion of asymptotic security, we have made two choices: we have identified efficient adversarial strategies with the class of *probabilistic, polynomial-time algorithms*, and have equated small chances of success with *negligible probabilities*. Both of these choices are—to some extent—arbitrary, and one could build a perfectly reasonable theory by defining, say, efficient strategies as those running in quadratic time, or small success probabilities as those bounded by 2^{-n} . Nevertheless, we briefly justify the choices we have made (which are the standard ones).

Those familiar with complexity theory or algorithms will recognize that the idea of equating efficient computation with (probabilistic) polynomial-time algorithms is not unique to cryptography. One advantage of using (probabilistic) polynomial time as our measure of efficiency is that this frees us from having to precisely specify our model of computation, since the extended Church–Turing thesis states that all “reasonable” models of computation are polynomially equivalent. Thus, we need not specify whether we use Turing machines, boolean circuits, or random-access machines; we can present algorithms in high-level pseudocode and be confident that if our analysis shows that these algorithms run in polynomial time, then any reasonable implementation will also.

Another advantage of (probabilistic) polynomial-time algorithms is that they satisfy desirable closure properties: in particular, an algorithm that makes polynomially many calls to a polynomial-time subroutine (and does only polynomial computation in addition) will itself run in polynomial time.

The most important feature of negligible probabilities is the closure property we have already seen in Proposition 3.6(2): any polynomial times a negligible function is still negligible. This means, in particular, that if an algorithm makes polynomially many calls to some subroutine that “fails” with negligible probability each time it is called, then the probability that any of the calls to that subroutine fail is still negligible.

Necessity of the Relaxations

Computational secrecy introduces two relaxations of perfect secrecy: first, security is guaranteed only against efficient adversaries; second, a small probability of success is allowed. Both these relaxations are essential for achieving practical encryption schemes, and in particular for bypassing the negative results for perfectly secret encryption. We informally discuss why this is the case. Assume we have an encryption scheme where the size of the key space $\mathcal{K}$ is much smaller than the size of the message space $\mathcal{M}$. (As shown in the previous chapter, this means the scheme cannot be perfectly secret.) Two attacks apply regardless of how the encryption scheme is constructed:

- Given a ciphertext c , an adversary can decrypt c using all keys $k \in \mathcal{K}$. This gives a list of all the messages to which c can possibly correspond. Since this list cannot contain all of $\mathcal{M}$ (because $|\mathcal{K}| < |\mathcal{M}|$), this attack leaks *some* information about the message that was encrypted.

Moreover, say the adversary carries out a known-plaintext attack and learns that ciphertexts $c_1, \dots, c_\ell$ correspond to the messages $m_1, \dots, m_\ell$, respectively. The adversary can again try decrypting each of these ciphertexts with all possible keys until it finds a key k for which $\text{Dec}_k(c_i) = m_i$ for all i . Later, given a ciphertext c that is the encryption of an unknown message m , it is almost surely the case that $\text{Dec}_k(c) = m$.

Exhaustive-search attacks like the above allow an adversary to succeed with probability essentially 1 in time linear in $|\mathcal{K}|$.

- Consider again the case where the adversary learns that ciphertexts $c_1, \dots, c_\ell$ correspond to messages $m_1, \dots, m_\ell$. The adversary can *guess* a uniform key $k \in \mathcal{K}$ and check to see whether $\text{Dec}_k(c_i) = m_i$ for all i . If so, then, as above, the attacker can use k to decrypt anything subsequently encrypted by the honest parties.

Here the adversary runs in essentially constant time and succeeds with nonzero (though very small) probability $1/|\mathcal{K}|$.

It follows that if we wish to encrypt many messages using a single short key, security can only be achieved if we limit the running time of the adversary (so the adversary does not have sufficient time to carry out a brute-force search) and are willing to allow a very small probability of success (so the second “attack” is ruled out).

3.2 Defining Computationally Secure Encryption

Given the background of the previous section, we are ready to present a definition of computational security for private-key encryption. First, we redefine the *syntax* of private-key encryption; this will be essentially the same as the syntax introduced in Chapter 2 except that we now explicitly take into account the security parameter n . We also allow the decryption algorithm to output an error message in case it is presented with an invalid ciphertext. Finally, by default, we let the message space be the set $\{0, 1\}^*$ of all (finite-length) binary strings.

DEFINITION 3.7 *A private-key encryption scheme is a tuple of probabilistic polynomial-time algorithms $(\text{Gen}, \text{Enc}, \text{Dec})$ such that:*

1. *The key-generation algorithm Gen takes as input 1^n (i.e., the security parameter written in unary) and outputs a key k ; we write $k \leftarrow \text{Gen}(1^n)$ (emphasizing that Gen is a randomized algorithm). We assume without loss of generality that any key k output by $\text{Gen}(1^n)$ satisfies $|k| \geq n$.*
2. *The encryption algorithm Enc takes as input a key k and a plaintext message $m \in \{0, 1\}^*$, and outputs a ciphertext c . Since Enc may be randomized, we write this as $c \leftarrow \text{Enc}_k(m)$.*
3. *The decryption algorithm Dec takes as input a key k and a ciphertext c , and outputs a message m or an error. We assume that Dec is deterministic, and so write $m := \text{Dec}_k(c)$ (assuming here that Dec does not return an error). We denote a generic error by the symbol $\perp$.*

It is required that for every n , every key k output by $\text{Gen}(1^n)$, and every $m \in \{0, 1\}^$, it holds that $\text{Dec}_k(\text{Enc}_k(m)) = m$.*

If $(\text{Gen}, \text{Enc}, \text{Dec})$ is such that for k output by $\text{Gen}(1^n)$, algorithm Enc_k is only defined for messages $m \in \{0, 1\}^{\ell(n)}$, then we say that $(\text{Gen}, \text{Enc}, \text{Dec})$ is a fixed-length private-key encryption scheme for messages of length $\ell(n)$.

Almost always, $\text{Gen}(1^n)$ simply outputs a uniform n -bit string as the key. When this is the case, we will omit Gen and simply define a private-key encryption scheme by a *pair* of algorithms (Enc, Dec) .

The above definition considers *stateless* schemes, in which each invocation of Enc (and Dec) is independent of all prior invocations. Later in the chapter, we will occasionally discuss *stateful* schemes in which the sender (and possibly the receiver) is required to maintain state across invocations. Unless explicitly noted otherwise, all our results assume stateless encryption/decryption.

3.2.1 The Basic Definition of Security

We begin by presenting the most basic notion of security for private-key encryption: security against a ciphertext-only attack where the adversary observes only a *single* ciphertext or, equivalently, security when a given key is used to encrypt just a *single* message. We consider stronger definitions of security later in the chapter.

Motivating the definition. As we have already discussed, any definition of security consists of two distinct components: a threat model (i.e., a specification of the assumed power of the adversary) and a security goal (usually specified by describing what constitutes a “break” of the scheme). We begin our definitional treatment by considering the simplest threat model, where we have an *eavesdropping adversary* who observes the encryption of a single message. This is exactly the threat model that was considered in the previous chapter with the exception that, as explained in the previous section, we are now interested only in adversaries that are computationally bounded and so limited to running in polynomial time.

Although we have made two assumptions about the adversary’s capabilities (namely, that it only eavesdrops, and that it runs in polynomial time), we make no assumptions whatsoever about the adversary’s *strategy* in trying to decipher the ciphertext it observes. This is crucial for obtaining meaningful notions of security; the definition ensures protection against *any* computationally bounded adversary, regardless of the algorithm it uses.

Correctly defining the security goal for encryption is not trivial, but we have already discussed this issue at length in Section 1.4.1 and in the previous chapter. We therefore just recall that the idea behind the definition is that the adversary should be unable to learn *any partial information* about the plaintext from the ciphertext. The definition of *semantic security* (cf. Section 3.2.2) exactly formalizes this notion, and was the first definition of computationally secure encryption to be proposed. Semantic security is complex and difficult to work with. Fortunately, there is an equivalent definition called *indistinguishability* that is much simpler.

The definition of indistinguishability is patterned on the alternative definition of perfect secrecy given as Definition 2.5. (This serves as further motivation that the definition of indistinguishability is a good one.) Recall that Definition 2.5 considers an experiment $\text{PrivK}_{\mathcal{A},\Pi}^{\text{eav}}$ in which an adversary $\mathcal{A}$ outputs two messages m_0 and m_1 , and is then given an encryption of one of those messages using a uniform key. The definition states that a scheme Π is secure if no adversary $\mathcal{A}$ can determine which of the messages m_0, m_1 was encrypted with probability any different from $1/2$, which is the probability that $\mathcal{A}$ is correct if it just makes a random guess.

Here, we keep the experiment $\text{PrivK}_{\mathcal{A},\Pi}^{\text{eav}}$ almost exactly the same (except for some technical differences discussed below), but introduce two key modifications in the definition itself:

1. We now consider only adversaries running in *polynomial time*, whereas Definition 2.5 considered even adversaries with unbounded running time.
2. We now concede that the adversary might determine the encrypted message with probability *negligibly better than* $1/2$.

As discussed extensively in the previous section, the above relaxations constitute the core elements of computational security.

As for the other differences, the most prominent is that we now parameterize the experiment by a security parameter n . We then measure both the running time of the adversary $\mathcal{A}$ as well as its success probability as functions of n . We write $\text{PrivK}_{\mathcal{A},\Pi}^{\text{eav}}(n)$ to denote the experiment being run with security parameter n , and write

$$\Pr[\text{PrivK}_{\mathcal{A},\Pi}^{\text{eav}}(n) = 1] \quad (3.1)$$

to denote the probability that the output of experiment $\text{PrivK}_{\mathcal{A},\Pi}^{\text{eav}}(n)$ is 1. Note that with $\mathcal{A}, \Pi$ fixed, Equation (3.1) is a function of n .

A second difference in experiment $\text{PrivK}_{\mathcal{A},\Pi}^{\text{eav}}$ is that we now explicitly require the adversary to output two messages m_0, m_1 of equal length. (In Definition 2.5 this requirement is implicit if the message space $\mathcal{M}$ only contains messages of some fixed length, as is the case for the one-time pad encryption scheme.) This means that, by default, we do not require a secure encryption scheme to hide the length of the plaintext. We revisit this point at the end of this section; see also Exercises 3.2 and 3.3.

Indistinguishability in the presence of an eavesdropper. We now give the formal definition, beginning with the experiment outlined above. The experiment is defined for any private-key encryption scheme $\Pi = (\text{Gen}, \text{Enc}, \text{Dec})$, any adversary $\mathcal{A}$, and any value n for the security parameter:

The adversarial indistinguishability experiment $\text{PrivK}_{\mathcal{A},\Pi}^{\text{eav}}(n)$:

1. The adversary $\mathcal{A}$ is given input 1^n , and outputs a pair of messages m_0, m_1 with $|m_0| = |m_1|$.
2. A key k is generated by running $\text{Gen}(1^n)$, and a uniform bit $b \in \{0, 1\}$ is chosen. Ciphertext $c \leftarrow \text{Enc}_k(m_b)$ is computed and given to $\mathcal{A}$. We refer to c as the challenge ciphertext.
3. $\mathcal{A}$ outputs a bit b' .
4. The output of the experiment is defined to be 1 if $b' = b$, and 0 otherwise. If $\text{PrivK}_{\mathcal{A},\Pi}^{\text{eav}}(n) = 1$, we say that $\mathcal{A}$ succeeds.

There is no limitation on the lengths of m_0 and m_1 , as long as they are the same. (Of course, if $\mathcal{A}$ runs in polynomial time, then m_0 and m_1 have length polynomial in n .) If Π is a fixed-length scheme for messages of length $\ell(n)$, the above experiment is modified by requiring $m_0, m_1 \in \{0, 1\}^{\ell(n)}$.

The fact that the adversary can only eavesdrop is implicit in the fact that its input is limited to a (single) ciphertext, and the adversary does not have any further interaction with the sender or the receiver. (As we will see later, allowing additional interaction makes the adversary significantly stronger.)

The definition of indistinguishability states that an encryption scheme is secure if no PPT adversary $\mathcal{A}$ succeeds in guessing which message was encrypted in the above experiment with probability significantly better than random guessing (which is correct with probability $1/2$):

DEFINITION 3.8 *A private-key encryption scheme $\Pi = (\text{Gen}, \text{Enc}, \text{Dec})$ has indistinguishable encryptions in the presence of an eavesdropper, or is EAV-secure, if for all probabilistic polynomial-time adversaries $\mathcal{A}$ there is a negligible function negl such that, for all n ,*

$$\Pr [\text{PrivK}_{\mathcal{A}, \Pi}^{\text{eav}}(n) = 1] \leq \frac{1}{2} + \text{negl}(n),$$

where the probability is taken over the randomness used by $\mathcal{A}$ and the randomness used in the experiment (for choosing the key and the bit b , as well as any randomness used by Enc).

Note: unless otherwise qualified, when we write “ $f(n) \leq g(n)$ ” we mean that inequality holds for all n .

It should be clear that Definition 3.8 is *weaker* than Definition 2.5, which is equivalent to perfect secrecy. Thus, any perfectly secret encryption scheme has indistinguishable encryptions in the presence of an eavesdropper. Our goal, therefore, will be to show that there exist encryption schemes satisfying the above in which the key is shorter than the message. That is, we will show schemes that satisfy Definition 3.8 but cannot satisfy Definition 2.5.

An equivalent formulation. Definition 3.8 requires that no PPT adversary can determine which of two messages was encrypted, with probability significantly better than $1/2$. An equivalent formulation is that every PPT adversary *behaves the same* whether it sees an encryption of m_0 or of m_1 . Since $\mathcal{A}$ outputs a single bit, “behaving the same” means it outputs 1 with almost the same probability in each case. To formalize this, define $\text{PrivK}_{\mathcal{A}, \Pi}^{\text{eav}}(n, b)$ as above except that the fixed bit b is used (rather than being chosen at random). Let $\text{out}_{\mathcal{A}}(\text{PrivK}_{\mathcal{A}, \Pi}^{\text{eav}}(n, b))$ denote the output bit b' of $\mathcal{A}$ in the experiment. The following essentially states that no $\mathcal{A}$ can determine whether it is running in experiment $\text{PrivK}_{\mathcal{A}, \Pi}^{\text{eav}}(n, 0)$ or experiment $\text{PrivK}_{\mathcal{A}, \Pi}^{\text{eav}}(n, 1)$.

DEFINITION 3.9 *A private-key encryption scheme $\Pi = (\text{Gen}, \text{Enc}, \text{Dec})$ has indistinguishable encryptions in the presence of an eavesdropper if for all PPT adversaries $\mathcal{A}$ there is a negligible function negl such that*

$$\left| \Pr[\text{out}_{\mathcal{A}}(\text{PrivK}_{\mathcal{A}, \Pi}^{\text{eav}}(n, 0)) = 1] - \Pr[\text{out}_{\mathcal{A}}(\text{PrivK}_{\mathcal{A}, \Pi}^{\text{eav}}(n, 1)) = 1] \right| \leq \text{negl}(n).$$

The fact that this is equivalent to Definition 3.8 is left as an exercise.

Encryption and Plaintext Length

The default notion of secure encryption does not require the encryption scheme to hide the plaintext length and, in fact, all commonly used encryption schemes reveal the plaintext length (or a close approximation thereof). The main reason for this is that it is *impossible* to support arbitrary-length messages while hiding all information about the plaintext length (cf. Exercise 3.2). In many cases this is inconsequential since the plaintext length is already public or is not sensitive. This is not always the case, however, and sometimes leaking the plaintext length is problematic. As examples:

- *Simple numeric/text data:* Say the encryption scheme being used reveals the plaintext length exactly. Then encrypted salary information would reveal whether someone makes a 5-figure or a 6-figure salary. Similarly, encryption of “yes”/“no” responses would leak the answer exactly.
- *Auto-suggestions:* Websites often include an “auto-complete” or “auto-suggestion” functionality by which the webserver suggests a list of potential words or phrases based on partial information the user has already typed. The *size* of this list can reveal information about the letters the user has typed so far. (For example, the number of auto-completions returned for “th” is far greater than the number for “zo.”)
- *Database searches:* Consider a user querying a database for all records matching some search term. The *number of records returned* can reveal a lot of information about what the user was searching for. This can be particularly damaging if the user is searching for medical information and the query reveals information about a disease the user has.
- *Compressed data:* If the plaintext is compressed before being encrypted, then information about the plaintext might be revealed even if only fixed-length data is ever encrypted. (Such an encryption scheme would therefore not satisfy Definition 3.8.) For example, a short compressed plaintext would indicate that the original (uncompressed) plaintext has a lot of redundancy. If an adversary can control a portion of what gets encrypted, this vulnerability can enable an adversary to learn additional information about the plaintext; it has been shown possible to use an attack of exactly this sort (the *CRIME attack*) against encrypted HTTP traffic to reveal secret session cookies.

When using encryption one should determine whether leaking the plaintext length is a concern and, if so, take steps to mitigate or prevent such leakage by padding all messages to some pre-determined length before encrypting them.

3.2.2 *Semantic Security

We motivated the definition of secure encryption by saying that it should be infeasible for an adversary to learn any partial information about the plaintext

from the ciphertext. However, the definition of indistinguishability looks very different. As we have mentioned, Definition 3.8 is equivalent to a definition called *semantic security* that formalizes the notion that partial information cannot be learned. We build up to that definition by discussing two weaker notions and showing that they are implied by indistinguishability.

We begin by showing that indistinguishability means that ciphertexts leak no information about individual bits of the plaintext. Formally, say encryption scheme (Enc, Dec) is EAV-secure (recall then when Gen is omitted, the key is a uniform n -bit string), and $m \in \{0, 1\}^\ell$ is uniform. Then we show that for any index i , it is infeasible to guess m^i from $\text{Enc}_k(m)$ (where, in this section, m^i denotes the i th bit of m) with probability much better than $1/2$.

THEOREM 3.10 *Let $\Pi = (\text{Enc}, \text{Dec})$ be a fixed-length private-key encryption scheme for messages of length ℓ that has indistinguishable encryptions in the presence of an eavesdropper. Then for all PPT adversaries $\mathcal{A}$ and any $i \in \{1, \dots, \ell\}$, there is a negligible function negl such that*

$$\Pr [\mathcal{A}(1^n, \text{Enc}_k(m)) = m^i] \leq \frac{1}{2} + \text{negl}(n),$$

where the probability is taken over uniform $m \in \{0, 1\}^\ell$ and $k \in \{0, 1\}^n$, the randomness of $\mathcal{A}$, and the randomness of Enc .

PROOF The idea behind the proof of this theorem is that if it were possible to guess the i th bit of m from $\text{Enc}_k(m)$, then it would also be possible to distinguish between encryptions of messages m_0 and m_1 whose i th bits differ. We formalize this via a *proof by reduction*, in which we show how to use any efficient adversary $\mathcal{A}$ to construct an efficient adversary $\mathcal{A}'$ such that if $\mathcal{A}$ violates the security notion of the theorem for Π , then $\mathcal{A}'$ violates the definition of indistinguishability for Π . (See Section 3.3.2.) Since Π has indistinguishable encryptions, it must also be secure in the sense of the theorem.

Fix an arbitrary PPT adversary $\mathcal{A}$ and $i \in \{1, \dots, \ell\}$. Let $I_0 \subset \{0, 1\}^\ell$ be the set of all strings whose i th bit is 0, and let $I_1 \subset \{0, 1\}^\ell$ be the set of all strings whose i th bit is 1. We have

$$\begin{aligned} \Pr [\mathcal{A}(1^n, \text{Enc}_k(m)) = m^i] \\ = \frac{1}{2} \cdot \Pr_{m_0 \leftarrow I_0} [\mathcal{A}(1^n, \text{Enc}_k(m_0)) = 0] + \frac{1}{2} \cdot \Pr_{m_1 \leftarrow I_1} [\mathcal{A}(1^n, \text{Enc}_k(m_1)) = 1]. \end{aligned}$$

Construct the following eavesdropping adversary $\mathcal{A}'$:

Adversary $\mathcal{A}'$:

1. Choose uniform $m_0 \in I_0$ and $m_1 \in I_1$. Output m_0, m_1 .
2. Upon observing a ciphertext c , invoke $\mathcal{A}(1^n, c)$. If $\mathcal{A}$ outputs 0, output $b' = 0$; otherwise, output $b' = 1$.

$\mathcal{A}'$ runs in polynomial time since $\mathcal{A}$ does.

By the definition of experiment $\text{PrivK}_{\mathcal{A}', \Pi}^{\text{eav}}(n)$, we have that $\mathcal{A}'$ succeeds if and only if $\mathcal{A}$ outputs b upon receiving $\text{Enc}_k(m_b)$. So

$$\begin{aligned} & \Pr [\text{PrivK}_{\mathcal{A}', \Pi}^{\text{eav}}(n) = 1] \\ &= \Pr [\mathcal{A}(1^n, \text{Enc}_k(m_b)) = b] \\ &= \frac{1}{2} \cdot \Pr_{m_0 \leftarrow I_0} [\mathcal{A}(1^n, \text{Enc}_k(m_0)) = 0] + \frac{1}{2} \cdot \Pr_{m_1 \leftarrow I_1} [\mathcal{A}(1^n, \text{Enc}_k(m_1)) = 1] \\ &= \Pr [\mathcal{A}(1^n, \text{Enc}_k(m)) = m^i]. \end{aligned}$$

By the assumption that (Enc, Dec) has indistinguishable encryptions in the presence of an eavesdropper, there is a negligible function negl such that $\Pr [\text{PrivK}_{\mathcal{A}', \Pi}^{\text{eav}}(n) = 1] \leq \frac{1}{2} + \text{negl}(n)$. We conclude that

$$\Pr [\mathcal{A}(1^n, \text{Enc}_k(m)) = m^i] \leq \frac{1}{2} + \text{negl}(n),$$

completing the proof. ■

We next claim, roughly, that indistinguishability means that no PPT adversary can learn *any* function of the plaintext given the ciphertext, regardless of the distribution of the message being sent. This is intended to capture the idea that no information about a plaintext is leaked by the resulting ciphertext. This requirement is, however, non-trivial to define formally. To see why, note that even for the case considered above, it is easy to compute the i th bit of m if m is chosen, say, uniformly from the set of all strings whose i th bit is 0 (rather than uniformly from $\{0, 1\}^\ell$). Thus, what we actually want to say is that if there exists any adversary who correctly computes $f(m)$ with some probability when given $\text{Enc}_k(m)$, then there exists an adversary that can correctly compute $f(m)$ with the same probability *without* being given the ciphertext at all (and only knowing the distribution of m). In what follows we focus on the case when m is chosen uniformly from some set $S \subseteq \{0, 1\}^\ell$.

THEOREM 3.11 *Let (Enc, Dec) be a fixed-length private-key encryption scheme for messages of length ℓ that has indistinguishable encryptions in the presence of an eavesdropper. Then for any PPT algorithm $\mathcal{A}$ there is a PPT algorithm $\mathcal{A}'$ such that for any $S \subseteq \{0, 1\}^\ell$ and any function $f : \{0, 1\}^\ell \rightarrow \{0, 1\}$, there is a negligible function negl such that:*

$$\left| \Pr [\mathcal{A}(1^n, \text{Enc}_k(m)) = f(m)] - \Pr [\mathcal{A}'(1^n) = f(m)] \right| \leq \text{negl}(n),$$

where the first probability is taken over uniform choice of $k \in \{0, 1\}^n$ and $m \in S$, the randomness of $\mathcal{A}$, and the randomness of Enc , and the second probability is taken over uniform choice of $m \in S$ and the randomness of $\mathcal{A}'$.

PROOF (Sketch) The fact that (Enc, Dec) is EAV-secure means that, for any $S \subseteq \{0, 1\}^\ell$, no PPT adversary can distinguish between $\text{Enc}_k(m)$ (for uniform $m \in S$) and $\text{Enc}_k(1^\ell)$. Consider now the probability that $\mathcal{A}$ successfully computes $f(m)$ given $\text{Enc}_k(m)$. We claim that $\mathcal{A}$ should successfully compute $f(m)$ given $\text{Enc}_k(1^\ell)$ with almost the same probability; otherwise, $\mathcal{A}$ could be used to distinguish between $\text{Enc}_k(m)$ and $\text{Enc}_k(1^\ell)$. The distinguisher is easily constructed: choose uniform $m \in S$, and output $m_0 = m$, $m_1 = 1^\ell$. When given a ciphertext c that is an encryption of either m_0 or m_1 , invoke $\mathcal{A}(1^n, c)$ and output 0 if and only if $\mathcal{A}$ outputs $f(m)$. If $\mathcal{A}$ outputs $f(m)$ when given an encryption of m with probability that is significantly different from the probability that it outputs $f(m)$ when given an encryption of 1^ℓ , then the described distinguisher violates Definition 3.9.

The above suggests the following algorithm $\mathcal{A}'$ that does not receive $c = \text{Enc}_k(m)$, yet computes $f(m)$ almost as well as $\mathcal{A}$ does: $\mathcal{A}'(1^n)$ chooses a uniform key $k \in \{0, 1\}^n$, invokes $\mathcal{A}$ on $c \leftarrow \text{Enc}_k(1^\ell)$, and outputs whatever $\mathcal{A}$ does. By the above, we have that $\mathcal{A}$ outputs $f(m)$ when run as a subroutine by $\mathcal{A}'$ with almost the same probability as when it receives $\text{Enc}_k(m)$. Thus, $\mathcal{A}'$ fulfills the property required by the claim. $\blacksquare$

Semantic security. The full definition of semantic security guarantees considerably more than the property considered in Theorem 3.11. The definition allows the length of the plaintext to depend on the security parameter, and allows for essentially arbitrary distributions over plaintexts. (Actually, we allow only *efficiently sampleable* distributions. This means that there is some probabilistic polynomial-time algorithm **Samp** such that $\text{Samp}(1^n)$ outputs messages according to the distribution.) The definition also takes into account arbitrary “external” information $h(m)$ about the plaintext that may be leaked to the adversary through other means (e.g., because the same message m is used for some other purpose as well).

DEFINITION 3.12 *A private-key encryption scheme (Enc, Dec) is semantically secure in the presence of an eavesdropper if for every PPT algorithm $\mathcal{A}$ there exists a PPT algorithm $\mathcal{A}'$ such that for any PPT algorithm **Samp** and polynomial-time computable functions f and h , the following is negligible:*

$$\left| \Pr[\mathcal{A}(1^n, \text{Enc}_k(m), h(m)) = f(m)] - \Pr[\mathcal{A}'(1^n, |m|, h(m)) = f(m)] \right|,$$

where the first probability is taken over uniform $k \in \{0, 1\}^n$, m output by $\text{Samp}(1^n)$, the randomness of $\mathcal{A}$, and the randomness of Enc , and the second probability is taken over m output by $\text{Samp}(1^n)$ and the randomness of $\mathcal{A}'$.

The adversary $\mathcal{A}$ is given the ciphertext $\text{Enc}_k(m)$ as well as the external information $h(m)$, and attempts to guess the value of $f(m)$. Algorithm $\mathcal{A}'$ also attempts to guess the value of $f(m)$, but is given *only* $h(m)$ and the

length of m . The security requirement states that $\mathcal{A}$'s probability of correctly guessing $f(m)$ is about the same as that of $\mathcal{A}'$. Intuitively, then, the ciphertext $\text{Enc}_k(m)$ does not reveal any additional information about the value of $f(m)$.

Definition 3.12 constitutes a very strong and convincing formulation of the security guarantees that should be provided by an encryption scheme. However, it is easier to work with the definition of indistinguishability (Definition 3.8). Fortunately, the definitions are *equivalent*:

THEOREM 3.13 *A private-key encryption scheme has indistinguishable encryptions in the presence of an eavesdropper if and only if it is semantically secure in the presence of an eavesdropper.*

Looking ahead, a similar equivalence between semantic security and indistinguishability is known for all the definitions that we present in this chapter as well as those in Chapter 11. We can therefore use indistinguishability as our working definition, while being assured that the guarantees achieved are those of semantic security.

3.3 Constructing Secure Encryption Schemes

Having defined what it means for an encryption scheme to be secure, the reader may expect us to turn immediately to constructions of secure encryption schemes. Before doing so, however, we need to introduce the notions of *pseudorandom generators* (PRGs) and *stream ciphers*, important building blocks for private-key encryption. These, in turn, will lead to a discussion of *pseudorandomness*, which plays a fundamental role in cryptography in general and private-key encryption in particular.

3.3.1 Pseudorandom Generators and Stream Ciphers

A pseudorandom generator G is an efficient, deterministic algorithm for transforming a short, uniform string called the *seed* into a longer, “uniform-looking” (or “pseudorandom”) output string. Stated differently, a pseudorandom generator uses a small amount of true randomness in order to generate a large amount of pseudorandomness. This is useful whenever a large number of random(-looking) bits are needed, since generating true random bits is difficult and slow. (See the discussion at the beginning of Chapter 2.) Indeed, pseudorandom generators have been studied since at least the 1940s when they were proposed for running statistical simulations. In that context, researchers proposed various statistical tests that a pseudorandom generator should pass in order to be considered “good.” As a simple example, the first

bit of the output of a pseudorandom generator should be equal to 1 with probability very close to $1/2$ (where the probability is taken over uniform choice of the seed), since the first bit of a uniform string is equal to 1 with probability exactly $1/2$. In fact, the parity of any fixed subset of the output bits should also be 1 with probability very close to $1/2$. More complex statistical tests can also be considered.

This historical approach to determining the quality of some candidate pseudorandom generator is ad hoc, and it is not clear when passing some set of statistical tests is sufficient to guarantee the soundness of using a candidate pseudorandom generator for some application. (In particular, there may be another statistical test that *does* successfully distinguish the output of the generator from true random bits.) The historical approach is even more problematic when using pseudorandom generators for cryptographic applications; in that setting, security may be compromised if an attacker is able to distinguish the output of a generator from uniform, and we do not know in advance what strategy an attacker might use.

The above considerations motivated a cryptographic approach to defining pseudorandom generators in the 1980s. The basic realization was that a good pseudorandom generator should pass *all* (efficient) statistical tests. That is, for *any* efficient statistical test (or *distinguisher*) D , the probability that D returns 1 when given the output of the pseudorandom generator should be close to the probability that D returns 1 when given a uniform string of the same length. Informally, then, the output of a pseudorandom generator should “look like” a uniform string to *any* efficient observer.

(We stress that, formally speaking, it does not make sense to say that any fixed string is “pseudorandom,” in the same way that it is meaningless to refer to any fixed string as “random.” Rather, pseudorandomness is a property of a *distribution* on strings. Nevertheless, we sometimes informally call a string sampled according to the uniform distribution a “uniform string,” and a string output by a pseudorandom generator a “pseudorandom string.”)

Another perspective is obtained by defining what it means for a distribution to be pseudorandom. Let Dist be a distribution on ℓ -bit strings. (This means that Dist assigns some probability to every string in $\{0, 1\}^\ell$; sampling from Dist means that we choose an ℓ -bit string according to this probability distribution.) Informally, Dist is *pseudorandom* if the experiment in which a string is sampled from Dist is indistinguishable from the experiment in which a uniform string of length ℓ is sampled. (Strictly speaking, since we are in an asymptotic setting we need to speak of the pseudorandomness of a *sequence* of distributions $\text{Dist} = \{\text{Dist}_n\}$, where distribution Dist_n is used for security parameter n . We ignore this point in our current discussion.) More precisely, it should be infeasible for any polynomial-time algorithm to tell (better than guessing) whether it is given a string sampled according to Dist , or whether it is given a uniform ℓ -bit string. This means that a *pseudorandom string is just as good as a uniform string*, as long as we consider only polynomial-time observers. Just as indistinguishability is a computational relaxation of

perfect secrecy, pseudorandomness is a computational relaxation of true randomness. (We will generalize this perspective when we discuss the notion of indistinguishability in Chapter 7.)

Now let $G : \{0, 1\}^n \rightarrow \{0, 1\}^\ell$ be a function, and define Dist to be the distribution on ℓ -bit strings obtained by choosing a uniform $s \in \{0, 1\}^n$ and outputting $G(s)$. Then G is a pseudorandom generator if and only if the distribution Dist is pseudorandom.

The formal definition. As discussed above, G is a pseudorandom generator if no efficient distinguisher can detect whether it is given a string output by G or a string chosen uniformly at random. As in Definition 3.9, this is formalized by requiring that every efficient algorithm outputs 1 with almost the same probability when given $G(s)$ (for uniform seed s) or a uniform string. (For an equivalent definition analogous to Definition 3.8, see Exercise 3.5.) We obtain a definition in the asymptotic setting by letting the security parameter n determine the length of the seed. We then insist that G be computable by an efficient algorithm. As a technicality, we also require that G 's output be longer than its input; otherwise, G is not very useful or interesting.

DEFINITION 3.14 *Let ℓ be a polynomial and let G be a deterministic polynomial-time algorithm such that for any n and any input $s \in \{0, 1\}^n$, the result $G(s)$ is a string of length $\ell(n)$. We say that G is a pseudorandom generator if the following conditions hold:*

1. **(Expansion:)** *For every n it holds that $\ell(n) > n$.*
2. **(Pseudorandomness:)** *For any PPT algorithm D , there is a negligible function negl such that*

$$|\Pr[D(G(s)) = 1] - \Pr[D(r) = 1]| \leq \text{negl}(n),$$

where the first probability is taken over uniform choice of $s \in \{0, 1\}^n$ and the randomness of D , and the second probability is taken over uniform choice of $r \in \{0, 1\}^{\ell(n)}$ and the randomness of D .

We call ℓ the expansion factor of G .

We give an example of an insecure pseudorandom generator to gain familiarity with the definition.

Example 3.15

Define $G(s)$ to output s followed by $\oplus_{i=1}^n s_i$, so the expansion factor of G is $\ell(n) = n + 1$. The output of G can easily be distinguished from uniform. Consider the following efficient distinguisher D : on input a string w , output 1 if and only if the final bit of w is equal to the XOR of all the preceding bits of w . Since this property holds for all strings output by G , we have

$\Pr[D(G(s)) = 1] = 1$. On the other hand, if w is uniform, the final bit of w is uniform and so $\Pr[D(w) = 1] = \frac{1}{2}$. The quantity $|\frac{1}{2} - 1|$ is constant, not negligible, and so this G is not a pseudorandom generator. (Note that D is not always “correct,” since it sometimes outputs 1 even when given a uniform string. This does not change the fact that D is a good distinguisher.) $\diamond$

Discussion. The distribution on the output of a pseudorandom generator G is far from uniform. To see this, consider the case that $\ell(n) = 2n$ and so G doubles the length of its input. Under the uniform distribution on $\{0, 1\}^{2n}$, each of the 2^{2n} possible strings is chosen with probability exactly 2^{-2n} . In contrast, consider the distribution of the output of G (when G is run on a uniform seed). When G receives an input of length n , the number of different strings in the range of G is at most 2^n . The fraction of strings of length $2n$ that are in the range of G is thus at most $2^n/2^{2n} = 2^{-n}$, and we see that the vast majority of strings of length $2n$ do not occur as outputs of G .

This in particular means that it is trivial to distinguish between a random string and a pseudorandom string *given an unlimited amount of time*. Let G be as above and consider the exponential-time distinguisher D that works as follows: $D(w)$ outputs 1 if and only if there exists an $s \in \{0, 1\}^n$ such that $G(s) = w$. (This computation is carried out in exponential time by exhaustively computing $G(s)$ for every $s \in \{0, 1\}^n$. Recall that by Kerckhoffs’ principle, the specification of G is known to D .) Now, if w were output by G , then D outputs 1 with probability 1. In contrast, if w is uniformly distributed in $\{0, 1\}^{2n}$, then the probability that there exists an s with $G(s) = w$ is at most 2^{-n} , and so D outputs 1 in this case with probability at most 2^{-n} . So

$$|\Pr[D(r) = 1] - \Pr[D(G(s)) = 1]| \geq 1 - 2^{-n},$$

which is large. This is just another example of a *brute-force attack*, and does not contradict the pseudorandomness of G since the attack is not efficient.

The seed and its length. The seed for a pseudorandom generator is analogous to the cryptographic key used by an encryption scheme, and the seed must be chosen uniformly and be kept secret from any adversary. Another important point, evident from the above discussion of brute-force attacks, is that s must be long enough so that it is not feasible to enumerate all possible seeds. In an asymptotic sense this is taken care of by setting the length of the seed equal to the security parameter, so that exhaustive search over all possible seeds requires exponential time. In practice, the seed must be long enough so that it is impossible to try all possible seeds within some specified time bound.

On the existence of pseudorandom generators. Do pseudorandom generators exist? They certainly seem difficult to construct, and one may rightly ask whether any algorithm satisfying Definition 3.14 exists. Although we do not know how to unconditionally prove the existence of pseudorandom generators, we have strong reasons to believe they exist. For one, they can be

constructed under the rather weak assumption that *one-way functions* exist (which is true if certain problems like factoring large numbers are hard); this will be discussed in detail in Chapter 7. We also have several practical constructions of candidate pseudorandom generators called *stream ciphers* for which no efficient distinguishers are known. (Later, we will introduce even stronger primitives called *block ciphers*.) We give a high-level overview of stream ciphers next, and discuss concrete stream ciphers in Chapter 6.

Stream Ciphers

Our definition of a pseudorandom generator is limited in two ways: the expansion factor is fixed, and the generator produces its entire output in “one shot.” *Stream ciphers*, used in practice to instantiate pseudorandom generators, work somewhat differently. The pseudorandom output bits of a stream cipher are produced gradually and on demand, so that an application can request exactly as many pseudorandom bits as needed. This improves efficiency (since an application can request fewer bits, if sufficient) and flexibility (since there is no upper bound on the number of bits that can be requested).

Formally, we view a stream cipher² as a pair of deterministic algorithms (Init, GetBits) where:

- Init takes as input a seed s and an optional *initialization vector* IV , and outputs an initial state st_0 .
- GetBits takes as input state information st_i , and outputs a bit y and updated state st_{i+1} . (In practice, y is a *block* of several bits; we treat y as a single bit here for generality and simplicity.)

Given a stream cipher and any desired expansion factor ℓ , we can define an algorithm G_ℓ mapping inputs of length n to outputs of length $\ell(n)$. The algorithm simply runs Init, and then repeatedly runs GetBits a total of ℓ times.

ALGORITHM 3.16

Constructing G_ℓ from (Init, GetBits)

Input: Seed s and optional initialization vector IV

Output: $y_1, \dots, y_\ell$

$st_0 := \text{Init}(s, IV)$

for $i = 1$ to ℓ :

$(y_i, st_i) := \text{GetBits}(st_{i-1})$

return $y_1, \dots, y_\ell$

²The terminology here is not completely standard, and beware that “stream cipher” is used by different people in different (but related) ways. For example, some use it to refer to G_ℓ (see below), while some use it to refer to Construction 3.17 when instantiated with G_ℓ .

A stream cipher is *secure* in the basic sense if it takes no IV and for any polynomial ℓ with $\ell(n) > n$, the function G_ℓ constructed above is a pseudo-random generator with expansion factor ℓ . We briefly discuss one possible security notion for stream ciphers that use an IV in Section 3.6.1.

3.3.2 Proofs by Reduction

If we wish to prove that a given construction is computationally secure, then we must rely on unproven assumptions³ (unless the scheme is information-theoretically secure). Our strategy will be to assume that some mathematical problem is hard, or that some *low-level* cryptographic primitive is secure, and then to *prove* that a given construction based on this problem/primitive is secure under this assumption. In Section 1.4.2 we have already explained in great detail why this approach is preferable so we do not repeat those arguments here.

The proof that a cryptographic construction is secure as long as some underlying problem is hard generally proceeds by presenting an explicit *reduction* showing how to transform any efficient adversary $\mathcal{A}$ that succeeds in “breaking” the construction into an efficient algorithm $\mathcal{A}'$ that solves the problem that was assumed to be hard. Since this is so important, we walk through a high-level outline of the steps of such a proof in detail. (We will see numerous concrete examples through the book, beginning with the proof of Theorem 3.18.) We begin with an assumption that some problem X cannot be solved (in some precisely defined sense) by any polynomial-time algorithm, except with negligible probability. We want to prove that some cryptographic construction Π is secure (again, in some sense that is precisely defined). A proof proceeds via the following steps (see also Figure 3.1):

1. Fix some efficient (i.e., probabilistic polynomial-time) adversary $\mathcal{A}$ attacking Π . Denote this adversary’s success probability by $\varepsilon(n)$.
2. Construct an efficient algorithm $\mathcal{A}'$, called the “reduction,” that attempts to solve problem X using adversary $\mathcal{A}$ as a subroutine. An important point here is that $\mathcal{A}'$ knows nothing about how $\mathcal{A}$ works; the only thing $\mathcal{A}'$ knows is that $\mathcal{A}$ is expecting to attack Π . So, given some input instance x of problem X , our algorithm $\mathcal{A}'$ will *simulate* for $\mathcal{A}$ an instance of Π such that:
 - (a) As far as $\mathcal{A}$ can tell, it is interacting with Π . That is, the view of $\mathcal{A}$ when run as a subroutine by $\mathcal{A}'$ should be distributed identically to (or at least close to) the view of $\mathcal{A}$ when it interacts with Π itself.
 - (b) If $\mathcal{A}$ succeeds in “breaking” the instance of Π that is being simulated by $\mathcal{A}'$, this should allow $\mathcal{A}'$ to solve the instance x it was given, at least with inverse polynomial probability $1/p(n)$.

³In particular, most of cryptography requires the unproven assumption that $\mathcal{P} \neq \mathcal{NP}$.

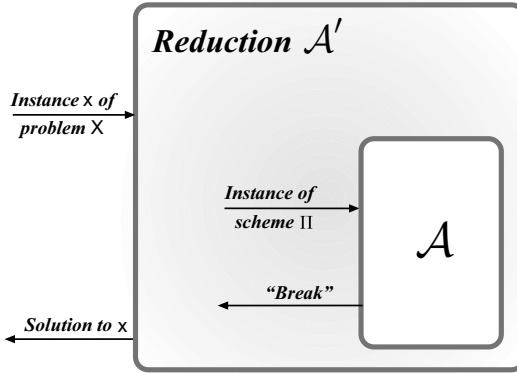


FIGURE 3.1: A high-level overview of a security proof by reduction.

3. Taken together, 2(a) and 2(b) imply that $\mathcal{A}'$ solves X with probability $\varepsilon(n)/p(n)$. If $\varepsilon(n)$ is not negligible, then neither is $\varepsilon(n)/p(n)$. Moreover, if $\mathcal{A}$ is efficient then we obtain an efficient algorithm $\mathcal{A}'$ solving X with non-negligible probability, contradicting the initial assumption.
4. Given our assumption regarding X , we conclude that *no* efficient adversary $\mathcal{A}$ can succeed in breaking Π with non-negligible probability. Stated differently, Π is computationally secure.

In the following section we will illustrate exactly the above idea: we will show how to use any pseudorandom generator G to construct an encryption scheme; we prove the encryption scheme secure by showing that any attacker who can “break” the encryption scheme can be used to distinguish the output of G from a uniform string. Under the assumption that G is a pseudorandom generator, then, the encryption scheme is secure.

3.3.3 A Secure Fixed-Length Encryption Scheme

A pseudorandom generator provides a natural way to construct a secure, fixed-length encryption scheme with a key shorter than the message. Recall that in the one-time pad (see Section 2.2), encryption is done by XORing a random pad with the message. The insight is that we can use a *pseudorandom* pad instead. Rather than sharing this long, pseudorandom pad, however, the sender and receiver can instead share a *seed* which is used to generate that pad when needed (see Figure 3.2); this seed will be shorter than the pad and hence shorter than the message. As for security, the intuition is that a pseudorandom string “looks random” to any polynomial-time adversary and so a computationally bounded eavesdropper cannot distinguish between a message encrypted using the one-time pad or a message encrypted using this “pseudo-”one-time pad encryption scheme.

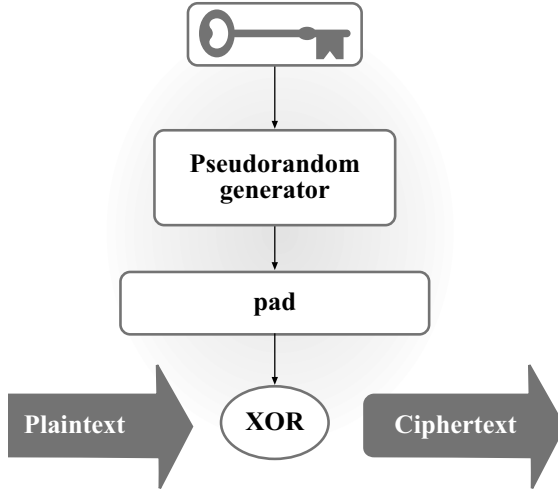


FIGURE 3.2: Encryption with a pseudorandom generator.

The encryption scheme. Fix some message length ℓ and let G be a pseudorandom generator with expansion factor ℓ (that is, $|G(s)| = \ell(|s|)$). Recall that an encryption scheme is defined by three algorithms: a key-generation algorithm Gen , an encryption algorithm Enc , and a decryption algorithm Dec . The key-generation algorithm is the trivial one: $\text{Gen}(1^n)$ simply outputs a uniform key k of length n . Encryption works by applying G to the key (which serves as a seed) in order to obtain a pad that is then XORed with the plaintext. Decryption applies G to the key and XORs the resulting pad with the ciphertext to recover the message. The scheme is described formally in Construction 3.17. In Section 3.6.1, we describe how stream ciphers are used to implement a variant of this scheme in practice.

CONSTRUCTION 3.17

Let G be a pseudorandom generator with expansion factor ℓ . Define a private-key encryption scheme for messages of length ℓ as follows:

- **Gen:** on input 1^n , choose uniform $k \in \{0, 1\}^n$ and output it as the key.
- **Enc:** on input a key $k \in \{0, 1\}^n$ and a message $m \in \{0, 1\}^{\ell(n)}$, output the ciphertext

$$c := G(k) \oplus m.$$
- **Dec:** on input a key $k \in \{0, 1\}^n$ and a ciphertext $c \in \{0, 1\}^{\ell(n)}$, output the message

$$m := G(k) \oplus c.$$

A private-key encryption scheme based on any pseudorandom generator.

THEOREM 3.18 *If G is a pseudorandom generator, then Construction 3.17 is a fixed-length private-key encryption scheme that has indistinguishable encryptions in the presence of an eavesdropper.*

PROOF Let Π denote Construction 3.17. We show that Π satisfies Definition 3.8. Namely, we show that for any probabilistic polynomial-time adversary $\mathcal{A}$ there is a negligible function negl such that

$$\Pr [\text{PrivK}_{\mathcal{A}, \Pi}^{\text{eav}}(n) = 1] \leq \frac{1}{2} + \text{negl}(n). \quad (3.2)$$

The intuition is that if Π used a uniform pad in place of the pseudorandom pad $G(k)$, then the resulting scheme would be identical to the one-time pad encryption scheme and $\mathcal{A}$ would be unable to correctly guess which message was encrypted with probability any better than $1/2$. Thus, if Equation (3.2) does *not* hold then $\mathcal{A}$ must implicitly be distinguishing the output of G from a random string. We make this explicit by showing a *reduction*; namely, by showing how to use $\mathcal{A}$ to construct an efficient distinguisher D , with the property that D 's ability to distinguish the output of G from a uniform string is directly related to $\mathcal{A}$'s ability to determine which message was encrypted by Π . Security of G then implies security of Π .

Let $\mathcal{A}$ be an arbitrary PPT adversary. We construct a distinguisher D that takes a string w as input, and whose goal is to determine whether w was chosen uniformly (i.e., w is a “random string”) or whether w was generated by choosing a uniform k and computing $w := G(k)$ (i.e., w is a “pseudorandom string”). We construct D so that it emulates the eavesdropping experiment for $\mathcal{A}$, as described below, and observes whether $\mathcal{A}$ succeeds or not. If $\mathcal{A}$ succeeds then D guesses that w must be a pseudorandom string, while if $\mathcal{A}$ does not succeed then D guesses that w is a random string. In detail:

Distinguisher D :

D is given as input a string $w \in \{0, 1\}^{\ell(n)}$. (We assume that n can be determined from $\ell(n)$.)

1. Run $\mathcal{A}(1^n)$ to obtain a pair of messages $m_0, m_1 \in \{0, 1\}^{\ell(n)}$.
2. Choose a uniform bit $b \in \{0, 1\}$. Set $c := w \oplus m_b$.
3. Give c to $\mathcal{A}$ and obtain output b' . Output 1 if $b' = b$, and output 0 otherwise.

D clearly runs in polynomial time (assuming $\mathcal{A}$ does).

Before analyzing the behavior of D , we define a modified encryption scheme $\tilde{\Pi} = (\widetilde{\text{Gen}}, \widetilde{\text{Enc}}, \widetilde{\text{Dec}})$ that is exactly the one-time pad encryption scheme, except that we now incorporate a security parameter that determines the length of the message to be encrypted. That is, $\widetilde{\text{Gen}}(1^n)$ outputs a uniform key k of length $\ell(n)$, and the encryption of message $m \in 2^{\ell(n)}$ using key $k \in \{0, 1\}^{\ell(n)}$

is the ciphertext $c := k \oplus m$. (Decryption can be performed as usual, but is inessential to what follows.) Perfect secrecy of the one-time pad implies

$$\Pr \left[\text{PrivK}_{\mathcal{A}, \Pi}^{\text{eav}}(n) = 1 \right] = \frac{1}{2}. \quad (3.3)$$

To analyze the behavior of D , the main observations are:

1. If w is chosen uniformly from $\{0, 1\}^{\ell(n)}$, then the view of $\mathcal{A}$ when run as a subroutine by D is distributed identically to the view of $\mathcal{A}$ in experiment $\text{PrivK}_{\mathcal{A}, \Pi}^{\text{eav}}(n)$. This is because when $\mathcal{A}$ is run as a subroutine by $D(w)$ in this case, $\mathcal{A}$ is given a ciphertext $c = w \oplus m_b$ where $w \in \{0, 1\}^{\ell(n)}$ is uniform. Since D outputs 1 exactly when $\mathcal{A}$ succeeds in its eavesdropping experiment, we therefore have (cf. Equation (3.3))

$$\Pr_{w \leftarrow \{0, 1\}^{\ell(n)}} [D(w) = 1] = \Pr \left[\text{PrivK}_{\mathcal{A}, \Pi}^{\text{eav}}(n) = 1 \right] = \frac{1}{2}. \quad (3.4)$$

(The subscript on the first probability just makes explicit that w is chosen uniformly from $\{0, 1\}^{\ell(n)}$ there.)

2. If w is instead generated by choosing uniform $k \in \{0, 1\}^n$ and then setting $w := G(k)$, the view of $\mathcal{A}$ when run as a subroutine by D is distributed identically to the view of $\mathcal{A}$ in experiment $\text{PrivK}_{\mathcal{A}, \Pi}^{\text{eav}}(n)$. This is because $\mathcal{A}$, when run as a subroutine by D , is now given a ciphertext $c = w \oplus m_b$ where $w = G(k)$ for a uniform $k \in \{0, 1\}^n$. Thus,

$$\Pr_{k \leftarrow \{0, 1\}^n} [D(G(k)) = 1] = \Pr \left[\text{PrivK}_{\mathcal{A}, \Pi}^{\text{eav}}(n) = 1 \right]. \quad (3.5)$$

Since G is a pseudorandom generator (and since D runs in polynomial time), we know there is a negligible function negl such that

$$\left| \Pr_{w \leftarrow \{0, 1\}^{\ell(n)}} [D(w) = 1] - \Pr_{k \leftarrow \{0, 1\}^n} [D(G(k)) = 1] \right| \leq \text{negl}(n).$$

Using Equations (3.4) and (3.5), we thus see that

$$\left| \frac{1}{2} - \Pr \left[\text{PrivK}_{\mathcal{A}, \Pi}^{\text{eav}}(n) = 1 \right] \right| \leq \text{negl}(n),$$

which implies $\Pr \left[\text{PrivK}_{\mathcal{A}, \Pi}^{\text{eav}}(n) = 1 \right] \leq \frac{1}{2} + \text{negl}(n)$. Since $\mathcal{A}$ was an arbitrary PPT adversary, this completes the proof that Π has indistinguishable encryptions in the presence of an eavesdropper. ■

It is easy to get lost in the details of the proof and wonder whether anything has been gained as compared to the one-time pad; after all, the one-time pad also encrypts an ℓ -bit message by XORing it with an ℓ -bit string! The point of the construction, of course, is that the ℓ -bit string $G(k)$ can be *much*

longer than the shared key k . In particular, using the above scheme it is possible to securely encrypt a 1 Mb file using only a 128-bit key. By relying on computational secrecy we have thus circumvented the impossibility result of Theorem 2.10, which states that any *perfectly* secret encryption scheme must use a key at least as long as the message.

Reductions—a discussion. We do *not* prove unconditionally that Construction 3.17 is secure. Rather, we prove that it is secure *under the assumption* that G is a pseudorandom generator. This approach of *reducing* the security of a higher-level construction to a lower-level primitive has a number of advantages (as discussed in Section 1.4.2). One of these advantages is that, in general, it is easier to design a lower-level primitive than a higher-level one; it is also easier, in general, to directly analyze an algorithm G with respect to a lower-level definition than to analyze a more complex scheme Π with respect to a higher-level definition. This does not mean that constructing a pseudorandom generator is “easy,” only that it is easier than constructing an encryption scheme from scratch. (In the present case the encryption scheme does nothing except XOR the output of a pseudorandom generator with the message and so this isn’t really true. However, we will see more complex constructions and in those cases the ability to reduce the task to a simpler one is of great importance.) Another advantage is that once an appropriate G has been constructed, it can be used as a component of various other schemes.

Concrete security. Although Theorem 3.18 and its proof are in an asymptotic setting, we can readily adapt the proof to bound the *concrete* security of the encryption scheme in terms of the concrete security of G . Fix some value of n for the remainder of this discussion, and let Π now denote Construction 3.17 using this value of n . Assume G is (t, ε) -pseudorandom (for the given value of n), in the sense that for all distinguishers D running in time at most t we have

$$|\Pr[D(r) = 1] - \Pr[D(G(s)) = 1]| \leq \varepsilon. \quad (3.6)$$

(Think of $t \approx 2^{80}$ and $\varepsilon \approx 2^{-60}$, though precise values are irrelevant for our discussion.) We claim that Π is $(t - c, \varepsilon)$ -secure for some (small) constant c , in the sense that for all $\mathcal{A}$ running in time at most $t - c$ we have

$$\Pr[\text{PrivK}_{\mathcal{A}, \Pi}^{\text{eav}} = 1] \leq \frac{1}{2} + \varepsilon. \quad (3.7)$$

(Note that the above are now fixed numbers, not functions of n , since we are not in an asymptotic setting here.) To see this, let $\mathcal{A}$ be an arbitrary adversary running in time at most $t - c$. Distinguisher D , as constructed in the proof of Theorem 3.18, has very little overhead besides running $\mathcal{A}$; setting c appropriately ensures that D runs in time at most t . Our assumption on the concrete security of G then implies Equation (3.6); proceeding exactly as in the proof of Theorem 3.18, we obtain Equation (3.7).

3.4 Stronger Security Notions

Until now we have considered a relatively weak definition of security in which the adversary only passively eavesdrops on a single ciphertext sent between the honest parties. In this section, we consider two stronger security notions. Recall that a security definition specifies a security goal and an attack model. In defining the first new security notion, we modify the security goal; for the second we strengthen the attack model.

3.4.1 Security for Multiple Encryptions

Definition 3.8 deals with the case where the communicating parties transmit a single ciphertext that is observed by an eavesdropper. It would be convenient, however, if the communicating parties could send multiple ciphertexts to each other—all generated using the same key—even if an eavesdropper might observe all of them. For such applications we need an encryption scheme secure for the encryption of multiple messages.

We begin with an appropriate definition of security for this setting. As in the case of Definition 3.8, we first introduce an appropriate experiment defined for any encryption scheme Π , adversary $\mathcal{A}$, and security parameter n :

The multiple-message eavesdropping experiment $\text{PrivK}_{\mathcal{A},\Pi}^{\text{mult}}(n)$:

1. The adversary $\mathcal{A}$ is given input 1^n , and outputs a pair of equal-length lists of messages $\vec{M}_0 = (m_{0,1}, \dots, m_{0,t})$ and $\vec{M}_1 = (m_{1,1}, \dots, m_{1,t})$, with $|m_{0,i}| = |m_{1,i}|$ for all i .
2. A key k is generated by running $\text{Gen}(1^n)$, and a uniform bit $b \in \{0, 1\}$ is chosen. For all i , the ciphertext $c_i \leftarrow \text{Enc}_k(m_{b,i})$ is computed and the list $\vec{C} = (c_1, \dots, c_t)$ is given to $\mathcal{A}$.
3. $\mathcal{A}$ outputs a bit b' .
4. The output of the experiment is defined to be 1 if $b' = b$, and 0 otherwise.

The definition of security is the same as before, except that it now refers to the above experiment.

DEFINITION 3.19 A private-key encryption scheme $\Pi = (\text{Gen}, \text{Enc}, \text{Dec})$ has indistinguishable multiple encryptions in the presence of an eavesdropper if for all probabilistic polynomial-time adversaries $\mathcal{A}$ there is a negligible function negl such that

$$\Pr \left[\text{PrivK}_{\mathcal{A},\Pi}^{\text{mult}}(n) = 1 \right] \leq \frac{1}{2} + \text{negl}(n),$$

where the probability is taken over the randomness used by $\mathcal{A}$ and the randomness used in the experiment.

Any scheme that has indistinguishable multiple encryptions in the presence of an eavesdropper clearly also satisfies Definition 3.8, since experiment $\text{PrivK}^{\text{eav}}$ corresponds to the special case of $\text{PrivK}^{\text{mult}}$ where the adversary outputs two lists containing only a single message each. In fact, our new definition is strictly stronger than Definition 3.8, as the following shows.

PROPOSITION 3.20 *There is a private-key encryption scheme that has indistinguishable encryptions in the presence of an eavesdropper, but not indistinguishable multiple encryptions in the presence of an eavesdropper.*

PROOF We do not have to look far to find an example of an encryption scheme satisfying the proposition. The one-time pad is perfectly secret, and so also has indistinguishable encryptions in the presence of an eavesdropper. We show that it is not secure in the sense of Definition 3.19. (We have discussed this attack in Chapter 2 already; here, we merely analyze the attack with respect to Definition 3.19.)

Concretely, consider the following adversary $\mathcal{A}$ attacking the scheme (in the sense defined by experiment $\text{PrivK}^{\text{mult}}$): $\mathcal{A}$ outputs $\vec{M}_0 = (0^\ell, 0^\ell)$ and $\vec{M}_1 = (0^\ell, 1^\ell)$. (The first contains the same plaintext twice, while the second contains two different messages.) Let $\vec{C} = (c_1, c_2)$ be the list of ciphertexts that $\mathcal{A}$ receives. If $c_1 = c_2$, then $\mathcal{A}$ outputs $b' = 0$; otherwise, $\mathcal{A}$ outputs $b' = 1$.

We now analyze the probability that $b' = b$. The crucial point is that the one-time pad is *deterministic*, so encrypting the same message twice (using the same key) yields the same ciphertext. Thus, if $b = 0$ then we must have $c_1 = c_2$ and $\mathcal{A}$ outputs 0 in this case. On the other hand, if $b = 1$ then a different message is encrypted each time; hence $c_1 \neq c_2$ and $\mathcal{A}$ outputs 1. We conclude that $\mathcal{A}$ correctly outputs $b' = b$ with probability 1, and so the encryption scheme is not secure with respect to Definition 3.19. ■

Necessity of probabilistic encryption. The above might appear to show that Definition 3.19 is impossible to achieve using *any* encryption scheme. But in fact this is true only if the encryption scheme is *deterministic* and so encrypting the same message multiple times (using the same key) always yields the same result. This is important enough to state as a theorem.

THEOREM 3.21 *If Π is a (stateless⁴) encryption scheme in which Enc is a deterministic function of the key and the message, then Π cannot have indistinguishable multiple encryptions in the presence of an eavesdropper.*

This should not be taken to mean that Definition 3.19 is too strong. Indeed,

⁴We will see in Section 3.6.1 that if the encryption scheme is stateful, then it is possible to securely encrypt multiple messages even if encryption is deterministic.

leaking to an eavesdropper the fact that two encrypted messages are the same can be a significant security breach. (Consider, e.g., a scenario in which a student encrypts a series of true/false answers!)

To construct a scheme secure for encrypting multiple messages, we must design a scheme in which encryption is *randomized* so that when the same message is encrypted multiple times, different ciphertexts can be produced. This may seem impossible since decryption must always be able to recover the message. However, we will soon see how to achieve it.

3.4.2 Chosen-Plaintext Attacks and CPA-Security

Chosen-plaintext attacks capture the ability of an adversary to exercise (partial) control over what the honest parties encrypt. We imagine a scenario in which two honest parties share a key k , and the attacker can influence these parties to encrypt messages $m_1, m_2, \dots$ (using k) and send the resulting ciphertexts over a channel that the attacker can observe. At some later point in time, the attacker observes a ciphertext corresponding to some *unknown* message m encrypted using the same key k ; let us even assume that the attacker knows that m is one of two possibilities m_0, m_1 . Security against chosen-plaintext attacks means that even in this case the attacker cannot tell which of these two messages was encrypted with probability significantly better than random guessing. (For now we revert back to the case where the eavesdropper is given only a single encryption of an unknown message. Shortly, we will return to consideration of the multiple-message case.)

Chosen-plaintext attacks in the real world. Are chosen-plaintext attacks a realistic concern? For starters, note that chosen-plaintext attacks also encompass *known-plaintext attacks*—in which the attacker knows what messages are being encrypted, even if it does not get to choose them—as a special case. Moreover, there are several real-world scenarios in which an adversary might have significant influence over what messages get encrypted. A simple example is given by an attacker typing on a terminal, which in turn encrypts and sends everything the adversary types using a key shared with a remote server (and unknown to the attacker). Here the attacker exactly controls what gets encrypted, but the encryption scheme should remain secure when it is used—with the same key—to encrypt data for another user.

Interestingly, chosen-plaintext attacks have also been used successfully as part of historical efforts to break military encryption schemes. For example, during World War II the British placed mines at certain locations, knowing that the Germans—when finding those mines—would encrypt the locations and send them back to headquarters. These encrypted messages were used by cryptanalysts at Bletchley Park to break the German encryption scheme.

Another example is given by the famous story involving the Battle of Midway. In May 1942, US Navy cryptanalysts intercepted an encrypted message from the Japanese which they were able to partially decode. The result in-

icated that the Japanese were planning an attack on AF, where AF was a ciphertext fragment that the US was unable to decode. For other reasons, the US believed that Midway Island was the target. Unfortunately, their attempts to convince Washington planners that this was the case were futile; the general belief was that Midway could not possibly be the target. The Navy cryptanalysts devised the following plan: They instructed US forces at Midway to send a fake message that their freshwater supplies were low. The Japanese intercepted this message and immediately reported to their superiors that “AF is low on water.” The Navy cryptanalysts now had their proof that AF corresponded to Midway, and the US dispatched three aircraft carriers to that location. The result was that Midway was saved, and the Japanese incurred significant losses. This battle was a turning point in the war between the US and Japan in the Pacific.

The Navy cryptanalysts here carried out a chosen-plaintext attack, as they were able to influence the Japanese (albeit in a roundabout way) to encrypt the word “Midway.” If the Japanese encryption scheme had been secure against chosen-plaintext attacks, this strategy by the US cryptanalysts would not have worked (and history may have turned out very differently)!

CPA-security. In the formal definition we model chosen-plaintext attacks by giving the adversary $\mathcal{A}$ access to an *encryption oracle*, viewed as a “black box” that encrypts messages of $\mathcal{A}$ ’s choice using a key k that is unknown to $\mathcal{A}$. That is, we imagine $\mathcal{A}$ has access to an “oracle” $\text{Enc}_k(\cdot)$; when $\mathcal{A}$ *queries* this oracle by providing it with a message m as input, the oracle returns a ciphertext $c \leftarrow \text{Enc}_k(m)$ as the reply. (When Enc is randomized, the oracle uses fresh randomness each time it answers a query.) The adversary is allowed to interact with the encryption oracle adaptively, as many times as it likes.

Consider the following experiment defined for any encryption scheme $\Pi = (\text{Gen}, \text{Enc}, \text{Dec})$, adversary $\mathcal{A}$, and value n for the security parameter:

The CPA indistinguishability experiment $\text{PrivK}_{\mathcal{A}, \Pi}^{\text{cpa}}(n)$:

1. A key k is generated by running $\text{Gen}(1^n)$.
2. The adversary $\mathcal{A}$ is given input 1^n and oracle access to $\text{Enc}_k(\cdot)$, and outputs a pair of messages m_0, m_1 of the same length.
3. A uniform bit $b \in \{0, 1\}$ is chosen, and then a ciphertext $c \leftarrow \text{Enc}_k(m_b)$ is computed and given to $\mathcal{A}$.
4. The adversary $\mathcal{A}$ continues to have oracle access to $\text{Enc}_k(\cdot)$, and outputs a bit b' .
5. The output of the experiment is defined to be 1 if $b' = b$, and 0 otherwise. In the former case, we say that $\mathcal{A}$ succeeds.

DEFINITION 3.22 A private-key encryption scheme $\Pi = (\text{Gen}, \text{Enc}, \text{Dec})$ has indistinguishable encryptions under a chosen-plaintext attack, or is CPA-secure, if for all probabilistic polynomial-time adversaries $\mathcal{A}$ there is a negligible function negl such that

$$\Pr \left[\text{PrivK}_{\mathcal{A}, \Pi}^{\text{cpa}}(n) = 1 \right] \leq \frac{1}{2} + \text{negl}(n),$$

where the probability is taken over the randomness used by $\mathcal{A}$, as well as the randomness used in the experiment.

CPA-Security for Multiple Encryptions

Definition 3.22 can be extended to the case of multiple encryptions in the same way that Definition 3.8 is extended to give Definition 3.19, i.e., by using lists of plaintexts. Here, we take a different approach that is somewhat simpler and has the advantage of modeling attackers that can *adaptively* choose plaintexts to be encrypted, even after observing previous ciphertexts. In the present definition, we give the attacker access to a “left-or-right” oracle $\text{LR}_{k,b}$ that, on input a pair of equal-length messages m_0, m_1 , computes the ciphertext $c \leftarrow \text{Enc}_k(m_b)$ and returns c . That is, if $b = 0$ then the adversary receives an encryption of the “left” plaintext, and if $b = 1$ then it receives an encryption of the “right” plaintext. Here, b is a random bit chosen at the beginning of the experiment, and as in previous definitions the goal of the attacker is to guess b . This generalizes the previous definition of multiple-message security (Definition 3.19) because instead of outputting the lists $(m_{0,1}, \dots, m_{0,t})$ and $(m_{1,1}, \dots, m_{1,t})$, one of whose messages will be encrypted, the attacker can now sequentially query $\text{LR}_{k,b}(m_{0,1}, m_{1,1}), \dots, \text{LR}_{k,b}(m_{0,t}, m_{1,t})$. This also encompasses the attacker’s access to an encryption oracle, since the attacker can simply query $\text{LR}_{k,b}(m, m)$ to obtain $\text{Enc}_k(m)$.

We now formally define this experiment, called the LR-oracle experiment. Let Π be an encryption scheme, $\mathcal{A}$ an adversary, and n the security parameter:

The LR-oracle experiment $\text{PrivK}_{\mathcal{A}, \Pi}^{\text{LR-cpa}}(n)$:

1. A key k is generated by running $\text{Gen}(1^n)$.
2. A uniform bit $b \in \{0, 1\}$ is chosen.
3. The adversary $\mathcal{A}$ is given input 1^n and oracle access to $\text{LR}_{k,b}(\cdot, \cdot)$, as defined above.
4. The adversary $\mathcal{A}$ outputs a bit b' .
5. The output of the experiment is defined to be 1 if $b' = b$, and 0 otherwise. In the former case, we say that $\mathcal{A}$ succeeds.

DEFINITION 3.23 *Private-key encryption scheme Π has indistinguishable multiple encryptions under a chosen-plaintext attack, or is CPA-secure for multiple encryptions, if for all probabilistic polynomial-time adversaries $\mathcal{A}$ there is a negligible function negl such that*

$$\Pr \left[\text{PrivK}_{\mathcal{A}, \Pi}^{\text{LR-cpa}}(n) = 1 \right] \leq \frac{1}{2} + \text{negl}(n),$$

where the probability is taken over the randomness used by $\mathcal{A}$ and the randomness used in the experiment.

Our earlier discussion shows that CPA-security for multiple encryptions is at least as strong as all our previous definitions. In particular, if a private-key encryption scheme is CPA-secure for multiple encryptions then it is clearly CPA-secure as well. Importantly, the converse also holds; that is, CPA-security implies CPA-security for multiple encryptions. (This stands in contrast to the case of eavesdropping adversaries; see Proposition 3.20.) We state the following theorem here without proof; a similar result in the public-key setting is proved in Section 11.2.2.

THEOREM 3.24 *Any private-key encryption scheme that is CPA-secure is also CPA-secure for multiple encryptions.*

This is a significant technical advantage of CPA-security: It suffices to prove that a scheme is CPA-secure (for a single encryption), and we then obtain “for free” that it is CPA-secure for multiple encryptions as well.

Security against chosen-plaintext attacks is nowadays the minimal notion of security an encryption scheme should satisfy, though it is becoming more common to require even the stronger security properties discussed in Section 4.5.

Fixed-length vs. arbitrary-length messages. Another advantage of working with the definition of CPA-security is that it allows us to treat fixed-length encryption schemes without loss of generality. In particular, given any CPA-secure *fixed-length* encryption scheme $\Pi = (\text{Gen}, \text{Enc}, \text{Dec})$, it is possible to construct a CPA-secure encryption scheme $\Pi' = (\text{Gen}', \text{Enc}', \text{Dec}')$ for *arbitrary-length* messages quite easily. For simplicity, say Π encrypts messages that are 1-bit long (though everything we say extends in the natural way regardless of the message length supported by Π). Leave Gen' the same as Gen . Define Enc'_k for any message m (having some arbitrary length ℓ) as $\text{Enc}'_k(m) = \text{Enc}_k(m_1), \dots, \text{Enc}_k(m_\ell)$, where m_i denotes the i th bit of m . Decryption is done in the natural way. Π' is CPA-secure if Π is; a proof follows from Theorem 3.24.

There are more efficient ways to encrypt messages of arbitrary length than by adapting a fixed-length encryption scheme in the above manner. We explore this further in Section 3.6.

3.5 Constructing CPA-Secure Encryption Schemes

Before constructing encryption schemes secure against chosen-plaintext attacks, we first introduce the important notion of *pseudorandom functions*.

3.5.1 Pseudorandom Functions and Block Ciphers

Pseudorandom functions (PRFs) generalize the notion of pseudorandom generators. Now, instead of considering “random-looking” *strings* we consider “random-looking” *functions*. As in our earlier discussion of pseudorandomness, it does not make much sense to say that any *fixed* function $f : \{0, 1\}^* \rightarrow \{0, 1\}^*$ is pseudorandom (in the same way that it makes little sense to say that any fixed function is random). Thus, we must instead refer to the pseudorandomness of a *distribution* on functions. Such a distribution is induced naturally by considering *keyed functions*, defined next.

A keyed function $F : \{0, 1\}^* \times \{0, 1\}^* \rightarrow \{0, 1\}^*$ is a two-input function, where the first input is called the *key* and denoted k . We say F is *efficient* if there is a polynomial-time algorithm that computes $F(k, x)$ given k and x . (We will only be interested in efficient keyed functions.) In typical usage a key k is chosen and fixed, and we are then interested in the single-input function $F_k : \{0, 1\}^* \rightarrow \{0, 1\}^*$ defined by $F_k(x) = F(k, x)$. The security parameter n dictates the key length, input length, and output length. That is, we associate with F three functions ℓ_{key} , ℓ_{in} , and ℓ_{out} ; for any key $k \in \{0, 1\}^{\ell_{key}(n)}$, the function F_k is only defined for inputs $x \in \{0, 1\}^{\ell_{in}(n)}$, in which case $F_k(x) \in \{0, 1\}^{\ell_{out}(n)}$. Unless stated otherwise, we assume for simplicity that F is *length-preserving*, meaning $\ell_{key}(n) = \ell_{in}(n) = \ell_{out}(n) = n$. That is, by fixing a key $k \in \{0, 1\}^n$ we obtain a function F_k mapping n -bit input strings to n -bit output strings.

A keyed function F induces a natural distribution on functions given by choosing a uniform key $k \in \{0, 1\}^n$ and then considering the resulting single-input function F_k . We call F *pseudorandom* if the function F_k (for a uniform key k) is indistinguishable from a function chosen uniformly at random from the set of all functions having the same domain and range; that is, if no efficient adversary can distinguish—in a sense we more carefully define below—whether it is interacting with F_k (for uniform k) or f (where f is chosen uniformly from the set of all functions mapping n -bit inputs to n -bit outputs).

Since choosing a function at random is less intuitive than choosing a string at random, it is worth spending a bit more time on this idea. Consider the set Func_n of all functions mapping n -bit strings to n -bit strings. This set is finite, and selecting a uniform function mapping n -bit strings to n -bit strings means choosing an element uniformly from this set. How large is Func_n ? A function f is specified by giving its value on each point in its domain. We can view any function (over a finite domain) as a large look-up table that stores

$f(x)$ in the row of the table labeled by x . For $f \in \text{Func}_n$, the look-up table for f has 2^n rows (one for each point of the domain $\{0,1\}^n$), with each row containing an n -bit string (since the range of f is $\{0,1\}^n$). Concatenating all the entries of the table, we see that any function in Func_n can be represented by a string of length $2^n \cdot n$. Moreover, this correspondence is one-to-one, as each string of length $2^n \cdot n$ (i.e., each table containing 2^n entries of length n) defines a unique function in Func_n . Thus, the size of Func_n is exactly the number of strings of length $n \cdot 2^n$, or $|\text{Func}_n| = 2^{n \cdot 2^n}$.

Viewing a function as a look-up table provides another useful way to think about selecting a uniform function $f \in \text{Func}_n$: It is exactly equivalent to choosing each row in the look-up table of f uniformly. This means, in particular, that the values $f(x)$ and $f(y)$ (for any two inputs $x \neq y$) are uniform and independent. We can view this look-up table being populated by random entries in advance, before f is evaluated on any input, or we can view entries of the table being chosen uniformly “on-the-fly,” as needed, whenever f is evaluated on a new input on which f has not been evaluated before.

Coming back to our discussion of pseudorandom functions, recall that a pseudorandom function is a keyed function F such that F_k (for $k \in \{0,1\}^n$ chosen uniformly at random) is indistinguishable from f (for $f \in \text{Func}_n$ chosen uniformly at random). The former is chosen from a distribution over (at most) 2^n distinct functions, whereas the latter is chosen from all $2^{n \cdot 2^n}$ functions in Func_n . Despite this, the “behavior” of these functions must look the same to any polynomial-time distinguisher.

A first attempt at formalizing the notion of a pseudorandom function would be to proceed in the same way as in Definition 3.14. That is, we could require that every polynomial-time distinguisher D that receives a description of the pseudorandom function F_k outputs 1 with “almost” the same probability as when it receives a description of a random function f . However, this definition is inappropriate since the description of a random function has *exponential length* (given by its look-up table of length $n \cdot 2^n$), while D is limited to running in polynomial time. So, D would not even have sufficient time to examine its entire input.

The definition therefore gives D access to an *oracle* $\mathcal{O}$ which is either equal to F_k (for uniform k) or f (for a uniform function f). The distinguisher D may query its oracle at any point x , in response to which the oracle returns $\mathcal{O}(x)$. We treat the oracle as a black box in the same way as when we provided the adversary with oracle access to the encryption algorithm in the definition of a chosen-plaintext attack. Here, however, the oracle computes a deterministic function and so returns the same result if queried twice on the same input. (For this reason, we may assume without loss of generality that D never queries the oracle twice on the same input.) D may interact freely with its oracle, choosing its queries adaptively based on all previous outputs. Since D runs in polynomial time, however, it can ask only polynomially many queries.

We now present the formal definition. (The definition assumes F is length-preserving for simplicity only.)

DEFINITION 3.25 Let $F : \{0, 1\}^* \times \{0, 1\}^* \rightarrow \{0, 1\}^*$ be an efficient, length-preserving, keyed function. F is a **pseudorandom function** if for all probabilistic polynomial-time distinguishers D , there is a negligible function negl such that:

$$\left| \Pr[D^{F_k(\cdot)}(1^n) = 1] - \Pr[D^{f(\cdot)}(1^n) = 1] \right| \leq \text{negl}(n),$$

where the first probability is taken over uniform choice of $k \in \{0, 1\}^n$ and the randomness of D , and the second probability is taken over uniform choice of $f \in \text{Func}_n$ and the randomness of D .

An important point is that D is *not* given the key k . It is meaningless to require that F_k be pseudorandom if k is known, since given k it is trivial to distinguish an oracle for F_k from an oracle for f . (All the distinguisher has to do is query the oracle at any point x to obtain the answer y , and compare this to the result $y' := F_k(x)$ that it computes itself using the known value k . An oracle for F_k will return $y = y'$, while an oracle for a random function will have $y = y'$ with probability only 2^{-n} .) This means that if k is revealed, any claims about the pseudorandomness of F_k no longer hold. To take a concrete example, if F is a pseudorandom function, then given oracle access to F_k (for uniform k) it must be hard to find an input x for which $F_k(x) = 0^n$ (since it would be hard to find such an input for a truly random function f). But if k is known, finding such an input may be easy.

Example 3.26

As usual, we can gain familiarity with the definition by looking at an insecure example. Define the keyed, length-preserving function F by $F(k, x) = k \oplus x$. For any input x , the value of $F_k(x)$ is uniformly distributed (when k is uniform). Nevertheless, F is not pseudorandom since its values on any *two* points are correlated. Concretely, consider the distinguisher D that queries its oracle $\mathcal{O}$ on arbitrary, distinct points x_1, x_2 to obtain values $y_1 = \mathcal{O}(x_1)$ and $y_2 = \mathcal{O}(x_2)$, and outputs 1 if and only if $y_1 \oplus y_2 = x_1 \oplus x_2$. If $\mathcal{O} = F_k$, for any k , then D outputs 1. On the other hand, if $\mathcal{O} = f$ for f chosen uniformly from Func_n , then the probability that $f(x_1) \oplus f(x_2) = x_1 \oplus x_2$ is exactly the probability that $f(x_2) = x_1 \oplus x_2 \oplus f(x_1)$, or 2^{-n} , and D outputs 1 with this probability. The difference is $|1 - 2^{-n}|$, which is not negligible. $\diamond$

Pseudorandom Permutations/Block Ciphers

Let Perm_n be the set of all permutations (i.e., bijections) on $\{0, 1\}^n$. Viewing any $f \in \text{Perm}_n$ as a look-up table as before, we now have the added constraint that the entries in any two distinct rows must be different. We have 2^n different choices for the entry in the first row of the table; once we fix this entry, we are left with only $2^n - 1$ choices for the second row, and so on. We thus see that the size of Perm_n is $(2^n)!$.

Let F be a keyed function. We call F a *keyed permutation* if $\ell_{in} = \ell_{out}$, and furthermore for all $k \in \ell_{key}(n)$ the function $F_k : \{0, 1\}^{\ell_{in}(n)} \rightarrow \{0, 1\}^{\ell_{in}(n)}$ is one-to-one (i.e., F_k is a permutation). We call ℓ_{in} the *block length* of F . As before, unless stated otherwise we assume F is *length-preserving* and so $\ell_{key}(n) = \ell_{in}(n) = n$. A keyed permutation is *efficient* if there is a polynomial-time algorithm for computing $F_k(x)$ given k and x , as well as a polynomial-time algorithm for computing $F_k^{-1}(y)$ given k and y . That is, F_k should be both efficiently computable and *efficiently invertible* given k .

The definition of what it means for an efficient, keyed permutation F to be a *pseudorandom permutation* is exactly analogous to Definition 3.25, with the only difference being that now we require F_k to be indistinguishable from a uniform *permutation* rather than a uniform function. That is, we require that no efficient algorithm can distinguish between access to F_k (for uniform key k) and access to f (for uniform $f \in \text{Perm}_n$). It turns out that this is merely an aesthetic choice since, whenever the block length is sufficiently long, a random permutation is itself indistinguishable from a random function. Intuitively this is due to the fact that a uniform function f looks identical to a uniform permutation unless distinct values x and y are found for which $f(x) = f(y)$, since in such a case the function cannot be a permutation. However, the probability of finding such values x, y using a polynomial number of queries is negligible. (This follows from the results of Appendix A.4.)

PROPOSITION 3.27 *If F is a pseudorandom permutation and additionally $\ell_{in}(n) \geq n$, then F is also a pseudorandom function.*

If F is a keyed permutation then cryptographic schemes based on F might require the honest parties to compute the inverse F_k^{-1} in addition to computing F_k itself. This potentially introduces new security concerns. In particular, it may now be necessary to impose the stronger requirement that F_k be indistinguishable from a uniform permutation *even if the distinguisher is additionally given oracle access to the inverse of the permutation*. If F has this property, we call it a *strong pseudorandom permutation*.

DEFINITION 3.28 *Let $F : \{0, 1\}^* \times \{0, 1\}^* \rightarrow \{0, 1\}^*$ be an efficient, length-preserving, keyed permutation. F is a **strong pseudorandom permutation** if for all probabilistic polynomial-time distinguishers D , there exists a negligible function negl such that:*

$$\left| \Pr[D^{F_k(\cdot), F_k^{-1}(\cdot)}(1^n) = 1] - \Pr[D^{f(\cdot), f^{-1}(\cdot)}(1^n) = 1] \right| \leq \text{negl}(n),$$

where the first probability is taken over uniform choice of $k \in \{0, 1\}^n$ and the randomness of D , and the second probability is taken over uniform choice of $f \in \text{Perm}_n$ and the randomness of D .

Of course, any strong pseudorandom permutation is also a pseudorandom permutation.

Block ciphers. In practice, *block ciphers* are designed to be secure instantiations of (strong) pseudorandom permutations with some *fixed* key length and block length. We discuss approaches for building block ciphers, and some popular candidate block ciphers, in Chapter 6. For the purposes of the present chapter the details of these constructions are unimportant, and for now we simply assume that (strong) pseudorandom permutations exist.

Pseudorandom functions and pseudorandom generators. As one might expect, there is a close relationship between pseudorandom functions and pseudorandom generators. It is fairly easy to construct a pseudorandom generator G from a pseudorandom function F by simply evaluating F on a series of different inputs; e.g., we can define $G(s) \stackrel{\text{def}}{=} F_s(1) \| F_s(2) \| \cdots \| F_s(\ell)$ for any desired ℓ . If F_s were replaced by a uniform function f , the output of G would be uniform; thus, when using F instead, the output is pseudorandom. You are asked to prove this formally in Exercise 3.14.

More generally, we can use the above idea to construct a stream cipher (`Init`, `GetBits`) that accepts an initialization vector IV . (See Section 3.3.1.) The only difference is that instead of evaluating F_s on the fixed input sequence $1, 2, 3, \dots$, we evaluate F on the inputs $IV + 1, IV + 2, \dots$

CONSTRUCTION 3.29

Let F be a pseudorandom function. Define a stream cipher (`Init`, `GetBits`), where each call to `GetBits` outputs n bits, as follows:

- `Init`: on input $s \in \{0, 1\}^n$ and $IV \in \{0, 1\}^n$, set $\text{st}_0 := (s, IV)$.
- `GetBits`: on input $\text{st}_i = (s, IV)$, compute $IV' := IV + 1$ and set $y := F_s(IV')$ and $\text{st}_{i+1} := (s, IV')$. Output (y, st_{i+1}) .

A stream cipher from any pseudorandom function/block cipher.

Although stream ciphers can be constructed from block ciphers, dedicated stream ciphers used in practice typically have better performance, especially in resource-constrained environments. On the other hand, stream ciphers appear to be less well understood (in practice) than block ciphers, and confidence in their security is lower. It is therefore recommended to use block ciphers (possibly by converting them to stream ciphers first) whenever possible.

Considering the other direction, a pseudorandom generator G immediately gives a pseudorandom function F *with small block length*. Specifically, say G has expansion factor $n \cdot 2^{t(n)}$. We can define the keyed function $F : \{0, 1\}^n \times \{0, 1\}^{t(n)} \rightarrow \{0, 1\}^n$ as follows: to compute $F_k(i)$, first compute $G(k)$ and interpret the result as a look-up table with $2^{t(n)}$ rows each containing n bits; output the i th row. This runs in polynomial time only if $t(n) = \mathcal{O}(\log n)$. It is possible, though more difficult, to construct pseudorandom functions *with large block length* from pseudorandom generators; this is

shown in Section 7.5. Pseudorandom generators, in turn, can be constructed based on certain mathematical problems conjectured to be hard. The existence of pseudorandom functions based on these hard mathematical problems represents one of the amazing contributions of modern cryptography.

3.5.2 CPA-Secure Encryption from Pseudorandom Functions

We focus here on constructing a CPA-secure, fixed-length encryption scheme. By what we have said at the end of Section 3.4.2, this implies the existence of a CPA-secure encryption scheme for arbitrary-length messages. In Section 3.6 we will discuss more efficient ways of encrypting messages of arbitrary length.

A naive attempt at constructing a secure encryption scheme from a pseudorandom permutation is to define $\text{Enc}_k(m) = F_k(m)$. Although we expect that this “reveals no information about m ” (since, if f is a uniform function, then $f(m)$ is simply a uniform n -bit string), this method of encryption is *deterministic* and so cannot possibly be CPA-secure. In particular, encrypting the same plaintext twice will yield the same ciphertext.

Our secure construction is *randomized*. Specifically, we encrypt by applying the pseudorandom function to a *random value* r (rather than the message) and XORing the result with the plaintext. (See Figure 3.3 and Construction 3.30.) This can again be viewed as an instance of XORing a pseudorandom pad with the plaintext, with the major difference being the fact that a *fresh* pseudorandom pad is used each time. (In fact, the pseudorandom pad is only “fresh” if the pseudorandom function is applied to a “fresh” value on which it has never been applied before. While it is possible that a random r will be equal to some r -value chosen previously, this happens with only negligible probability.)

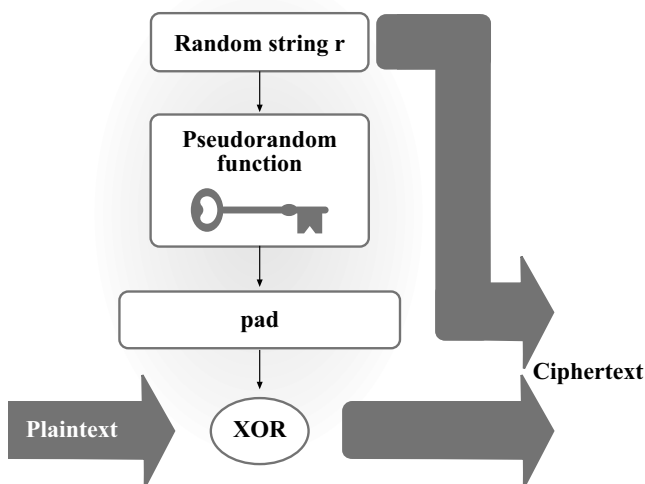


FIGURE 3.3: Encryption with a pseudorandom function.

Proofs of security based on pseudorandom functions. Before turning to the proof that the above construction is CPA-secure, we highlight a common template that is used by most proofs of security (even outside the context of encryption) for constructions based on pseudorandom functions. The first step of such proofs is to consider a hypothetical version of the construction in which the pseudorandom function is replaced with a random function. It is then argued—using a proof by reduction—that this modification does not significantly affect the attacker’s success probability. We are then left with analyzing a scheme that uses a completely random function. At this point the rest of the proof typically relies on probabilistic analysis and does not rely on any computational assumptions. We will utilize this proof template several times in this and the next chapter.

CONSTRUCTION 3.30

Let F be a pseudorandom function. Define a private-key encryption scheme for messages of length n as follows:

- **Gen:** on input 1^n , choose uniform $k \in \{0, 1\}^n$ and output it.
- **Enc:** on input a key $k \in \{0, 1\}^n$ and a message $m \in \{0, 1\}^n$, choose uniform $r \in \{0, 1\}^n$ and output the ciphertext

$$c := \langle r, F_k(r) \oplus m \rangle.$$

- **Dec:** on input a key $k \in \{0, 1\}^n$ and a ciphertext $c = \langle r, s \rangle$, output the plaintext message

$$m := F_k(r) \oplus s.$$

A CPA-secure encryption scheme from any pseudorandom function.

THEOREM 3.31 *If F is a pseudorandom function, then Construction 3.30 is a CPA-secure private-key encryption scheme for messages of length n .*

PROOF Let $\widetilde{\Pi} = (\widetilde{\text{Gen}}, \widetilde{\text{Enc}}, \widetilde{\text{Dec}})$ be an encryption scheme that is exactly the same as $\Pi = (\text{Gen}, \text{Enc}, \text{Dec})$ from Construction 3.30, except that a truly random function f is used in place of F_k . That is, $\widetilde{\text{Gen}}(1^n)$ chooses a uniform function $f \in \text{Func}_n$, and $\widetilde{\text{Enc}}$ encrypts just like Enc except that f is used instead of F_k . (This modified encryption scheme is not efficient. But we can still define it as a hypothetical encryption scheme for the sake of the proof.)

Fix an arbitrary PPT adversary $\mathcal{A}$, and let $q(n)$ be an upper bound on the number of queries that $\mathcal{A}(1^n)$ makes to its encryption oracle. (Note that q must be upper-bounded by some polynomial.) As the first step of the proof, we show that there is a negligible function negl such that

$$\left| \Pr \left[\text{PrivK}_{\mathcal{A}, \Pi}^{\text{cpa}}(n) = 1 \right] - \Pr \left[\text{PrivK}_{\mathcal{A}, \widetilde{\Pi}}^{\text{cpa}}(n) = 1 \right] \right| \leq \text{negl}(n). \quad (3.8)$$

We prove this by reduction. We use $\mathcal{A}$ to construct a distinguisher D for the pseudorandom function F . The distinguisher D is given oracle access to some function $\mathcal{O}$, and its goal is to determine whether this function is “pseudorandom” (i.e., equal to F_k for uniform $k \in \{0, 1\}^n$) or “random” (i.e., equal to f for uniform $f \in \text{Func}_n$). To do this, D emulates experiment $\text{PrivK}^{\text{cpa}}_{\mathcal{A}}$ for $\mathcal{A}$ in the manner described below, and observes whether $\mathcal{A}$ succeeds or not. If $\mathcal{A}$ succeeds then D guesses that its oracle must be a pseudorandom function, whereas if $\mathcal{A}$ does not succeed then D guesses that its oracle must be a random function. In detail:

Distinguisher D :

D is given input 1^n and access to an oracle $\mathcal{O} : \{0, 1\}^n \rightarrow \{0, 1\}^n$.

1. Run $\mathcal{A}(1^n)$. Whenever $\mathcal{A}$ queries its encryption oracle on a message $m \in \{0, 1\}^n$, answer this query in the following way:
 - (a) Choose uniform $r \in \{0, 1\}^n$.
 - (b) Query $\mathcal{O}(r)$ and obtain response y .
 - (c) Return the ciphertext $\langle r, y \oplus m \rangle$ to $\mathcal{A}$.
2. When $\mathcal{A}$ outputs messages $m_0, m_1 \in \{0, 1\}^n$, choose a uniform bit $b \in \{0, 1\}$ and then:
 - (a) Choose uniform $r \in \{0, 1\}^n$.
 - (b) Query $\mathcal{O}(r)$ and obtain response y .
 - (c) Return the challenge ciphertext $\langle r, y \oplus m_b \rangle$ to $\mathcal{A}$.
3. Continue answering encryption-oracle queries of $\mathcal{A}$ as before until $\mathcal{A}$ outputs a bit b' . Output 1 if $b' = b$, and 0 otherwise.

D runs in polynomial time since $\mathcal{A}$ does. The key points are as follows:

1. If D 's oracle is a pseudorandom function, then the view of $\mathcal{A}$ when run as a subroutine by D is distributed identically to the view of $\mathcal{A}$ in experiment $\text{PrivK}^{\text{cpa}}_{\mathcal{A}, \Pi}(n)$. This is because, in this case, a key k is chosen uniformly at random and then every encryption is carried out by choosing a uniform r , computing $y := F_k(r)$, and setting the ciphertext equal to $\langle r, y \oplus m \rangle$, exactly as in Construction 3.30. Thus,

$$\Pr_{k \leftarrow \{0, 1\}^n} \left[D^{F_k(\cdot)}(1^n) = 1 \right] = \Pr \left[\text{PrivK}^{\text{cpa}}_{\mathcal{A}, \Pi}(n) = 1 \right], \quad (3.9)$$

where we emphasize that k is chosen uniformly on the left-hand side.

2. If D 's oracle is a random function, then the view of $\mathcal{A}$ when run as a subroutine by D is distributed identically to the view of $\mathcal{A}$ in experiment $\text{PrivK}^{\text{cpa}}_{\mathcal{A}, \tilde{\Pi}}(n)$. This can be seen exactly as above, with the only difference being that a uniform function $f \in \text{Func}_n$ is used instead of F_k . Thus,

$$\Pr_{f \leftarrow \text{Func}_n} \left[D^{f(\cdot)}(1^n) = 1 \right] = \Pr \left[\text{PrivK}^{\text{cpa}}_{\mathcal{A}, \tilde{\Pi}}(n) = 1 \right], \quad (3.10)$$

where f is chosen uniformly from Func_n on the left-hand side.

By the assumption that F is a pseudorandom function (and since D is efficient), there exists a negligible function negl for which

$$\left| \Pr \left[D^{F_k(\cdot)}(1^n) = 1 \right] - \Pr \left[D^{f(\cdot)}(1^n) = 1 \right] \right| \leq \text{negl}(n).$$

Combining the above with Equations (3.9) and (3.10) gives Equation (3.8).

For the second part of the proof, we show that

$$\Pr \left[\text{PrivK}_{\mathcal{A}, \Pi}^{\text{cpa}}(n) = 1 \right] \leq \frac{1}{2} + \frac{q(n)}{2^n}. \quad (3.11)$$

(Recall that $q(n)$ is a bound on the number of encryption queries made by $\mathcal{A}$. The above holds even if we place no computational restrictions on $\mathcal{A}$.) To see that Equation (3.11) holds, observe that every time a message m is encrypted in $\text{PrivK}_{\mathcal{A}, \Pi}^{\text{cpa}}(n)$ (either by the encryption oracle or when the challenge ciphertext is computed), a uniform $r \in \{0, 1\}^n$ is chosen and the ciphertext is set equal to $\langle r, f(r) \oplus m \rangle$. Let r^* denote the random string used when generating the challenge ciphertext $\langle r^*, f(r^*) \oplus m_b \rangle$. There are two possibilities:

1. *The value r^* is never used when answering any of $\mathcal{A}$'s encryption-oracle queries:* In this case, $\mathcal{A}$ learns nothing about $f(r^*)$ from its interaction with the encryption oracle (since f is a truly random function). This means that, as far as $\mathcal{A}$ is concerned, the value $f(r^*)$ that is XORed with m_b is uniformly distributed and independent of the rest of the experiment, and so the probability that $\mathcal{A}$ outputs $b' = b$ in this case is exactly $1/2$ (as in the case of the one-time pad).
2. *The value r^* is used when answering at least one of $\mathcal{A}$'s encryption-oracle queries:* In this case, $\mathcal{A}$ may easily determine whether m_0 or m_1 was encrypted. This is so because if the encryption oracle ever returns a ciphertext $\langle r^*, s \rangle$ in response to a request to encrypt the message m , the adversary learns that $f(r^*) = s \oplus m$.

However, since $\mathcal{A}$ makes at most $q(n)$ queries to its encryption oracle (and thus at most $q(n)$ values of r are used when answering $\mathcal{A}$'s encryption-oracle queries), and since r^* is chosen uniformly from $\{0, 1\}^n$, the probability of this event is at most $q(n)/2^n$.

Let Repeat denote the event that r^* is used by the encryption oracle when answering at least one of $\mathcal{A}$'s queries. As just discussed, the probability of Repeat is at most $q(n)/2^n$, and the probability that $\mathcal{A}$ succeeds in $\text{PrivK}_{\mathcal{A}, \Pi}^{\text{cpa}}$ if Repeat does not occur is exactly $1/2$. Therefore:

$$\begin{aligned} & \Pr[\text{PrivK}_{\mathcal{A}, \Pi}^{\text{cpa}}(n) = 1] \\ &= \Pr[\text{PrivK}_{\mathcal{A}, \Pi}^{\text{cpa}}(n) = 1 \wedge \text{Repeat}] + \Pr[\text{PrivK}_{\mathcal{A}, \Pi}^{\text{cpa}}(n) = 1 \wedge \overline{\text{Repeat}}] \\ &\leq \Pr[\text{Repeat}] + \Pr[\text{PrivK}_{\mathcal{A}, \Pi}^{\text{cpa}}(n) = 1 \mid \overline{\text{Repeat}}] \leq \frac{q(n)}{2^n} + \frac{1}{2}. \end{aligned}$$

Combining the above with Equation (3.8), we see that there is a negligible function negl such that $\Pr[\text{PrivK}_{\mathcal{A},\Pi}^{\text{cpa}}(n) = 1] \leq \frac{1}{2} + \frac{q(n)}{2^n} + \text{negl}(n)$. Since q is polynomial, $\frac{q(n)}{2^n}$ is negligible. In addition, the sum of two negligible functions is negligible, and thus there exists a negligible function negl' such that $\Pr[\text{PrivK}_{\mathcal{A},\Pi}^{\text{cpa}}(n) = 1] \leq \frac{1}{2} + \text{negl}'(n)$, completing the proof. ■

Concrete security. The above proof shows that

$$\Pr[\text{PrivK}_{\mathcal{A},\Pi}^{\text{cpa}}(n) = 1] \leq \frac{1}{2} + \frac{q(n)}{2^n} + \text{negl}(n)$$

for some negligible function negl . The final term depends on the security of F as a pseudorandom function; it is a bound on the distinguishing probability of algorithm D (which has roughly the same running time as the adversary $\mathcal{A}$). The term $\frac{q(n)}{2^n}$ represents a bound on the probability that the value r^* used to encrypt the challenge ciphertext was used to encrypt some other message.

3.6 Modes of Operation

Modes of operation provide a way to securely (and efficiently) encrypt long messages using stream or block ciphers.

3.6.1 Stream-Cipher Modes of Operation

Construction 3.17 provides a way to construct an encryption scheme using a pseudorandom generator. That scheme has two main drawbacks. First, as presented, the length of the message to be encrypted must be fixed and known in advance. Second, the scheme is only EAV-secure, not CPA-secure.

Stream ciphers, which can be viewed as flexible pseudorandom generators, can be used to address these drawbacks. In practice, stream ciphers are used for encryption in two ways: *synchronized mode* and *unsynchronized mode*.

Synchronized mode. In this stateful encryption scheme, the sender and receiver must be *synchronized* in the sense that they know how much plaintext has been encrypted (resp., decrypted) so far. Synchronized mode is typically used in a single communication session between parties (see Section 4.5.3), where statefulness is acceptable and messages are received in order without being lost. The intuition here is that a long pseudorandom stream is generated, and a different part of it is used to encrypt each message. Synchronization is needed to ensure correct decryption (i.e., so the receiver knows which part of the stream was used to encrypt the next message), and to ensure that no portion of the stream is re-used. We describe this mode in detail next.

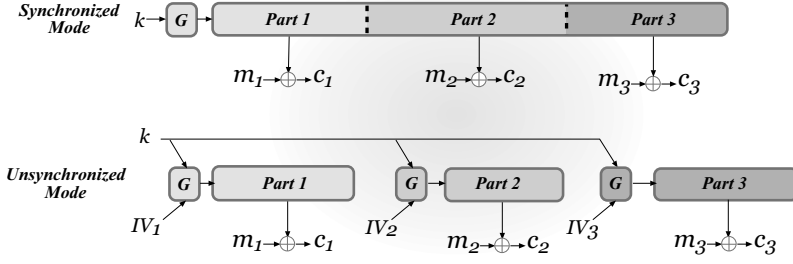


FIGURE 3.4: Synchronized mode and unsynchronized mode.

We have seen in Algorithm 3.16 that a stream cipher can be used to construct a pseudorandom generator G_ℓ with any desired expansion factor ℓ . We can easily modify that algorithm to obtain a pseudorandom generator G_∞ with *variable* output length. G_∞ takes two inputs: a seed s and a desired output length 1^ℓ (we specify this in unary since G_∞ will run in time polynomial in ℓ). As in Algorithm 3.16, $G_\infty(s, 1^\ell)$ runs $\text{Init}(s)$ and then repeatedly runs GetBits a total of ℓ times.

We can use G_∞ in Construction 3.17 to handle encryption of arbitrary-length messages: encryption of a message m using the key k is done by computing the ciphertext $c := G_\infty(k, 1^{|m|}) \oplus m$; decryption of a ciphertext c using the key k is carried out by computing the message $m := G_\infty(k, 1^{|c|}) \oplus c$. A minor modification of the proof of Theorem 3.18 shows that if the stream cipher is secure then this encryption scheme has indistinguishable encryptions in the presence of an eavesdropper.

A little thought shows that if the communicating parties are willing to maintain state, then they can use the same key to encrypt *multiple* messages. (See Figure 3.4.) The conceptual insight is that the parties can treat multiple messages $m_1, m_2, \dots$ as a single, long message; furthermore, Construction 3.17 (as well as the modified version in the previous paragraph) has the property that initial portions of a message can be encrypted and transmitted even if the rest of the message is not yet known. Concretely, the parties share a key k and both begin by computing $\text{st}_0 := \text{Init}(k)$. To encrypt the first message m_1 of length ℓ_1 , the sender repeatedly runs GetBits a total of ℓ_1 times, beginning at st_0 , to obtain a stream of bits $\text{pad}_1 \stackrel{\text{def}}{=} y_1, \dots, y_{\ell_1}$ along with updated state st_{ℓ_1} ; it then sends $c_1 := \text{pad}_1 \oplus m_1$. Upon receiving c_1 , the other party repeatedly runs GetBits a total of ℓ_1 times to obtain the same values pad_1 and st_{ℓ_1} ; it uses pad_1 to recover $m_1 := \text{pad}_1 \oplus c_1$. Later, to encrypt a second message m_2 of length ℓ_2 , the sender will repeatedly run GetBits a total of ℓ_2 times, beginning at st_{ℓ_1} , to obtain $\text{pad}_2 \stackrel{\text{def}}{=} y_{\ell_1+1}, \dots, y_{\ell_1+\ell_2}$ and updated state $\text{st}_{\ell_1+\ell_2}$, and then compute the ciphertext $c_2 := \text{pad}_2 \oplus m_2$, and so on. This can continue indefinitely, allowing the parties to send an unlimited number of messages of arbitrary length. We remark that in this mode, the stream cipher does not need to use an IV .

This method of encrypting multiple messages requires the communicating parties to maintain synchronized state, explaining the terminology “synchronized mode.” For that reason, the method is appropriate when two parties are communicating within a single “session,” but it does not work well for sporadic communication or when a party might, over time, communicate from different devices. (It is relatively easy to keep copies of a fixed key in different locations; it is harder to maintain synchronized state across multiple locations.) Furthermore, if the parties ever get out of sync (e.g., because one of the transmissions between the parties is dropped), decryption will return an incorrect result. Resynchronization is possible, but adds additional overhead.

Unsynchronized mode. For stream ciphers whose `Init` function accepts an initialization vector as input, we can achieve *stateless* CPA-secure encryption for messages of arbitrary length. Here we modify G_∞ to accept three inputs: a seed s , an initialization vector IV , and a desired output length 1^ℓ . Now this algorithm first computes $\text{st}_0 := \text{Init}(s, IV)$ before repeatedly running `GetBits` a total of ℓ times. Encryption can then be carried out using a variant of Construction 3.30: the encryption of a message m using the key k is done by choosing a uniform initialization vector $IV \in \{0, 1\}^n$ and computing the ciphertext $\langle IV, G_\infty(s, IV, 1^{|m|}) \oplus m \rangle$; decryption is performed in the natural way. (See Figure 3.4.) This scheme is CPA-secure if the stream cipher now has the stronger property that, for any polynomial ℓ , the function F defined by $F_k(IV) \stackrel{\text{def}}{=} G_\infty(k, IV, 1^\ell)$ is a pseudorandom function. (In fact, F need only be pseudorandom when evaluated on *uniform* inputs. Keyed functions with this weaker property are called *weak* pseudorandom functions.)

3.6.2 Block-Cipher Modes of Operation

We have already seen a construction of a CPA-secure encryption scheme based on pseudorandom functions/block ciphers. But Construction 3.30 (and its extension to arbitrary-length messages as discussed at the end of Section 3.4.2) have the drawback that the length of the ciphertext is *double* the length of the plaintext. Block-cipher modes of operation provide a way of encrypting arbitrary-length messages using shorter ciphertexts.

In this section, let F be a block cipher with block length n . We assume here that all messages m being encrypted have length a multiple of n , and write $m = m_1, m_2, \dots, m_\ell$ where each $m_i \in \{0, 1\}^n$ represents a block of the plaintext. Messages that are not a multiple of n can always be unambiguously padded to have length a multiple of n by appending a 1 followed by sufficiently many 0s, and so this assumption is without much loss of generality.

Several block-cipher modes of operation are known; we present four of the most common ones and discuss their security.

Electronic Code Book (ECB) mode. This is a naive mode of operation in which the ciphertext is obtained by direct application of the block cipher to each plaintext block. That is, $c := \langle F_k(m_1), F_k(m_2), \dots, F_k(m_\ell) \rangle$; see

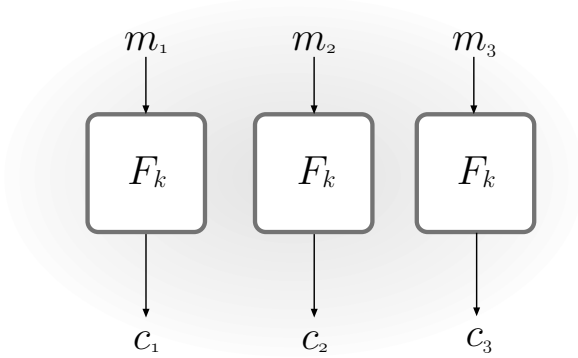


FIGURE 3.5: Electronic Code Book (ECB) mode.

Figure 3.5. Decryption is done in the obvious way, using the fact that F_k^{-1} is efficiently computable.

ECB mode is *deterministic* and therefore cannot be CPA-secure. Worse, ECB-mode encryption does not even have indistinguishable encryptions in the presence of an eavesdropper. This is because if a block is repeated in the plaintext, it will result in a repeating block in the ciphertext. Thus, it is easy to distinguish an encryption of a plaintext that consists of two identical blocks from an encryption of a plaintext that consists of two different blocks. This is not just a theoretical problem. Consider encrypting an image in which small groups of pixels correspond to a plaintext block. Encrypting using ECB mode may reveal a significant amount of information about patterns in the image, something that should not happen when using a secure encryption scheme. Figure 3.6 demonstrates this.

For these reasons, ECB mode should never be used. (We include it only because of its historical significance.)

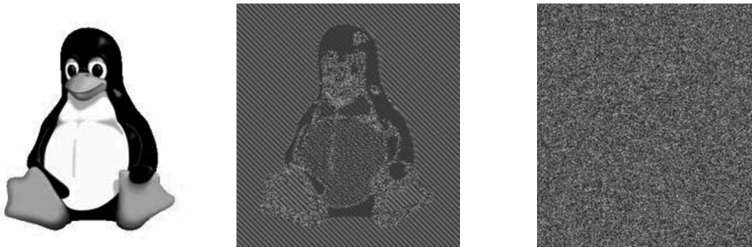


FIGURE 3.6: An illustration of the dangers of using ECB mode. The middle figure is an encryption of the image on the left using ECB mode; the figure on the right is an encryption of the same image using a secure mode. (Taken from <http://en.wikipedia.org> and derived from images created by Larry Ewing (lewing@isc.tamu.edu) using The GIMP.)

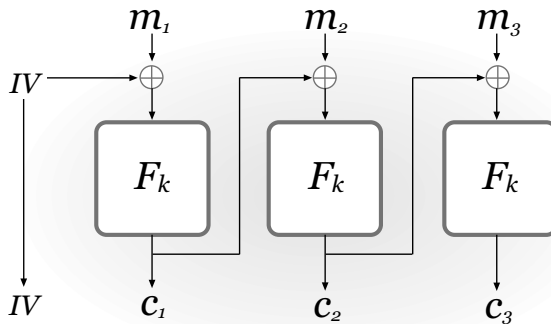


FIGURE 3.7: Cipher Block Chaining (CBC) mode.

Cipher Block Chaining (CBC) mode. To encrypt using this mode, a uniform initialization vector (IV) of length n is first chosen. Then, ciphertext blocks are generated by applying the block cipher to the XOR of the current plaintext block and the previous ciphertext block. That is, set $c_0 := IV$ and then, for $i = 1$ to ℓ , set $c_i := F_k(c_{i-1} \oplus m_i)$. The final ciphertext is $\langle c_0, c_1, \dots, c_\ell \rangle$. (See Figure 3.7.) Decryption of a ciphertext $c_0, \dots, c_\ell$ is done by computing $m_i := F_k^{-1}(c_i) \oplus c_{i-1}$ for $i = 1, \dots, \ell$. We stress that the IV is included in the ciphertext; this is crucial so decryption can be carried out.

Importantly, encryption in CBC mode is probabilistic and it has been proven that if F is a pseudorandom permutation then CBC-mode encryption is CPA-secure. The main drawback of this mode is that encryption must be carried out sequentially because the ciphertext block c_{i-1} is needed in order to encrypt the plaintext block m_i . Thus, if parallel processing is available, CBC-mode encryption may not be the most efficient choice.

One may be tempted to think that it suffices to use a *distinct* IV (rather than a random IV) for every encryption, e.g., to first use $IV = 1$ and then increment the IV by one each time a message is encrypted. In Exercise 3.20, we ask you to show that this variant of CBC-mode encryption is not secure.

One might also consider a stateful variant of CBC-mode encryption—called *chained CBC mode*—in which the last block of the previous ciphertext is used as the IV when encrypting the next message. This reduces the bandwidth, as the IV need not be sent each time. See Figure 3.8, where an initial message m_1, m_2, m_3 is encrypted using a random IV , and then subsequently a second message m_4, m_5 is encrypted using c_3 as the IV . (In contrast, encryption using stateless CBC mode would generate a fresh IV when encrypting the second message.) Chained CBC mode is used in SSL 3.0 and TLS 1.0.

It may appear that chained CBC mode is as secure as CBC mode, since the chained-CBC encryption of m_1, m_2, m_3 followed by encryption of m_4, m_5 yields the same ciphertext blocks as CBC-mode encryption of the (single) message m_1, m_2, m_3, m_4, m_5 . Nevertheless, chained CBC mode is vulnerable to a chosen-plaintext attack. The basis of the attack is that the adversary knows in

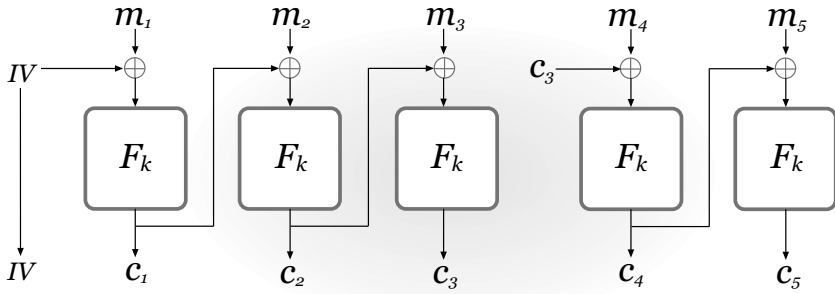


FIGURE 3.8: Chained CBC.

advance the “initialization vector” that will be used for the second encrypted message. We describe the attack informally, based on Figure 3.8. Assume the attacker knows that $m_1 \in \{m_1^0, m_1^1\}$, and observes the first ciphertext IV, c_1, c_2, c_3 . The attacker then requests an encryption of a second message m_4, m_5 with $m_4 = IV \oplus m_1^0 \oplus c_3$, and observes a second ciphertext c_4, c_5 . One can verify that $m_1 = m_1^0$ if and only if $c_4 = c_1$. This example should serve as a strong warning against making any modifications to cryptographic schemes, even if those modifications seem benign.

Output Feedback (OFB) mode. The third mode we present can be viewed as an unsynchronized stream-cipher mode, where the stream cipher is constructed in a specific way from the underlying block cipher. We describe the mode directly. First, a uniform $IV \in \{0, 1\}^n$ is chosen. Then, a pseudorandom stream is generated from IV in the following way: Define $y_0 := IV$, and set the i th block y_i of the stream to be $y_i := F_k(y_{i-1})$. Each block of the plaintext is encrypted by XORing it with the appropriate block of the stream; that is, $c_i := y_i \oplus m_i$. (See Figure 3.9.) As in CBC mode, the IV is included as part of the ciphertext to enable decryption. However, in contrast to CBC mode, here it is not required that F be invertible. (In fact, it need

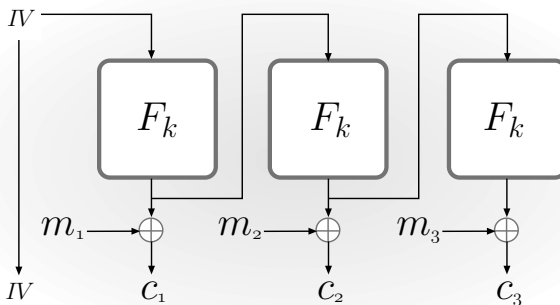


FIGURE 3.9: Output Feedback (OFB) mode.

not even be a permutation.) Furthermore, as in stream-cipher modes of operation, here it is not necessary for the plaintext length to be a multiple of the block length. Instead, the generated stream can be truncated to exactly the plaintext length. Another advantage of OFB mode is that its stateful variant (in which the final value y_ℓ used to encrypt some message is used as the IV for encrypting the next message) *is* secure. This stateful variant is equivalent to a *synchronized* stream-cipher mode, with the stream cipher constructed from the block cipher in the specific manner just described.

OFB mode can be shown to be CPA-secure if F is a pseudorandom function. Although both encryption and decryption must be carried out sequentially, this mode has the advantage relative to CBC mode that the bulk of the computation (namely, computation of the pseudorandom stream) can be done independently of the actual message to be encrypted. So, it is possible to generate a pseudorandom stream ahead of time using preprocessing, after which point encryption of the plaintext (once it is known) is incredibly fast.

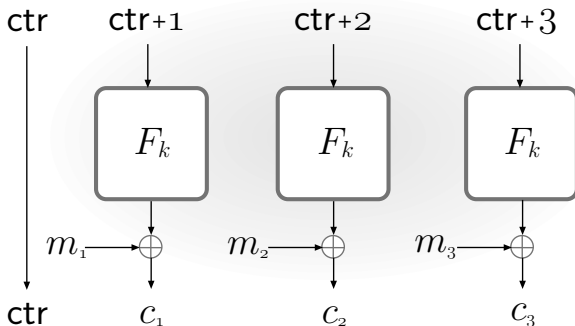


FIGURE 3.10: Counter (CTR) mode.

Counter (CTR) mode. Counter mode can also be viewed as an unsynchronized stream-cipher mode, where the stream cipher is constructed from the block cipher as in Construction 3.29. We give a self-contained description here. To encrypt using CTR mode, a uniform value $\text{ctr} \in \{0, 1\}^n$ is first chosen. Then, a pseudorandom stream is generated by computing $y_i := F_k(\text{ctr} + i)$, where ctr and i are viewed as integers and addition is done modulo 2^n . The i th ciphertext block is $c_i := y_i \oplus m_i$, and the IV is again sent as part of the ciphertext; see Figure 3.10. Note again that decryption does not require F to be invertible, or even a permutation. As with OFB mode, another “stream-cipher” mode, the generated stream can be truncated to exactly the plaintext length, preprocessing can be used to generate the pseudorandom stream before the message is known, and the stateful variant of CTR mode is secure.

In contrast to all the secure modes discussed previously, CTR mode has the advantage that encryption and decryption can be fully *parallelized*, since all the blocks of the pseudorandom stream can be computed independently of each other. In contrast to OFB, it is also possible to decrypt the i th block of

the ciphertext using only a single evaluation of F . These features make CTR mode an attractive choice.

CTR mode is also fairly simple to analyze:

THEOREM 3.32 *If F is a pseudorandom function, then CTR mode is CPA-secure.*

PROOF We follow the same template as in the proof of Theorem 3.31: we first replace F with a random function and then analyze the resulting scheme.

Let $\Pi = (\text{Gen}, \text{Enc}, \text{Dec})$ be the (stateless) CTR-mode encryption scheme, and let $\tilde{\Pi} = (\tilde{\text{Gen}}, \tilde{\text{Enc}}, \tilde{\text{Dec}})$ be the encryption scheme that is identical to Π except that a truly random function is used in place of F_k . That is, $\tilde{\text{Gen}}(1^n)$ chooses a uniform function $f \in \text{Func}_n$, and $\tilde{\text{Enc}}$ encrypts just like Enc except that f is used instead of F_k . (Once again, neither $\tilde{\text{Gen}}$ nor $\tilde{\text{Enc}}$ is efficient but this does not matter for the purposes of defining an experiment involving $\tilde{\Pi}$.)

Fix an arbitrary PPT adversary $\mathcal{A}$, and let $q(n)$ be a polynomial upper-bound on the number of encryption-oracle queries made by $\mathcal{A}(1^n)$ as well as the maximum number of blocks in any such query and the maximum number of blocks in m_0 and m_1 . As the first step of the proof, we claim that there is a negligible function negl such that

$$\left| \Pr \left[\text{PrivK}_{\mathcal{A}, \Pi}^{\text{cpa}}(n) = 1 \right] - \Pr \left[\text{PrivK}_{\mathcal{A}, \tilde{\Pi}}^{\text{cpa}}(n) = 1 \right] \right| \leq \text{negl}(n). \quad (3.12)$$

This is proved by reduction in a way similar to the analogous step in the proof of Theorem 3.31, and is left as an exercise for the reader.

We next claim that

$$\Pr \left[\text{PrivK}_{\mathcal{A}, \tilde{\Pi}}^{\text{cpa}}(n) = 1 \right] < \frac{1}{2} + \frac{2q(n)^2}{2^n}. \quad (3.13)$$

Combined with Equation (3.12) this means that

$$\Pr \left[\text{PrivK}_{\mathcal{A}, \Pi}^{\text{cpa}}(n) = 1 \right] < \frac{1}{2} + \frac{2q(n)^2}{2^n} + \text{negl}(n).$$

Since q is polynomial, $\frac{2q(n)^2}{2^n}$ is negligible and this completes the proof.

We now prove Equation (3.13). Fix some value n for the security parameter. Let $\ell^* \leq q(n)$ denote the length (in blocks) of the messages m_0, m_1 output by $\mathcal{A}$ in experiment $\text{PrivK}_{\mathcal{A}, \tilde{\Pi}}^{\text{cpa}}(n)$, and let ctr^* denote the initial value used when generating the challenge ciphertext. Similarly, let $\ell_i \leq q(n)$ be the length (in blocks) of the i th encryption-oracle query made by $\mathcal{A}$, and let ctr_i denote the initial value used when answering this query. When the i th encryption-oracle query is answered, f is applied to the values $\text{ctr}_i + 1, \dots, \text{ctr}_i + \ell_i$. When the challenge ciphertext is encrypted, f is applied to $\text{ctr}^* + 1, \dots, \text{ctr}^* + \ell^*$, and the i th ciphertext block is computed by XORing $f(\text{ctr}^* + i)$ with the i th message block. There are two cases:

1. *There do not exist any $i, j, j^* \geq 1$ (with $j \leq \ell_i$ and $j^* \leq \ell^*$) for which $\text{ctr}_i + j = \text{ctr}^* + j^*$:* In this case, the values $f(\text{ctr}^* + 1), \dots, f(\text{ctr}^* + \ell^*)$ used when encrypting the challenge ciphertext are uniformly distributed and independent of the rest of the experiment since f was not applied to any of these inputs when encrypting the adversary's oracle queries. This means that the challenge ciphertext is computed by XORing a stream of uniform bits with the message m_b , and so the probability that $\mathcal{A}$ outputs $b' = b$ is exactly $1/2$ (as in the case of the one-time pad).
2. *There exist $i, j, j^* \geq 1$ (with $j \leq \ell_i$ and $j^* \leq \ell^*$) for which $\text{ctr}_i + j = \text{ctr}^* + j^*$:* We denote this event by **Overlap**. In this case $\mathcal{A}$ may potentially determine which of its messages was encrypted to give the challenge ciphertext, since $\mathcal{A}$ can deduce the value of $f(\text{ctr}_i + j) = f(\text{ctr}^* + j^*)$ from the answer to its i th oracle query.

Let us analyze the probability that **Overlap** occurs. The probability is maximized if ℓ^* and each ℓ_i are as large as possible, so assume $\ell^* = \ell_i = q(n)$ for all i . Let **Overlap _{i}** denote the event that the sequence $\text{ctr}_i + 1, \dots, \text{ctr}_i + q(n)$ overlaps the sequence $\text{ctr}^* + 1, \dots, \text{ctr}^* + q(n)$; then **Overlap** is the event that **Overlap _{i}** occurs for some i . Since there are at most $q(n)$ oracle queries, a union bound (cf. Proposition A.7) gives

$$\Pr[\text{Overlap}] \leq \sum_{i=1}^{q(n)} \Pr[\text{Overlap}_i]. \quad (3.14)$$

Fixing ctr^* , event **Overlap _{i}** occurs exactly when ctr_i satisfies

$$\text{ctr}^* - q(n) + 1 \leq \text{ctr}_i \leq \text{ctr}^* + q(n) - 1.$$

Since there are $2q(n) - 1$ values of ctr_i for which **Overlap _{i}** occurs, and ctr_i is chosen uniformly from $\{0, 1\}^n$, we see that

$$\Pr[\text{Overlap}_i] = \frac{2q(n) - 1}{2^n} < \frac{2q(n)}{2^n}.$$

Combined with Equation (3.14), this gives $\Pr[\text{Overlap}] < 2q(n)^2/2^n$.

Given the above, we can easily bound the success probability of $\mathcal{A}$:

$$\begin{aligned} \Pr[\text{PrivK}_{\mathcal{A}, \tilde{\Pi}}^{\text{cpa}}(n) = 1] &= \Pr[\text{PrivK}_{\mathcal{A}, \tilde{\Pi}}^{\text{cpa}}(n) = 1 \wedge \overline{\text{Overlap}}] \\ &\quad + \Pr[\text{PrivK}_{\mathcal{A}, \tilde{\Pi}}^{\text{cpa}}(n) = 1 \wedge \text{Overlap}] \\ &\leq \Pr[\text{PrivK}_{\mathcal{A}, \tilde{\Pi}}^{\text{cpa}}(n) = 1 \mid \overline{\text{Overlap}}] + \Pr[\text{Overlap}] \\ &< \frac{1}{2} + \frac{2q(n)^2}{2^n}. \end{aligned}$$

This proves Equation (3.13) and completes the proof. ■

Modes of operation and message tampering. In many texts, modes of operation are also compared based on how well they protect against adversarial modification of the ciphertext. We do *not* include such a comparison here because the issue of *message integrity* or *message authentication* should be dealt with separately from encryption, and we do so in the next chapter. None of the above modes achieves message integrity in the sense we will define there. We defer further discussion to the next chapter.

With regard to the behavior of different modes in the presence of “benign” (i.e., non-adversarial) transmission errors, see Exercises 3.21 and 3.22. We remark, however, that in general such errors can be addressed using standard techniques (e.g., error correction or re-transmission).

Block length and concrete security. CBC, OFB, and CTR modes all use a random IV . This has the effect of randomizing the encryption process, and ensures that (with high probability) the underlying block cipher is always evaluated on *fresh* (i.e., new) inputs. This is important because, as we have seen in the proofs of Theorem 3.31 and Theorem 3.32, if an input to the block cipher is used more than once then security can be violated.

The block length of a block cipher thus has a significant impact on the concrete security of encryption schemes based on that cipher. Consider, e.g., CTR mode using a block cipher F with block length ℓ . The IV is then a uniform ℓ -bit string, and we expect an IV to repeat after encrypting about $2^{\ell/2}$ messages (see Appendix A.4). If ℓ is too short, then—even if F is secure as a pseudorandom permutation—the resulting concrete-security bound will be too weak for practical applications. Concretely, if $\ell = 64$ as is the case for DES (a block cipher we will study in Chapter 6), then after $2^{32} \approx 4,300,000,000$ encryptions—or roughly 34 gigabytes of plaintext—a repeated IV is expected to occur. Although this may seem like a lot of data, it is smaller than the capacity of modern hard drives.

IV misuse. In our description and discussion of the various (secure) modes, we have assumed a random IV is chosen each time a message is encrypted. What if this assumption goes wrong due, e.g., to poor randomness generation or a mistaken implementation? Certainly we can no longer guarantee security in the sense of Definition 3.22. From a practical point of view, though, the “stream-cipher modes” (OFB and CTR) are much worse than CBC mode. If an IV repeats when using the former, an attacker can XOR the two resulting ciphertexts and learn a lot of information about the *entire* contents of both encrypted messages (as we have seen previously in the context of the one-time pad if the key is re-used). With CBC mode, however, it is likely that after only a few blocks the inputs to the block cipher will “diverge” and the attacker will be unable to learn any information beyond the first few message blocks.

One way to address potential IV misuse is to use stateful encryption, as discussed in the context of OFB and CTR modes. If stateful encryption is not possible, and there is concern about potential IV misuse, then CBC mode is recommended for the reasons described above.

3.7 Chosen-Ciphertext Attacks

3.7.1 Defining CCA-Security

Until now we have defined security against two types of attacks: passive eavesdropping and chosen-plaintext attacks. A *chosen-ciphertext attack* is even more powerful. In a chosen-ciphertext attack, the adversary has the ability not only to obtain encryptions of messages of its choice (as in a chosen-plaintext attack), but also to obtain the *decryption* of ciphertexts of its choice (with one exception discussed later). Formally, we give the adversary access to a *decryption oracle* in addition to an encryption oracle. We present the formal definition and defer further discussion.

Consider the following experiment for any private-key encryption scheme $\Pi = (\text{Gen}, \text{Enc}, \text{Dec})$, adversary $\mathcal{A}$, and value n for the security parameter.

The CCA indistinguishability experiment $\text{PrivK}_{\mathcal{A}, \Pi}^{\text{cca}}(n)$:

1. A key k is generated by running $\text{Gen}(1^n)$.
2. Adversary $\mathcal{A}$ is given input 1^n and oracle access to $\text{Enc}_k(\cdot)$ and $\text{Dec}_k(\cdot)$. It outputs a pair of messages m_0, m_1 of the same length.
3. A uniform bit $b \in \{0, 1\}$ is chosen, and then a ciphertext $c \leftarrow \text{Enc}_k(m_b)$ is computed and given to $\mathcal{A}$. We call c the challenge ciphertext.
4. The adversary $\mathcal{A}$ continues to have oracle access to $\text{Enc}_k(\cdot)$ and $\text{Dec}_k(\cdot)$, but is not allowed to query the latter on the challenge ciphertext itself. Eventually, $\mathcal{A}$ outputs a bit b' .
5. The output of the experiment is defined to be 1 if $b' = b$, and 0 otherwise. If the output of the experiment is 1, we say that $\mathcal{A}$ succeeds.

DEFINITION 3.33 A private-key encryption scheme Π has indistinguishable encryptions under a chosen-ciphertext attack, or is CCA-secure, if for all probabilistic polynomial-time adversaries $\mathcal{A}$ there is a negligible function negl such that:

$$\Pr[\text{PrivK}_{\mathcal{A}, \Pi}^{\text{cca}}(n) = 1] \leq \frac{1}{2} + \text{negl}(n),$$

where the probability is taken over all randomness used in the experiment.

For completeness, we remark that the natural analogue of Theorem 3.24 holds for CCA-security as well. (Namely, if a scheme has indistinguishable encryptions under a chosen-ciphertext attack then it has indistinguishable *multiple* encryptions under a chosen-ciphertext attack, defined appropriately.)

In the experiment above, the adversary's access to the decryption oracle is unlimited *except* for the restriction that the adversary may not request decryption of the challenge ciphertext itself. This restriction is necessary or else there is clearly no hope for any encryption scheme to satisfy the definition.

At this point you may be wondering if chosen-ciphertext attacks realistically model any real-world attack. As in the case of chosen-plaintext attacks, we do not expect honest parties to decrypt arbitrary ciphertexts of an adversary's choice. Nevertheless, there may be scenarios where an adversary might be able to *influence* what gets decrypted and learn some partial information about the result. For example:

1. In the Midway example from Section 3.4.2, it is conceivable that the US cryptanalysts might also have tried to send encrypted messages to the Japanese and then monitor their behavior. Such behavior (e.g., movement of forces and the like) could have provided important information about the underlying plaintext.
2. Imagine a user sending encrypted messages to their bank. An adversary may be able to send ciphertexts on behalf of that user; the bank will decrypt those ciphertexts and the adversary may learn something about the result. For example, if a ciphertext corresponds to an *ill-formed* plaintext (e.g., an unintelligible message, or simply one that is not formatted correctly), the adversary may be able to deduce this from the bank's reaction (i.e., the pattern of subsequent communication). A practical attack of this type is presented in Section 3.7.2 below.
3. Encryption is often used in higher-level protocols; e.g., an encryption scheme might be used as part of an authentication protocol where one party sends a ciphertext to the other, who decrypts it and returns the result. In this case, one of the honest parties behaves exactly like a decryption oracle.

Insecurity of the schemes we have studied. None of the encryption schemes we have seen thus far is CCA-secure. We demonstrate this for Construction 3.30, where encryption is computed as $\text{Enc}_k(m) = \langle r, F_k(r) \oplus m \rangle$. Consider an adversary $\mathcal{A}$ running in the CCA indistinguishability experiment who chooses $m_0 = 0^n$ and $m_1 = 1^n$. Then, upon receiving a ciphertext $c = \langle r, s \rangle$, the adversary can flip the first bit of s and ask for a decryption of the resulting ciphertext c' . Since $c' \neq c$, this query is allowed and the decryption oracle answers with either 10^{n-1} (in which case it is clear that $b = 0$) or 01^{n-1} (in which case $b = 1$). This example demonstrates that CCA-security is quite stringent. Any encryption scheme that allows ciphertexts to be “manipulated” in any controlled way cannot be CCA-secure. Thus, CCA-security implies a very important property called *non-malleability*. Loosely speaking, a non-malleable encryption scheme has the property that if the adversary tries to modify a given ciphertext, the result is either an invalid ciphertext or one

that decrypts to a plaintext having no relation to the original one. This is a very useful property for schemes used in complex cryptographic protocols.

Constructing a CCA-secure encryption scheme. We show how to construct a CCA-secure encryption scheme in Section 4.5.4. The construction is presented there because it uses tools that we develop in Chapter 4.

3.7.2 Padding-Oracle Attacks

The chosen-ciphertext attack on Construction 3.30 described in the previous section is a bit contrived, since it assumes the attacker can obtain the complete decryption of a modified ciphertext. While this sort of attack is allowed under Definition 3.33, it is not clear that it represents a serious concern in practice. Here we show a chosen-ciphertext attack on a natural and widely used encryption scheme; moreover, the attack only requires the ability of an attacker to determine whether or not a modified ciphertext decrypts correctly. Such information is frequently very easy to obtain since, for example, a server might request a retransmission or terminate a session if it ever receives a ciphertext that does not decrypt correctly, and either of these events would generate a noticeable change in the observed traffic. The attack has been shown to work in practice on various deployed protocols; we give one concrete example at the end of this section.

As mentioned previously, when using CBC mode the length of the plaintext must be a multiple of the block length; if a plaintext does not satisfy this property, it must be padded before being encrypted. We refer to the original plaintext as the *message*, and the result after padding as *encoded data*. The padding scheme we use must allow the receiver to unambiguously determine from the encoded data where the message ends. One popular and standardized approach is to use PKCS #5 padding. Assume the original message has an integral number of bytes, and let L denote the block length (in bytes) of the block cipher being used. Let b denote the number of bytes that need to be appended to the message in order to make the total length of the resulting encoded data a multiple of the block length. Here, b is an integer between 1 and L , inclusive. (We cannot have $b = 0$ since this would lead to ambiguous padding. Thus, if the message length is already a multiple of the block length, L bytes of padding are appended.) Then we append to the message the string containing the integer b (represented in 1 byte, or 2 hexadecimal digits) repeated b times. That is, if 1 byte of padding is needed then the string 0x01 (written in hexadecimal) is appended; if 4 bytes of padding are needed then the hexadecimal string 0x04040404 is appended; etc. The encoded data is then encrypted using regular CBC-mode encryption.

When decrypting, the receiver first applies CBC-mode decryption as usual to recover the encoded data, and then checks that the encoded data is correctly padded. (This is easily done: simply read the value b of the final byte and then verify that the final b bytes of the result all have value b .) If so,

then the padding is stripped off and the original message returned. Otherwise, the standard procedure is to return a “bad padding” error (e.g., in Java the standard exception is called `javax.crypto.BadPaddingException`). The presence of such an error message provides an adversary with a *partial* decryption oracle. That is, an adversary can send any ciphertext to the server and learn (based on whether a “bad padding” error is returned or not) whether the underlying encoded data is padded in the correct format. Although this may seem like meaningless information, we show that it enables an adversary to completely recover the original message for any ciphertext of its choice.

We describe the attack on a 3-block ciphertext for simplicity. Let IV, c_1, c_2 be a ciphertext observed by the attacker, and let m_1, m_2 be the underlying encoded data (unknown to the attacker) which corresponds to the padded message, as discussed above. (Each block is L bytes long.) Note that

$$m_2 = F_k^{-1}(c_2) \oplus c_1, \quad (3.15)$$

where k is the key (which is, of course, not known to the attacker) being used by the honest parties. The second block m_2 ends in $\underbrace{0xb \cdots 0xb}_{b \text{ times}}$, where we let

$0xb$ denote the 1-byte representation of the integer b . The key property used in the attack is that certain changes to the ciphertext yield predictable changes in the underlying encoded data after CBC-mode decryption. Specifically, let c'_1 be identical to c_1 except for a modification in the final byte. Consider decryption of the modified ciphertext IV, c'_1, c_2 . This will result in encoded data m'_1, m'_2 where $m'_2 = F_k^{-1}(c_2) \oplus c'_1$. Comparing to Equation (3.15) we see that m'_2 will be identical to m_2 except for a modification in the final byte. (The value of m'_1 is unpredictable, but this will not adversely affect the attack.) Similarly, if c'_1 is the same as c_1 except for a change in its i th byte, then decryption of IV, c'_1, c_2 will result in m'_1, m'_2 where m'_2 is the same as m_2 except for a change in its i th byte. More generally, if $c'_1 = c_1 \oplus \Delta$ for any string Δ , then decryption of IV, c'_1, c_2 yields m'_1, m'_2 where $m'_2 = m_2 \oplus \Delta$. The upshot is that the attacker can exercise significant control over the final block of the encoded data.

As a warmup, let us see how the adversary can exploit this to learn b , the amount of padding. (This reveals the length of the original message.) Recall that upon decryption, the receiver looks at the value b of the final byte of the second block of the encoded data, and then verifies that the final b bytes all have the same value. The attacker begins by modifying the first byte of c_1 and sending the resulting ciphertext IV, c'_1, c_2 to the receiver. If decryption fails (i.e., the receiver returns an error) then it must be the case that the receiver is checking all L bytes of m'_2 , and therefore $b = L$! Otherwise, the attacker learns that $b < L$, and it can then repeat the process with the second byte, and so on. The left-most modified byte for which decryption fails reveals exactly the left-most byte being checked by the receiver, and so reveals exactly b .

With b known, the attacker can proceed to learn the bytes of the message one-by-one. We illustrate the idea for the final byte of the message, which we

denote by B . The attacker knows that m_2 ends in $0xB0xb \cdots 0xb$ (with $0xb$ repeated b times) and wishes to learn B . Define

$$\begin{aligned} \Delta_i \stackrel{\text{def}}{=} & 0x00 \cdots 0x00 \ 0xi \ \overbrace{0x(b+1) \cdots 0x(b+1)}^{b \text{ times}} \\ & \oplus \ 0x00 \cdots 0x00 \ 0x00 \ \overbrace{0xb \cdots 0xb}^{b \text{ times}} \end{aligned}$$

for $0 \leq i < 2^8$; i.e., the final $b+1$ bytes of Δ_i contain the integer i (represented in hexadecimal) followed by the value $(b+1) \oplus b$ (in hexadecimal) repeated b times. If the attacker submits the ciphertext $IV, c_1 \oplus \Delta_i, c_2$ to the receiver then, after CBC-mode decryption, the resulting encoded data will equal $0x(B \oplus i)0x(b+1) \cdots 0x(b+1)$ (with $0x(b+1)$ repeated b times). Decryption will fail unless $0x(B \oplus i) = 0x(b+1)$. The attacker simply needs to try at most 2^8 values $\Delta_0, \dots, \Delta_{2^8-1}$ until decryption succeeds for some Δ_i , at which point it can deduce that $B = 0x(b+1) \oplus i$. We leave a full description of how to extend this attack so as to learn the entire plaintext—and not just the final block—as an exercise.

A padding-oracle attack on CAPTCHAs. A CAPTCHA is a distorted image of, say, an English word that is easy for humans to read, but hard for a computer to process. CAPTCHAs are used in order to ensure that a human user—and not some automated software—is interacting with a webpage. CAPTCHAs are used, e.g., by online email services to ensure that humans open accounts; this is important to prevent spammers from automatically opening thousands of accounts and using them to send spam.

One way that CAPTCHAs can be configured is as a separate service run on an independent server. In order to see how this works, we denote a webserver by $\mathcal{S}_W$, the CAPTCHA server by $\mathcal{S}_C$, and the user by $\mathcal{U}$. When the user $\mathcal{U}$ loads a webpage served by $\mathcal{S}_W$, the following events occur: $\mathcal{S}_W$ encrypts a random English word w using a key k that was initially shared between $\mathcal{S}_W$ and $\mathcal{S}_C$, and sends the resulting ciphertext to the user (along with the webpage). $\mathcal{U}$ forwards the ciphertext to $\mathcal{S}_C$, who decrypts it, obtains w , and renders a distorted image of w to $\mathcal{U}$. Finally, $\mathcal{U}$ sends the word w back to $\mathcal{S}_W$ for verification. The interesting observation here is that $\mathcal{S}_C$ decrypts any ciphertext it receives from $\mathcal{U}$ and will issue a “bad padding” error message if decryption fails as described earlier. This presents $\mathcal{U}$ with an opportunity to carry out a padding-oracle attack as described above, and thus to solve the CAPTCHA (i.e., to determine w) automatically *without any human involvement*, rendering the CAPTCHA ineffective. While ad hoc measures can be used (e.g., having $\mathcal{S}_C$ return a random image instead of a decryption error), what is really needed is to use an encryption scheme that is CCA-secure.

References and Additional Reading

The modern computational approach to cryptography was initiated in a groundbreaking paper by Goldwasser and Micali [80]. That paper introduced the notion of semantic security, and showed how this goal could be achieved in the setting of public-key encryption (see Chapters 10 and 11). That paper also proposed the notion of indistinguishability (i.e., Definition 3.8), and showed that it implies semantic security. The converse was shown in [125]. The reader is referred to [76] for further discussion of semantic security.

Blum and Micali [37] introduced the notion of pseudorandom generators and proved their existence based on a specific number-theoretic assumption. In the same work, Blum and Micali also pointed out the connection between pseudorandom generators and private-key encryption as in Construction 3.17. The definition of pseudorandom generators given by Blum and Micali is different from the definition we use in this book (Definition 3.14); the latter definition originates in the work of Yao [179], who showed equivalence of the two formulations. Yao also showed constructions of pseudorandom generators based on general assumptions; we will explore this result in Chapter 7.

Formal definitions of security against chosen-plaintext attacks were given by Luby [115] and Bellare et al. [15]. Chosen-ciphertext attacks (in the context of public-key encryption) were first formally defined by Naor and Yung [129] and Rackoff and Simon [147], and were considered also in [61] and [15]. See [101] for other notions of security for private-key encryption.

Pseudorandom functions were defined and constructed by Goldreich et al. [78], and their application to encryption was demonstrated in subsequent work by the same authors [77]. Pseudorandom permutations and strong pseudorandom permutations were studied by Luby and Rackoff [116]. A proof of Proposition 3.27 (the “switching lemma”) can be found in [24]. Block ciphers had been used for many years before they began to be studied in the theoretical sense initiated by the above works. Practical stream ciphers and block ciphers are studied in detail in Chapter 6.

The ECB, CBC, and OFB modes of operation (as well as CFB, a mode of operation not covered here) were standardized along with the DES block cipher [130]. CTR mode was standardized by NIST in 2001. CBC and CTR mode were proven CPA-secure in [15]. For more recent modes of operation, see <http://csrc.nist.gov/groups/ST/toolkit/BCM/index.html>.

The attack on chained CBC was first described by Rogaway (unpublished), and was used to attack SSL/TLS in the so-called “BEAST attack” by Rizzo and Duong. The padding-oracle attack we describe here originated in the work of Vaudenay [171].

Exercises

3.1 Prove Proposition 3.6.

3.2 Prove that Definition 3.8 cannot be satisfied if Π can encrypt arbitrary-length messages and the adversary is *not* restricted to output equal-length messages in experiment $\text{PrivK}_{\mathcal{A}, \Pi}^{\text{eav}}$.

Hint: Let $q(n)$ be a polynomial upper-bound on the length of the ciphertext when Π is used to encrypt a single bit. Then consider an adversary who outputs $m_0 \in \{0, 1\}$ and a uniform $m_1 \in \{0, 1\}^{q(n)+2}$.

3.3 Say $\Pi = (\text{Gen}, \text{Enc}, \text{Dec})$ is such that for $k \in \{0, 1\}^n$, algorithm Enc_k is only defined for messages of length at most $\ell(n)$ (for some polynomial ℓ). Construct a scheme satisfying Definition 3.8 even when the adversary is *not* restricted to outputting equal-length messages in $\text{PrivK}_{\mathcal{A}, \Pi}^{\text{eav}}$.

3.4 Prove the equivalence of Definition 3.8 and Definition 3.9.

3.5 Let $|G(s)| = \ell(|s|)$ for some ℓ . Consider the following experiment:

The PRG indistinguishability experiment $\text{PRG}_{\mathcal{A}, G}(n)$:

- (a) A uniform bit $b \in \{0, 1\}$ is chosen. If $b = 0$ then choose a uniform $r \in \{0, 1\}^{\ell(n)}$; if $b = 1$ then choose a uniform $s \in \{0, 1\}^n$ and set $r := G(s)$.
- (b) The adversary $\mathcal{A}$ is given r , and outputs a bit b' .
- (c) The output of the experiment is defined to be 1 if $b' = b$, and 0 otherwise.

Provide a definition of a pseudorandom generator based on this experiment, and prove that your definition is equivalent to Definition 3.14. (That is, show that G satisfies your definition if and only if it satisfies Definition 3.14.)

3.6 Let G be a pseudorandom generator with expansion factor $\ell(n) > 2n$. In each of the following cases, say whether G' is necessarily a pseudorandom generator. If yes, give a proof; if not, show a counterexample.

- (a) Define $G'(s) \stackrel{\text{def}}{=} G(s_1 \cdots s_{\lfloor n/2 \rfloor})$, where $s = s_1 \cdots s_n$.
- (b) Define $G'(s) \stackrel{\text{def}}{=} G(0^{|s|} \| s)$.
- (c) Define $G'(s) \stackrel{\text{def}}{=} G(s) \| G(s+1)$.

3.7 Prove the converse of Theorem 3.18. Namely, show that if G is not a pseudorandom generator then Construction 3.17 does not have indistinguishable encryptions in the presence of an eavesdropper.

- 3.8 (a) Define a notion of indistinguishability for the encryption of multiple *distinct* messages, in which a scheme need not hide whether the same message is encrypted twice.
- (b) Show that Construction 3.17 does not satisfy your definition.
- (c) Give a construction of a *deterministic* (stateless) encryption scheme that satisfies your definition.

- 3.9 Prove *unconditionally* the existence of a pseudorandom function $F : \{0, 1\}^* \times \{0, 1\}^* \rightarrow \{0, 1\}^*$ with $\ell_{key}(n) = n$ and $\ell_{in}(n) = O(\log n)$.

Hint: Implement a uniform function with logarithmic input length.

- 3.10 Let F be a length-preserving pseudorandom function. For the following constructions of a keyed function $F' : \{0, 1\}^n \times \{0, 1\}^{n-1} \rightarrow \{0, 1\}^{2n}$, state whether F' is a pseudorandom function. If yes, prove it; if not, show an attack.

- (a) $F'_k(x) \stackrel{\text{def}}{=} F_k(0\|x) \parallel F_k(1\|x)$.
- (b) $F'_k(x) \stackrel{\text{def}}{=} F_k(0\|x) \parallel F_k(x\|1)$.

- 3.11 Assuming the existence of pseudorandom functions, prove that there is an encryption scheme that has indistinguishable multiple encryptions in the presence of an eavesdropper (i.e., satisfies Definition 3.19), but is not CPA-secure (i.e., does not satisfy Definition 3.22).

Hint: The scheme need not be “natural.” You will need to use the fact that in a chosen-plaintext attack the adversary can choose its queries to the encryption oracle *adaptively*.

- 3.12 Let F be a keyed function and consider the following experiment:

The PRF indistinguishability experiment $\text{PRF}_{\mathcal{A}, F}(n)$:

- (a) A uniform bit $b \in \{0, 1\}$ is chosen. If $b = 1$ then choose uniform $k \in \{0, 1\}^n$.
- (b) $\mathcal{A}$ is given 1^n for input. If $b = 0$ then $\mathcal{A}$ is given access to a uniform function $f \in \text{Func}_n$. If $b = 1$ then $\mathcal{A}$ is instead given access to $F_k(\cdot)$.
- (c) $\mathcal{A}$ outputs a bit b' .
- (d) The output of the experiment is defined to be 1 if $b' = b$, and 0 otherwise.

Define pseudorandom functions using this experiment, and prove that your definition is equivalent to Definition 3.25.

- 3.13 Consider the following keyed function F : For security parameter n , the key is an $n \times n$ boolean matrix A and an n -bit boolean vector b . Define $F_{A,b} : \{0, 1\}^n \rightarrow \{0, 1\}^n$ by $F_{A,b}(x) \stackrel{\text{def}}{=} Ax + b$, where all operations are done modulo 2. Show that F is not a pseudorandom function.

- 3.14 Prove that if F is a length-preserving pseudorandom function, then $G(s) \stackrel{\text{def}}{=} F_s(1) \| F_s(2) \| \cdots \| F_s(\ell)$ is a pseudorandom generator with expansion factor $\ell \cdot n$.
- 3.15 Define a notion of perfect secrecy under a chosen-plaintext attack by adapting Definition 3.22. Show that the definition cannot be achieved.
- 3.16 Prove Proposition 3.27.

Hint: Use the results of Appendix A.4.

- 3.17 Assume pseudorandom permutations exist. Show that there exists a function F' that is a pseudorandom permutation but is *not* a strong pseudorandom permutation.

Hint: Construct F' such that $F'_k(k) = 0^{|k|}$.

- 3.18 Let F be a pseudorandom permutation, and define a fixed-length encryption scheme (**Enc**, **Dec**) as follows: On input $m \in \{0, 1\}^{n/2}$ and key $k \in \{0, 1\}^n$, algorithm **Enc** chooses a uniform string $r \in \{0, 1\}^{n/2}$ of length $n/2$ and computes $c := F_k(r \| m)$.

Show how to decrypt, and prove that this scheme is CPA-secure for messages of length $n/2$. (If you are looking for a real challenge, prove that this scheme is CCA-secure if F is a *strong* pseudorandom permutation.)

- 3.19 Let F be a pseudorandom function and G be a pseudorandom generator with expansion factor $\ell(n) = n + 1$. For each of the following encryption schemes, state whether the scheme has indistinguishable encryptions in the presence of an eavesdropper and whether it is CPA-secure. (In each case, the shared key is a uniform $k \in \{0, 1\}^n$.) Explain your answer.

- (a) To encrypt $m \in \{0, 1\}^{n+1}$, choose uniform $r \in \{0, 1\}^n$ and output the ciphertext $\langle r, G(r) \oplus m \rangle$.
- (b) To encrypt $m \in \{0, 1\}^n$, output the ciphertext $m \oplus F_k(0^n)$.
- (c) To encrypt $m \in \{0, 1\}^{2n}$, parse m as $m_1 \| m_2$ with $|m_1| = |m_2|$, then choose uniform $r \in \{0, 1\}^n$ and send $\langle r, m_1 \oplus F_k(r), m_2 \oplus F_k(r+1) \rangle$.

- 3.20 Consider a stateful variant of CBC-mode encryption where the sender simply increments the IV by 1 each time a message is encrypted (rather than choosing IV at random each time). Show that the resulting scheme is *not* CPA-secure.
- 3.21 What is the effect of a single-bit error in the ciphertext when using the CBC, OFB, and CTR modes of operation?
- 3.22 What is the effect of a dropped ciphertext block (e.g., if the transmitted ciphertext $c_1, c_2, c_3, \dots$ is received as $c_1, c_3, \dots$) when using the CBC, OFB, and CTR modes of operation?

- 3.23 Say CBC-mode encryption is used with a block cipher having a 256-bit key and 128-bit block length to encrypt a 1024-bit message. What is the length of the resulting ciphertext?
- 3.24 Give the details of the proof by reduction for Equation (3.12).
- 3.25 Let F be a pseudorandom function such that for $k \in \{0, 1\}^n$ the function F_k maps $\ell_{in}(n)$ -bit inputs to $\ell_{out}(n)$ -bit outputs.
- (a) Consider implementing CTR-mode encryption using F . For which functions ℓ_{in}, ℓ_{out} is the resulting encryption scheme CPA-secure?
 - (b) Consider implementing CTR-mode encryption using F , but only for *fixed-length* messages of length $\ell(n)$ (which is an integer multiple of $\ell_{out}(n)$). For which $\ell_{in}, \ell_{out}, \ell$ does the scheme have indistinguishable encryptions in the presence of an eavesdropper?
- 3.26 For any function $g : \{0, 1\}^n \rightarrow \{0, 1\}^n$, define $g^\$ (\cdot)$ to be a *probabilistic* oracle that, on input 1^n , chooses uniform $r \in \{0, 1\}^n$ and returns $\langle r, g(r) \rangle$. A keyed function F is a *weak pseudorandom function* if for all PPT algorithms D , there exists a negligible function negl such that:

$$\left| \Pr[D^{F_k^\$(\cdot)}(1^n) = 1] - \Pr[D^{f^\$(\cdot)}(1^n) = 1] \right| \leq \text{negl}(n),$$

where $k \in \{0, 1\}^n$ and $f \in \text{Func}_n$ are chosen uniformly.

- (a) Prove that if F is pseudorandom then it is weakly pseudorandom.
- (b) Let F' be a pseudorandom function, and define

$$F_k(x) \stackrel{\text{def}}{=} \begin{cases} F'_k(x) & \text{if } x \text{ is even} \\ F'_k(x+1) & \text{if } x \text{ is odd.} \end{cases}$$

Prove that F is weakly pseudorandom, but *not* pseudorandom.

- (c) Is CTR-mode encryption using a weak pseudorandom function necessarily CPA-secure? Does it necessarily have indistinguishable encryptions in the presence of an eavesdropper? Prove your answers.
 - (d) Prove that Construction 3.30 is CPA-secure if F is a weak pseudorandom function.
- 3.27 Let F be a pseudorandom permutation. Consider the mode of operation in which a uniform value $\text{ctr} \in \{0, 1\}^n$ is chosen, and the i th ciphertext block c_i is computed as $c_i := F_k(\text{ctr} + i + m_i)$. Show that this scheme does not have indistinguishable encryptions in the presence of an eavesdropper.
- 3.28 Show that the CBC, OFB, and CTR modes of operation do not yield CCA-secure encryption schemes (regardless of F).

- 3.29 Let $\Pi_1 = (\text{Enc}_1, \text{Dec}_1)$ and $\Pi_2 = (\text{Enc}_2, \text{Dec}_2)$ be two encryption schemes for which it is known that at least one is CPA-secure (but you don't know which one). Show how to construct an encryption scheme Π that is guaranteed to be CPA-secure as long as at least one of Π_1 or Π_2 is CPA-secure. Provide a full proof of your solution.

Hint: Generate two plaintext messages from the original plaintext so that knowledge of either one reveals nothing about the original plaintext, but knowledge of both enables the original plaintext to be computed.

- 3.30 Write pseudocode for obtaining the entire plaintext via a padding-oracle attack on CBC-mode encryption using PKCS #5 padding, as described in the text.
- 3.31 Describe a padding-oracle attack on CTR-mode encryption (assuming PKCS #5 padding is used to pad messages to a multiple of the block length before encrypting).

Chapter 4

Message Authentication Codes

4.1 Message Integrity

4.1.1 Secrecy vs. Integrity

One of the most basic goals of cryptography is to enable parties to communicate over an *open communication channel* in a secure way. But what does “secure communication” entail? In Chapter 3 we showed that it is possible to obtain *secret* communication over an open channel. That is, we showed how encryption can be used to prevent an eavesdropper (or possibly a more active adversary) from learning anything about the content of messages sent over an unprotected communication channel. However, not all security concerns are related to secrecy. In many cases, it is of equal or greater importance to guarantee *message integrity* (or *message authentication*) in the sense that each party should be able to identify when a message it receives was sent by the party claiming to send it, and was not modified in transit. We look at two canonical examples.

Consider the case of a user communicating with their bank over the Internet. When the bank receives a request to transfer \$1,000 from the user’s account to the account of some other user X , the bank has to consider the following:

1. Is the request authentic? That is, did the user in question really issue this request, or was the request issued by an adversary (perhaps X itself) who is impersonating the legitimate user?
2. Assuming a transfer request was issued by the legitimate user, are the details of the request as received exactly those intended by the legitimate user? Or was, e.g., the transfer amount modified?

Note that standard error-correction techniques do not suffice for the second concern. Error-correcting codes are only intended to detect and recover from “random” errors that affect only a small portion of the transmission, but they do nothing to protect against a malicious adversary who can choose exactly where to introduce an arbitrary number of errors.

A second scenario where the need for message integrity arises in practice is with regard to web cookies. The HTTP protocol used for web traffic is *stateless*, and so when a client and server communicate in some session (e.g., when

a user [client] shops at a merchant's [server's] website), any state generated as part of that session (e.g., the contents of the user's shopping cart) is often placed in a "cookie" that is stored by the client and sent from the client to the server as part of each message the client sends. Assume that the cookie stored by some user includes the items in the user's shopping cart along with a price for each item, as might be done if the merchant offers different prices to different users (reflecting, e.g., discounts and promotions, or user-specific pricing). It should be infeasible here for the user to modify the cookie that it stores so as to alter the prices of the items in its cart. The merchant thus needs a technique to ensure the integrity of the cookie that it stores at the user. Note that the contents of the cookie (namely, the items and their prices) are not secret and, in fact, must be known by the user. The problem here is thus purely one of integrity.

In general, one cannot assume the integrity of communication without taking specific measures to ensure it. Indeed, any unprotected online purchase order, online banking operation, email, or SMS message cannot, in general, be trusted to have originated from the claimed source and to have been unmodified in transit. Unfortunately, people are in general trusting and thus information like the caller-ID or an email return address are taken to be "proofs of origin" in many cases, even though they are relatively easy to forge. This leaves the door open to potentially damaging attacks.

In this chapter we will show how to achieve message integrity by using cryptographic techniques to prevent the *undetected* tampering of messages sent over an open communication channel. Note that we cannot hope to prevent adversarial tampering of messages altogether, as that can only be defended against at the physical level. Instead, what we will guarantee is that any such tampering will be detected by the honest parties.

4.1.2 Encryption vs. Message Authentication

Just as the goals of secrecy and message integrity are different, so are the techniques and tools for achieving them. Unfortunately, secrecy and integrity are often confused and unnecessarily intertwined, so let us be clear up front: encryption does *not* (in general) provide any integrity, and encryption should never be used with the intent of achieving message authentication unless it is specifically designed with that purpose in mind (something we will return to in Section 4.5).

One might mistakenly think that encryption solves the problem of message authentication. (In fact, this is a common error.) This is due to the fuzzy, and incorrect, reasoning that since a ciphertext completely hides the contents of the message, an adversary cannot possibly modify an encrypted message in any meaningful way. Despite its intuitive appeal, this reasoning is completely false. We illustrate this point by showing that all the encryption schemes we have seen thus far do not provide message integrity.

Encryption using stream ciphers. Consider the simple encryption scheme in which $\text{Enc}_k(m)$ computes the ciphertext $c := G(k) \oplus m$, where G is a pseudorandom generator. Ciphertexts in this case are very easy to manipulate: flipping any bit in the ciphertext c results in the same bit being flipped in the message that is recovered upon decryption. Thus, given a ciphertext c that encrypts a (possibly unknown) message m , it is possible to generate a modified ciphertext c' such that $m' := \text{Dec}_k(c')$ is the same as m but with one (or more) of the bits flipped. This simple attack can have severe consequences. As an example, consider the case of a user encrypting some dollar amount he wants to transfer from his bank account, where the amount is represented in binary. Flipping the least significant bit has the effect of changing this amount by only \$1, but flipping the 11th least significant bit changes the amount by more than \$1,000! (Interestingly, the adversary in this example does not necessarily learn whether it is increasing or decreasing the initial amount, i.e., whether it is flipping a 0 to a 1 or vice versa. But if the adversary has some partial knowledge about the amount—say, that it is less than \$1,000 to begin with—then the modifications it introduces can have a predictable effect.) We stress that this attack does not contradict the *secrecy* of the encryption scheme (in the sense of Definition 3.8). In fact, the exact same attack applies to the one-time pad encryption scheme, showing that even perfect secrecy is not sufficient to ensure the most basic level of message integrity.

Encryption using block ciphers. The attack described above utilizes the fact that flipping a single bit in a ciphertext keeps the underlying plaintext unchanged except for the corresponding bit (which is also flipped). The same attack applies to the OFB- and CTR-mode encryption schemes, which also encrypt messages by XORing them with a pseudorandom stream (albeit one that changes each time a message is encrypted). We thus see that even using CPA-secure encryption is not enough to prevent message tampering.

One may hope that attacking ECB- or CBC-mode encryption would be more difficult since decryption in these cases involves inverting a (strong) pseudorandom permutation F , and we expect that $F_k^{-1}(x)$ and $F_k^{-1}(x')$ will be completely uncorrelated even if x and x' differ in only a single bit. (Of course, ECB mode does not even guarantee the most basic notion of secrecy, but that does not matter for the present discussion.) Nevertheless, single-bit modifications of a ciphertext still cause partially predictable changes in the plaintext. For example, when using ECB mode, flipping a bit in the i th block of the ciphertext affects *only* the i th block of the plaintext—all other blocks remain unchanged. Although the effect on the i th block of the plaintext may be impossible to predict, changing that one block (while leaving everything else unchanged) may represent a harmful attack. Moreover, the order of plaintext blocks can be changed (without garbling any block) by simply changing the order of the corresponding ciphertext blocks, and the message can be truncated by just dropping ciphertext blocks.

Similarly, when using CBC mode, flipping the j th bit of the IV changes only

the j th bit of the first message block m_1 (since $m_1 := F_k^{-1}(c_1) \oplus IV'$, where IV' is the modified IV); all plaintext blocks other than the first remain unchanged (since the i th block of the plaintext is computed as $m_i := F_k^{-1}(c_i) \oplus c_{i-1}$, and blocks c_i and c_{i-1} have not been modified). Therefore, the first block of a CBC-encrypted message can be changed arbitrarily. This is a serious concern in practice since the first block often contains important header information.

Finally, note that all encryption schemes we have seen thus far have the property that *every* ciphertext (perhaps satisfying some length constraint) corresponds to *some* valid message. So it is trivial for an adversary to “spoof” a message on behalf of one of the communicating parties—by sending some arbitrary ciphertext—even if the adversary has no idea what the underlying message will be. As we will see when we formally define authenticated encryption in Section 4.5, even an attack of this sort should be ruled out.

4.2 Message Authentication Codes – Definitions

We have seen that, in general, encryption does not solve the problem of message integrity. Rather, an additional mechanism is needed that will enable the communicating parties to know whether or not a message was tampered with. The right tool for this task is a *message authentication code* (MAC).

The aim of a message authentication code is to prevent an adversary from modifying a message sent by one party to another, or from injecting a new message, without the receiver detecting that the message did not originate from the intended party. As in the case of encryption, this is only possible if the communicating parties share some secret that the adversary does not know (otherwise nothing can prevent an adversary from impersonating the party sending the message). Here, we will continue to consider the private-key setting where the parties share the same secret key.¹

The Syntax of a Message Authentication Code

Before formally defining security of a message authentication code, we first define what a MAC is and how it is used. Two users who wish to communicate in an authenticated manner begin by generating and sharing a secret key k in advance of their communication. When one party wants to send a message m to the other, she computes a MAC tag (or simply a tag) t based on the message and the shared key, and sends the message m and the tag t to the other party. The tag is computed using a *tag-generation algorithm* denoted by Mac ; thus, rephrasing what we have already said, the sender of a message m computes $t \leftarrow \text{Mac}_k(m)$ and transmits (m, t) to the receiver. Upon receiving (m, t) , the

¹In the web-cookie example discussed earlier, the merchant is (in effect) communicating “with itself” with the user acting as a communication channel. In that setting, the server alone needs to know the key since it is acting as both sender and receiver.

second party *verifies* whether t is a valid tag on the message m (with respect to the shared key) or not. This is done by running a *verification algorithm* Vrfy that takes as input the shared key as well as a message m and a tag t , and indicates whether the given tag is valid. Formally:

DEFINITION 4.1 A message authentication code (or MAC) consists of three probabilistic polynomial-time algorithms $(\text{Gen}, \text{Mac}, \text{Vrfy})$ such that:

1. The key-generation algorithm Gen takes as input the security parameter 1^n and outputs a key k with $|k| \geq n$.
2. The tag-generation algorithm Mac takes as input a key k and a message $m \in \{0, 1\}^*$, and outputs a tag t . Since this algorithm may be randomized, we write this as $t \leftarrow \text{Mac}_k(m)$.
3. The deterministic verification algorithm Vrfy takes as input a key k , a message m , and a tag t . It outputs a bit b , with $b = 1$ meaning valid and $b = 0$ meaning invalid. We write this as $b := \text{Vrfy}_k(m, t)$.

It is required that for every n , every key k output by $\text{Gen}(1^n)$, and every $m \in \{0, 1\}^*$, it holds that $\text{Vrfy}_k(m, \text{Mac}_k(m)) = 1$.

If there is a function ℓ such that for every k output by $\text{Gen}(1^n)$, algorithm Mac_k is only defined for messages $m \in \{0, 1\}^{\ell(n)}$, then we call the scheme a fixed-length MAC for messages of length $\ell(n)$.

As with private-key encryption, $\text{Gen}(1^n)$ almost always simply chooses a uniform key $k \in \{0, 1\}^n$, and we omit Gen in that case.

Canonical verification. For deterministic message authentication codes (that is, where Mac is a deterministic algorithm), the canonical way to perform verification is to simply re-compute the tag and check for equality. In other words, $\text{Vrfy}_k(m, t)$ first computes $\tilde{t} := \text{Mac}_k(m)$ and then outputs 1 if and only if $\tilde{t} = t$. Even for deterministic MACs, however, it is useful to define a separate Vrfy algorithm in order to explicitly distinguish the semantics of *authenticating* a message vs. *verifying* its authenticity.

Security of Message Authentication Codes

We now define the default notion of security for message authentication codes. The intuitive idea behind the definition is that no efficient adversary should be able to generate a valid tag on any “new” message that was not previously sent (and authenticated) by one of the communicating parties.

As with any security definition, to formalize this notion we have to define both the adversary’s power as well as what should be considered a “break.” As usual, we consider only probabilistic polynomial-time adversaries² and

²See Section 4.6 for a discussion of information-theoretic message authentication, where no computational restrictions are placed on the adversary.

so the real question is how we model the adversary's interaction with the communicating parties. In the setting of message authentication, an adversary observing the communication between the honest parties may be able to see all the messages sent by these parties along with their corresponding MAC tags. The adversary may also be able to influence the *content* of these messages, whether directly or indirectly (if, e.g., external actions of the adversary affect the messages sent by the parties). This is true, for example, in the web cookie example from earlier, where the user's own actions influence the contents of the cookie being stored on his computer.

To formally model the above we allow the adversary to request MAC tags for *any* messages of its choice. Formally, we give the adversary access to a *MAC oracle* $\text{Mac}_k(\cdot)$; the adversary can repeatedly submit any message m of its choice to this oracle, and is given in return a tag $t \leftarrow \text{Mac}_k(m)$. (For a fixed-length MAC, only messages of the correct length can be submitted.)

We will consider it a “break” of the scheme if the adversary is able to output any message m along with a tag t such that: (1) t is a valid tag on the message m (i.e., $\text{Vrfy}_k(m, t) = 1$), and (2) the adversary had not previously requested a MAC tag on the message m (i.e., from its oracle). The first condition means that if the adversary were to send (m, t) to one of the honest parties, then this party would be mistakenly fooled into thinking that m originated from the legitimate party since $\text{Vrfy}_k(m, t) = 1$. The second condition is required because it is always possible for the adversary to just copy a message and MAC tag that were previously sent by one of the legitimate parties (and, of course, these would be accepted as valid). Such a *replay attack* is not considered a “break” of the message authentication code. This does *not* mean that replay attacks are not a security concern; they are, and we will have more to say about them below.

A MAC satisfying the level of security specified above is said to be *existentially unforgeable under an adaptive chosen-message attack*. “Existential unforgeability” refers to the fact that the adversary must not be able to forge a valid tag on *any* message, and “adaptive chosen-message attack” refers to the fact that the adversary is able to obtain MAC tags on arbitrary messages chosen adaptively during its attack.

Toward the formal definition, consider the following experiment for a message authentication code $\Pi = (\text{Gen}, \text{Mac}, \text{Vrfy})$, an adversary $\mathcal{A}$, and value n for the security parameter:

The message authentication experiment $\text{Mac-forge}_{\mathcal{A}, \Pi}(n)$:

1. A key k is generated by running $\text{Gen}(1^n)$.
2. The adversary $\mathcal{A}$ is given input 1^n and oracle access to $\text{Mac}_k(\cdot)$. The adversary eventually outputs (m, t) . Let $\mathcal{Q}$ denote the set of all queries that $\mathcal{A}$ asked its oracle.
3. $\mathcal{A}$ succeeds if and only if (1) $\text{Vrfy}_k(m, t) = 1$ and (2) $m \notin \mathcal{Q}$. In that case the output of the experiment is defined to be 1.

A MAC is secure if no efficient adversary can succeed in the above experiment with non-negligible probability:

DEFINITION 4.2 *A message authentication code $\Pi = (\text{Gen}, \text{Mac}, \text{Vrfy})$ is existentially unforgeable under an adaptive chosen-message attack, or just secure, if for all probabilistic polynomial-time adversaries $\mathcal{A}$, there is a negligible function negl such that:*

$$\Pr[\text{Mac-forge}_{\mathcal{A}, \Pi}(n) = 1] \leq \text{negl}(n).$$

Is the definition too strong? The above definition is rather strong in two respects. First, the adversary is allowed to request MAC tags for *any* messages of its choice. Second, the adversary is considered to have “broken” the scheme if it can output a valid tag on *any* previously unauthenticated message. One might object that both these components of the definition are unrealistic and overly strong: in “real-world” usage of a MAC, the honest parties would only authenticate “meaningful” messages (over which the adversary might have only limited control), and similarly it should only be considered a breach of security if the adversary can forge a valid tag on a “meaningful” message. Why not tailor the definition to capture this?

The crucial point is that what constitutes a meaningful message is entirely *application dependent*. While some applications of a MAC may only ever authenticate English-text messages, other applications may authenticate spreadsheet files, others database entries, and others raw data. Protocols may also be designed where *anything* will be authenticated—in fact, certain protocols for entity authentication do exactly this. By making the definition of security for MACs as strong as possible, we ensure that secure MACs are broadly applicable for a wide range of purposes, without having to worry about compatibility of the MAC with the semantics of the application.

Replay attacks. We emphasize that the above definition, and message authentication codes on their own, offer no protection against *replay attacks* whereby a previously sent message (and its MAC tag) are replayed to an honest party. Nevertheless, replay attacks are a serious concern! Consider again the scenario where a user (say, Alice) sends a request to her bank to transfer \$1,000 from her account to some other user (say, Bob). In doing so, Alice can compute a MAC tag and append it to the request so the bank knows the request is authentic. If the MAC is secure, Bob will be unable to intercept the request and change the amount to \$10,000 because this would involve forging a valid tag on a previously unauthenticated message. However, nothing prevents Bob from intercepting Alice’s message and replaying it *ten times* to the bank. If the bank accepts each of these messages, the net effect is that \$10,000 will be transferred to Bob’s account rather than the desired \$1,000.

Despite the real threat that replay attacks represent, a MAC by itself *cannot* protect against such attacks since the definition of a MAC (Definition 4.1)

does not incorporate any notion of *state* into the verification algorithm (and so every time a valid pair (m, t) is presented to the verification algorithm, it will always output 1). Rather, protection against replay attacks—if such protection is necessary at all—must be handled by some higher-level application. The reason the definition of a MAC is structured this way is, once again, because we are unwilling to assume any semantics regarding applications that use MACs; in particular, the decision as to whether or not a replayed message should be treated as “valid” may be application dependent.

Two common techniques for preventing replay attacks are to use *sequence numbers* (also known as *counters*) or *time-stamps*. The first approach, described (in a more general context) in Section 4.5.3, requires the communicating users to maintain (synchronized) state, and can be problematic when users communicate over a lossy channel where messages are occasionally dropped (though this problem can be mitigated). In the second approach, using time-stamps, the sender prepends the current time T (say, to the nearest millisecond) to the message before authenticating, and sends T along with the message and the resulting tag t . When the receiver obtains T, m, t , it verifies that t is a valid tag on $T\|m$ and that T is within some acceptable clock skew from the current time T' at the receiver. This method has certain drawbacks as well, including the need for the sender and receiver to maintain closely synchronized clocks, and the possibility that a replay attack can still take place if it is done quickly enough (specifically, within the acceptable time window).

Strong MACs. As defined, a secure MAC ensures that an adversary cannot generate a valid tag on a new message that was never previously authenticated. But it does not rule out the possibility that an attacker might be able to generate a new *tag* on a previously authenticated message. That is, a MAC guarantees that if an attacker learns tags $t_1, \dots$ on messages $m_1, \dots$, then it will not be able to forge a valid tag t on any message $m \notin \{m_1, \dots\}$. However, it may be possible for an adversary to “forge” a *different* valid tag $t'_1 \neq t_1$ on the message m_1 . In general, this type of adversarial behavior is not a concern. Nevertheless, in some settings it is useful to consider a stronger definition of security for MACs where such behavior is ruled out.

Formally, we consider a modified experiment **Mac-sforge** that is defined in exactly the same way as **Mac-forge**, except that now the set $\mathcal{Q}$ contains *pairs* of oracle queries and their associated responses. (That is, $(m, t) \in \mathcal{Q}$ if $\mathcal{A}$ queried $\text{Mac}_k(m)$ and received in response the tag t .) The adversary $\mathcal{A}$ succeeds (and experiment **Mac-sforge** evaluates to 1) if and only if $\mathcal{A}$ outputs (m, t) such that $\text{Vrfy}_k(m, t) = 1$ and $(m, t) \notin \mathcal{Q}$.

DEFINITION 4.3 A message authentication code $\Pi = (\text{Gen}, \text{Mac}, \text{Vrfy})$ is strongly secure, or a strong MAC, if for all probabilistic polynomial-time adversaries $\mathcal{A}$, there is a negligible function negl such that:

$$\Pr[\text{Mac-sforge}_{\mathcal{A}, \Pi}(n) = 1] \leq \text{negl}(n).$$

It is not hard to see that if a secure MAC uses canonical verification then it is also strongly secure. This is important since all real-world MACs use canonical verification. We leave the proof of the following as an exercise.

PROPOSITION 4.4 *Let $\Pi = (\text{Gen}, \text{Mac}, \text{Vrfy})$ be a secure MAC that uses canonical verification. Then Π is a strong MAC.*

Verification Queries

Definitions 4.2 and 4.3 give the adversary access to a MAC oracle, which corresponds to a real-world adversary who can influence an honest sender to generate a tag for some message m . One could also consider an adversary who interacts with an honest receiver, sending m', t' to the receiver to learn whether $\text{Vrfy}_k(m', t') = 1$. Such an adversary could be captured formally in the natural way by giving the adversary in the above definitions access to a verification oracle as well.

A definition that incorporates a verification oracle in this way is, perhaps, the “right” way to define security for message authentication codes. It turns out, however, that for MACs that use canonical verification it makes no difference: any such MAC that satisfies Definition 4.2 also satisfies the definitional variant in which verification queries are allowed. Similarly, any strong MAC automatically also remains secure even in a setting where verification queries are possible. (This, in fact, serves as one motivation for the definition of strong security for MACs.) For MACs that do not use canonical verification, however, allowing verification queries can make a difference; see Exercises 4.2 and 4.3. Since most MACs covered in this book (as well as MACs used in practice) use canonical verification, we use the traditional definitions that omit access to a verification oracle.

A potential timing attack. One issue not addressed by the above is the possibility of carrying out a *timing attack* on MAC verification. Here, we consider an adversary who can send message/tag pairs to the receiver—thus using the receiver as a verification oracle—and learn not only whether the receiver accepts or rejects, but also the *time* it takes for the receiver to make this decision. We show that if such an attack is possible then a natural implementation of MAC verification leads to an easily exploitable vulnerability.

(Note that in our usual cryptographic definitions of security, the attacker learns only the output of the oracles it has access to, but nothing else. The attack we describe here, which is an example of a *side-channel attack*, shows that certain real-world attacks are not captured by the usual definitions.)

Concretely, assume a MAC using canonical verification. To verify a tag t on a message m , the receiver computes $t' := \text{Mac}_k(m)$ and then compares t' to t , outputting 1 if and only if t' and t are equal. Assume this comparison is implemented using a standard routine (like `strcmp` in C) that compares t and t' one byte at a time, and rejects as soon as the first unequal byte is

encountered. The observation is that, when implemented in this way, the *time* to reject differs depending on the *position* of the first unequal byte.

It is possible to use this seemingly inconsequential information to forge a tag on any desired message m . The basic idea is this: say the attacker knows the first i bytes of the correct tag for m . (At the outset, $i = 0$.) The attacker will learn the next byte of the correct tag by sending $(m, t_0), \dots, (m, t_{255})$ to the receiver, where t_j is the string with the first i bytes set correctly, the $(i + 1)$ st byte equal to j (in hexadecimal), and the remaining bytes set to $0x00$. All of these tags will likely be rejected (if not, then the attacker succeeds anyway); however, for exactly one of these tags the first $(i + 1)$ bytes will match the correct tag and rejection will take slightly longer than the rest. If t_j is the tag that caused rejection to take the longest, the attacker learns that the $(i + 1)$ st byte of the correct tag is j . In this way, the attacker learns each byte of the correct tag using at most 256 queries to the verification oracle. For a 16-byte tag, this attack requires only 4096 queries in the worst case.

One might wonder whether this attack is realistic, as it requires access to a verification oracle as well as the ability to measure the difference in time taken to compare i vs. $i + 1$ bytes. In fact, exactly such attacks have been carried out against real systems! As just one example, MACs were used to verify code updates in the Xbox 360, and the implementation of MAC verification used there had a difference of 2.2 milliseconds between rejection times. Attackers were able to exploit this and load pirated games onto the hardware.

Based on the above, we conclude that MAC verification should use *time-independent* string comparison that always compares *all* bytes.

4.3 Constructing Secure Message Authentication Codes

4.3.1 A Fixed-Length MAC

Pseudorandom functions are a natural tool for constructing secure message authentication codes. Intuitively, if the MAC tag t is obtained by applying a pseudorandom function to the message m , then forging a tag on a previously unauthenticated message requires the adversary to correctly guess the value of the pseudorandom function at a “new” input point. The probability of guessing the value of a *random* function on a new point is 2^{-n} (if the output length of the function is n). The probability of guessing such a value for a *pseudorandom* function can be only negligibly greater.

The above idea, shown in Construction 4.5, works for constructing a secure *fixed-length* MAC for messages of length n (since our pseudorandom functions by default have n -bit block length). This is useful, but falls short of our goal. In Section 4.3.2, we show how to extend this to handle messages of arbitrary length. We explore more efficient constructions of MACs for arbitrary-length messages in Sections 4.4 and 5.3.2.

CONSTRUCTION 4.5

Let F be a pseudorandom function. Define a fixed-length MAC for messages of length n as follows:

- **Mac**: on input a key $k \in \{0, 1\}^n$ and a message $m \in \{0, 1\}^n$, output the tag $t := F_k(m)$. (If $|m| \neq |k|$ then output nothing.)
- **Vrfy**: on input a key $k \in \{0, 1\}^n$, a message $m \in \{0, 1\}^n$, and a tag $t \in \{0, 1\}^n$, output 1 if and only if $t \stackrel{?}{=} F_k(m)$. (If $|m| \neq |k|$, then output 0.)

A fixed-length MAC from any pseudorandom function.

THEOREM 4.6 *If F is a pseudorandom function, then Construction 4.5 is a secure fixed-length MAC for messages of length n .*

PROOF As in previous uses of pseudorandom functions, this proof follows the paradigm of first analyzing the security of the scheme using a truly random function, and then considering the result of replacing the truly random function with a pseudorandom one.

Let $\mathcal{A}$ be a probabilistic polynomial-time adversary. Consider the message authentication code $\tilde{\Pi} = (\widetilde{\text{Gen}}, \widetilde{\text{Mac}}, \widetilde{\text{Vrfy}})$ which is the same as $\Pi = (\text{Mac}, \text{Vrfy})$ in Construction 4.5 except that a truly random function f is used instead of the pseudorandom function F_k . That is, $\widetilde{\text{Gen}}(1^n)$ works by choosing a uniform function $f \in \text{Func}_n$, and $\widetilde{\text{Mac}}$ computes a tag just as Mac does except that f is used instead of F_k . It is immediate that

$$\Pr[\text{Mac-forge}_{\mathcal{A}, \tilde{\Pi}}(n) = 1] \leq 2^{-n} \quad (4.1)$$

because for any message $m \notin \mathcal{Q}$, the value $t = f(m)$ is uniformly distributed in $\{0, 1\}^n$ from the point of view of the adversary $\mathcal{A}$.

We next show that there is a negligible function negl such that

$$\left| \Pr[\text{Mac-forge}_{\mathcal{A}, \Pi}(n) = 1] - \Pr[\text{Mac-forge}_{\mathcal{A}, \tilde{\Pi}}(n) = 1] \right| \leq \text{negl}(n); \quad (4.2)$$

combined with Equation (4.1), this shows that

$$\Pr[\text{Mac-forge}_{\mathcal{A}, \Pi}(n) = 1] \leq 2^{-n} + \text{negl}(n),$$

proving the theorem.

To prove Equation (4.2), we construct a polynomial-time distinguisher D that is given oracle access to some function, and whose goal is to determine whether this function is pseudorandom (i.e., equal to F_k for uniform $k \in \{0, 1\}^n$) or random (i.e., equal to f for uniform $f \in \text{Func}_n$). To do this, D emulates the message authentication experiment for $\mathcal{A}$ and observes whether $\mathcal{A}$ succeeds in outputting a valid tag on a “new” message. If so, D guesses that its oracle is a pseudorandom function; otherwise, D guesses that its oracle is a random function. In detail:

Distinguisher D :

D is given input 1^n and access to an oracle $\mathcal{O} : \{0, 1\}^n \rightarrow \{0, 1\}^n$, and works as follows:

1. Run $\mathcal{A}(1^n)$. Whenever $\mathcal{A}$ queries its MAC oracle on a message m (i.e., whenever $\mathcal{A}$ requests a tag on a message m), answer this query in the following way:
 - Query $\mathcal{O}$ with m and obtain response t ; return t to $\mathcal{A}$.
2. When $\mathcal{A}$ outputs (m, t) at the end of its execution, do:
 - (a) Query $\mathcal{O}$ with m and obtain response $\hat{t}$.
 - (b) If (1) $\hat{t} = t$ and (2) $\mathcal{A}$ never queried its MAC oracle on m , then output 1; otherwise, output 0.

It is clear that D runs in polynomial time.

Notice that if D 's oracle is a pseudorandom function, then the view of $\mathcal{A}$ when run as a sub-routine by D is distributed identically to the view of $\mathcal{A}$ in experiment $\text{Mac-forge}_{\mathcal{A}, \Pi}(n)$. Furthermore, D outputs 1 exactly when $\text{Mac-forge}_{\mathcal{A}, \Pi}(n) = 1$. Therefore

$$\Pr \left[D^{F_k(\cdot)}(1^n) = 1 \right] = \Pr \left[\text{Mac-forge}_{\mathcal{A}, \Pi}(n) = 1 \right],$$

where $k \in \{0, 1\}^n$ is chosen uniformly in the above. If D 's oracle is a random function, then the view of $\mathcal{A}$ when run as a sub-routine by D is distributed identically to the view of $\mathcal{A}$ in experiment $\text{Mac-forge}_{\mathcal{A}, \tilde{\Pi}}(n)$, and again D outputs 1 exactly when $\text{Mac-forge}_{\mathcal{A}, \tilde{\Pi}}(n) = 1$. Thus,

$$\Pr \left[D^{f(\cdot)}(1^n) = 1 \right] = \Pr \left[\text{Mac-forge}_{\mathcal{A}, \tilde{\Pi}}(n) = 1 \right],$$

where $f \in \text{Func}_n$ is chosen uniformly.

Since F is a pseudorandom function and D runs in polynomial time, there exists a negligible function negl such that

$$\left| \Pr \left[D^{F_k(\cdot)}(1^n) = 1 \right] - \Pr \left[D^{f(\cdot)}(1^n) = 1 \right] \right| \leq \text{negl}(n).$$

This implies Equation (4.2), completing the proof of the theorem. ■

4.3.2 Domain Extension for MACs

Construction 4.5 is important in that it shows a general paradigm for constructing secure message authentication codes from pseudorandom functions. Unfortunately, the construction is only capable of handling *fixed-length* messages that are furthermore rather short.³ These limitations are unacceptable

³Given a pseudorandom function taking arbitrary-length inputs, Construction 4.5 would yield a secure MAC for messages of arbitrary length. Likewise, a pseudorandom function

in most applications. We show here how a general MAC, handling arbitrary-length messages, can be constructed from any fixed-length MAC for messages of length n . The construction we show is not very efficient and is unlikely to be used in practice. Indeed, far more efficient constructions of secure MACs are known, as we discuss in Sections 4.4 and 5.3.2. We include the present construction for its simplicity and generality.

Let $\Pi' = (\text{Mac}', \text{Vrfy}')$ be a secure fixed-length MAC for messages of length n . Before presenting the construction of a MAC for arbitrary-length messages based on Π' , we rule out some simple ideas and describe some canonical attacks that must be prevented. Below, we parse the message m to be authenticated as a sequence of blocks $m_1, \dots, m_d$; note that, since our aim is to handle messages of arbitrary length, d can vary from message to message.

1. A natural first idea is to simply authenticate each block separately, i.e., compute $t_i := \text{Mac}'_k(m_i)$ for all i , and output $\langle t_1, \dots, t_d \rangle$ as the tag. This prevents an adversary from sending any previously unauthenticated block without being detected. However, it does not prevent a *block re-ordering attack* in which the attacker shuffles the order of blocks in an authenticated message. Specifically, if $\langle t_1, t_2 \rangle$ is a valid tag on the message m_1, m_2 (with $m_1 \neq m_2$), then $\langle t_2, t_1 \rangle$ is a valid tag on the (different) message m_2, m_1 (something that is not allowed by Definition 4.2).
2. We can prevent the previous attack by authenticating a block index along with each block. That is, we now compute $t_i = \text{Mac}'_k(i \| m_i)$ for all i , and output $\langle t_1, \dots, t_d \rangle$ as the tag. (Note that the block length $|m_i|$ will have to change.) This does not prevent a *truncation attack* whereby an attacker simply drops blocks from the end of the message (and drops the corresponding blocks of the tag as well).
3. The truncation attack can be thwarted by additionally authenticating the message length along with each block. (Authenticating the message length as a separate block does not work. Do you see why?) That is, compute $t_i = \text{Mac}'_k(\ell \| i \| m_i)$ for all i , where ℓ denotes the length of the message in bits. (Once again, the block length $|m_i|$ will need to decrease.) This scheme is vulnerable to a *“mix-and-match” attack* where the adversary combines blocks from different messages. For example, if the adversary obtains tags $\langle t_1, \dots, t_d \rangle$ and $\langle t'_1, \dots, t'_d \rangle$ on messages $m = m_1, \dots, m_d$ and $m' = m'_1, \dots, m'_d$, respectively, it can output the valid tag $\langle t_1, t'_2, t_3, t'_4, \dots \rangle$ on the message $m_1, m'_2, m_3, m'_4, \dots$.

We can prevent this last attack by also including a random “message identifier” along with each block that prevents blocks from different messages from being combined. This leads us to Construction 4.7.

with a larger domain would yield a secure MAC for longer messages. However, existing *practical* pseudorandom functions (i.e., block ciphers) take short, fixed-length inputs.

CONSTRUCTION 4.7

Let $\Pi' = (\text{Mac}', \text{Vrfy}')$ be a fixed-length MAC for messages of length n . Define a MAC as follows:

- **Mac**: on input a key $k \in \{0, 1\}^n$ and a message $m \in \{0, 1\}^*$ of (nonzero) length $\ell < 2^{n/4}$, parse m into d blocks $m_1, \dots, m_d$, each of length $n/4$. (The final block is padded with 0s if necessary.) Choose a uniform identifier $r \in \{0, 1\}^{n/4}$.
For $i = 1, \dots, d$, compute $t_i \leftarrow \text{Mac}'_k(r \parallel \ell \parallel i \parallel m_i)$, where i, ℓ are encoded as strings of length $n/4$.[†] Output the tag $t := \langle r, t_1, \dots, t_d \rangle$.
- **Vrfy**: on input a key $k \in \{0, 1\}^n$, a message $m \in \{0, 1\}^*$ of length $\ell < 2^{n/4}$, and a tag $t = \langle r, t_1, \dots, t_{d'} \rangle$, parse m into d blocks $m_1, \dots, m_d$, each of length $n/4$. (The final block is padded with 0s if necessary.) Output 1 if and only if $d' = d$ and $\text{Vrfy}'_k(r \parallel \ell \parallel i \parallel m_i, t_i) = 1$ for $1 \leq i \leq d$.

[†] Note that i and ℓ can be encoded using $n/4$ bits because $i, \ell < 2^{n/4}$.

A MAC for arbitrary-length messages from any fixed-length MAC.

(Technically, the scheme only handles messages of length less than $2^{n/4}$. Asymptotically, since this is an exponential bound, honest parties will not authenticate messages that long and any polynomial-time adversary could not submit messages that long to its MAC oracle. In practice, when a concrete value of n is fixed, one must ensure that this bound is acceptable.)

THEOREM 4.8 *If Π' is a secure fixed-length MAC for messages of length n , then Construction 4.7 is a secure MAC (for arbitrary-length messages).*

PROOF The intuition is that as long as Π' is secure, an adversary cannot introduce a new block with a valid tag. Furthermore, the extra information included in each block prevents the various attacks (dropping blocks, re-ordering blocks, etc.) sketched earlier. We will prove security by essentially showing that these attacks are the only ones possible.

Let Π be the MAC given by Construction 4.7, and let $\mathcal{A}$ be a probabilistic polynomial-time adversary. We show that $\Pr[\text{Mac-forge}_{\mathcal{A}, \Pi}(n) = 1]$ is negligible. We first introduce some notation that will be used in the proof. Let **Repeat** denote the event that the same random identifier appears in two of the tags returned by the MAC oracle in experiment $\text{Mac-forge}_{\mathcal{A}, \Pi}(n)$. Letting $(m, t = \langle r, t_1, \dots \rangle)$ denote the final output of $\mathcal{A}$, where $m = m_1, \dots$ has length ℓ , we let **NewBlock** be the event that at least one of the blocks $r \parallel \ell \parallel i \parallel m_i$ was never previously authenticated by Mac' in the course of answering $\mathcal{A}$'s **Mac** queries. (Note that, by construction of Π , it is easy to tell exactly which blocks are authenticated by Mac'_k when computing $\text{Mac}_k(m)$.) Informally, **NewBlock** is the event that $\mathcal{A}$ tries to output a valid tag on a block that was

never authenticated by the underlying fixed-length MAC Π' .

We have

$$\begin{aligned}
 \Pr[\text{Mac-forge}_{\mathcal{A},\Pi}(n) = 1] &= \Pr[\text{Mac-forge}_{\mathcal{A},\Pi}(n) = 1 \wedge \text{Repeat}] \\
 &\quad + \Pr[\text{Mac-forge}_{\mathcal{A},\Pi}(n) = 1 \wedge \overline{\text{Repeat}} \wedge \text{NewBlock}] \\
 &\quad + \Pr[\text{Mac-forge}_{\mathcal{A},\Pi}(n) = 1 \wedge \overline{\text{Repeat}} \wedge \overline{\text{NewBlock}}] \\
 &\leq \Pr[\text{Repeat}] \\
 &\quad + \Pr[\text{Mac-forge}_{\mathcal{A},\Pi}(n) = 1 \wedge \text{NewBlock}] \\
 &\quad + \Pr[\text{Mac-forge}_{\mathcal{A},\Pi}(n) = 1 \wedge \overline{\text{Repeat}} \wedge \overline{\text{NewBlock}}].
 \end{aligned} \tag{4.3}$$

We show that the first two terms of Equation (4.3) are negligible, and the final term is 0. This implies $\Pr[\text{Mac-forge}_{\mathcal{A},\Pi}(n) = 1]$ is negligible, as desired.

CLAIM 4.9 $\Pr[\text{Repeat}]$ is negligible.

PROOF Let $q(n)$ be the number of MAC oracle queries made by $\mathcal{A}$. To answer the i th oracle query of $\mathcal{A}$, the oracle chooses r_i uniformly from a set of size $2^{n/4}$. The probability of event **Repeat** is exactly the probability that $r_i = r_j$ for some $i \neq j$. Applying the “birthday bound” (Lemma A.15), we have that $\Pr[\text{Repeat}] \leq \frac{q(n)^2}{2^{n/4}}$. Since $\mathcal{A}$ makes only polynomially many queries, this value is negligible. $\blacksquare$

We next consider the final term on the right-hand side of Equation (4.3). We argue that if $\text{Mac-forge}_{\mathcal{A},\Pi}(n) = 1$, but **Repeat** did not occur, then it must be the case that **NewBlock** occurred. That is, $\text{Mac-forge}_{\mathcal{A},\Pi}(n) = 1 \wedge \overline{\text{Repeat}}$ implies **NewBlock**, and so

$$\Pr[\text{Mac-forge}_{\mathcal{A},\Pi}(n) = 1 \wedge \overline{\text{Repeat}} \wedge \overline{\text{NewBlock}}] = 0.$$

This is, in some sense, the heart of the proof.

Again let $q = q(n)$ denote the number of MAC oracle queries made by $\mathcal{A}$, and let r_i denote the random identifier used to answer the i th oracle query of $\mathcal{A}$. If **Repeat** does not occur then the values $r_1, \dots, r_q$ are distinct. Let $(m, t = \langle r, t_1, \dots \rangle)$ be the output of $\mathcal{A}$, with $m = m_1, \dots$. If $r \notin \{r_1, \dots, r_q\}$, then **NewBlock** clearly occurs. If not, then $r = r_j$ for some unique j , and the blocks $r \parallel \ell \parallel 1 \parallel m_1, \dots$ could then not possibly have been authenticated during the course of answering any **Mac** queries other than the j th such query. Let $m^{(j)}$ be the message that was used by $\mathcal{A}$ for its j th oracle query, and let ℓ_j be its length. There are two cases to consider:

Case 1: $\ell \neq \ell_j$. The blocks authenticated when answering the j th **Mac** query all have $\ell_j \neq \ell$ in the second position. So $r \parallel \ell \parallel 1 \parallel m_1$, in particular, was never authenticated in the course of answering the j th **Mac** query, and **NewBlock** occurs.

Case 2: $\ell = \ell_j$. If $\text{Mac-forge}_{\mathcal{A}, \Pi}(n) = 1$, then we must have $m \neq m^{(j)}$. Let $m^{(j)} = m_1^{(j)}, \dots$. Since m and $m^{(j)}$ have equal length, there must be at least one index i for which $m_i \neq m_i^{(j)}$. The block $r\|\ell\|i\|m_i$ was then never authenticated in the course of answering the j th **Mac** query. (Because i is included in the third position of the block, the block $r\|\ell\|i\|m_i$ could only possibly have been authenticated if $r\|\ell\|i\|m_i = r_j\|\ell_j\|i\|m_i^{(j)}$, but this is not true since $m_i \neq m_i^{(j)}$.)

To complete the proof of the theorem, we bound the second term on the right-hand side of Equation (4.3):

CLAIM 4.10 $\Pr[\text{Mac-forge}_{\mathcal{A}, \Pi}(n) = 1 \wedge \text{NewBlock}]$ is negligible.

The claim relies on security of Π' . We construct a PPT adversary $\mathcal{A}'$ who attacks the fixed-length MAC Π' and succeeds in outputting a valid forgery on a previously unauthenticated message with probability

$$\Pr[\text{Mac-forge}_{\mathcal{A}', \Pi'}(n) = 1] \geq \Pr[\text{Mac-forge}_{\mathcal{A}, \Pi}(n) = 1 \wedge \text{NewBlock}]. \quad (4.4)$$

Security of Π' means that the left-hand side is negligible, proving the claim.

The construction of $\mathcal{A}'$ is the obvious one and so we describe it briefly. $\mathcal{A}'$ runs $\mathcal{A}$ as a sub-routine, and answers the request by $\mathcal{A}$ for a tag on m by choosing $r \leftarrow \{0, 1\}^{n/4}$ itself, parsing m appropriately, and making the necessary queries to its own MAC oracle $\text{Mac}'_k(\cdot)$. When $\mathcal{A}$ outputs $(m, t = \langle r, t_1, \dots \rangle)$, then $\mathcal{A}'$ checks whether **NewBlock** occurs (this is easy to do since $\mathcal{A}'$ can keep track of all the queries it makes to its own oracle). If so, then $\mathcal{A}'$ finds the first block $r\|\ell\|i\|m_i$ that was never previously authenticated by Mac' and outputs $(r\|\ell\|i\|m_i, t_i)$. (If not, $\mathcal{A}'$ outputs nothing.)

The view of $\mathcal{A}$ when run as a sub-routine by $\mathcal{A}'$ is distributed identically to the view of $\mathcal{A}$ in experiment $\text{Mac-forge}_{\mathcal{A}, \Pi}(n)$, and so the probabilities of events $\text{Mac-forge}_{\mathcal{A}, \Pi}(n) = 1$ and **NewBlock** do not change. If **NewBlock** occurs then $\mathcal{A}'$ outputs a block $r\|\ell\|i\|m_i$ that was never previously authenticated by its own MAC oracle; if $\text{Mac-forge}_{\mathcal{A}, \Pi}(n) = 1$ then the tag on every block is valid (with respect to Π'), and so in particular this is true for the block output by $\mathcal{A}'$. This means that whenever $\text{Mac-forge}_{\mathcal{A}, \Pi}(n) = 1$ and **NewBlock** occur we have $\text{Mac-forge}_{\mathcal{A}', \Pi'}(n) = 1$, proving Equation (4.4). ■

4.4 CBC-MAC

Theorems 4.6 and 4.8 show that it is possible to construct a secure message authentication code for arbitrary-length messages from a pseudorandom

function taking inputs of fixed length n . This demonstrates, in principle, that secure MACs can be constructed from block ciphers. Unfortunately, the resulting construction is extremely inefficient: to compute a tag on a message of length dn , the block cipher is evaluated $4d$ times; the tag is more than $4dn$ bits long. Fortunately, far more efficient constructions are available. We explore one such construction here that relies solely on block ciphers, and another in Section 5.3.2 that uses an additional cryptographic primitive.

4.4.1 The Basic Construction

CBC-MAC is a standardized message authentication code used widely in practice. A basic version of CBC-MAC, secure when authenticating messages of any *fixed* length, is given as Construction 4.11. (See also Figure 4.1.) We caution that this basic scheme is *not* secure in the general case when messages of different lengths may be authenticated; see further discussion below.

CONSTRUCTION 4.11

Let F be a pseudorandom function, and fix a length function $\ell > 0$. The basic CBC-MAC construction is as follows:

- **Mac:** on input a key $k \in \{0, 1\}^n$ and a message m of length $\ell(n) \cdot n$, do the following (we set $\ell = \ell(n)$ in what follows):
 1. Parse m as $m = m_1, \dots, m_\ell$ where each m_i is of length n .
 2. Set $t_0 := 0^n$. Then, for $i = 1$ to ℓ :
 Set $t_i := F_k(t_{i-1} \oplus m_i)$.

Output t_ℓ as the tag.
- **Vrfy:** on input a key $k \in \{0, 1\}^n$, a message m , and a tag t , do: If m is not of length $\ell(n) \cdot n$ then output 0. Otherwise, output 1 if and only if $t \stackrel{?}{=} \text{Mac}_k(m)$.

Basic CBC-MAC (for fixed-length messages).

THEOREM 4.12 *Let ℓ be a polynomial. If F is a pseudorandom function, then Construction 4.11 is a secure MAC for messages of length $\ell(n) \cdot n$.*

The proof of Theorem 4.12 is somewhat involved. In the following section we will prove a more general result from which the above theorem follows.

Although Construction 4.11 can be extended in the obvious way to handle messages whose length is an arbitrary multiple of n , the construction is only secure when the length of the messages being authenticated is fixed and agreed upon in advance by the sender and receiver. (See Exercise 4.13.)

The advantage of this construction over Construction 4.5, which also gives a fixed-length MAC, is that the present construction can authenticate longer messages. Compared to Construction 4.7, CBC-MAC is much more efficient, requiring only d block-cipher evaluations for a message of length dn , and with a tag of length n only.

CBC-MAC vs. CBC-mode encryption. CBC-MAC is similar to the CBC mode of operation. There are, however, some important differences:

1. CBC-mode encryption uses a *random IV* and this is crucial for security. In contrast, CBC-MAC uses no *IV* (alternately, it can be viewed as using the fixed value $IV = 0^n$) and this is also crucial for security. Specifically, CBC-MAC using a random *IV* is not secure.
2. In CBC-mode encryption all intermediate values t_i (called c_i in the case of CBC-mode encryption) are output by the encryption algorithm as part of the ciphertext, whereas in CBC-MAC only the final block is output as the tag. If CBC-MAC is modified to output all the $\{t_i\}$ obtained during the course of the computation then it is no longer secure.

In Exercise 4.14 you are asked to verify that the modifications of CBC-MAC discussed above are insecure. These examples illustrate the fact that harmless-looking modifications to cryptographic constructions can render them insecure. One should always implement a cryptographic construction exactly as specified and not introduce any variations (unless the variations themselves can be proven secure). Furthermore, it is essential to understand the construction being used. In many cases a cryptographic library provides a programmer with a “CBC function,” but does not distinguish between the use of this function for encryption or message authentication.

Secure CBC-MAC for arbitrary-length messages. We briefly describe two ways Construction 4.11 can be modified, in a provably secure fashion, to handle arbitrary-length messages. (Here for simplicity we assume that all messages being authenticated have length a multiple of n , and that Vrfy rejects

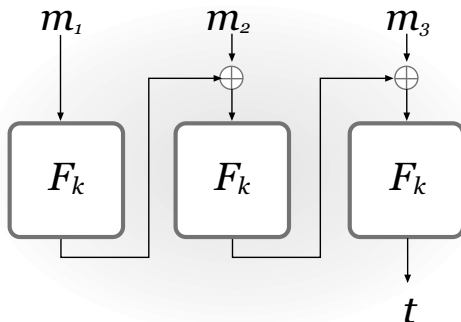


FIGURE 4.1: Basic CBC-MAC (for fixed-length messages).

any message whose length is not a multiple of n . In the following section we treat the more general case where messages can have arbitrary length.)

1. *Prepend* the message m with its length $|m|$ (encoded as an n -bit string), and then compute basic CBC-MAC on the result; see Figure 4.2. Security of this variant follows from the results proved in the next section.

Note that appending $|m|$ to the *end* of the message and then computing the basic CBC-MAC is *not* secure.

2. Change the scheme so that key generation chooses two independent, uniform keys $k_1 \in \{0,1\}^n$ and $k_2 \in \{0,1\}^n$. Then to authenticate a message m , first compute the basic CBC-MAC of m using k_1 and let t be the result; output the tag $\hat{t} := F_{k_2}(t)$.

The second option has the advantage of not needing to know the message length in advance (i.e., when beginning to compute the tag). However, it has the drawback of using two keys for F . Note that, at the expense of two additional applications of the pseudorandom function, it is possible to store a single key k and then derive keys $k_1 := F_k(1)$ and $k_2 := F_k(2)$ at the beginning of the computation. Despite this, in practice, the operation of initializing a key for a block cipher is considered relatively expensive. Therefore, requiring two different keys—even if they are derived on the fly—is less desirable.

4.4.2 *Proof of Security

In this section we prove security of different variants of CBC-MAC. We begin by summarizing the results, and then give the details of the proof. Before beginning, we remark that the proof in this section is quite involved, and is intended for advanced readers.

Throughout this section, fix a keyed function F that, for security parameter n , maps n -bit keys and n -bit inputs to n -bit outputs. We define a keyed

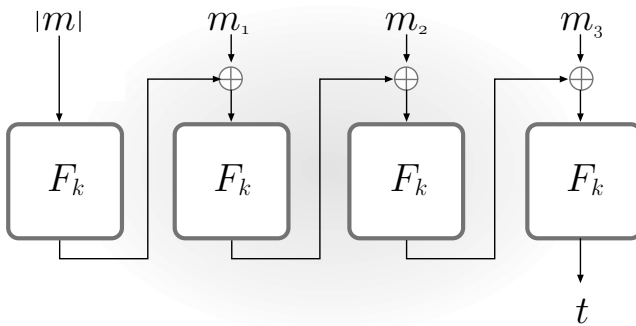


FIGURE 4.2: A variant of CBC-MAC secure for authenticating arbitrary-length messages.

function CBC that, for security parameter n , maps n -bit keys and inputs in $(\{0,1\}^n)^*$ (i.e., strings whose length is a multiple of n) to n -bit outputs. This function is defined as

$$\text{CBC}_k(x_1, \dots, x_\ell) \stackrel{\text{def}}{=} F_k(F_k(\dots F_k(F_k(x_1) \oplus x_2) \oplus \dots) \oplus x_\ell),$$

where $|x_1| = \dots = |x_\ell| = |k| = n$. (We leave CBC_k undefined on the empty string.) Note that CBC is exactly basic CBC-MAC, although here we consider inputs of different lengths.

A set of strings $P \subset (\{0,1\}^n)^*$ is *prefix-free* if it does not contain the empty string, and no string $X \in P$ is a prefix of any other string $X' \in P$. We show:

THEOREM 4.13 *If F is a pseudorandom function, then CBC is a pseudorandom function as long as the set of inputs on which it is queried is prefix-free. Formally, for all probabilistic polynomial-time distinguishers D that query their oracle on a prefix-free set of inputs, there is a negligible function negl such that*

$$\left| \Pr[D^{\text{CBC}_k(\cdot)}(1^n) = 1] - \Pr[D^{f(\cdot)}(1^n) = 1] \right| \leq \text{negl}(n),$$

where k is chosen uniformly from $\{0,1\}^n$ and f is chosen uniformly from the set of functions mapping $(\{0,1\}^n)^*$ to $\{0,1\}^n$ (i.e., the value of f at each input is uniform and independent of the values of f at all other inputs).

Thus, we can convert a pseudorandom function F for fixed-length inputs into a pseudorandom function CBC for arbitrary-length inputs (subject to a constraint on which inputs can be queried)! To use this for message authentication, we adapt the idea of Construction 4.5 as follows: to authenticate a message m , first apply some *encoding function* encode to obtain a (nonempty) string $\text{encode}(m) \in (\{0,1\}^n)^*$; then output the tag $\text{CBC}_k(\text{encode}(m))$. For this to be secure (cf. the proof of Theorem 4.6), the encoding needs to be *prefix-free*, namely, to have the property that for any distinct (legal) messages m_1, m_2 , the string $\text{encode}(m_1)$ is not a prefix of $\text{encode}(m_2)$. This implies that for any set of (legal) messages $\{m_1, \dots\}$, the set of encoded messages $\{\text{encode}(m_1), \dots\}$ is prefix-free.

We now examine two concrete applications of this idea:

- Fix ℓ , and let the set of legal messages be $\{0,1\}^{\ell(n) \cdot n}$. Then we can take the trivial encoding $\text{encode}(m) = m$, which is prefix-free since one string cannot be a prefix of a different string of the same length. This is exactly basic CBC-MAC, and what we have said above implies that basic CBC-MAC is secure for messages of any fixed length (cf. Theorem 4.12).
- One way of handling arbitrary-length (nonempty) messages (technically, messages of length less than 2^n) is to encode a string $m \in \{0,1\}^*$ by prepending its length $|m|$ (encoded as an n -bit string), and then appending as many 0s as needed to make the length of the resulting

string a multiple of n . (This is essentially what is shown in Figure 4.2.) This encoding is prefix-free, and we therefore obtain a secure MAC for arbitrary-length messages.

The rest of this section is devoted to a proof of Theorem 4.13. In proving the theorem, we analyze CBC when it is “keyed” with a *random function* g rather than a random key k for some underlying pseudorandom function F . That is, we consider the keyed function CBC_g defined as

$$\text{CBC}_g(x_1, \dots, x_\ell) \stackrel{\text{def}}{=} g(g(\dots g(g(x_1) \oplus x_2) \oplus \dots) \oplus x_\ell)$$

where, for security parameter n , the function g maps n -bit inputs to n -bit outputs, and $|x_1| = \dots = |x_\ell| = n$. Note that CBC_g as defined here is not efficient (since the representation of g requires space exponential in n); nevertheless, it is still a well-defined, keyed function.

We show that if g is chosen uniformly from Func_n , then CBC_g is indistinguishable from a random function mapping $(\{0, 1\}^n)^*$ to n -bit strings, as long as a prefix-free set of inputs is queried. More precisely:

CLAIM 4.14 *Fix any $n \geq 1$. For all distinguishers D that query their oracle on a prefix-free set of q inputs, where the longest such input contains ℓ blocks, it holds that:*

$$\left| \Pr[D^{\text{CBC}_g(\cdot)}(1^n) = 1] - \Pr[D^{f(\cdot)}(1^n) = 1] \right| \leq \frac{q^2 \ell^2}{2^n},$$

where g is chosen uniformly from Func_n , and f is chosen uniformly from the set of functions mapping $(\{0, 1\}^n)^*$ to $\{0, 1\}^n$.

(The claim is unconditional, and does not impose any constraints on the running time of D . Thus we may take D to be deterministic.) The above implies Theorem 4.13 using standard techniques that we have already seen. In particular, for any D running in polynomial time we must have $q(n), \ell(n) = \text{poly}(n)$ and so $q(n)^2 \ell(n)^2 \cdot 2^{-n}$ is negligible.

PROOF (of Claim 4.14) Fix some $n \geq 1$. The proof proceeds in two steps: We first define a notion of *smoothness* and prove that CBC is smooth; we then show that smoothness implies the claim.

Let $P = \{X_1, \dots, X_q\}$ be a prefix-free set of q inputs, where each X_i is in $(\{0, 1\}^n)^*$ and the longest string in P contains ℓ blocks (i.e., each $X_i \in P$ contains at most ℓ blocks of length n). Note that for any $t_1, \dots, t_q \in \{0, 1\}^n$ it holds that $\Pr[\forall i : f(X_i) = t_i] = 2^{-nq}$, where the probability is over uniform choice of the function f from the set of functions mapping $(\{0, 1\}^n)^*$ to $\{0, 1\}^n$. We say that CBC is (q, ℓ, δ) -smooth if for every prefix-free set $P = \{X_1, \dots, X_q\}$ as above and every $t_1, \dots, t_q \in \{0, 1\}^n$, it holds that

$$\Pr[\forall i : \text{CBC}_g(X_i) = t_i] \geq (1 - \delta) \cdot 2^{-nq},$$

where the probability is over uniform choice of $g \in \text{Func}_n$.

In words, CBC is smooth if for every fixed set of input/output pairs $\{(X_i, t_i)\}$, where the $\{X_i\}$ form a prefix-free set, the probability that $\text{CBC}_g(X_i) = t_i$ for all i is δ -close to the probability that $f(X_i) = t_i$ for all i (where g is a random function from $\{0, 1\}^n$ to $\{0, 1\}^n$, and f is a random function from $(\{0, 1\}^n)^*$ to $\{0, 1\}^n$).

CLAIM 4.15 CBC_g is (q, ℓ, δ) -smooth, for $\delta = q^2 \ell^2 \cdot 2^{-n}$.

PROOF For any $X \in (\{0, 1\}^n)^*$, with $X = x_1, \dots$ and $x_i \in \{0, 1\}^n$, let $\mathcal{C}_g(X)$ denote the set of inputs on which g is evaluated during the computation of $\text{CBC}_g(X)$; i.e., if $X \in (\{0, 1\}^n)^m$ then

$$\mathcal{C}_g(X) \stackrel{\text{def}}{=} (x_1, \text{CBC}_g(x_1) \oplus x_2, \dots, \text{CBC}_g(x_1, \dots, x_{m-1}) \oplus x_m).$$

For $X \in (\{0, 1\}^n)^m$ and $X' \in (\{0, 1\}^n)^{m'}$, with $\mathcal{C}_g(X) = (I_1, \dots, I_m)$ and $\mathcal{C}_g(X') = (I'_1, \dots, I'_{m'})$, say there is a *non-trivial collision* in X if $I_i = I_j$ for some $i \neq j$, and say that there is a *non-trivial collision between X and X'* if $I_i = I'_j$ but $(x_1, \dots, x_i) \neq (x'_1, \dots, x'_j)$ (in this latter case i may equal j). We say that there is a non-trivial collision in P if there is a non-trivial collision in some $X \in P$ or between some pair of strings $X, X' \in P$. Let Coll be the event that there is a non-trivial collision in P .

We prove the claim in two steps. First, we show that conditioned on there not being a non-trivial collision in P , the probability that $\text{CBC}_g(X_i) = t_i$ for all i is exactly 2^{-nq} . Next, we show that the probability that there is a non-trivial collision in P is less than $\delta = q^2 \ell^2 \cdot 2^{-n}$.

Consider choosing a uniform g by choosing, one-by-one, uniform values for the outputs of g on different inputs. Determining whether there is a non-trivial collision between two strings $X, X' \in P$ can be done by first choosing the values of $g(I_1)$ and $g(I'_1)$ (if $I'_1 = I_1$, these values are the same), then choosing values for $g(I_2)$ and $g(I'_2)$ (note that $I_2 = g(I_1) \oplus x_2$ and $I'_2 = g(I'_1) \oplus x'_2$ are defined once $g(I_1), g(I'_1)$ have been fixed), and continuing in this way until we choose values for $g(I_{m-1})$ and $g(I'_{m'-1})$. In particular, the values of $g(I_m), g(I'_{m'})$ need not be chosen in order to determine whether there is a non-trivial collision between X and X' . Continuing this line of reasoning, it is possible to determine whether Coll occurs by choosing the values of g on all but the final entries of each of $\mathcal{C}_g(X_1), \dots, \mathcal{C}_g(X_q)$.

Assume Coll has not occurred after fixing the values of g on various inputs as described above. Consider the final entries in each of $\mathcal{C}_g(X_1), \dots, \mathcal{C}_g(X_q)$. These entries are all distinct (this is immediate from the fact that Coll has not occurred), and we claim that the value of g on each of those points has not yet been fixed. Indeed, the only way the value of g could already be fixed on any of those points is if the final entry I_m of some $\mathcal{C}_g(X)$ is equal to a non-final entry I_j of some $\mathcal{C}_g(X')$. But since Coll has not occurred, this can

only happen if $X \neq X'$ and $(x'_1, \dots, x'_j) = (x_1, \dots, x_m)$. But then X would be a prefix of X' , violating the assumption that P is prefix-free.

Since g is a random function, the above means that $\text{CBC}_g(X_1), \dots, \text{CBC}_g(X_q)$ are *uniform* and *independent* of each other as well as all the other values of g that have already been fixed. (This is because $\text{CBC}_g(X_i)$ is the value of g when evaluated at the final entry of $\mathcal{C}_g(X_i)$, an input value which is different from all the other inputs at which g has already been fixed.) Thus, for any $t_1, \dots, t_q \in \{0, 1\}^n$ we have:

$$\Pr [\forall i : \text{CBC}_g(X_i) = t_i \mid \overline{\text{Coll}}] = 2^{-nq}. \quad (4.5)$$

We next show that $\overline{\text{Coll}}$ occurs with high probability by upper-bounding $\Pr[\text{Coll}]$. For distinct $X_i, X_j \in P$, let $\text{Coll}_{i,j}$ denote the event that there is a non-trivial collision in X or X' , or a non-trivial collision between X and X' . We have $\text{Coll} = \bigvee_{i,j} \text{Coll}_{i,j}$ and so a union bound gives

$$\Pr[\text{Coll}] \leq \sum_{i,j: i < j} \Pr[\text{Coll}_{i,j}] = \binom{q}{2} \cdot \Pr[\text{Coll}_{i,j}] \leq \frac{q^2}{2} \cdot \Pr[\text{Coll}_{i,j}]. \quad (4.6)$$

Fixing distinct $X = X_i$ and $X' = X_j$ in P , we now bound $\text{Coll}_{i,j}$. As will be clear from the analysis, the probability is maximized when X and X' are both as long as possible, and thus we assume they are each ℓ blocks long. Let $X = (x_1, \dots, x_\ell)$ and $X' = (x'_1, \dots, x'_\ell)$, and let t be the largest integer such that $(x_1, \dots, x_t) = (x'_1, \dots, x'_t)$. (Note that $t < \ell$ or else $X = X'$.) We assume $t > 0$, but the analysis below can be easily modified, giving the same result, if $t = 0$. We continue to let $I_1, I_2, \dots$ (resp., $I'_1, I'_2, \dots$) denote the inputs to g during the course of computing $\text{CBC}_g(X)$ (resp., $\text{CBC}_g(X')$); note that $(I'_1, \dots, I'_t) = (I_1, \dots, I_t)$. Consider choosing g by choosing uniform values for the outputs of g , one-by-one. We do this in $2\ell - 2$ steps as follows:

Steps 1 through $t - 1$ (if $t > 1$): In each step i , choose a uniform value for $g(I_i)$, thus defining I_{i+1} and I'_{i+1} (which are equal).

Step t : Choose a uniform value for $g(I_t)$, thus defining I_{t+1} and I'_{t+1} .

Steps $t + 1$ to $\ell - 1$ (if $t < \ell - 1$): Choose, in turn, uniform values for each of $g(I_{t+1}), g(I_{t+2}), \dots, g(I_{\ell-1})$, thus defining $I_{t+2}, I_{t+3}, \dots, I_\ell$.

Steps ℓ to $2\ell - 2$ (if $t < \ell - 1$): Choose, in turn, uniform values for each of $g(I'_{t+1}), g(I'_{t+2}), \dots, g(I'_{\ell-1})$, thus defining $I'_{t+2}, I'_{t+3}, \dots, I'_\ell$.

Let $\text{Coll}(k)$ be the event that a non-trivial collision occurs by step k . Then

$$\Pr[\text{Coll}_{i,j}] = \Pr[\bigvee_k \text{Coll}(k)] \leq \Pr[\text{Coll}(1)] + \sum_{k=2}^{2\ell-2} \Pr[\text{Coll}(k) \mid \overline{\text{Coll}(k-1)}], \quad (4.7)$$

using Proposition A.9. For $k < t$, we claim $\Pr[\text{Coll}(k) \mid \overline{\text{Coll}(k-1)}] = k/2^n$: indeed, if no non-trivial collision has occurred by step $k - 1$, the value of

$g(I_k)$ is chosen uniformly in step k ; a non-trivial collision occurs only if it happens that $I_{k+1} = g(I_k) \oplus x_{k+1}$ is equal to one of $\{I_1, \dots, I_k\}$ (which are all distinct, since $\text{Coll}(k-1)$ has not occurred). By similar reasoning, we have $\Pr[\text{Coll}(t) \mid \overline{\text{Coll}(t-1)}] \leq 2t/2^n$ (here there are two values I_{t+1}, I'_{t+1} to consider; note that they cannot be equal to each other). Finally, arguing as before, for $k > t$ we have $\Pr[\text{Coll}(k) \mid \overline{\text{Coll}(k-1)}] = (k+1)/2^n$. Using Equation (4.7), we thus have

$$\begin{aligned} \Pr[\text{Coll}_{i,j}] &\leq 2^{-n} \cdot \left(\sum_{k=1}^{t-1} k + 2t + \sum_{k=t+1}^{2\ell-2} (k+1) \right) \\ &= 2^{-n} \cdot \sum_{k=2}^{2\ell-1} k = 2^{-n} \cdot (2\ell+1) \cdot (\ell-1) < 2\ell^2 \cdot 2^{-n}. \end{aligned}$$

From Equation (4.6) we get $\Pr[\text{Coll}] < q^2 \ell^2 \cdot 2^{-n} = \delta$. Finally, using Equation (4.5) we see that

$$\begin{aligned} \Pr[\forall i : \text{CBC}_g(X_i) = t_i] &\geq \Pr[\forall i : \text{CBC}_g(X_i) = t_i \mid \overline{\text{Coll}}] \cdot \Pr[\overline{\text{Coll}}] \\ &= 2^{-nq} \cdot \Pr[\overline{\text{Coll}}] \geq (1 - \delta) \cdot 2^{-nq}, \end{aligned}$$

as claimed. ■

We now show that smoothness implies the theorem. Assume without loss of generality that D always makes q (distinct) queries, each containing at most ℓ blocks. D may choose its queries adaptively (i.e., depending on the answers to previous queries), but the set of D 's queries must be prefix-free.

For distinct $X_1, \dots, X_q \in \{0, 1\}^n$ and arbitrary $t_1, \dots, t_q \in \{0, 1\}^n$, define $\alpha(X_1, \dots, X_q; t_1, \dots, t_q)$ to be 1 if and only if D outputs 1 when making queries $X_1, \dots, X_q$ and getting responses $t_1, \dots, t_q$. (If, say, D does not make query X_1 as its first query, then $\alpha(X_1, \dots, \dots) = 0$.) Letting $\vec{X} = (X_1, \dots, X_q)$ and $\vec{t} = (t_1, \dots, t_q)$, we then have

$$\begin{aligned} \Pr[D^{\text{CBC}_g(\cdot)}(1^n) = 1] &= \sum_{\vec{X} \text{ prefix-free}; \vec{t}} \alpha(\vec{X}, \vec{t}) \cdot \Pr[\forall i : \text{CBC}_g(X_i) = t_i] \\ &\geq \sum_{\vec{X} \text{ prefix-free}; \vec{t}} \alpha(\vec{X}, \vec{t}) \cdot (1 - \delta) \cdot \Pr[\forall i : f(X_i) = t_i] \\ &= (1 - \delta) \cdot \Pr[D^{f(\cdot)}(1^n) = 1], \end{aligned}$$

where, above, g is chosen uniformly from Func_n , and f is chosen uniformly from the set of functions mapping $\{0, 1\}^n$ to $\{0, 1\}^n$. This implies

$$\Pr[D^{f(\cdot)}(1^n) = 1] - \Pr[D^{\text{CBC}_g(\cdot)}(1^n) = 1] \leq \delta \cdot \Pr[D^{f(\cdot)}(1^n) = 1] \leq \delta.$$

A symmetric argument for when D outputs 0 completes the proof. ■

4.5 Authenticated Encryption

In Chapter 3, we studied how it is possible to obtain *secrecy* in the private-key setting using encryption. In this chapter, we have shown how to ensure *integrity* using message authentication codes. One might naturally want to achieve both goals simultaneously, and this is the problem we turn to now.

It is best practice to *always ensure secrecy and integrity by default* in the private-key setting. Indeed, in many applications where secrecy is required it turns out that integrity is essential also. Moreover, a lack of integrity can sometimes lead to a breach of secrecy.

4.5.1 Definitions

We begin, as usual, by defining precisely what we wish to achieve. At an abstract level, our goal is to realize an “ideally secure” communication channel that provides both secrecy and integrity. Pursuing a definition of this sort is beyond the scope of this book. Instead, we provide a simpler set of definitions that treat secrecy and integrity separately. These definitions and our subsequent analysis suffice for understanding the key issues at hand. (We caution the reader, however, that—in contrast to encryption and message authentication codes—the field has not yet settled on standard terminology and definitions for authenticated encryption.)

Let $\Pi = (\text{Gen}, \text{Enc}, \text{Dec})$ be a private-key encryption scheme. As mentioned already, we define security by separately defining secrecy and integrity. The notion of secrecy we consider is one we have seen before: we require that Π be secure against chosen-ciphertext attacks, i.e., that it be CCA-secure. (Refer to Section 3.7 for a discussion and definition of CCA-security.) We are concerned about chosen-ciphertext attacks here because we are explicitly considering an active adversary who can modify the data sent from one honest party to the other. Our notion of integrity will be essentially that of existential unforgeability under an adaptive chosen-message attack. Since Π does not satisfy the syntax of a message authentication code, however, we introduce a definition specific to this case. Consider the following experiment defined for a private-key encryption scheme $\Pi = (\text{Gen}, \text{Enc}, \text{Dec})$, adversary $\mathcal{A}$, and value n for the security parameter:

The unforgeable encryption experiment $\text{Enc-Forge}_{\mathcal{A}, \Pi}(n)$:

1. Run $\text{Gen}(1^n)$ to obtain a key k .
2. The adversary $\mathcal{A}$ is given input 1^n and access to an encryption oracle $\text{Enc}_k(\cdot)$. The adversary outputs a ciphertext c .
3. Let $m := \text{Dec}_k(c)$, and let $\mathcal{Q}$ denote the set of all queries that $\mathcal{A}$ asked its encryption oracle. The output of the experiment is 1 if and only if (1) $m \neq \perp$ and (2) $m \notin \mathcal{Q}$.

DEFINITION 4.16 A private-key encryption scheme Π is *unforgeable* if for all probabilistic polynomial-time adversaries $\mathcal{A}$, there is a negligible function negl such that:

$$\Pr[\text{Enc-Forge}_{\mathcal{A},\Pi}(n) = 1] \leq \text{negl}(n).$$

Paralleling our discussion about verification queries following Definition 4.2, here one could also consider a stronger definition in which $\mathcal{A}$ is additionally given access to a decryption oracle. One can verify that the secure construction we present below also satisfies that stronger definition.

We now define a (secure) *authenticated encryption* scheme.

DEFINITION 4.17 A private-key encryption scheme is an *authenticated encryption scheme* if it is CCA-secure and unforgeable.

4.5.2 Generic Constructions

It may be tempting to think that any reasonable combination of a secure encryption scheme and a secure message authentication code should result in an authenticated encryption scheme. In this section we show that this is not the case. This demonstrates that even secure cryptographic tools can be combined in such a way that the result is insecure, and highlights once again the importance of definitions and proofs of security. On the positive side, we show how encryption and message authentication can be combined properly to achieve joint secrecy and integrity.

Throughout, let $\Pi_E = (\text{Enc}, \text{Dec})$ be a CPA-secure encryption scheme and let $\Pi_M = (\text{Mac}, \text{Vrfy})$ denote a message authentication code, where key generation in both schemes simply involves choosing a uniform n -bit key. There are three natural approaches to combining encryption and message authentication using independent keys⁴ k_E and k_M for Π_E and Π_M , respectively:

1. *Encrypt-and-authenticate*: In this method, encryption and message authentication are computed independently in parallel. That is, given a plaintext message m , the sender transmits the ciphertext $\langle c, t \rangle$ where:

$$c \leftarrow \text{Enc}_{k_E}(m) \text{ and } t \leftarrow \text{Mac}_{k_M}(m).$$

The receiver decrypts c to recover m ; assuming no error occurred, it then verifies the tag t . If $\text{Vrfy}_{k_M}(m, t) = 1$, the receiver outputs m ; otherwise, it outputs an error.

⁴*Independent* cryptographic keys should always be used when different schemes are combined. We return to this point at the end of this section.

2. *Authenticate-then-encrypt*: Here a MAC tag t is first computed, and then the message and tag are encrypted together. That is, given a message m , the sender transmits the ciphertext c computed as:

$$t \leftarrow \text{Mac}_{k_M}(m) \text{ and } c \leftarrow \text{Enc}_{k_E}(m||t).$$

The receiver decrypts c to obtain $m||t$; assuming no error occurs, it then verifies the tag t . As before, if $\text{Vrfy}_{k_M}(m, t) = 1$ the receiver outputs m ; otherwise, it outputs an error.

3. *Encrypt-then-authenticate*: In this case, the message m is first encrypted and then a MAC tag is computed over the result. That is, the ciphertext is the pair $\langle c, t \rangle$ where:

$$c \leftarrow \text{Enc}_{k_E}(m) \text{ and } t \leftarrow \text{Mac}_{k_M}(c).$$

(See also Construction 4.18.) If $\text{Vrfy}_{k_M}(c, t) = 1$, then the receiver decrypts c and outputs the result; otherwise, it outputs an error.

We analyze each of the above approaches when they are instantiated with “generic” secure components, i.e., an *arbitrary* CPA-secure encryption scheme and an *arbitrary* (strongly) secure MAC. We want an approach that provides joint secrecy and integrity when using *any* (secure) components, and we will therefore reject as “unsafe” any approach for which there exists even a single counterexample of a secure encryption scheme/MAC for which the combination is insecure. This “all-or-nothing” approach reduces the likelihood of implementation flaws. Specifically, an authenticated encryption scheme might be implemented by making calls to an “encryption subroutine” and a “message authentication subroutine,” and the implementation of those subroutines may be changed at some later point in time. (This commonly occurs when cryptographic libraries are updated, or when standards are modified.) Implementing an approach whose security depends on how its components are implemented (rather than on the security they provide) is therefore dangerous.

We stress that if an approach is rejected this does not mean that it is insecure for all possible instantiations of the components; it does, however, mean that any instantiation of the approach must be analyzed and proven secure before it is used.

Encrypt-and-authenticate. Recall that in this approach encryption and message authentication are carried out independently. Given a message m , the transmitted value is $\langle c, t \rangle$ where

$$c \leftarrow \text{Enc}_{k_E}(m) \text{ and } t \leftarrow \text{Mac}_{k_M}(m).$$

This approach may not achieve even the most basic level of secrecy. To see this, note that a secure MAC does not guarantee *any* secrecy and so it is possible for the tag $\text{Mac}_{k_M}(m)$ to leak information about m to an eavesdropper.

(As a trivial example, consider a secure MAC where the first bit of the tag is always equal to the first bit of the message.) So the encrypt-and-authenticate approach may yield a scheme that does not even have indistinguishable encryptions in the presence of an eavesdropper.

In fact, the encrypt-and-authenticate approach is likely to be insecure against chosen-plaintext attacks even when instantiated with standard components (unlike the contrived counterexample in the previous paragraph). In particular, if a *deterministic* MAC like CBC-MAC is used, then the tag computed on a message (for some fixed key k_M) is the same every time. This allows an eavesdropper to identify when the same message is sent twice, and so the scheme is not CPA-secure. Most MACs used in practice are deterministic, so this represents a real concern.

Authenticate-then-encrypt. Here, a MAC tag $t \leftarrow \text{Mac}_{k_M}(m)$ is first computed; then $m||t$ is encrypted and the resulting value $\text{Enc}_{k_E}(m||t)$ is transmitted. We show that this combination also does not necessarily yield an authenticated encryption scheme.

Actually, we have already encountered a CPA-secure encryption scheme for which this approach is insecure: the CBC-mode-with-padding scheme discussed in Section 3.7.2. (We assume in what follows that the reader is familiar with that section.) Recall that this scheme works by first padding the plaintext (which in our case will be $m||t$) in a specific way so the result is a multiple of the block length, and then encrypting the result using CBC mode. During decryption, if an error in the padding is detected after performing the CBC-mode decryption, then a “bad padding” error is returned. With regard to authenticate-then-encrypt, this means there are now *two* sources of potential decryption failure: the padding may be incorrect, or the MAC tag may not verify. Schematically, the decryption algorithm Dec' in the combined scheme works as follows:

$\text{Dec}'_{k_E, k_M}(c)$:

1. Compute $\tilde{m} := \text{Dec}_{k_E}(c)$. If an error in the padding is detected, return “bad padding” and stop.
2. Parse $\tilde{m}$ as $m||t$. If $\text{Vrfy}_{k_M}(m, t) = 1$ return m ; else, output “authentication failure.”

Assuming the attacker can distinguish between the two error messages, the attacker can apply the same chosen-ciphertext attack described in Section 3.7.2 to the above scheme to recover the entire original plaintext from a given ciphertext. (This is due to the fact that the padding-oracle attack shown in Section 3.7.2 relies only on the ability to learn whether or not there was a padding error, something that is revealed by this scheme.) This type of attack has been carried out successfully in the real world in various settings, e.g., in configurations of IPsec that use authenticate-then-encrypt.

One way to fix the above scheme would be to ensure that only a *single* error message is returned, regardless of the source of decryption failure. This is an unsatisfying solution for several reasons: (1) there may be legitimate reasons (e.g., usability, debugging) to have multiple error messages; (2) forcing the error messages to be the same means that the combination is no longer truly generic, i.e., it requires the implementer of the authenticate-then-encrypt approach to be aware of what error messages are returned by the underlying CPA-secure encryption scheme; (3) most of all, it is extraordinarily hard to ensure that the different errors cannot be distinguished since, e.g., even a difference in the time to return each of these errors may be used by an adversary to distinguish between them (cf. our earlier discussion of timing attacks at the end of Section 4.2). Some versions of SSL tried using only a single error message in conjunction with an authenticate-then-encrypt approach, but a padding-oracle attack was still successfully carried out using timing information of this sort. We conclude that authenticate-then-encrypt does not provide authenticated encryption in general, and should not be used.

Encrypt-then-authenticate. In this approach, the message is first encrypted and then a MAC is computed over the result. That is, the message is the pair $\langle c, t \rangle$ where

$$c \leftarrow \text{Enc}_{k_E}(m) \text{ and } t \leftarrow \text{Mac}_{k_M}(c).$$

Decryption of $\langle c, t \rangle$ is done by outputting $\perp$ if $\text{Vrfy}_{k_M}(c, t) \neq 1$, and otherwise outputting $\text{Dec}_{k_E}(c)$. See Construction 4.18 for a formal description.

CONSTRUCTION 4.18

Let $\Pi_E = (\text{Enc}, \text{Dec})$ be a private-key encryption scheme and let $\Pi_M = (\text{Mac}, \text{Vrfy})$ be a message authentication code, where in each case key generation is done by simply choosing a uniform n -bit key. Define a private-key encryption scheme $(\text{Gen}', \text{Enc}', \text{Dec}')$ as follows:

- **Gen'**: on input 1^n , choose independent, uniform $k_E, k_M \in \{0, 1\}^n$ and output the key (k_E, k_M) .
- **Enc'**: on input a key (k_E, k_M) and a plaintext message m , compute $c \leftarrow \text{Enc}_{k_E}(m)$ and $t \leftarrow \text{Mac}_{k_M}(c)$. Output the ciphertext $\langle c, t \rangle$.
- **Dec'**: on input a key (k_E, k_M) and a ciphertext $\langle c, t \rangle$, first check whether $\text{Vrfy}_{k_M}(c, t) \stackrel{?}{=} 1$. If yes, then output $\text{Dec}_{k_E}(c)$; if no, then output $\perp$.

A generic construction of an authenticated encryption scheme.

This approach *is* sound, as long as the MAC is *strongly secure*, as in Definition 4.3. As intuition for the security of this approach, say a ciphertext $\langle c, t \rangle$ is *valid* if t is a valid MAC tag on c . Strong security of the MAC ensures that an

adversary will be unable to generate *any* valid ciphertext that it did not receive from its encryption oracle. This immediately implies that Construction 4.18 is unforgeable. As for CCA-security, the MAC computed over the ciphertext has the effect of rendering the decryption oracle useless since for every ciphertext $\langle c, t \rangle$ the adversary submits to its decryption oracle, the adversary either already knows the decryption (if it received $\langle c, t \rangle$ from its own encryption oracle) or else can expect the result to be an error (since the adversary cannot generate any new, valid ciphertexts). This means that CCA-security of the combined scheme reduces to the CPA-security of Π_E . Observe also that the MAC is verified before decryption takes place; thus, MAC verification cannot leak anything about the plaintext (in contrast to the padding-oracle attack we saw for the authenticate-then-encrypt approach). We now formalize the above arguments.

THEOREM 4.19 *Let Π_E be a CPA-secure private-key encryption scheme, and let Π_M be a strongly secure message authentication code. Then Construction 4.18 is an authenticated encryption scheme.*

PROOF Let Π' denote the scheme resulting from Construction 4.18. We need to show that Π' is unforgeable, and that it is CCA-secure. Following the intuition given above, say a ciphertext $\langle c, t \rangle$ is *valid* (with respect to some fixed secret key (k_E, k_M)) if $\text{Vrfy}_{k_M}(c, t) = 1$. We show that strong security of Π_M implies that (except with negligible probability) any “new” ciphertexts the adversary submits to the decryption oracle will be invalid. As discussed already, this immediately implies unforgeability. (In fact, it is stronger than unforgeability.) This fact also renders the decryption oracle useless, and means that CCA-security of $\Pi' = (\text{Gen}', \text{Enc}', \text{Dec}')$ reduces to the CPA-security of Π_E .

In more detail, let $\mathcal{A}$ be a probabilistic polynomial-time adversary attacking Construction 4.18 in a chosen-ciphertext attack (cf. Definition 3.33). Say a ciphertext $\langle c, t \rangle$ is *new* if $\mathcal{A}$ did not receive $\langle c, t \rangle$ from its encryption oracle or as the challenge ciphertext. Let **ValidQuery** be the event that $\mathcal{A}$ submits a new ciphertext $\langle c, t \rangle$ to its decryption oracle which is valid, i.e., for which $\text{Vrfy}_{k_M}(c, t) = 1$. We prove:

CLAIM 4.20 $\Pr[\text{ValidQuery}]$ is negligible.

PROOF Intuitively, this is due to the fact that if **ValidQuery** occurs then the adversary has forged a new, valid pair (c, t) in the **Mac-sforge** experiment. Formally, let $q(\cdot)$ be a polynomial upper bound on the number of decryption-oracle queries made by $\mathcal{A}$, and consider the following adversary $\mathcal{A}_M$ attacking the message authentication code Π_M (i.e., running in experiment $\text{Mac-sforge}_{\mathcal{A}_M, \Pi_M}(n)$):

Adversary $\mathcal{A}_M$:

$\mathcal{A}_M$ is given input 1^n and has access to a MAC oracle $\text{Mac}_{k_M}(\cdot)$.

1. Choose uniform $k_E \in \{0, 1\}^n$ and $i \in \{1, \dots, q(n)\}$.
2. Run $\mathcal{A}$ on input 1^n . When $\mathcal{A}$ makes an encryption-oracle query for the message m , answer it as follows:
 - (i) Compute $c \leftarrow \text{Enc}_{k_E}(m)$.
 - (ii) Query c to the MAC oracle and receive t in response. Return $\langle c, t \rangle$ to $\mathcal{A}$.

The challenge ciphertext is prepared in the exact same way (with a uniform bit $b \in \{0, 1\}$ chosen to select the message m_b that gets encrypted).

When $\mathcal{A}$ makes a decryption-oracle query for the ciphertext $\langle c, t \rangle$, answer it as follows: If this is the i th decryption-oracle query, output (c, t) . Otherwise:

- (i) If $\langle c, t \rangle$ was a response to a previous encryption-oracle query for a message m , return m .
- (ii) Otherwise, return $\perp$.

In essence, $\mathcal{A}_M$ is “guessing” that the i th decryption-oracle query of $\mathcal{A}$ will be the first new, valid query $\mathcal{A}$ makes. In that case, $\mathcal{A}_M$ outputs a valid forgery on a message c that it had never previously submitted to its own MAC oracle.

Clearly $\mathcal{A}_M$ runs in probabilistic polynomial time. We now analyze the probability that $\mathcal{A}_M$ produces a good forgery. The key point is that the view of $\mathcal{A}$ when run as a subroutine by $\mathcal{A}_M$ is distributed identically to the view of $\mathcal{A}$ in experiment $\text{PrivK}_{\mathcal{A}, \Pi'}^{\text{cca}}(n)$ until event **ValidQuery** occurs. To see this, note that the encryption-oracle queries of $\mathcal{A}$ (as well as computation of the challenge ciphertext) are simulated perfectly by $\mathcal{A}_M$. As for the decryption-oracle queries of $\mathcal{A}$, until **ValidQuery** occurs these are all simulated properly. In case (i) this is obvious. As for case (ii), if the ciphertext $\langle c, t \rangle$ submitted to the decryption oracle is new, then as long as **ValidQuery** has not yet occurred the correct answer to the decryption-oracle query is indeed $\perp$. (Note that case (i) is exactly the case that $\langle c, t \rangle$ is not new, and case (ii) is exactly the case that $\langle c, t \rangle$ is new.) Recall that $\mathcal{A}$ is disallowed from submitting the challenge ciphertext to the decryption oracle.

Because the view of $\mathcal{A}$ when run as a subroutine by $\mathcal{A}_M$ is distributed identically to the view of $\mathcal{A}$ in experiment $\text{PrivK}_{\mathcal{A}, \Pi'}^{\text{cca}}(n)$ until event **ValidQuery** occurs, the probability of event **ValidQuery** in experiment $\text{Mac-forg}_{\mathcal{A}_M, \Pi_M}(n)$ is the same as the probability of that event in experiment $\text{PrivK}_{\mathcal{A}, \Pi'}^{\text{cca}}(n)$.

If $\mathcal{A}_M$ correctly guesses the first index i when **ValidQuery** occurs, then $\mathcal{A}_M$ outputs (c, t) for which $\text{Vrfy}_{k_M}(c, t) = 1$ (since $\langle c, t \rangle$ is valid) and for which it was never given tag t in response to the query $\text{Mac}_{k_M}(c)$ (since $\langle c, t \rangle$ is new). In this case, then, $\mathcal{A}_M$ succeeds in experiment $\text{Mac-sforge}_{\mathcal{A}_M, \Pi_M}(n)$.

The probability that $\mathcal{A}_M$ guesses i correctly is $1/q(n)$. Therefore

$$\Pr[\text{Mac-sforge}_{\mathcal{A}_M, \Pi_M}(n) = 1] \geq \Pr[\text{ValidQuery}]/q(n).$$

Since Π_M is a strongly secure MAC and q is polynomial, we conclude that $\Pr[\text{ValidQuery}]$ is negligible. $\blacksquare$

We use Claim 4.20 to prove security of Π' . The easier case is to prove that Π' is unforgeable. This follows immediately from the claim, and so we just provide informal reasoning rather than a formal proof. Observe first that the adversary $\mathcal{A}'$ in the unforgeable encryption experiment is a restricted version of the adversary in the chosen-ciphertext experiment (in the former, the adversary only has access to an encryption oracle). When $\mathcal{A}'$ outputs a ciphertext $\langle c, t \rangle$ at the end of its experiment, it “succeeds” only if $\langle c, t \rangle$ is valid and new. But the previous claim shows precisely that the probability of such an event is negligible.

It is slightly more involved to prove that Π' is CCA-secure. Let $\mathcal{A}$ again be a probabilistic polynomial-time adversary attacking Π' in a chosen-ciphertext attack. We have

$$\begin{aligned} \Pr[\text{PrivK}_{\mathcal{A}, \Pi'}^{\text{cca}}(n) = 1] \\ \leq \Pr[\text{ValidQuery}] + \Pr[\text{PrivK}_{\mathcal{A}, \Pi'}^{\text{cca}}(n) = 1 \wedge \overline{\text{ValidQuery}}]. \end{aligned} \quad (4.8)$$

We have already shown that $\Pr[\text{ValidQuery}]$ is negligible. The following claim thus completes the proof of the theorem.

CLAIM 4.21 *There exists a negligible function negl such that*

$$\Pr[\text{PrivK}_{\mathcal{A}, \Pi'}^{\text{cca}}(n) = 1 \wedge \overline{\text{ValidQuery}}] \leq \frac{1}{2} + \text{negl}(n).$$

To prove this claim, we rely on CPA-security of Π_E . Consider the following adversary $\mathcal{A}_E$ attacking Π_E in a chosen-plaintext attack:

Adversary $\mathcal{A}_E$:

$\mathcal{A}_E$ is given input 1^n and has access to $\text{Enc}_{k_E}(\cdot)$.

1. Choose uniform $k_M \in \{0, 1\}^n$.
2. Run $\mathcal{A}$ on input 1^n . When $\mathcal{A}$ makes an encryption-oracle query for the message m , answer it as follows:
 - (i) Query m to $\text{Enc}_{k_E}(\cdot)$ and receive c in response.
 - (ii) Compute $t \leftarrow \text{Mac}_{k_M}(c)$ and return $\langle c, t \rangle$ to $\mathcal{A}$.

When $\mathcal{A}$ makes a decryption-oracle query for the ciphertext $\langle c, t \rangle$, answer it as follows:

- If $\langle c, t \rangle$ was a response to a previous encryption-oracle query for a message m , return m . Otherwise, return $\perp$.
- 3. When $\mathcal{A}$ outputs messages (m_0, m_1) , output these same messages and receive a challenge ciphertext c in response. Compute $t \leftarrow \text{Mac}_{k_M}(c)$, and return $\langle c, t \rangle$ as the challenge ciphertext for $\mathcal{A}$. Continue answering $\mathcal{A}$'s oracle queries as above.
- 4. Output the same bit b' that is output by $\mathcal{A}$.

Notice that $\mathcal{A}_E$ does not need a decryption oracle because it simply assumes that any decryption query by $\mathcal{A}$ that was not the result of a previous encryption-oracle query is invalid.

Clearly, $\mathcal{A}_E$ runs in probabilistic polynomial time. Furthermore, the view of $\mathcal{A}$ when run as a subroutine by $\mathcal{A}_E$ is distributed identically to the view of $\mathcal{A}$ in experiment $\text{PrivK}_{\mathcal{A}, \Pi'}^{\text{cca}}(n)$ as long as event **ValidQuery** never occurs. Therefore, the probability that $\mathcal{A}_E$ succeeds when **ValidQuery** does not occur is the same as the probability that $\mathcal{A}$ succeeds when **ValidQuery** does not occur; i.e.,

$$\Pr[\text{PrivK}_{\mathcal{A}_E, \Pi_E}^{\text{cpa}}(n) = 1 \wedge \overline{\text{ValidQuery}}] = \Pr[\text{PrivK}_{\mathcal{A}, \Pi'}^{\text{cca}}(n) = 1 \wedge \overline{\text{ValidQuery}}],$$

implying that

$$\begin{aligned} \Pr[\text{PrivK}_{\mathcal{A}_E, \Pi_E}^{\text{cpa}}(n) = 1] &\geq \Pr[\text{PrivK}_{\mathcal{A}_E, \Pi_E}^{\text{cpa}}(n) = 1 \wedge \overline{\text{ValidQuery}}] \\ &= \Pr[\text{PrivK}_{\mathcal{A}, \Pi'}^{\text{cca}}(n) = 1 \wedge \overline{\text{ValidQuery}}]. \end{aligned}$$

Since Π_E is CPA-secure, there exists a negligible function negl such that $\Pr[\text{PrivK}_{\mathcal{A}_E, \Pi_E}^{\text{cpa}}(n) = 1] \leq \frac{1}{2} + \text{negl}(n)$. This proves the claim. ■

The need for independent keys. We conclude this section by stressing a basic principle of cryptography: *different instances of cryptographic primitives should always use independent keys*. To illustrate this here, consider what can happen to the encrypt-then-authenticate methodology when the same key k is used for both encryption and authentication. Let F be a strong pseudorandom permutation. It follows that F^{-1} is a strong pseudorandom permutation also. Define $\text{Enc}_k(m) = F_k(m||r)$ for $m \in \{0, 1\}^{n/2}$ and a uniform $r \in \{0, 1\}^{n/2}$, and define $\text{Mac}_k(c) = F_k^{-1}(c)$. It can be shown that this encryption scheme is CPA-secure (in fact, it is even CCA-secure; see Exercise 4.25), and we know that the given message authentication code is a secure MAC. However, the encrypt-then-authenticate combination using the same key k as applied to the message m yields:

$$\text{Enc}_k(m), \text{Mac}_k(\text{Enc}_k(m)) = F_k(m||r), F_k^{-1}(F_k(m||r)) = F_k(m||r), m||r,$$

and the message m is revealed in the clear! This does not in any way contradict Theorem 4.19, since Construction 4.18 expressly requires that k_M, k_E are chosen (uniformly and) independently. We encourage the reader to examine where this independence is used in the proof of Theorem 4.19.

4.5.3 Secure Communication Sessions

We briefly describe the application of authenticated encryption to the setting of two parties who wish to communicate “securely”—namely, with joint secrecy and integrity—over the course of a communication session. (For the purposes of this section, a *communication session* is simply a period of time during which the communicating parties maintain state.) In our treatment here we are deliberately informal; a formal definition is quite involved, and this topic arguably lies more in the area of network security than cryptography.

Let $\Pi = (\text{Enc}, \text{Dec})$ be an authenticated encryption scheme. Consider two parties A and B who share a key k and wish to use this key to secure their communication over the course of a session. The obvious thing to do here is to use Π : Whenever, say, A wants to transmit a message m to B , it computes $c \leftarrow \text{Enc}_k(m)$ and sends c to B ; in turn, B decrypts c to recover the result (ignoring the result if decryption returns $\perp$). Likewise, the same procedure is followed when B wants to send a message to A . This simple approach, however, does not suffice, as there are various potential attacks:

Re-ordering attack An attacker can swap the order of messages. For example, if A transmits c_1 (an encryption of m_1) and subsequently transmits c_2 (an encryption of m_2), an attacker who has some control over the network can deliver c_2 before c_1 and thus cause B to output the messages in the wrong order. This causes a mismatch between the two parties’ views of their communication session.

Replay attack An attacker can *replay* a (valid) ciphertext c sent previously by one of the parties. Again, this causes a mismatch between what is sent by one party and received by the other.

Reflection attack An attacker can take a ciphertext c sent from A to B and send it back to A . This again can cause a mismatch between the two parties’ transcripts of their communication session: A may output a message m , even though B never sent such a message.

Fortunately, the above attacks are easy to prevent using *counters* to address the first two and a *directionality bit* to prevent the third.⁵ We describe these in tandem. Each party maintains two counters $\text{ctr}_{A,B}$ and $\text{ctr}_{B,A}$ keeping track of the number of messages sent from A to B (resp., B to A) during the session. These counters are initialized to 0 and incremented each time a party sends or receives a (valid) message. The parties also agree on a bit $b_{A,B}$, and define $b_{B,A}$ to be its complement. (One way to do this is to set $b_{A,B} = 0$ iff the identity of A is lexicographically smaller than the identity of B .)

⁵In practice, the issue of directionality is often solved by simply having separate keys for each direction (i.e., the parties use a key k_A for messages sent from A to B , and a different key k_B for messages sent from B to A).

When A wants to transmit a message m to B , she computes the ciphertext $c \leftarrow \text{Enc}_k(b_{A,B} \parallel \text{ctr}_{A,B} \parallel m)$ and sends c ; she then increments $\text{ctr}_{A,B}$. Upon receiving c , party B decrypts; if the result is $\perp$, he immediately rejects. Otherwise, he parses the decrypted message as $b \parallel \text{ctr} \parallel m$. If $b = b_{A,B}$ and $\text{ctr} = \text{ctr}_{A,B}$, then B outputs m and increments $\text{ctr}_{A,B}$; otherwise, B rejects. The above steps, *mutatis mutandis*, are applied when B sends a message to A .

We remark that since the parties are anyway maintaining state (namely, the counters $\text{ctr}_{A,B}$ and $\text{ctr}_{B,A}$), the parties could easily use a *stateful* authenticated encryption scheme Π .

4.5.4 CCA-Secure Encryption

It follows directly from the definition that any authenticated encryption scheme is also secure against chosen-ciphertext attacks. Can there be CCA-secure private-key encryption schemes that are *not* unforgeable? Indeed, there are; see Exercise 4.25.

One can imagine applications where CCA-security is needed but authenticated encryption is not. One example might be when private-key encryption is used for *key transport*. As a concrete example, say a server gives a tamper-proof hardware token to a user, where embedded in the token is a long-term key k . The server can upload a fresh, short-term key k' to this token by giving the user $\text{Enc}_k(k')$; the user is supposed to give this ciphertext to the token, which will decrypt it and use k' for the next time period. A chosen-ciphertext attack in this setting could allow the user to learn k' , something the user is not supposed to be able to do. (Note that here a padding-oracle attack, which only relies on the user's ability to determine whether a decryption failure occurs, could potentially be carried out rather easily.) On the other hand, not much harm is done if the user can generate a “valid” ciphertext that causes the token to use an arbitrary (unrelated) key k'' for the next time period. (Of course, this depends on what the token does with this short-term key.)

Notwithstanding the above, for private-key encryption most “natural” constructions of CCA-secure schemes that we know anyway satisfy the stronger definition of authenticated encryption. Put differently, there is no real reason to ever use a CCA-secure scheme that is *not* an authenticated encryption scheme, simply because we don't really have any constructions satisfying the former that are more efficient than constructions achieving the latter.

From a *conceptual* point of view, however, it is important to keep the notions of CCA-security and authenticated encryption distinct. With regard to CCA-security we are not interested in message integrity *per se*; rather, we wish to ensure privacy even against an active adversary who can interfere with the communication as it goes from sender to receiver. In contrast, with regard to authenticated encryption we are interested in the twin goals of secrecy and integrity. We stress this here because in the public-key setting that we study later in the book, the difference between authenticated encryption and CCA-security is more pronounced.

4.6 *Information-Theoretic MACs

In previous sections we have explored message authentication codes with *computational* security, i.e., where bounds on the attacker's running time are assumed. Recalling the results of Chapter 2, it is natural to ask whether message authentication in the presence of an *unbounded* adversary is possible. In this section, we show under which conditions *information-theoretic* (as opposed to computational) security is attainable.

A first observation is that it is impossible to achieve “perfect” security in this context: namely, we cannot hope to have a message authentication code for which the probability that an adversary outputs a valid tag on a previously unauthenticated message is 0. The reason is that an adversary can simply guess a valid tag t on any message and the guess will be correct with probability (at least) $1/2^{|t|}$, where $|t|$ denotes the tag length of the scheme.

The above example tells us what we *can* hope to achieve: a MAC with tags of length $|t|$ where the probability of forgery is at most $1/2^{|t|}$, even for unbounded adversaries. We will see that this is achievable, but only under restrictions on how many messages are authenticated by the honest parties.

We first define information-theoretic security for message authentication codes. A starting point is to take experiment $\text{Mac-forge}_{\mathcal{A},\Pi}(n)$ that is used to define security for computationally secure MACs (cf. Definition 4.2), but to drop the security parameter n and require that $\Pr[\text{Mac-forge}_{\mathcal{A},\Pi} = 1]$ should be “small” for *all* adversaries $\mathcal{A}$ (and not just adversaries running in polynomial time). As mentioned above (and as will be proved formally in Section 4.6.2), however, such a definition is impossible to achieve. Rather, information-theoretic security can be achieved only if we place some bound on the number of messages authenticated by the honest parties. We look here at the most basic setting, where the parties authenticate just a *single* message. We refer to this as *one-time message authentication*. The following experiment modifies $\text{Mac-forge}_{\mathcal{A},\Pi}(n)$ following the above discussion:

The one-time message authentication experiment $\text{Mac-forge}_{\mathcal{A},\Pi}^{1\text{-time}}$:

1. A key k is generated by running Gen .
2. The adversary $\mathcal{A}$ outputs a message m' , and is given in return a tag $t' \leftarrow \text{Mac}_k(m')$.
3. $\mathcal{A}$ outputs (m, t) .
4. The output of the experiment is defined to be 1 if and only if
(1) $\text{Vrfy}_k(m, t) = 1$ and (2) $m \neq m'$.

DEFINITION 4.22 A message authentication code $\Pi = (\text{Gen}, \text{Mac}, \text{Vrfy})$ is *one-time ε -secure*, or just *ε -secure*, if for all (even unbounded) adversaries $\mathcal{A}$:

$$\Pr[\text{Mac-forge}_{\mathcal{A},\Pi}^{1\text{-time}} = 1] \leq \varepsilon.$$

4.6.1 Constructing Information-Theoretic MACs

In this section we show how to build an ε -secure MAC based on any *strongly universal function*.⁶ We then show a simple construction of the latter.

Let $h : \mathcal{K} \times \mathcal{M} \rightarrow \mathcal{T}$ be a keyed function whose first input is a key $k \in \mathcal{K}$ and whose second input is taken from some domain $\mathcal{M}$. As usual, we write $h_k(m)$ instead of $h(k, m)$. Then h is *strongly universal* (or *pairwise-independent*) if for any two distinct inputs m, m' the values $h_k(m)$ and $h_k(m')$ are uniformly and independently distributed in $\mathcal{T}$ when k is a uniform key. This is equivalent to saying that the probability that $h_k(m), h_k(m')$ take on any particular values t, t' is exactly $1/|\mathcal{T}|^2$. That is:

DEFINITION 4.23 A function $h : \mathcal{K} \times \mathcal{M} \rightarrow \mathcal{T}$ is *strongly universal* if for all distinct $m, m' \in \mathcal{M}$ and all $t, t' \in \mathcal{T}$ it holds that

$$\Pr[h_k(m) = t \wedge h_k(m') = t'] = \frac{1}{|\mathcal{T}|^2},$$

where the probability is taken over uniform choice of $k \in \mathcal{K}$.

The above should motivate the construction of a one-time message authentication code from any strongly universal function h . The tag t on a message m is obtained by computing $h_k(m)$, where the key k is uniform; see Construction 4.24. Intuitively, even after an adversary observes the tag $t' := h_k(m')$ for any message m' , the correct tag $h_k(m)$ for any *other* message m is still uniformly distributed in $\mathcal{T}$ from the adversary's point of view. Thus, the adversary can do nothing more than blindly guess the tag, and this guess will be correct only with probability $1/|\mathcal{T}|$.

CONSTRUCTION 4.24

Let $h : \mathcal{K} \times \mathcal{M} \rightarrow \mathcal{T}$ be a strongly universal function. Define a MAC for messages in $\mathcal{M}$ as follows:

- **Gen**: choose uniform $k \in \mathcal{K}$ and output it as the key.
- **Mac**: on input a key $k \in \mathcal{K}$ and a message $m \in \mathcal{M}$, output the tag $t := h_k(m)$.
- **Vrfy**: on input a key $k \in \mathcal{K}$, a message $m \in \mathcal{M}$, and a tag $t \in \mathcal{T}$, output 1 if and only if $t \stackrel{?}{=} h_k(m)$. (If $m \notin \mathcal{M}$, then output 0.)

A MAC from any strongly universal function.

⁶These are often called strongly universal *hash* functions, but in cryptographic contexts the term “hash” has another meaning that we will see later in the book.

The above construction can be viewed as analogous to Construction 4.5. This is because a strongly universal function h is identical to a random function, as long as it is only evaluated twice.

THEOREM 4.25 *Let $h : \mathcal{K} \times \mathcal{M} \rightarrow \mathcal{T}$ be a strongly universal function. Then Construction 4.24 is a $1/|\mathcal{T}|$ -secure MAC for messages in $\mathcal{M}$.*

PROOF Let $\mathcal{A}$ be an adversary. As usual in the information-theoretic setting, we may assume $\mathcal{A}$ is deterministic without loss of generality. So the message m' that $\mathcal{A}$ outputs at the outset of the experiment is fixed. Furthermore, the pair (m, t) that $\mathcal{A}$ outputs at the end of the experiment is a deterministic function of the tag t' on m' that $\mathcal{A}$ receives. We thus have

$$\begin{aligned} \Pr[\text{Mac-forge}_{\mathcal{A}, \Pi}^{1\text{-time}} = 1] &= \sum_{t' \in \mathcal{T}} \Pr[\text{Mac-forge}_{\mathcal{A}, \Pi}^{1\text{-time}} = 1 \wedge h_k(m') = t'] \\ &= \sum_{\substack{t' \in \mathcal{T} \\ (m, t) := \mathcal{A}(t')}} \Pr[h_k(m) = t \wedge h_k(m') = t'] \\ &= \sum_{\substack{t' \in \mathcal{T} \\ (m, t) := \mathcal{A}(t')}} \frac{1}{|\mathcal{T}|^2} = \frac{1}{|\mathcal{T}|}. \end{aligned}$$

This proves the theorem. ■

We now turn to a classical construction of a strongly universal function. We assume some basic knowledge about arithmetic modulo a prime number; readers may refer to Sections 8.1.1 and 8.1.2 for the necessary background. Fix some prime p , and let $\mathbb{Z}_p \stackrel{\text{def}}{=} \{0, \dots, p-1\}$. We take as our message space $\mathcal{M} = \mathbb{Z}_p$; the space of possible tags will also be $\mathcal{T} = \mathbb{Z}_p$. A key (a, b) consists of a pair of elements from $\mathbb{Z}_p$; thus, $\mathcal{K} = \mathbb{Z}_p \times \mathbb{Z}_p$. Define h as

$$h_{a,b}(m) \stackrel{\text{def}}{=} [a \cdot m + b \bmod p],$$

where the notation $[X \bmod p]$ refers to the reduction of the integer X modulo p (and so $[X \bmod p] \in \mathbb{Z}_p$ always).

THEOREM 4.26 *For any prime p , the function h is strongly universal.*

PROOF Fix any distinct $m, m' \in \mathbb{Z}_p$ and any $t, t' \in \mathbb{Z}_p$. For which keys (a, b) does it hold that both $h_{a,b}(m) = t$ and $h_{a,b}(m') = t'$? This holds only if

$$a \cdot m + b = t \bmod p \quad \text{and} \quad a \cdot m' + b = t' \bmod p.$$

We thus have two linear equations in the two unknowns a, b . These two equations are both satisfied exactly when $a = [(t - t') \cdot (m - m')^{-1} \bmod p]$ and $b = [t - a \cdot m \bmod p]$; note that $[(m - m')^{-1} \bmod p]$ exists because $m \neq m'$ and so $m - m' \not\equiv 0 \bmod p$. Restated, this means that for any m, m', t, t' as above there is a *unique* key (a, b) with $h_{a,b}(m) = t$ and $h_{a,b}(m') = t'$. Since there are $|\mathcal{T}|^2$ keys, we conclude that the probability (over choice of the key) that $h_{a,b}(m) = t$ and $h_{a,b}(m') = t'$ is exactly $1/|\mathcal{K}| = 1/|\mathcal{T}|^2$ as required. ■

Parameters of Construction 4.24. We briefly discuss the parameters of Construction 4.24 when instantiated with the strongly universal function described above, ignoring the fact that p is not a power of 2. The construction is a $1/|\mathcal{T}|$ -secure MAC with tags of length $\log |\mathcal{T}|$; the tag length is optimal for the level of security achieved.

Let $\mathcal{M}$ be some fixed message space for which we want to construct a one-time secure MAC. The construction above gives a $1/|\mathcal{M}|$ -secure MAC with keys that are twice the length of the messages. The reader may notice two problems here, at opposite ends of the spectrum: First, if $|\mathcal{M}|$ is small then a $1/|\mathcal{M}|$ probability of forgery may be unacceptably large. On the flip side, if $|\mathcal{M}|$ is large then a $1/|\mathcal{M}|$ probability of forgery may be overkill; one might be willing to accept a (somewhat) larger probability of forgery if that level of security can be achieved with shorter keys. The first problem (when $|\mathcal{M}|$ is small) is easy to deal with by simply embedding $\mathcal{M}$ into a larger message space $\mathcal{M}'$ by, for example, padding messages with 0s. The second problem can be addressed as well; see the references at the end of this chapter.

4.6.2 Limitations on Information-Theoretic MACs

In this section we explore limitations on information-theoretic message authentication. We show that any 2^{-n} -secure MAC must have keys of length at least $2n$. An extension of the proof shows that any ℓ -time 2^{-n} -secure MAC (where security is defined via a natural modification of Definition 4.23) requires keys of length at least $(\ell + 1) \cdot n$. A corollary is that *no* MAC with bounded-length keys can provide information-theoretic security when authenticating an unbounded number of messages.

In the following, we assume the message space contains at least two messages; if not, there is no point in communicating, let alone authenticating.

THEOREM 4.27 *Let $\Pi = (\text{Gen}, \text{Mac}, \text{Vrfy})$ be a 2^{-n} -secure MAC where all keys output by Gen are the same length. Then the keys output by Gen must have length at least $2n$.*

PROOF Fix two distinct messages m_0, m_1 in the message space. The intuition for the proof is that there must be at least 2^n possibilities for the tag

of m_0 (or else the adversary could guess it with probability better than 2^{-n}); furthermore, even conditioned on the value of the tag for m_0 , there must be 2^n possibilities for the tag of m_1 (or else the adversary could forge a tag on m_1 with probability better than 2^{-n}). Since each key defines tags for m_0 and m_1 , this means there must be at least $2^n \times 2^n$ keys. We make this formal below.

Let $\mathcal{K}$ denote the key space (i.e., the set of all possible keys that can be output by **Gen**). For any possible tag t_0 , let $\mathcal{K}(t_0)$ denote the set of keys for which t_0 is a valid tag on m_0 ; i.e.,

$$\mathcal{K}(t_0) \stackrel{\text{def}}{=} \{k \mid \text{Vrfy}_k(m_0, t_0) = 1\}.$$

For any t_0 we must have $|\mathcal{K}(t_0)| \leq 2^{-n} \cdot |\mathcal{K}|$. Otherwise the adversary could simply output (m_0, t_0) as its forgery; this would be a valid forgery with probability at least $|\mathcal{K}(t_0)|/|\mathcal{K}| > 2^{-n}$, contradicting the claimed security.

Consider now the adversary $\mathcal{A}$ who requests a tag on the message m_0 , receives in return a tag t_0 , chooses a uniform key $k \in \mathcal{K}(t_0)$, and outputs $(m_1, \text{Mac}_k(m_1))$ as its forgery. The probability that $\mathcal{A}$ outputs a valid forgery is at least

$$\begin{aligned} \sum_{t_0} \Pr[\text{Mac}_k(m_0) = t_0] \cdot \frac{1}{|\mathcal{K}(m_0, t_0)|} &\geq \sum_{t_0} \Pr[\text{Mac}_k(m_0) = t_0] \cdot \frac{2^n}{|\mathcal{K}|} \\ &= \frac{2^n}{|\mathcal{K}|}. \end{aligned}$$

By the claimed security of the scheme, the probability that the adversary can output a valid forgery is at most 2^{-n} . Thus, we must have $|\mathcal{K}| \geq 2^{2n}$. Since all keys have the same length, each key must have length at least $2n$. ■

References and Additional Reading

The definition of security for message authentication codes was adapted by Bellare et al. [18] from the definition of security for digital signatures given by Goldwasser et al. [81] (see Chapter 12). For more on the definitional variant where verification queries are allowed, see [17].

The paradigm of using pseudorandom functions for message authentication (as in Construction 4.5) was introduced by Goldreich et al. [77]. Construction 4.7 is due to Goldreich [76].

CBC-MAC was standardized in the early 1980s [94, 178] and is still used widely today. Basic CBC-MAC was proven secure (for authenticating fixed-length messages) by Bellare et al. [18]. Bernstein [26, 27] gives a more direct (though perhaps less intuitive) proof, and also discusses some generalized

versions of CBC-MAC. As noted in this chapter, basic CBC-MAC is insecure when used to authenticate messages of different lengths. One way to fix this is to prepend the length to the message. This has the disadvantage of not being able to cope with *streaming* data, where the length of the message is not known in advance. Petrank and Rackoff [137] suggest an alternate, “on-line” approach addressing this issue. Further improvements were given by Black and Rogaway [32] and Iwata and Kurosawa [95]; these led to a new proposed standard called CMAC.

The importance of authenticated encryption was first explicitly highlighted in [100, 19], who propose definitions similar to what we have given here. Bellare and Namprempre [19] analyze the three generic approaches discussed here, although the idea of using encrypt-then-authenticate for achieving CCA-security goes back at least to the work of Dolev et al. [61]. Krawczyk [108] examines other methods for achieving secrecy and authentication, and also analyzes the authenticate-then-encrypt approach used by SSL. Degabriele and Paterson [54] show an attack on IPsec when configured to authenticate-then-encrypt (the default for authenticated encryption is actually encrypt-then-authenticate; however it is possible to achieve authenticate-then-encrypt in some configurations). Several nongeneric schemes for authenticated encryption have also been proposed; see [110] for a detailed comparison.

Information-theoretic MACs were first studied by Gilbert et al. [73]. Wegman and Carter [177] introduced the notion of strongly universal functions, and noted their application to one-time message authentication. They also show how to reduce the key length for this task by using an *almost* strongly universal function. Specifically, the construction we give here achieves 2^{-n} -security for messages of length n with keys of length $O(n)$; Wegman and Carter show how to construct a 2^{-n} -secure MAC for messages of length ℓ with keys of (essentially) length $O(n \cdot \log \ell)$. The simple construction of a strongly universal function that we give here is (with minor differences) due to Carter and Wegman [42]. The reader interested in learning more about information-theoretic MACs is referred to the paper by Stinson [166], the survey by Simmons [162], or the first edition of Stinson’s textbook [167, Chapter 10].

Exercises

- 4.1 Say $\Pi = (\text{Gen}, \text{Mac}, \text{Vrfy})$ is a secure MAC, and for $k \in \{0, 1\}^n$ the tag-generation algorithm Mac_k always outputs tags of length $t(n)$. Prove that t must be super-logarithmic or, equivalently, that if $t(n) = O(\log n)$ then Π cannot be a secure MAC.

Hint: Consider the probability of randomly guessing a valid tag.

- 4.2 Consider an extension of the definition of secure message authentication where the adversary is provided with both a **Mac** and a **Vrfy** oracle.
- Provide a formal definition of security for this case.
 - Assume Π is a deterministic MAC using canonical verification that satisfies Definition 4.2. Prove that Π also satisfies your definition from part (a).
- 4.3 Assume secure MACs exist. Give a construction of a MAC that is secure with respect to Definition 4.2 but that is not secure when the adversary is additionally given access to a **Vrfy** oracle (cf. the previous exercise).
- 4.4 Prove Proposition 4.4.
- 4.5 Assume secure MACs exist. Prove that there exists a MAC that is secure (by Definition 4.2) but is *not* strongly secure (by Definition 4.3).
- 4.6 Consider the following MAC for messages of length $\ell(n) = 2n - 2$ using a pseudorandom function F : On input a message $m_0 \| m_1$ (with $|m_0| = |m_1| = n - 1$) and key $k \in \{0, 1\}^n$, algorithm **Mac** outputs $t = F_k(0 \| m_0) \| F_k(1 \| m_1)$. Algorithm **Vrfy** is defined in the natural way. Is $(\text{Gen}, \text{Mac}, \text{Vrfy})$ secure? Prove your answer.
- 4.7 Let F be a pseudorandom function. Show that each of the following MACs is insecure, even if used to authenticate fixed-length messages. (In each case **Gen** outputs a uniform $k \in \{0, 1\}^n$. Let $\langle i \rangle$ denote an $n/2$ -bit encoding of the integer i .)
- To authenticate a message $m = m_1, \dots, m_\ell$, where $m_i \in \{0, 1\}^n$, compute $t := F_k(m_1) \oplus \dots \oplus F_k(m_\ell)$.
 - To authenticate a message $m = m_1, \dots, m_\ell$, where $m_i \in \{0, 1\}^{n/2}$, compute $t := F_k(\langle 1 \rangle \| m_1) \oplus \dots \oplus F_k(\langle \ell \rangle \| m_\ell)$.
 - To authenticate a message $m = m_1, \dots, m_\ell$, where $m_i \in \{0, 1\}^{n/2}$, choose uniform $r \leftarrow \{0, 1\}^n$, compute

$$t := F_k(r) \oplus F_k(\langle 1 \rangle \| m_1) \oplus \dots \oplus F_k(\langle \ell \rangle \| m_\ell),$$
 and let the tag be $\langle r, t \rangle$.
- 4.8 Let F be a pseudorandom function. Show that the following MAC for messages of length $2n$ is insecure: **Gen** outputs a uniform $k \in \{0, 1\}^n$. To authenticate a message $m_1 \| m_2$ with $|m_1| = |m_2| = n$, compute the tag $F_k(m_1) \| F_k(F_k(m_2))$.
- 4.9 Given any *deterministic* MAC $(\text{Mac}, \text{Vrfy})$, we may view **Mac** as a keyed function. In both Constructions 4.5 and 4.11, **Mac** is a pseudorandom function. Give a construction of a secure, deterministic MAC in which **Mac** is *not* a pseudorandom function.

- 4.10 Is Construction 4.5 necessarily secure when instantiated using a weak pseudorandom function (cf. Exercise 3.26)? Explain.
- 4.11 Prove that Construction 4.7 is secure even when the adversary is additionally given access to a Vrfy oracle (cf. Exercise 4.2).
- 4.12 Prove that Construction 4.7 is secure if it is changed as follows: Set $t_i := F_k(r \| b \| i \| m_i)$ where b is a single bit such that $b = 0$ in all blocks but the last one, and $b = 1$ in the last block. (Assume for simplicity that the length of all messages being authenticated is always an integer multiple of $n/2 - 1$.) What is the advantage of this modification?
- 4.13 We explore what happens when the basic CBC-MAC construction is used with messages of different lengths.
 - (a) Say the sender and receiver do not agree on the message length in advance (and so $\text{Vrfy}_k(m, t) = 1$ iff $t \stackrel{?}{=} \text{Mac}_k(m)$, regardless of the length of m), but the sender is careful to only authenticate messages of length $2n$. Show that an adversary can forge a valid tag on a message of length $4n$.
 - (b) Say the receiver only accepts 3-block messages (so $\text{Vrfy}_k(m, t) = 1$ only if m has length $3n$ and $t \stackrel{?}{=} \text{Mac}_k(m)$), but the sender authenticates messages of any length a multiple of n . Show that an adversary can forge a valid tag on a new message.
- 4.14 Prove that the following modifications of basic CBC-MAC do not yield a secure MAC (even for fixed-length messages):
 - (a) Mac outputs all blocks $t_1, \dots, t_\ell$, rather than just t_ℓ . (Verification only checks whether t_ℓ is correct.)
 - (b) A random initial block is used each time a message is authenticated. That is, choose uniform $t_0 \in \{0, 1\}^n$, run basic CBC-MAC over the “message” $t_0, m_1, \dots, m_\ell$, and output the tag $\langle t_0, t_\ell \rangle$. Verification is done in the natural way.
- 4.15 Show that appending the message length to the *end* of the message before applying basic CBC-MAC does not result in a secure MAC for arbitrary-length messages.
- 4.16 Show that the encoding for arbitrary-length messages described in Section 4.4.2 is prefix-free.
- 4.17 Consider the following encoding that handles messages whose length is less than $n \cdot 2^n$: We encode a string $m \in \{0, 1\}^*$ by first appending as many 0s as needed to make the length of the resulting string $\hat{m}$ a nonzero multiple of n . Then we prepend the number of *blocks* in $\hat{m}$ (equivalently, prepend the integer $\lceil \hat{m} \rceil / n$), encoded as an n -bit string. Show that this encoding is *not* prefix-free.

- 4.18 Prove that the following modification of basic CBC-MAC gives a secure MAC for arbitrary-length messages (for simplicity, assume all messages have length a multiple of the block length). $\text{Mac}_k(m)$ first computes $k_\ell = F_k(\ell)$, where ℓ is the length of m . The tag is then computed using basic CBC-MAC with key k_ℓ . Verification is done in the natural way.
- 4.19 Let F be a keyed function that is a secure (deterministic) MAC for messages of length n . (Note that F need not be a pseudorandom permutation.) Show that basic CBC-MAC is not necessarily a secure MAC (even for fixed-length messages) when instantiated with F .
- 4.20 Show that Construction 4.7 is strongly secure.
- 4.21 Show that Construction 4.18 might not be CCA-secure if it is instantiated with a secure MAC that is *not* strongly secure.
- 4.22 Prove that Construction 4.18 is unforgeable when instantiated with any encryption scheme (even if not CPA-secure) and any secure MAC (even if the MAC is not strongly secure).
- 4.23 Consider a strengthened version of unforgeability (Definition 4.16) where $\mathcal{A}$ is additionally given access to a decryption oracle.
- Write a formal definition for this version of unforgeability.
 - Prove that Construction 4.18 satisfies this stronger definition if Π_M is a strongly secure MAC.
 - Show by counterexample that Construction 4.18 need not satisfy this stronger definition if Π_M is a secure MAC that is not strongly secure. (Compare to the previous exercise.)
- 4.24 Prove that the authenticate-then-encrypt approach, instantiated with any CPA-secure encryption scheme and any secure MAC, yields a CPA-secure encryption scheme that is unforgeable.
- 4.25 Let F be a strong pseudorandom permutation, and define the following fixed-length encryption scheme: On input a message $m \in \{0, 1\}^{n/2}$ and key $k \in \{0, 1\}^n$, algorithm Enc chooses a uniform $r \in \{0, 1\}^{n/2}$ and computes $c := F_k(m \| r)$. (See Exercise 3.18.) Prove that this scheme is CCA-secure, but is not an authenticated encryption scheme.
- 4.26 Show a CPA-secure private-key encryption scheme that is unforgeable but is not CCA-secure.
- 4.27 Fix $\ell > 0$ and a prime p . Let $\mathcal{K} = \mathbb{Z}_p^{\ell+1}$, $\mathcal{M} = \mathbb{Z}_p^\ell$, and $\mathcal{T} = \mathbb{Z}_p$. Define $h : \mathcal{K} \times \mathcal{M} \rightarrow \mathcal{T}$ as

$$h_{k_0, k_1, \dots, k_\ell}(m_1, \dots, m_\ell) = [k_0 + \sum_i k_i m_i \bmod p].$$

Prove that h is strongly universal.

- 4.28 Fix $\ell, n > 0$. Let $\mathcal{K} = \{0, 1\}^{\ell \times n} \times \{0, 1\}^\ell$ (interpreted as a boolean $\ell \times n$ matrix and an ℓ -dimensional vector), let $\mathcal{M} = \{0, 1\}^n$, and let $\mathcal{T} = \{0, 1\}^\ell$. Define $h : \mathcal{K} \times \mathcal{M} \rightarrow \mathcal{T}$ as $h_{K,v}(m) = K \cdot m \oplus v$, where all operations are performed modulo 2. Prove that h is strongly universal.
- 4.29 A *Toeplitz matrix* K is a matrix in which $K_{i,j} = K_{i-1,j-1}$ when $i, j > 1$; i.e., the values along any diagonal are equal. So an $\ell \times n$ Toeplitz matrix (for $\ell > n$) has the form

$$\begin{bmatrix} K_{1,1} & K_{1,2} & K_{1,3} & \cdots & K_{1,n} \\ K_{2,1} & K_{1,1} & K_{1,2} & \cdots & K_{1,n-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ K_{\ell,1} & K_{\ell-1,1} & K_{\ell-2,1} & \cdots & K_{\ell-n+1,1} \end{bmatrix}.$$

Let $\mathcal{K} = T^{\ell \times n} \times \{0, 1\}^\ell$ (where $T^{\ell \times n}$ denotes the set of $\ell \times n$ Toeplitz matrices), let $\mathcal{M} = \{0, 1\}^n$, and let $\mathcal{T} = \{0, 1\}^\ell$. Define $h : \mathcal{K} \times \mathcal{M} \rightarrow \mathcal{T}$ as $h_{K,v}(m) = K \cdot m \oplus v$, where all operations are performed modulo 2. Prove that h is strongly universal. What is the advantage here as compared to the construction in the previous exercise?

- 4.30 Define an appropriate notion of a *two-time* ε -secure MAC, and give a construction that meets your definition.
- 4.31 Let $\{h_n : \mathcal{K}_n \times \{0, 1\}^{10 \cdot n} \rightarrow \{0, 1\}^n\}_{n \in \mathbb{N}}$ be such that h_n is strongly universal for all n , and let F be a pseudorandom function. (When $K \in \mathcal{K}_n$ we write $h_K(\cdot)$ instead of $h_{n,K}(\cdot)$.) Consider the following MAC: $\text{Gen}(1^n)$ chooses uniform $K \in \mathcal{K}_n$ and $k \in \{0, 1\}^n$, and outputs (K, k) . To authenticate a message $m \in \{0, 1\}^{10 \cdot n}$, choose uniform $r \in \{0, 1\}^n$ and output $\langle r, h_K(m) \oplus F_k(r) \rangle$. Verification is done in the natural way. Prove that this gives a (computationally) secure MAC for messages of length $10n$.

Chapter 5

Hash Functions and Applications

In this chapter we introduce *cryptographic hash functions* and explore a few of their applications. At the most basic level, a hash function provides a way to map a long input string to a shorter output string sometimes called a *digest*. The primary requirement is to avoid *collisions*, or two inputs that map to the same digest. Collision-resistant hash functions have numerous uses. One example that we will see here is another approach—standardized as HMAC—for achieving domain extension for message authentication codes.

Beyond that, hash functions have become ubiquitous in cryptography, and they are often used in scenarios that require properties much stronger than collision resistance. It has become common to model cryptographic hash functions as being “completely unpredictable” (a.k.a., *random oracles*), and we discuss this framework—and the controversy that surrounds it—in detail later in the chapter. We touch on only a few applications of the random-oracle model here, but will encounter it again when we turn to the setting of public-key cryptography.

Hash functions are intriguing in that they can be viewed as lying between the worlds of private- and public-key cryptography. On the one hand, as we will see in Chapter 6, they are (in practice) constructed using symmetric-key techniques, and many of the canonical applications of hash functions are in the symmetric-key setting. From a theoretical point of view, however, the existence of collision-resistant hash functions appears to represent a qualitatively stronger assumption than the existence of pseudorandom functions (yet a weaker assumption than the existence of public-key encryption).

5.1 Definitions

Hash functions are simply functions that take inputs of some length and *compress* them into short, fixed-length outputs. The classic use of hash functions is in data structures, where they can be used to build hash tables that enable $\mathcal{O}(1)$ lookup time when storing a set of elements. Specifically, if the range of the hash function H is of size N , then element x is stored in row $H(x)$ of a table of size N . To retrieve x , it suffices to compute $H(x)$ and probe

that row of the table for the elements stored there. A “good” hash function for this purpose is one that yields few *collisions*, where a collision is a pair of distinct items x and x' for which $H(x) = H(x')$; in this case we also say that x and x' *collide*. (When a collision occurs, two elements end up being stored in the same cell, increasing the lookup time.)

Collision-resistant hash functions are similar in spirit. Again, the goal is to avoid collisions. However, there are fundamental differences. For one, the desire to minimize collisions in the setting of data structures becomes a *requirement* to avoid collisions in the setting of cryptography. Furthermore, in the context of data structures we can assume that the set of data elements is chosen independently of the hash function and without any intention to cause collisions. In the context of cryptography, in contrast, we are faced with an adversary who may select elements with the explicit goal of causing collisions. This means that collision-resistant hash functions are much harder to design.

5.1.1 Collision Resistance

Informally, a function H is *collision resistant* if it is infeasible for any probabilistic polynomial-time algorithm to find a collision in H . We will only be interested in hash functions whose domain is larger than their range. In this case collisions must *exist*, but such collisions should be hard to find.

Formally, we consider *keyed* hash functions. That is, H is a two-input function that takes as input a key s and a string x , and outputs a string $H^s(x) \stackrel{\text{def}}{=} H(s, x)$. The requirement is that it must be hard to find a collision in H^s for a randomly generated key s . There are at least two differences between keys in this context and keys as we have used them until now. First, not all strings necessarily correspond to valid keys (i.e., H^s may not be defined for certain s), and therefore the key s will typically be generated by an algorithm Gen rather than being chosen uniformly. Second, and perhaps more importantly, this key s is (generally) not kept secret, and collision resistance is required even when the adversary is given s . In order to emphasize this, we superscript the key and write H^s rather than H_s .

DEFINITION 5.1 A hash function (with output length ℓ) is a pair of probabilistic polynomial-time algorithms (Gen, H) satisfying the following:

- Gen is a probabilistic algorithm which takes as input a security parameter 1^n and outputs a key s . We assume that 1^n is implicit in s .
- H takes as input a key s and a string $x \in \{0, 1\}^*$ and outputs a string $H^s(x) \in \{0, 1\}^{\ell(n)}$ (where n is the value of the security parameter implicit in s).

If H^s is defined only for inputs $x \in \{0, 1\}^{\ell'(n)}$ and $\ell'(n) > \ell(n)$, then we say that (Gen, H) is a fixed-length hash function for inputs of length ℓ' . In this case, we also call H a compression function.

In the fixed-length case we require that ℓ' be greater than ℓ . This ensures that the function *compresses* its input. In the general case the function takes as input strings of arbitrary length. Thus, it also compresses (albeit only strings of length greater than $\ell(n)$). Note that without compression, collision resistance is trivial (since one can just take the identity function $H^s(x) = x$).

We now proceed to define security. As usual, we first define an experiment for a hash function $\Pi = (\text{Gen}, H)$, an adversary $\mathcal{A}$, and a security parameter n :

The collision-finding experiment $\text{Hash-coll}_{\mathcal{A}, \Pi}(n)$:

1. A key s is generated by running $\text{Gen}(1^n)$.
2. The adversary $\mathcal{A}$ is given s and outputs x, x' . (If Π is a fixed-length hash function for inputs of length $\ell'(n)$, then we require $x, x' \in \{0, 1\}^{\ell'(n)}$.)
3. The output of the experiment is defined to be 1 if and only if $x \neq x'$ and $H^s(x) = H^s(x')$. In such a case we say that $\mathcal{A}$ has found a collision.

The definition of collision resistance states that no efficient adversary can find a collision in the above experiment except with negligible probability.

DEFINITION 5.2 A hash function $\Pi = (\text{Gen}, H)$ is collision resistant if for all probabilistic polynomial-time adversaries $\mathcal{A}$ there is a negligible function negl such that

$$\Pr[\text{Hash-coll}_{\mathcal{A}, \Pi}(n) = 1] \leq \text{negl}(n).$$

For simplicity, we sometimes refer to H or H^s as a “collision-resistant hash function,” even though technically we should only say that (Gen, H) is. This should not cause any confusion.

Cryptographic hash functions are designed with the explicit goal of being collision resistant (among other things). We will discuss some common real-world hash functions in Chapter 6. In Section 8.4.2 we will see how it is possible to construct hash functions with proven collision resistance based on an assumption about the hardness of a certain number-theoretic problem.

Unkeyed hash functions. Cryptographic hash functions used in practice generally have a fixed output length (just as block ciphers have a fixed key length) and are usually *unkeyed*, meaning that the hash function is just a fixed function $H : \{0, 1\}^* \rightarrow \{0, 1\}^\ell$. This is problematic from a theoretical standpoint since for any such function there is always a constant-time algorithm that outputs a collision in H : the algorithm simply outputs a colliding pair (x, x') hardcoded into the algorithm itself. Using keyed hash functions solves this technical issue since it is impossible to hardcode a colliding pair for every possible key using a reasonable amount of space (and in an asymptotic setting, it would be impossible to hardcode a colliding pair for every value of the security parameter).

Notwithstanding the above, the (unkeyed) cryptographic hash functions used in the real world are collision resistant for all practical purposes since colliding pairs are unknown (and computationally difficult to find) even though they must exist. Proofs of security for some construction based on collision resistance of a hash function are meaningful even when an unkeyed hash function H is used, as long as the proof shows that any efficient adversary “breaking” the primitive can be used to efficiently find a collision in H . (All the proofs in this book satisfy this condition.) In this case, the interpretation of the proof of security is that if an adversary can break the scheme in practice, then it can be used to find a collision in practice, something that we believe is hard to do.

5.1.2 Weaker Notions of Security

In some applications it suffices to rely on security requirements weaker than collision resistance. These include:

- *Second-preimage or target-collision resistance*: Informally, a hash function is second preimage resistant if given s and a uniform x it is infeasible for a PPT adversary to find $x' \neq x$ such that $H^s(x') = H^s(x)$.
- *Preimage resistance*: Informally, a hash function is preimage resistant if given s and a uniform y it is infeasible for a PPT adversary to find a value x such that $H^s(x) = y$. (Looking ahead to Chapter 7, this essentially means that H^s is *one-way*.)

Any hash function that is collision resistant is also second preimage resistant. This holds since if, given a uniform x , an adversary can find $x' \neq x$ for which $H^s(x') = H^s(x)$, then it can clearly find a colliding pair x and x' . Likewise, any hash function that is second preimage resistant is also preimage resistant. This is due to the fact that if it were possible, given y , to find an x such that $H^s(x) = y$, then one could also take a given input x' , compute $y := H^s(x')$, and then obtain an x with $H^s(x) = y$. With high probability $x' \neq x$ (relying on the fact that H compresses, and so multiple inputs map to the same output), in which case a second preimage has been found.

We do not formally define the above notions or prove the above implications, since they are not used in the rest of the book. You are asked to formalize the above in Exercise 5.1.

5.2 Domain Extension: The Merkle-Damgård Transform

Hash functions are often constructed by first designing a collision-resistant compression function handling fixed-length inputs, and then using *domain*

extension to handle arbitrary-length inputs. In this section, we show one solution to the problem of domain extension. We return to the question of designing collision-resistant compression functions in Section 6.3.

The *Merkle–Damgård transform* is a common approach for extending a compression function to a full-fledged hash function, while maintaining the collision-resistance property of the former. It is used extensively in practice for hash functions including MD5 and the SHA family (see Section 6.3). The existence of this transform means that when designing collision-resistant hash functions, we can restrict our attention to the fixed-length case. This, in turn, makes the job of designing collision-resistant hash functions much easier. The Merkle–Damgård transform is also interesting from a theoretical point of view since it implies that compressing by a single bit is as easy (or as hard) as compressing by an arbitrary amount.

For concreteness, assume the compression function (Gen, h) compresses its input by half; say its input length is $2n$ and its output length is n . (The construction works regardless of the input/output lengths, as long as h compresses.) We construct a collision-resistant hash function (Gen, H) that maps inputs of *arbitrary* length to outputs of length n . (Gen remains unchanged.) The Merkle–Damgård transform is defined in Construction 5.3 and depicted in Figure 5.1. The value z_0 used in step 2 of the construction, called the *initialization vector* or *IV*, is arbitrary and can be replaced by any constant.

CONSTRUCTION 5.3

Let (Gen, h) be a fixed-length hash function for inputs of length $2n$ and with output length n . Construct hash function (Gen, H) as follows:

- **Gen**: remains unchanged.
- **H**: on input a key s and a string $x \in \{0, 1\}^*$ of length $L < 2^n$, do the following:
 1. Set $B := \lceil \frac{L}{n} \rceil$ (i.e., the number of blocks in x). Pad x with zeros so its length is a multiple of n . Parse the padded result as the sequence of n -bit blocks $x_1, \dots, x_B$. Set $x_{B+1} := L$, where L is encoded as an n -bit string.
 2. Set $z_0 := 0^n$. (This is also called the *IV*.)
 3. For $i = 1, \dots, B + 1$, compute $z_i := h^s(z_{i-1} \| x_i)$.
 4. Output z_{B+1} .

The Merkle–Damgård transform.

THEOREM 5.4 *If (Gen, h) is collision resistant, then so is (Gen, H) .*

PROOF We show that for any s , a collision in H^s yields a collision in h^s . Let x and x' be two different strings of length L and L' , respectively, such that $H^s(x) = H^s(x')$. Let $x_1, \dots, x_B$ be the B blocks of the padded x , and

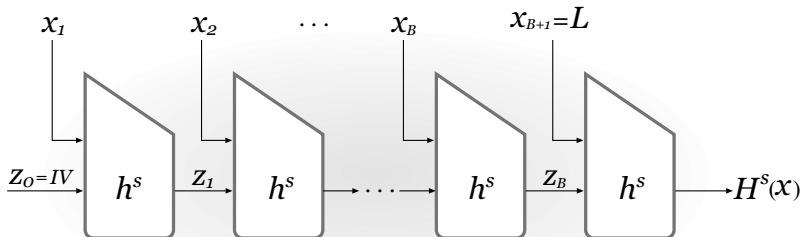


FIGURE 5.1: The Merkle–Damgård transform.

let $x'_1, \dots, x'_{B'}$ be the B' blocks of the padded x' . Recall that $x_{B+1} = L$ and $x'_{B'+1} = L'$. There are two cases to consider:

1. *Case 1:* $L \neq L'$. In this case, the last step of the computation of $H^s(x)$ is $z_{B+1} := h^s(z_B \| L)$, and the last step of the computation of $H^s(x')$ is $z'_{B'+1} := h^s(z'_{B'} \| L')$. Since $H^s(x) = H^s(x')$ it follows that $h^s(z_B \| L) = h^s(z'_{B'} \| L')$. However, $L \neq L'$ and so $z_B \| L$ and $z'_{B'} \| L'$ are two different strings that collide under h^s .
2. *Case 2:* $L = L'$. This means that $B = B'$. Let $z_0, \dots, z_{B+1}$ be the values defined during the computation of $H^s(x)$, let $I_i \stackrel{\text{def}}{=} z_{i-1} \| x_i$ denote the i th input to h^s , and set $I_{B+2} \stackrel{\text{def}}{=} z_{B+1}$. Define $I'_1, \dots, I'_{B+2}$ analogously with respect to x' . Let N be the *largest* index for which $I_N \neq I'_N$. Since $|x| = |x'|$ but $x \neq x'$, there is an i with $x_i \neq x'_i$ and so such an N certainly exists. Because

$$I_{B+2} = z_{B+1} = H^s(x) = H^s(x') = z'_{B+1} = I'_{B+2},$$

we have $N \leq B + 1$. By maximality of N , we have $I_{N+1} = I'_{N+1}$ and in particular $z_N = z'_N$. But this means that I_N, I'_N are a collision in h^s .

We leave it as an exercise to turn the above into a formal reduction. ■

5.3 Message Authentication Using Hash Functions

In the previous chapter, we presented two constructions of message authentication codes for arbitrary-length messages. The first approach was generic, but inefficient. The second, CBC-MAC, was based on pseudorandom functions. Here we will see another approach, which we call “hash-and-MAC,” that relies on collision-resistant hashing along with any message authentication code. We then discuss a standardized and widely used construction called HMAC that can be viewed as a specific instantiation of this approach.

5.3.1 Hash-and-MAC

The idea behind the hash-and-MAC approach is simple. First, an arbitrarily long message m is hashed down to a fixed-length string $H^s(m)$ using a collision-resistant hash function. Then, a (fixed-length) MAC is applied to the result. See Construction 5.5 for a formal description.

CONSTRUCTION 5.5

Let $\Pi = (\text{Mac}, \text{Vrfy})$ be a MAC for messages of length $\ell(n)$, and let $\Pi_H = (\text{Gen}_H, H)$ be a hash function with output length $\ell(n)$. Construct a MAC $\Pi' = (\text{Gen}', \text{Mac}', \text{Vrfy}')$ for arbitrary-length messages as follows:

- **Gen'**: on input 1^n , choose uniform $k \in \{0, 1\}^n$ and run $\text{Gen}_H(1^n)$ to obtain s ; the key is $k' := \langle k, s \rangle$.
- **Mac'**: on input a key $\langle k, s \rangle$ and a message $m \in \{0, 1\}^*$, output $t \leftarrow \text{Mac}_k(H^s(m))$.
- **Vrfy'**: on input a key $\langle k, s \rangle$, a message $m \in \{0, 1\}^*$, and a MAC tag t , output 1 if and only if $\text{Vrfy}_k(H^s(m), t) \stackrel{?}{=} 1$.

The hash-and-MAC paradigm.

Construction 5.5 is secure if Π is a secure MAC for fixed-length messages and (Gen, H) is collision resistant. Intuitively, since the hash function is collision resistant, authenticating $H^s(m)$ is as good as authenticating m itself: if the sender can ensure that the receiver obtains the correct value $H^s(m)$, collision resistance guarantees that the attacker cannot find a different message m' that hashes to the same value. A bit more formally, say a sender uses Construction 5.5 to authenticate some set of messages $\mathcal{Q}$, and an attacker $\mathcal{A}$ is then able to forge a valid tag on a new message $m^* \notin \mathcal{Q}$. There are two possible cases:

Case 1: *there is a message $m \in \mathcal{Q}$ such that $H^s(m^*) = H^s(m)$.* Then $\mathcal{A}$ has found a collision in H^s , contradicting the collision resistance of (Gen, H) .

Case 2: *for every message $m \in \mathcal{Q}$ it holds that $H^s(m^*) \neq H^s(m)$.* Let $H^s(\mathcal{Q}) \stackrel{\text{def}}{=} \{H^s(m) \mid m \in \mathcal{Q}\}$. Then $H^s(m^*) \notin H^s(\mathcal{Q})$. In this case, $\mathcal{A}$ has forged a valid tag on the “new message” $H^s(m^*)$ with respect to the *fixed-length* message authentication code Π . This contradicts the assumption that Π is a secure MAC.

We now turn the above into a formal proof.

THEOREM 5.6 *If Π is a secure MAC for messages of length ℓ and Π_H is collision resistant, then Construction 5.5 is a secure MAC (for arbitrary-length messages).*

PROOF Let Π' denote Construction 5.5, and let $\mathcal{A}'$ be a PPT adversary attacking Π' . In an execution of experiment $\text{Mac-forge}_{\mathcal{A}', \Pi'}(n)$, let $k' = \langle k, s \rangle$ denote the MAC key, let $\mathcal{Q}$ denote the set of messages whose tags were requested by $\mathcal{A}'$, and let (m^*, t) be the final output of $\mathcal{A}'$. We assume without loss of generality that $m^* \notin \mathcal{Q}$. Define coll to be the event that, in experiment $\text{Mac-forge}_{\mathcal{A}', \Pi'}(n)$, there is an $m \in \mathcal{Q}$ for which $H^s(m^*) = H^s(m)$. We have

$$\begin{aligned} & \Pr[\text{Mac-forge}_{\mathcal{A}', \Pi'}(n) = 1] \\ &= \Pr[\text{Mac-forge}_{\mathcal{A}', \Pi'}(n) = 1 \wedge \text{coll}] + \Pr[\text{Mac-forge}_{\mathcal{A}', \Pi'}(n) = 1 \wedge \overline{\text{coll}}] \\ &\leq \Pr[\text{coll}] + \Pr[\text{Mac-forge}_{\mathcal{A}', \Pi'}(n) = 1 \wedge \overline{\text{coll}}]. \end{aligned} \quad (5.1)$$

We show that both terms in Equation (5.1) are negligible, thus completing the proof. Intuitively, the first term is negligible by collision resistance of Π_H , and the second term is negligible by security of Π .

Consider the following algorithm $\mathcal{C}$ for finding a collision in Π_H :

Algorithm $\mathcal{C}$:

The algorithm is given s as input (with n implicit).

- Choose uniform $k \in \{0, 1\}^n$.
- Run $\mathcal{A}'(1^n)$. When $\mathcal{A}'$ requests a tag on the i th message $m_i \in \{0, 1\}^*$, compute $t_i \leftarrow \text{Mac}_k(H^s(m_i))$ and give t_i to $\mathcal{A}'$.
- When $\mathcal{A}'$ outputs (m^*, t) , then if there exists an i for which $H^s(m^*) = H^s(m_i)$, output (m^*, m_i) .

It is clear that $\mathcal{C}$ runs in polynomial time. Let us analyze its behavior. When the input to $\mathcal{C}$ is generated by running $\text{Gen}_H(1^n)$ to obtain s , the view of $\mathcal{A}'$ when run as a subroutine by $\mathcal{C}$ is distributed identically to the view of $\mathcal{A}'$ in experiment $\text{Mac-forge}_{\mathcal{A}', \Pi'}(n)$. In particular, the tags given to $\mathcal{A}'$ by $\mathcal{C}$ have the same distribution as the tags that $\mathcal{A}'$ receives in $\text{Mac-forge}_{\mathcal{A}', \Pi'}(n)$. Since $\mathcal{C}$ outputs a collision exactly when coll occurs, we have

$$\Pr[\text{Hash-coll}_{\mathcal{C}, \Pi_H}(n) = 1] = \Pr[\text{coll}].$$

Because Π_H is collision resistant, we conclude that $\Pr[\text{coll}]$ is negligible.

We now proceed to prove that the second term in Equation (5.1) is negligible. Consider the following adversary $\mathcal{A}$ attacking Π in $\text{Mac-forge}_{\mathcal{A}, \Pi}(n)$:

Adversary $\mathcal{A}$:

The adversary is given access to a MAC oracle $\text{Mac}_k(\cdot)$.

- Compute $\text{Gen}_H(1^n)$ to obtain s .
- Run $\mathcal{A}'(1^n)$. When $\mathcal{A}'$ requests a tag on the i th message $m_i \in \{0, 1\}^*$, then: (1) compute $\hat{m}_i := H^s(m_i)$; (2) obtain a tag t_i on $\hat{m}_i$ from the MAC oracle; and (3) give t_i to $\mathcal{A}'$.
- When $\mathcal{A}'$ outputs (m^*, t) , then output $(H^s(m^*), t)$.

Clearly $\mathcal{A}$ runs in polynomial time. Consider experiment $\text{Mac-forge}_{\mathcal{A},\Pi}(n)$. In that experiment, the view of $\mathcal{A}'$ when run as a subroutine by $\mathcal{A}$ is distributed identically to its view in experiment $\text{Mac-forge}_{\mathcal{A}',\Pi'}(n)$. Furthermore, whenever both $\text{Mac-forge}_{\mathcal{A}',\Pi'}(n) = 1$ and coll do not occur, $\mathcal{A}$ outputs a valid forgery. (In that case t is a valid tag on $H^s(m^*)$ in scheme Π with respect to k . The fact that coll did not occur means that $H^s(m^*)$ was never asked by $\mathcal{A}$ to its own MAC oracle and so this is indeed a forgery.) Therefore,

$$\Pr[\text{Mac-forge}_{\mathcal{A},\Pi}(n) = 1] = \Pr[\text{Mac-forge}_{\mathcal{A}',\Pi'}(n) \wedge \overline{\text{coll}}],$$

and security of Π implies that the former probability is negligible. This concludes the proof of the theorem. $\blacksquare$

5.3.2 HMAC

All the constructions of message authentication codes we have seen so far are ultimately based on some block cipher. Is it possible to construct a secure MAC (for arbitrary-length messages) based directly on a hash function? A first thought might be to define $\text{Mac}_k(m) = H(k\|m)$; we might expect that if H is a “good” hash function then it should be difficult for an attacker to predict the value of $H(k\|m')$ given the value of $H(k\|m)$, for any $m' \neq m$, assuming k is chosen at random (and unknown to the attacker). Unfortunately, if H is constructed using the Merkle–Damgård transform—as most real-world hash functions are—then a MAC designed in this way is completely insecure, as you are asked to show in Exercise 5.10.

Instead, we can try using *two* layers of hashing. See Construction 5.7 for a standardized scheme called *HMAC* based on this idea.

CONSTRUCTION 5.7

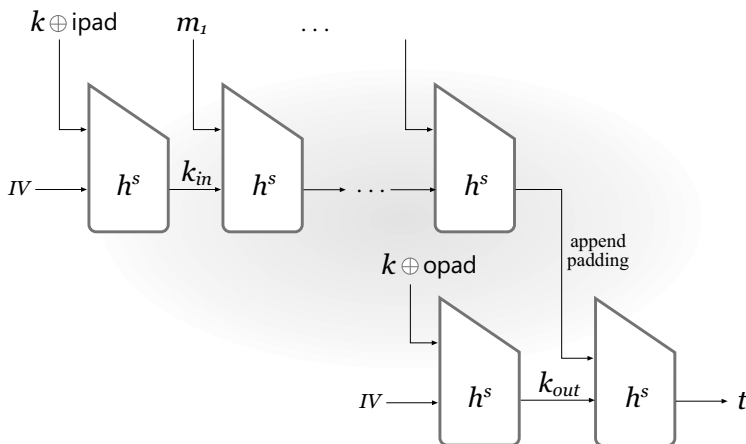
Let (Gen_H, H) be a hash function constructed by applying the Merkle–Damgård transform to a compression function (Gen_H, h) taking inputs of length $n + n'$. (See text.) Let opad and ipad be fixed constants of length n' . Define a MAC as follows:

- **Gen**: on input 1^n , run $\text{Gen}_H(1^n)$ to obtain a key s . Also choose uniform $k \in \{0, 1\}^{n'}$. Output the key $\langle s, k \rangle$.
- **Mac**: on input a key $\langle s, k \rangle$ and a message $m \in \{0, 1\}^*$, output

$$t := H^s\left((k \oplus \text{opad}) \parallel H^s((k \oplus \text{ipad}) \parallel m)\right).$$

- **Vrfy**: on input a key $\langle s, k \rangle$, a message $m \in \{0, 1\}^*$, and a tag t , output 1 if and only if $t \stackrel{?}{=} H^s((k \oplus \text{opad}) \parallel H^s((k \oplus \text{ipad}) \parallel m))$.

HMAC.

**FIGURE 5.2:** HMAC, pictorially.

Why should we have any confidence that HMAC is secure? One reason is that we can view HMAC as a specific instantiation of the hash-and-MAC paradigm from the previous section. To see this, we will look “under the hood” at what happens when a message is authenticated; see Figure 5.2. We must also specify parameters more carefully and go into a bit more detail regarding the way the Merkle–Damgård transform is implemented in practice.

Say (Gen_H, H) is constructed based on a compression function (Gen_H, h) in which h maps inputs of length $n + n'$ to outputs of length n (where, formally, n' is a function of n). When we described the Merkle–Damgård transform in Section 5.2, we assumed $n' = n$, but that need not always be the case. We also said that the length of the message being hashed was encoded as an extra message block that is appended to the message. In practice, the length is instead encoded in a *portion* of a block using $\ell < n'$ bits. That is, computation of $H^s(x)$ begins by padding x with zeros to a string of length exactly ℓ less than a multiple of n' ; it then appends the length $L = |x|$, encoded using exactly ℓ bits. The hash of the resulting sequence of n' -bit blocks $x_1, \dots$ is then computed as in Construction 5.3. We will assume that $n + \ell \leq n'$. This means, in particular, that if we hash an input x of length $n' + n$ then the padded result (including the length) will be exactly $2n'$ bits long. The proof of Theorem 5.4, showing that (Gen_H, H) is collision resistant if (Gen_H, h) is collision resistant, remains unchanged.

Coming back to HMAC, and looking at Figure 5.2, we can see that the general form of HMAC involves hashing an arbitrary-length message down to a short string $y \stackrel{\text{def}}{=} H^s((k \oplus \text{ipad}) \parallel m)$, and then computing the (secretly keyed) function $H^s((k \oplus \text{opad}) \parallel y)$ of the result. But we can say more than this. Note first that the “inner” computation

$$\tilde{H}^s(m) \stackrel{\text{def}}{=} H^s((k \oplus \text{ipad}) \parallel m)$$

is collision resistant (assuming h is), for any value of $k \oplus \text{ipad}$. Moreover, the first step in the “outer” computation $H^s((k \oplus \text{opad}) \parallel y)$ is to compute a value $k_{out} \stackrel{\text{def}}{=} h^s(IV \parallel (k \oplus \text{opad}))$. Then, we evaluate $h^s(k_{out} \parallel \hat{y})$ where $\hat{y}$ refers to the padded value of y (i.e., including the length of $(k \oplus \text{opad}) \parallel y$, which is always $n' + n$ bits, encoded using exactly ℓ bits). Thus, if we treat k_{out} as uniform—we will be more formal about this below—and assume that

$$\widetilde{\text{Mac}}_k(y) \stackrel{\text{def}}{=} h^s(k \parallel \hat{y}) \quad (5.2)$$

is a secure *fixed-length* MAC, then HMAC can be viewed as an instantiation of the hash-and-MAC approach with

$$\text{HMAC}_{s,k}(m) = \widetilde{\text{Mac}}_{k_{out}}(\tilde{H}^s(m)) \quad (5.3)$$

(where $k_{out} = h^s(IV \parallel (k \oplus \text{opad}))$). Because of the way the compression function h is typically designed (see Section 6.3.1), the assumption that $\widetilde{\text{Mac}}$ is a secure fixed-length MAC is a reasonable one.

The roles of *ipad* and *opad*. Given the above, one might wonder why it is necessary to incorporate k in the “inner” computation $H^s((k \oplus \text{ipad}) \parallel m)$. (In particular, for the hash-and-MAC approach to be secure we require collision resistance in the first step, which does not require any secret key.) The reason is that this allows security of HMAC to be based on the potentially weaker assumption that (Gen_H, H) is *weakly* collision resistant, where weak collision resistance is defined by the following experiment: a key s is generated using Gen_H and a uniform *secret* $k_{in} \in \{0, 1\}^n$ is chosen. Then the adversary is allowed to interact with a “hash oracle” that returns $H_{k_{in}}^s(m)$ in response to the query m , where $H_{k_{in}}^s$ refers to computation of H^s using the Merkle–Damgård transform applied to h^s , but using the secret value k_{in} as the *IV*. (Refer again to Figure 5.2.) The adversary succeeds if it can output distinct values m, m' such that $H_{k_{in}}^s(m) = H_{k_{in}}^s(m')$, and we say that (Gen_H, H) is *weakly collision resistant* if every PPT $\mathcal{A}$ succeeds in this experiment with only negligible probability. If (Gen_H, H) is collision resistant then it is clearly weakly collision resistant; the latter, however, is a weaker condition that is potentially easier to satisfy. This is a good example of sound security engineering. This defensive design strategy paid off when it was discovered that the hash function MD5 (see Section 6.3.2) was *not* collision resistant. The collision-finding attacks on MD5 did not violate weak collision resistance, and HMAC-MD5 was not broken even though MD5 was. This gave developers time to replace MD5 in HMAC implementations, without immediate fear of attack. (Despite this, HMAC-MD5 should no longer be used now that weaknesses in MD5 are known.)

The above discussion suggests that *independent* keys should be used in the outer and inner computations. For reasons of efficiency, a single key k is used for HMAC, but the key is used in combination with *ipad* and *opad* to derive

two other keys. Define

$$G^s(k) \stackrel{\text{def}}{=} h^s(IV \parallel (k \oplus \text{opad})) \parallel h^s(IV \parallel (k \oplus \text{ipad})) = k_{out} \parallel k_{in}. \quad (5.4)$$

If we assume that G^s is a pseudorandom generator for any s , then k_{out} and k_{in} can be treated as independent and uniform keys when k is uniform. Security of HMAC then reduces to the security of the following construction:

$$\text{Mac}_{s,k_{in},k_{out}}(m) = h^s(k_{out} \parallel H_{k_{in}}^s(m)).$$

(Compare to Equation (5.3).) As noted earlier, this construction can be proven secure (using a variant of the proof for the hash-and-MAC approach) if H is weakly collision resistant and the MAC defined in Equation (5.2) is a secure fixed-length MAC.

THEOREM 5.8 *Assume G^s as defined in Equation (5.4) is a pseudorandom generator for any s , the MAC defined in Equation (5.2) is a secure fixed-length MAC for messages of length n , and (Gen_H, H) is weakly collision resistant. Then HMAC is a secure MAC (for arbitrary-length messages).*

HMAC in practice. HMAC is an industry standard and is widely used in practice. It is highly efficient and easy to implement, and is supported by a proof of security based on assumptions that are believed to hold for practical hash functions. The importance of HMAC is partially due to the timeliness of its appearance. Before the introduction of HMAC, many practitioners refused to use CBC-MAC (with the claim that it was “too slow”) and instead used heuristic constructions that were insecure. HMAC provided a standardized, secure way of doing message authentication based on hash functions.

5.4 Generic Attacks on Hash Functions

What is the best security we can hope for a hash function H to provide? We explore this question by showing two attacks that are *generic* in the sense that they apply to *arbitrary* hash functions. The existence of these attacks implies lower bounds on the output length of H needed to achieve some desired level of security, and therefore has important practical ramifications.

5.4.1 Birthday Attacks for Finding Collisions

Let $H : \{0, 1\}^* \rightarrow \{0, 1\}^\ell$ be a hash function. (Here and in the rest of the chapter, we drop explicit mention of the hash key s since it is not directly relevant. One can also view s as being generated and fixed before these

algorithms are applied.) There is a trivial collision-finding attack running in time $\mathcal{O}(2^\ell)$: simply evaluate H on $2^\ell + 1$ distinct inputs; by the pigeonhole principle, two of the outputs must be equal. Can we do better?

Generalizing the above algorithm, say we choose q distinct inputs $x_1, \dots, x_q$, compute $y_i := H(x_i)$, and check whether any of the two y_i values are equal. What is the probability that this algorithm finds a collision? As we have just said, if $q > 2^\ell$ then a collision occurs with probability 1. What is the probability of a collision when q is smaller? It is somewhat difficult to analyze this probability exactly, and so we will instead analyze an idealized case in which H is treated as a random function.¹ That is, for each i we assume that the value $y_i = H(x_i)$ is uniformly distributed in $\{0, 1\}^\ell$ and independent of any of the previous output values $\{y_j\}_{j < i}$ (recall we assume all $\{x_i\}$ are distinct). We have thus reduced our problem to the following one: if we choose values $y_1, \dots, y_q \in \{0, 1\}^\ell$ uniformly at random, what is the probability that there exist distinct i, j with $y_i = y_j$?

This problem has been extensively studied, and is related to the so-called *birthday problem* discussed in detail in Appendix A.4. For this reason, the collision-finding algorithm we have described is often called a *birthday attack*. The birthday problem is the following: if q people are in a room, what is the probability that two of them have the same birthday? (Assume birthdays are uniformly and independently distributed among the 365 days of a non-leap year.) This is exactly analogous to our problem: if y_i represents the birthday of person i , then we have $y_1, \dots, y_q \in \{1, \dots, 365\}$ chosen uniformly, and matching birthdays correspond to distinct i, j with $y_i = y_j$ (i.e., matching birthdays correspond to collisions).

In Appendix A.4 we show that for $y_1, \dots, y_q$ chosen uniformly in $\{1, \dots, N\}$, the probability of a collision is roughly $1/2$ when $q = \Theta(N^{1/2})$. In the case of birthdays, once there are only 23 people the probability that some two of them have the same birthday is greater than $1/2$. In our setting, this means that when the hash function has output length ℓ (and so the range is of size 2^ℓ), then taking $q = \Theta(2^{\ell/2})$ yields a collision with probability roughly $1/2$.

From a concrete-security perspective, the above means that for a hash function to resist collision-finding attacks that run in time T (where we take the time to evaluate H as our unit of time), the output length of the hash function needs to be at least $2 \log T$ bits (since $2^{(2 \log T)/2} = T$). Taking specific parameters, this means that if we want finding collisions to be as difficult as an exhaustive search over 128-bit keys, then we need the output length of the hash function to be at least 256 bits. We stress that having an output this long is only a *necessary* condition, not a sufficient one. We also note that birthday attacks work only for finding collisions. There are no generic attacks

¹It can be shown that this is (essentially) the worst case, and collisions occur with higher probability if H deviates from random and the $\{x_i\}$ are chosen uniformly.

for second preimage resistance or preimage resistance of a hash functions H that require fewer than 2^ℓ evaluations of H (though see Section 5.4.3).

Finding meaningful collisions. The birthday attack just described gives a collision that is not necessarily very useful. But the same idea can be used to find “meaningful” collisions as well. Assume Alice wishes to find two messages x and x' such that $H(x) = H(x')$, and furthermore x should be a letter from her employer explaining why she was fired from work, while x' should be a flattering letter of recommendation. (This might allow Alice to forge an appropriate tag on a letter of recommendation if the hash-and-MAC approach is being used by her employer to authenticate messages.) The observation is that the birthday attack only requires the hash inputs $x_1, \dots, x_q$ to be distinct; they do not need to be random. Alice can carry out a birthday-type attack by generating $q = \Theta(2^{\ell/2})$ messages of the first type and q messages of the second type, and then looking for collisions between messages of the two types. A small change to the analysis from Appendix A.4 shows that this gives a collision between messages of different types with probability roughly $1/2$. A little thought shows that it is easy to write the same message in many different ways. For example, consider the following:

It is *hard/difficult/challenging/impossible* to *imagine/believe* that we will *find/locate/hire* another *employee/person* having similar *abilities/skills/character* as Alice. She has done a *great/super* job.

Any combination of the italicized words is possible, and expresses the same point. Thus, the sentence can be written in $4 \cdot 2 \cdot 3 \cdot 2 \cdot 3 \cdot 2 = 288$ different ways. This is just one sentence and so it is actually easy to generate a message that can be rewritten in 2^{64} different ways—all that is needed are 64 words with one synonym each. Alice can prepare $2^{\ell/2}$ letters explaining why she was fired and another $2^{\ell/2}$ letters of recommendation; with good probability, a collision between the two types of letters will be found.

5.4.2 Small-Space Birthday Attacks

The birthday attacks described above require a large amount of memory; specifically, they require the attacker to store all $\mathcal{O}(q) = \mathcal{O}(2^{\ell/2})$ values $\{y_i\}$, because the attacker does not know in advance which pair of values will yield a collision. This is a significant drawback because memory is, in general, a scarcer resource than time. It is arguably more difficult to allocate and manage storage for 2^{60} bytes than to execute 2^{60} CPU instructions. Furthermore, one can always let a computation run indefinitely, whereas the memory requirements of an algorithm must be satisfied as soon as that amount of memory is needed.

We show here a better birthday attack with drastically reduced memory requirements. In fact, it has similar time complexity and success probability as before, but uses only *constant* memory. The attack begins by choosing a

random value x_0 and then computing $x_i := H(x_{i-1})$ and $x_{2i} := H(H(x_{i-1}))$ for $i = 1, 2, \dots$ (Note that $x_i = H^{(i)}(x_0)$ for all i , where $H^{(i)}$ refers to i -fold iteration of H .) In each step the values x_i and x_{2i} are compared; if they are equal then there is a collision somewhere in the sequence $x_0, x_1, \dots, x_{2i-1}$. The algorithm then finds the least value of j for which $x_j = x_{j+i}$ (note that $j \leq i$ since $j = i$ works), and outputs x_{j-1}, x_{j+i-1} as a collision. This attack, described formally as Algorithm 5.9 and analyzed below, only requires storage of two hash values in each iteration.

ALGORITHM 5.9

A small-space birthday attack

Input: A hash function $H : \{0, 1\}^* \rightarrow \{0, 1\}^\ell$

Output: Distinct x, x' with $H(x) = H(x')$

$x_0 \leftarrow \{0, 1\}^{\ell+1}$

$x' := x := x_0$

for $i = 1, 2, \dots$ **do**:

$x := H(x)$

$x' := H(H(x'))$

 // now $x = H^{(i)}(x_0)$ and $x' = H^{(2i)}(x_0)$

if $x = x'$ **break**

$x' := x, x := x_0$

for $j = 1$ to i :

if $H(x) = H(x')$ **return** x, x' and **halt**

else $x := H(x), x' := H(x')$

 // now $x = H^{(j)}(x_0)$ and $x' = H^{(i+j)}(x_0)$

How many iterations of the first loop do we expect before $x' = x$? Consider the sequence of values $x_1, x_2, \dots$, where $x_i = H^{(i)}(x_0)$ as defined before. If we model H as a random function, then each of these values is uniformly and independently distributed in $\{0, 1\}^\ell$ until the first repeat occurs. Thus, we expect a repeat to occur with probability $1/2$ in the first $q = \Theta(2^{\ell/2})$ terms of the sequence. We show that when there is a repeat in the first q elements, the algorithm finds a repeat in at most q iterations of the first loop:

CLAIM 5.10 *Let $x_1, \dots, x_q$ be a sequence of values with $x_m = H(x_{m-1})$. If $x_I = x_J$ with $1 \leq I < J \leq q$, then there is an $i < J$ such that $x_i = x_{2i}$.*

PROOF The sequence $x_I, x_{I+1}, \dots$ repeats with period $\Delta \stackrel{\text{def}}{=} J - I$. That is, for all $i \geq I$ and $k \geq 0$ it holds that $x_i = x_{i+k \cdot \Delta}$. Let i be the smallest multiple of Δ that is also greater than or equal to I . We have $i < J$ since the sequence of Δ values $I, I+1, \dots, I+(\Delta-1) = J-1$ contains a multiple of Δ . Since $i \geq I$ and $2i - i = i$ is a multiple of Δ , it follows that $x_i = x_{2i}$. ■

Thus, if there is a repeated value in the sequence $x_1, \dots, x_q$, then there is some $i < q$ for which $x_i = x_{2i}$. But then in iteration i of our algorithm, we have $x = x'$ and the algorithm breaks out of the first loop. At that point in the algorithm, we know that $x_i = x_{i+i}$. The algorithm then sets $x' := x (= x_i)$ and $x := x_0$, and proceeds to find the *smallest* $j \geq 0$ for which $x_j = x_{j+i}$. (Note $j \neq 0$ because $|x_0| = \ell + 1$.) It outputs x_{j-1}, x_{j+i-1} as a collision.

Finding meaningful collisions. The algorithm just described may not seem amenable to finding meaningful collisions since it has no control over the elements sampled. Nevertheless, we show how finding meaningful collisions is possible. The trick is to find a collision in the right function!

Assume, as before, that Alice wishes to find a collision between messages of two different “types,” e.g., a letter explaining why Alice was fired and a flattering letter of recommendation that both hash to the same value. Then, Alice writes each message so that there are $\ell - 1$ interchangeable words in each; i.e., there are $2^{\ell-1}$ messages of each type. Define the one-to-one function $g : \{0, 1\}^\ell \rightarrow \{0, 1\}^*$ such that the ℓ th bit of the input selects between messages of type 0 or type 1, and the i th bit (for $1 \leq i \leq \ell - 1$) selects between options for the i th interchangeable word in messages of the appropriate type. For example, consider the sentences:

- 0: Bob is a *good/hardworking* and *honest/trustworthy* worker/employee.
- 1: Bob is a *difficult/problematic* and *taxing/irritating* worker/employee.

Define a function g that takes 4-bit inputs, where the last bit determines the type of sentence output, and the initial three bits determine the choice of words in that sentence. For example:

- $g(0000)$ = Bob is a good and honest worker.
- $g(0001)$ = Bob is a difficult and taxing worker.
- $g(1010)$ = Bob is a hardworking and honest employee.
- $g(1011)$ = Bob is a problematic and taxing employee.

Now define $f : \{0, 1\}^\ell \rightarrow \{0, 1\}^\ell$ by $f(x) \stackrel{\text{def}}{=} H(g(x))$. Alice can find a collision in f using the small-space birthday attack shown earlier. The point here is that any collision x, x' in f yields two messages $g(x), g(x')$ that collide under H . If x, x' is a random collision then we expect that with probability $1/2$ the colliding messages $g(x), g(x')$ will be of different types (since x and x' differ in their final bit with that probability). If the colliding messages are not of different types, the process can be repeated again from scratch.

5.4.3 *Time/Space Tradeoffs for Inverting Functions

In this section we consider the question of preimage resistance, i.e., we are interested in algorithms for the problem of function inversion. Here, an algorithm is given $y = H(x)$ for uniform x , and the goal is to find any x' such

that $H(x') = y$. We begin by assuming that the input and output lengths of H are equal, and briefly consider the more general case at the end.

Let $H : \{0, 1\}^\ell \rightarrow \{0, 1\}^\ell$ be a function. Without exploiting any weaknesses of H , finding a preimage of a point y can be done in time $\mathcal{O}(2^\ell)$ via an exhaustive search over the domain. We show that with significant preprocessing, and a relatively large amount of memory, it is possible to do better.

To be clear: we view preprocessing as a one-time operation and we will not be overly concerned with its cost. We are instead interested in the *on-line* time required to invert H at a point y , after the preprocessing has been done. This is justified if the cost of preprocessing can be amortized over the inversion of many points, or if we are willing to invest computational resources for preprocessing before y is known for the benefit of faster inversion afterwards.

In fact, it is trivial to use preprocessing to enable function inversion in very little time. All we need to do is evaluate H on every point during the preprocessing phase, and then store the pairs $\{(x, H(x))\}$ in a table, sorted by their second entry. Upon receiving any point y , a preimage of y can be found easily by searching the table for a pair with second entry y . The drawback here is that we need to allocate space for storing $\mathcal{O}(2^\ell)$ pairs in the table, which can be prohibitive, if not impossible for large ℓ (e.g., $\ell = 80$).

The initial brute-force attack uses constant memory and $\mathcal{O}(2^\ell)$ time, while the attack just described stores $\mathcal{O}(2^\ell)$ points and enables inversion in essentially constant time. We now present an approach that allows an attacker to *trade off* time and memory. Specifically, we show how to store $\mathcal{O}(2^{2\ell/3})$ points and find preimages in time $\mathcal{O}(2^{2\ell/3})$; other tradeoffs are possible.

A warmup. We begin by considering a simple case where the function H defines a cycle, meaning that $x, H(x), H(H(x)), \dots$ covers all of $\{0, 1\}^\ell$ for any starting point x (note that most functions do not define a cycle, but we assume this in order to demonstrate the idea in a very simple case). For clarity, let $N = 2^\ell$ denote the domain size.

In the preprocessing phase, the attacker simply exhausts the entire cycle, beginning at an arbitrary starting point x_0 and computing $x_1 := H(x_0)$, $x_2 := H(H(x_0))$, up to $x_N = H^{(N)}(x_0)$, where $H^{(i)}$ refers to i -fold evaluation of H . Let $x_i \stackrel{\text{def}}{=} H^{(i)}(x_0)$. We imagine partitioning the cycle into $\sqrt{N}$ segments of length $\sqrt{N}$ each, and having the attacker store the points at the beginning and end of each such segment. That is, the attacker stores in a table pairs of the form $(x_{i \cdot \sqrt{N}}, x_{(i+1) \cdot \sqrt{N}})$, for $i = 0$ to $\sqrt{N} - 1$, sorted by the second component of each pair. The resulting table contains $\mathcal{O}(\sqrt{N})$ points.

When the attacker is given a point y to invert in the on-line phase, it checks which of $y, H(y), H^{(2)}(y), \dots$ corresponds to the endpoint of a segment. (Each check just involves a table lookup on the second component of the stored pairs.) Since y lies in some segment, it is guaranteed to hit an endpoint within $\sqrt{N}$ steps. Once an endpoint $x = x_{(i+1) \cdot \sqrt{N}}$ is identified, the attacker takes the starting point $x' = x_{i \cdot \sqrt{N}}$ of the corresponding segment

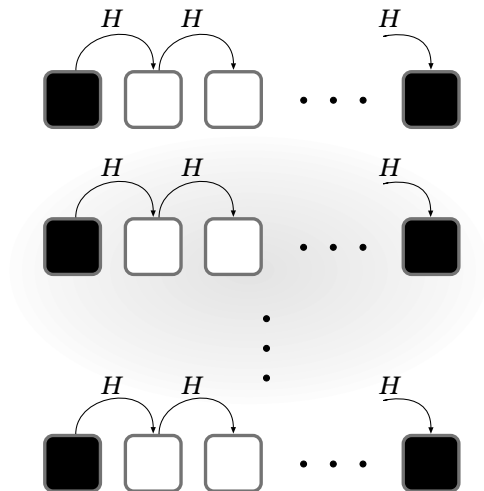


FIGURE 5.3: Table generation. Only the (SP_i, EP_i) pairs are stored.

and computes $H(x')$, $H^{(2)}(x')$, $\dots$ until y is reached; this immediately gives the desired preimage. Observe that this takes at most $\sqrt{N}$ evaluations of H .

In summary, this attack stores $\mathcal{O}(\sqrt{N})$ points and finds preimages with probability 1 using $\mathcal{O}(\sqrt{N})$ hash computations.

Hellman's time/space tradeoff. Martin Hellman introduced a more general time/space tradeoff applicable to an arbitrary function H (though the analysis treats H as a random function). Hellman's attack still stores the starting point and endpoint of several segments, but in this case the segments are “independent” rather than being part of one large cycle. In more detail: let s, t be parameters we will set later. The attack first chooses s uniform starting points $SP_1, \dots, SP_s \in \{0, 1\}^\ell$. For each such point SP_i , it computes a corresponding endpoint $EP_i := H^{(t)}(SP_i)$ using t -fold application of H . (See Figure 5.3.) The attacker then stores the values $\{(SP_i, EP_i)\}_{i=1}^s$ in a table, sorted by the second entry of each pair.

Upon receiving a value y to invert, the attack proceeds as in the simple case discussed earlier. Specifically, it checks if any of $y, H(y), \dots, H^{(t-1)}(y)$ is equal to the endpoint of some segment (stopping as soon as the first such match is found). It is possible that none of these values is equal to an endpoint (as we discuss below). However, if $H^{(j)}(y) = EP_i = H^{(t)}(SP_i)$ for some i, j , then the attacker computes $H^{(t-j-1)}(SP_i)$ and checks whether this is a preimage of y . The entire process requires at most t evaluations of H .

This seems to work, but there are several subtleties we have ignored. First, it may happen that none of $y, H(y), \dots, H^{(t-1)}(y)$ is the endpoint of a segment. This can happen if y is not in the collection of $s \cdot t$ values (not counting the starting points) obtained during the initial process of generating the table. We can set $s \cdot t \geq N$ in an attempt to include every ℓ -bit string in the

table, but this does not solve the problem since there can be collisions in the table itself—in fact, for $s \cdot t \geq N^{1/2}$ our previous analysis of the birthday problem tells us that collisions are likely—which will reduce the number of distinct points in the collection of values. A second problem, which arises even if y is in the table, is that even if we find a matching endpoint, and so $H^{(j)}(y) = EP_i = H^{(t)}(SP_i)$ for some i, j , this does not guarantee that $H^{(t-j-1)}(SP_i)$ is a preimage of y . The issue here is that the segment $y, H(y), \dots, H^{(t-1)}(y)$ might collide with the i th segment even though y itself is not in that segment; see Figure 5.4. (Even if y lies in some segment, the first matching endpoint may not be in that segment.) We call this a *false positive*. One might think this is unlikely to occur if H is collision resistant; again, however, we are dealing with a situation where more than $\sqrt{N}$ points are involved and so collisions actually become likely.

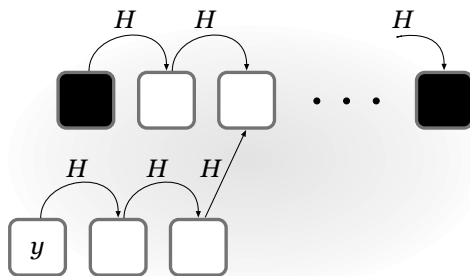


FIGURE 5.4: Colliding in the on-line phase.

The problem of false positives can be addressed by modifying the algorithm so that it always computes the entire sequence $y, H(y), \dots, H^{(t-1)}(y)$, and checks whether $H^{(t-j-1)}(SP_i)$ is a preimage of y for *every* i, j such that $H^{(j)}(y) = EP_i$. This is guaranteed to find a preimage as long as y is in the collection of values (not including the starting points) generated during preprocessing. A concern now is that the running time of the algorithm might increase, since each false positive incurs an additional $\mathcal{O}(t)$ hash evaluations. One can show that the expected number of false positives is $\mathcal{O}(st^2/N)$. (There are t values in the sequence $y, H(y), \dots, H^{(t-1)}(y)$ and at most st distinct points in the table. Treating H as a random function, the probability that any point in the sequence equals some point in the table is $1/N$. The expected number of false positives is thus $t \cdot st \cdot 1/N = st^2/N$.) Thus, as long as $st^2 \approx N$, which we will ensure for other reasons below, the expected number of false positives is constant and dealing with false positives is expected to require only $\mathcal{O}(t)$ additional hash computations.

Given the above modification, the probability of inverting $y = H(x)$ is at least the probability that x is in the collection of points (not including the endpoints) generated during preprocessing. We now lower bound this probability, taken over the randomness of the preprocessing stage as well as

uniform choice of x , treating H as a random function in the analysis. We first compute the expected number of distinct points in the table. Consider what happens when the i th row of the table is generated. The starting point SP_i is uniform and there are at most $(i-1) \cdot t$ distinct points (not including the endpoints) in the table already, so the probability that SP_i is “new” (i.e., not equal to any previous value) is at least $1 - (i-1) \cdot t/N$. What is the probability that $H(SP_i)$ is new? If SP_i is not new, then almost surely neither is $H(SP_i)$. On the other hand, if SP_i is new then $H(SP_i)$ is uniform (because we treat H as a random function) and so is new with probability at least $1 - ((i-1) \cdot t + 1)/N$. (We now have the additional point SP_i .) Thus, the probability that $H(SP_i)$ is new is at least

$$\begin{aligned} & \Pr[SP_i \text{ is new}] \cdot \Pr[H(SP_i) \text{ is new} \mid SP_i \text{ is new}] \\ & \geq \left(1 - \frac{(i-1) \cdot t}{N}\right) \cdot \left(1 - \frac{(i-1) \cdot t + 1}{N}\right) \\ & > \left(1 - \frac{(i-1) \cdot t + 1}{N}\right)^2. \end{aligned}$$

Continuing in this way, the probability that $H^{(t-1)}(SP_i)$ is new is at least

$$\left(1 - \frac{i \cdot t}{N}\right)^t = \left[\left(1 - \frac{i \cdot t}{N}\right)^{\frac{N}{i \cdot t}}\right]^{\frac{i \cdot t^2}{N}} \approx e^{-it^2/N}.$$

The thing to notice here is that when $it^2 \leq N/2$, this probability is at least $1/2$; on the other hand, once $it^2 > N$ the probability is rather small. Considering the last row, when $i = s$, this means that we will not gain much additional coverage if $st^2 > N$. A good setting of the parameters is thus $st^2 = N/2$. Assuming this, the expected number of distinct points in the table is

$$\sum_{i=1}^s \sum_{j=0}^{t-1} \Pr[H^{(j)}(SP_i) \text{ is new}] \geq \sum_{i=1}^s \sum_{j=0}^{t-1} \frac{1}{2} = \frac{st}{2}.$$

The probability that x is “covered” is then at least $\frac{st}{2N} = \frac{1}{4t}$.

This gives a weak time/space tradeoff, in which we can use more space (and consequently less time) at the expense of decreasing the probability of inverting y . But we can do better by generating $T = 4t$ “independent” tables. (This increases both the space and time by a factor of T .) As long as we can treat the probabilities of x being in each of the associated tables as independent, the probability that *at least one* of these tables contains x is

$$1 - \Pr[\text{no table contains } x] = 1 - \left(1 - \frac{1}{4t}\right)^{4t} \approx 1 - e^{-1} = 0.63.$$

The only remaining question is how to generate an independent table. (Note that generating a table exactly as before is the same as adding s additional

rows to our original table, which we have already seen does not help.) We can do this for the i th such table by applying some function F_i after every evaluation of H , where $F_1, \dots, F_T$ are all distinct. (A good choice might be to set $F_i(x) = x \oplus c_i$ for some fixed constant c_i that is different in each table.) Let $H_i \stackrel{\text{def}}{=} F_i \circ H$, i.e., $H_i(x) = F_i(H(x))$. Then for the i th table we again choose s random starting points, but for each such point we now compute $H_i(SP)$, $H_i^{(2)}(SP)$, and so on. Upon receiving a value $y = H(x)$ to invert, the attacker first computes $y' = F_i(y)$ and then checks if any of y' , $H_i(y')$, $\dots$, $H_i^{(t-1)}(y')$ corresponds to an endpoint in the i th table; this is repeated for $i = 1, \dots, T$. (We omit further details.) While it is difficult to argue independence formally, this approach leads to good results in practice.

Choosing parameters. Summarizing the above discussion, we see that as long as $st^2 = N/2$ we have an algorithm that stores $\mathcal{O}(s \cdot T) = \mathcal{O}(s \cdot t) = \mathcal{O}(N/t)$ points during a preprocessing phase, and can then invert y with constant probability in time $\mathcal{O}(t \cdot T) = \mathcal{O}(t^2)$. One setting of the parameters is $t = N^{1/3} = 2^{\ell/3}$, in which case we have an algorithm storing $\mathcal{O}(2^{2\ell/3})$ points that finds preimages using $\mathcal{O}(2^{2\ell/3})$ hash computations. If a hash function with 80 bits of output is used, then this is feasible in practice.

Handling different domain and range. In practice, it is common to be faced with a situation in which the domain and range of H are different. One example is in the context of password cracking (see Section 5.6.3), where an attacker has $H(pw)$ but $|pw| \ll \ell$. In the general case, say x is chosen from some domain D which may be larger or smaller than $\{0, 1\}^\ell$. While it is, of course, possible to artificially expand the domain/range to make them match, this will *not* be useful for the attack described above. To see why, consider the password example. For the attack to succeed we want pw to be in some table of values generated during preprocessing. If we generate each row of the table by simply computing $H(SP)$, $H^{(2)}(SP)$, $\dots$, for $SP \in D$, then *none* of these values (except possibly SP itself) will be equal to pw .

We can address this by applying a function F_i , as before, between each evaluation of H , though now we choose F_i mapping $\{0, 1\}^\ell$ to D . This solves the above issue, since $F_i(H(SP))$, $(F_i \circ H)^{(2)}(SP)$, $\dots$ now all lie in D .

Applications to key-recovery attacks. Time/space tradeoffs give attacks on cryptographic primitives other than hash functions. One canonical application—in fact, the application originally considered by Hellman—is an attack on an arbitrary block cipher F that leads to recovery of the key. Define $H(k) \stackrel{\text{def}}{=} F_k(x)$ where x is some arbitrary, but fixed, input that will be used for building the table. If an attacker can obtain $F_k(x)$ for an unknown key k —either via a chosen-plaintext attack or by choosing x such that $F_k(x)$ is likely to be obtained in a known-plaintext attack—then by inverting H the attacker learns (a candidate value for) k . Note that it is possible for the key length of F to differ from its block length, but in this case we can use the technique just described for handling H with different domain and range.

5.5 The Random-Oracle Model

There are several examples of constructions based on cryptographic hash functions that cannot be proven secure based only on the assumption that the hash function is collision or preimage resistant. (We will see some in the following section.) In many cases, there appears to be *no* simple and reasonable assumption regarding the hash function that would be sufficient for proving the construction secure.

Faced with this situation, there are several options. One is to look for schemes that *can* be proven secure based on some reasonable assumption about the underlying hash function. This is a good approach, but it leaves open the question of what to do until such schemes are found. Also, provably secure constructions may be significantly less efficient than other approaches that have not been proven secure. (This is a prominent issue that we will encounter in the setting of public-key cryptography.)

Another possibility, of course, is to use an existing cryptosystem even if it has no justification for its security other than, perhaps, the fact that the designers tried to attack it and were unsuccessful. This flies in the face of everything we have said about the importance of the rigorous, modern approach to cryptography, and it should be clear that this is unacceptable.

An approach that has been hugely successful in practice, and which offers a “middle ground” between a fully rigorous proof of security on the one hand and no proof whatsoever on the other, is to introduce an idealized model in which to prove the security of cryptographic schemes. Although the idealization may not be an accurate reflection of reality, we can at least derive some measure of confidence in the soundness of a scheme’s design from a proof within the idealized model. As long as the model is reasonable, such proofs are certainly better than no proofs at all.

The most popular example of this approach is the *random-oracle model*, which treats a cryptographic hash function H as a truly random function. (We have already seen an example of this in our discussion of time/space tradeoffs, although there we were analyzing an attack rather than a construction.) More specifically, the random-oracle model posits the existence of a public, random function H that can be evaluated *only* by “querying” an oracle—which can be thought of as a “black box”—that returns $H(x)$ when given input x . (We will discuss how this is to be interpreted in the following section.) To differentiate things, the model we have been using until now (where no random oracle is present) is often called the “standard model.”

No one claims that a random oracle exists, although there have been suggestions that a random oracle could be implemented in practice using a trusted party (i.e., some server on the Internet). Rather, the random-oracle model provides a formal *methodology* that can be used to design and validate cryptographic schemes using the following two-step approach:

1. First, a scheme is designed and proven secure in the random-oracle model. That is, we assume the world contains a random oracle, and construct and analyze a cryptographic scheme within this model. Standard cryptographic assumptions of the type we have seen until now may be utilized in the proof of security as well.
2. When we want to implement the scheme in the real world, a random oracle is not available. Instead, the random oracle is *instantiated* with an appropriately designed cryptographic hash function $\hat{H}$. (We return to this point at the end of this section.) That is, at each point where the scheme dictates that a party should query the oracle for the value $H(x)$, the party instead computes $\hat{H}(x)$ on its own.

The hope is that the cryptographic hash function used in the second step is “sufficiently good” at emulating a random oracle, so that the security proof given in the first step will carry over to the real-world instantiation of the scheme. The difficulty here is that there is no theoretical justification for this hope, and in fact there are (contrived) schemes that can be proven secure in the random-oracle model but are insecure *no matter how the random oracle is instantiated* in the second step. Furthermore, it is not clear (mathematically or heuristically) what it means for a hash function to be “sufficiently good” at emulating a random oracle, nor is it clear that this is an achievable goal. In particular, no concrete instantiation $\hat{H}$ can ever behave like a random function, since $\hat{H}$ is deterministic and fixed. For these reasons, a proof of security in the random-oracle model should be viewed as providing evidence that a scheme has no “inherent design flaws,” but is *not* a rigorous proof that any real-world instantiation of the scheme is secure. Further discussion on how to interpret proofs in the random-oracle model is given in Section 5.5.2.

5.5.1 The Random-Oracle Model in Detail

Before continuing, let us pin down exactly what the random-oracle model entails. A good way to think about the random-oracle model is as follows: The “oracle” is simply a box that takes a binary string as input and returns a binary string as output. The internal workings of the box are unknown and inscrutable. Everyone—honest parties as well as the adversary—can interact with the box, where such interaction consists of feeding in a binary string x as input and receiving a binary string y as output; we refer to this as *querying the oracle on x* , and call x itself a *query* made to the oracle. Queries to the oracle are assumed to be private so that if some party queries the oracle on input x then no one else learns x , or even learns that this party queried the oracle at all. This makes sense, because calls to the oracle correspond (in the real-world instantiation) to local evaluations of a cryptographic hash function.

An important property of this “box” is that it is *consistent*. That is, if the box ever outputs y for a particular input x , then it always outputs the same answer y when given the same input x again. This means that we can view the

box as implementing a well-defined function H ; i.e., we define the function H in terms of the input/output characteristics of the box. For convenience, we thus speak of “querying H ” rather than querying the box. No one “knows” the entire function H (except the box itself); at best, all that is known are the values of H on the strings that have been explicitly queried thus far.

We have already discussed in Chapter 3 what it means to choose a random function H . We only reiterate here that there are two equivalent ways to think about the uniform selection of H : either picture H being chosen “in one shot” uniformly from the set of all functions on some specified domain and range, or imagine generating outputs for H “on-the-fly,” as needed. Specifically, in the second case we can view the function as being defined by a table that is initially empty. When the oracle receives a query x it first checks whether $x = x_i$ for some pair (x_i, y_i) in the table; if so, the corresponding value y_i is returned. Otherwise, a *uniform* string $y \in \{0, 1\}^\ell$ is chosen (for some specified ℓ), the answer y is returned, and the oracle stores (x, y) in its table. This second viewpoint is often conceptually easier to reason about, and is also technically easier to deal with if H is defined over an infinite domain (e.g., $\{0, 1\}^*$).

When we defined pseudorandom functions in Section 3.5.1, we also considered algorithms having oracle access to a random function. Lest there be any confusion, we note that the usage of a random function there is very different from the usage of a random function here. There, a random function was used *as a way of defining* what it means for a (concrete) keyed function to be pseudorandom. In the random-oracle model, in contrast, the random function is used *as part of a construction itself* and must somehow be instantiated in the real world if we want a concrete realization of the construction. A pseudorandom function is not a random oracle because it is only pseudorandom if the key is *secret*. However, in the random-oracle model all parties need to be able to compute the function; thus there can be no secret key.

Definitions and Proofs in the Random-Oracle Model

Definitions in the random-oracle model are slightly different from their counterparts in the standard model because the probability spaces considered in each case are not the same. In the standard model a scheme Π is secure if for all PPT adversaries $\mathcal{A}$ the probability of some event is below some threshold, where *this probability is taken over the random choices of the parties running Π and those of the adversary $\mathcal{A}$* . Assuming the honest parties who use Π in the real world make random choices as directed by the scheme, satisfying a definition of this sort guarantees security for real-world usage of Π .

In the random-oracle model, in contrast, a scheme Π may rely on an oracle H . As before, Π is secure if for all PPT adversaries $\mathcal{A}$ the probability of some event is below some threshold, but now *this probability is taken over random choice of H* as well as the random choices of the parties running Π and those of the adversary $\mathcal{A}$. When using Π in the real world, some (instantiation of) H must be fixed. Unfortunately, security of Π is not guaranteed

for any *particular* choice of H . This indicates one reason why it is difficult to argue that any concrete instantiation of the oracle H by a deterministic function yields a secure scheme. (An additional, technical, difficulty is that once a concrete function H is fixed, the adversary $\mathcal{A}$ is no longer restricted to querying H as an oracle but can instead look at and use the *code* of H in the course of its attack.)

Proofs in the random-oracle model can exploit the fact that H is chosen at random, and that the only way to evaluate $H(x)$ is to explicitly query x to H . Three properties in particular are especially useful; we sketch them informally here, and show some simple applications of them below and in the next section, but caution that a full understanding will likely have to wait until we present formal proofs in the random-oracle model in later chapters.

The first useful property of the random-oracle model is:

*If x has not been queried to H , then the value of $H(x)$ is **uniform**.*

This may seem superficially similar to the guarantee provided by a pseudorandom generator, but is actually much stronger. If G is a pseudorandom generator then $G(x)$ is pseudorandom to an observer *assuming x is chosen uniformly at random and is completely unknown to the observer*. If H is a random oracle, however, then $H(x)$ is truly uniform to an observer as long as the observer has not queried x . This is true even if x is known, or if x is not uniform but *is* hard to guess. (For example, if x is an n -bit string where the first half of x is known and the last half is random then $G(x)$ might be easy to distinguish from random but $H(x)$ will not be.)

The remaining two properties relate explicitly to *proofs by reduction* in the random-oracle model. (It may be helpful here to review Section 3.3.2.) As part of the reduction, the random oracle that the adversary $\mathcal{A}$ interacts with must be simulated. That is: $\mathcal{A}$ will submit queries to, and receive answers from, what it believes to be the oracle, but the reduction itself must now answer these queries. This turns out to give a lot of power. For starters:

*If $\mathcal{A}$ queries x to H , the reduction can **see this query** and learn x .*

This is sometimes called “extractability.” (This does not contradict the fact, mentioned earlier, that queries to the random oracle are “private.” While that is true in the random-oracle model itself, here we are using $\mathcal{A}$ as a subroutine within a reduction that is simulating the random oracle for $\mathcal{A}$.) Finally:

*The reduction can **set** the value of $H(x)$ (i.e., the response to query x) to a value of its choice, as long as this value is correctly distributed, i.e., uniform.*

This is called “programmability.” There is no counterpart to extractability or programmability once H is instantiated with any concrete function.

Simple Illustrations of the Random-Oracle Model

At this point some examples may be helpful. The examples given here are relatively simple, and do not use the full power that the random-oracle model affords. Rather, these examples are presented merely to provide a gentle introduction to the model. In what follows, we assume a random oracle mapping ℓ_{in} -bit inputs to ℓ_{out} -bit outputs, where $\ell_{in}, \ell_{out} > n$, the security parameter (so ℓ_{in}, ℓ_{out} are functions of n).

A random oracle as a pseudorandom generator. We first show that, for $\ell_{out} > \ell_{in}$, a random oracle can be used as a pseudorandom generator. (We do not say that a random oracle *is* a pseudorandom generator, since a random oracle is not a fixed function.) Formally, we claim that for any PPT adversary $\mathcal{A}$, there is a negligible function negl such that

$$\left| \Pr[\mathcal{A}^{H(\cdot)}(y) = 1] - \Pr[\mathcal{A}^{H(\cdot)}(H(x)) = 1] \right| \leq \text{negl}(n),$$

where in the first case the probability is taken over uniform choice of H , uniform choice of $y \in \{0, 1\}^{\ell_{out}(n)}$, and the randomness of $\mathcal{A}$, and in the second case the probability is taken over uniform choice of H , uniform choice of $x \in \{0, 1\}^{\ell_{in}(n)}$, and the randomness of $\mathcal{A}$. We have explicitly indicated that $\mathcal{A}$ has oracle access to H in each case; once H has been chosen then $\mathcal{A}$ can freely make queries to it.

Let S denote the set of points on which $\mathcal{A}$ has queried H ; of course, $|S|$ is polynomial in n . Observe that in the second case, the probability that $x \in S$ is negligible. This holds since $\mathcal{A}$ starts with no information about x (note that $H(x)$ by itself reveals nothing about x because H is a random function), and because S is exponentially smaller than $\{0, 1\}^{\ell_{in}}$. Moreover, conditioned on $x \notin S$ in the second case, $\mathcal{A}$'s input in each case is a uniform string that is independent of the answers to $\mathcal{A}$'s queries.

A random oracle as a collision-resistant hash function. If $\ell_{out} < \ell_{in}$, a random oracle is collision resistant. That is, the success probability of any PPT adversary $\mathcal{A}$ in the following experiment is negligible:

1. A random function H is chosen.
2. $\mathcal{A}$ succeeds if it outputs distinct x, x' with $H(x) = H(x')$.

To see this, assume without loss of generality that $\mathcal{A}$ only outputs values x, x' that it had previously queried to the oracle, and that $\mathcal{A}$ never makes the same query to the oracle twice. Letting the oracle queries of $\mathcal{A}$ be $x_1, \dots, x_q$, with $q = \text{poly}(n)$, it is clear that the probability that $\mathcal{A}$ succeeds is upper-bounded by the probability that $H(x_i) = H(x_j)$ for some $i \neq j$. But this is exactly equal to the probability that if we pick q strings $y_1, \dots, y_q \in \{0, 1\}^{\ell_{out}}$ independently and uniformly at random, we have $y_i = y_j$ for some $i \neq j$. This is exactly the birthday problem, and so using the results of Appendix A.4 we have that $\mathcal{A}$ succeeds with negligible probability $\mathcal{O}(q^2/2^{\ell_{out}})$.

Constructing a pseudorandom function from a random oracle. It is also rather easy to construct a pseudorandom function in the random-oracle model. Suppose $\ell_{in}(n) = 2n$ and $\ell_{out}(n) = n$, and define

$$F_k(x) \stackrel{\text{def}}{=} H(k\|x),$$

where $|k| = |x| = n$. In Exercise 5.11 you are asked to show that this is a pseudorandom function, namely, that for any polynomial-time $\mathcal{A}$ the success probability of $\mathcal{A}$ in the following experiment is not more than $1/2$ plus a negligible function:

1. A function H and values $k \in \{0, 1\}^n$ and $b \in \{0, 1\}$ are chosen uniformly.
2. If $b = 0$, the adversary $\mathcal{A}$ is given access to an oracle for $F_k(\cdot) = H(k\|\cdot)$.
If $b = 1$, then $\mathcal{A}$ is given access to a random function mapping n -bit inputs to n -bit outputs. (This random function is *independent* of H .)
3. $\mathcal{A}$ outputs a bit b' , and succeeds if $b' = b$.

In step 2, $\mathcal{A}$ can access H in addition to the function oracle provided to it by the experiment. (A pseudorandom function in the random-oracle model must be indistinguishable from a random function that is independent of H .)

An interesting aspect of all the above claims is that they make no computational assumptions; they hold even for computationally unbounded adversaries as long as those adversaries are limited to making polynomially many queries to the oracle. This has no counterpart in the real world, where we have seen that computational assumptions are necessary.

5.5.2 Is the Random-Oracle Methodology Sound?

Schemes designed in the random-oracle model are implemented in the real world by instantiating H with some concrete function. With the mechanics of the random-oracle model behind us, we turn to a more fundamental question:

What do proofs of security in the random-oracle model guarantee as far as security of any real-world instantiation?

This question does not have a definitive answer: there is currently debate within the cryptographic community regarding how to interpret proofs in the random-oracle model, and an active area of research is to determine what, precisely, a proof of security in the random-oracle model implies vis-a-vis the real world. We can only hope to give a flavor of both sides of the debate.

Objections to the random-oracle model. The starting point for arguments against using random oracles is simple: as we have already noted, there is no formal or rigorous justification for believing that a proof of security for some scheme Π in the random-oracle model says anything about the security of Π in the real world, once the random oracle H has been instantiated with

any particular hash function $\hat{H}$. This is more than just theoretical uneasiness. A little thought shows that *no* concrete hash function can ever act as a “true” random oracle. For example, in the random-oracle model the value $H(x)$ is “completely random” if x was not explicitly queried. The counterpart would be to require that $\hat{H}(x)$ is random (or pseudorandom) if $\hat{H}$ was not explicitly evaluated on x . How are we to interpret this in the real world? It is not even clear what it means to “explicitly evaluate” $\hat{H}$: what if an adversary knows some shortcut for computing $\hat{H}$ that does not involve running the actual code for $\hat{H}$? Moreover, $\hat{H}(x)$ cannot possibly be random (or even pseudorandom) since once the adversary learns the description of $\hat{H}$, the value of that function on *all* inputs is immediately determined.

Limitations of the random-oracle model become clearer once we examine the proof techniques introduced earlier. Recall that one proof technique is to use the fact that a reduction can “see” the queries that an adversary $\mathcal{A}$ makes to the random oracle. If we replace the random oracle by a particular hash function $\hat{H}$, this means we must provide a description of $\hat{H}$ to the adversary at the beginning of the experiment. But then $\mathcal{A}$ can evaluate $\hat{H}$ on its own, without making any *explicit* queries, and so a reduction will no longer have the ability to “see” any queries made by $\mathcal{A}$. (In fact, as noted previously, the notion of $\mathcal{A}$ performing explicit evaluations of $\hat{H}$ may not be true and certainly cannot be formally defined.) Likewise, proofs of security in the random-oracle model allow the reduction to choose the outputs of H as it wishes, something that is clearly not possible when a concrete function is used.

Even if we are willing to overlook the above theoretical concerns, a practical problem is that we do not currently have a very good understanding of what it means for a concrete hash function to be “sufficiently good” at instantiating a random oracle. For concreteness, say we want to instantiate the random oracle using some appropriate modification of SHA-1 (SHA-1 is a cryptographic hash function discussed in Section 6.3.3). While for some particular scheme Π it might be reasonable to assume that Π is secure when instantiated using SHA-1, it is much less reasonable to assume that SHA-1 can take the place of a random oracle in *every* scheme designed in the random-oracle model. Indeed, as we have said earlier, we *know* that SHA-1 is not a random oracle. And it is not hard to design a scheme that is secure in the random-oracle model, but is insecure when the random oracle is replaced by SHA-1.

We emphasize that an assumption of the form “SHA-1 acts like a random oracle” is qualitatively different from assumptions such as “SHA-1 is collision resistant” or “AES is a pseudorandom function.” The problem lies partly with the fact that there is no satisfactory *definition* of what the first statement means, while we do have such definitions for the latter two statements.

Because of this, using the random-oracle model to prove security of a scheme is *qualitatively* different from, e.g., introducing a new cryptographic assumption in order to prove a scheme secure in the standard model. Therefore, proofs of security in the random-oracle model are less satisfying than proofs of security in the standard model.

Support for the random-oracle model. Given all the problems with the random-oracle model, why use it at all? More to the point: why has the random-oracle model been so influential in the development of modern cryptography (especially current practical usage of cryptography), and why does it continue to be so widely used? As we will see, the random-oracle model enables the design of substantially more efficient schemes than those we know how to construct in the standard model. As such, there are few (if any) public-key cryptosystems used today having proofs of security in the standard model, while there are numerous deployed schemes having proofs of security in the random-oracle model. In addition, proofs in the random-oracle model are almost universally recognized as lending confidence to the security of schemes being considered for standardization.

The fundamental reason for this is the belief that:

A proof of security in the random-oracle model is significantly better than no proof at all.

Although some disagree, we offer the following in support of this assertion:

- A proof of security for a scheme in the random-oracle model indicates that the scheme’s design is “sound,” in the sense that the only possible attacks on a real-world instantiation of the scheme are those that arise due to a weakness in the hash function used to instantiate the random oracle. Thus, if a “good enough” hash function is used to instantiate the random oracle, we should have confidence in the security of the scheme. Moreover, if a given instantiation of the scheme *is* successfully attacked, we can simply replace the hash function being used with a “better” one.
- Importantly, *there have been no successful real-world attacks on schemes proven secure in the random-oracle model*, when the random oracle was instantiated properly. (We do not include here attacks on “contrived” schemes, but remark that great care must be taken in instantiating the random oracle, as indicated by Exercise 5.10.) This gives evidence to the usefulness of the random-oracle model in designing practical schemes.

Nevertheless, the above ultimately represent only intuitive speculation as to the usefulness of proofs in the random-oracle model and—all else being equal—proofs without random oracles are preferable.

Instantiating the Random Oracle

Properly instantiating a random oracle is subtle, and a full discussion is beyond the scope of this book. Here we only alert the reader that using an “off-the-shelf” cryptographic hash function without modification is not, generally speaking, a sound approach. For one thing, most cryptographic hash functions are constructed using the Merkle–Damgård paradigm (cf. Section 5.2), which can be distinguished easily from a random oracle when variable-length inputs

are allowed. (See Exercise 5.10.) Also, in some constructions it is necessary for the output of the random oracle to lie in a certain range (e.g., the oracle should output elements of some group), which results in additional complications.

5.6 Additional Applications of Hash Functions

We conclude this chapter with a brief discussion of some additional applications of cryptographic hash functions in cryptography and computer security.

5.6.1 Fingerprinting and Deduplication

When using a collision-resistant hash function H , the hash (or *digest*) of a file serves as a unique identifier for that file. (If any other file is found to have the same identifier, this implies a collision in H). The hash $H(x)$ of a file x is like a fingerprint, and one can check whether two files are equal by comparing their digests. This simple idea has many applications.

- *Virus fingerprinting:* Virus scanners identify viruses and block or quarantine them. One of the most basic steps toward this goal is to store a database containing the hashes of known viruses, and then to look up the hash of a downloaded application or email attachment in this database. Since only a short string needs to be recorded (and/or distributed) for each virus, the overhead involved is feasible.
- *Deduplication:* Data deduplication is used to eliminate duplicate copies of data, especially in the context of cloud storage where multiple users rely on a single cloud service to store their data. The observation here is that if multiple users wish to store the same file (e.g., a popular video), then the file only needs to be stored once and need not be uploaded separately by each user. Deduplication can be achieved by first having a user upload a hash of the new file they want to store; if a file with this hash is already stored in the cloud, then the cloud-storage provider can simply add a pointer to the existing file to indicate that this specific user has also stored this file. This saves both communication and storage, and the soundness of the methodology follows from the collision resistance of the hash function.
- *Peer-to-peer (P2P) file sharing:* In P2P file-sharing systems, tables are held by servers to provide a file-lookup service. These tables contain the hashes of the available files, once again providing a unique identifier without using much memory.

It may be surprising that a small digest can uniquely identify every file in the world. But this is the guarantee provided by collision-resistant hash functions, which makes them useful in the settings above.

5.6.2 Merkle Trees

Consider a client who uploads a file x to a server. When the client later retrieves x , it wants to make sure that the server returns the original, unmodified file. The client could simply store x and check that the retrieved file is equal to x , but that defeats the purpose of using the server in the first place. We are looking for a solution in which the storage of the client is small.

A natural solution is to use the “fingerprinting” approach described above. The client can locally store the short digest $h := H(x)$; when the server returns a candidate file x' the client need only check that $H(x') \stackrel{?}{=} h$.

What happens if we want to extend this solution to *multiple* files $x_1, \dots, x_t$? There are two obvious ways of doing this. One is to simply hash each file independently; the client will locally store the digests $h_1, \dots, h_t$, and verify retrieved files as before. This has the disadvantage that the client’s storage grows linearly in t . Another possibility is to hash all the files together. That is, the client can compute $h := H(x_1, \dots, x_t)$ and store only h . The drawback now is that when the client wants to retrieve and verify correctness of the i th file x_i , it needs to retrieve *all* the files in order to recompute the digest.

Merkle trees, introduced by Ralph Merkle, give a tradeoff between these extremes. A Merkle tree computed over input values $x_1, \dots, x_t$ is simply a binary tree of depth $\log t$ in which the inputs are placed at the leaves, and the value of each internal node is the hash of the values of its two children; see Figure 5.5. (We assume t is a power of 2; if not, then we can fix some input values to null or use an incomplete binary tree, depending on the application.)

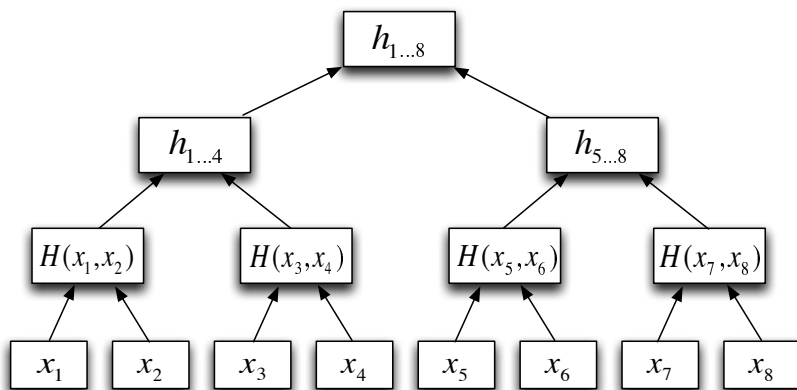


FIGURE 5.5: A Merkle tree.

Fixing some hash function H , we denote by $\mathcal{MT}_t$ the function that takes t input values $x_1, \dots, x_t$, computes the resulting Merkle tree, and outputs the value of the root of the tree. (A keyed hash function yields a keyed function $\mathcal{MT}_t$ in the obvious way.) We have:

THEOREM 5.11 *Let (Gen_H, H) be collision resistant. Then $(\text{Gen}_H, \mathcal{MT}_t)$ is also collision resistant for any fixed t .*

Merkle trees thus provide an alternative to the Merkle–Damgård transform for achieving domain extension for collision-resistant hash functions. (As described, however, Merkle trees are *not* collision resistant if the number of input values t is allowed to vary.)

Merkle trees provide an efficient solution to our original problem, since they allow verification of any of the original t inputs using $\mathcal{O}(\log t)$ communication. The client computes $h := \mathcal{MT}_t(x_1, \dots, x_t)$, uploads $x_1, \dots, x_t$ to the server, and stores h (along with the number of files t) locally. When the client retrieves the i th file, the server sends x_i along with a “proof” π_i that this is the correct value. This proof consists of the values of the nodes in the Merkle tree adjacent to the path from x_i to the root. From these values the client can recompute the value of the root and verify that it is equal to the stored value h . As an example, consider the Merkle tree in Figure 5.5. The client computes $h_{1\dots 8} := \mathcal{MT}_8(x_1, \dots, x_8)$, uploads $x_1, \dots, x_8$ to the server, and stores $h_{1\dots 8}$ locally. When the client retrieves x_3 , the server sends x_3 along with x_4 , $h_{1\dots 2} = H(x_1, x_2)$, and $h_{5\dots 8} = H(H(x_5, x_6), H(x_7, x_8))$. (If files are large we may wish to avoid sending any file other than the one the client has requested. That can easily be done if we define the Merkle tree over the *hashes* of the files rather than the files themselves. We omit the details.) The client computes $h'_{1\dots 4} := H(h_{1\dots 2}, H(x_3, x_4))$ and $h'_{1\dots 8} := H(h'_{1\dots 4}, h_{5\dots 8})$, and then verifies that $h_{1\dots 8} \stackrel{?}{=} h'_{1\dots 8}$.

If H is collision resistant, it is infeasible for the server to send an incorrect file (and any proof) that will cause verification to succeed. Using this approach, the client’s local storage is *constant* (independent of the number of files t), and the communication from server to client is proportional to $\log t$.

5.6.3 Password Hashing

One of the most common and important uses of hash functions in computer security is for password protection. Consider a user typing in a password before using their laptop. To authenticate the user, some form of the user’s password must be stored somewhere on their laptop. If the user’s password is stored in the clear, then an adversary who steals the laptop can read the user’s password off the hard drive and then login as that user. (It may seem pointless to try to hide one’s password from an attacker who can already read the contents of the hard drive. However, files on the hard drive may be

encrypted with a key derived from the user's password, and would thus only be accessible after the password is entered. In addition, the user is likely to use the same password at other sites.)

This risk can be mitigated by storing a *hash of the password* instead of the password itself. That is, the hard drive stores the value $h_{pw} = H(pw)$ in a password file; later, when the user enters its password pw , the operating system checks whether $H(pw) \stackrel{?}{=} h_{pw}$ before granting access. The same basic approach is also used for password-based authentication on the web. Now, if an attacker steals the hard drive (or breaks into a web server), all it obtains is the hash of the password and not the password itself.

If the password is chosen from some relatively small space D of possibilities (e.g., D might be a dictionary of English words, in which case $|D| \approx 80,000$), an attacker can enumerate all possible passwords $pw_1, pw_2, \dots \in D$ and, for each candidate pw_i , check whether $H(pw_i) = h_{pw}$. We would like to claim that an attacker can do no better than this. (This would also ensure that the adversary could not learn the password of any user who chose a *strong* password from a large space.) Unfortunately, preimage resistance (i.e., one-wayness) of H is not sufficient to imply what we want. For one thing, preimage resistance only says that $H(x)$ is hard to invert when x is chosen uniformly from a large domain like $\{0, 1\}^n$. It says nothing about the hardness of inverting H when x is chosen from some other space, or when x is chosen according to some other distribution. Moreover, preimage resistance says nothing about the *concrete* amount of time needed to find a preimage. For example, a hash function H for which computing $x \in \{0, 1\}^n$ given $H(x)$ requires time $2^{n/2}$ could still qualify as preimage resistant, yet this would mean that a 30-bit password could be recovered in only 2^{15} time.

If we model H as a random oracle, then we can formally prove the security we want, namely, recovering pw from h_{pw} (assuming pw is chosen uniformly from D) requires $|D|/2$ evaluations of H , on average.

The above discussion assumes no preprocessing is done by the attacker. As we have seen in Section 5.4.3, though, preprocessing can be used to generate large tables that enable inversion (even of a random function!) faster than exhaustive search. This is a significant concern in practice: even if a user chooses their password as a random combination of 8 alphanumeric characters—giving a password space of size $N = 62^8 \approx 2^{47.6}$ —there is an attack using time and space $N^{2/3} \approx 2^{32}$ that will be highly effective. The tables only need to be generated once, and can be used to crack hundreds of thousands of passwords in case of a server breach. Such attacks are routinely carried out in practice.

Mitigation. We briefly describe two mechanisms used to mitigate the threat of password cracking; further discussion can be found in texts on computer security. One technique is to use “slow” hash functions, or to slow down existing hash functions by using multiple iterations (i.e., computing $H^{(I)}(pw)$ for $I \gg 1$). This has the effect of slowing down legitimate users by a factor of I , which is not a problem if I is set to some “moderate” value (e.g., 1,000).

On the other hand, it has a significant impact on an adversary attempting to crack thousands of passwords at once.

A second mechanism is to introduce a *salt*. When a user registers their password, the laptop/server will generate a long random value s (a “salt”) unique to that user, and store $(s, h_{pw} = H(s, pw))$ instead of merely storing $H(pw)$ as before. Since s is unknown to the attacker in advance, preprocessing is ineffective and the best an attacker can do is to wait until it obtains the password file and then do a linear-time exhaustive search over the domain D as discussed before. Note also that since a different salt is used for each stored password, a separate brute-force search is needed to recover each password.

5.6.4 Key Derivation

All the symmetric-key cryptosystems we have seen require a *uniformly distributed* bit-string for the secret key. Often, however, it is more convenient for two parties to rely on shared information such as a password or biometric data that is *not* uniformly distributed. (Jumping ahead, in Chapter 10 we will see how parties can interact to generate a high-entropy shared secret that is *not* uniformly distributed.) The parties could try to use their shared information directly as a secret key, but in general this will not be secure (since, e.g., private-key schemes all assume a uniformly distributed key). Moreover, the shared data may not even have the correct format to be used as a secret key (it may be too long, for example).

Truncating the shared secret, or mapping it in some other ad hoc way to a string of the correct length, may lose a significant amount of entropy. (We define one notion of entropy more formally below, but for now one can think of entropy as the logarithm of the space of possible shared secrets.) For example, imagine two parties share a password composed of 28 random upper-case letters, and want to use a cryptosystem with a 128-bit key. Since there are 26 possibilities for each character, there are $26^{28} > 2^{130}$ possible passwords. If the password is shared in ASCII format, each character is stored using 8 bits, and so the total length of the password is 224 bits. If the parties truncate their password to the first 128 bits, they will be using only the first 16 characters of their password. However, this will not be a uniformly distributed 128-bit string! In fact, the ASCII representations of the letters A–Z lie between 0x41 and 0x5A; in particular, the first 3 bits of every byte are always 010. This means that *37.5% of the bits of the resulting key will be fixed*, and the 128-bit key the parties derive will have only about 75 bits of entropy (i.e., there are only 2^{75} or so possibilities for the key).

What we need is a generic solution for deriving a key from a high-entropy (but not necessarily uniform) shared secret. Before continuing, we define the notion of entropy we consider here.

DEFINITION 5.12 A probability distribution $\mathcal{X}$ has m bits of min-entropy if for every fixed value x it holds that $\Pr_{X \leftarrow \mathcal{X}}[X = x] \leq 2^{-m}$. That is, even the most likely outcome occurs with probability at most 2^{-m} .

The uniform distribution over a set of size S has min-entropy $\log S$. A distribution in which one element occurs with probability $1/10$ and 90 elements each occur with probability $1/100$ has min-entropy $\log 10 \approx 3.3$. The min-entropy of a distribution measures the probability with which an attacker can guess a value sampled from that distribution; the attacker's best strategy is to guess the most likely value, and so if the distribution has min-entropy m the attacker guesses correctly with probability at most 2^{-m} . This explains why min-entropy (rather than other notions of entropy) is useful in our context. An extension of min-entropy, called *computational min-entropy*, is defined as above except that the distribution is only required to be *computationally indistinguishable* from a distribution with the given min-entropy. (The notion of computational indistinguishability is formally defined in Section 7.8.)

A *key-derivation function* provides a way to obtain a uniformly distributed string from any distribution with high (computational) min-entropy. It is not hard to see that if we model a hash function H as a random oracle, then H serves as a good key-derivation function. Consider an attacker's uncertainty about $H(X)$, where X is sampled from a distribution with min-entropy m (as a technical point, we require the distribution to be independent of H). Each of the attacker's queries to H can be viewed as a "guess" for the value of X ; by assumption on the min-entropy of the distribution, an attacker making q queries to H will query $H(X)$ with probability at most $q \cdot 2^{-m}$. If the attacker does not query X to H , then $H(X)$ is a uniform string.

It is also possible to design key-derivation functions, without relying on the random-oracle model, using keyed hash functions called (*strong*) *extractors*. The key for the extractor must be uniform, but need not be kept secret. One standard for this is called HKDF; see the references at the end of the chapter.

5.6.5 Commitment Schemes

A *commitment scheme* allows one party to "commit" to a message m by sending a commitment value com , while obtaining the following seemingly contradictory properties:

- *Hiding*: the commitment reveals nothing about m .
- *Binding*: it is infeasible for the committer to output a commitment com that it can later "open" as two different messages m, m' . (In this sense, com truly "commits" the committer to some well-defined value.)

A commitment scheme can be seen as a digital envelope: sealing a message in an envelope and handing it over to another party provides privacy (until the

envelope is opened) and binding (since the envelope is sealed).

Formally, a (non-interactive) commitment scheme is defined by a randomized algorithm Gen that outputs public parameters params and an algorithm Com that takes params and a message $m \in \{0, 1\}^n$ and outputs a commitment com ; we will make the randomness used by Com explicit, and denote it by r . A sender commits to m by choosing uniform r , computing $\text{com} := \text{Com}(\text{params}, m; r)$, and sending it to a receiver. The sender can later decommit com and reveal m by sending m, r to the receiver; the receiver verifies this by checking that $\text{Com}(\text{params}, m; r) \stackrel{?}{=} \text{com}$.

Hiding, informally, means that com reveals nothing about m ; binding means that it is impossible to output a commitment com that can be opened two different ways. We define these properties formally now.

The commitment hiding experiment $\text{Hiding}_{\mathcal{A}, \text{Com}}(n)$:

1. Parameters $\text{params} \leftarrow \text{Gen}(1^n)$ are generated.
2. The adversary $\mathcal{A}$ is given input params , and outputs a pair of messages $m_0, m_1 \in \{0, 1\}^n$.
3. A uniform $b \in \{0, 1\}$ is chosen and $\text{com} \leftarrow \text{Com}(\text{params}, m_b; r)$ is computed.
4. The adversary $\mathcal{A}$ is given com and outputs a bit b' .
5. The output of the experiment is 1 if and only if $b' = b$.

The commitment binding experiment $\text{Binding}_{\mathcal{A}, \text{Com}}(n)$:

1. Parameters $\text{params} \leftarrow \text{Gen}(1^n)$ are generated.
2. $\mathcal{A}$ is given input params and outputs $(\text{com}, m, r, m', r')$.
3. The output of the experiment is defined to be 1 if and only if $m \neq m'$ and $\text{Com}(\text{params}, m; r) = \text{com} = \text{Com}(\text{params}, m'; r')$.

DEFINITION 5.13 A commitment scheme Com is secure if for all PPT adversaries $\mathcal{A}$ there is a negligible function negl such that

$$\Pr [\text{Hiding}_{\mathcal{A}, \text{Com}}(n) = 1] \leq \frac{1}{2} + \text{negl}(n)$$

and

$$\Pr [\text{Binding}_{\mathcal{A}, \text{Com}}(n) = 1] \leq \text{negl}(n).$$

It is easy to construct a secure commitment scheme from a random oracle H . To commit to a message m , the sender chooses uniform $r \in \{0, 1\}^n$ and outputs $\text{com} := H(m \| r)$. (In the random-oracle model, Gen and params are not needed since H , in effect, serves as the public parameters of the scheme.) Intuitively, hiding follows from the fact that an adversary queries $H(\star \| r)$ with

only negligible probability (since r is a uniform n -bit string); if it never makes a query of this form then $H(m||r)$ reveals nothing about m . Binding follows from the fact that H is collision resistant.

Commitment schemes can be constructed without random oracles (in fact, from one-way functions), but the details are beyond the scope of this book.

References and Additional Reading

Collision-resistant hash functions were formally defined by Damgård [52]. Additional discussion regarding notions of security for hash functions besides collision resistance can be found in [120, 150]. The Merkle–Damgård transform was introduced independently by Damgård and Merkle [53, 123]

HMAC was introduced by Bellare et al. [14] and later standardized [131].

The small-space birthday attack described in Section 5.4.2 relies on a cycle-finding algorithm of Floyd. Related algorithms and results are described at http://en.wikipedia.org/wiki/Cycle_detection. The idea for finding meaningful collisions using the small-space attack is by Yuval [180]. The possibility of parallelizing collision-finding attacks, which can offer significant speedups in practice, is discussed in [170]. Time/space tradeoffs for function inversion were introduced by Hellman [87], with practical improvements—not discussed here—given by Rivest (unpublished) and by Oechslin [134].

The first formal treatment of the random-oracle model was given by Bellare and Rogaway [21], although the idea of using a “random-looking” function in cryptographic applications had been suggested previously, most notably by Fiat and Shamir [65]. Proper instantiation of a random oracle based on concrete cryptographic hash functions is discussed in [21, 22, 23, 48]. The seminal negative result concerning the random-oracle model is that of Canetti et al. [41], who show (contrived) schemes that are secure in the random-oracle model but are insecure for *any* concrete instantiation of the random oracle.

Merkle trees were introduced in [121]. Key-derivation functions used in practice include HKDF, PBKDF2, and bcrypt. See [109] for a formal treatment of the problem and an analysis of HKDF.

Exercises

- 5.1 Provide formal definitions for second preimage resistance and preimage resistance. Prove that any hash function that is collision resistant is second preimage resistant, and any hash function that is second preimage resistant is preimage resistant.

- 5.2 Let (Gen_1, H_1) and (Gen_2, H_2) be two hash functions. Define (Gen, H) so that Gen runs Gen_1 and Gen_2 to obtain keys s_1 and s_2 , respectively. Then define $H^{s_1, s_2}(x) = H_1^{s_1}(x) \| H_2^{s_2}(x)$.
- (a) Prove that if at least one of (Gen_1, H_1) and (Gen_2, H_2) is collision resistant, then (Gen, H) is collision resistant.
 - (b) Determine whether an analogous claim holds for second preimage resistance and preimage resistance, respectively. Prove your answer in each case.
- 5.3 Let (Gen, H) be a collision-resistant hash function. Is $(\text{Gen}, \hat{H})$ defined by $\hat{H}^s(x) \stackrel{\text{def}}{=} H^s(H^s(x))$ necessarily collision resistant?
- 5.4 Provide a formal proof of Theorem 5.4 (i.e., describe the reduction).
- 5.5 Generalize the Merkle–Damgård transform (Construction 5.3) for the case when the fixed-length hash function h has input length $n + \kappa$ (with $\kappa > 0$) and output length n , and the length of the input to H should be encoded as an ℓ -bit value (as discussed in Section 5.3.2). Prove collision resistance of (Gen, H) , assuming collision resistance of (Gen, h) .
- 5.6 For each of the following modifications to the Merkle–Damgård transform (Construction 5.3), determine whether the result is collision resistant. If yes, provide a proof; if not, demonstrate an attack.
- (a) Modify the construction so that the input length is not included at all (i.e., output z_B and not $z_{B+1} = h^s(z_B \| L)$). (Assume the resulting hash function is only defined for inputs whose length is an integer multiple of the block length.)
 - (b) Modify the construction so that instead of outputting $z = h^s(z_B \| L)$, the algorithm outputs $z_B \| L$.
 - (c) Instead of using an IV , just start the computation from x_1 . That is, define $z_1 := x_1$ and then compute $z_i := h^s(z_{i-1} \| x_i)$ for $i = 2, \dots, B+1$ and output z_{B+1} as before.
 - (d) Instead of using a fixed IV , set $z_0 := L$ and then compute $z_i := h^s(z_{i-1} \| x_i)$ for $i = 1, \dots, B$ and output z_B .
- 5.7 Assume collision-resistant hash functions exist. Show a construction of a fixed-length hash function (Gen, h) that is *not* collision resistant, but such that the hash function (Gen, H) obtained from the Merkle–Damgård transform to (Gen, h) as in Construction 5.3 *is* collision resistant.
- 5.8 Prove or disprove: if (Gen, h) is preimage resistant, then so is the hash function (Gen, H) obtained by applying the Merkle–Damgård transform to (Gen, h) as in Construction 5.3.

- 5.9 Prove or disprove: if (Gen, h) is second preimage resistant, then so is the hash function (Gen, H) obtained by applying the Merkle–Damgård transform to (Gen, h) as in Construction 5.3.
- 5.10 Before HMAC, it was common to define a MAC for arbitrary-length messages by $\text{Mac}_{s,k}(m) = H^s(k||m)$ where H is a collision-resistant hash function.
- (a) Show that this is never a secure MAC when H is constructed via the Merkle–Damgård transform. (Assume the hash key s is known to the attacker, and only k is kept secret.)
 - (b) Prove that this is a secure MAC if H is modeled as a random oracle.
- 5.11 Prove that the construction of a pseudorandom function given in Section 5.5.1 is secure in the random-oracle model.
- 5.12 Prove Theorem 5.11.
- 5.13 Show how to find a collision in the Merkle tree construction if t is not fixed. Specifically, show how to find two sets of inputs $x_1, \dots, x_t$ and $x'_1, \dots, x'_{2t}$ such that $\mathcal{MT}_t(x_1, \dots, x_t) = \mathcal{MT}_{2t}(x'_1, \dots, x'_{2t})$.
- 5.14 Consider the scenario introduced in Section 5.6.2 in which a client stores files on a server and wants to verify that files are returned unmodified.
- (a) Provide a formal definition of security for this setting.
 - (b) Formalize the construction based on Merkle trees as discussed in Section 5.6.2.
 - (c) Prove that your construction is secure relative to your definition under the assumption that (Gen_H, H) is collision resistant.
- 5.15 Prove that the commitment scheme discussed in Section 5.6.5 is secure in the random-oracle model.

Chapter 6

Practical Constructions of Symmetric-Key Primitives

In previous chapters we have demonstrated how secure encryption schemes and message authentication codes can be constructed from cryptographic primitives such as pseudorandom generators, pseudorandom permutations, and hash functions. One question we have not yet addressed, however, is how these cryptographic primitives are constructed in the first place, or even whether they exist at all! In the next chapter we will study this question from a theoretical vantage point, and show constructions of pseudorandom generators and pseudorandom permutations based on quite weak assumptions. (It turns out that hash functions are more difficult to construct, and appear to require stronger assumptions. We will see a provably secure construction of hash functions in Section 8.4.2.) In this chapter, our focus will be on comparatively heuristic, but far more efficient, constructions of these primitives that are widely used in practice.

As just mentioned, the constructions that we will explore in this chapter are heuristic in the sense that they cannot be proven secure based on any weaker assumption. These constructions are, however, based on a number of design principles, some of which can be justified by theoretical analysis. Perhaps more importantly, many of these constructions have withstood years of public scrutiny and attempted cryptanalysis and, given this, it is quite reasonable to assume the security of these constructions.

In some sense there is no fundamental difference between assuming, say, that factoring is hard and assuming that AES (a block cipher we will study in detail later in this chapter) is a pseudorandom permutation. There is, however, a significant *qualitative* difference between these assumptions.¹ The primary difference is that the former assumption is more believable since it seemingly relates to a weaker requirement: the assumption that large integers are hard to factor is arguably more natural than the assumption that AES with a uniform key is indistinguishable from a random permutation. Other relevant differences between the assumptions are that factoring has been studied much longer than the problem of distinguishing AES from a random permutation,

¹It should be clear that the discussion in this paragraph is informal, as we cannot formally argue about any of this when we cannot even prove that factoring is hard in the first place!

and was recognized as a hard problem well before the advent of cryptographic schemes based on it.

To summarize, it is reasonable to assume that the recommended constructions described in this chapter are secure, and people are comfortable relying on such assumptions in practice. Still, it would be preferable to base security of cryptographic primitives on weaker and more long-standing assumptions. As we will see in Chapter 7, this is (in principle) possible; unfortunately, the constructions we will see there are orders of magnitude less efficient than the constructions described here, and as such are not useful in practice.

The Aim of This Chapter

The main aims of this chapter are (1) to present some design principles used in the construction of modern cryptographic primitives, and (2) to introduce the reader to some popular constructions used extensively in the real world. We caution that:

- It is *not* the aim of this chapter to teach readers how to design new cryptographic primitives. On the contrary, we believe that the design of new primitives requires significant expertise and effort, and is not something to be attempted lightly. Those who are interested in developing additional expertise in this area are advised to read the more advanced references included at the end of the chapter.
- It is *not* our intent to present all the low-level details of the various primitives we discuss here, and our descriptions should not be relied upon for implementation. In fact, our descriptions are sometimes purposefully inaccurate, as we omit certain details that are not relevant to the broader conceptual point we are trying to emphasize.

6.1 Stream Ciphers

Recall from Section 3.3.1 that a stream cipher is defined by two deterministic algorithms (`Init`, `GetBits`). The `Init` algorithm takes as input a key k and an (optional) initialization vector IV and returns some initial state st . The `GetBits` algorithm can be used to generate an infinite stream of bits $y_1, y_2, \dots$ based on st . The main requirement of a stream cipher is that it should behave like a pseudorandom generator, namely, when k is chosen uniformly at random the resulting sequence $y_1, y_2, \dots$ should be indistinguishable from a sequence of uniform and independent bits by any computationally bounded attacker. (In Section 3.6.1 we noted that stream ciphers must sometimes satisfy stronger security requirements. We do not explicitly address this here.)

We have already pointed out (see the end of Section 3.5.1) that stream ciphers can be constructed easily from block ciphers, which are a stronger primitive. The primary motivation for using the dedicated stream-cipher constructions introduced in this section is efficiency, especially in resource-constrained environments (e.g., in hardware where there may be a desire to keep the number of gates small). Attacks have been shown against several recent constructions of stream ciphers, however, and their security appears much more tenuous than is the case for block ciphers. We therefore recommend using block ciphers (possibly in stream-cipher mode) when possible.

6.1.1 Linear-Feedback Shift Registers

We begin by discussing *linear-feedback shift registers* (LFSRs). These have been used historically for pseudorandom-number generation, as they are extremely efficient to implement in hardware, and generate output having good statistical properties. By themselves, however, they do *not* give cryptographically strong pseudorandom generators, and in fact we will show an easy key-recovery attack on LFSRs. Nevertheless, LFSRs can be used as a component in building stream ciphers with better security.

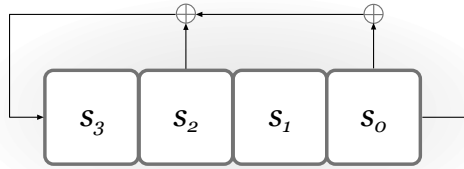


FIGURE 6.1: A linear-feedback shift register.

An LFSR consists of an array of n registers $s_{n-1}, \dots, s_0$ along with a feedback loop specified by a set of n *feedback coefficients* $c_{n-1}, \dots, c_0$. (See Figure 6.1.) The size of the array is called the *degree* of the LFSR. Each register stores a single bit, and the state $\mathbf{st}$ of the LFSR at any point in time is simply the set of bits contained in the registers. The state of the LFSR is updated in each of a series of “clock ticks” by shifting the values in all the registers to the right, and setting the new value of the left-most register equal to the XOR of some subset of the current registers, with the subset determined by the feedback coefficients. That is, if the state at some time t is $s_{n-1}^{(t)}, \dots, s_0^{(t)}$, then the state after the next clock tick is $s_{n-1}^{(t+1)}, \dots, s_0^{(t+1)}$ with

$$\begin{aligned} s_i^{(t+1)} &:= s_{i+1}^{(t)}, & i = 0, \dots, n-2 \\ s_{n-1}^{(t+1)} &:= \bigoplus_{i=0}^{n-1} c_i s_i^{(t)}. \end{aligned}$$

Figure 6.1 shows a degree-4 LFSR with $c_0 = c_2 = 1$ and $c_1 = c_3 = 0$.

At each clock tick, the LFSR outputs the value of the right-most register s_0 . If the initial state of the LFSR is $s_{n-1}^{(0)}, \dots, s_0^{(0)}$, the first n bits of the output stream are exactly $s_0^{(0)}, \dots, s_{n-1}^{(0)}$. The next output bit is $s_{n-1}^{(1)} = \bigoplus_{i=0}^{n-1} c_i s_i^{(0)}$. If we denote the output bits by $y_1, y_2, \dots$, where $y_i = s_0^{(i-1)}$, then

$$\begin{aligned} y_i &= s_{i-1}^{(0)}, & i &= 1, \dots, n \\ y_i &= \bigoplus_{j=0}^{n-1} c_j y_{i-n+j-1} & i &> n. \end{aligned}$$

As an example using the LFSR from Figure 6.1, if the initial state is $(0, 0, 1, 1)$ then the states for the first few time periods are

$$\begin{aligned} (0, 0, 1, 1) \\ (1, 0, 0, 1) \\ (1, 1, 0, 0) \\ (1, 1, 1, 0) \\ (1, 1, 1, 1) \end{aligned}$$

and the output (which can be read off the right-most column of the above) is the stream of bits $1, 1, 0, 0, 1, \dots$

The state of the LFSR consists of n bits; thus, the LFSR can cycle through at most 2^n possible states before repeating. When the states repeat the output bits repeat, and this means that the output sequence will begin repeating after at most 2^n output bits have been generated. A *maximum-length* LFSR cycles through all $2^n - 1$ nonzero states before repeating. (Note that if the all-0 state is ever realized then the LFSR remains in that state forever, which is why we exclude it.) Whether an LFSR is maximal length or not depends only on the feedback coefficients; if it is maximal length then, once it is initialized in any nonzero state, it will cycle through all $2^n - 1$ nonzero states. It is well understood how to set the feedback coefficients to obtain a maximal-length LFSR, although the details are beyond the scope of this book.

Reconstruction attacks. The output of a maximal-length LFSR of degree n has good statistical properties; for example, every n -bit string occurs with roughly equal frequency in the output stream of the LFSR. Nevertheless, LFSRs are not good pseudorandom generators for cryptographic purposes because their output is predictable. This follows from the fact that an attacker can reconstruct the entire state of a degree- n LFSR after observing at most $2n$ output bits. To see this, assume both the initial state and the feedback coefficients of some LFSR are unknown. The first n output bits $y_1, \dots, y_n$ of the LFSR exactly reveal the initial state. Given the next n output bits $y_{n+1}, \dots, y_{2n}$, the attacker can set up a system of n linear equations in the n

unknowns $c_0, \dots, c_{n-1}$:

$$\begin{aligned} y_{n+1} &= c_{n-1} y_n \oplus \dots \oplus c_0 y_1 \\ &\vdots \\ y_{2n} &= c_{n-1} y_{2n-1} \oplus \dots \oplus c_0 y_n. \end{aligned}$$

One can show that the above equations are linearly independent (modulo 2) for a maximal-length LFSR, and so uniquely determine the feedback coefficients. (The solution can be found efficiently using linear algebra.) With the feedback coefficients known, all subsequent output bits of the LFSR can be easily computed.

6.1.2 Adding Nonlinearity

The linear relationships between the outputs bits of an LFSR are exactly what enable an easy attack. To thwart such attacks, we must introduce some nonlinearity, i.e., some operations other than XORs. There are several different approaches to doing so, and we only explore some of them here.

Nonlinear feedback. One obvious way to modify LFSRs is to make the feedback loop nonlinear. A nonlinear-feedback shift register (FSR) will again consist of an array of registers, each containing a single bit. As before, the state of the FSR is updated in each of a series of clock ticks by shifting the values in all the registers to the right; now, however, the new value of the left-most register is a *nonlinear* function of the current registers. In other words, if the state at some time t is $s_0^{(t)}, \dots, s_{n-1}^{(t)}$ then the state after the next clock tick is $s_0^{(t+1)}, \dots, s_{n-1}^{(t+1)}$ with

$$\begin{aligned} s_i^{(t+1)} &:= s_{i+1}^{(t)}, & i &= 0, \dots, n-2 \\ s_{n-1}^{(t+1)} &:= g(s_0^{(t)}, \dots, s_{n-1}^{(t)}) \end{aligned}$$

for some nonlinear function g . As before, the FSR outputs the value of the right-most register s_0 at each clock tick.

It is possible to design nonlinear FSRs with maximal length and such that the output has good statistical properties.

Nonlinear combination generators. Another approach is to introduce nonlinearity in the output sequence. In the most basic case, we could have an LFSR as before (where the new value of the left-most register is again computed as a linear function of the current registers), but where the output at each clock tick is a nonlinear function g of all the current registers, rather than just the right-most register. It is important here that g be *balanced* in the sense that $\Pr[g(s_0, \dots, s_{n-1}) = 1] \approx 1/2$ (where the probability is over uniform choice of $s_0, \dots, s_{n-1}$); otherwise, although it might be difficult to

reconstruct the entire state of the LFSR based on the output, the output stream will be biased and hence easily distinguishable from uniform.

A variant of the above is to use several LFSRs (with each individual output stream computed, as before, by simply taking the value of the right-most register of each LFSR), and to generate the actual output stream by combining the output of the individual LFSRs in some nonlinear way. This yields what is known as a (nonlinear) *combination generator*. The individual LFSRs need not have the same degree, and in fact the cycle length of the combination generator will be maximized if they do *not* have the same degree. Here, care must be taken to ensure that the output stream of the combination generator is not too highly correlated with any of the output streams of the individual LFSRs; high correlation can lead to attacks on the individual LFSRs, thereby defeating the purpose of using several LFSRs in the construction.

6.1.3 Trivium

To illustrate the ideas from the previous section, we briefly describe the stream cipher *Trivium*. This stream cipher was selected as part of the portfolio of the eSTREAM project, a European effort completed in 2008 whose goal was to identify new stream ciphers. Trivium was designed to have a simple description and to admit a compact hardware implementation.

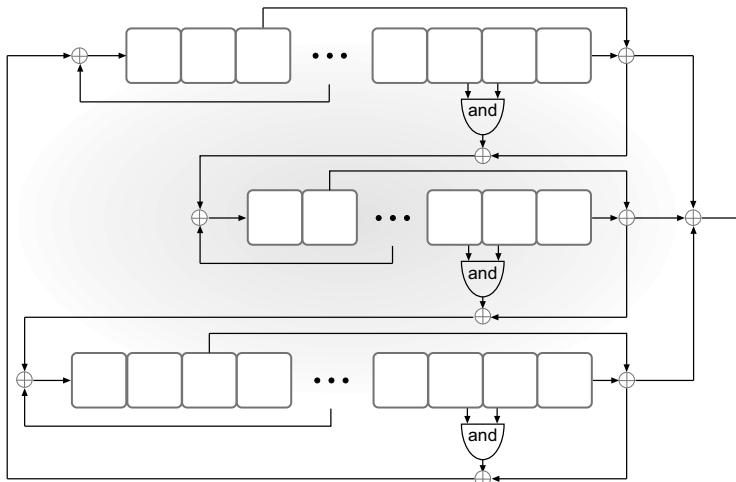


FIGURE 6.2: A schematic illustration of Trivium with (from top to bottom) three coupled, nonlinear FSRs A , B , and C .

Trivium uses three coupled, nonlinear FSRs denoted by A , B , and C and having degree 93, 84, and 111, respectively. (See Figure 6.2.) The state $\mathbf{st}$ of Trivium is simply the 288 bits comprising the values in all the registers

of these FSRs. The **GetBits** algorithm for Trivium works as follows: At each clock tick, the output of each FSR is the XOR of its right-most register and one additional register; the output of Trivium is the XOR of the output bits of the three FSRs. The FSRs are *coupled*: at each clock tick, the new value of the left-most register of each FSR is computed as a function of one of the registers in the same FSR and a subset of the registers from a second FSR. (The feedback function for A depends on one register of A and four registers of C ; the feedback function for B depends on one register of B and four registers of A ; and the feedback function for C depends on one register of C and four registers of B .) The feedback function in each case is nonlinear.

The **Init** algorithm of Trivium accepts an 80-bit key and an 80-bit IV . The key is loaded into the 80 left-most registers of A , and the IV is loaded into the 80 left-most registers of B . The remaining registers are set to 0, except for the three right-most registers of C , which are set to 1. Then **GetBits** is run $4 \cdot 288$ times (with output discarded), and the resulting state is taken as st_0 .

To date, no cryptanalytic attacks better than exhaustive key search are known against the full Trivium cipher.

6.1.4 RC4

LFSRs are efficient when implemented in hardware, but have poor performance in software. For this reason, alternate designs of stream ciphers have been explored. A prominent example is RC4, which was designed by Ron Rivest in 1987. RC4 is remarkable for its speed and simplicity, and resisted serious attack for several years. It is widely used today, and we discuss it for this reason; we caution the reader, however, that recent attacks have shown serious cryptographic weaknesses in RC4 and it should no longer be used.

ALGORITHM 6.1

Init algorithm for RC4

Input: 16-byte key k

Output: Initial state (S, i, j)

(Note: All addition is done modulo 256)

for $i = 0$ to 255:

$S[i] := i$

$k[i] := k[i \bmod 16]$

$j := 0$

for $i = 0$ to 255:

$j := j + S[i] + k[i]$

Swap $S[i]$ and $S[j]$

$i := 0, j := 0$

return (S, i, j)

The state of RC4 is a 256-byte array S , which always contains a permutation of the elements $0, \dots, 255$, along with two values $i, j \in \{0, \dots, 255\}$. The **Init**

algorithm for RC4 is presented as Algorithm 6.1. For simplicity we assume a 16-byte (128-bit) key k , although the algorithm can handle keys between 1 byte and 256 bytes long. We index the bytes of S as $S[0], \dots, S[255]$, and the bytes of the key as $k[0], \dots, k[15]$.

During initialization, S is first set to the identity permutation (i.e., with $S[i] = i$ for all i) and k is expanded to 256 bytes by repetition. Then each entry of S is swapped at least once with another entry of S in a “pseudorandom” location. The indices i, j are set to 0, and (S, i, j) is output as the initial state.

The state is then used to generate a sequence of output bits, as shown in Algorithm 6.2. The index i is simply incremented (modulo 256), and j is changed in some “pseudorandom” way. Entries $S[i]$ and $S[j]$ are swapped, and the value of S at position $S[i] + S[j]$ (again computed modulo 256) is output. Note that each entry of S is swapped with some other entry of S (possibly itself) at least once every 256 iterations, ensuring good “mixing” of the permutation S .

ALGORITHM 6.2

GetBits algorithm for RC4

Input: Current state (S, i, j)

Output: Output byte y ; updated state (S, i, j)

(Note: All addition is done modulo 256)

$i := i + 1$

$j := j + S[i]$

Swap $S[i]$ and $S[j]$

$t := S[i] + S[j]$

$y := S[t]$

return $(S, i, j), y$

RC4 was not designed to take an IV as input; however, in practice an IV is often incorporated by simply concatenating it with the actual key k' before initialization. That is, a random IV of the desired length is chosen, k is set to be equal to the concatenation of IV and k' (this can be done by either prepending or appending IV), and then `Init` is run as in Algorithm 6.1 to generate an initial state. Output bits are then produced using Algorithm 6.2 exactly as before. Assuming RC4 is being used in unsynchronized mode (see Section 3.6.1), the IV would then be sent in the clear to the receiver—who presumably already has the actual key k' —thus enabling them to generate the same initial state and hence the same output stream. This method of incorporating an IV is used in the *Wired Equivalent Privacy* (WEP) encryption standard for protecting communications in 802.11 wireless networks.

One should be concerned by this relatively ad hoc way of modifying RC4 to accept an IV . Even if RC4 were a secure stream cipher when using (only) a key as originally designed, there is no reason to believe that it should be

secure when modified to use an *IV* in this way. Indeed, contrary to the key, the *IV* is revealed to an attacker (since it is sent in the clear); furthermore, using different *IV*s with the same fixed key k' —as would be done when using RC4 in unsynchronized mode—means that *related* values k are being used to initialize the state of RC4. As we will see below, both of these issues lead to attacks when RC4 is used in this fashion.

Attacks on RC4. Although RC4 is pervasive in modern systems, various attacks on RC4 have been known for several years. Due to this, RC4 should no longer be used; instead, a more modern stream cipher or block cipher should be used in its place.

We begin by demonstrating a simple statistical attack on RC4 that does not rely on the honest parties' using an *IV*. The attack exploits the fact that the second output byte of RC4 is (slightly) *biased* toward 0. Let S_t denote the state of the array S after t iterations of `GetBits`, with S_0 denoting the initial state. Treating S_0 (heuristically) as a uniform permutation of $\{0, \dots, 255\}$, with probability $1/256 \cdot (1 - 1/255) \approx 1/256$, it holds that $S_0[2] = 0$ and $X \stackrel{\text{def}}{=} S_0[1] \neq 2$. Assume for a moment that this is the case. In the first iteration of `GetBits`, the value of i is incremented to 1, and j is set equal to $S_0[i] = S_0[1] = X$. Then $S_0[1]$ and $S_0[X]$ are swapped, so that at the end of the iteration we have $S_1[X] = S_0[1] = X$. In the second iteration, i is incremented to 2 and j is assigned the value

$$j + S_1[i] = X + S_1[2] = X + S_0[2] = X,$$

since $S_0[2] = 0$. Then $S_1[2]$ and $S_1[X]$ are swapped, so that $S_2[X] = S_1[2] = S_0[2] = 0$ and $S_2[2] = S_1[X] = X$. Finally, the value of S_2 at position $S_2[i] + S_2[j] = S_2[2] + S_2[X] = X$ is output; this is exactly the value $S_2[X] = 0$.

When $S_0[2] \neq 0$ the second output byte is uniformly distributed. Overall, then, the probability that the second output byte is 0 is

$$\begin{aligned} \Pr[S_0[2] = 0 \text{ and } S_0[1] \neq 2] + \frac{1}{256} \cdot \left(1 - \Pr[S_0[2] = 0 \text{ and } S_0[1] \neq 2]\right) \\ = \frac{1}{256} + \frac{1}{256} \cdot \left(1 - \frac{1}{256}\right) \approx \frac{2}{256}, \end{aligned}$$

or twice what would be expected for a uniform value.

By itself the above might not be viewed as a particularly serious attack, although it does seem to indicate underlying structural problems with RC4. A more serious attack against RC4 is possible when an *IV* is incorporated by prepending it to the key. This attack can be used to recover the key, regardless of its length, and is thus more serious than a distinguishing attack such as the one described above. Importantly, this attack can be used to completely break the WEP encryption standard mentioned earlier, and was influential in getting the standard replaced.

The core of the attack is a way to extend knowledge of the first n bytes of k to knowledge of the first $(n + 1)$ bytes of k . Note that when an *IV* is

prepended to the actual key k' (so $k = IV \| k'$), the first few bytes of k are given to the attacker for free! If the IV is n bytes long, then the adversary can use this attack to first recover the $(n + 1)$ st byte of k (which is the first byte of the real key k'), then the next byte of k , and so on, until it deduces the entire key.

Assume the IV is 3 bytes long, as is the case for WEP. The attacker waits until the first two bytes of the IV have a specific form. The attack can be carried out with several possibilities for the first two bytes of the IV , but we look at the case where the IV takes the form $IV = (3, 255, X)$ for X an arbitrary byte. This means, of course, that $k[0] = 3$, $k[1] = 255$, and $k[2] = X$ in Algorithm 6.1. One can check that after the first four iterations of the second loop of `Init`, we have

$$S[0] = 3, \quad S[1] = 0, \quad S[3] = X + 6 + k[3]. \quad (6.1)$$

In the next 252 iterations of the `Init` algorithm, i is always greater than 3. So the values of $S[0]$, $S[1]$, and $S[3]$ are not subsequently modified as long as j never takes on the values 0, 1, or 3. If we (heuristically) treat j as taking on a uniform value in each iteration, this means that $S[0]$, $S[1]$, and $S[3]$ are not subsequently modified with probability $(253/256)^{252} \approx 0.05$, or 5% of the time. Assuming this is the case, the first byte output by `GetBits` will be $S[3] = X + 6 + k[3]$; since X is known, this reveals $k[3]$.

So, the attacker knows that 5% of the time the first byte of the output is correlated with $k[3]$ as described above. (This is much better than random guessing, which is correct $1/256 = 0.4\%$ of the time.) Thus, by collecting sufficiently many samples of the first byte of the output—for several IV s having the correct form—the attacker gets a high-confidence estimate for $k[3]$.

6.2 Block Ciphers

Recall from Section 3.5.1 that a block cipher is an efficient, keyed permutation $F : \{0, 1\}^n \times \{0, 1\}^\ell \rightarrow \{0, 1\}^\ell$. This means the function F_k defined by $F_k(x) \stackrel{\text{def}}{=} F(k, x)$ is a bijection (i.e., a permutation), and moreover F_k and its inverse F_k^{-1} are efficiently computable given k . We refer to n as the *key length* and ℓ as the *block length* of F , and here we explicitly allow them to differ. The key length and block length are now fixed constants, whereas in Chapter 3 they were viewed as functions of a security parameter. This puts us in the setting of concrete security rather than asymptotic security.² The concrete

²Although a block cipher with fixed key length has no “security parameter” to speak of, we still view security as depending on the length of the key and thus denote this value by n .

security requirements for block ciphers are quite stringent, and a block cipher is generally only considered “good” if the best known attack (without pre-processing) has time complexity roughly equivalent to a brute-force search for the key. Thus, if a cipher with key length $n = 256$ can be broken in time 2^{128} , the cipher is (generally) considered insecure even though a 2^{128} -time attack is still infeasible. In contrast, in an asymptotic setting an attack of complexity $2^{n/2}$ is not considered efficient since it requires exponential time (and thus a cipher where such an attack is possible might still satisfy the definition of being a pseudorandom permutation). In the concrete setting, however, we must worry about the actual complexity of the attack (rather than its asymptotic behavior). Furthermore, there is a concern that existence of such an attack may indicate some more fundamental weakness in the design of the cipher.

Block ciphers are designed to behave, at a minimum, as (strong) pseudorandom permutations; see Definition 3.28. Modeling block ciphers as pseudorandom permutations allows proofs of security for constructions based on block ciphers, and also makes explicit the necessary requirements of a block cipher. A solid understanding of what block ciphers are supposed to achieve is instrumental in their design. The view that block ciphers should be modeled as pseudorandom permutations has, at least in the recent past, served as a major influence in their design. As an example, the call for proposals for the recent Advanced Encryption Standard (AES) that we will encounter later in this chapter stated the following evaluation criterion:

The security provided by an algorithm is the most important factor. . . . Algorithms will be judged on the following factors. . .

- *The extent to which the algorithm output is indistinguishable from a random permutation . . .*

Modern block ciphers are suitable for all the constructions using pseudorandom permutations (or pseudorandom functions) we have seen in this book.

Often, block ciphers are designed (and assumed) to satisfy even stronger security properties, as we discuss briefly in Section 6.3.1.

Notwithstanding the fact that block ciphers are not, on their own, encryption schemes, the standard terminology for attacks on a block cipher F is:

- In a *known-plaintext attack*, the attacker is given pairs of inputs/outputs $\{(x_i, F_k(x_i))\}$ (for an unknown key k), with the $\{x_i\}$ outside the attacker’s control.
- In a *chosen-plaintext attack*, the attacker is given $\{F_k(x_i)\}$ (again, for an unknown key k) for a series of inputs $\{x_i\}$ chosen by the attacker.
- In a *chosen-ciphertext attack*, the attacker is given $\{F_k(x_i)\}$ for $\{x_i\}$ chosen by the attacker, as well as $\{F_k^{-1}(y_i)\}$ for chosen $\{y_i\}$.

Viewing the key length as a parameter makes sense when comparing block ciphers with different key lengths, or when using a block cipher that supports keys of different lengths.

Besides using the above to distinguish F_k from a uniform permutation, we will also be interested in *key-recovery attacks* in which the attacker is able to recover the key k after interacting with F_k . (This is stronger than being able to distinguish F_k from uniform.)

With respect to this taxonomy, a pseudorandom permutation cannot be distinguished from a uniform permutation under a chosen-plaintext attack, while a strong pseudorandom permutation cannot be so distinguished even under a chosen-ciphertext attack.

6.2.1 Substitution-Permutation Networks

A block cipher must behave like a random permutation. There are $2^\ell!$ permutations on ℓ -bit strings, so representing an arbitrary permutation with an ℓ -bit block length requires $\log(2^\ell!) \approx \ell \cdot 2^\ell$ bits. This is impractical for $\ell > 20$ and infeasible for $\ell > 50$. (Looking ahead, modern block ciphers have block lengths $\ell \geq 128$.) The challenge when designing a block cipher is to construct a set of permutations with a *concise* description (namely, a short key) that behaves like a random permutation. In particular, just as evaluating a random permutation at two inputs that differ in only a single bit should yield two (almost) independent outputs (they are not completely independent since they cannot be equal), so too changing one bit of the input to $F_k(\cdot)$, where k is uniform and unknown to an attacker, should yield an (almost) independent result. This implies that a one-bit change in the input should “affect” every bit of the output. (Note that this does not mean that all the output bits will be changed—that would be different behavior than one would expect for a random permutation. Rather, we just mean informally that each bit of the output is changed with probability roughly half.) This takes some work to achieve.

The confusion-diffusion paradigm. In addition to his work on perfect secrecy, Shannon also introduced a basic paradigm for constructing concise, random-looking permutations. The basic idea is to construct a random-looking permutation F with a large block length from many smaller random (or random-looking) permutations $\{f_i\}$ with small block length. Let us see how this works on the most basic level. Say we want F to have a block length of 128 bits. We can define F as follows: the key k for F will specify 16 permutations $f_1, \dots, f_{16}$ that each have an 8-bit (1-byte) block length.³ Given an input $x \in \{0, 1\}^{128}$, we parse it as 16 bytes $x_1 \cdots x_{16}$ and then set

$$F_k(x) = f_1(x_1) \parallel \cdots \parallel f_{16}(x_{16}). \quad (6.2)$$

These round functions $\{f_i\}$ are said to introduce *confusion* into F .

³An arbitrary permutation on 8 bits can be represented using $\log(2^8!) \approx 1600$ bits, so the length of the key for F is about $16 \cdot 1600$ bits, or 3 kbytes. This is much smaller than the $\approx 128 \cdot 2^{128}$ bits that would be required to specify an arbitrary permutation on 128 bits.

It should be immediately clear, however, that F as defined above will *not* be pseudorandom. Specifically, if x and x' differ only in their first bit then $F_k(x)$ and $F_k(x')$ will differ only in their first byte (regardless of the key k). In contrast, if F were a truly random permutation then changing the first bit of the input would be expected to affect all bytes of the output.

For this reason, a *diffusion* step is introduced whereby the bits of the output are permuted, or “mixed,” using a *mixing permutation*. This has the effect of spreading a local change (e.g., a change in the first byte) throughout the entire block. The confusion/diffusion steps—together called a *round*—are repeated multiple times. This helps ensure that changing a single bit of the input will affect all the bits of the output.

As an example, a two-round block cipher following this approach would operate as follows. First, confusion is introduced by computing the intermediate result $f_1(x_1) \parallel \cdots \parallel f_{16}(x_{16})$ as in Equation (6.2). The bits of the result are then “shuffled,” or re-ordered, to give x' . Then $f'_1(x'_1) \parallel \cdots \parallel f'_{16}(x'_{16})$ is computed (where $x' = x'_1 \cdots x'_{16}$), using possibly different functions f'_i , and the bits of the result are permuted to give output x'' . The $\{f_i\}$, $\{f'_i\}$, and the mixing permutation(s) could be random and dependent on the key, as we have described above. In practice, however, they are specially designed and fixed, and the key is incorporated in a different way, as we will describe below.

Substitution-permutation networks. A substitution-permutation network (SPN) can be viewed as a direct implementation of the confusion-diffusion paradigm. The difference is that now the round functions have a particular form rather than being chosen from the set of all possible permutations on some domain. Specifically, rather than having (a portion of) the key k specify an arbitrary permutation f , we instead fix a public “substitution function” (i.e., permutation) S called an *S-box*, and then let k define the function f given by $f(x) = S(k \oplus x)$.

To see how this works concretely, consider an SPN with a 64-bit block length based on a collection of 8-bit (1-byte) *S*-boxes $S_1, \dots, S_8$. (See Figure 6.3.) Evaluating the cipher proceeds in a series of rounds, where in each round we apply the following sequence of operations to the 64-bit input x of that round (the input to the first round is just the input to the cipher):

1. *Key mixing:* Set $x := x \oplus k$, where k is the current-round sub-key;
2. *Substitution:* Set $x := S_1(x_1) \parallel \cdots \parallel S_8(x_8)$, where x_i is the i th byte of x ;
3. *Permutation:* Permute the bits of x to obtain the output of the round.

The output of each round is fed as input to the next round. After the last round there is a *final key-mixing step*, and the result is the output of the cipher. (By Kerckhoffs’ principle, we assume the *S*-boxes and the mixing permutation(s) are public and known to any attacker. This means that without a final key-mixing step, the last substitution and permutation steps would offer no additional security since they do not depend on the key.) Figure 6.4 shows

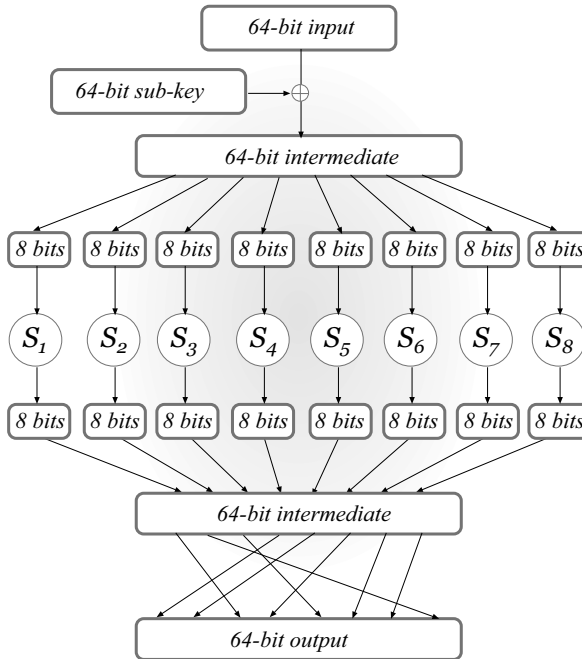


FIGURE 6.3: A single round of a substitution-permutation network.

the high-level structure of an SPN with a 16-bit block length and a different set of 4-bit S -boxes used in each round.

Different *sub-keys* (or *round keys*) are used in each round. The actual key of the block cipher is sometimes called the *master key*. The round sub-keys are derived from the master key according to a *key schedule*. The key schedule is often simple and may work by just taking different subsets of the bits of the master key, although more complex schedules can also be defined. An r -round SPN has r (full) rounds of key mixing, S -box substitution, and application of a mixing permutation, followed by a final key-mixing step. (This means that in an r -round SPN, $r + 1$ sub-keys are used.)

Any SPN is invertible (given the key). To see this, we show that given the output of the SPN and the key it is possible to recover the input. It suffices to show that a single round can be inverted; this implies the entire SPN can be inverted by working from the final round back to the beginning. But inverting a single round is easy: the mixing permutation can easily be inverted since it is just a re-ordering of bits. Since the S -boxes are permutations (i.e., one-to-one), these too can be inverted. The result can then be XORed with the appropriate sub-key to obtain the original input. Therefore:

PROPOSITION 6.3 *Let F be a keyed function defined by an SPN in which the S -boxes are all permutations. Then regardless of the key schedule and the number of rounds, F_k is a permutation for any k .*

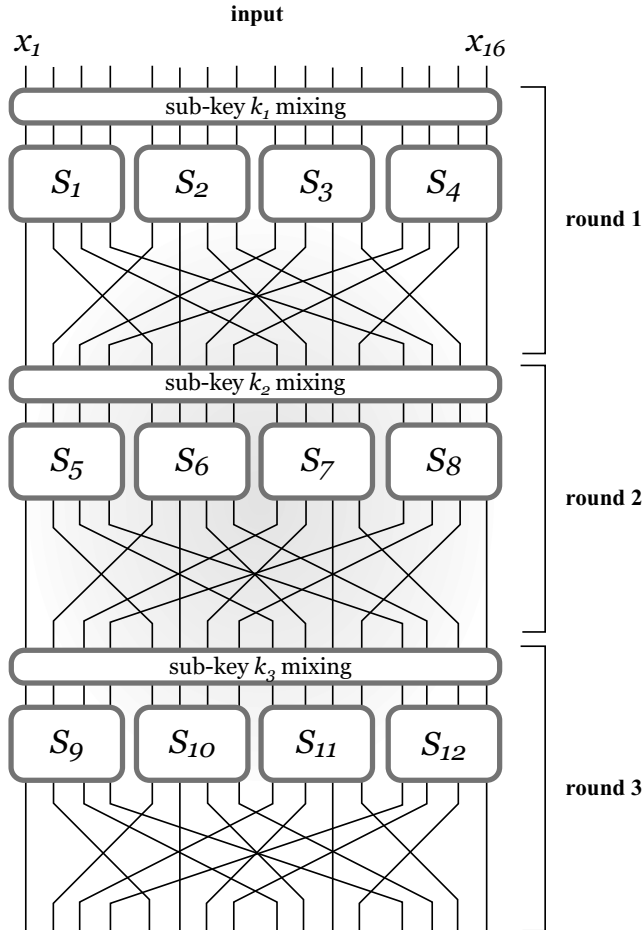


FIGURE 6.4: A substitution-permutation network.

The number of rounds, along with the exact choices of the S -boxes, mixing permutations, and key schedule, are what ultimately determine whether a given block cipher is trivially breakable or highly secure. We now discuss a basic principle behind the design of the S -boxes and mixing permutations.

The avalanche effect. As noted repeatedly, an important property in any block cipher is that a small change in the input must “affect” every bit of the output. We refer to this as the *avalanche effect*. One way to induce the avalanche effect in a substitution-permutation network is to ensure that the following two properties hold (and sufficiently many rounds are used):

1. The S -boxes are designed so that changing a single bit of the input to an S -box changes at least *two bits* in the output of the S -box.
2. The mixing permutations are designed so that the output bits of any given S -box are used as input to *multiple* S -boxes in the next round.

To see how this yields the avalanche effect, at least heuristically, assume that the S -boxes are all such that changing a single bit of the input of the S -box results in a change in exactly two bits of the output of the S -box, and that the mixing permutations are chosen as required above. For concreteness, assume the S -boxes have input/output size of 8 bits, and that the block length of the cipher is 128 bits. Consider now what happens when the block cipher is applied to two inputs that differ in a single bit:

1. After the first round, the intermediate values differ in exactly two bit-positions. This is because XORing the current sub-key maintains the 1-bit difference in the intermediate values, and so the inputs to all the S -boxes except one are identical. In the one S -box where the inputs differ, the output of the S -box causes a 2-bit difference. The mixing permutation applied to the results changes the positions of these differences, but maintains a 2-bit difference.
2. The mixing permutation applied at the end of the first round spreads the two bit-positions where the intermediate results differ into two *different* S -boxes in the second round. This remains true even after the appropriate sub-key is XORed with the result of the previous round. So, in the second round there are now *two* S -boxes that receive inputs differing by a single bit. Thus, at the end of the second round the intermediate values differ in 4 bits.
3. Continuing the same argument, we expect 8 bits of the intermediate value to be affected after the 3rd round, 16 bits to be affected after the 4th round, and all 128 bits of the output to be affected at the end of the 7th round.

The last point is not quite precise and it is certainly possible that there will be fewer differences than expected at the end of some round. (In fact, this must be the case because the outputs should not differ in all their bits, either.) For this reason, it is customary to use many more than 7 rounds. However, the above analysis gives a *lower bound* on the number of rounds: if fewer than 7 rounds are used then there must be some set of output bits that are not affected by a single-bit change in the input, implying that it will be possible to distinguish the cipher from a random permutation.

One might expect that the “best” way to design S -boxes would be to choose them at random (subject to the restriction that they are permutations). Interestingly, this turns out not to be the case, at least if we want to satisfy the design criterion mentioned earlier. Consider the case of an S -box operating on 4-bit inputs and let x and x' be two distinct values. Let $y = S(x)$, and now consider choosing uniform $y' \neq y$ as the value of $S(x')$. There are 4 strings that differ from y in only 1 bit, and so with probability $4/15$ we will choose y' that does *not* differ from y in two or more bits. The problem is compounded when we consider all pairs of inputs that differ in a single bit.

We conclude based on this example that, as a general rule, the S -boxes must be designed carefully rather than being chosen blindly at random. Random S -boxes are also not good for defending against attacks like the ones we will show in Section 6.2.6.

If a block cipher should also be *strongly* pseudorandom, then the avalanche effect must also apply to its *inverse*. That is, changing a single bit of the output should affect every bit of the input. For this it is useful if the S -boxes are designed so that changing a single bit of the output of an S -box changes at least two bits of the input to the S -box. Achieving the avalanche effect in both directions is another reason for further increasing the number of rounds.

Attacking Reduced-Round SPNs

Experience, along with many years of cryptanalytic effort, indicate that substitution-permutation networks are a good choice for constructing pseudorandom permutations as long as care is taken in the choice of the S -boxes, the mixing permutations, and the key schedule. The Advanced Encryption Standard, described in Section 6.2.5, is similar in structure to the substitution-permutation network described above, and is widely believed to be a strong pseudorandom permutation.

The strength of a cipher F constructed in this way depends heavily on the number of rounds. In order to obtain more of an insight into substitution-permutation networks, we will demonstrate attacks on SPNs having very few rounds. These attacks are straightforward, but are worth seeing as they demonstrate conclusively why a large number of rounds is needed.

A trivial case. We first consider a trivial case where F consists of one full round and no final key-mixing step. We show that an adversary given only a *single* input/output pair (x, y) can easily learn the secret key k for which $y = F_k(x)$. The adversary begins with the output value y and then inverts the mixing permutation and the S -boxes. It can do this, as noted before, because the full specification of the mixing permutation and the S -boxes is public. The intermediate value that the adversary computes is exactly $x \oplus k$ (assuming, without loss of generality, that the master key is used as the sub-key in the only round of the network). Since the adversary also knows the input x , it can immediately derive the secret key k . This is therefore a complete break.

Although this is a trivial attack, it demonstrates that in any substitution-permutation network there is no security gained by performing S -box substitution or applying a mixing permutation after the final sub-key mixing.

Attacking a one-round SPN. Now we have one full round followed by a key-mixing step. For concreteness, we assume a 64-bit block length and S -boxes with 8-bit (1-byte) input/output length. We assume independent 64-bit sub-keys k_1, k_2 are used for the two key-mixing steps, and so the master key $k_1 || k_2$ of the SPN is 128 bits long.

A first observation is that we can extend the attack from the trivial case above to give a key-recovery attack here using much less than 2^{128} work. The idea is as follows: Given a single input/output pair (x, y) as before, the attacker enumerates over all possible values for the second-round sub-key k_2 . For each such value, the attacker can invert the final key-mixing step to get a candidate intermediate value y' . We have seen above that given an input x and an output y' of a (full) SPN round, a unique possible sub-key k_1 can be easily identified. Thus, for each possible choice of k_2 the attacker derives a unique corresponding k_1 for which $k_1 \| k_2$ can be the master key. In this way, the attacker obtains (in 2^{64} time) a list of 2^{64} possibilities for the master key. These can be narrowed down using additional input/output pairs in roughly 2^{64} additional time; see also below.

A better attack is possible by noting that individual bits of the output depend on only part of the master key. Fix some given input/output pair (x, y) as before. Now, the adversary will enumerate over all possible values for the *first byte* of k_2 . It can XOR each such value with the first byte of y to obtain a candidate value for the output of the first S -box. Inverting this S -box, the attacker learns a candidate value for the *input* to that S -box. Since the input to that S -box is the XOR of 8 bits of x and 8 bits of k_1 (where the positions of those bits depend on the first-round mixing permutation and are known to the attacker), this yields a candidate value for 8 bits of k_1 .

To summarize: for each candidate value of the first byte of k_2 , there is a *unique* possible corresponding value for some 8 bits of k_1 . Put differently, this means that for some 16 bits of the master key, the attacker has reduced the number of possible values for those bits from 2^{16} to 2^8 . The attacker can tabulate all those feasible values in 2^8 time. This can be repeated for each byte of k_2 , giving 8 lists—each containing 2^8 values—that together characterize the possible values of the entire master key. The attacker has thus reduced the number of possible master keys to $(2^8)^8 = 2^{64}$, as in the earlier attack. The total time to do this, however, is now $8 \cdot 2^8 = 2^{11}$, a dramatic improvement.

The attacker can use additional input/output pairs to further reduce the space of possible keys. Consider the list of 2^8 feasible values for some set of 16 bits of the master key. The attacker knows that the correct value from that list must be consistent with any additional input/output pairs the attacker learns. Heuristically, any *incorrect* value from the list is consistent with some additional input/output pair (x', y') with probability no better than random guessing; since each 16-bit value from the table can be used to compute 1 byte of the output given the input x' , we expect that an incorrect value will be consistent with the actual output y' with probability 2^{-8} . A small number of additional input/output pairs will thus suffice to narrow down *all* the tables to just a single value each, at which point the entire master key is known.

There is an important lesson to be learned here. The attack is possible since different parts of the key can be isolated from other parts. Thus, further *diffusion* is needed to make sure that all the bits of the key affect all of the bits of the output. Multiple rounds are needed for this to take place.

Attacking a two-round SPN. It is possible to extend the above ideas to give a better-than-brute-force attack on a two-round SPN using independent sub-keys in each round; we leave this as an exercise.

Instead, we simply note that a two-round SPN will not be a good pseudo-random permutation. Here we rely on the fact, mentioned earlier, that the avalanche effect does not occur after only two rounds (of course, this depends on the block length of the cipher and the input/output length of the S -boxes, but with reasonable parameters this will be the case). An attacker can distinguish a two-round SPN from a uniform permutation if it learns the result of evaluating the SPN on two inputs that differ in a single bit: in a two-round SPN many bits of the two outputs will be the same, something not expected to occur for a random permutation.)

6.2.2 Feistel Networks

Feistel networks offer another approach for constructing block ciphers. An advantage of Feistel networks over substitution-permutation networks is that the underlying functions used in a Feistel network—in contrast to the S -boxes used in SPNs—need not be invertible. *A Feistel network thus gives a way to construct an invertible function from non-invertible components.* This is important because a good block cipher should have “unstructured” behavior (so it looks random), yet requiring all the components of a construction to be invertible inherently introduces structure. Requiring invertibility also introduces an additional constraint on S -boxes, making them harder to design.

A Feistel network operates in a series of rounds. In each round, a keyed *round function* is applied in the manner described below. Round functions need not be invertible. They will typically be constructed from components like S -boxes and mixing permutations, but a Feistel network can deal with *any* round functions irrespective of their design.

In a balanced Feistel network (the only type we will consider), the i th round function $\hat{f}_i$ takes as input a sub-key k_i and an $\ell/2$ -bit string and outputs an $\ell/2$ -bit string. As in the case of SPNs, a master key k is used to derive sub-keys for each round. When some master key is chosen, thereby determining each sub-key k_i , we define $f_i : \{0, 1\}^{\ell/2} \rightarrow \{0, 1\}^{\ell/2}$ via $f_i(R) \stackrel{\text{def}}{=} \hat{f}_i(k_i, R)$. Note that the round functions $\hat{f}_i$ are fixed and publicly known, but the f_i depend on the master key and so are not known to the attacker.

The i th round of a Feistel network operates as follows. The input to the round is divided into two halves denoted L_{i-1} and R_{i-1} (the “left” and “right” halves, respectively). If the block length of the cipher is ℓ bits, then L_{i-1} and R_{i-1} each has length $\ell/2$. The output (L_i, R_i) of the round is

$$L_i := R_{i-1} \quad \text{and} \quad R_i := L_{i-1} \oplus f_i(R_{i-1}). \quad (6.3)$$

In an r -round Feistel network, the ℓ -bit input to the network is parsed as (L_0, R_0) , and the output is the ℓ -bit value (L_r, R_r) obtained after applying all r rounds. A three-round Feistel network is shown in Figure 6.5.

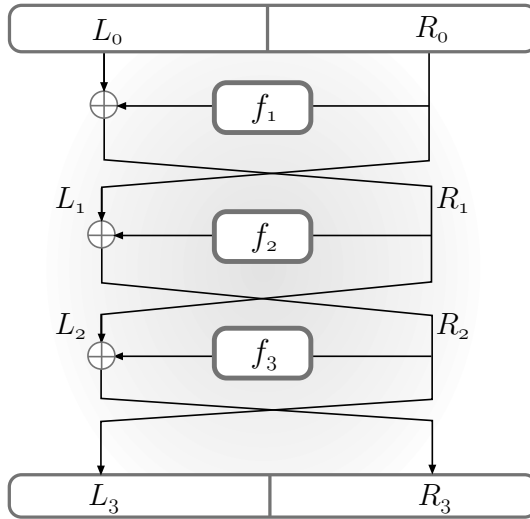


FIGURE 6.5: A three-round Feistel network.

Inverting a Feistel network. A Feistel network is invertible *regardless of the* $\{f_i\}$ (and thus regardless of the round functions $\{\hat{f}_i\}$). To show this we need only show that each round of the network can be inverted if the $\{f_i\}$ are known. Given the output (L_i, R_i) of the i th round, we can compute (L_{i-1}, R_{i-1}) as follows: first set $R_{i-1} := L_i$. Then compute

$$L_{i-1} := R_i \oplus f_i(R_{i-1}).$$

This gives the value (L_{i-1}, R_{i-1}) that was the input of this round (i.e., it computes the inverse of Equation (6.3)). Note that f_i is evaluated only in the forward direction, so it need not be invertible. We thus have:

PROPOSITION 6.4 *Let F be a keyed function defined by a Feistel network. Then regardless of the round functions $\{\hat{f}_i\}$ and the number of rounds, F_k is an efficiently invertible permutation for all k .*

As in the case of substitution-permutation networks, attacks on Feistel networks are possible when the number of rounds is too low. We will see such attacks when we discuss DES in the next section. Theoretical results concerning the security of Feistel networks are discussed in Section 7.6.

6.2.3 DES – The Data Encryption Standard

The Data Encryption Standard, or DES, was developed in the 1970s by IBM (with help from the National Security Agency) and adopted in 1977 as

a Federal Information Processing Standard for the US. In its basic form, DES is no longer considered secure due to its short key length of 56 bits, which makes it vulnerable to brute-force attacks. Nevertheless, it remains in wide use today in the strengthened form of triple-DES, described in Section 6.2.4.

DES is of great historical significance. It has undergone intensive scrutiny within the cryptographic community, arguably more than any other cryptographic algorithm in history. The common consensus is that, apart from its key length, DES is an extremely well-designed cipher. Indeed, even after many years, the best known attack on DES *in practice* is an exhaustive search over all 2^{56} possible keys. (As we will see, there are important theoretical attacks on DES that require less computation; however, these attacks assume certain conditions that seem difficult to realize in practice.)

In this section, we provide a high-level overview of the main components of DES. We stress that we will not provide a full specification that is correct in every detail, and some parts of the design will be omitted from our description. Our aim is to present the basic ideas underlying the construction of DES, and not all the low-level details; the reader interested in such details can consult the references at the end of this chapter.

The Design of DES

The DES block cipher is a 16-round Feistel network with a block length of 64 bits and a key length of 56 bits. The same round function $\hat{f}$ is used in each of the 16 rounds. The round function takes a 48-bit sub-key and, as expected for a (balanced) Feistel network, a 32-bit input (namely, half a block). The *key schedule* of DES is used to derive a sequence of 48-bit sub-keys $k_1, \dots, k_{16}$ from the 56-bit master key. The key schedule of DES is relatively simple, with each sub-key k_i being a permuted subset of 48 bits of the master key. For our purposes, it suffices to note that the 56 bits of the master key are divided into two halves—a “left half” and a “right half”—each containing 28 bits. (This division occurs after an initial permutation is applied to the key, but we ignore this in our description.) In each round, the left-most 24 bits of the sub-key are taken as some subset of the 28 bits in the left half of the master key, and the right-most 24 bits of the round sub-key are taken as some subset of the 28 bits in the right half of the master key. We stress that the entire key schedule (including the manner in which bits are divided into the left and right halves, and which bits are used in forming each sub-key k_i) is fixed and public, and the only secret is the master key itself.

The DES round function. The DES round function $\hat{f}$ —sometimes called the *DES mangler function*—is constructed using a paradigm we have previously analyzed: it is (essentially) just a substitution-permutation network! In more detail, computation of $\hat{f}(k_i, R)$ with $k_i \in \{0, 1\}^{48}$ and $R \in \{0, 1\}^{32}$ proceeds as follows: first, R is *expanded* to a 48-bit value R' . This is carried out by simply duplicating half the bits of R ; we denote this by $R' := E(R)$ where

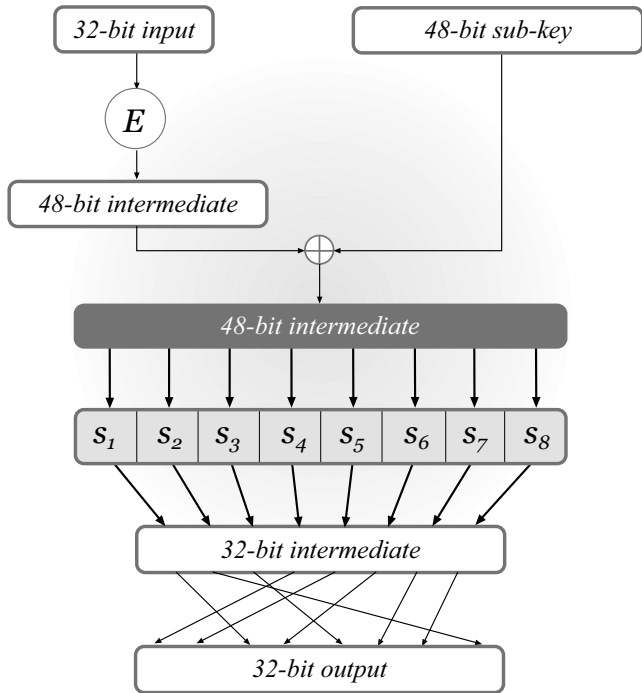


FIGURE 6.6: The DES mangler function.

E is called the *expansion function*. Following this, computation proceeds exactly as in our earlier discussion of SPNs: The expanded value R' is XORed with k_i , which is also 48 bits long, and the resulting value is divided into 8 blocks, each of which is 6 bits long. Each block is passed through a (different) S -box that takes a 6-bit input and yields a 4-bit output; concatenating the output from the 8 S -boxes gives a 32-bit result. A mixing permutation is then applied to the bits of this result to obtain the final output. See Figure 6.6.

One difference as compared to our original discussion of SPNs is that the S -boxes here are *not* invertible; indeed, they cannot be invertible since their inputs are longer than their outputs. Further discussion regarding the structural details of the S -boxes is given below.

We stress once again that everything in the above description (including the S -boxes themselves as well as the mixing permutation) is *publicly known*. The only secret is the master key which is used to derive all the sub-keys.

The S -boxes and the mixing permutation. The eight S -boxes that form the “core” of $\hat{f}$ are a crucial element of the DES construction and were very carefully designed. Studies of DES have shown that if the S -boxes were slightly modified, DES would have been much more vulnerable to attack.

This should serve as a warning to anyone who wishes to design a block cipher: seemingly arbitrary choices are not arbitrary at all, and if not made correctly may render the entire construction insecure.

Recall that each S -box maps a 6-bit input to a 4-bit output. Each S -box can be viewed as a table with 4 rows and 16 columns, where each cell of the table contains a 4-bit entry. A 6-bit input can be viewed as indexing one of the $2^6 = 64 = 4 \times 16$ cells of the table in the following way: The first and last input bits are used to choose the table row, and bits 2–5 are used to choose the table column. The 4-bit entry at some position of the table represents the output value for the input associated with that position.

The DES S -boxes have the following properties (among others):

1. Each S -box is a 4-to-1 function. (That is, exactly 4 inputs are mapped to each possible output.) This follows from the properties below.
2. Each row in the table contains each of the 16 possible 4-bit strings exactly once.
3. Changing *one bit* of any input to an S -box always changes at least *two bits* of the output.

The mixing permutation was also designed carefully. In particular it has the property that the four output bits from any S box will affect the input to six S -boxes in the next round. (This is possible because of the expansion function that is applied in the next round before the S -boxes are computed.)

The DES avalanche effect. The design of the mangler function ensures that DES exhibits a strong avalanche effect. In order to see this, we will trace the difference between the intermediate values in a DES computation of two inputs that differ by just a single bit. Let us denote the two inputs to the cipher by (L_0, R_0) and (L'_0, R'_0) , where we assume that $R_0 = R'_0$ and so the single-bit difference occurs in the left half of the inputs (it may help to refer to Equation (6.3) and Figure 6.6 in what follows). After the first round the intermediate values (L_1, R_1) and (L'_1, R'_1) still differ by only a single bit, although now this difference is in the right half. In the second round of DES, the right half of each input is run through $\hat{f}$. Assuming that the bit where R_1 and R'_1 differ is not duplicated in the expansion step, the intermediate values before applying the S -boxes still differ by only a single bit. By property 3 of the S -boxes, the intermediate values *after* the S -box computation differ in at least *two* bits. The result is that the intermediate values (L_2, R_2) and (L'_2, R'_2) differ in *three* bits: there is a 1-bit difference between L_2 and L'_2 (carried over from the difference between R_1 and R'_1) and a 2-bit difference between R_2 and R'_2 .

The mixing permutation spreads the two-bit difference between R_2 and R'_2 such that, in the following round, each of the two bits is used as input to a *different* S -box, resulting in a difference of at least 4 bits in the right halves of the intermediate values. (If either or both of the two bits in which R_2 and R'_2

differ are duplicated by E , the difference may be even greater.) There is also now a 2-bit difference in the left halves. As with a substitution-permutation network, we have an exponential effect and so after 7 rounds we expect all 32 bits in the right half to be affected (and after 8 rounds all 32 bits in the left half will be affected as well).

DES has 16 rounds, and so the avalanche effect occurs very early in the computation. This ensures that the computation of DES on similar inputs yields independent-looking outputs.

Attacks on Reduced-Round DES

A useful exercise for understanding more about the DES construction and its security is to look at the behavior of DES with only a few rounds. We will show attacks on one-, two-, and three-round variants of DES (recall that the real DES has 16 rounds). DES with three rounds or fewer cannot be a pseudorandom function because three rounds are not enough for the avalanche effect to occur. Thus, we will be interested in demonstrating more difficult (and more damaging) key-recovery attacks which compute the key k using only a relatively small number of input/output pairs computed using that key. Some of the attacks are similar to those we have seen in the context of substitution-permutation networks; here, however, we will see how they are applied to a concrete block cipher rather than to an abstract design.

The attacks below will be known-plaintext attacks in which the adversary knows some plaintext/ciphertext pairs $\{(x_i, y_i)\}$ with $y_i = DES_k(x_i)$ for some secret key k . When we describe the attacks, we will focus on a particular input/output pair (x, y) and will describe the information about the key that the adversary can derive from this pair. Continuing to use the notation developed earlier, we denote the left and right halves of the input x as L_0 and R_0 , respectively, and let L_i, R_i denote the left and right halves after the i th round. Recall that E denotes the DES expansion function, k_i denotes the sub-key used in round i , and $f_i(R) = \hat{f}(k_i, R)$ denotes the actual function being applied in the Feistel network in the i th round.

One-round DES. Say we are given an input/output pair (x, y) . In one-round DES, we have $y = (L_1, R_1)$, where $L_1 = R_0$ and $R_1 = L_0 \oplus f_1(R_0)$. We therefore know an input/output pair for f_1 ; specifically, we know that $f_1(R_0) = R_1 \oplus L_0$. By applying the inverse of the mixing permutation to the output $R_1 \oplus L_0$, we obtain the intermediate value consisting of the outputs from all the S -boxes, where the first 4 bits are the output from the first S -box, the next 4 bits are the output from the second S -box, and so on.

Consider the (known) 4-bit output of the first S -box. Since each S -box is a 4-to-1 function, this means there are exactly four possible inputs to this S -box that would result in the given output, and similarly for all the other S -boxes; each such input is 6 bits long. The input to the S -boxes is simply the XOR of $E(R_0)$ with the sub-key k_1 . Since R_0 , and hence $E(R_0)$, is known, we can

compute a set of four possible values for each 6-bit portion of k_1 . This means we have reduced the number of possible keys k_1 from 2^{48} to $4^{48/6} = 4^8 = 2^{16}$ (since there are four possibilities for each of the eight 6-bit portions of k_1). This is already a small number and so we can just try all the possibilities on a different input/output pair (x', y') to find the right key. We thus obtain the key using only two known plaintexts in time roughly 2^{16} .

Two-round DES. In two-round DES, the output y is equal to (L_2, R_2) where

$$\begin{aligned} L_1 &= R_0 \\ R_1 &= L_0 \oplus f_1(R_0) \\ L_2 &= R_1 = L_0 \oplus f_1(R_0) \\ R_2 &= L_1 \oplus f_2(R_1). \end{aligned}$$

L_0, R_0, L_2 , and R_2 are known from the given input/output pair (x, y) , and thus we also know $L_1 = R_0$ and $R_1 = L_2$. This means that we know the input/output of both f_1 and f_2 , and so the same method used in the attack on one-round DES can be used here to determine both k_1 and k_2 in time roughly $2 \cdot 2^{16}$. This attack works even if k_1 and k_2 are completely independent keys, although in fact the key schedule of DES ensures that many of the bits of k_1 and k_2 are equal (which can be used to further speed up the attack).

Three-round DES. Referring to Figure 6.5, the output value y is now equal to (L_3, R_3) . Since $L_1 = R_0$ and $R_2 = L_3$, the only unknown values in the figure are R_1 and L_2 (which are equal).

Now we no longer have the input/output to any round function f_i . For example, the output value of f_2 is equal to $L_1 \oplus R_2$, where both of these values are known. However, we do *not* know the value R_1 that is input to f_2 . Similarly, we can determine the inputs to f_1 and f_3 but not the outputs of those functions. Thus, the attack we used to break one-round and two-round DES will not work here.

Instead of relying on full knowledge of the input and output of one of the round functions, we will use knowledge of a certain relation between the inputs and outputs of f_1 and f_3 . Observe that the output of f_1 is equal to $L_0 \oplus R_1 = L_0 \oplus L_2$, and the output of f_3 is equal to $L_2 \oplus R_3$. Therefore,

$$f_1(R_0) \oplus f_3(R_2) = (L_0 \oplus L_2) \oplus (L_2 \oplus R_3) = L_0 \oplus R_3,$$

where both L_0 and R_3 are known. That is, *the XOR of the outputs of f_1 and f_3 is known*. Furthermore, the input to f_1 is R_0 and the input to f_3 is L_3 , both of which are known. We conclude that we can determine the inputs to f_1 and f_3 , and the XOR of their outputs. We now describe an attack that finds the secret key based on this information.

Recall that the key schedule of DES has the property that the master key is divided into a “left half,” which we denote by k_L , and a “right half” k_R , each containing 28 bits. Furthermore, the 24 left-most bits of the sub-key used in

each round are taken only from k_L , and the 24 right-most bits of each sub-key are taken only from k_R . This means that k_L affects only the inputs to the first four S -boxes in any round, while k_R affects only the inputs to the last four S -boxes. Since the mixing permutation is known, we also know which bits of the output of each round function come out of each S -box.

The idea behind the attack is to separately traverse the key space for each half of the master key, giving an attack with complexity roughly $2 \cdot 2^{28}$ rather than complexity 2^{56} . Such an attack will be possible if we can verify a guess of half the master key, and we now show how this can be done. Say we guess some value for k_L , the left half of the master key. We know the input R_0 of f_1 , and so using our guess of k_L we can compute the input to the first four S -boxes. This means that we can compute half the output bits of f_1 (the mixing permutation spreads out the bits we know, but since the mixing permutation is known we know exactly which bits these are). Likewise, we can compute the same locations in the output of f_3 by using the known input L_3 to f_3 and the same guess for k_L . Finally, we can compute the XOR of these output values and check whether they match the appropriate bits in the known value of the XOR of the outputs of f_1 and f_3 . If they are not equal, then our guess for k_L is incorrect. A correct guess for k_L will always pass this test, and so will not be eliminated, but an incorrect guess is expected to pass this test only with probability roughly 2^{-16} (since we check equality of 16 bits in two computed values). There are 2^{28} possible values for k_L , so if each incorrect value remains a viable candidate with probability 2^{-16} then we expect to be left with only $2^{28} \cdot 2^{-16} = 2^{12}$ possibilities for k_L after the above.

By performing the above for each half of the master key, we obtain in time $2 \cdot 2^{28}$ approximately 2^{12} candidates for the left half and 2^{12} candidates for the right half. Since each combination of the left half and right half is possible, we have 2^{24} candidate keys overall and can run a brute-force search over this set using an additional input/output pair (x', y') . (An alternative approach which is more efficient is to simply repeat the previous attack using the 2^{12} remaining candidates for each half-key.) The time complexity of the attack is roughly $2 \cdot 2^{28} + 2^{24} < 2^{30}$, which is much less than 2^{56} .

Security of DES

After almost 30 years of intensive study, the best known practical attack on DES is still an exhaustive search through its key space. (We discuss some important theoretical attacks in Section 6.2.6. These attacks require a large number of input/output pairs, which would be difficult to obtain in an attack on any real-world system using DES.) Unfortunately, the 56-bit key length of DES is short enough that an exhaustive search through all 2^{56} possible keys is now feasible. Already in the late 1970s there were strong objections to the choice of such a short key for DES. Back then, the objection was theoretical, as the computational power needed to search through that many

keys was generally unavailable.⁴ The practicality of a brute-force attack on DES, however, was demonstrated in 1997 when the first of a set of DES challenges set up by RSA Security was solved by the DESCHALL project using thousands of computers coordinated across the Internet; the computation took 96 days. A second challenge was broken the following year in just 41 days by the **distributed.net** project. A significant breakthrough came in 1998 when the third challenge was solved in just *56 hours*. This impressive feat was achieved via a special-purpose DES-breaking machine called *Deep Crack* that was built by the Electronic Frontier Foundation at a cost of \$250,000. In 1999, a DES challenge was solved in just over 22 hours as a combined effort of Deep Crack and **distributed.net**. The current state-of-the-art is the DES cracking box by PICO Computing, which uses 48 FPGAs and can find a DES key in approximately 23 hours.

The time/space tradeoffs discussed in Section 5.4.3 show that exhaustive key-search attacks can be accelerated using pre-computation and additional memory. Due to the short key length of DES, time/space tradeoffs can be especially effective. Specifically, using pre-processing it is possible to generate a table a few terabytes large that then enables recovery of a DES key with high probability from a single input/output pair using approximately 2^{38} DES evaluations (which can be computed in mere minutes). The bottom line is that DES has a key that is far too short, and cannot be considered secure for any serious application today.

A secondary cause for concern is the relatively short block length of DES. A short block length is problematic because the concrete security of many constructions based on block ciphers depends on the block length—even if the cipher used is “perfect.” For example, the proof of security for CTR mode (cf. Theorem 3.32) shows that even when a completely random function is used an attacker can break the security of this encryption scheme with probability $2q^2/2^\ell$ if it obtains q plaintext/ciphertext pairs. In the case of DES where $\ell = 64$, this means that if an attacker obtains only $q = 2^{30}$ plaintext/ciphertext pairs, security is compromised with high probability. Obtaining plaintext/ciphertext pairs is relatively easy if an adversary eavesdrops on the encryption of messages containing known headers, redundancies, etc.

The insecurity of DES has nothing to do with its design *per se*, but rather is due to its short key length (and, to a lesser extent, its short block length). This is a great tribute to the designers of DES, who seem to have succeeded in constructing an almost “perfect” block cipher (besides its too-short key). Since DES itself seems not to have significant structural weaknesses, it makes sense to use DES as a building block for constructing block ciphers with longer keys. We discuss this further in Section 6.2.4.

⁴In 1977, it was estimated that a computer that could crack DES in one day would cost \$20 million to build.

The replacement for DES—the *Advanced Encryption Standard* (AES), covered later in this chapter—was explicitly designed to address concerns regarding the short key length and block length of DES. AES supports 128-, 192-, or 256-bit keys, and a block length of 128 bits.

Better-than-brute-force attacks on DES were first shown in the early 1990s by Biham and Shamir, who developed a technique called *differential cryptanalysis*. Their attack takes time 2^{37} and requires 2^{47} chosen plaintexts. While the attack was a breakthrough from a theoretical standpoint, it does not appear to be of much practical concern since it is hard to imagine a realistic scenario where an adversary can obtain this many encryptions of chosen plaintexts.

Interestingly, the work of Biham and Shamir indicated that the DES *S*-boxes had been specifically designed to be resistant to differential cryptanalysis, suggesting that the technique of differential cryptanalysis was known (but not publicly revealed) by the designers of DES. After Biham and Shamir announced their result, this suspicion was confirmed.

Linear cryptanalysis was developed by Matsui in the mid-1990s and was also applied successfully to DES. The advantage of this attack is that it uses *known* plaintexts rather than *chosen* plaintexts. Nevertheless, the number of plaintext/ciphertext pairs required—about 2^{43} —is still huge.

We briefly describe differential and linear cryptanalysis in Section 6.2.6.

6.2.4 3DES: Increasing the Key Length of a Block Cipher

The main weakness of DES is its short key. It thus makes sense to try to design a block cipher with a larger key length using DES as a building block. Some approaches to doing so are discussed in this section. Although we refer to DES frequently throughout the discussion, and DES is the most prominent block cipher to which these techniques have been applied, everything we say here applies generically to *any* block cipher.

Internal modifications vs. “black-box” constructions. There are two general approaches one could take to constructing another cipher based on DES. The first approach would be to somehow modify the *internal structure* of DES, while increasing the key length. For example, one could leave the round function untouched and simply use a 128-bit master key with a different key schedule (still choosing a 48-bit sub-key in each round). Or, one could change the *S*-boxes themselves and use a larger sub-key in each round. The disadvantage of such approaches is that by modifying DES—in even the smallest way—we lose the confidence we have gained in DES by virtue of the fact that it has remained resistant to attack for so many years. Cryptographic constructions are very sensitive, and even mild, seemingly insignificant changes can render a construction completely insecure.⁵ Changing the internals of a block cipher is therefore not recommended.

⁵In fact, various results to this effect have been shown for DES; e.g., changing the *S*-boxes or the mixing permutation can make DES much more vulnerable to attack.

An alternative approach that does not suffer from the above problem is to use DES as a “black box” and not touch its internal structure at all. In this approach we treat DES as a “perfect” block cipher with a 56-bit key, and construct a new block cipher that only invokes the original, unmodified DES. Since DES itself is not tampered with, this is a much more prudent approach, and is the one we will pursue here.

Double Encryption

Let F be a block cipher with an n -bit key length and ℓ -bit block length. Then a new block cipher F' with a key of length $2n$ can be defined by

$$F'_{k_1, k_2}(x) \stackrel{\text{def}}{=} F_{k_2}(F_{k_1}(x)),$$

where k_1 and k_2 are independent keys. For the case where F is DES, we obtain a cipher F' called 2DES that takes a 112-bit key; if exhaustive key search were the best available attack, a key length of 112 bits would be sufficient since an attack requiring time 2^{112} is completely out of reach. Unfortunately, we now show an attack on F' that runs in time roughly 2^n , significantly less than the 2^{2n} time one would hope would be necessary to carry out an exhaustive search for a $2n$ -bit key. This means that the new block cipher is essentially no better than the old one, even though it has a key that is twice as long.⁶

The attack is called a “meet-in-the-middle attack,” for reasons that will soon become clear. Say the adversary is given a single input/output pair (x, y) , where $y = F'_{k_1^*, k_2^*}(x) = F_{k_2^*}(F_{k_1^*}(x))$ for unknown k_1^*, k_2^* . The adversary can narrow down the set of possible keys in the following way:

1. For each $k_1 \in \{0, 1\}^n$, compute $z := F_{k_1}(x)$ and store (z, k_1) in a list L .
2. For each $k_2 \in \{0, 1\}^n$, compute $z := F_{k_2}^{-1}(y)$ and store (z, k_2) in a list L' .
3. Entries $(z_1, k_1) \in L$ and $(z_2, k_2) \in L'$ are a *match* if $z_1 = z_2$. For each such match, add (k_1, k_2) to a set S . (Matches can be found easily after sorting L and L' by their first components.)

See Figure 6.7 for a graphical depiction of the attack.

The attack takes time $\mathcal{O}(n \cdot 2^n)$, and requires space $\mathcal{O}((n + \ell) \cdot 2^n)$. The set S output by this algorithm contains exactly those values (k_1, k_2) for which

$$F_{k_1}(x) = F_{k_2}^{-1}(y) \tag{6.4}$$

or, equivalently, for which $y = F'_{k_1, k_2}(x)$. In particular, $(k_1^*, k_2^*) \in S$. On the other hand, a pair $(k_1, k_2) \neq (k_1^*, k_2^*)$ is (heuristically) expected to satisfy Equation (6.4) with probability $2^{-\ell}$ if we treat $F_{k_1}(x)$ and $F_{k_2}^{-1}(y)$ as uniform

⁶This is not quite true since a brute-force attack on F can be carried out in time 2^n and constant memory, whereas the attack we show on F' requires 2^n time and 2^n memory. Nevertheless, the attack illustrates that F' does not achieve the desired level of security.

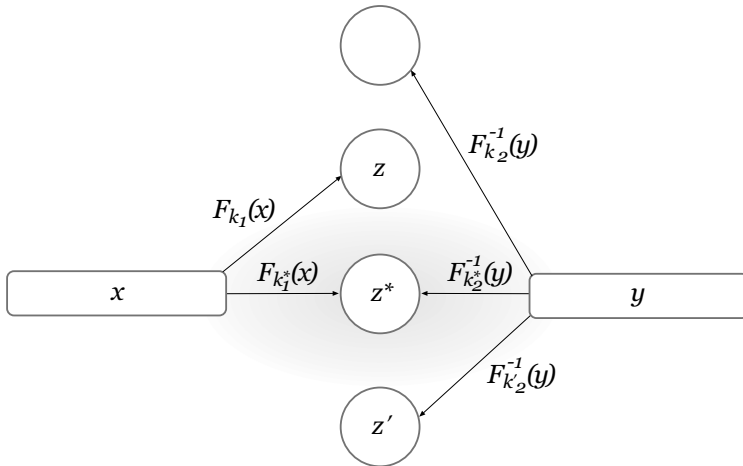


FIGURE 6.7: A meet-in-the-middle attack.

ℓ -bit strings, and so the expected size of S is $2^{2n} \cdot 2^{-\ell} = 2^{2n-\ell}$. Using another few input/output pairs, and taking the intersection of the sets that are obtained, the correct (k_1^*, k_2^*) can be identified with very high probability.

Triple Encryption

The obvious generalization of the preceding approach is to apply the block cipher *three* times in succession. Two variants of this approach are common:

Variant 1: three keys. Choose three independent keys k_1, k_2, k_3 and define

$$F''_{k_1, k_2, k_3}(x) \stackrel{\text{def}}{=} F_{k_3}(F_{k_2}^{-1}(F_{k_1}(x))).$$

Variant 2: two keys. Choose two independent keys k_1, k_2 and then define

$$F''_{k_1, k_2}(x) \stackrel{\text{def}}{=} F_{k_1}(F_{k_2}^{-1}(F_{k_1}(x))).$$

Before comparing the security of the two alternatives we note that the middle invocation of F is reversed. If F is a sufficiently good cipher this makes no difference as far as security is concerned, since if F is a strong pseudorandom permutation then F^{-1} must be too. The reason for reversing the second application of F is to obtain backward compatibility: if one sets $k_1 = k_2 = k_3$, the resulting function is equivalent to a single invocation of F using key k_1 .

Security of the first variant. The key length of the first variant is $3n$, and so we might hope that the best attack on this cipher would require time 2^{3n} . However, the cipher is susceptible to a meet-in-the-middle attack just as in the case of double encryption, though the attack now takes time 2^{2n} . This is the best known attack. Thus, although this variant is not as secure as one might have hoped, it obtains sufficient security for all practical purposes even for $n = 56$ (assuming, of course, the original cipher F has no weaknesses).

Security of the second variant. The key length of this variant is $2n$ and so the best we can hope for is security against attacks running in time 2^{2n} . There is no known attack with better time complexity when the adversary is given only a small number of input/output pairs. (See Exercise 6.13 for an attack using 2^n chosen plaintexts.) Thus, two-key triple encryption is a reasonable choice in practice.

Triple-DES (3DES). Triple-DES (or 3DES) is based on a triple invocation of DES using two or three keys, as described above. 3DES was standardized in 1999, and is widely used today. Its main drawbacks are its relatively small block length and the fact that it is relatively slow since it requires 3 full block-cipher operations. Since the minimum recommended key length nowadays is 128 bits, 2-key 3DES is no longer recommended (due to its key length of only 112 bits). These drawbacks have led to the replacement of DES/triple-DES by the Advanced Encryption Standard, presented in the next section.

6.2.5 AES – The Advanced Encryption Standard

In January 1997, the United States National Institute of Standards and Technology (NIST) announced that it would hold a competition to select a new block cipher—to be called the *Advanced Encryption Standard*, or AES—to replace DES. The competition began with an open call for teams to submit candidate block ciphers for evaluation. A total of 15 different algorithms were submitted from all over the world, including contributions from many of the best cryptographers and cryptanalysts. Each team’s candidate cipher was intensively analyzed by members of NIST, the public, and (especially) the other teams. Two workshops were held, one in 1998 and one in 1999, to discuss and analyze the various submissions. Following the second workshop, NIST narrowed the field down to 5 “finalists” and the second round of the competition began. A third AES workshop was held in April 2000, inviting additional scrutiny on the five finalists. In October 2000, NIST announced that the winning algorithm was Rijndael (a block cipher designed by the Belgian cryptographers Vincent Rijmen and Joan Daemen), although NIST conceded that any of the 5 finalists would have made an excellent choice. In particular, no serious security vulnerabilities were found in any of the 5 finalists, and the selection of a “winner” was based in part on properties such as efficiency, performance in hardware, flexibility, etc.

The process of selecting AES was ingenious because any group that submitted an algorithm, and was therefore interested in having its algorithm adopted, had strong motivation to find attacks on the other submissions. In this way, the world’s best cryptanalysts focused their attention on finding even the slightest weaknesses in the candidate ciphers submitted to the competition. After only a few years each candidate algorithm was already subjected to intensive study, thus increasing our confidence in the security of the winning algorithm. Of course, the longer the algorithm is used and studied without

being broken, the more our confidence will continue to grow. Today, AES is widely used and no significant security weaknesses have been discovered.

The AES construction. In this section, we present the high-level structure of Rijndael/AES. (Technically speaking, Rijndael and AES are not the same thing but the differences are unimportant for our discussion here.) As with DES, we will not present a full specification and our description should not be used as a basis for implementation. Our aim is only to provide a general idea of how the algorithm works.

The AES block cipher has a 128-bit block length and can use 128-, 192-, or 256-bit keys. The length of the key affects the key schedule (i.e., the sub-key that is used in each round) as well as the number of rounds, but does not affect the high-level structure of each round.

In contrast to DES, which uses a Feistel structure, AES is essentially a substitution-permutation network. During computation of the AES algorithm, a 4-by-4 array of bytes called the *state* is modified in a series of rounds. The state is initially set equal to the input to the cipher (note that the input is 128 bits, which is exactly 16 bytes). The following operations are then applied to the state in a series of four stages during each round:

Stage 1 – AddRoundKey: In every round of AES, a 128-bit sub-key is derived from the master key, and is interpreted as a 4-by-4 array of bytes. The state array is updated by XORing it with this sub-key.

Stage 2 – SubBytes: In this step, each byte of the state array is replaced by another byte according to a single fixed lookup table S . This substitution table (or S -box) is a bijection over $\{0, 1\}^8$.

Stage 3 – ShiftRows: In this step, the bytes in each row of the state array are shifted to the left as follows: the first row of the array is untouched, the second row is shifted one place to the left, the third row is shifted two places to the left, and the fourth row is shifted three places to the left. All shifts are cyclic so that, e.g., in the second row the first byte becomes the fourth byte.

Stage 4 – MixColumns: In this step, an invertible transformation is applied to the four bytes in each column. (Technically speaking, this is a linear transformation—i.e., matrix multiplication—over an appropriate field.) This transformation has the property that if two inputs differ in $b > 0$ bytes, then applying the transformation yields two outputs differing in at least $5 - b$ bytes.

In the final round, MixColumns is replaced with AddRoundKey. This prevents an adversary from simply inverting the last three stages, which do not depend on the key.

By viewing stages 3 and 4 together as a “mixing” step, we see that each round of AES has the structure of a substitution-permutation network: the

round sub-key is first XORed with the input to the current round; next, a small, invertible function is applied to “chunks” of the resulting value; finally, the bits of the result are mixed in order to obtain diffusion. The only difference is that, unlike our previous description of substitution-permutation networks, here the mixing step does not consist of a simple shuffling of the bits but is instead carried out using a shuffling plus an invertible linear transformation. (Simplifying things a bit and looking at a trivial 3-bit example, shuffling the bits of $x = x_1\|x_2\|x_3$ might, e.g., map x to $x' = x_2\|x_1\|x_3$. An invertible linear transformation might map x to $x_1 \oplus x_2\|x_2 \oplus x_3\|x_1 \oplus x_2 \oplus x_3$.)

The number of rounds depends on the key length. Ten rounds are used for a 128-bit key, 12 rounds for a 192-bit key, and 14 rounds for a 256-bit key.

Security of AES. As we have mentioned, the AES cipher was subject to intense scrutiny during the selection process and has continued to be studied ever since. To date, there are no practical cryptanalytic attacks that are significantly better than an exhaustive search for the key.

We conclude that, as of today, AES constitutes an excellent choice for any cryptographic scheme that requires a (strong) pseudorandom permutation. It is free, standardized, efficient, and highly secure.

6.2.6 *Differential and Linear Cryptanalysis

Block ciphers are relatively complicated, and as such are difficult to analyze. Nevertheless, one should not be fooled into thinking that a complicated cipher is difficult to break. On the contrary, it is very hard to construct a secure block cipher and surprisingly easy to find attacks on most constructions (no matter how complicated they appear). This should serve as a warning that non-experts should not try to construct new ciphers. Given the availability of triple-DES and AES, it is hard to justify using anything else.

In this section we describe two tools that are now a standard part of the cryptanalyst’s toolbox. Our goal here is give a taste of some advanced cryptanalysis, as well as to reinforce the idea that designing a secure block cipher involves careful choice of its components.

Differential cryptanalysis. This technique, which can lead to a chosen-plaintext attack on a block cipher, was first presented in the late 1980s by Biham and Shamir, who used it to attack DES in 1993. The basic idea behind the attack is to tabulate *specific differences in the input* that lead to *specific differences in the output* with probability greater than would be expected for a random permutation. Specifically, say *the differential* (Δ_x, Δ_y) *occurs in some keyed permutation* F' *with probability* p if for uniform inputs x_1 and x_2 satisfying $x_1 \oplus x_2 = \Delta_x$, and uniform choice of key k , the probability that $F'_k(x_1) \oplus F'_k(x_2) = \Delta_y$ is p . For any fixed (Δ_x, Δ_y) and x_1, x_2 satisfying $x_1 \oplus x_2 = \Delta_x$, if we choose a uniform function $f : \{0, 1\}^\ell \rightarrow \{0, 1\}^\ell$, we have $\Pr[f(x_1) \oplus f(x_2) = \Delta_y] = 2^{-\ell}$. In a weak block cipher, however, there may

be differentials that occur with significantly higher probability. This can be leveraged to give a full key-recovery attack, as we now show for SPNs.

We describe the basic idea, and then work through a concrete example. Let F be a block cipher with ℓ -bit block length which is an r -round SPN, and let $F'_k(x)$ denote the intermediate result in a computation of $F_k(x)$ after applying the key-mixing step of round r . (That is, F' excludes the S -box substitution and mixing permutation of the last round, as well as the final key-mixing step.) Say there is a differential (Δ_x, Δ_y) in F' that occurs with probability $p \gg 2^{-\ell}$. It is possible to exploit this high-probability differential to learn bits of the final mixing sub-key k_{r+1} . The high-level idea is as follows: let $\{(x_1^i, x_2^i)\}_{i=1}^L$ be a collection of L pairs of random inputs with differential Δ_x , i.e., with $x_1^i \oplus x_2^i = \Delta_x$ for all i . Using a chosen-plaintext attack, obtain the values $y_1^i = F_k(x_1^i)$ and $y_2^i = F_k(x_2^i)$ for all i . Now, for all possible bitstrings $k^* \in \{0, 1\}^\ell$, compute $\tilde{y}_1^i = F'_k(x_1^i)$ and $\tilde{y}_2^i = F'_k(x_2^i)$, assuming the value of the final sub-key k_{r+1} is k^* . This is done by inverting the final key-mixing step using k^* , and then inverting the mixing permutation and S -boxes of round r , which do not depend on the master key. When $k^* = k_{r+1}$, we expect that a p -fraction of the pairs will satisfy $\tilde{y}_1^i \oplus \tilde{y}_2^i = \Delta_y$. On the other hand, when $k^* \neq k_{r+1}$ we may heuristically expect only a $2^{-\ell}$ -fraction of the pairs to yield this differential. By setting L large enough, the correct value of the final sub-key k_{r+1} can be determined.

This works, but is not very efficient since in each step we enumerate over 2^ℓ possible values. We can do better by guessing portions of k_{r+1} at a time. More concretely, assume the S -boxes in F have 1-byte input/output length, and focus on the first byte of Δ_y , which we assume is nonzero. It is possible to verify if the differential holds in that byte by guessing only 8 bits of k_{r+1} , namely, the 8 bits that correspond (after the round- r mixing permutation) to the output of the first S -box. Thus, proceeding as above, we can learn these 8 bits by enumerating over all possible values for those bits, and seeing which value yields the desired differential in the first byte with the highest probability. Incorrect guesses for those 8 bits yield the expected differential in that byte with (heuristic) probability 2^{-8} , but the correct guess will give the expected differential with probability roughly $p + 2^{-8}$; this is because with probability p the differential holds on the entire block (so in particular for the first byte), and when this is not the case then we can treat the differential in the first byte as random. Note that different differentials may be needed to learn different portions of k_{r+1} .

In practice, various optimizations are performed to improve the effectiveness of the above test or, more specifically, to increase the gap between the probability that an incorrect guess yields the differential vs. the probability that a correct guess does. One optimization is to use a *low-weight* differential in which Δ_y has many zero bytes in the positions that enter the S -boxes in round r . Any pairs $\tilde{y}_1, \tilde{y}_2$ satisfying such a differential have equal values entering many of the S -boxes in round r , and so will result in output values y_1, y_2 that are equal in the corresponding bit-positions (depending on the final

mixing permutation). This means that when performing the test described earlier, one can simply discard any pairs (y_1^i, y_2^i) that do not agree in those bit-positions (since the corresponding intermediate values $(\tilde{y}_1, \tilde{y}_2)$ cannot possibly satisfy the differential, for any choice of the final sub-key). This significantly improves the effectiveness of the attack.

Once k_{r+1} is known, the attacker can “peel off” the final key-mixing step, as well as the mixing permutation and S -box substitution steps of round r (since these do not depend on the master key), and then apply the same attack—using a different differential—to find the r th-round sub-key k_r , and so on, until it learns all sub-keys (or, equivalently, the entire master key).

A worked example. We work through a “toy” example, illustrating also how a good differential can be found. We use a four-round SPN with a block length of 16 bits, based on a single S -box with 4-bit input/output length. The S -box is defined as follows (the table shows how each 4-bit input is mapped to a 4-bit output):

Input:	0000	0001	0010	0011	0100	0101	0110	0111
Output:	0000	1011	0101	0001	0110	1000	1101	0100
Input:	1000	1001	1010	1011	1100	1101	1110	1111
Output:	1111	0111	0010	1100	1001	0011	1110	1010

The mixing permutation, showing where each bit is moved for each of the 16 bits in a block, is as follows:

In:	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Out:	7	2	3	8	12	5	11	9	10	1	14	13	4	6	16	15

x	S(x)	x ⊕ 1111	S(x ⊕ 1111)	S(x) ⊕ S(x ⊕ 1111)
0000	0000	1111	1010	1010
0001	1011	1110	1110	0101
0010	0101	1101	0011	0110
0011	0001	1100	1001	1000
0100	0110	1011	1100	1010
0101	1000	1010	0010	1010
0110	1101	1001	0111	1010
0111	0100	1000	1111	1011
1000	1111	0111	0100	1011
1001	0111	0110	1101	1010
1010	0010	0101	1000	1010
1011	1100	0100	0110	1010
1100	1001	0011	0001	1000
1101	0011	0010	0101	0110
1110	1110	0001	1011	0101
1111	1010	0000	0000	1010

FIGURE 6.8: The effect of the input difference $\Delta_x = 1111$ in our S -box.

We first find a differential in the S -box. Let $S(x)$ denote the output of the S -box on input x . Consider the differential $\Delta_x = 1111$. Then, for example, we have $S(0000) \oplus S(1111) = 0000 \oplus 1010 = 1010$ and so in this case a difference of 1111 in the inputs leads to a difference of 1010 in the outputs. Let us see if this relation holds frequently. We have $S(0001) = 1011$ and $S(0001 \oplus 1111) = S(1110) = 1110$, and so here a difference of 1111 in the inputs does *not* lead to a difference of 1010 in the outputs. However, $S(0100) = 0110$ and $S(0100 \oplus 1111) = S(1011) = 1100$ and so in this case, a difference of 1111 in the inputs yields a difference of 1010 in the outputs. In Figure 6.8 we have tabulated results for all possible inputs. We see that *half* the time a difference of 1111 in the inputs yields a difference of 1010 in the outputs. Thus, (1111, 1010) is a differential in S that occurs with probability $1/2$.

		Output Difference Δ_y															
		0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
Input Difference Δ_x	0	16	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	1	0	0	0	0	4	0	0	0	2	2	2	2	0	0	4	0
	2	0	0	0	0	0	2	0	2	0	2	2	4	2	2	0	0
	3	0	2	2	4	0	4	0	0	0	0	0	0	0	2	2	0
	4	0	0	0	2	2	2	6	0	2	0	0	0	2	0	0	0
	5	0	2	2	0	0	0	0	0	4	0	0	0	4	2	2	0
	6	0	2	0	2	0	0	0	0	0	2	0	2	0	4	0	4
	7	0	2	0	0	2	4	2	2	0	2	0	0	0	2	0	0
	8	0	0	0	2	0	0	0	2	0	0	0	2	2	2	2	4
	9	0	2	0	2	2	2	0	4	0	2	2	0	0	0	0	0
	A	0	0	4	0	2	0	2	4	2	0	2	0	0	0	0	0
	B	0	0	2	0	0	0	2	0	0	2	0	0	4	2	4	0
	C	0	0	0	0	0	0	0	0	4	4	0	4	0	0	0	4
	D	0	4	2	2	0	0	2	2	0	0	0	0	0	0	0	4
	E	0	2	4	2	4	0	0	0	0	0	0	0	2	0	2	0
	F	0	0	0	0	0	2	2	0	2	0	8	2	0	0	0	0

FIGURE 6.9: Differentials in our S -box.

This same process can be carried out for all 2^4 input differences Δ_x to calculate the probability of every differential. Namely, for each pair (Δ_x, Δ_y) we tabulate the number of 4-bit inputs x for which $S(x) \oplus S(x \oplus \Delta_x) = \Delta_y$. We have done this for our example S -box in Figure 6.9. (For conciseness we represent (Δ_x, Δ_y) using hexadecimal notation.) The table should be read as follows: entry (i, j) counts how many inputs with difference i map to outputs with difference j . Observe, for example, that there are 8 inputs with difference $0xF = 1111$ that map to output $0xA = 1010$, as we have shown above. This is the highest-probability differential (apart from the trivial differential $(0, 0)$). But there are other differentials of interest: an input difference of $0x4 = 0100$ maps to an output difference of $0x6 = 0110$ with probability $6/16 = 3/8$, and there are several differentials with probability $4/16 = 1/4$.

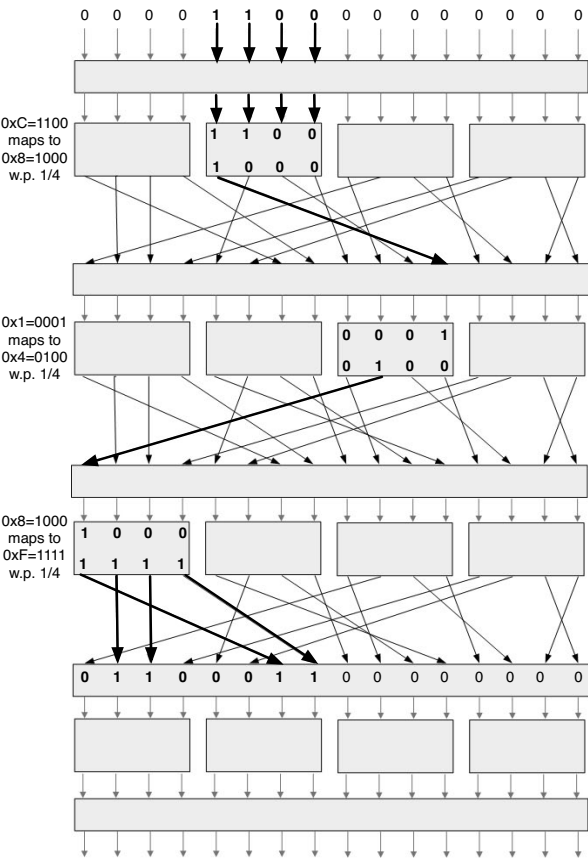


FIGURE 6.10: Tracing differentials through a four-round SPN that uses the S -box and mixing permutation given in the text.

We now extend this to find a good differential for the first three rounds of the SPN. Consider evaluating the SPN on two inputs that have a differential of 0000 1100 0000 0000, and tracing the differential between the intermediate values at each step of this evaluation. (Refer to Figure 6.10, which shows the four full rounds of the SPN plus the final key-mixing step. For clarity, the figure omits the mixing permutation in the 4th round; that mixing permutation just has the effect of shuffling the bits of the differential, and so can easily be taken into account in the attack.) The key-mixing step in the first round does not affect the differential, and so the inputs to the second S -box in the first round have differential 1100. We see from Figure 6.9 that a difference of 0xC = 1100 in the inputs to the S -box yields a difference of 0x8 = 1000 in the outputs of the S -box with probability 1/4. So with probability 1/4 the differential in the output of the 2nd S -box after round 1 is a single bit which is

moved by the mixing permutation from the 5th position to the 12th position. (The inputs to the other S -boxes are equal, so their outputs are equal and the differential of the outputs is 0000.) Assuming this to be the case, the input difference to the third S -box in the second round is $0x1 = 0001$ (once again, the key-mixing step in the second round does not affect the differential); using Figure 6.9 we have that with probability $1/4$ the output difference from that S -box is $0x4 = 0100$. Thus, once again there is just a single output bit that is different, and it is moved from the 10th position to the first position by the mixing permutation. Finally, consulting Figure 6.9 yet again, we see that an input difference of $0x8 = 1000$ to the S -box results in an output difference of $0xF = 1111$ with probability $1/4$. The bits in positions 1, 2, 3, and 4 are then moved by the mixing permutation to positions 7, 2, 3, and 8.

Overall, then, we see that an input difference of $\Delta_x = 0000\ 1100\ 0000\ 0000$ yields the output difference $\Delta_y = 0110\ 0011\ 0000\ 0000$ after three rounds with probability at least $\frac{1}{4} \cdot \frac{1}{4} \cdot \frac{1}{4} = \frac{1}{64}$.⁷ (We multiply the probabilities since we heuristically assume independence of each round.) For a random function, the probability that any given differential occurs is just $2^{-16} = 1/65536$. Thus, the differential we have found occurs with probability significantly higher than what would be expected for a random function. Observe also that we have found a low-weight differential.

We can use this differential to find the first 8 bits of the final sub-key k_5 . As discussed earlier, we begin by letting $\{(x_1^i, x_2^i)\}_{i=1}^L$ be a set of L pairs of random inputs with differential Δ_x . Using a chosen-plaintext attack, we then obtain the values $y_1^i = F_k(x_1^i)$ and $y_2^i = F_k(x_2^i)$ for all i . Now, for all possible values for the initial 8 bits of k_5 , we compute the initial 8 bits of $\tilde{y}_1^i, \tilde{y}_2^i$, the intermediate values after the key-mixing step of the 4th round. (We can do this because we only need to invert the two left-most S -boxes of the 4th round in order to derive those 8 bits.) When we guess the correct value for the initial 8 bits of k_5 , we expect the 8-bit differential 0110 0011 to occur with probability at least $1/64$. Heuristically, an incorrect guess yields the expected differential only with probability $2^{-8} = 1/256$. By setting L large enough, we can (with high probability) identify the correct value.

Differential attacks in practice. Differential cryptanalysis is very powerful, and has been used to attack real ciphers. A prominent example is FEAL-8, which was proposed as an alternative to DES in 1987. A differential attack on FEAL-8 was found that requires just 1,000 chosen plaintexts. In 1991, it took less than 2 minutes using this attack to find the entire key. Today, any proposed cipher is tested for resistance to differential cryptanalysis.

A differential attack was the first attack on DES to require less time than a simple brute-force search. While an interesting theoretical result, the attack is not of significant concern in practice since it requires 2^{47} chosen plaintexts. It

⁷This is a lower bound on the probability of the differential, since there may be other differences in the intermediate values that result in the same difference in the outputs.

is very difficult for an attacker to obtain this many chosen plaintext/ciphertext pairs in most real-world applications. Interestingly, small modifications to the S -boxes of DES make the cipher much more vulnerable to differential attacks. Personal testimony of the DES designers (after differential attacks were discovered in the outside world) has confirmed that the S -boxes of DES were designed specifically to thwart differential attacks.

Linear cryptanalysis. Linear cryptanalysis was developed by Matsui in the early 1990s. We will only describe the technique at a high level. The basic idea is to consider linear relationships between the input and output that hold with higher probability than would be expected for a random function. In more detail, say that bit positions $i_1, \dots, i_{in}$ and $i'_1, \dots, i'_{out}$ have *linear bias* ε if, for uniform x and k , and $y \stackrel{\text{def}}{=} F_k(x)$, it holds that

$$\left| \Pr[x_{i_1} \oplus \dots \oplus x_{i_{in}} \oplus y_{i'_1} \oplus \dots \oplus y_{i'_{out}} = 0] - \frac{1}{2} \right| = \varepsilon,$$

where x_i, y_i denote the i th bits of x and y . For a random function and any fixed set of bit positions, we expect the bias to be close to 0. Matsui showed how to use a large enough bias in a cipher F to find the secret key. Besides giving another method for attacking ciphers, an important feature of this attack is that it does not require *chosen* plaintexts, but rather *known* plaintexts suffice. This is very significant, since an encrypted file can provide a huge amount of known plaintext, whereas gathering encryptions of chosen plaintexts is much more difficult. Matsui showed that DES can be broken with just 2^{43} known plaintext/ciphertext pairs.

Impact on block-cipher design. Modern block ciphers are designed and evaluated based, in part, on their resistance to differential and linear cryptanalysis. When constructing a block cipher, designers choose S -boxes and other components so as to minimize differential probabilities and linear biases. We remark that it is not possible to eliminate *all* high-probability differentials in an S -box: any S -box will have *some* differential that occurs more frequently than others. Still, these deviations can be minimized. Moreover, increasing the number of rounds (and choosing the mixing permutation carefully) can both reduce the differential probabilities as well as make it more difficult for cryptanalysts to find any differentials to exploit.

6.3 Hash Functions

Recall from Chapter 5 that the primary security requirement for a hash function H is *collision resistance*; that is, it should be difficult to find a collision, or distinct inputs x, x' such that $H(x) = H(x')$. (We drop mention

of any key here, since real-world hash functions are generally unkeyed.) If the hash function has ℓ -bit output length, then the best we can hope for is that it should be infeasible to find a collision using substantially fewer than $2^{\ell/2}$ invocations of H . (See Section 5.4.1.) We would also like the hash function to achieve (second) preimage resistance against attacks running in time much less than 2^ℓ , although we do not consider such attacks in our discussion here.

Hash functions are generally constructed in two steps. First, a *compression function* (i.e., a fixed-length hash function) h is designed; next, some mechanism is used to extend h so as to handle arbitrary input lengths. In Section 5.2 we have already shown one approach—the Merkle–Damgård transform—for the second step. Here, we explore a technique for designing the underlying compression function. We also discuss some hash functions used in practice. A theoretical construction of a compression function based on a number-theoretic assumption is given in Section 8.4.2.

6.3.1 Hash Functions from Block Ciphers

Perhaps surprisingly, it is possible to build a collision-resistant compression function from a block cipher that satisfies certain additional properties. There are several ways to do this; one of the most common is via the *Davies–Meyer construction*. Let F be a block cipher with n -bit key length and ℓ -bit block length. We can then define the compression function $h : \{0, 1\}^{n+\ell} \rightarrow \{0, 1\}^\ell$ by $h(k, x) \stackrel{\text{def}}{=} F_k(x) \oplus x$. (See Figure 6.11.)

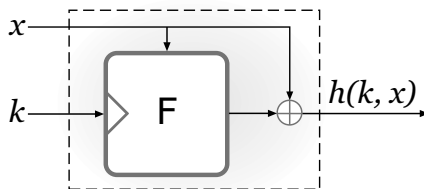


FIGURE 6.11: The Davies–Meyer construction.

We do not know how to prove collision resistance of the resulting compression function based only on the assumption that F is a strong pseudorandom permutation, and in fact there are reasons to believe such a proof is not possible. We *can*, however, prove collision resistance if we are willing to model F as an *ideal cipher*. The ideal-cipher model is a strengthening of the random-oracle model (see Section 5.5), in which we posit that all parties have access to an oracle for a random keyed permutation $F : \{0, 1\}^n \times \{0, 1\}^\ell \rightarrow \{0, 1\}^\ell$ as well as its inverse F^{-1} (i.e., such that $F^{-1}(k, F(k, x)) = x$ for all k, x). Another way to think of this is that each key $k \in \{0, 1\}^n$ specifies an independent, uniform permutation $F(k, \cdot)$ on ℓ -bit strings. As in the random-oracle model, the *only* way to compute F (or F^{-1}) is to explicitly query the oracle with (k, x) and receive back $F(k, x)$ (or $F^{-1}(k, x)$).

Analyzing constructions in the ideal-cipher model comes with all the advantages and disadvantages of working in the random-oracle model, as discussed at length in Section 5.5. We only add here that the ideal-cipher model implies the absence of *related-key attacks*, in the sense that (as we have just said) the permutations $F(k, \cdot)$ and $F(k', \cdot)$ must behave independently even if, for example, k and k' differ in only a single bit. In addition, there can be no “weak keys” k (say, the all-0 key) for which $F(k, \cdot)$ is easily distinguishable from random. It also means that $F(k, \cdot)$ should “behave randomly” *even when k is known*. For any real-world cipher F , these properties do *not* necessarily hold (and are not even well defined) even if F is a strong pseudorandom permutation, and the reader may note that we have not discussed these properties in any of our analysis of real-world block-cipher constructions. (In fact, DES and triple-DES do not satisfy these properties.) Any block cipher being used to instantiate an ideal cipher must be evaluated with respect to these more stringent requirements.

We prove the following theorem in a concrete setting, but the proof could be adapted easily for the asymptotic setting as well.

THEOREM 6.5 *If F is modeled as an ideal cipher, then the Davies–Meyer construction yields a collision-resistant compression function. Concretely, any attacker making $q < 2^{\ell/2}$ queries to its ideal-cipher oracles finds a collision with probability at most $q^2/2^\ell$.*

PROOF To be clear, we consider here the probabilistic experiment in which F is sampled at random (more precisely, for each $k \in \{0, 1\}^n$ the function $F(k, \cdot) : \{0, 1\}^\ell \rightarrow \{0, 1\}^\ell$ is chosen uniformly from the set Perm_ℓ of permutations on ℓ -bit strings) and then the attacker is given oracle access to F and F^{-1} . The attacker then tries to find a colliding pair $(k, x), (k', x')$, i.e., for which $F(k, x) \oplus x = F(k', x') \oplus x'$. No computational bounds are placed on the attacker other than bounding the number of oracle queries it makes. We assume that if the attacker outputs a colliding pair $(k, x), (k', x')$ then it has previously made the oracle queries necessary to compute the values $F(k, x)$ and $F(k', x')$. We also assume the attacker never makes the same query more than once, and never queries $F^{-1}(k, y)$ once it has learned that $y = F(k, x)$ (and vice versa). All these assumptions are without loss of generality.

Consider the i th query the attacker makes to its oracles. A query (k_i, x_i) to F reveals only the hash value $h_i \stackrel{\text{def}}{=} h(k_i, x_i) = F(k_i, x_i) \oplus x_i$; similarly, a query to F^{-1} giving the result $x_i = F^{-1}(k_i, y_i)$ yields only the hash value $h_i \stackrel{\text{def}}{=} h(k_i, x_i) = y_i \oplus F^{-1}(k_i, y_i)$. The attacker does not obtain a collision unless $h_i = h_j$ for some $i \neq j$.

Fix i, j with $i > j$ and consider the probability that $h_i = h_j$. At the time of the i th query, the value of h_j is fixed. A collision between h_i and h_j is obtained on the i th query only if the attacker queries (k_i, x_i) to F and obtains the result $F(k_i, x_i) = h_j \oplus x_i$, or queries (k_i, y_i) to F^{-1} and obtains

the result $F^{-1}(k_i, y_i) = h_j \oplus y_i$. Either event occurs with probability at most $1/(2^\ell - (i - 1))$ since, for example, $F(k_i, x_i)$ is uniform over $\{0, 1\}^\ell$ except that it cannot be equal to any value $F(k_i, x)$ already defined by the attacker's (at most) $i - 1$ previous oracle queries using key k_i . Since $i \leq q < 2^{\ell/2}$, the probability that $h_i = h_j$ is at most $2/2^\ell$.

Taking a union bound over all $\binom{q}{2} < q^2/2$ distinct pairs i, j gives the result stated in the theorem. ■

Davies–Meyer and DES. As we have mentioned above, one must take care when instantiating the Davies–Meyer construction with any concrete block cipher, since the cipher must satisfy additional properties (beyond being a strong pseudorandom permutation) in order for the resulting construction to be secure. In Exercise 6.21 we explore what goes wrong when DES is used in the Davies–Meyer construction.

This should serve as a *warning* that the proof of security for the Davies–Meyer construction in the ideal-cipher model does not necessarily translate into real security when instantiated with a real cipher. Nevertheless, as we will describe below, this paradigm has been used to construct practical hash functions that have resisted attack (but specifically when the block cipher at the center of the construction was designed specifically for this purpose).

In conclusion, the Davies–Meyer construction is a useful paradigm for constructing collision-resistant compression functions. However, it should *not* be applied to block ciphers designed for encryption, like DES and AES.

6.3.2 MD5

MD5 is a hash function with a 128-bit output length. It was designed in 1991 and for some time was believed to be collision resistant. Over a period of several years, various weaknesses began to be found in MD5 but these did not appear to lead to any easy way to find collisions. Shockingly, in 2004 a team of Chinese cryptanalysts presented a new method for finding collisions in MD5; they were easily able to convince others that their approach was correct by demonstrating an explicit collision! Since then, the attack has been improved and today collisions can be found in under a minute on a desktop PC. In addition, the attacks have been extended so that even “controlled collisions” (e.g., two postscript files generating arbitrary viewable content) can be found.

Due to these attacks, MD5 should not be used anywhere cryptographic security is needed. We mention MD5 only because it is still found in legacy code.

6.3.3 SHA-0, SHA-1, and SHA-2

The *Secure Hash Algorithm* (SHA) refers to a series of cryptographic hash functions standardized by NIST. Perhaps the most well known of these is SHA-1, which was introduced in 1995. This algorithm has a 160-bit output length and supplanted a predecessor called SHA-0, which was withdrawn due to unspecified flaws discovered in that algorithm.

At the time of this writing, an explicit collision has yet to be found in SHA-1. However, theoretical analysis over the past few years indicates that collisions in SHA-1 can be found using significantly fewer than the 2^{80} hash-function evaluations that would be necessary using a birthday attack, and it is conjectured that a collision will be found soon. It is therefore recommended to migrate to SHA-2, which does not currently appear to have the same weaknesses. SHA-2 is comprised of two related functions: SHA-256 and SHA-512, with 256- and 512-bit output lengths, respectively.

All hash functions in the SHA family are constructed using the same basic design, which incorporates components we have already seen: A compression function is first defined by applying the Davies–Meyer construction to a block cipher, and this is then extended to support arbitrary length inputs using the Merkle–Damgård transform. One interesting thing here is that the block cipher in each case was designed specifically for building the compression function. In fact, it was only retroactively that the underlying components in the compression functions were isolated and analyzed as the block ciphers SHACAL-1 (for SHA-1) and SHACAL-2 (for SHA-2). These ciphers are themselves intriguing, as they have large block lengths (160 and 256 bits, respectively) and 512-bit keys.

6.3.4 SHA-3 (Keccak)

In the aftermath of the collision attack on MD5 and the theoretical weaknesses found in SHA-1, NIST announced in late 2007 a public competition to design a new cryptographic hash function to be called SHA-3. Submitted algorithms were required to support at least 256- and 512-bit output lengths. As in the case of the AES competition from roughly 10 years earlier, the competition was completely open and transparent; anyone could submit an algorithm for consideration, and the public was invited to submit their opinions on any of the candidates. The 51 first-round candidates were narrowed down to 14 in December 2008, and these were further reduced to five finalists in 2010. The remaining candidates were subject to intense scrutiny by the cryptographic community over the next two years. In October 2012, NIST announced the selection of *Keccak* as the winner of the competition. As of the time of this writing, this algorithm is undergoing standardization as the next-generation replacement for SHA-2.

Keccak is unusual in several respects. (Interestingly, one of the reasons Keccak was chosen is because its structure is very different from that of SHA-1 and SHA-2.) At its core, it is based on an *unkeyed* permutation f with a large block length of 1600 bits; this is radically different from, e.g., the Davies–Meyer construction, which relies on a *keyed* permutation. Furthermore, Keccak does not use the Merkle–Damgård transform to handle arbitrary input lengths. Instead, it uses a newer approach called the *sponge construction*. Keccak—and the sponge construction more generally—can be analyzed in the *random-permutation model* in which we postulate that parties

have access to an oracle for a random permutation $f : \{0, 1\}^\ell \rightarrow \{0, 1\}^\ell$ (and possibly its inverse). This is weaker than the ideal-cipher model; indeed, we can easily obtain a random permutation in the ideal-cipher model by simply fixing the key to the cipher to be any constant value.

It will be fascinating to watch the new hash standard evolve, and to see how quickly developers adapt from SHA-1/SHA-2 to the newer SHA-3.

References and Additional Reading

Additional information on LFSRs and stream ciphers can be found in the *Handbook of Applied Cryptography* [120] or the more recent text by Paar and Pelzl [135]. Further details regarding eSTREAM, as well as a detailed specification of Trivium, can be found at <http://www.ecrypt.eu.org/stream>. See the work of AlFardan et al. [9] for a recent survey of attacks on RC4.

The confusion-diffusion paradigm and substitution-permutation networks were introduced by Shannon [154] and Feistel [64]. See the thesis of Heys [90] for further information regarding SPN design. Better generic attacks on three-round SPNs than what we have shown here are known [31]. Miles and Viola [126] give a theoretical analysis of SPNs.

Feistel networks were first described in [64]. A theoretical analysis of Feistel networks was given by Luby and Rackoff [116]; see Chapter 7.

More details on DES, AES, and block-cipher constructions in general can be found in the text by Knudsen and Robshaw [106]. The meet-in-the-middle attack on double encryption is due to Diffie and Hellman [59]. The attack on two-key triple encryption mentioned in the text (and explored in Exercise 6.13) is by Merkle and Hellman [124]. Theoretical analysis of the security of double and triple encryption can be found in [6, 24].

DESX is another technique for increasing the effective key length of DES. The secret key consists of values $k_i, k_o \in \{0, 1\}^{64}$, and $k \in \{0, 1\}^{56}$, and the cipher is defined by

$$DESX_{k_i, k, k_o}(x) \stackrel{\text{def}}{=} k_o \oplus DES_k(x \oplus k_i).$$

This methodology was first studied by Even and Mansour [63] in a slightly different context. Its application to DES was proposed in unpublished work by Rivest, and its security was later analyzed by Kilian and Rogaway [105, 149].

Differential cryptanalysis was introduced by Biham and Shamir [29] and its application to DES is described in a book by those authors [30]. Copper-smith [45] describes design principles of the DES S -boxes in light of the public discovery of differential cryptanalysis. Linear cryptanalysis was discovered by Matsui [118], who shows its application to DES there. For more information on these advanced cryptanalytic techniques, we refer the reader to the tutorial

on differential and linear cryptanalysis by Heys [91] or to the aforementioned book by Knudsen and Robshaw [106].

For further information about MD5 and SHA-1 see [120]. Note, however, that their treatment pre-dates the attacks by Wang et al. [175, 174]. Constructions of compression functions from block ciphers are analyzed in [143, 33]. The sponge construction is described and analyzed by Bertoni et al. [28]. For additional details about the SHA-3 competition, see the NIST webpage at <http://csrc.nist.gov/groups/ST/hash/sha-3/index.html>.

Exercises

- 6.1 Assume a degree-6 LFSR with $c_0 = c_5 = 1$ and $c_1 = c_2 = c_3 = c_4 = 0$.
 - (a) What are the first 10 bits output by this LFSR if it starts in initial state $(1, 1, 1, 1, 1, 1)$?
 - (b) Is this LFSR maximal length?
- 6.2 In this question we consider a nonlinear combination generator, where we have a degree- n LFSR but the output at each time step is not s_0 but instead $g(s_0, \dots, s_{n-1})$ for some nonlinear function g . Assume the feedback coefficients of the LFSR are known, but its initial state is not. Show that each of the following choices of g does not yield a good pseudorandom generator:
 - (a) $g(s_0, \dots, s_{n-1}) = s_0 \wedge s_1$.
 - (b) $g(s_0, \dots, s_{n-1}) = (s_0 \wedge s_1) \oplus s_2$.
- 6.3 Let F be a block cipher with n -bit key length and block length. Say there is a key-recovery attack on F that succeeds with probability 1 using n chosen plaintexts and minimal computational effort. Prove formally that F cannot be a pseudorandom permutation.
- 6.4 In our attack on a one-round SPN, we considered a block length of 64 bits and 16 S -boxes that each take a 4-bit input. Repeat the analysis for the case of 8 S -boxes, each taking an 8-bit input. What is the complexity of the attack now? Repeat the analysis again with a 128-bit block length and 16 S -boxes that each take an 8-bit input.
- 6.5 Consider a modified SPN where instead of carrying out the key-mixing, substitution, and permutation steps in alternating order for r (full) rounds, the cipher instead first applies r rounds of key mixing, then carries out r rounds of substitution, and finally applies r mixing permutations. Analyze the security of this construction.

6.6 In this question we assume a two-round SPN with 64-bit block length.

- (a) Assume independent 64-bit sub-keys are used in each round, so the master key is 192 bits long. Show a key-recovery attack using much less than 2^{192} time.
- (b) Assume the first and third sub-keys are equal, and the second sub-key is independent, so the master key is 128 bits long. Show a key-recovery attack using much less than 2^{128} time.

6.7 What is the output of an r -round Feistel network when the input is (L_0, R_0) in each of the following two cases:

- (a) Each round function outputs all 0s, regardless of the input.
- (b) Each round function is the identity function.

6.8 Let $\text{Feistel}_{f_1, f_2}(\cdot)$ denote a two-round Feistel network using functions f_1 and f_2 (in that order). Show that if $\text{Feistel}_{f_1, f_2}(L_0, R_0) = (L_2, R_2)$, then $\text{Feistel}_{f_2, f_1}(R_2, L_2) = (R_0, L_0)$.

6.9 For this exercise, rely on the description of DES given in this chapter, but use the fact that in the actual construction of DES the two halves of the output of the final round of the Feistel network are swapped. That is, if the output of the final round of the Feistel network is (L_{16}, R_{16}) , then the output of DES is (R_{16}, L_{16}) .

- (a) Show that the only difference between computation of DES_k and DES_k^{-1} is the order in which sub-keys are used. (Rely on the previous exercise.)
- (b) Show that for $k = 0^{56}$ it holds that $DES_k(DES_k(x)) = x$.

Hint: Consider the sub-keys generated from this key.

- (c) Find three other DES keys with the same property. These keys are known as *weak keys* for DES. (Note: the keys you find will differ from the actual weak keys of DES because of differences in our presentation.)
- (d) Do these 4 weak keys represent a serious vulnerability in the use of triple-DES as a pseudorandom permutation? Explain.

6.10 Show that DES has the property that $DES_k(x) = \overline{DES_{\bar{k}}(\bar{x})}$ for every key k and input x (where $\bar{z}$ denotes the bitwise complement of z). (This is called the *complementarity property* of DES.) Does this represent a serious vulnerability in the use of triple-DES as a pseudorandom permutation? Explain.

6.11 Describe attacks on the following modifications to DES:

- (a) Each sub-key is 32 bits long, and the round function simply XORs the sub-key with the input to the round (i.e., $\hat{f}(k, R) = k_i \oplus R$). For this question, the key schedule is unimportant and you can treat the sub-keys k_i as independent keys.
- (b) Instead of using different sub-keys in every round, the same 48-bit sub-key is used in every round. Show how to distinguish the cipher from a random permutation in $\ll 2^{48}$ time.

Hint: Exercises 6.8 and 6.9 may help. . .

6.12 (This exercise relies on Exercise 6.9.) Our goal is to show that for any weak key k of DES, it is easy to find an input x such that $DES_k(x) = x$.

- (a) Assume we evaluate DES_k on input (L_0, R_0) , and the output after 8 rounds of the Feistel network is (L_8, R_8) with $L_8 = R_8$. Show that the output of $DES_k(L_0, R_0)$ is (L_0, R_0) . (Recall from Exercise 6.9 that DES swaps the two halves of the 16th round of the Feistel network before outputting the result.)
- (b) Show how to find an input (L_0, R_0) with the property in part (a).

6.13 This question illustrates an attack on two-key triple encryption. Let F be a block cipher with n -bit block length and key length, and set $F'_{k_1, k_2}(x) \stackrel{\text{def}}{=} F_{k_1}(F_{k_2}^{-1}(F_{k_1}(x)))$.

- (a) Assume that given a pair (m_1, m_2) it is possible to find in *constant* time all keys k_2 such that $m_2 = F_{k_2}^{-1}(m_1)$. Show how to recover the entire key for F' (with high probability) in time roughly 2^n using three known input/output pairs.
- (b) In general, it will *not* be possible to find k_2 as above in constant time. However, show that by using a preprocessing step taking 2^n time it is possible, given m_2 , to find in (essentially) constant time all keys k_2 such that $m_2 = F_{k_2}^{-1}(0^n)$.
- (c) Assume k_1 is known and that the pre-processing step above has already been run. Show how to use the value $y = F'_{k_1, k_2}(x)$ for a single *chosen* plaintext x to determine k_2 in constant time.
- (d) Put the above components together to devise an attack that recovers the entire key of F' by running in roughly 2^n time and requesting the encryption of roughly 2^n chosen inputs.

6.14 Say the key schedule of DES is modified as follows: the left half of the master key is used to derive all the sub-keys in rounds 1–8, while the right half of the master key is used to derive all the sub-keys in rounds 9–16. Show an attack on this modified scheme that recovers the entire key in time roughly 2^{28} .

- 6.15 Let $f : \{0, 1\}^m \times \{0, 1\}^\ell \rightarrow \{0, 1\}^\ell$ and $g : \{0, 1\}^n \times \{0, 1\}^\ell \rightarrow \{0, 1\}^\ell$ be secure block ciphers with $m > n$, and define $F_{k_1, k_2}(x) = f_{k_1}(g_{k_2}(x))$. Show a key-recovery attack on F using time $\mathcal{O}(2^m)$ and space $\mathcal{O}(\ell \cdot 2^n)$.
- 6.16 Define $DESY_{k, k'}(x) = DES_k(x \oplus k')$. The key length of $DESY$ is 120 bits. Show a key-recovery attack on $DESY$ taking time and space $\approx 2^{64}$.
- 6.17 Choose random S -boxes and mixing permutations for SPNs of different sizes, and develop differential attacks against them. We recommend trying five-round SPNs with 16-bit and 24-bit block lengths, using S -boxes with 4-bit input/output. Write code to compute the differential tables, and to carry out the attack.
- 6.18 Implement the time/space tradeoff for 40-bit DES (i.e., fix the first 16 bits of the key of DES to 0). Calculate the time and memory needed, and empirically estimate the probability of success. Experimentally verify the increase in success probability as the number of tables is increased. (Warning: this is a big project!)
- 6.19 For each of the following constructions of a compression function h from a block cipher F , either show an attack or prove collision resistance in the ideal-cipher model:
- (a) $h(k, x) = F_k(x)$.
 - (b) $h(k, x) = F_k(x) \oplus k \oplus x$.
 - (c) $h(k, x) = F_k(x) \oplus k$.
- 6.20 Consider using DES to construct a compression function in the following way: Define $h : \{0, 1\}^{112} \rightarrow \{0, 1\}^{64}$ as $h(x_1, x_2) \stackrel{\text{def}}{=} DES_{x_1}(DES_{x_2}(0^{64}))$ where $|x_1| = |x_2| = 56$.
- (a) Write down an explicit collision in h .
Hint: Use Exercise 6.9(a–b).
 - (b) Show how to find a preimage of an arbitrary value y (that is, x_1, x_2 such that $h(x_1 || x_2) = y$) in roughly 2^{56} time.
 - (c) Show a more clever preimage attack that runs in roughly 2^{32} time and succeeds with high probability.
Hint: Rely on the results of Appendix A.4.
- 6.21 Let F be a block cipher for which it is easy to find fixed points for some key: namely, there is a key k for which it is easy to find inputs x for which $F_k(x) = x$. Find a collision in the Davies–Meyer construction when applied to F . (Consider this in light of Exercise 6.12.)

Chapter 7

**Theoretical Constructions of Symmetric-Key Primitives*

In Chapter 3 we introduced the notion of pseudorandomness and defined some basic cryptographic primitives including pseudorandom generators, functions, and permutations. We showed in Chapters 3 and 4 that these primitives serve as the building blocks for all of private-key cryptography. As such, it is of great importance to understand these primitives from a theoretical point of view. In this chapter we formally introduce the concept of *one-way functions*—functions that are, informally, easy to compute but hard to invert—and show how pseudorandom generators, functions, and permutations can be constructed under the sole assumption that one-way functions exist.¹ Moreover, we will see that one-way functions are necessary for “non-trivial” private-key cryptography. That is: *the existence of one-way functions is equivalent to the existence of all (non-trivial) private-key cryptography*. This is one of the major contributions of modern cryptography.

The constructions we show in this chapter should be viewed as complementary to the constructions of stream ciphers and block ciphers discussed in the previous chapter. The focus of the previous chapter was on how various cryptographic primitives are currently realized in practice, and the intent of that chapter was to introduce some basic approaches and design principles that are used. Somewhat disappointing, though, was the fact that none of the constructions we showed could be *proven* secure based on any weaker (i.e., more reasonable) assumptions. In contrast, in the present chapter we will prove that it is possible to construct pseudorandom permutations starting from the very mild assumption that one-way functions exist. This assumption is more palatable than assuming, say, that AES is a pseudorandom permutation, both because it is a qualitatively weaker assumption and also because we have a number of candidate, number-theoretic one-way functions that have been studied for many years, even before the advent of cryptography. (See the very beginning of Chapter 6 for further discussion of this point.) The downside, however, is that the constructions we show here are all far less efficient than those of Chapter 6, and thus are not actually used. It remains an important challenge for cryptographers to “bridge this gap” and develop provably

¹This is not quite true since we are for the most part going to rely on one-way *permutations* in this chapter. But it is known that one-way functions suffice.

secure constructions of pseudorandom generators, functions, and permutations whose efficiency is comparable to the best available stream ciphers and block ciphers.

Collision-resistant hash functions. In contrast to the previous chapter, here we do not consider collision-resistant hash functions. The reason is that constructions of such hash functions from one-way functions are unknown and, in fact, there is evidence suggesting that such constructions are impossible. We will turn to provable constructions of collision-resistant hash functions—based on specific, number-theoretic assumptions—in Section 8.4.2.

A note regarding this chapter. The material in this chapter is somewhat more advanced than the material in the rest of this book. This material is not used explicitly elsewhere, and so this chapter can be skipped if desired. Having said this, we have tried to present the material in such a way that it is understandable (with effort) to an advanced undergraduate or beginning graduate student. We encourage all readers to peruse Sections 7.1 and 7.2, which introduce one-way functions and provide an overview of the rest of this chapter. We believe that familiarity with at least some of the topics covered here is important enough to warrant the effort.

7.1 One-Way Functions

In this section we formally define one-way functions, and then briefly discuss some candidates that are widely believed to satisfy this definition. (We will see more examples of conjectured one-way functions in Chapter 8.) We next introduce the notion of *hard-core predicates*, which can be viewed as encapsulating the hardness of inverting a one-way function and will be used extensively in the constructions that follow in subsequent sections.

7.1.1 Definitions

A one-way function $f : \{0, 1\}^* \rightarrow \{0, 1\}^*$ is easy to compute, yet hard to invert. The first condition is easy to formalize: we will simply require that f be computable in polynomial time. Since we are ultimately interested in building cryptographic schemes that are hard for a probabilistic polynomial-time adversary to break except with negligible probability, we will formalize the second condition by requiring that it be infeasible for any probabilistic polynomial-time algorithm to invert f —that is, to find a preimage of a given value y —except with negligible probability. A technical point is that this probability is taken over an experiment in which y is generated by choosing a uniform element x of the domain of f and then setting $y := f(x)$ (rather than

choosing y uniformly from the range of f). The reason for this should become clear from the constructions we will see in the remainder of the chapter.

Let $f : \{0, 1\}^* \rightarrow \{0, 1\}^*$ be a function. Consider the following experiment defined for any algorithm $\mathcal{A}$ and any value n for the security parameter:

The inverting experiment $\text{Invert}_{\mathcal{A},f}(n)$

1. Choose uniform $x \in \{0, 1\}^n$, and compute $y := f(x)$.
2. $\mathcal{A}$ is given 1^n and y as input, and outputs x' .
3. The output of the experiment is defined to be 1 if $f(x') = y$, and 0 otherwise.

We stress that $\mathcal{A}$ need not find the original preimage x ; it suffices for $\mathcal{A}$ to find any value x' for which $f(x') = y = f(x)$. We give the security parameter 1^n to $\mathcal{A}$ in the second step to stress that $\mathcal{A}$ may run in time polynomial in the security parameter n , regardless of the length of y .

We can now define what it means for a function f to be one-way.

DEFINITION 7.1 A function $f : \{0, 1\}^* \rightarrow \{0, 1\}^*$ is one-way if the following two conditions hold:

1. **(Easy to compute:)** There exists a polynomial-time algorithm M_f computing f ; that is, $M_f(x) = f(x)$ for all x .
2. **(Hard to invert:)** For every probabilistic polynomial-time algorithm $\mathcal{A}$, there is a negligible function negl such that

$$\Pr[\text{Invert}_{\mathcal{A},f}(n) = 1] \leq \text{negl}(n).$$

Notation. In this chapter we will often make the probability space more explicit by subscripting (part of) it in the probability notation. For example, we can succinctly express the second requirement in the definition above as follows: For every probabilistic polynomial-time algorithm $\mathcal{A}$, there exists a negligible function negl such that

$$\Pr_{x \leftarrow \{0,1\}^n} [\mathcal{A}(1^n, f(x)) \in f^{-1}(f(x))] \leq \text{negl}(n).$$

(Recall that $x \leftarrow \{0, 1\}^n$ means that x is chosen uniformly from $\{0, 1\}^n$.) The probability above is also taken over the randomness used by $\mathcal{A}$, which here is left implicit.

Successful inversion of one-way functions. A function that is *not* one-way is not necessarily easy to invert all the time (or even “often”). Rather, the converse of the second condition of Definition 7.1 is that there exists a probabilistic polynomial-time algorithm $\mathcal{A}$ and a non-negligible function γ

such that $\mathcal{A}$ inverts $f(x)$ with probability at least $\gamma(n)$ (where the probability is taken over uniform choice of $x \in \{0, 1\}^n$ and the randomness of $\mathcal{A}$). This means, in turn, that there exists a positive polynomial $p(\cdot)$ such that for *infinitely many values of n* , algorithm $\mathcal{A}$ inverts f with probability at least $1/p(n)$. Thus, if there exists an $\mathcal{A}$ that inverts f with probability n^{-10} for all even values of n (but always fails to invert f when n is odd), then f is not one-way—even though $\mathcal{A}$ only succeeds on half the values of n , and only succeeds with probability n^{-10} (for values of n where it succeeds at all).

Exponential-time inversion. Any one-way function can be inverted at any point y in exponential time, by simply trying all values $x \in \{0, 1\}^n$ until a value x is found such that $f(x) = y$. Thus, the existence of one-way functions is inherently an assumption about *computational complexity* and *computational hardness*. That is, it concerns a problem that can be solved in principle but is assumed to be hard to solve efficiently.

One-way permutations. We will often be interested in one-way functions with additional structural properties. We say a function f is *length-preserving* if $|f(x)| = |x|$ for all x . A one-way function that is length-preserving and one-to-one is called a *one-way permutation*. If f is a one-way permutation, then any value y has a unique preimage $x = f^{-1}(y)$. Nevertheless, it is still hard to find x in polynomial time.

One-way function/permutation families. The above definitions of one-way functions and permutations are convenient in that they consider a single function over an infinite domain and range. However, most candidate one-way functions and permutations do not fit neatly into this framework. Instead, there is an algorithm that generates some set of parameters I which define a function f_I ; one-wayness here means essentially that f_I should be one-way with all but negligible probability over choice of I . Because each value of I defines a different function, we now refer to *families* of one-way functions (resp., permutations). We give the definition now, and refer the reader to the next section for a concrete example. (See also Section 8.4.1.)

DEFINITION 7.2 A tuple $\Pi = (\text{Gen}, \text{Samp}, f)$ of probabilistic polynomial-time algorithms is a function family if the following hold:

1. The parameter-generation algorithm Gen , on input 1^n , outputs parameters I with $|I| \geq n$. Each value of I output by Gen defines sets $\mathcal{D}_I$ and $\mathcal{R}_I$ that constitute the domain and range, respectively, of a function f_I .
2. The sampling algorithm Samp , on input I , outputs a uniformly distributed element of $\mathcal{D}_I$.
3. The deterministic evaluation algorithm f , on input I and $x \in \mathcal{D}_I$, outputs an element $y \in \mathcal{R}_I$. We write this as $y := f_I(x)$.

Π is a permutation family if for each value of I output by $\text{Gen}(1^n)$, it holds that $\mathcal{D}_I = \mathcal{R}_I$ and the function $f_I : \mathcal{D}_I \rightarrow \mathcal{D}_I$ is a bijection.

Let Π be a function family. What follows is the natural analogue of the experiment introduced previously.

The inverting experiment $\text{Invert}_{\mathcal{A}, \Pi}(n)$:

1. $\text{Gen}(1^n)$ is run to obtain I , and then $\text{Samp}(I)$ is run to obtain a uniform $x \in \mathcal{D}_I$. Finally, $y := f_I(x)$ is computed.
2. $\mathcal{A}$ is given I and y as input, and outputs x' .
3. The output of the experiment is 1 if $f_I(x') = y$.

DEFINITION 7.3 A function/permutation family $\Pi = (\text{Gen}, \text{Samp}, f)$ is one-way if for all probabilistic polynomial-time algorithms $\mathcal{A}$ there exists a negligible function negl such that

$$\Pr[\text{Invert}_{\mathcal{A}, \Pi}(n) = 1] \leq \text{negl}(n).$$

Throughout this chapter we work with one-way functions/permutations over an infinite domain (as in Definition 7.1), rather than working with families of one-way functions/permutations. This is primarily for convenience, and does not significantly affect any of the results. (See Exercise 7.7.)

7.1.2 Candidate One-Way Functions

One-way functions are of interest only if they exist. We do not know how to prove they exist unconditionally (this would be a major breakthrough in complexity theory), so we must conjecture or assume their existence. Such a conjecture is based on the fact that several natural computational problems have received much attention, yet still have no polynomial-time algorithm for solving them. Perhaps the most famous such problem is *integer factorization*, i.e., finding the prime factors of a large integer. It is easy to multiply two numbers and obtain their product, but difficult to take a number and find its factors. This leads us to define the function $f_{\text{mult}}(x, y) = x \cdot y$. If we do not place any restriction on the lengths of x and y , then f_{mult} is easy to invert: with high probability $x \cdot y$ will be *even*, in which case $(2, xy/2)$ is an inverse. This issue can be addressed by restricting the domain of f_{mult} to equal-length *primes* x and y . We return to this idea in Section 8.2.

Another candidate one-way function, not relying directly on number theory, is based on the *subset-sum problem* and is defined by

$$f_{\text{ss}}(x_1, \dots, x_n, J) = \left(x_1, \dots, x_n, \left[\sum_{j \in J} x_j \bmod 2^n \right] \right),$$

where each x_i is an n -bit string interpreted as an integer, and J is an n -bit string interpreted as specifying a subset of $\{1, \dots, n\}$. Inverting f_{ss} on an output $(x_1, \dots, x_n, y)$ requires finding a subset $J' \subseteq \{1, \dots, n\}$ such that $\sum_{j \in J'} x_j = y \bmod 2^n$. Students who have studied $\mathcal{NP}$ -completeness may recall that this problem is $\mathcal{NP}$ -complete. But even $\mathcal{P} \neq \mathcal{NP}$ does not imply that f_{ss} is one-way: $\mathcal{P} \neq \mathcal{NP}$ would mean that every polynomial-time algorithm fails to solve the subset-sum problem on *at least one* input, whereas for f_{ss} to be a one-way function it is required that every polynomial-time algorithm fails to solve the subset-sum problem (at least for certain parameters) *almost always*. Thus, our belief that the function above is one-way is based on the lack of known algorithms to solve this problem even with “small” probability on random inputs, and not merely on the fact that the problem is $\mathcal{NP}$ -complete.

We conclude by showing a family of *permutations* that is believed to be one-way. Let **Gen** be a probabilistic polynomial-time algorithm that, on input 1^n , outputs an n -bit prime p along with a special element $g \in \{2, \dots, p-1\}$. (The element g should be a *generator* of $\mathbb{Z}_p^*$; see Section 8.3.3.) Let **Samp** be an algorithm that, given p and g , outputs a uniform integer $x \in \{1, \dots, p-1\}$. Finally, define

$$f_{p,g}(x) = [g^x \bmod p].$$

(The fact that $f_{p,g}$ can be computed efficiently follows from the results in Appendix B.2.3.) It can be shown that this function is one-to-one, and thus a permutation. The presumed difficulty of inverting this function is based on the conjectured hardness of the *discrete-logarithm problem*; we will have much more to say about this in Section 8.3.

Finally, we remark that very efficient one-way functions can be obtained from practical cryptographic constructions such as SHA-1 or AES under the assumption that they are collision resistant or a pseudorandom permutation, respectively; see Exercises 7.4 and 7.5. (Technically speaking, they cannot satisfy the definition of one-wayness since they have fixed-length input/output and so we cannot look at their asymptotic behavior. Nevertheless, it is plausible to conjecture that they are one-way in a concrete sense.)

7.1.3 Hard-Core Predicates

By definition, a one-way function is hard to invert. Stated differently: given $y = f(x)$, the value x cannot be computed *in its entirety* by any polynomial-time algorithm (except with negligible probability; we ignore this here). One might get the impression that nothing about x can be determined from $f(x)$ in polynomial time. This is *not* necessarily the case. Indeed, it is possible for $f(x)$ to “leak” a lot of information about x even if f is one-way. For a trivial example, let g be a one-way function and define $f(x_1, x_2) \stackrel{\text{def}}{=} (x_1, g(x_2))$, where $|x_1| = |x_2|$. It is easy to show that f is also a one-way function (this is left as an exercise), even though it reveals half its input.

For our applications, we will need to identify a specific piece of information about x that is “hidden” by $f(x)$. This motivates the notion of a *hard-core predicate*. A hard-core predicate $\text{hc} : \{0, 1\}^* \rightarrow \{0, 1\}$ of a function f has the property that $\text{hc}(x)$ is hard to compute with probability significantly better than $1/2$ given $f(x)$. (Since hc is a boolean function, it is always possible to compute $\text{hc}(x)$ with probability $1/2$ by random guessing.) Formally:

DEFINITION 7.4 *A function $\text{hc} : \{0, 1\}^* \rightarrow \{0, 1\}$ is a hard-core predicate of a function f if hc can be computed in polynomial time, and for every probabilistic polynomial-time algorithm $\mathcal{A}$ there is a negligible function negl such that*

$$\Pr_{x \leftarrow \{0,1\}^n} [\mathcal{A}(1^n, f(x)) = \text{hc}(x)] \leq \frac{1}{2} + \text{negl}(n),$$

where the probability is taken over the uniform choice of x in $\{0, 1\}^n$ and the randomness of $\mathcal{A}$.

We stress that $\text{hc}(x)$ is efficiently computable given x (since the function hc can be computed in polynomial time); the definition requires that $\text{hc}(x)$ is hard to compute given $f(x)$. The above definition does not require f to be one-way; if f is a permutation, however, then it cannot have a hard-core predicate unless it is one-way. (See Exercise 7.13.)

Simple ideas don’t work. Consider the predicate $\text{hc}(x) \stackrel{\text{def}}{=} \bigoplus_{i=1}^n x_i$ where $x_1, \dots, x_n$ denote the bits of x . One might hope that this is a hard-core predicate of any one-way function f : if f cannot be inverted, then $f(x)$ must hide at least one of the bits x_i of its preimage x , which would seem to imply that the exclusive-or of all of the bits of x is hard to compute. Despite its appeal, this argument is incorrect. To see this, let g be a one-way function and define $f(x) \stackrel{\text{def}}{=} (g(x), \bigoplus_{i=1}^n x_i)$. It is not hard to show that f is one-way. However, it is clear that $f(x)$ does not hide the value of $\text{hc}(x) = \bigoplus_{i=1}^n x_i$ because this is part of its output; therefore, $\text{hc}(x)$ is not a hard-core predicate of f . Extending this, one can show that for any fixed predicate hc , there is a one-way function f for which hc is not a hard-core predicate of f .

Trivial hard-core predicates. Some functions have “trivial” hard-core predicates. For example, let f be the function that drops the last bit of its input (i.e., $f(x_1 \cdots x_n) = x_1 \cdots x_{n-1}$). It is hard to determine x_n given $f(x)$ since x_n is independent of the output; thus, $\text{hc}(x) = x_n$ is a hard-core predicate of f . However, f is not one-way. When we use hard-core predicates for our constructions, it will become clear why trivial hard-core predicates of this sort are of no use.

7.2 From One-Way Functions to Pseudorandomness

The goal of this chapter is to show how to construct pseudorandom generators, functions, and permutations based on any one-way function/permutation. In this section, we give an overview of these constructions. Details are given in the sections that follow.

A hard-core predicate from any one-way function. The first step is to show that a hard-core predicate exists for any one-way function. Actually, it remains open whether this is true; we show something weaker that suffices for our purposes. Namely, we show that given a one-way function f we can construct a *different* one-way function g along with a hard-core predicate of g .

THEOREM 7.5 (Goldreich–Levin theorem) *Assume one-way functions (resp., permutations) exist. Then there exists a one-way function (resp., permutation) g and a hard-core predicate hc of g .*

Let f be a one-way function. Functions g and hc are constructed as follows: set $g(x, r) \stackrel{\text{def}}{=} (f(x), r)$, for $|x| = |r|$, and define

$$\text{hc}(x, r) \stackrel{\text{def}}{=} \bigoplus_{i=1}^n x_i \cdot r_i,$$

where x_i (resp., r_i) denotes the i th bit of x (resp., r). Notice that if r is uniform, then $\text{hc}(x, r)$ outputs the exclusive-or of a *random subset* of the bits of x . (When $r_i = 1$ the bit x_i is included in the XOR, and otherwise it is not.) The Goldreich–Levin theorem essentially states that if f is a one-way function then $f(x)$ hides the exclusive-or of a *random subset* of the bits of x .

Pseudorandom generators from one-way permutations. The next step is to show how a hard-core predicate of a one-way *permutation* can be used to construct a pseudorandom generator. (It is known that a hard-core predicate of a one-way *function* suffices, but the proof is extremely complicated and well beyond the scope of this book.) Specifically, we show:

THEOREM 7.6 *Let f be a one-way permutation and let hc be a hard-core predicate of f . Then, $G(s) \stackrel{\text{def}}{=} f(s) \parallel \text{hc}(s)$ is a pseudorandom generator with expansion factor $\ell(n) = n + 1$.*

As intuition for why G as defined in the theorem constitutes a pseudorandom generator, note first that the initial n bits of the output of $G(s)$ (i.e., the bits of $f(s)$) are *truly* uniformly distributed when s is uniformly distributed, by virtue of the fact that f is a permutation. Next, the fact that hc is a hard-core predicate of f means that $\text{hc}(s)$ “looks random”—i.e., is *pseudo-*

random—even given $f(s)$ (assuming again that s is uniform). Putting these observations together, we see that the entire output of G is pseudorandom.

Pseudorandom generators with arbitrary expansion. The existence of a pseudorandom generator that stretches its seed by even a single bit (as we have just seen) is already highly non-trivial. But for applications (e.g., for efficient encryption of large messages as in Section 3.3), we need a pseudorandom generator with much larger expansion. Fortunately, we can obtain any polynomial expansion factor we want:

THEOREM 7.7 *If there exists a pseudorandom generator with expansion factor $\ell(n) = n + 1$, then for any polynomial poly there exists a pseudorandom generator with expansion factor $\text{poly}(n)$.*

We conclude that pseudorandom generators with arbitrary (polynomial) expansion can be constructed from any one-way permutation.

Pseudorandom functions/permutations from pseudorandom generators. Pseudorandom generators suffice for constructing EAV-secure private-key encryption schemes. For achieving CPA-secure private-key encryption (not to mention message authentication codes), however, we relied on pseudorandom functions. The following result shows that the latter can be constructed from the former:

THEOREM 7.8 *If there exists a pseudorandom generator with expansion factor $\ell(n) = 2n$, then there exists a pseudorandom function.*

In fact, we can do even more:

THEOREM 7.9 *If there exists a pseudorandom function, then there exists a strong pseudorandom permutation.*

Combining all the above theorems, as well as the results of Chapters 3 and 4, we have the following corollaries:

COROLLARY 7.10 *Assuming the existence of one-way permutations, there exist pseudorandom generators with any polynomial expansion factor, pseudorandom functions, and strong pseudorandom permutations.*

COROLLARY 7.11 *Assuming the existence of one-way permutations, there exist CCA-secure private-key encryption schemes and secure message authentication codes.*

As noted earlier, it is possible to obtain all these results based solely on the existence of one-way functions.

7.3 Hard-Core Predicates from One-Way Functions

In this section, we prove Theorem 7.5 by showing the following:

THEOREM 7.12 *Let f be a one-way function and define g by $g(x, r) \stackrel{\text{def}}{=} (f(x), r)$, where $|x| = |r|$. Define $\text{gl}(x, r) \stackrel{\text{def}}{=} \bigoplus_{i=1}^n x_i \cdot r_i$, where $x = x_1 \cdots x_n$ and $r = r_1 \cdots r_n$. Then gl is a hard-core predicate of g .*

Due to the complexity of the proof, we prove three successively stronger results culminating in what is claimed in the theorem.

7.3.1 A Simple Case

We first show that if there exists a polynomial-time adversary $\mathcal{A}$ that *always* correctly computes $\text{gl}(x, r)$ given $g(x, r) = (f(x), r)$, then it is possible to invert f in polynomial time. Given the assumption that f is a one-way function, it follows that no such adversary $\mathcal{A}$ exists.

PROPOSITION 7.13 *Let f and gl be as in Theorem 7.12. If there exists a polynomial-time algorithm $\mathcal{A}$ such that $\mathcal{A}(f(x), r) = \text{gl}(x, r)$ for all n and all $x, r \in \{0, 1\}^n$, then there exists a polynomial-time algorithm $\mathcal{A}'$ such that $\mathcal{A}'(1^n, f(x)) = x$ for all n and all $x \in \{0, 1\}^n$.*

PROOF We construct $\mathcal{A}'$ as follows. $\mathcal{A}'(1^n, y)$ computes $x_i := \mathcal{A}(y, e^i)$ for $i = 1, \dots, n$, where e^i denotes the n -bit string with 1 in the i th position and 0 everywhere else. Then $\mathcal{A}'$ outputs $x = x_1 \cdots x_n$. Clearly $\mathcal{A}'$ runs in polynomial time.

In the execution of $\mathcal{A}'(1^n, f(\hat{x}))$, the value x_i computed by $\mathcal{A}'$ satisfies

$$x_i = \mathcal{A}(f(\hat{x}), e^i) = \text{gl}(\hat{x}, e^i) = \bigoplus_{j=1}^n \hat{x}_j \cdot e_j^i = \hat{x}_i.$$

Thus, $x_i = \hat{x}_i$ for all i and so $\mathcal{A}'$ outputs the correct inverse $x = \hat{x}$. ■

If f is one-way, it is impossible for any probabilistic polynomial-time algorithm to invert f with non-negligible probability. Thus, we conclude that there is no polynomial-time algorithm that always correctly computes $\text{gl}(x, r)$ from $(f(x), r)$. This is a rather weak result that is very far from our ultimate goal of showing that $\text{gl}(x, r)$ cannot be computed (with probability significantly better than $1/2$) given $(f(x), r)$.

7.3.2 A More Involved Case

We now show that it is hard for any probabilistic polynomial-time algorithm $\mathcal{A}$ to compute $\text{gl}(x, r)$ from $(f(x), r)$ with probability significantly better than $3/4$. We will again show that any such $\mathcal{A}$ would imply the existence of a polynomial-time algorithm $\mathcal{A}'$ that inverts f with non-negligible probability. Notice that the strategy in the proof of Proposition 7.13 fails here because it may be that $\mathcal{A}$ *never* succeeds when $r = e^i$ (although it may succeed, say, on all other values of r). Furthermore, in the present case $\mathcal{A}'$ does not know if the result $\mathcal{A}(f(x), r)$ is equal to $\text{gl}(x, r)$ or not; the only thing $\mathcal{A}'$ knows is that with high probability, algorithm $\mathcal{A}$ is correct. This further complicates the proof.

PROPOSITION 7.14 *Let f and gl be as in Theorem 7.12. If there exists a probabilistic polynomial-time algorithm $\mathcal{A}$ and a polynomial $p(\cdot)$ such that*

$$\Pr_{x, r \leftarrow \{0,1\}^n} [\mathcal{A}(f(x), r) = \text{gl}(x, r)] \geq \frac{3}{4} + \frac{1}{p(n)}$$

for infinitely many values of n , then there exists a probabilistic polynomial-time algorithm $\mathcal{A}'$ such that

$$\Pr_{x \leftarrow \{0,1\}^n} [\mathcal{A}'(1^n, f(x)) \in f^{-1}(f(x))] \geq \frac{1}{4 \cdot p(n)}$$

for infinitely many values of n .

PROOF The main observation underlying the proof of this proposition is that for every $r \in \{0, 1\}^n$, the values $\text{gl}(x, r \oplus e^i)$ and $\text{gl}(x, r)$ together can be used to compute the i th bit of x . (Recall that e^i denotes the n -bit string with 0s everywhere except the i th position.) This is true because

$$\begin{aligned} & \text{gl}(x, r) \oplus \text{gl}(x, r \oplus e^i) \\ &= \left(\bigoplus_{j=1}^n x_j \cdot r_j \right) \oplus \left(\bigoplus_{j=1}^n x_j \cdot (r_j \oplus e_j^i) \right) = x_i \cdot r_i \oplus (x_i \cdot \bar{r}_i) = x_i, \end{aligned}$$

where $\bar{r}_i$ is the complement of r_i , and the second equality is due to the fact that for $j \neq i$, the value $x_j \cdot r_j$ appears in both sums and so is canceled out.

The above demonstrates that if $\mathcal{A}$ answers correctly on both $(f(x), r)$ and $(f(x), r \oplus e^i)$, then $\mathcal{A}'$ can correctly compute x_i . Unfortunately, $\mathcal{A}'$ does not know when $\mathcal{A}$ answers correctly and when it does not; $\mathcal{A}'$ knows only that $\mathcal{A}$ answers correctly with “high” probability. For this reason, $\mathcal{A}'$ will use multiple random values of r , using each one to obtain an estimate of x_i , and will then take the estimate occurring a majority of the time as its final guess for x_i .

As a preliminary step, we show that for many x 's the probability that $\mathcal{A}$ answers correctly for both $(f(x), r)$ and $(f(x), r \oplus e^i)$, when r is uniform, is sufficiently high. This allows us to fix x and then focus solely on uniform choice of r , which makes the analysis easier.

CLAIM 7.15 *Let n be such that*

$$\Pr_{x, r \leftarrow \{0,1\}^n} [\mathcal{A}(f(x), r) = \text{gl}(x, r)] \geq \frac{3}{4} + \frac{1}{p(n)}.$$

Then there exists a set $S_n \subseteq \{0,1\}^n$ of size at least $\frac{1}{2p(n)} \cdot 2^n$ such that for every $x \in S_n$ it holds that

$$\Pr_{r \leftarrow \{0,1\}^n} [\mathcal{A}(f(x), r) = \text{gl}(x, r)] \geq \frac{3}{4} + \frac{1}{2p(n)}.$$

PROOF Let $\varepsilon(n) = 1/p(n)$, and define $S_n \subseteq \{0,1\}^n$ to be the set of all x 's for which

$$\Pr_{r \leftarrow \{0,1\}^n} [\mathcal{A}(f(x), r) = \text{gl}(x, r)] \geq \frac{3}{4} + \frac{\varepsilon(n)}{2}.$$

We have

$$\begin{aligned} \Pr_{x, r \leftarrow \{0,1\}^n} [\mathcal{A}(f(x), r) = \text{gl}(x, r)] &= \frac{1}{2^n} \sum_{x \in \{0,1\}^n} \Pr_{r \leftarrow \{0,1\}^n} [\mathcal{A}(f(x), r) = \text{gl}(x, r)] \\ &= \frac{1}{2^n} \sum_{x \in S_n} \Pr_{r \leftarrow \{0,1\}^n} [\mathcal{A}(f(x), r) = \text{gl}(x, r)] \\ &\quad + \frac{1}{2^n} \sum_{x \notin S_n} \Pr_{r \leftarrow \{0,1\}^n} [\mathcal{A}(f(x), r) = \text{gl}(x, r)] \\ &\leq \frac{|S_n|}{2^n} + \frac{1}{2^n} \cdot \sum_{x \notin S_n} \left(\frac{3}{4} + \frac{\varepsilon(n)}{2} \right) \\ &\leq \frac{|S_n|}{2^n} + \left(\frac{3}{4} + \frac{\varepsilon(n)}{2} \right). \end{aligned}$$

Since $\frac{3}{4} + \varepsilon(n) \leq \Pr_{x, r \leftarrow \{0,1\}^n} [\mathcal{A}(f(x), r) = \text{gl}(x, r)]$, straightforward algebra gives $|S_n| \geq \frac{\varepsilon(n)}{2} \cdot 2^n$. ■

The following now follows as an easy consequence.

CLAIM 7.16 *Let n be such that*

$$\Pr_{x, r \leftarrow \{0,1\}^n} [\mathcal{A}(f(x), r) = \text{gl}(x, r)] \geq \frac{3}{4} + \frac{1}{p(n)}.$$

Then there exists a set $S_n \subseteq \{0, 1\}^n$ of size at least $\frac{1}{2p(n)} \cdot 2^n$ such that for every $x \in S_n$ and every i it holds that

$$\Pr_{r \leftarrow \{0,1\}^n} \left[\mathcal{A}(f(x), r) = \text{gl}(x, r) \bigwedge \mathcal{A}(f(x), r \oplus e^i) = \text{gl}(x, r \oplus e^i) \right] \geq \frac{1}{2} + \frac{1}{p(n)}.$$

PROOF Let $\varepsilon(n) = 1/p(n)$, and take S_n to be the set guaranteed by the previous claim. We know that for any $x \in S_n$ we have

$$\Pr_{r \leftarrow \{0,1\}^n} [\mathcal{A}(f(x), r) \neq \text{gl}(x, r)] \leq \frac{1}{4} - \frac{\varepsilon(n)}{2}.$$

Fix $i \in \{1, \dots, n\}$. If r is uniformly distributed then so is $r \oplus e^i$; thus

$$\Pr_{r \leftarrow \{0,1\}^n} [\mathcal{A}(f(x), r \oplus e^i) \neq \text{gl}(x, r \oplus e^i)] \leq \frac{1}{4} - \frac{\varepsilon(n)}{2}.$$

We are interested in lower-bounding the probability that $\mathcal{A}$ outputs the correct answer for *both* $\text{gl}(x, r)$ and $\text{gl}(x, r \oplus e^i)$; equivalently, we want to upper-bound the probability that $\mathcal{A}$ fails to output the correct answer in *either* of these cases. Note that r and $r \oplus e^i$ are not independent, so we cannot just multiply the probabilities of failure. However, we can apply the union bound (see Proposition A.7) and sum the probabilities of failure. That is, the probability that $\mathcal{A}$ is *incorrect* on either $\text{gl}(x, r)$ or $\text{gl}(x, r \oplus e^i)$ is at most

$$\left(\frac{1}{4} - \frac{\varepsilon(n)}{2} \right) + \left(\frac{1}{4} - \frac{\varepsilon(n)}{2} \right) = \frac{1}{2} - \varepsilon(n),$$

and so $\mathcal{A}$ is correct on *both* $\text{gl}(x, r)$ and $\text{gl}(x, r \oplus e^i)$ with probability *at least* $1/2 + \varepsilon(n)$. This proves the claim. ■

For the rest of the proof we set $\varepsilon(n) = 1/p(n)$ and consider only those values of n for which

$$\Pr_{x, r \leftarrow \{0,1\}^n} [\mathcal{A}(f(x), r) = \text{gl}(x, r)] \geq \frac{3}{4} + \varepsilon(n). \quad (7.1)$$

The previous claim states that for an $\varepsilon(n)/2$ fraction of inputs x , and any i , algorithm $\mathcal{A}$ answers correctly on both $(f(x), r)$ and $(f(x), r \oplus e^i)$ with probability at least $1/2 + \varepsilon(n)$ over uniform choice of r , and from now on we focus only on such values of x . We construct a probabilistic polynomial-time algorithm $\mathcal{A}'$ that inverts $f(x)$ with probability at least $1/2$ when $x \in S_n$. This suffices to prove Proposition 7.14 since then, for infinitely many values of n ,

$$\begin{aligned} & \Pr_{x \leftarrow \{0,1\}^n} [\mathcal{A}'(1^n, f(x)) \in f^{-1}(f(x))] \\ & \geq \Pr_{x \leftarrow \{0,1\}^n} [\mathcal{A}'(1^n, f(x)) \in f^{-1}(f(x)) \mid x \in S_n] \cdot \Pr_{x \leftarrow \{0,1\}^n} [x \in S_n] \\ & \geq \frac{1}{2} \cdot \frac{\varepsilon(n)}{2} = \frac{1}{4p(n)}. \end{aligned}$$

Algorithm $\mathcal{A}'$, given as input 1^n and y , works as follows:

1. For $i = 1, \dots, n$ do:
 - Repeatedly choose a uniform $r \in \{0, 1\}^n$ and compute $\mathcal{A}(y, r) \oplus \mathcal{A}(y, r \oplus e^i)$ as an “estimate” for the i th bit of the preimage of y . After doing this sufficiently many times (see below), let x_i be the “estimate” that occurs a majority of the time.
2. Output $x = x_1 \cdots x_n$.

We sketch an analysis of the probability that $\mathcal{A}'$ correctly inverts its given input y . (We allow ourselves to be a bit laconic, since a full proof for a more difficult case is given in the following section.) Say $y = f(\hat{x})$ and recall that we assume here that n is such that Equation (7.1) holds and $\hat{x} \in S_n$. Fix some i . The previous claim implies that the estimate $\mathcal{A}(y, r) \oplus \mathcal{A}(y, r \oplus e^i)$ is equal to $\text{gl}(\hat{x}, e^i)$ with probability at least $\frac{1}{2} + \varepsilon(n)$ over choice of r . By obtaining sufficiently many estimates and letting x_i be the majority value, $\mathcal{A}'$ can ensure that x_i is equal to $\text{gl}(\hat{x}, e^i)$ with probability at least $1 - \frac{1}{2n}$. Of course, we need to make sure that polynomially many estimates are enough. Fortunately, since $\varepsilon(n) = 1/p(n)$ for some polynomial p and an independent value of r is used for obtaining each estimate, the *Chernoff bound* (cf. Proposition A.14) shows that polynomially many estimates suffice.

Summarizing, we have that for each i the value x_i computed by $\mathcal{A}'$ is incorrect with probability at most $\frac{1}{2n}$. A union bound thus shows that $\mathcal{A}'$ is incorrect for *some* i with probability at most $n \cdot \frac{1}{2n} = \frac{1}{2}$. That is, $\mathcal{A}'$ is correct for all i —and thus correctly inverts y —with probability at least $1 - \frac{1}{2} = \frac{1}{2}$. This completes the proof of Proposition 7.14. ■

A corollary of Proposition 7.14 is that if f is a one-way function, then for any polynomial-time algorithm $\mathcal{A}$ the probability that $\mathcal{A}$ correctly guesses $\text{gl}(x, r)$ when given $(f(x), r)$ is at most negligibly more than $3/4$.

7.3.3 The Full Proof

We assume familiarity with the simplified proofs in the previous sections, and build on the ideas developed there. We rely on some terminology and standard results from probability theory discussed in Appendix A.3.

We prove the following proposition, which implies Theorem 7.12:

PROPOSITION 7.17 *Let f and gl be as in Theorem 7.12. If there exists a probabilistic polynomial-time algorithm $\mathcal{A}$ and a polynomial $p(\cdot)$ such that*

$$\Pr_{x, r \leftarrow \{0, 1\}^n} [\mathcal{A}(f(x), r) = \text{gl}(x, r)] \geq \frac{1}{2} + \frac{1}{p(n)}$$

for infinitely many values of n , then there exists a probabilistic polynomial-time algorithm $\mathcal{A}'$ and a polynomial $p'(\cdot)$ such that

$$\Pr_{x \leftarrow \{0,1\}^n} [\mathcal{A}'(1^n, f(x)) \in f^{-1}(f(x))] \geq \frac{1}{p'(n)}$$

for infinitely many values of n .

PROOF Once again we set $\varepsilon(n) = 1/p(n)$ and consider only those values of n for which $\Pr_{x,r \leftarrow \{0,1\}^n} [\mathcal{A}(f(x), r) = \text{gl}(x, r)] \geq \frac{1}{2} + \frac{1}{p(n)}$. The following is analogous to Claim 7.15 and is proved in the same way.

CLAIM 7.18 *Let n be such that*

$$\Pr_{x,r \leftarrow \{0,1\}^n} [\mathcal{A}(f(x), r) = \text{gl}(x, r)] \geq \frac{1}{2} + \varepsilon(n).$$

Then there exists a set $S_n \subseteq \{0,1\}^n$ of size at least $\frac{\varepsilon(n)}{2} \cdot 2^n$ such that for every $x \in S_n$ it holds that

$$\Pr_{r \leftarrow \{0,1\}^n} [\mathcal{A}(f(x), r) = \text{gl}(x, r)] \geq \frac{1}{2} + \frac{\varepsilon(n)}{2}. \quad (7.2)$$

If we start by trying to prove an analogue of Claim 7.16, the best we can claim here is that when $x \in S_n$ we have

$$\Pr_{r \leftarrow \{0,1\}^n} [\mathcal{A}(f(x), r) = \text{gl}(x, r) \bigwedge \mathcal{A}(f(x), r \oplus e^i) = \text{gl}(x, r \oplus e^i)] \geq \varepsilon(n)$$

for any i . Thus, if we try to use $\mathcal{A}(f(x), r) \oplus \mathcal{A}(f(x), r \oplus e^i)$ as an estimate for x_i , all we can claim is that this estimate will be correct with probability at least $\varepsilon(n)$, which may not be any better than taking a random guess! We cannot claim that flipping the result gives a good estimate, either.

Instead, we design $\mathcal{A}'$ so that it computes $\text{gl}(x, r)$ and $\text{gl}(x, r \oplus e^i)$ by invoking $\mathcal{A}$ only once. We do this by having $\mathcal{A}'$ run $\mathcal{A}(x, r \oplus e^i)$, and having $\mathcal{A}'$ simply “guess” the value $\text{gl}(x, r)$ itself. The naive way to do this would be to choose the r ’s independently, as before, and to have $\mathcal{A}'$ make an independent guess of $\text{gl}(x, r)$ for each value of r . But then the probability that all such guesses are correct—which, as we will see, is necessary if $\mathcal{A}'$ is to output the correct inverse—would be negligible because polynomially many r ’s are used.

The crucial observation of the present proof is that $\mathcal{A}'$ can generate the r ’s in a *pairwise-independent* manner and make its guesses in a particular way so that with non-negligible probability all its guesses are correct. Specifically, in order to generate m values of r , we have $\mathcal{A}'$ select $\ell = \lceil \log(m+1) \rceil$ independent and uniformly distributed strings $s^1, \dots, s^\ell \in \{0,1\}^n$. Then, for every nonempty subset $I \subseteq \{1, \dots, \ell\}$, we set $r^I := \oplus_{i \in I} s^i$. Since there are $2^\ell - 1$

nonempty subsets, this defines a collection of $2^{\lceil \log(m+1) \rceil} - 1 \geq m$ strings. Each such string is uniformly distributed. The strings are not independent, but they *are* pairwise independent. To see this, notice that for every two subsets $I \neq J$ there is an index $j \in I \cup J$ such that $j \notin I \cap J$. Without loss of generality, assume $j \notin I$. Then the value of s^j is uniform and independent of the value of r^I . Since s^j is included in the XOR that defines r^J , this implies that r^J is uniform and independent of r^I as well.

We now have the following two important observations:

1. Given $\text{gl}(x, s^1), \dots, \text{gl}(x, s^\ell)$, it is possible to compute $\text{gl}(x, r^I)$ for every subset $I \subseteq \{1, \dots, \ell\}$. This is because

$$\text{gl}(x, r^I) = \text{gl}(x, \oplus_{i \in I} s^i) = \oplus_{i \in I} \text{gl}(x, s^i).$$

2. If $\mathcal{A}'$ simply guesses the values of $\text{gl}(x, s^1), \dots, \text{gl}(x, s^\ell)$ by choosing a uniform bit for each, then *all* these guesses will be correct with probability $1/2^\ell$. If m is polynomial in the security parameter n , then $1/2^\ell$ is not negligible, and so with *non-negligible probability* $\mathcal{A}'$ correctly guesses all the values $\text{gl}(x, s^1), \dots, \text{gl}(x, s^\ell)$.

Combining the above yields a way of obtaining $m = \text{poly}(n)$ uniform and pairwise-independent strings $\{r^I\}$ along with *correct* values for $\{\text{gl}(x, r^I)\}$ with non-negligible probability. These values can then be used to compute x_i in the same way as in the proof of Proposition 7.14. Details follow.

The inversion algorithm $\mathcal{A}'$. We now provide a full description of an algorithm $\mathcal{A}'$ that receives inputs $1^n, y$ and tries to compute an inverse of y . The algorithm proceeds as follows:

1. Set $\ell := \lceil \log(2n/\varepsilon(n)^2 + 1) \rceil$.
2. Choose uniform, independent $s^1, \dots, s^\ell \in \{0, 1\}^n$ and $\sigma^1, \dots, \sigma^\ell \in \{0, 1\}$.
3. For every nonempty subset $I \subseteq \{1, \dots, \ell\}$, compute $r^I := \oplus_{i \in I} s^i$ and $\sigma^I := \oplus_{i \in I} \sigma^i$.
4. For $i = 1, \dots, n$ do:

- (a) For every nonempty subset $I \subseteq \{1, \dots, \ell\}$, set

$$x_i^I := \sigma^I \oplus \mathcal{A}(y, r^I \oplus e^i).$$

- (b) Set $x_i := \text{majority}_I\{x_i^I\}$ (i.e., take the bit that appeared a majority of the time in the previous step).

5. Output $x = x_1 \cdots x_n$.

It remains to compute the probability that $\mathcal{A}'$ outputs $x \in f^{-1}(y)$. As in the proof of Proposition 7.14, we focus only on n as in Claim 7.18 and assume $y = f(\hat{x})$ for some $\hat{x} \in S_n$. Each σ^i represents a “guess” for the value of $\text{gl}(\hat{x}, s^i)$. As noted earlier, with non-negligible probability all these guesses are correct; we show that conditioned on this event, $\mathcal{A}'$ outputs $x = \hat{x}$ with probability at least $1/2$.

Assume $\sigma^i = \text{gl}(\hat{x}, s^i)$ for all i . Then $\sigma^I = \text{gl}(\hat{x}, r^I)$ for all I . Fix an index $i \in \{1, \dots, n\}$ and consider the probability that $\mathcal{A}'$ obtains the correct value $x_i = \hat{x}_i$. For any nonempty I we have $\mathcal{A}(y, r^I \oplus e^i) = \text{gl}(\hat{x}, r^I \oplus e^i)$ with probability at least $\frac{1}{2} + \varepsilon(n)/2$ over choice of r ; this follows because $\hat{x} \in S_n$ and $r^I \oplus e^i$ is uniformly distributed. Thus, for any nonempty subset I we have $\Pr[x_i^I = \hat{x}_i] \geq \frac{1}{2} + \varepsilon(n)/2$. Moreover, the $\{x_i^I\}_{I \subseteq \{1, \dots, \ell\}}$ are pairwise independent because the $\{r^I\}_{I \subseteq \{1, \dots, \ell\}}$ (and hence the $\{r^I \oplus e^i\}_{I \subseteq \{1, \dots, \ell\}}$) are pairwise independent. Since x_i is defined to be the value that occurs a majority of the time among the $\{x_i^I\}_{I \subseteq \{1, \dots, \ell\}}$, we can apply Proposition A.13 to obtain

$$\begin{aligned} \Pr[x_i \neq \hat{x}_i] &\leq \frac{1}{4 \cdot (\varepsilon(n)/2)^2 \cdot (2^\ell - 1)} \\ &\leq \frac{1}{4 \cdot (\varepsilon(n)/2)^2 \cdot (2n/\varepsilon(n)^2)} \\ &= \frac{1}{2n}. \end{aligned}$$

The above holds for all i , so by applying a union bound we see that the probability that $x_i \neq \hat{x}_i$ for *some* i is at most $1/2$. That is, $x_i = \hat{x}_i$ for *all* i (and hence $x = \hat{x}$) with probability at least $1/2$.

Putting everything together: Let n be as in Claim 7.18 and $y = f(\hat{x})$. With probability at least $\varepsilon(n)/2$ we have $\hat{x} \in S_n$. All the guesses σ^i are correct with probability at least

$$\frac{1}{2^\ell} \geq \frac{1}{2 \cdot (2n/\varepsilon(n)^2 + 1)} > \frac{\varepsilon(n)^2}{5n}$$

for n sufficiently large. Conditioned on both the above, $\mathcal{A}'$ outputs $x = \hat{x}$ with probability at least $1/2$. The overall probability with which $\mathcal{A}'$ inverts its input is thus at least $\varepsilon(n)^3/20n = 1/(20np(n)^3)$ for infinitely many n . Since $20np(n)^3$ is polynomial in n , this proves Proposition 7.17. $\blacksquare$

7.4 Constructing Pseudorandom Generators

We first show how to construct pseudorandom generators that stretch their input by a single bit, under the assumption that one-way *permutations* exist. We then show how to extend this to obtain any polynomial expansion factor.

7.4.1 Pseudorandom Generators with Minimal Expansion

Let f be a one-way permutation with hard-core predicate hc . This means that $\text{hc}(s)$ “looks random” given $f(s)$, when s is uniform. Furthermore, since f is a permutation, $f(s)$ itself is uniformly distributed. (Applying a permutation to a uniformly distributed value yields a uniformly distributed value.) So if s is a uniform n -bit string, the $(n+1)$ -bit string $f(s)\|\text{hc}(s)$ consists of a uniform n -bit string plus on additional bit that looks uniform even conditioned on the initial n bits; in other words, this $(n+1)$ -bit string is *pseudorandom*. Thus, the algorithm G defined by $G(s) = f(s)\|\text{hc}(s)$ is a pseudorandom generator.

THEOREM 7.19 *Let f be a one-way permutation with hard-core predicate hc . Then algorithm G defined by $G(s) = f(s)\|\text{hc}(s)$ is a pseudorandom generator with expansion factor $\ell(n) = n+1$.*

PROOF Let D be a probabilistic polynomial-time algorithm. We prove that there is a negligible function negl such that

$$\Pr_{r \leftarrow \{0,1\}^{n+1}}[D(r) = 1] - \Pr_{s \leftarrow \{0,1\}^n}[D(G(s)) = 1] \leq \text{negl}(n). \quad (7.3)$$

A similar argument shows that there is a negligible function negl' for which

$$\Pr_{s \leftarrow \{0,1\}^n}[D(G(s)) = 1] - \Pr_{r \leftarrow \{0,1\}^{n+1}}[D(r) = 1] \leq \text{negl}'(n),$$

which completes the proof.

Observe first that

$$\begin{aligned} \Pr_{r \leftarrow \{0,1\}^{n+1}}[D(r) = 1] &= \Pr_{r \leftarrow \{0,1\}^n, r' \leftarrow \{0,1\}}[D(r\|r') = 1] \\ &= \Pr_{s \leftarrow \{0,1\}^n, r' \leftarrow \{0,1\}}[D(f(s)\|r') = 1] \\ &= \frac{1}{2} \cdot \Pr_{s \leftarrow \{0,1\}^n}[D(f(s)\|\text{hc}(s)) = 1] \\ &\quad + \frac{1}{2} \cdot \Pr_{s \leftarrow \{0,1\}^n}[D(f(s)\|\overline{\text{hc}}(s)) = 1], \end{aligned}$$

using the fact that f is a permutation for the second equality, and that a uniform bit r' is equal to $\text{hc}(s)$ with probability exactly $1/2$ for the third equality. Since

$$\Pr_{s \leftarrow \{0,1\}^n}[D(G(s)) = 1] = \Pr_{s \leftarrow \{0,1\}^n}[D(f(s)\|\text{hc}(s)) = 1]$$

(by definition of G), this means that Equation (7.3) is equivalent to

$$\frac{1}{2} \cdot \left(\Pr_{s \leftarrow \{0,1\}^n}[D(f(s)\|\overline{\text{hc}}(s)) = 1] - \Pr_{s \leftarrow \{0,1\}^n}[D(f(s)\|\text{hc}(s)) = 1] \right) \leq \text{negl}(n).$$

Consider the following algorithm $\mathcal{A}$ that is given as input a value $y = f(s)$ and tries to predict the value of $\text{hc}(s)$:

1. Choose uniform $r' \in \{0, 1\}$.
2. Run $D(y\|r')$. If D outputs 0, output r' ; otherwise output $\bar{r}'$.

Clearly $\mathcal{A}$ runs in polynomial time. By definition of $\mathcal{A}$, we have

$$\begin{aligned}
 & \Pr_{s \leftarrow \{0,1\}^n} [\mathcal{A}(f(s)) = \text{hc}(s)] \\
 &= \frac{1}{2} \cdot \Pr_{s \leftarrow \{0,1\}^n} [\mathcal{A}(f(s)) = \text{hc}(s) \mid r' = \text{hc}(s)] \\
 &\quad + \frac{1}{2} \cdot \Pr_{s \leftarrow \{0,1\}^n} [\mathcal{A}(f(s)) = \text{hc}(s) \mid r' \neq \text{hc}(s)] \\
 &= \frac{1}{2} \cdot \left(\Pr_{s \leftarrow \{0,1\}^n} [D(f(s)\|\text{hc}(s)) = 0] + \Pr_{s \leftarrow \{0,1\}^n} [D(f(s)\|\bar{\text{hc}}(s)) = 1] \right) \\
 &= \frac{1}{2} \cdot \left(\left(1 - \Pr_{s \leftarrow \{0,1\}^n} [D(f(s)\|\text{hc}(s)) = 1] \right) + \Pr_{s \leftarrow \{0,1\}^n} [D(f(s)\|\bar{\text{hc}}(s)) = 1] \right) \\
 &= \frac{1}{2} + \frac{1}{2} \cdot \left(\Pr_{s \leftarrow \{0,1\}^n} [D(f(s)\|\bar{\text{hc}}(s)) = 1] - \Pr_{s \leftarrow \{0,1\}^n} [D(f(s)\|\text{hc}(s)) = 1] \right).
 \end{aligned}$$

Since hc is a hard-core predicate of f , it follows that there exists a negligible function negl for which

$$\frac{1}{2} \cdot \left(\Pr_{s \leftarrow \{0,1\}^n} [D(f(s)\|\bar{\text{hc}}(s)) = 1] - \Pr_{s \leftarrow \{0,1\}^n} [D(f(s)\|\text{hc}(s)) = 1] \right) \leq \text{negl}(n),$$

as desired. ■

7.4.2 Increasing the Expansion Factor

We now show that the expansion factor of a pseudorandom generator can be increased by any desired (polynomial) amount. This means that the previous construction, with expansion factor $\ell(n) = n + 1$, suffices for constructing a pseudorandom generator with arbitrary (polynomial) expansion factor.

THEOREM 7.20 *If there exists a pseudorandom generator G with expansion factor $n + 1$, then for any polynomial poly there exists a pseudorandom generator $\hat{G}$ with expansion factor $\text{poly}(n)$.*

PROOF We first consider constructing a pseudorandom generator $\hat{G}$ that outputs $n + 2$ bits. $\hat{G}$ works as follows: Given an initial seed $s \in \{0, 1\}^n$, it

computes $t_1 := G(s)$ to obtain $n + 1$ pseudorandom bits. The initial n bits of t_1 are then used again as a seed for G ; the resulting $n + 1$ bits, concatenated with the final bit of t_1 , yield the $(n + 2)$ -bit output. (See Figure 7.1.) The second application of G uses a pseudorandom seed rather than a random one. The proof of security we give next shows that this does not impact the pseudorandomness of the output.

We now prove that $\hat{G}$ is a pseudorandom generator. Define three sequences of distributions $\{H_n^0\}_{n=1,\dots}$, $\{H_n^1\}_{n=1,\dots}$, and $\{H_n^2\}_{n=1,\dots}$, where each of H_n^0 , H_n^1 , and H_n^2 is a distribution on strings of length $n + 2$. In distribution H_n^0 , a uniform string $t_0 \in \{0, 1\}^n$ is chosen and the output is $\hat{G}(t_0)$. In distribution H_n^1 , a uniform string $t_1 \in \{0, 1\}^{n+1}$ is chosen and parsed as $s_1 \parallel \sigma_1$ (where s_1 are the initial n bits of t_1 and σ_1 is the final bit). The output is $t_2 := G(s_1) \parallel \sigma_1$. In distribution H_n^2 , the output is a uniform string $t_2 \in \{0, 1\}^{n+2}$. We denote by $t_2 \leftarrow H_n^i$ the process of generating an $(n + 2)$ -bit string t_2 according to distribution H_n^i .

Fix an arbitrary probabilistic polynomial-time distinguisher D . We first claim that there is a negligible function negl' such that

$$\left| \Pr_{t_2 \leftarrow H_n^0} [D(t_2) = 1] - \Pr_{t_2 \leftarrow H_n^1} [D(t_2) = 1] \right| \leq \text{negl}'(n). \quad (7.4)$$

To see this, consider the polynomial-time distinguisher D' that, on input $t_1 \in \{0, 1\}^{n+1}$, parses t_1 as $s_1 \parallel \sigma_1$ with $|s_1| = n$, computes $t_2 := G(s_1) \parallel \sigma_1$, and outputs $D(t_2)$. Clearly D' runs in polynomial time. Observe that:

1. If t_1 is uniform, the distribution on t_2 generated by D' is exactly that of distribution H_n^1 . Thus,

$$\Pr_{t_1 \leftarrow \{0,1\}^{n+1}} [D'(t_1) = 1] = \Pr_{t_2 \leftarrow H_n^1} [D(t_2) = 1].$$

2. If $t_1 = G(s)$ for uniform $s \in \{0, 1\}^n$, the distribution on t_2 generated by D' is exactly that of distribution H_n^0 . That is,

$$\Pr_{s \leftarrow \{0,1\}^n} [D'(G(s)) = 1] = \Pr_{t_2 \leftarrow H_n^0} [D(t_2) = 1].$$

Pseudorandomness of G implies that there is a negligible function negl' with

$$\left| \Pr_{s \leftarrow \{0,1\}^n} [D'(G(s)) = 1] - \Pr_{t_1 \leftarrow \{0,1\}^{n+1}} [D'(t_1) = 1] \right| \leq \text{negl}'(n).$$

Equation (7.4) follows.

We next claim that there is a negligible function negl'' such that

$$\left| \Pr_{t_2 \leftarrow H_n^1} [D(t_2) = 1] - \Pr_{t_2 \leftarrow H_n^2} [D(t_2) = 1] \right| \leq \text{negl}''(n). \quad (7.5)$$

To see this, consider the polynomial-time distinguisher D'' that, on input $w \in \{0, 1\}^{n+1}$, chooses uniform $\sigma_1 \in \{0, 1\}$, sets $t_2 := w \| \sigma_1$, and outputs $D(t_2)$. If w is uniform then so is t_2 ; thus,

$$\Pr_{w \leftarrow \{0,1\}^{n+1}} [D''(w) = 1] = \Pr_{t_2 \leftarrow H_n^2} [D(t_2) = 1].$$

On the other hand, if $w = G(s)$ for uniform $s \in \{0, 1\}^n$, then t_2 is distributed exactly according to H_n^1 and so

$$\Pr_{s \leftarrow \{0,1\}^n} [D''(G(s)) = 1] = \Pr_{t_2 \leftarrow H_n^1} [D(t_2) = 1].$$

As before, pseudorandomness of G implies Equation (7.5).

Putting everything together, we have

$$\begin{aligned} & \left| \Pr_{s \leftarrow \{0,1\}^n} [D(\hat{G}(s)) = 1] - \Pr_{r \leftarrow \{0,1\}^{n+2}} [D(r) = 1] \right| \\ &= \left| \Pr_{t_2 \leftarrow H_n^0} [D(t_2) = 1] - \Pr_{t_2 \leftarrow H_n^2} [D(t_2) = 1] \right| \\ &\leq \left| \Pr_{t_2 \leftarrow H_n^0} [D(t_2) = 1] - \Pr_{t_2 \leftarrow H_n^1} [D(t_2) = 1] \right| \\ &\quad + \left| \Pr_{t_2 \leftarrow H_n^1} [D(t_2) = 1] - \Pr_{t_2 \leftarrow H_n^2} [D(t_2) = 1] \right| \\ &\leq \text{negl}'(n) + \text{negl}''(n), \end{aligned} \tag{7.6}$$

using Equations (7.4) and (7.5). Since D was an arbitrary polynomial-time distinguisher, this proves that $\hat{G}$ is a pseudorandom generator.

The general case. The same idea as above can be iteratively applied to generate as many pseudorandom bits as desired. Formally, say we wish to construct a pseudorandom generator $\hat{G}$ with expansion factor $n + p(n)$, for some polynomial p . On input $s \in \{0, 1\}^n$, algorithm $\hat{G}$ does (cf. Figure 7.1):

1. Set $t_0 := s$. For $i = 1, \dots, p(n)$ do:
 - (a) Let s_{i-1} be the first n bits of t_{i-1} , and let σ_{i-1} denote the remaining $i-1$ bits. (When $i = 1$, $s_0 = t_0$ and σ_0 is the empty string.)
 - (b) Set $t_i := G(s'_{i-1}) \| \sigma_{i-1}$.
2. Output $t_{p(n)}$.

We show that $\hat{G}$ is a pseudorandom generator. The proof uses a common technique known as a *hybrid argument*. (Actually, even the case of $p(n) = 2$, above, used a simple hybrid argument.) The main difference with respect to the previous proof is a technical one. Previously, we could define and explicitly work with three sequences of distributions $\{H_n^0\}$, $\{H_n^1\}$, and $\{H_n^2\}$. Here that is not possible since the number of distributions to consider grows with n .

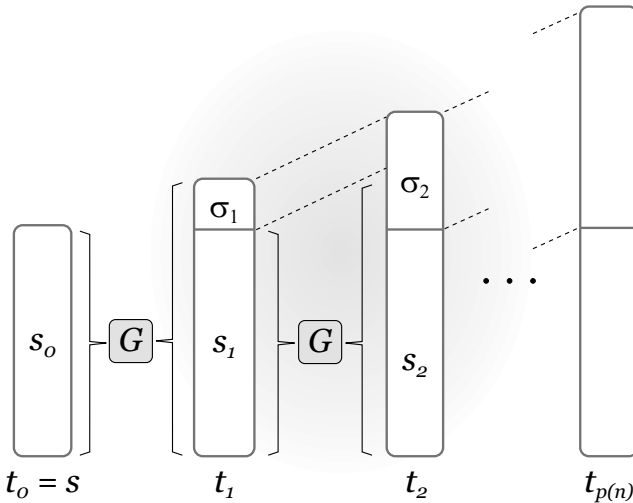


FIGURE 7.1: Increasing the expansion of a pseudorandom generator.

For any n and $0 \leq j \leq p(n)$, let H_n^j be the distribution on strings of length $n + p(n)$ defined as follows: choose uniform $t_j \in \{0, 1\}^{n+j}$, then run $\hat{G}$ starting from iteration $j + 1$ and output $t_{p(n)}$. (When $j = p(n)$ this means we simply choose uniform $t_{p(n)} \in \{0, 1\}^{n+p(n)}$ and output it.) The crucial observation is that H_n^0 corresponds to outputting $\hat{G}(s)$ for uniform $s \in \{0, 1\}^n$, while $H_n^{p(n)}$ corresponds to outputting a uniform $(n + p(n))$ -bit string. Fixing any polynomial-time distinguisher D , this means that

$$\begin{aligned} & \left| \Pr_{s \leftarrow \{0, 1\}^n} [D(\hat{G}(s)) = 1] - \Pr_{r \leftarrow \{0, 1\}^{n+p(n)}} [D(r) = 1] \right| \\ &= \left| \Pr_{t \leftarrow H_n^0} [D(t) = 1] - \Pr_{t \leftarrow H_n^{p(n)}} [D(t) = 1] \right|. \end{aligned} \quad (7.7)$$

We prove the above is negligible, hence $\hat{G}$ is a pseudorandom generator.

Fix D as above, and consider the distinguisher D' that does the following when given a string $w \in \{0, 1\}^{n+1}$ as input:

1. Choose uniform $j \in \{1, \dots, p(n)\}$.
2. Choose uniform $\sigma'_j \in \{0, 1\}^{j-1}$. (When $j = 1$ then σ'_j is the empty string.)
3. Set $t_j := w \parallel \sigma'_j$. Then run $\hat{G}$ starting from iteration $j + 1$ to compute $t_{p(n)} \in \{0, 1\}^{n+p(n)}$. Output $D(t_{p(n)})$.

Clearly D' runs in polynomial time. Analyzing the behavior of D' is more complicated than before, although the underlying ideas are the same. Fix n

and say D' chooses $j = j^*$. If w is uniform, then t_{j^*} is uniform and so the distribution on $t \stackrel{\text{def}}{=} t_{p(n)}$ is exactly that of distribution $H_n^{j^*}$. That is,

$$\Pr_{w \leftarrow \{0,1\}^{n+1}} [D'(w) = 1 \mid j = j^*] = \Pr_{t \leftarrow H_n^{j^*}} [D(t) = 1].$$

Since each value for j is chosen with equal probability,

$$\begin{aligned} \Pr_{w \leftarrow \{0,1\}^{n+1}} [D'(w) = 1] &= \frac{1}{p(n)} \cdot \sum_{j^*=1}^{p(n)} \Pr_{w \leftarrow \{0,1\}^{n+1}} [D'(w) = 1 \mid j = j^*] \\ &= \frac{1}{p(n)} \cdot \sum_{j^*=1}^{p(n)} \Pr_{t \leftarrow H_n^{j^*}} [D(t) = 1]. \end{aligned} \quad (7.8)$$

On the other hand, say D' chooses $j = j^*$ and $w = G(s)$ for uniform $s \in \{0,1\}^n$. Defining $t_{j^*-1} = s \parallel \sigma'_{j^*}$, we see that t_{j^*-1} is uniform and so the experiment involving D' is equivalent to running $\hat{G}$ from iteration j^* to compute $t_{p(n)}$. That is, the distribution on $t \stackrel{\text{def}}{=} t_{p(n)}$ is now exactly that of distribution $H_n^{j^*-1}$, and so

$$\Pr_{s \leftarrow \{0,1\}^n} [D'(G(s)) = 1 \mid j = j^*] = \Pr_{t \leftarrow H_n^{j^*-1}} [D(t) = 1].$$

Therefore,

$$\begin{aligned} \Pr_{s \leftarrow \{0,1\}^n} [D'(\hat{G}(s)) = 1] &= \frac{1}{p(n)} \cdot \sum_{j^*=1}^{p(n)} \Pr_{s \leftarrow \{0,1\}^n} [D'(G(s)) = 1 \mid j = j^*] \\ &= \frac{1}{p(n)} \cdot \sum_{j^*=1}^{p(n)} \Pr_{t \leftarrow H_n^{j^*-1}} [D(t) = 1] \\ &= \frac{1}{p(n)} \cdot \sum_{j^*=0}^{p(n)-1} \Pr_{t \leftarrow H_n^{j^*}} [D(t) = 1]. \end{aligned} \quad (7.9)$$

We can now analyze how well D' distinguishes outputs of G from random:

$$\begin{aligned} &\left| \Pr_{s \leftarrow \{0,1\}^n} [D'(G(s)) = 1] - \Pr_{w \leftarrow \{0,1\}^{n+1}} [D'(w) = 1] \right| \quad (7.10) \\ &= \frac{1}{p(n)} \cdot \left| \sum_{j^*=0}^{p(n)-1} \Pr_{t \leftarrow H_n^{j^*}} [D(t) = 1] - \sum_{j^*=1}^{p(n)} \Pr_{t \leftarrow H_n^{j^*}} [D(t) = 1] \right| \\ &= \frac{1}{p(n)} \cdot \left| \Pr_{t \leftarrow H_n^0} [D(t) = 1] - \Pr_{t \leftarrow H_n^{p(n)}} [D(t) = 1] \right|, \end{aligned}$$

relying on Equations (7.8) and (7.9) for the first equality. (The second equality holds because the same terms are included in each sum, except for the first term of the left sum and the last term of the right sum.) Since G is a pseudorandom generator, the term on the left-hand side of Equation (7.10) is negligible; because p is polynomial, this implies that Equation (7.7) is negligible, completing the proof that $\hat{G}$ is a pseudorandom generator. ■

Putting it all together. Let f be a one-way permutation. Taking the pseudorandom generator with expansion factor $n + 1$ from Theorem 7.19, and increasing the expansion factor to $n + \ell$ using the approach from the proof of Theorem 7.20, we obtain the following pseudorandom generator $\hat{G}$:

$$\hat{G}(s) = f^{(\ell)}(s) \parallel \text{hc}(f^{(\ell-1)}(s)) \parallel \cdots \parallel \text{hc}(s),$$

where $f^{(i)}(s)$ refers to i -fold iteration of f . Note that $\hat{G}$ uses ℓ evaluations of f , and generates one pseudorandom bit per evaluation using the hard-core predicate hc .

Connection to stream ciphers. Recall from Section 3.3.1 that a stream cipher (without an IV) is defined by algorithms (Init , GetBits), where Init takes a seed $s \in \{0, 1\}^n$ and returns initial state st , and GetBits takes as input the current state st and outputs a bit σ and updated state st' . The construction $\hat{G}$ from the preceding proof fits nicely into this paradigm: take Init to be the trivial algorithm that outputs $\text{st} = s$, and define $\text{GetBits}(\text{st})$ to compute $G(\text{st})$, parse the result as $\text{st}' \parallel \sigma$ with $|\text{st}'| = n$, and output the bit σ and updated state st' . (If we use this stream cipher to generate $p(n)$ output bits starting from seed s , then we get exactly the final $p(n)$ bits of $\hat{G}(s)$ in reverse order.) The preceding proof shows that this yields a pseudorandom generator.

Hybrid arguments. A *hybrid argument* is a basic tool for proving indistinguishability when a basic primitive is (or several different primitives are) applied multiple times. Somewhat informally, the technique works by defining a series of intermediate “hybrid distributions” that bridge between two “extreme distributions” that we wish to prove indistinguishable. (In the proof above, these extreme distributions correspond to the output of $\hat{G}$ and a random string.) To apply the proof technique, three conditions should hold. First, the extreme distributions should match the original cases of interest. (In the proof above, H_n^0 was equal to the distribution induced by $\hat{G}$, while $H_n^{p(n)}$ was the uniform distribution.) Second, it must be possible to translate the capability of distinguishing consecutive hybrid distributions into breaking some underlying assumption. (Above, we essentially showed that distinguishing H_n^i from H_n^{i+1} was equivalent to distinguishing the output of G from random.) Finally, the number of hybrid distributions should be polynomial. See also Theorem 7.32.

7.5 Constructing Pseudorandom Functions

We now show how to construct a pseudorandom function from any (length-doubling) pseudorandom generator. Recall that a pseudorandom function is an efficiently computable, keyed function F that is indistinguishable from a truly random function in the sense described in Section 3.5.1. For simplicity, we restrict our attention here to the case where F is length preserving, meaning that for $k \in \{0, 1\}^n$ the function F_k maps n -bit inputs to n -bit outputs. A (length-preserving) pseudorandom function can be viewed, informally, as a pseudorandom generator with expansion factor $n \cdot 2^n$; given such a pseudorandom generator G we could define $F_k(i)$ (for $0 \leq i < 2^n$) to be the i th n -bit block of $G(k)$. The reason this does not work is that F must be efficiently computable; there are exponentially many blocks, and we need a way to compute the i th block without having to compute all other blocks.

We will do this by computing “blocks” of the output by walking down a binary tree. We exemplify the construction by first showing a pseudorandom function taking 2-bit inputs. Let G be a pseudorandom generator with expansion factor $2n$. If we use G as in the proof of Theorem 7.20 we can obtain a pseudorandom generator $\hat{G}$ with expansion factor $4n$ that uses three invocations of G . (We produce n additional pseudorandom bits each time G is applied.) If we define $F'_k(i)$ (where $0 \leq i < 4$ and i is encoded as a 2-bit binary string) to be the i th block of $\hat{G}(k)$, then computation of $F'_k(3)$ would require computing all of $\hat{G}$ and hence three invocations of G . We show how to construct a pseudorandom function F using only two invocations of G on any input.

Let G_0 and G_1 be functions denoting the first and second halves of the output of G ; i.e., $G(k) = G_0(k) \parallel G_1(k)$ where $|G_0(k)| = |G_1(k)| = |k|$. Define F as follows:

$$\begin{aligned} F_k(00) &= G_0(G_0(k)) & F_k(10) &= G_0(G_1(k)) \\ F_k(01) &= G_1(G_0(k)) & F_k(11) &= G_1(G_1(k)). \end{aligned}$$

We claim that the four strings above are pseudorandom *even when viewed together*. (This suffices to prove that F is pseudorandom.) Intuitively, this is because $G_0(k) \parallel G_1(k) = G(k)$ is pseudorandom and hence indistinguishable from a uniform $2n$ -bit string $k_0 \parallel k_1$. But then

$$G_0(G_0(k)) \parallel G_1(G_0(k)) \parallel G_0(G_1(k)) \parallel G_1(G_1(k))$$

is indistinguishable from

$$G_0(k_0) \parallel G_1(k_0) \parallel G_0(k_1) \parallel G_1(k_1) = G(k_0) \parallel G(k_1).$$

Since G is a pseudorandom generator, the above is indistinguishable from a uniform $4n$ -bit string. A formal proof uses a hybrid argument.

Generalizing this idea, we can obtain a pseudorandom function on n -bit inputs by defining

$$F_k(x) = G_{x_n}(\cdots G_{x_1}(k) \cdots),$$

where $x = x_1 \cdots x_n$; see Construction 7.21. The intuition for why this function is pseudorandom is the same as before, but the formal proof is complicated by the fact that there are now exponentially many inputs to consider.

CONSTRUCTION 7.21

Let G be a pseudorandom generator with expansion factor $\ell(n) = 2n$, and define G_0, G_1 as in the text. For $k \in \{0, 1\}^n$, define the function $F_k : \{0, 1\}^n \rightarrow \{0, 1\}^n$ as:

$$F_k(x_1x_2 \cdots x_n) = G_{x_n}(\cdots (G_{x_2}(G_{x_1}(k))) \cdots).$$

A pseudorandom function from a pseudorandom generator.

It is useful to view this construction as defining, for each key $k \in \{0, 1\}^n$, a complete binary tree of depth n in which each node contains an n -bit value. (See Figure 7.2, in which $n = 3$.) The root has value k , and for every internal node with value k' its left child has value $G_0(k')$ and its right child has value $G_1(k')$. The result $F_k(x)$ for $x = x_1 \cdots x_n$ is then defined to be the value on the leaf node reached by traversing the tree according to the bits of x , where $x_i = 0$ means “go left” and $x_i = 1$ means “go right.” (The function is only defined for inputs of length n , and thus only values on the leaves are ever output.) The size of the tree is exponential in n . Nevertheless, to compute $F_k(x)$ the entire tree need not be constructed or stored; only n evaluations of G are needed.

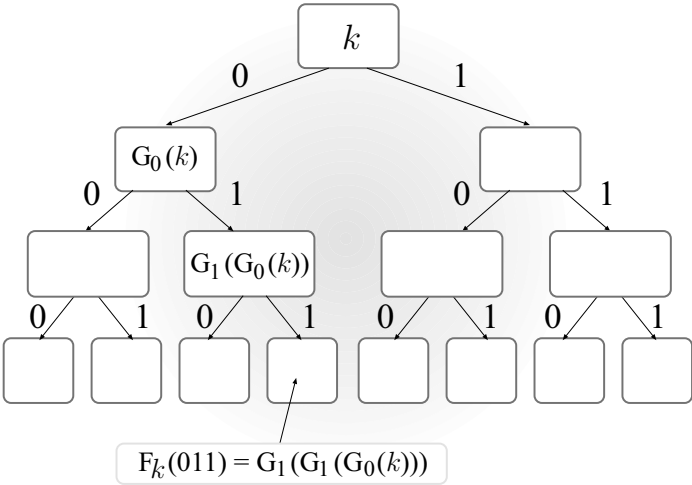


FIGURE 7.2: Constructing a pseudorandom function.

THEOREM 7.22 *If G is a pseudorandom generator with expansion factor $\ell(n) = 2n$, then Construction 7.21 is a pseudorandom function.*

PROOF We first show that for any polynomial t it is infeasible to distinguish $t(n)$ uniform $2n$ -bit strings from $t(n)$ pseudorandom strings; i.e., for any polynomial t and any PPT algorithm A , the following is negligible:

$$\left| \Pr [A(r_1 \| \cdots \| r_{t(n)}) = 1] - \Pr [A(G(s_1) \| \cdots \| G(s_{t(n)})) = 1] \right|,$$

where the first probability is over uniform choice of $r_1, \dots, r_{t(n)} \in \{0, 1\}^{2n}$, and the second probability is over uniform choice of $s_1, \dots, s_{t(n)} \in \{0, 1\}^n$.

The proof is by a hybrid argument. Fix a polynomial t and a PPT algorithm A , and consider the following algorithm A' :

Distinguisher A' :

A' is given as input a string $w \in \{0, 1\}^{2n}$.

1. Choose uniform $j \in \{1, \dots, t(n)\}$.
2. Choose uniform, independent values $r_1, \dots, r_{j-1} \in \{0, 1\}^{2n}$ and $s_{j+1}, \dots, s_{t(n)} \in \{0, 1\}^n$.
3. Output $A(r_1 \| \cdots \| r_{j-1} \| w \| G(s_{j+1}) \| \cdots \| G(s_{t(n)}))$.

For any n and $0 \leq i \leq t(n)$, let G_n^i denote the distribution on strings of length $2n \cdot t(n)$ in which the first i “blocks” of length $2n$ are uniform and the remaining $t(n) - i$ blocks are pseudorandom. Note that $G_n^{t(n)}$ corresponds to the distribution in which all $t(n)$ blocks are uniform, while G_n^0 corresponds to the distribution in which all $t(n)$ blocks are pseudorandom. That is,

$$\begin{aligned} & \left| \Pr_{y \leftarrow G_n^{t(n)}} [A(y) = 1] - \Pr_{y \leftarrow G_n^0} [A(y) = 1] \right| \\ &= \left| \Pr [A(r_1 \| \cdots \| r_{t(n)}) = 1] - \Pr [A(G(s_1) \| \cdots \| G(s_{t(n)})) = 1] \right| \end{aligned} \quad (7.11)$$

Say A' chooses $j = j^*$. If its input w is a uniform $2n$ -bit string, then A is run on an input distributed according to $G_n^{j^*}$. If, on the other hand, $w = G(s)$ for uniform s , then A is run on an input distributed according to $G_n^{j^*-1}$. This means that

$$\Pr_{r \leftarrow \{0,1\}^{2n}} [A'(r) = 1] = \frac{1}{t(n)} \cdot \sum_{j=1}^{t(n)} \Pr_{y \leftarrow G_n^j} [A(y) = 1]$$

and

$$\Pr_{s \leftarrow \{0,1\}^n} [A'(G(s)) = 1] = \frac{1}{t(n)} \cdot \sum_{j=0}^{t(n)-1} \Pr_{y \leftarrow G_n^j} [A(y) = 1].$$

Therefore,

$$\begin{aligned} & \left| \Pr_{r \leftarrow \{0,1\}^{2n}} [A'(r) = 1] - \Pr_{s \leftarrow \{0,1\}^n} [A'(G(s)) = 1] \right| \\ &= \frac{1}{t(n)} \cdot \left| \Pr_{y \leftarrow G_n^{t(n)}} [A(y) = 1] - \Pr_{y \leftarrow G_n^0} [A(y) = 1] \right|. \end{aligned} \quad (7.12)$$

Since G is a pseudorandom generator and A' runs in polynomial time, we know that the left-hand side of Equation (7.12) must be negligible; because $t(n)$ is polynomial, this implies that the left-hand side of Equation (7.11) is negligible as well.

Turning to the crux of the proof, we now show that F as in Construction 7.21 is a pseudorandom function. Let D be an arbitrary PPT distinguisher that is given 1^n as input. We show that D cannot distinguish between the case when it is given oracle access to a function that is equal to F_k for a uniform k , or a function chosen uniformly from Func_n . (See Section 3.5.1.) To do so, we use another hybrid argument. Here, we define a sequence of distributions over the values at the leaves of a complete binary tree of depth n . By associating each leaf with a string of length n as in Construction 7.21, we can equivalently view these as distributions over functions mapping n -bit inputs to n -bit outputs. For any n and $0 \leq i \leq n$, let H_n^i be the following distribution over the values at the leaves of a binary tree of depth n : first choose values for the nodes at level i independently and uniformly from $\{0,1\}^n$. Then for every node at level i or below with value k , its left child is given value $G_0(k)$ and its right child is given value $G_1(k)$. Note that H_n^n corresponds to the distribution in which all values at the leaves are chosen uniformly and independently, and thus corresponds to choosing a uniform function from Func_n , whereas H_n^0 corresponds to choosing a uniform key k in Construction 7.21 since in that case only the root (at level 0) is chosen uniformly. That is,

$$\begin{aligned} & \left| \Pr_{k \leftarrow \{0,1\}^n} [D^{F_k(\cdot)}(1^n) = 1] - \Pr_{f \leftarrow \text{Func}_n} [D^{f(\cdot)}(1^n) = 1] \right| \\ &= \left| \Pr_{f \leftarrow H_n^0} [D^{f(\cdot)}(1^n) = 1] - \Pr_{f \leftarrow H_n^n} [D^{f(\cdot)}(1^n) = 1] \right|. \end{aligned} \quad (7.13)$$

We show that Equation (7.13) is negligible, completing the proof.

Let $t = t(n)$ be a polynomial upper bound on the number of queries D makes to its oracle on input 1^n . Define a distinguisher A that tries to distinguish $t(n)$ uniform $2n$ -bit strings from $t(n)$ pseudorandom strings, as follows:

Distinguisher A :

A is given as input a $2n \cdot t(n)$ -bit string $w_1 \| \cdots \| w_{t(n)}$.

1. Choose uniform $j \in \{0, \dots, n-1\}$. In what follows, A (implicitly) maintains a binary tree of depth n with n -bit values at (a subset of the) internal nodes at depth $j+1$ and below.

2. Run $D(1^n)$. When D makes oracle query $x = x_1 \cdots x_n$, look at the prefix $x_1 \cdots x_j$. There are two cases:

- If D has never made a query with this prefix before, then use $x_1 \cdots x_j$ to reach a node v on the j th level of the tree. Take the next unused $2n$ -bit string w and set the value of the left child of node v to the left half of w , and the value of the right child of v to the right half of w .
- If D has made a query with prefix $x_1 \cdots x_j$ before, then node $x_1 \cdots x_{j+1}$ has already been assigned a value.

Using the value at node $x_1 \cdots x_{j+1}$, compute the value at the leaf corresponding to $x_1 \cdots x_n$ as in Construction 7.21, and return this value to D .

3. When execution of D is done, output the bit returned by D .

A runs in polynomial time. It is important here that A does not need to store the entire binary tree of exponential size. Instead, it “fills in” the values of at most $2t(n)$ nodes in the tree. Say A chooses $j = j^*$. Observe that:

1. If A ’s input is a uniform $2n \cdot t(n)$ -bit string, then the answers it gives to D are distributed exactly as if D were interacting with a function chosen from distribution $H_n^{j^*+1}$. This holds because the values of the nodes at level $j^* + 1$ of the tree are uniform and independent.
2. If A ’s input consists of $t(n)$ pseudorandom strings—i.e., $w_i = G(s_i)$ for uniform seed s_i —then the answers it gives to D are distributed exactly as if D were interacting with a function chosen from distribution $H_n^{j^*}$. This holds because the values of the nodes at level j^* of the tree (namely, the s -values) are uniform and independent. (These s -values are unknown to A , but this makes no difference.)

Proceeding as before, one can show that

$$\begin{aligned} & \left| \Pr [A(r_1 \| \cdots \| r_{t(n)}) = 1] - \Pr [A(G(s_1) \| \cdots \| G(s_{t(n)})) = 1] \right| \quad (7.14) \\ &= \frac{1}{n} \cdot \left| \Pr_{f \leftarrow H_n^0} [D^{f(\cdot)}(1^n) = 1] - \Pr_{f \leftarrow H_n^n} [D^{f(\cdot)}(1^n) = 1] \right|. \end{aligned}$$

We have shown earlier that Equation (7.14) must be negligible. The above thus implies that Equation (7.13) must be negligible as well. ■

7.6 Constructing (Strong) Pseudorandom Permutations

We next show how pseudorandom permutations and strong pseudorandom permutations can be constructed from any pseudorandom function. Recall

from Section 3.5.1 that a pseudorandom permutation is a pseudorandom function that is also efficiently invertible, while a *strong* pseudorandom permutation is additionally hard to distinguish from a random permutation even by an adversary given oracle access to both the permutation *and its inverse*.

Feistel networks revisited. A Feistel network, introduced in Section 6.2.2, provides a way of constructing an invertible function from an arbitrary set of functions. A Feistel network operates in a series of rounds. The input to the i th round is a string of length $2n$, divided into two n -bit halves L_{i-1} and R_{i-1} (the “left half” and the “right half,” respectively). The output of the i th round is the $2n$ -bit string (L_i, R_i) where

$$L_i := R_{i-1} \quad \text{and} \quad R_i := L_{i-1} \oplus f_i(R_{i-1})$$

for some efficiently computable (but not necessarily invertible) function f_i mapping n -bit inputs to n -bit outputs. We denote by $\text{Feistel}_{f_1, \dots, f_r}$ the r -round Feistel network using functions $f_1, \dots, f_r$. (That is, $\text{Feistel}_{f_1, \dots, f_r}(L_0, R_0)$ outputs the $2n$ -bit string (L_r, R_r) .) We saw in Section 6.2.2 that $\text{Feistel}_{f_1, \dots, f_r}$ is an efficiently invertible permutation regardless of the $\{f_i\}$.

We can define a keyed permutation by using a Feistel network in which the $\{f_i\}$ depend on a key. For example, let $F : \{0, 1\}^n \times \{0, 1\}^n \rightarrow \{0, 1\}^n$ be a pseudorandom function, and define the keyed *permutation* $F^{(1)}$ as

$$F_k^{(1)}(x) \stackrel{\text{def}}{=} \text{Feistel}_{F_k}(x).$$

(Note that $F_k^{(1)}$ has an n -bit key and maps $2n$ -bit inputs to $2n$ -bit outputs.) Is $F^{(1)}$ pseudorandom? A little thought shows that it is decidedly *not*. For any key $k \in \{0, 1\}^n$, the first n bits of the output of $F_k^{(1)}$ (that is, L_1) are equal to the last n bits of the input (i.e., R_0), something that occurs with only negligible probability for a random function.

Trying again, define $F^{(2)} : \{0, 1\}^{2n} \times \{0, 1\}^{2n} \rightarrow \{0, 1\}^{2n}$ as follows:

$$F_{k_1, k_2}^{(2)}(x) \stackrel{\text{def}}{=} \text{Feistel}_{F_{k_1}, F_{k_2}}(x). \quad (7.15)$$

(Note that k_1 and k_2 are independent keys.) Unfortunately, $F^{(2)}$ is not pseudorandom either, as you are asked to show in Exercise 7.16.

Given this, it may be somewhat surprising that a *three*-round Feistel network *is* pseudorandom. Define the keyed permutation $F^{(3)}$, taking a key of length $3n$ and mapping $2n$ -bit inputs to $2n$ -bit outputs, as follows:

$$F_{k_1, k_2, k_3}^{(3)}(x) \stackrel{\text{def}}{=} \text{Feistel}_{F_{k_1}, F_{k_2}, F_{k_3}}(x) \quad (7.16)$$

where, once again, k_1, k_2 , and k_3 are independent. We have:

THEOREM 7.23 *If F is a pseudorandom function, then $F^{(3)}$ is a pseudorandom permutation.*

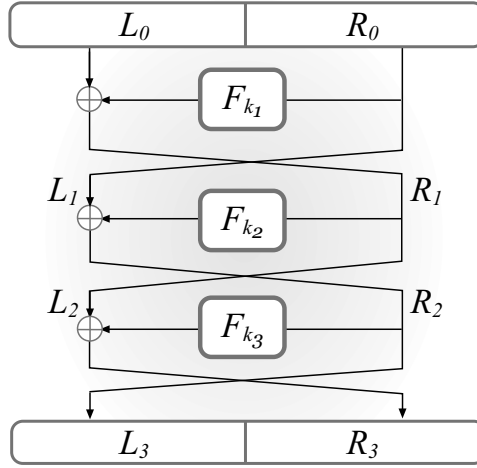


FIGURE 7.3: A three-round Feistel network, as used to construct a pseudorandom permutation from a pseudorandom function.

PROOF In the standard way, we can replace the pseudorandom functions used in the construction of $F^{(3)}$ with functions chosen uniformly at random instead. Pseudorandomness of F implies that this has only a negligible effect on the output of any probabilistic polynomial-time distinguisher interacting with $F^{(3)}$ as an oracle. We leave the details as an exercise.

Let D be a probabilistic polynomial-time distinguisher. In the remainder of the proof, we show the following is negligible:

$$\left| \Pr[D^{\text{Feistel}_{f_1, f_2, f_3}(\cdot)}(1^n) = 1] - \Pr[D^{\pi(\cdot)}(1^n) = 1] \right|,$$

where the first probability is taken over uniform and independent choice of f_1, f_2, f_3 from Func_n , and the second probability is taken over uniform choice of π from Perm_{2n} . Fix some value for the security parameter n , and let $q = q(n)$ denote a polynomial upper bound on the number of oracle queries made by D . We assume without loss of generality that D never makes the same oracle query twice. Focusing on D 's interaction with $\text{Feistel}_{f_1, f_2, f_3}(\cdot)$, let (L_0^i, R_0^i) denote the i th query D makes to its oracle, and let (L_1^i, R_1^i) , (L_2^i, R_2^i) , and (L_3^i, R_3^i) denote the intermediate values after rounds 1, 2, and 3, respectively, that result from that query. (See Figure 7.3.) Note that D chooses (L_0^i, R_0^i) and sees the result (L_3^i, R_3^i) , but does not directly observe (L_1^i, R_1^i) or (L_2^i, R_2^i) .

We say there is a *collision* at R_1 if $R_1^i = R_1^j$ for some distinct i, j . We first prove that a collision at R_1 occurs with only negligible probability. Consider any fixed, distinct i, j . If $R_0^i = R_0^j$ then $L_0^i \neq L_0^j$, but then

$$R_1^i = L_0^i \oplus f_1(R_0^i) \neq L_0^j \oplus f_1(R_0^j) = R_1^j.$$

If $R_0^i \neq R_0^j$ then $f_1(R_0^i)$ and $f_1(R_0^j)$ are uniform and independent, so

$$\Pr \left[L_0^i \oplus f_1(R_0^i) = L_0^j \oplus f_1(R_0^j) \right] = \Pr \left[f_1(R_0^j) = L_0^i \oplus f_1(R_0^i) \oplus L_0^j \right] = 2^{-n}.$$

Taking a union bound over all distinct i, j shows that the probability of a collision at R_1 is at most $q^2/2^n$.

Say there is a *collision* at R_2 if $R_2^i = R_2^j$ for some distinct i, j . We prove that conditioned on no collision at R_1 , the probability of a collision at R_2 is negligible. The analysis is as above: consider any fixed i, j , and note that if there is no collision at R_1 then $R_1^i \neq R_1^j$. Thus $f_2(R_1^i)$ and $f_2(R_1^j)$ are uniform and independent, and therefore

$$\Pr \left[L_1^i \oplus f_2(R_1^i) = L_1^j \oplus f_2(R_1^j) \mid \text{no collision at } R_1 \right] = 2^{-n}.$$

(Note that f_2 is independent of f_1 , making the above calculation easy.) Taking a union bound over all distinct i, j gives

$$\Pr[\text{collision at } R_2 \mid \text{no collision at } R_1] \leq q^2/2^n.$$

Note that $L_3^i = R_2^i = L_1^i \oplus f_2(R_1^i)$; so, conditioned on there being no collision at R_1 , the values $L_3^1, \dots, L_3^q$ are all independent and uniformly distributed in $\{0, 1\}^n$. If we additionally condition on the event that there is no collision at R_2 , then the values $L_3^1, \dots, L_3^q$ are uniformly distributed among all sequences of q *distinct* values in $\{0, 1\}^n$. Similarly, $R_3^i = L_2^i \oplus f_3(R_2^i)$; thus, conditioned on there being no collision at R_2 , the values $R_3^1, \dots, R_3^q$ are all uniformly distributed in $\{0, 1\}^n$, independent of each other as well as $L_3^1, \dots, L_3^q$.

To summarize: when querying $F^{(3)}$ (with uniform round functions) on a series of q distinct inputs, except with negligible probability the output values $(L_3^1, R_3^1), \dots, (L_3^q, R_3^q)$ are distributed such that the $\{L_3^i\}$ are uniform and independent, but distinct, n -bit values, and the $\{R_3^i\}$ are uniform and independent n -bit values. In contrast, when querying a random permutation on a series of q distinct inputs, the output values $(L_3^1, R_3^1), \dots, (L_3^q, R_3^q)$ are uniform and independent, but distinct, $2n$ -bit values. The best distinguishing attack for D , then, is to guess that it is interacting with a random permutation if $L_3^i = L_3^j$ for some distinct i, j . But that event occurs with negligible probability even in that case. This can be turned into a formal proof. ■

$F^{(3)}$ is not a *strong* pseudorandom permutation, as you are asked to demonstrate in Exercise 7.17. Fortunately, adding a fourth round *does* yield a strong pseudorandom permutation. The details are given as Construction 7.24.

THEOREM 7.25 *If F is a pseudorandom function, then Construction 7.24 is a strong pseudorandom permutation that maps $2n$ -bit inputs to $2n$ -bit outputs (and uses a $4n$ -bit key).*

CONSTRUCTION 7.24

Let F be a keyed, length-preserving function. Define the keyed permutation $F^{(4)}$ as follows:

- **Inputs:** A key $k = (k_1, k_2, k_3, k_4)$ with $|k_i| = n$, and an input $x \in \{0, 1\}^{2n}$ parsed as (L_0, R_0) with $|L_0| = |R_0| = n$.
- **Computation:**
 1. Compute $L_1 := R_0$ and $R_1 := L_0 \oplus F_{k_1}(R_0)$.
 2. Compute $L_2 := R_1$ and $R_2 := L_1 \oplus F_{k_2}(R_1)$.
 3. Compute $L_3 := R_2$ and $R_3 := L_2 \oplus F_{k_3}(R_2)$.
 4. Compute $L_4 := R_3$ and $R_4 := L_3 \oplus F_{k_4}(R_3)$.
 5. Output (L_4, R_4) .

A strong pseudorandom permutation from any pseudorandom function.

7.7 Assumptions for Private-Key Cryptography

We have shown that (1) if there exist one-way permutations, then there exist pseudorandom generators; (2) if there exist pseudorandom generators, then there exist pseudorandom functions; and (3) if there exist pseudorandom functions, then there exist (strong) pseudorandom permutations. Although we did not prove it here, it is possible to construct pseudorandom generators from one-way *functions*. We thus have the following fundamental theorem:

THEOREM 7.26 *If one-way functions exist, then so do pseudorandom generators, pseudorandom functions, and strong pseudorandom permutations.*

All the private-key schemes we have studied in Chapters 3 and 4 can be constructed from pseudorandom generators/functions. We therefore have:

THEOREM 7.27 *If one-way functions exist, then so do CCA-secure private-key encryption schemes and secure message authentication codes.*

That is, *one-way functions are sufficient for all private-key cryptography*. Here, we show that one-way functions are also *necessary*.

Pseudorandomness implies one-way functions. We begin by showing that pseudorandom generators imply the existence of one-way functions:

PROPOSITION 7.28 *If a pseudorandom generator exists, then so does a one-way function.*

PROOF Let G be a pseudorandom generator with expansion factor $\ell(n) = 2n$. (By Theorem 7.20, we know that the existence of a pseudorandom generator implies the existence of one with this expansion factor.) We show that G itself is one-way. Efficient computability is straightforward (since G can be computed in polynomial time). We show that the ability to invert G can be translated into the ability to distinguish the output of G from uniform. Intuitively, this holds because the ability to invert G implies the ability to find the seed used by the generator.

Let $\mathcal{A}$ be an arbitrary probabilistic polynomial-time algorithm. We show that $\Pr[\text{Invert}_{\mathcal{A},G}(n) = 1]$ is negligible (cf. Definition 7.1). To see this, consider the following PPT distinguisher D : on input a string $w \in \{0, 1\}^{2n}$, run $\mathcal{A}(w)$ to obtain output s . If $G(s) = w$ then output 1; otherwise, output 0.

We now analyze the behavior of D . First consider the probability that D outputs 1 when its input string w is uniform. Since there are at most 2^n values in the range of G (namely, the values $\{G(s)\}_{s \in \{0,1\}^n}$), the probability that w is in the range of G is at most $2^n/2^{2n} = 2^{-n}$. When w is not in the range of G , it is impossible for $\mathcal{A}$ to compute an inverse of w and thus impossible for D to output 1. We conclude that

$$\Pr_{w \leftarrow \{0,1\}^{2n}}[D(w) = 1] \leq 2^{-n}.$$

On the other hand, if $w = G(s)$ for a seed $s \in \{0, 1\}^n$ chosen uniformly at random then, by definition, $\mathcal{A}$ computes a correct inverse (and so D outputs 1) with probability exactly equal to $\Pr[\text{Invert}_{\mathcal{A},G}(n) = 1]$. Thus,

$$\left| \Pr_{w \leftarrow \{0,1\}^{2n}}[D(w) = 1] - \Pr_{s \leftarrow \{0,1\}^n}[D(G(s)) = 1] \right| \geq \Pr[\text{Invert}_{\mathcal{A},G}(n) = 1] - 2^{-n}.$$

Since G is a pseudorandom generator, the above must be negligible. Since 2^{-n} is negligible, this implies that $\Pr[\text{Invert}_{\mathcal{A},G}(n) = 1]$ is negligible as well and so G is one-way. ■

Non-trivial private-key encryption implies one-way functions. Proposition 7.28 does not imply that one-way functions are needed for constructing secure private-key encryption schemes, since it may be possible to construct the latter without relying on a pseudorandom generator. Furthermore, it is possible to construct perfectly secret encryption schemes (see Chapter 2), as long as the plaintext is no longer than the key. Thus, a proof that secure private-key encryption implies one-way functions requires more care.

PROPOSITION 7.29 *If there exists an EAV-secure private-key encryption scheme that encrypts messages twice as long as its key, then a one-way function exists.*

PROOF Let $\Pi = (\text{Enc}, \text{Dec})$ be a private-key encryption scheme that has indistinguishable encryptions in the presence of an eavesdropper and encrypts messages of length $2n$ when the key has length n . (We assume for simplicity that the key is chosen uniformly.) Say that when an n -bit key is used, Enc uses at most $\ell(n)$ bits of randomness. Denote the encryption of a message m using key k and randomness r by $\text{Enc}_k(m; r)$.

Define the following function f :

$$f(k, m, r) \stackrel{\text{def}}{=} \text{Enc}_k(m; r) \parallel m,$$

where $|k| = n$, $|m| = 2n$, and $|r| = \ell(n)$. We claim that f is a one-way function. Clearly it can be efficiently computed; we show that it is hard to invert. Letting $\mathcal{A}$ be an arbitrary PPT algorithm, we show that $\Pr[\text{Invert}_{\mathcal{A}, f}(n) = 1]$ is negligible (cf. Definition 7.1).

Consider the following probabilistic polynomial-time adversary $\mathcal{A}'$ attacking private-key encryption scheme Π (i.e., in experiment $\text{PrivK}_{\Pi, \mathcal{A}'}^{\text{eav}}(n)$):

Adversary $\mathcal{A}'(1^n)$

1. Choose uniform $m_0, m_1 \leftarrow \{0, 1\}^{2n}$ and output them. Receive in return a challenge ciphertext c .
2. Run $\mathcal{A}(c \parallel m_0)$ to obtain (k', m', r') . If $f(k', m', r') = c \parallel m_0$, output 0; else, output 1.

We now analyze the behavior of $\mathcal{A}'$. When c is an encryption of m_0 , then $c \parallel m_0$ is distributed exactly as $f(k, m_0, r)$ for uniform k, m_0 , and r . Therefore, $\mathcal{A}$ outputs a valid inverse of $c \parallel m_0$ (and hence $\mathcal{A}'$ outputs 0) with probability exactly equal to $\Pr[\text{Invert}_{\mathcal{A}, f}(n) = 1]$.

On the other hand, when c is an encryption of m_1 then c is independent of m_0 . For any fixed value of the challenge ciphertext c , there are at most 2^n possible messages (one for each possible key) to which c can correspond. Since m_0 is a uniform $2n$ -bit string, this means the probability there *exists* some key k for which $\text{Dec}_k(c) = m_0$ is at most $2^n / 2^{2n} = 2^{-n}$. This gives an upper bound on the probability with which $\mathcal{A}$ can possibly output a valid inverse of $c \parallel m_0$ under f , and hence an upper bound on the probability with which $\mathcal{A}'$ outputs 0 in that case.

Putting the above together, we have:

$$\begin{aligned} & \Pr[\text{PrivK}_{\Pi, \mathcal{A}'}^{\text{eav}}(n) = 1] \\ &= \frac{1}{2} \cdot \Pr[\mathcal{A}' \text{ outputs } 0 \mid b = 0] + \frac{1}{2} \cdot \Pr[\mathcal{A}' \text{ outputs } 1 \mid b = 1] \\ &\geq \frac{1}{2} \cdot \Pr[\text{Invert}_{\mathcal{A}, f}(n) = 1] + \frac{1}{2} \cdot (1 - 2^{-n}) \\ &= \frac{1}{2} + \frac{1}{2} \cdot (\Pr[\text{Invert}_{\mathcal{A}, f}(n) = 1] - 2^{-n}) . \end{aligned}$$

Security of Π means that $\Pr[\text{PrivK}_{\Pi, \mathcal{A}'}^{\text{eav}}(n) = 1] \leq \frac{1}{2} + \text{negl}(n)$ for some negligible function negl . This, in turn, implies that $\Pr[\text{Invert}_{\mathcal{A}, f}(n) = 1]$ is negligible, completing the proof that f is one-way. ■

Message authentication codes imply one-way functions. It is also true that message authentication codes satisfying Definition 4.2 imply the existence of one-way functions. As in the case of private-key encryption, a proof of this fact is somewhat subtle because unconditional message authentication codes *do* exist when there is an *a priori* bound on the number of messages that will be authenticated. (See Section 4.6.) Thus, a proof relies on the fact that Definition 4.2 requires security even when the adversary sees the authentication tags of an *arbitrary* (polynomial) number of messages. The proof is somewhat involved, so we do not give it here.

Discussion. We conclude that the existence of one-way functions is necessary and sufficient for all (non-trivial) private-key cryptography. In other words, one-way functions are a *minimal* assumption as far as private-key cryptography is concerned. Interestingly, this appears not to be the case for hash functions and public-key encryption, where one-way functions are known to be necessary but are not known (or believed) to be sufficient.

7.8 Computational Indistinguishability

The notion of computational indistinguishability is central to the theory of cryptography, and it underlies much of what we have seen in Chapter 3 and this chapter. Informally, two probability distributions are *computationally indistinguishable* if no efficient algorithm can tell them apart (or *distinguish* them). In more detail, consider two distributions X and Y over strings of some length ℓ ; that is, X and Y each assigns some probability to every string in $\{0, 1\}^\ell$. When we say that some algorithm D cannot distinguish these two distributions, we mean that D cannot tell whether it is given a string sampled according to distribution X or whether it is given a string sampled according to distribution Y . Put differently, if we imagine D outputting “0” when it believes its input was sampled according to X and outputting “1” if it thinks its input was sampled according to Y , then the probability that D outputs “1” should be roughly the same regardless of whether D is provided with a sample from X or from Y . In other words, we want

$$\left| \Pr_{s \leftarrow X}[D(s) = 1] - \Pr_{s \leftarrow Y}[D(s) = 1] \right|$$

to be small.

This should be reminiscent of the way we defined pseudorandom generators and, indeed, we will soon formally redefine the notion of a pseudorandom generator using this terminology.

The formal definition of computational indistinguishability refers to *probability ensembles*, which are infinite sequences of probability distributions. (This formalism is necessary for a meaningful asymptotic approach.) Although the notion can be generalized, for our purposes we consider probability ensembles in which the underlying distributions are indexed by natural numbers. If for every natural number n we have a distribution X_n , then $\mathcal{X} = \{X_n\}_{n \in \mathbb{N}}$ is a probability ensemble. It is often the case that $X_n = Y_{t(n)}$ for some function t , in which case we write $\{Y_{t(n)}\}_{n \in \mathbb{N}}$ in place of $\{X_n\}_{n \in \mathbb{N}}$.

We will only be interested in *efficiently sampleable* probability ensembles. An ensemble $\mathcal{X} = \{X_n\}_{n \in \mathbb{N}}$ is efficiently sampleable if there is a probabilistic polynomial-time algorithm S such that the random variables $S(1^n)$ and X_n are identically distributed. That is, algorithm S is an efficient way of sampling $\mathcal{X}$.

We can now formally define what it means for two ensembles to be computationally indistinguishable.

DEFINITION 7.30 Two probability ensembles $\mathcal{X} = \{X_n\}_{n \in \mathbb{N}}$ and $\mathcal{Y} = \{Y_n\}_{n \in \mathbb{N}}$ are computationally indistinguishable, denoted $\mathcal{X} \stackrel{c}{\equiv} \mathcal{Y}$, if for every probabilistic polynomial-time distinguisher D there exists a negligible function negl such that:

$$\left| \Pr_{x \leftarrow X_n} [D(1^n, x) = 1] - \Pr_{y \leftarrow Y_n} [D(1^n, y) = 1] \right| \leq \text{negl}(n).$$

In the definition, D is given the unary input 1^n so it can run in time polynomial in n . This is important when the outputs of X_n and Y_n may have length less than n . As shorthand in probability expressions, we will sometimes write X as a placeholder for a random sample from distribution X . That is, we would write $\Pr[D(1^n, X_n) = 1]$ in place of $\Pr_{x \leftarrow X_n} [D(1^n, x) = 1]$.

The relation of computational indistinguishability is transitive: if $\mathcal{X} \stackrel{c}{\equiv} \mathcal{Y}$ and $\mathcal{Y} \stackrel{c}{\equiv} \mathcal{Z}$, then $\mathcal{X} \stackrel{c}{\equiv} \mathcal{Z}$.

Pseudorandomness and pseudorandom generators. Pseudorandomness is just a special case of computational indistinguishability. For any integer ℓ , let U_ℓ denote the uniform distribution over $\{0, 1\}^\ell$. We can define a pseudorandom generator as follows:

DEFINITION 7.31 Let $\ell(\cdot)$ be a polynomial and let G be a (deterministic) polynomial-time algorithm where for all s it holds that $|G(s)| = \ell(|s|)$. We say that G is a pseudorandom generator if the following two conditions hold:

1. **(Expansion:)** For every n it holds that $\ell(n) > n$.

2. **(Pseudorandomness:)** The ensemble $\{G(U_n)\}_{n \in \mathbb{N}}$ is computationally indistinguishable from the ensemble $\{U_{\ell(n)}\}_{n \in \mathbb{N}}$.

Many of the other definitions and assumptions in this book can also be cast as special cases or variants of computational indistinguishability.

Multiple samples. An important theorem regarding computational indistinguishability is that polynomially many samples of (efficiently sampleable) computationally indistinguishable ensembles are also computationally indistinguishable.

THEOREM 7.32 *Let $\mathcal{X}$ and $\mathcal{Y}$ be efficiently sampleable probability ensembles that are computationally indistinguishable. Then, for every polynomial p , the ensemble $\overline{\mathcal{X}} = \{(X_n^{(1)}, \dots, X_n^{(p(n))})\}_{n \in \mathbb{N}}$ is computationally indistinguishable from the ensemble $\overline{\mathcal{Y}} = \{(Y_n^{(1)}, \dots, Y_n^{(p(n))})\}_{n \in \mathbb{N}}$.*

For example, let G be a pseudorandom generator with expansion factor $2n$, in which case the ensembles $\{G(U_n)\}_{n \in \mathbb{N}}$ and $\{U_{2n}\}_{n \in \mathbb{N}}$ are computationally indistinguishable. In the proof of Theorem 7.22 we showed that for any polynomial t the ensembles

$$\underbrace{\{(G(U_n), \dots, G(U_n))\}_{n \in \mathbb{N}}}_{t(n)} \quad \text{and} \quad \underbrace{\{(U_{2n}, \dots, U_{2n})\}_{n \in \mathbb{N}}}_{t(n)}$$

are also computationally indistinguishable. Theorem 7.32 is proved by a hybrid argument in exactly the same way.

References and Additional Reading

The notion of a one-way function was first proposed by Diffie and Hellman [58] and later formalized by Yao [179]. Hard-core predicates were introduced by Blum and Micali [37], and the fact that there exists a hard-core predicate for every one-way function was proved by Goldreich and Levin [79].

The first construction of pseudorandom generators (under a specific number-theoretic hardness assumption) was given by Blum and Micali [37]. The construction of a pseudorandom generator from any one-way permutation was given by Yao [179], and the result that pseudorandom generators can be constructed from any one-way function was shown by Håstad et al. [85]. Pseudorandom functions were defined and constructed by Goldreich, Goldwasser and Micali [78] and their extension to (strong) pseudorandom permutations was shown by Luby and Rackoff [116]. The fact that one-way functions are a

necessary assumption for most of private-key cryptography was shown in [93]. The proof of Proposition 7.29 is from [72].

Our presentation is heavily influenced by Goldreich's book [75], which is highly recommended for those interested in exploring the topics of this chapter in greater detail.

Exercises

- 7.1 Prove that if there exists a one-way function, then there exists a one-way function f such that $f(0^n) = 0^n$ for every n . Note that for infinitely many values y , it is easy to compute $f^{-1}(y)$. Why does this not contradict one-wayness?
- 7.2 Prove that if f is a one-way function, then the function g defined by $g(x_1, x_2) \stackrel{\text{def}}{=} (f(x_1), x_2)$, where $|x_1| = |x_2|$, is also a one-way function. Observe that g reveals half of its input, but is nevertheless one-way.
- 7.3 Prove that if there exists a one-way function, then there exists a length-preserving one-way function.

Hint: Let f be a one-way function and let $p(\cdot)$ be a polynomial such that $|f(x)| \leq p(|x|)$. (Justify the existence of such a p .) Define $f'(x) \stackrel{\text{def}}{=} f(x) \| 10^{p(|x|) - |f(x)|}$. Further modify f' to get a length-preserving function that remains one-way.

- 7.4 Let (Gen, H) be a collision-resistant hash function, where H maps strings of length $2n$ to strings of length n . Prove that the function family $(\text{Gen}, \text{Samp}, H)$ is one-way (cf. Definition 7.3), where **Samp** is the trivial algorithm that samples a uniform string of length $2n$.

Hint: Choosing uniform $x \in \{0, 1\}^{2n}$ and finding an inverse of $y = H^s(x)$ does not guarantee a collision. But it does yield a collision most of the time...

- 7.5 Let F be a (length-preserving) pseudorandom permutation.
 - (a) Show that the function $f(x, y) = F_x(y)$ is not one-way.
 - (b) Show that the function $f(y) = F_{0^n}(y)$ (where $n = |y|$) is not one-way.
 - (c) Prove that the function $f(x) = F_x(0^n)$ (where $n = |x|$) is one-way.
- 7.6 Let f be a length-preserving one-way function, and let hc be a hardcore predicate of f . Define G as $G(x) = f(x) \| \text{hc}(x)$. Is G necessarily a pseudorandom generator? Prove your answer.

- 7.7 Prove that there exist one-way functions if and only if there exist one-way function families. Discuss why your proof does not carry over to the case of one-way permutations.
- 7.8 Let f be a one-way function. Is $g(x) \stackrel{\text{def}}{=} f(f(x))$ necessarily a one-way function? What about $g'(x) \stackrel{\text{def}}{=} f(x) \| f(f(x))$? Prove your answers.
- 7.9 Let $\Pi = (\text{Gen}, \text{Samp}, f)$ be a function family. A function $\text{hc} : \{0, 1\}^* \rightarrow \{0, 1\}$ is a *hard-core predicate* of Π if it is efficiently computable and if for every PPT algorithm $\mathcal{A}$ there is a negligible function negl such that

$$\Pr_{I \leftarrow \text{Gen}(1^n), x \leftarrow \text{Samp}(I)} [\mathcal{A}(I, f_I(x)) = \text{hc}(I, x)] \leq \frac{1}{2} + \text{negl}(n).$$

Prove a version of the Goldreich–Levin theorem for this setting, namely, if a one-way function (resp., permutation) family Π exists, then there exists a one-way function (resp., permutation) family Π' and a hard-core predicate hc of Π' .

- 7.10 Show a construction of a pseudorandom generator from any one-way permutation family. You may use the result of the previous exercise.
- 7.11 This exercise is for students who have taken a course in complexity theory or are otherwise familiar with $\mathcal{NP}$ completeness.
- (a) Show that the existence of one-way functions implies $\mathcal{P} \neq \mathcal{NP}$.
 - (b) Assume that $\mathcal{P} \neq \mathcal{NP}$. Show that there exists a function f that is: (1) computable in polynomial time, (2) hard to invert *in the worst case* (i.e., for all probabilistic polynomial-time $\mathcal{A}$, $\Pr_{x \leftarrow \{0, 1\}^n} [f(\mathcal{A}(f(x))) = f(x)] \neq 1$), but (3) is *not* one-way.
- 7.12 Let $x \in \{0, 1\}^n$ and denote $x = x_1 \cdots x_n$. Prove that if there exists a one-way function, then there exists a one-way function f such that for every i there is an algorithm A_i such that

$$\Pr_{x \leftarrow \{0, 1\}^n} [A_i(f(x)) = x_i] \geq \frac{1}{2} + \frac{1}{2n}.$$

(This exercise demonstrates that it is not possible to claim that every one-way function hides at least one *specific* bit of the input.)

- 7.13 Show that if a one-to-one function f has a hard-core predicate, then f is one-way.
- 7.14 Show that if Construction 7.21 is modified in the natural way so that $F_k(x)$ is defined for every nonempty string x of length at most n , then the construction is no longer a pseudorandom function.

- 7.15 Prove that if there exists a pseudorandom function that, using a key of length n , maps n -bit inputs to single-bit outputs, then there exists a pseudorandom function that maps n -bit inputs to n -bit outputs.

Hint: Use a key of length n^2 , and prove your construction secure using a hybrid argument.

- 7.16 Prove that a two-round Feistel network using pseudorandom round functions (as in Equation (7.15)) is not pseudorandom.

- 7.17 Prove that a three-round Feistel network using pseudorandom round functions (as in Equation (7.16)) is not *strongly* pseudorandom.

Hint: This is significantly more difficult than the previous exercise. Use a distinguisher that makes two queries to the permutation and one query to its inverse.

- 7.18 Consider the keyed permutation F^* defined by

$$F_k^*(x) \stackrel{\text{def}}{=} \text{Feistel}_{F_k, F_k, F_k}(x).$$

(Note that the same key is used in each round.) Show that F^* is not pseudorandom.

- 7.19 Let G be a pseudorandom generator with expansion factor $\ell(n) = n + 1$. Prove that G is a one-way function.

- 7.20 Let $\mathcal{X}, \mathcal{Y}, \mathcal{Z}$ be probability ensembles. Prove that if $\mathcal{X} \stackrel{c}{\equiv} \mathcal{Y}$ and $\mathcal{Y} \stackrel{c}{\equiv} \mathcal{Z}$, then $\mathcal{X} \stackrel{c}{\equiv} \mathcal{Z}$.

- 7.21 Prove Theorem 7.32.

- 7.22 Let $\mathcal{X} = \{X_n\}_{n \in \mathbb{N}}$ and $\mathcal{Y} = \{Y_n\}_{n \in \mathbb{N}}$ be computationally indistinguishable probability ensembles. Prove that for any probabilistic polynomial-time algorithm $\mathcal{A}$, the ensembles $\{\mathcal{A}(X_n)\}_{n \in \mathbb{N}}$ and $\{\mathcal{A}(Y_n)\}_{n \in \mathbb{N}}$ are computationally indistinguishable.

Part III

Public-Key (Asymmetric) Cryptography

Chapter 8

Number Theory and Cryptographic Hardness Assumptions

Modern cryptosystems are invariably based on an assumption that *some* problem is hard. In Chapters 3 and 4, for example, we saw that private-key cryptography—both encryption schemes and message authentication codes—can be based on the assumption that pseudorandom permutations (a.k.a. block ciphers) exist. Recall, roughly, this means that there exists some keyed permutation F for which it is hard to distinguish in polynomial time between interactions with F_k (for a uniform, unknown key k) and interactions with a truly random permutation.

On the face of it, the assumption that pseudorandom permutations exist seems quite strong and unnatural, and it is reasonable to ask whether this assumption is true or whether there is any evidence to support it. In Chapter 6 we explored how block ciphers are constructed in practice. The fact that existing constructions are resistant to attack serves as an indication that the existence of pseudorandom permutations is plausible. Still, it may be difficult to believe that there are *no* efficient distinguishing attacks on existing block ciphers. Moreover, the current state of our theory is such that we do not know how to prove the pseudorandomness of any of the existing practical constructions relative to any “simpler” or “more reasonable” assumption. All in all, this is not entirely a satisfying state of affairs.

In contrast, as mentioned in Chapter 3 (and investigated in detail in Chapter 7) it is possible to *prove* that pseudorandom permutations exist based on the much milder assumption that one-way functions exist. (Informally, a function is *one-way* if it is easy to compute but hard to invert; see Section 8.4.1.) Apart from a brief discussion in Section 7.1.2, however, we have not seen concrete examples of functions believed to be one-way.

One goal of this chapter is to introduce various problems believed to be “hard,” and to present conjectured one-way functions based on those problems.¹ This chapter can thus be viewed as a culmination of a “top down” approach to private-key cryptography. (See Figure 8.1.) That is, in Chapters 3 and 4 we have shown that private-key cryptography can be based on pseudorandom functions and permutations. We have then seen that the latter

¹Recall we currently do not know how to *prove* that one-way functions exist, so the best we can do is base one-way functions on assumptions regarding the hardness of certain problems.

can be instantiated in practice using block ciphers, as explored in Chapter 6, or can be constructed in a rigorous fashion from any one-way function, as shown in Chapter 7. Here, we take this one step further and show how one-way functions can be based on certain hard mathematical problems.

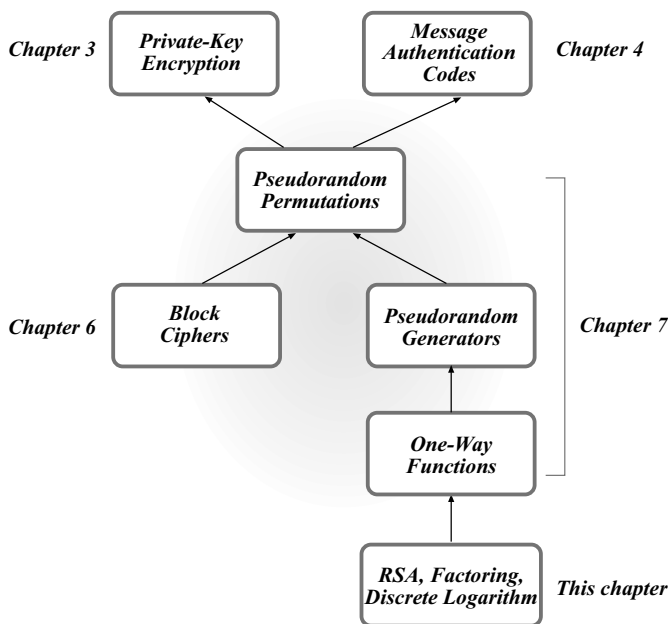


FIGURE 8.1: Private-key cryptography: a top-down approach.

The examples we explore are *number theoretic* in nature, and we therefore begin with a short introduction to number theory and group theory. Because we are also interested in problems that can be solved efficiently (even a one-way function needs to be easy to compute in one direction, and a cryptographic scheme must admit efficient algorithms for the honest parties), we also initiate a study of *algorithmic* number theory. Even the reader who is familiar with number theory or group theory is encouraged to read this chapter, since algorithmic aspects are typically ignored in a purely mathematical treatment of these topics.

A second goal of this chapter is to develop the material needed for *public-key cryptography*, whose study we will begin in Chapter 10. Strikingly, although in the private-key setting there exist efficient constructions of the necessary primitives (block ciphers and hash functions) without invoking any number theory, in the public-key setting *all known constructions rely on hard number-theoretic problems*. The material in this chapter thus serves both as a culmination of what we have studied so far with regard to private-key cryptography, as well as the foundation for public-key cryptography.

8.1 Preliminaries and Basic Group Theory

We begin with a review of prime numbers and basic modular arithmetic. Even the reader who has seen these topics before should skim the next two sections since some of the material may be new and we include proofs for most of the stated results.

8.1.1 Primes and Divisibility

The set of integers is denoted by $\mathbb{Z}$. For $a, b \in \mathbb{Z}$, we say that a *divides* b , written $a \mid b$, if there exists an integer c such that $ac = b$. If a does not divide b , we write $a \nmid b$. (We are primarily interested in the case where a, b , and c are all positive, although the definition makes sense even when one or more of these is negative or zero.) A simple observation is that if $a \mid b$ and $a \mid c$ then $a \mid (Xb + Yc)$ for any $X, Y \in \mathbb{Z}$.

If $a \mid b$ and a is positive, we call a a *divisor* of b . If in addition $a \notin \{1, b\}$ then a is called a *nontrivial* divisor, or a *factor*, of b . A positive integer $p > 1$ is *prime* if it has no factors; i.e., it has only two divisors: 1 and itself. A positive integer greater than 1 that is not prime is called *composite*. By convention, the number 1 is neither prime nor composite.

A fundamental theorem of arithmetic is that every integer greater than 1 can be expressed *uniquely* (up to ordering) as a product of primes. That is, any positive integer $N > 1$ can be written as $N = \prod_i p_i^{e_i}$, where the $\{p_i\}$ are distinct primes and $e_i \geq 1$ for all i ; furthermore, the $\{p_i\}$ (and $\{e_i\}$) are uniquely determined up to ordering.

We are familiar with the process of *division with remainder* from elementary school. The following proposition formalizes this notion.

PROPOSITION 8.1 *Let a be an integer and let b be a positive integer. Then there exist unique integers q, r for which $a = qb + r$ and $0 \leq r < b$.*

Furthermore, given integers a and b as in the proposition it is possible to compute q and r in polynomial time; see Appendix B.1. (An algorithm's running time is measured as a function of the length(s) of its input(s). An important point in the context of algorithmic number theory is that integer inputs are always assumed to be represented in binary. The running time of an algorithm taking as input an integer N is therefore measured in terms of $\|N\|$, the *length of the binary representation of N* . Note that $\|N\| = \lfloor \log N \rfloor + 1$.)

The *greatest common divisor* of two integers a, b , written $\gcd(a, b)$, is the largest integer c such that $c \mid a$ and $c \mid b$. (We leave $\gcd(0, 0)$ undefined.) The notion of greatest common divisor makes sense when either or both of a, b are negative but we will typically have $a, b \geq 1$; anyway, $\gcd(a, b) = \gcd(|a|, |b|)$.

Note that $\gcd(b, 0) = \gcd(0, b) = b$; also, if p is prime then $\gcd(a, p)$ is either equal to 1 or p . If $\gcd(a, b) = 1$ we say that a and b are *relatively prime*.

The following is a useful result:

PROPOSITION 8.2 *Let a, b be positive integers. Then there exist integers X, Y such that $Xa + Yb = \gcd(a, b)$. Furthermore, $\gcd(a, b)$ is the smallest positive integer that can be expressed in this way.*

PROOF Consider the set $I \stackrel{\text{def}}{=} \{\hat{X}a + \hat{Y}b \mid \hat{X}, \hat{Y} \in \mathbb{Z}\}$. Note that $a, b \in I$, and so I certainly contains some positive integers. Let d be the smallest positive integer in I . We show that $d = \gcd(a, b)$; since d can be written as $d = Xa + Yb$ for some $X, Y \in \mathbb{Z}$ (because $d \in I$), this proves the theorem.

To show this, we must prove that $d \mid a$ and $d \mid b$, and that d is the largest integer with this property. In fact, we can show that d divides every element in I . To see this, take an arbitrary $c \in I$ and write $c = X'a + Y'b$ with $X', Y' \in \mathbb{Z}$. Using division with remainder (Proposition 8.1) we have that $c = qd + r$ with q, r integers and $0 \leq r < d$. Then

$$r = c - qd = X'a + Y'b - q(Xa + Yb) = (X' - qX)a + (Y' - qY)b \in I.$$

If $r \neq 0$, this contradicts our choice of d as the *smallest* positive integer in I (because $r < d$). So, $r = 0$ and hence $d \mid c$. This shows that d divides every element of I .

Since $a \in I$ and $b \in I$, the above shows that $d \mid a$ and $d \mid b$ and so d is a common divisor of a and b . It remains to show that it is the largest common divisor. Assume there is an integer $d' > d$ such that $d' \mid a$ and $d' \mid b$. Then by the observation made earlier, $d' \mid Xa + Yb$. Since the latter is equal to d , this means $d' \mid d$. But this is impossible if d' is larger than d . We conclude that d is the largest integer dividing both a and b , and hence $d = \gcd(a, b)$. ■

Given a and b , the *Euclidean algorithm* can be used to compute $\gcd(a, b)$ in polynomial time. The *extended Euclidean algorithm* can be used to compute X, Y (as in the above proposition) in polynomial time as well. See Appendix B.1.2 for details.

The preceding proposition is very useful in proving additional results about divisibility. We show two examples now.

PROPOSITION 8.3 *If $c \mid ab$ and $\gcd(a, c) = 1$, then $c \mid b$. Thus, if p is prime and $p \mid ab$ then either $p \mid a$ or $p \mid b$.*

PROOF Since $c \mid ab$ we have $\gamma c = ab$ for some integer γ . If $\gcd(a, c) = 1$ then, by the previous proposition, we know there exist integers X, Y such that

$1 = Xa + Yc$. Multiplying both sides by b , we obtain

$$b = Xab + Ycb = X\gamma c + Ycb = c \cdot (X\gamma + Yb).$$

Since $(X\gamma + Yb)$ is an integer, it follows that $c \mid b$.

The second part of the proposition follows from the fact that if $p \nmid a$ then $\gcd(a, p) = 1$. ■

PROPOSITION 8.4 *If $a \mid N$, $b \mid N$, and $\gcd(a, b) = 1$, then $ab \mid N$.*

PROOF Write $ac = N$, $bd = N$, and (using Proposition 8.2) $1 = Xa + Yb$, where c, d, X, Y are all integers. Multiplying both sides of the last equation by N we obtain

$$N = XaN + YbN = Xabd + Ybac = ab(Xd + Yc),$$

showing that $ab \mid N$. ■

8.1.2 Modular Arithmetic

Let $a, b, N \in \mathbb{Z}$ with $N > 1$. We use the notation $[a \bmod N]$ to denote the remainder of a upon division by N . In more detail: by Proposition 8.1 there exist unique q, r with $a = qN + r$ and $0 \leq r < N$, and we define $[a \bmod N]$ to be equal to this r . Note therefore that $0 \leq [a \bmod N] < N$. We refer to the process of mapping a to $[a \bmod N]$ as *reduction modulo N* .

We say that a and b are *congruent modulo N* , written $a = b \bmod N$, if $[a \bmod N] = [b \bmod N]$, i.e., if the remainder when a is divided by N is the same as the remainder when b is divided by N . Note that $a = b \bmod N$ if and only if $N \mid (a - b)$. By way of notation, in an expression such as

$$a = b = c = \cdots = z \bmod N,$$

the understanding is that *every* equal sign in this sequence (and not just the last) refers to congruence modulo N .

Note that $a = [b \bmod N]$ implies $a = b \bmod N$, but not vice versa. For example, $36 = 21 \bmod 15$ but $36 \neq [21 \bmod 15] = 6$. (On the other hand, $[a \bmod N] = [b \bmod N]$ if and only if $a = b \bmod N$.)

Congruence modulo N is an equivalence relation: i.e., it is reflexive ($a = a \bmod N$ for all a), symmetric ($a = b \bmod N$ implies $b = a \bmod N$), and transitive (if $a = b \bmod N$ and $b = c \bmod N$, then $a = c \bmod N$). Congruence modulo N also obeys the standard rules of arithmetic with respect to addition, subtraction, and multiplication; so, for example, if $a = a' \bmod N$ and $b = b' \bmod N$ then $(a + b) = (a' + b') \bmod N$ and $ab = a'b' \bmod N$. A consequence is that we can “reduce and then add/multiply” instead of having to “add/multiply and then reduce,” which can often simplify calculations.

Example 8.5

Let us compute $[1093028 \cdot 190301 \bmod 100]$. Since $1093028 = 28 \bmod 100$ and $190301 = 1 \bmod 100$, we have

$$\begin{aligned} 1093028 \cdot 190301 &= [1093028 \bmod 100] \cdot [190301 \bmod 100] \bmod 100 \\ &= 28 \cdot 1 = 28 \bmod 100. \end{aligned}$$

The alternate way of calculating the answer (i.e., computing the product $1093028 \cdot 190301$ and then reducing the result modulo 100) is less efficient. $\diamond$

Congruence modulo N does *not* (in general) respect division. That is, if $a = a' \bmod N$ and $b = b' \bmod N$ then it is not necessarily true that $a/b = a'/b' \bmod N$; in fact, the expression “ $a/b \bmod N$ ” is not always well-defined. As a specific example that often causes confusion, $ab = cb \bmod N$ does *not* necessarily imply that $a = c \bmod N$.

Example 8.6

Take $N = 24$. Then $3 \cdot 2 = 6 = 15 \cdot 2 \bmod 24$, but $3 \neq 15 \bmod 24$. $\diamond$

In certain cases, however, we can define a meaningful notion of division. If for a given integer b there exists an integer c such that $bc = 1 \bmod N$, we say that b is *invertible* modulo N and call c a (multiplicative) *inverse* of b modulo N . Clearly, 0 is never invertible. It is also not difficult to show that if c is a multiplicative inverse of b modulo N then so is $[c \bmod N]$. Furthermore, if c' is another multiplicative inverse of b then $[c \bmod N] = [c' \bmod N]$. When b is invertible we can therefore simply let b^{-1} denote the *unique* multiplicative inverse of b that lies in the range $\{1, \dots, N-1\}$.

When b is invertible modulo N , we define division by b modulo N as multiplication by b^{-1} (i.e., we define $[a/b \bmod N] \stackrel{\text{def}}{=} [ab^{-1} \bmod N]$). We stress that division by b is *only defined* when b is invertible. If $ab = cb \bmod N$ and b is invertible, then we may divide each side of the equation by b (or, really, multiply each side by b^{-1}) to obtain

$$(ab) \cdot b^{-1} = (cb) \cdot b^{-1} \bmod N \quad \Rightarrow \quad a = c \bmod N.$$

We see that in this case, division works “as expected.” Invertible integers modulo N are therefore “nicer” to work with, in some sense.

The natural question is: which integers are invertible modulo a given modulus N ? We can fully answer this question using Proposition 8.2:

PROPOSITION 8.7 *Let b, N be integers, with $b \geq 1$ and $N > 1$. Then b is invertible modulo N if and only if $\gcd(b, N) = 1$.*

PROOF Assume b is invertible modulo N , and let c denote its inverse. Since $bc = 1 \bmod N$, this implies that $bc - 1 = \gamma N$ for some $\gamma \in \mathbb{Z}$. Equiv-

alently, $bc - \gamma N = 1$. Since, by Proposition 8.2, $\gcd(b, N)$ is the smallest positive integer that can be expressed in this way, and there is no positive integer smaller than 1, this implies that $\gcd(b, N) = 1$.

Conversely, if $\gcd(b, N) = 1$ then by Proposition 8.2 there exist integers X, Y such that $Xb + YN = 1$. Reducing each side of this equation modulo N gives $Xb = 1 \pmod{N}$, and we see that X is a multiplicative inverse of b . (In fact, this gives an efficient algorithm to compute inverses.) ■

Example 8.8

Let $b = 11$ and $N = 17$. Then $(-3) \cdot 11 + 2 \cdot 17 = 1$, and so $14 = [-3 \pmod{17}]$ is the inverse of 11. One can verify that $14 \cdot 11 = 1 \pmod{17}$. ◇

Addition, subtraction, multiplication, and computation of inverses (when they exist) modulo N can all be carried out in polynomial time; see Appendix B.2. Exponentiation (i.e., computing $[a^b \pmod{N}]$ for $b > 0$ an integer) can also be computed in polynomial time; see Appendix B.2.3.

8.1.3 Groups

Let $\mathbb{G}$ be a set. A *binary operation* $\circ$ on $\mathbb{G}$ is simply a function $\circ(\cdot, \cdot)$ that takes as input two elements of $\mathbb{G}$. If $g, h \in \mathbb{G}$ then instead of using the cumbersome notation $\circ(g, h)$, we write $g \circ h$.

We now introduce the important notion of a *group*.

DEFINITION 8.9 A group is a set $\mathbb{G}$ along with a binary operation $\circ$ for which the following conditions hold:

- **(Closure:)** For all $g, h \in \mathbb{G}$, $g \circ h \in \mathbb{G}$.
- **(Existence of an identity:)** There exists an identity $e \in \mathbb{G}$ such that for all $g \in \mathbb{G}$, $e \circ g = g = g \circ e$.
- **(Existence of inverses:)** For all $g \in \mathbb{G}$ there exists an element $h \in \mathbb{G}$ such that $g \circ h = e = h \circ g$. Such an h is called an inverse of g .
- **(Associativity:)** For all $g_1, g_2, g_3 \in \mathbb{G}$, $(g_1 \circ g_2) \circ g_3 = g_1 \circ (g_2 \circ g_3)$.

When $\mathbb{G}$ has a finite number of elements, we say $\mathbb{G}$ is **finite** and let $|\mathbb{G}|$ denote the **order** of the group (that is, the number of elements in $\mathbb{G}$).

A group $\mathbb{G}$ with operation $\circ$ is **abelian** if the following holds:

- **(Commutativity:)** For all $g, h \in \mathbb{G}$, $g \circ h = h \circ g$.

When the binary operation is understood, we simply call the set $\mathbb{G}$ a group.

We will always deal with finite, abelian groups. We will be careful to specify, however, when a result requires these assumptions.

Associativity implies that we do not need to include parentheses when writing long expressions; that is, the notation $g_1 \circ g_2 \circ \cdots \circ g_n$ is unambiguous since it does not matter in what order we evaluate the operation $\circ$.

One can show that the identity element in a group $\mathbb{G}$ is *unique*, and so we can therefore refer to *the* identity of a group. One can also show that each element g of a group has a *unique* inverse. See Exercise 8.1.

If $\mathbb{G}$ is a group, a set $\mathbb{H} \subseteq \mathbb{G}$ is a *subgroup* of $\mathbb{G}$ if $\mathbb{H}$ itself forms a group under the same operation associated with $\mathbb{G}$. To check that $\mathbb{H}$ is a subgroup, we need to verify closure, existence of identity and inverses, and associativity as per Definition 8.9. (In fact, associativity—as well as commutativity if $\mathbb{G}$ is abelian—is inherited automatically from $\mathbb{G}$.) Every group $\mathbb{G}$ always has the trivial subgroups $\mathbb{G}$ and $\{1\}$. We call $\mathbb{H}$ a *strict* subgroup of $\mathbb{G}$ if $\mathbb{H} \neq \mathbb{G}$.

In general, we will not use the notation $\circ$ to denote the group operation. Instead, we will use either *additive* notation or *multiplicative* notation depending on the group under discussion. *This does not imply that the group operation corresponds to integer addition or multiplication; it is merely useful notation.* When using additive notation, the group operation applied to two elements g, h is denoted $g + h$; the identity is denoted by 0; the inverse of an element g is denoted by $-g$; and we write $h - g$ in place of $h + (-g)$. When using multiplicative notation, the group operation applied to g, h is denoted by $g \cdot h$ or simply gh ; the identity is denoted by 1; the inverse of an element g is denoted by g^{-1} ; and we sometimes write h/g in place of hg^{-1} .

At this point, it may be helpful to see some examples.

Example 8.10

A set may be a group under one operation, but not another. For example, the set of integers $\mathbb{Z}$ is an abelian group under addition: the identity is the element 0, and every integer g has inverse $-g$. On the other hand, it is not a group under multiplication since, for example, the integer 2 does not have a multiplicative inverse in the integers. $\diamond$

Example 8.11

The set of real numbers $\mathbb{R}$ is not a group under multiplication, since 0 does not have a multiplicative inverse. The set of *nonzero* real numbers, however, is an abelian group under multiplication with identity 1. $\diamond$

The following example introduces the group $\mathbb{Z}_N$ that we will use frequently.

Example 8.12

Let $N > 1$ be an integer. The set $\{0, \dots, N - 1\}$ with respect to addition modulo N (i.e., where $a + b \stackrel{\text{def}}{=} [a + b \bmod N]$) is an abelian group of order N . Closure is obvious; associativity and commutativity follow from the fact that the integers satisfy these properties; the identity is 0; and, since $a + (N - a) = 0 \bmod N$, it follows that the inverse of any element a is $[(N - a) \bmod N]$. We

denote this group by $\mathbb{Z}_N$. (We will also sometimes use $\mathbb{Z}_N$ to denote the set $\{0, \dots, N-1\}$ without regard to any particular group operation.) $\diamond$

We end this section with an easy lemma that formalizes a “cancellation law” for groups.

LEMMA 8.13 *Let $\mathbb{G}$ be a group and $a, b, c \in \mathbb{G}$. If $ac = bc$, then $a = b$. In particular, if $ac = c$ then a is the identity in $\mathbb{G}$.*

PROOF We know $ac = bc$. Multiplying both sides by the unique inverse c^{-1} of c , we obtain $a = b$. In detail:

$$\begin{aligned} ac = bc &\Rightarrow (ac)c^{-1} = (bc) \cdot c^{-1} \Rightarrow a(cc^{-1}) = b(cc^{-1}) \Rightarrow a \cdot 1 = b \cdot 1 \\ &\Rightarrow a = b. \end{aligned}$$

■

Compare the above proof to the discussion (preceding Proposition 8.7) regarding a cancellation law for division modulo N . As indicated by the similarity, the *invertible* elements modulo N form a group under multiplication modulo N . We will return to this example in more detail shortly.

Group Exponentiation

It is often useful to be able to describe the group operation applied m times to a fixed element g , where m is a positive integer. When using additive notation, we express this as $m \cdot g$ or mg ; that is,

$$mg = m \cdot g \stackrel{\text{def}}{=} \underbrace{g + \dots + g}_{m \text{ times}}.$$

Note that m is an *integer*, while g is a *group element*. So mg does *not* represent the group operation applied to m and g (indeed, we are working in a group where the group operation is written additively). Thankfully, however, the notation “behaves as it should”; so, for example, if $g \in \mathbb{G}$ and m, m' are integers then $(mg) + (m'g) = (m + m')g$, $m(m'g) = (mm')g$, and $1 \cdot g = g$. In an abelian group $\mathbb{G}$ with $g, h \in \mathbb{G}$, $(mg) + (mh) = m(g + h)$.

When using multiplicative notation, we express application of the group operation m times to an element g by g^m . That is,

$$g^m \stackrel{\text{def}}{=} \underbrace{g \cdots g}_{m \text{ times}}.$$

The familiar rules of exponentiation hold: $g^m \cdot g^{m'} = g^{m+m'}$, $(g^m)^{m'} = g^{mm'}$, and $g^1 = g$. Also, if $\mathbb{G}$ is an abelian group and $g, h \in \mathbb{G}$ then $g^m \cdot h^m = (gh)^m$.

All these are simply “translations” of the results from the previous paragraph to the setting of groups written multiplicatively rather than additively.

The above notation is extended in the natural way to the case when m is zero or a negative integer. When using additive notation we define $0 \cdot g \stackrel{\text{def}}{=} 0$ and $(-m) \cdot g \stackrel{\text{def}}{=} m \cdot (-g)$ for m a positive integer. (Note that in the equation ‘ $0 \cdot g = 0$ ’ the 0 on the left-hand side is the integer 0 while the 0 on the right-hand side is the identity element in the group.) Observe that $-g$ is the inverse of g and, as one would expect, $(-m) \cdot g = -(mg)$. When using multiplicative notation, $g^0 \stackrel{\text{def}}{=} 1$ and $g^{-m} \stackrel{\text{def}}{=} (g^{-1})^m$. Again, g^{-1} is the inverse of g , and we have $g^{-m} = (g^m)^{-1}$.

Let $g \in \mathbb{G}$ and $b \geq 0$ be an integer. Then the exponentiation g^b can be computed using a polynomial number of underlying group operations in $\mathbb{G}$. Thus, if the group operation can be computed in polynomial time then so can exponentiation. This is discussed in Appendix B.2.3.

We now know enough to prove the following remarkable result:

THEOREM 8.14 *Let $\mathbb{G}$ be a finite group with $m = |\mathbb{G}|$, the order of the group. Then for any element $g \in \mathbb{G}$, $g^m = 1$.*

PROOF We prove the theorem only when $\mathbb{G}$ is abelian (although it holds for any finite group). Fix arbitrary $g \in \mathbb{G}$, and let $g_1, \dots, g_m$ be the elements of $\mathbb{G}$. We claim that

$$g_1 \cdot g_2 \cdots g_m = (gg_1) \cdot (gg_2) \cdots (gg_m).$$

To see this, note that $gg_i = gg_j$ implies $g_i = g_j$ by Lemma 8.13. So each of the m elements in parentheses on the right-hand side is distinct. Because there are exactly m elements in $\mathbb{G}$, the m elements being multiplied together on the right-hand side are simply all elements of $\mathbb{G}$ in some permuted order. Since $\mathbb{G}$ is abelian, the order in which elements are multiplied does not matter, and so the right-hand side is equal to the left-hand side.

Again using the fact that $\mathbb{G}$ is abelian, we can “pull out” all occurrences of g and obtain

$$g_1 \cdot g_2 \cdots g_m = (gg_1) \cdot (gg_2) \cdots (gg_m) = g^m \cdot (g_1 \cdot g_2 \cdots g_m).$$

Appealing once again to Lemma 8.13, this implies $g^m = 1$. ■

An important corollary of the above is that we can work “modulo the group order” in the exponent:

COROLLARY 8.15 *Let $\mathbb{G}$ be a finite group with $m = |\mathbb{G}| > 1$. Then for any $g \in \mathbb{G}$ and any integer x , we have $g^x = g^{[x \bmod m]}$.*

PROOF Say $x = qm + r$, where q, r are integers and $r = [x \bmod m]$. Then

$$g^x = g^{qm+r} = g^{qm} \cdot g^r = (g^m)^q \cdot g^r = 1^q \cdot g^r = g^r$$

(using Theorem 8.14), as claimed. ■

Example 8.16

Written additively, the above corollary says that if g is an element in a group of order m , then $x \cdot g = [x \bmod m] \cdot g$. As an example, consider the group $\mathbb{Z}_{15}$ of order $m = 15$, and take $g = 11$. The corollary says that

$$152 \cdot 11 = [152 \bmod 15] \cdot 11 = 2 \cdot 11 = 11 + 11 = 22 = 7 \bmod 15.$$

The above agrees with the fact (cf. Example 8.5) that we can “reduce and then multiply” rather than having to “multiply and then reduce.” ◇

Another corollary that will be extremely useful for cryptographic applications is the following:

COROLLARY 8.17 *Let $\mathbb{G}$ be a finite group with $m = |\mathbb{G}| > 1$. Let $e > 0$ be an integer, and define the function $f_e : \mathbb{G} \rightarrow \mathbb{G}$ by $f_e(g) = g^e$. If $\gcd(e, m) = 1$, then f_e is a permutation (i.e., a bijection). Moreover, if $d = e^{-1} \bmod m$ then f_d is the inverse of f_e . (Note by Proposition 8.7, $\gcd(e, m) = 1$ implies e is invertible modulo m .)*

PROOF Since $\mathbb{G}$ is finite, the second part of the claim implies the first; thus, we need only show that f_d is the inverse of f_e . This is true because for any $g \in \mathbb{G}$ we have

$$f_d(f_e(g)) = f_d(g^e) = (g^e)^d = g^{ed} = g^{[ed \bmod m]} = g^1 = g,$$

where the fourth equality follows from Corollary 8.15. ■

8.1.4 The Group $\mathbb{Z}_N^*$

As discussed in Example 8.12, the set $\mathbb{Z}_N = \{0, \dots, N-1\}$ is a group under addition modulo N . Can we define a group with respect to *multiplication* modulo N ? In doing so, we will have to eliminate those elements in $\mathbb{Z}_N$ that are not invertible; e.g., we will have to eliminate 0 since it has no multiplicative inverse. Nonzero elements may also fail to be invertible (cf. Proposition 8.7).

Which elements $b \in \{1, \dots, N-1\}$ are invertible modulo N ? Proposition 8.7 says that these are exactly those elements b for which $\gcd(b, N) = 1$. We have

also seen in Section 8.1.2 that whenever b is invertible, it has an inverse lying in the range $\{1, \dots, N-1\}$. This leads us to define, for any $N > 1$, the set

$$\mathbb{Z}_N^* \stackrel{\text{def}}{=} \{b \in \{1, \dots, N-1\} \mid \gcd(b, N) = 1\};$$

i.e., $\mathbb{Z}_N^*$ consists of integers in the set $\{1, \dots, N-1\}$ that are relatively prime to N . The group operation is multiplication modulo N ; i.e., $ab \stackrel{\text{def}}{=} [ab \bmod N]$.

We claim that $\mathbb{Z}_N^*$ is an abelian group with respect to this operation. Since 1 is always in $\mathbb{Z}_N^*$, the set clearly contains an identity element. The discussion above shows that each element in $\mathbb{Z}_N^*$ has a multiplicative inverse in the same set. Commutativity and associativity follow from the fact that these properties hold over the integers. To show that closure holds, let $a, b \in \mathbb{Z}_N^*$; then $[ab \bmod N]$ has inverse $[b^{-1}a^{-1} \bmod N]$, which means that $\gcd([ab \bmod N], N) = 1$ and so $ab \in \mathbb{Z}_N^*$. Summarizing:

PROPOSITION 8.18 *Let $N > 1$ be an integer. Then $\mathbb{Z}_N^*$ is an abelian group under multiplication modulo N .*

Define $\phi(N) \stackrel{\text{def}}{=} |\mathbb{Z}_N^*|$, the order of the group $\mathbb{Z}_N^*$. (ϕ is called the *Euler phi function*.) What is the value of $\phi(N)$? First consider the case when $N = p$ is prime. Then *all* elements in $\{1, \dots, p-1\}$ are relatively prime to p , and so $\phi(p) = |\mathbb{Z}_p^*| = p-1$. Next consider the case that $N = pq$, where p, q are distinct primes. If an integer $a \in \{1, \dots, N-1\}$ is not relatively prime to N , then either $p \mid a$ or $q \mid a$ (a cannot be divisible by both p and q since this would imply $pq \mid a$ but $a < N = pq$). The elements in $\{1, \dots, N-1\}$ divisible by p are exactly the $(q-1)$ elements $p, 2p, 3p, \dots, (q-1)p$, and the elements divisible by q are exactly the $(p-1)$ elements $q, 2q, \dots, (p-1)q$. The number of elements remaining (i.e., those that are neither divisible by p nor q) is therefore given by

$$(N-1) - (q-1) - (p-1) = pq - p - q + 1 = (p-1)(q-1).$$

We have thus proved that $\phi(N) = (p-1)(q-1)$ when N is the product of two distinct primes p and q .

You are asked to prove the following general result (used only rarely in the rest of the book) in Exercise 8.4:

THEOREM 8.19 *Let $N = \prod_i p_i^{e_i}$, where the $\{p_i\}$ are distinct primes and $e_i \geq 1$. Then $\phi(N) = \prod_i p_i^{e_i-1}(p_i-1)$.*

Example 8.20

Take $N = 15 = 5 \cdot 3$. Then $\mathbb{Z}_{15}^* = \{1, 2, 4, 7, 8, 11, 13, 14\}$ and $|\mathbb{Z}_{15}^*| = 8 = 4 \cdot 2 = \phi(15)$. The inverse of 8 in $\mathbb{Z}_{15}^*$ is 2, since $8 \cdot 2 = 16 = 1 \bmod 15$. $\diamond$

We have shown that $\mathbb{Z}_N^*$ is a group of order $\phi(N)$. The following are now easy corollaries of Theorem 8.14 and Corollary 8.17:

COROLLARY 8.21 *Take arbitrary integer $N > 1$ and $a \in \mathbb{Z}_N^*$. Then*

$$a^{\phi(N)} = 1 \bmod N.$$

For the specific case that $N = p$ is prime and $a \in \{1, \dots, p-1\}$, we have

$$a^{p-1} = 1 \bmod p.$$

COROLLARY 8.22 *Fix $N > 1$. For integer $e > 0$ define $f_e : \mathbb{Z}_N^* \rightarrow \mathbb{Z}_N^*$ by $f_e(x) = [x^e \bmod N]$. If e is relatively prime to $\phi(N)$ then f_e is a permutation. Moreover, if $d = e^{-1} \bmod \phi(N)$ then f_d is the inverse of f_e .*

8.1.5 *Isomorphisms and the Chinese Remainder Theorem

Two groups are *isomorphic* if they have the same underlying structure. From a mathematical point of view, an isomorphism of a group $\mathbb{G}$ provides an alternate, but equivalent, way of thinking about $\mathbb{G}$. From a computational perspective, an isomorphism provides a different way to *represent* elements in $\mathbb{G}$, which can often have a significant impact on algorithmic efficiency.

DEFINITION 8.23 *Let $\mathbb{G}, \mathbb{H}$ be groups with respect to the operations $\circ_{\mathbb{G}}, \circ_{\mathbb{H}}$, respectively. A function $f : \mathbb{G} \rightarrow \mathbb{H}$ is an isomorphism from $\mathbb{G}$ to $\mathbb{H}$ if:*

1. *f is a bijection, and*
2. *For all $g_1, g_2 \in \mathbb{G}$ we have $f(g_1 \circ_{\mathbb{G}} g_2) = f(g_1) \circ_{\mathbb{H}} f(g_2)$.*

If there exists an isomorphism from $\mathbb{G}$ to $\mathbb{H}$ then we say that these groups are isomorphic and write $\mathbb{G} \simeq \mathbb{H}$.

In essence, an isomorphism from $\mathbb{G}$ to $\mathbb{H}$ is just a *renaming* of elements of $\mathbb{G}$ as elements of $\mathbb{H}$. Note that if $\mathbb{G}$ is finite and $\mathbb{G} \simeq \mathbb{H}$, then $\mathbb{H}$ must be finite and of the same size as $\mathbb{G}$. Also, if there exists an isomorphism f from $\mathbb{G}$ to $\mathbb{H}$ then f^{-1} is an isomorphism from $\mathbb{H}$ to $\mathbb{G}$. It is possible, however, that f is efficiently computable while f^{-1} is not (or vice versa).

The aim of this section is to use the language of isomorphisms to better understand the group structure of $\mathbb{Z}_N$ and $\mathbb{Z}_N^*$ when $N = pq$ is a product of two distinct primes. We first need to introduce the notion of a *direct product* of groups. Given groups $\mathbb{G}, \mathbb{H}$ with group operations $\circ_{\mathbb{G}}, \circ_{\mathbb{H}}$, respectively, we define a new group $\mathbb{G} \times \mathbb{H}$ (the *direct product* of $\mathbb{G}$ and $\mathbb{H}$) as follows. The elements of $\mathbb{G} \times \mathbb{H}$ are ordered pairs (g, h) with $g \in \mathbb{G}$ and $h \in \mathbb{H}$; thus, if $\mathbb{G}$

has n elements and $\mathbb{H}$ has n' elements, $\mathbb{G} \times \mathbb{H}$ has $n \cdot n'$ elements. The group operation $\circ$ on $\mathbb{G} \times \mathbb{H}$ is applied component-wise; that is:

$$(g, h) \circ (g', h') \stackrel{\text{def}}{=} (g \circ_{\mathbb{G}} g', h \circ_{\mathbb{H}} h').$$

We leave it to Exercise 8.8 to verify that $\mathbb{G} \times \mathbb{H}$ is indeed a group. The above notation can be extended to direct products of more than two groups in the natural way, although we will not need this for what follows.

We may now state and prove the *Chinese remainder theorem*.

THEOREM 8.24 (Chinese remainder theorem) *Let $N = pq$ where $p, q > 1$ are relatively prime. Then*

$$\mathbb{Z}_N \simeq \mathbb{Z}_p \times \mathbb{Z}_q \quad \text{and} \quad \mathbb{Z}_N^* \simeq \mathbb{Z}_p^* \times \mathbb{Z}_q^*.$$

Moreover, let f be the function mapping elements $x \in \{0, \dots, N-1\}$ to pairs (x_p, x_q) with $x_p \in \{0, \dots, p-1\}$ and $x_q \in \{0, \dots, q-1\}$ defined by

$$f(x) \stackrel{\text{def}}{=} ([x \bmod p], [x \bmod q]).$$

Then f is an isomorphism from $\mathbb{Z}_N$ to $\mathbb{Z}_p \times \mathbb{Z}_q$, and the restriction of f to $\mathbb{Z}_N^*$ is an isomorphism from $\mathbb{Z}_N^*$ to $\mathbb{Z}_p^* \times \mathbb{Z}_q^*$.

PROOF For any $x \in \mathbb{Z}_N$ the output $f(x)$ is a pair of elements (x_p, x_q) with $x_p \in \mathbb{Z}_p$ and $x_q \in \mathbb{Z}_q$. We claim that if $x \in \mathbb{Z}_N^*$, then $(x_p, x_q) \in \mathbb{Z}_p^* \times \mathbb{Z}_q^*$. Indeed, if $x_p \notin \mathbb{Z}_p^*$ then this means that $\gcd([x \bmod p], p) \neq 1$. But then $\gcd(x, p) \neq 1$. This implies $\gcd(x, N) \neq 1$, contradicting the assumption that $x \in \mathbb{Z}_N^*$. (An analogous argument holds if $x_q \notin \mathbb{Z}_q^*$.)

We now show that f is an isomorphism from $\mathbb{Z}_N$ to $\mathbb{Z}_p \times \mathbb{Z}_q$. (The proof that it is an isomorphism from $\mathbb{Z}_N^*$ to $\mathbb{Z}_p^* \times \mathbb{Z}_q^*$ is similar.) Let us start by proving that f is one-to-one. Say $f(x) = (x_p, x_q) = f(x')$. Then $x = x_p = x' \bmod p$ and $x = x_q = x' \bmod q$. This in turn implies that $(x - x')$ is divisible by both p and q . Since $\gcd(p, q) = 1$, Proposition 8.4 says that $pq = N$ divides $(x - x')$. But then $x = x' \bmod N$. For $x, x' \in \mathbb{Z}_N$, this means that $x = x'$ and so f is indeed one-to-one. Since $|\mathbb{Z}_N| = N = p \cdot q = |\mathbb{Z}_p| \cdot |\mathbb{Z}_q|$, the sizes of $\mathbb{Z}_N$ and $\mathbb{Z}_p \times \mathbb{Z}_q$ are the same. This in combination with the fact that f is one-to-one implies that f is bijective.

In the following paragraph, let $+_N$ denote addition modulo N , and let $\boxplus$ denote the group operation in $\mathbb{Z}_p \times \mathbb{Z}_q$ (i.e., addition modulo p in the first component and addition modulo q in the second component). To conclude the proof that f is an isomorphism from $\mathbb{Z}_N$ to $\mathbb{Z}_p \times \mathbb{Z}_q$, we need to show that for all $a, b \in \mathbb{Z}_N$ it holds that $f(a +_N b) = f(a) \boxplus f(b)$.

To see that this is true, note that

$$\begin{aligned} f(a +_N b) &= \left([(a +_N b) \bmod p], [(a +_N b) \bmod q] \right) \\ &= \left([(a + b) \bmod p], [(a + b) \bmod q] \right) \\ &= \left([a \bmod p], [a \bmod q] \right) \boxplus \left([b \bmod p], [b \bmod q] \right) = f(a) \boxplus f(b). \end{aligned}$$

(For the second equality, above, we use the fact that $[(X \bmod N) \bmod p] = [(X \bmod p) \bmod p]$ when $p \mid N$; see Exercise 8.9.) $\blacksquare$

An extension of the Chinese remainder theorem says that if $p_1, p_2, \dots, p_\ell$ are pairwise relatively prime (i.e., $\gcd(p_i, p_j) = 1$ for all $i \neq j$) and $N \stackrel{\text{def}}{=} \prod_{i=1}^{\ell} p_i$, then

$$\mathbb{Z}_N \simeq \mathbb{Z}_{p_1} \times \cdots \times \mathbb{Z}_{p_\ell} \quad \text{and} \quad \mathbb{Z}_N^* \simeq \mathbb{Z}_{p_1}^* \times \cdots \times \mathbb{Z}_{p_\ell}^*.$$

An isomorphism in each case is obtained by a natural extension of the one used in the theorem above.

By way of notation, with N understood and $x \in \{0, 1, \dots, N-1\}$ we write $x \leftrightarrow (x_p, x_q)$ for $x_p = [x \bmod p]$ and $x_q = [x \bmod q]$. That is, $x \leftrightarrow (x_p, x_q)$ if and only if $f(x) = (x_p, x_q)$, where f is as in the theorem above. One way to think about this notation is that it means “ x (in $\mathbb{Z}_N$) corresponds to (x_p, x_q) (in $\mathbb{Z}_p \times \mathbb{Z}_q$).” The same notation is used when dealing with $x \in \mathbb{Z}_N^*$.

Example 8.25

Take $15 = 5 \cdot 3$, and consider $\mathbb{Z}_{15}^* = \{1, 2, 4, 7, 8, 11, 13, 14\}$. The Chinese remainder theorem says this group is isomorphic to $\mathbb{Z}_5^* \times \mathbb{Z}_3^*$. We can compute

$$\begin{array}{cccc} 1 \leftrightarrow (1, 1) & 2 \leftrightarrow (2, 2) & 4 \leftrightarrow (4, 1) & 7 \leftrightarrow (2, 1) \\ 8 \leftrightarrow (3, 2) & 11 \leftrightarrow (1, 2) & 13 \leftrightarrow (3, 1) & 14 \leftrightarrow (4, 2) \end{array},$$

where each pair (a, b) with $a \in \mathbb{Z}_5^*$ and $b \in \mathbb{Z}_3^*$ appears exactly once. $\diamond$

Using the Chinese Remainder Theorem

If two groups are isomorphic, then they both serve as representations of the same underlying “algebraic structure.” Nevertheless, the choice of which representation to use can affect the *computational efficiency* of group operations. We discuss this abstractly, and then in the specific context of $\mathbb{Z}_N$ and $\mathbb{Z}_N^*$.

Let $\mathbb{G}, \mathbb{H}$ be groups with operations $\circ_{\mathbb{G}}, \circ_{\mathbb{H}}$, respectively, and say f is an isomorphism from $\mathbb{G}$ to $\mathbb{H}$ where both f and f^{-1} can be computed efficiently. Then for $g_1, g_2 \in \mathbb{G}$ we can compute $g = g_1 \circ_{\mathbb{G}} g_2$ in two ways: either by directly computing the group operation in $\mathbb{G}$, or via the following steps:

1. Compute $h_1 = f(g_1)$ and $h_2 = f(g_2)$;

2. Compute $h = h_1 \circ_{\mathbb{H}} h_2$ using the group operation in $\mathbb{H}$;
3. Compute $g = f^{-1}(h)$.

The above extends in the natural way when we want to compute multiple group operations in $\mathbb{G}$ (e.g., to compute g^x for some integer x). Which method is better depends on the relative efficiency of computing the group operation in each group, as well as the efficiency of computing f and f^{-1} .

We now turn to the specific case of computations modulo N , when $N = pq$ is a product of distinct primes. The Chinese remainder theorem shows that addition, multiplication, or exponentiation (which is just repeated multiplication) modulo N can be “transformed” to analogous operations modulo p and q . Building on Example 8.25, we show some simple examples with $N = 15$.

Example 8.26

Say we want to compute the product $14 \cdot 13$ modulo 15 (i.e., in $\mathbb{Z}_{15}^*$). Example 8.25 gives $14 \leftrightarrow (4, 2)$ and $13 \leftrightarrow (3, 1)$. In $\mathbb{Z}_5^* \times \mathbb{Z}_3^*$, we have

$$(4, 2) \cdot (3, 1) = ([4 \cdot 3 \bmod 5], [2 \cdot 1 \bmod 3]) = (2, 2).$$

Note $(2, 2) \leftrightarrow 2$, which is the correct answer since $14 \cdot 13 = 2 \bmod 15$. ◇

Example 8.27

Say we want to compute $11^{53} \bmod 15$. Example 8.25 gives $11 \leftrightarrow (1, 2)$. Notice that $2 = -1 \bmod 3$ and so

$$(1, 2)^{53} = ([1^{53} \bmod 5], [(-1)^{53} \bmod 3]) = (1, [-1 \bmod 3]) = (1, 2).$$

Thus, $11^{53} \bmod 15 = 11$. ◇

Example 8.28

Say we want to compute $[29^{100} \bmod 35]$. We first compute the correspondence $29 \leftrightarrow ([29 \bmod 5], [29 \bmod 7]) = ([-1 \bmod 5], 1)$. Using the Chinese remainder theorem, we have

$$([-1 \bmod 5], 1)^{100} = ([-1^{100} \bmod 5], [1^{100} \bmod 7]) = (1, 1),$$

and it is immediate that $(1, 1) \leftrightarrow 1$. We conclude that $[29^{100} \bmod 35] = 1$. ◇

Example 8.29

Say we want to compute $[18^{25} \bmod 35]$. We have $18 \leftrightarrow (3, 4)$ and so

$$18^{25} \bmod 35 \leftrightarrow (3, 4)^{25} = ([3^{25} \bmod 5], [4^{25} \bmod 7]).$$

Since $\mathbb{Z}_5^*$ is a group of order 4, we can “work modulo 4 in the exponent” (cf. Corollary 8.15) and see that

$$3^{25} = 3^{[25 \bmod 4]} = 3^1 = 3 \bmod 5.$$

Similarly,

$$4^{25} = 4^{[25 \bmod 6]} = 4^1 = 4 \bmod 7.$$

Thus, $([3^{25} \bmod 5], [4^{25} \bmod 7]) = (3, 4) \leftrightarrow 18$ and so $[18^{25} \bmod 35] = 18$. $\diamond$

One thing we have not yet discussed is how to convert back and forth between the representation of an element modulo N and its representation modulo p and q . The conversion can be carried out efficiently provided the factorization of N is known. Assuming p and q are known, it is easy to map an element x modulo N to its corresponding representation modulo p and q : the element x corresponds to $([x \bmod p], [x \bmod q])$, and both the modular reductions can be carried out efficiently (cf. Appendix B.2).

For the other direction, we make use of the following observation: an element with representation (x_p, x_q) can be written as

$$(x_p, x_q) = x_p \cdot (1, 0) + x_q \cdot (0, 1).$$

So, if we can find elements $1_p, 1_q \in \{0, \dots, N-1\}$ such that $1_p \leftrightarrow (1, 0)$ and $1_q \leftrightarrow (0, 1)$, then (appealing to the Chinese remainder theorem) we know that

$$(x_p, x_q) \leftrightarrow [(x_p \cdot 1_p + x_q \cdot 1_q) \bmod N].$$

Since p, q are distinct primes, $\gcd(p, q) = 1$. We can use the extended Euclidean algorithm (cf. Appendix B.1.2) to find integers X, Y such that

$$Xp + Yq = 1.$$

Note that $Yq = 0 \bmod q$ and $Yq = 1 - Xp = 1 \bmod p$. This means that $[Yq \bmod N] \leftrightarrow (1, 0)$; i.e., $[Yq \bmod N] = 1_p$. Similarly, $[Xp \bmod N] = 1_q$.

In summary, we can convert an element represented as (x_p, x_q) to its representation modulo N in the following way (assuming p and q are known):

1. Compute X, Y such that $Xp + Yq = 1$.
2. Set $1_p := [Yq \bmod N]$ and $1_q := [Xp \bmod N]$.
3. Compute $x := [(x_p \cdot 1_p + x_q \cdot 1_q) \bmod N]$.

If many such conversions will be performed, then $1_p, 1_q$ can be computed once-and-for-all in a preprocessing phase.

Example 8.30

Take $p = 5$, $q = 7$, and $N = 5 \cdot 7 = 35$. Say we are given the representation $(4, 3)$ and want to convert this to the corresponding element of $\mathbb{Z}_{35}$. Using the extended Euclidean algorithm, we compute

$$3 \cdot 5 - 2 \cdot 7 = 1.$$

Thus, $1_p = [-2 \cdot 7 \bmod 35] = 21$ and $1_q = [3 \cdot 5 \bmod 35] = 15$. (We can check that these are correct: e.g., for $1_p = 21$ we can verify that $[21 \bmod 5] = 1$ and $[21 \bmod 7] = 0$.) Using these values, we can then compute

$$\begin{aligned}(4, 3) &= 4 \cdot (1, 0) + 3 \cdot (0, 1) \\ &\leftrightarrow [4 \cdot 1_p + 3 \cdot 1_q \bmod 35] \\ &= [4 \cdot 21 + 3 \cdot 15 \bmod 35] = 24.\end{aligned}$$

Since $24 = 4 \bmod 5$ and $24 = 3 \bmod 7$, this is indeed the correct result. $\diamond$

8.2 Primes, Factoring, and RSA

In this section, we show the first examples of number-theoretic problems that are conjectured to be “hard.” We begin with a discussion of one of the oldest problems: *integer factorization* or just *factoring*.

Given a composite integer N , the factoring problem is to find integers $p, q > 1$ such that $pq = N$. Factoring is a classic example of a hard problem, both because it is so simple to describe and since it has been recognized as a hard computational problem for a long time (even before its use in cryptography). The problem can be solved in *exponential* time $\mathcal{O}(\sqrt{N} \cdot \text{polylog}(N))$ using *trial division*: that is, by exhaustively checking whether p divides N for $p = 2, \dots, \lfloor \sqrt{N} \rfloor$. (This method requires $\sqrt{N}$ divisions, each one taking $\text{polylog}(N) = \|N\|^c$ time for some constant c .) This always succeeds because although the *largest* prime factor of N may be as large as $N/2$, the *smallest* prime factor of N can be at most $\lfloor \sqrt{N} \rfloor$. Although algorithms with better running time are known (see Chapter 9), no *polynomial-time* algorithm for factoring has been demonstrated despite many years of effort.

Consider the following experiment for a given algorithm $\mathcal{A}$ and parameter n :

The weak factoring experiment $\text{w-Factor}_{\mathcal{A}}(n)$:

1. Choose two uniform n -bit integers x_1, x_2 .
2. Compute $N := x_1 \cdot x_2$.
3. $\mathcal{A}$ is given N , and outputs $x'_1, x'_2 > 1$.
4. The output of the experiment is defined to be 1 if $x'_1 \cdot x'_2 = N$, and 0 otherwise.

We have just said that the factoring problem is believed to be hard. Does this mean that

$$\Pr[\text{w-Factor}_{\mathcal{A}}(n) = 1] \leq \text{negl}(n)$$

is negligible for every PPT algorithm $\mathcal{A}$? Not at all. For starters, the number N in the above experiment is *even* with probability $3/4$ (this occurs when

either x_1 or x_2 is even); it is, of course, easy for $\mathcal{A}$ to factor N in this case. While we can make $\mathcal{A}$'s job more difficult by requiring $\mathcal{A}$ to output integers x'_1, x'_2 of length n , it remains the case that x_1 or x_2 (and hence N) might have small prime factors that can still be easily found. For cryptographic applications, we will need to prevent this.

As this discussion indicates, the “hardest” numbers to factor are those having only large prime factors. This suggests redefining the above experiment so that x_1, x_2 are random n -bit *primes* rather than random n -bit *integers*, and in fact such an experiment will be used when we formally define the factoring assumption in Section 8.2.3. For this experiment to be useful in a cryptographic setting, however, it is necessary to be able to generate random n -bit primes *efficiently*. This is the topic of the next two sections.

8.2.1 Generating Random Primes

A natural approach to generating a random n -bit prime is to repeatedly choose random n -bit integers until we find one that is prime; we repeat this at most t times or until we are successful. See Algorithm 8.31 for a high-level description of the process.

ALGORITHM 8.31

Generating a random prime – high-level outline

Input: Length n ; parameter t

Output: A uniform n -bit prime

for $i = 1$ to t :

$p' \leftarrow \{0, 1\}^{n-1}$

$p := 1 \parallel p'$

if p is prime **return** p

return fail

Note that the algorithm forces the output to be an integer of length *exactly* n (rather than length *at most* n) by fixing the high-order bit of p to “1.” Our convention throughout this book is that an “integer of length n ” means an integer whose binary representation *with most significant bit equal to 1* is exactly n bits long.

Given a way to determine whether or not a given integer p is prime, the above algorithm outputs a *uniform* n -bit prime conditioned on the event that it does not output fail. The probability that the algorithm outputs fail depends on t , and for our purposes we will want to set t so as to obtain a failure probability that is negligible in n . To show that Algorithm 8.31 leads to an *efficient* (i.e., polynomial-time in n) algorithm for generating primes, we need a better understanding of two issues: (1) the probability that a uniform n -bit integer is prime and (2) how to efficiently test whether a given integer

p is prime. We discuss these issues briefly now, and defer a more in-depth exploration of the second topic to the following section.

The distribution of primes. The *prime number theorem*, an important result in mathematics, gives fairly precise bounds on the fraction of integers of a given length that are prime. For our purposes, we need only a weak, one-sided version of that result that we do not prove here:

THEOREM 8.32 (Bertrand’s postulate) *For any $n > 1$, the fraction of n -bit integers that are prime is at least $1/3n$.*

Returning to the approach for generating primes described above, this implies that if we set $t = 3n^2$ then the probability that a prime is *not* chosen in all t iterations of the algorithm is at most

$$\left(1 - \frac{1}{3n}\right)^t = \left(\left(1 - \frac{1}{3n}\right)^{3n}\right)^n \leq (e^{-1})^n = e^{-n}$$

(using Inequality A.2), which is negligible in n . Thus, using $\text{poly}(n)$ iterations we obtain an algorithm for which the probability of outputting *fail* is negligible in n . (Tighter results than Theorem 8.32 are known, and so in practice even fewer iterations are needed.)

Testing primality. The problem of efficiently determining whether a given number is prime has a long history. In the 1970s the first efficient algorithms for testing primality were developed. These algorithms were *probabilistic* and had the following guarantee: if the input p were a prime number, the algorithm would always output “prime.” On the other hand, if p were composite, then the algorithm would almost always output “composite,” but might output the wrong answer (“prime”) with probability negligible in the length of p . Put differently, if the algorithm outputs “composite” then p is definitely composite, but if the output is “prime” then it is very likely that p is prime but it is also possible that a mistake has occurred (and p is really composite).²

When using a randomized primality test of this sort in Algorithm 8.31 (the prime-generation algorithm shown earlier), the output of the algorithm is a uniform prime of the desired length so long as the algorithm does not output *fail* *and* the randomized primality test did not err during the execution of the algorithm. This means that an additional source of error (besides the possibility of outputting *fail*) is introduced, and the algorithm may now output a composite number by mistake. Since we can ensure that this happens with only negligible probability, this remote possibility is of no practical concern and we can safely ignore it.

²There also exist probabilistic primality tests that work in the opposite way: they always correctly identify composite numbers but sometimes make a mistake when given a prime as input. We will not consider algorithms of this type.

A *deterministic* polynomial-time algorithm for testing primality was demonstrated in a breakthrough result in 2002. That algorithm, although running in polynomial time, is slower than the probabilistic tests mentioned above. For this reason, probabilistic primality tests are still used exclusively in practice for generating large prime numbers.

In Section 8.2.2 we describe and analyze one of the most commonly used probabilistic primality tests: the *Miller–Rabin* algorithm. This algorithm takes two inputs: an integer p and a parameter t (in unary) that determines the error probability. The Miller–Rabin algorithm runs in time polynomial in $\|p\|$ and t , and satisfies:

THEOREM 8.33 *If p is prime, then the Miller–Rabin test always outputs “prime.” If p is composite, the algorithm outputs “composite” except with probability at most 2^{-t} .*

Putting it all together. Given the preceding discussion, we can now describe a polynomial-time prime-generation algorithm that, on input n , outputs an n -bit prime except with probability negligible in n ; moreover, conditioned on the output p being prime, p is a uniformly distributed n -bit prime. The full procedure is described in Algorithm 8.34.

ALGORITHM 8.34

Generating a random prime

Input: Length n

Output: A uniform n -bit prime

for $i = 1$ to $3n^2$:

$p' \leftarrow \{0, 1\}^{n-1}$

$p := 1\|p'$

run the Miller–Rabin test on input p and parameter 1^n

if the output is “prime,” **return** p

return fail

Generating primes of a particular form. It is sometimes desirable to generate a random n -bit prime p of a particular form, for example, satisfying $p = 3 \bmod 4$ or such that $p = 2q + 1$ where q is also prime (p of the latter type are called *strong primes*). In this case, appropriate modifications of the prime-generation algorithm shown above can be used. (For example, in order to obtain a prime of the form $p = 2q + 1$, modify the algorithm to generate a random prime q , compute $p := 2q + 1$, and then output p if it too is prime.) While these modified algorithms work well in practice, rigorous proofs that they run in polynomial time and fail with only negligible probability are more complex (and, in some cases, rely on unproven number-theoretic conjectures

regarding the density of primes of a particular form). A detailed exploration of these issues is beyond the scope of this book, and we will simply assume the existence of appropriate prime-generation algorithms when needed.

8.2.2 *Primality Testing

We now describe the Miller–Rabin primality test and prove Theorem 8.33. (We rely on the material presented in Section 8.1.5.) This material is not used directly in the rest of the book.

The key to the Miller–Rabin algorithm is to find a property that distinguishes primes and composites. Let N denote the input number to be tested. We start with the following observation: if N is prime then $|\mathbb{Z}_N^*| = N - 1$, and so for any $a \in \{1, \dots, N - 1\}$ we have $a^{N-1} = 1 \pmod N$ by Theorem 8.14. This suggests testing whether N is prime by choosing a uniform element a and checking whether $a^{N-1} \stackrel{?}{=} 1 \pmod N$. If $a^{N-1} \neq 1 \pmod N$, then N cannot be prime. Conversely, we might hope that if N is not prime then there is a reasonable chance that we will pick a with $a^{N-1} \neq 1 \pmod N$, and so by repeating this test many times we can determine whether N is prime or not with high confidence. The above approach is shown as Algorithm 8.35. (Recall that exponentiation modulo N and computation of greatest common divisors can be carried out in polynomial time. Choosing a uniform element of $\{1, \dots, N - 1\}$ can also be done in polynomial time. See Appendix B.2.)

ALGORITHM 8.35

Primality testing – first attempt

Input: Integer N and parameter 1^t

Output: A decision as to whether N is prime or composite

for $i = 1$ **to** t :

$a \leftarrow \{1, \dots, N - 1\}$

if $a^{N-1} \neq 1 \pmod N$ **return** “composite”

return “prime”

If N is prime the algorithm always outputs “prime.” If N is composite, the algorithm outputs “composite” if in any iteration it finds an $a \in \{1, \dots, N - 1\}$ such that $a^{N-1} \neq 1 \pmod N$. Observe that if $a \notin \mathbb{Z}_N^*$ then $a^{N-1} \neq 1 \pmod N$. (If $\gcd(a, N) \neq 1$ then $\gcd(a^{N-1}, N) \neq 1$ and so $[a^{N-1} \pmod N]$ cannot equal 1.) For now, we therefore restrict our attention to $a \in \mathbb{Z}_N^*$. We refer to any such a with $a^{N-1} \neq 1 \pmod N$ as a *witness that N is composite*, or simply a *witness*. We might hope that when N is composite there are *many* witnesses, and thus the algorithm finds such a witness with “high” probability. This intuition is correct *provided there is at least one witness*. Before proving this, we need two group-theoretic lemmas.

PROPOSITION 8.36 *Let $\mathbb{G}$ be a finite group, and $\mathbb{H} \subseteq \mathbb{G}$. Assume $\mathbb{H}$ is nonempty, and for all $a, b \in \mathbb{H}$ we have $ab \in \mathbb{H}$. Then $\mathbb{H}$ is a subgroup of $\mathbb{G}$.*

PROOF We need to verify that $\mathbb{H}$ satisfies all the conditions of Definition 8.9. By assumption, $\mathbb{H}$ is closed under the group operation. Associativity in $\mathbb{H}$ is inherited automatically from $\mathbb{G}$. Let $m = |\mathbb{G}|$ (here is where we use the fact that $\mathbb{G}$ is finite), and consider an arbitrary element $a \in \mathbb{H}$. Closure of $\mathbb{H}$ means that $\mathbb{H}$ contains $a^{m-1} = a^{-1}$ as well as $a^m = 1$. Thus, $\mathbb{H}$ contains the inverse of each of its elements, as well as the identity. ■

LEMMA 8.37 *Let $\mathbb{H}$ be a strict subgroup of a finite group $\mathbb{G}$ (i.e., $\mathbb{H} \neq \mathbb{G}$). Then $|\mathbb{H}| \leq |\mathbb{G}|/2$.*

PROOF Let $\bar{h}$ be an element of $\mathbb{G}$ that is *not* in $\mathbb{H}$; since $\mathbb{H} \neq \mathbb{G}$, we know such an $\bar{h}$ exists. Consider the set $\bar{\mathbb{H}} \stackrel{\text{def}}{=} \{\bar{h}h \mid h \in \mathbb{H}\}$. We show that (1) $|\bar{\mathbb{H}}| = |\mathbb{H}|$, and (2) every element of $\bar{\mathbb{H}}$ lies outside of $\mathbb{H}$; i.e., the intersection of $\mathbb{H}$ and $\bar{\mathbb{H}}$ is empty. Since both $\mathbb{H}$ and $\bar{\mathbb{H}}$ are subsets of $\mathbb{G}$, these imply $|\mathbb{G}| \geq |\mathbb{H}| + |\bar{\mathbb{H}}| = 2|\mathbb{H}|$, proving the lemma.

For any $h_1, h_2 \in \mathbb{H}$, if $\bar{h}h_1 = \bar{h}h_2$ then, multiplying by $\bar{h}^{-1}$ on each side, we have $h_1 = h_2$. This shows that every distinct element $h \in \mathbb{H}$ corresponds to a distinct element $\bar{h}h \in \bar{\mathbb{H}}$, proving (1).

Assume toward a contradiction that $\bar{h}h \in \mathbb{H}$ for some h . This means $\bar{h}h = h'$ for some $h' \in \mathbb{H}$, and so $\bar{h} = h'h^{-1}$. Now, $h'h^{-1} \in \mathbb{H}$ since $\mathbb{H}$ is a subgroup and $h', h^{-1} \in \mathbb{H}$. But this means that $\bar{h} \in \mathbb{H}$, in contradiction to the way $\bar{h}$ was chosen. This proves (2) and completes the proof of the lemma. ■

The following theorem will enable us to analyze the algorithm given earlier.

THEOREM 8.38 *Fix N . Say there exists a witness that N is composite. Then at least half the elements of $\mathbb{Z}_N^*$ are witnesses that N is composite.*

PROOF Let Bad be the set of elements in $\mathbb{Z}_N^*$ that are *not* witnesses; that is, $a \in \text{Bad}$ means $a^{N-1} \neq 1 \pmod{N}$. Clearly, $1 \in \text{Bad}$. If $a, b \in \text{Bad}$, then $(ab)^{N-1} = a^{N-1} \cdot b^{N-1} \neq 1 \cdot 1 = 1 \pmod{N}$ and hence $ab \in \text{Bad}$. By Lemma 8.36, we conclude that Bad is a subgroup of $\mathbb{Z}_N^*$. Since (by assumption) there is at least one witness, Bad is a *strict* subgroup of $\mathbb{Z}_N^*$. Lemma 8.37 then shows that $|\text{Bad}| \leq |\mathbb{Z}_N^*|/2$, showing that at least half the elements of $\mathbb{Z}_N^*$ are *not* in Bad (and hence are witnesses). ■

Let N be composite. If there exists a witness that N is composite, then there are at least $|\mathbb{Z}_N^*|/2$ witnesses. The probability that we find either a witness or an element not in $\mathbb{Z}_N^*$ in any given iteration of the algorithm is

thus at least $1/2$, and so the probability that the algorithm does not find a witness in any of the t iterations (and hence the probability that the algorithm mistakenly outputs “prime”) is at most 2^{-t} .

The above, unfortunately, does not give a complete solution since there are infinitely many composite numbers N that do not have *any* witnesses that they are composite! Such values N are known as *Carmichael numbers*; a detailed discussion is beyond the scope of this book.

Happily, a refinement of the above test can be shown to work for all N . Let $N - 1 = 2^r u$, where u is odd and $r \geq 1$. (It is easy to compute r and u given N . Also, restricting to $r \geq 1$ means that N is odd, but testing primality is easy when N is even!) The algorithm shown previously tests only whether $a^{N-1} = a^{2^r u} = 1 \bmod N$. A more refined algorithm looks at the *sequence* of $r + 1$ values $a^u, a^{2u}, \dots, a^{2^r u}$ (all modulo N). Each term in this sequence is the square of the preceding term; thus, if some value is equal to ± 1 then all subsequent values will be equal to 1.

Say that $a \in \mathbb{Z}_N^*$ is a *strong witness that N is composite* (or simply a *strong witness*) if (1) $a^u \neq \pm 1 \bmod N$ and (2) $a^{2^i u} \neq -1 \bmod N$ for all $i \in \{1, \dots, r - 1\}$. Note that when an element a is *not* a strong witness then the sequence $(a^u, a^{2u}, \dots, a^{2^r u})$ (all taken modulo N) takes one of the following forms:

$$(\pm 1, 1, \dots, 1) \quad \text{or} \quad (\star, \dots, \star, -1, 1, \dots, 1),$$

where $\star$ is an arbitrary term. If a is *not* a strong witness then we have $a^{2^{r-1}u} = \pm 1 \bmod N$ and

$$a^{N-1} = a^{2^r u} = \left(a^{2^{r-1}u}\right)^2 = 1 \bmod N,$$

and so a is not a witness that N is composite, either. Put differently, if a is a witness then it is also a strong witness and so there can only possibly be *more* strong witnesses than witnesses.

We first show that if N is prime then there does not exist a strong witness that N is composite. In doing so, we rely on the following easy lemma (which is a special case of Proposition 13.16 proved subsequently in Chapter 13):

LEMMA 8.39 *Say $x \in \mathbb{Z}_N^*$ is a square root of 1 modulo N if $x^2 = 1 \bmod N$. If N is an odd prime then the only square roots of 1 modulo N are $[\pm 1 \bmod N]$.*

PROOF Say $x^2 = 1 \bmod N$ with $x \in \{1, \dots, N - 1\}$. Then $0 = x^2 - 1 = (x + 1)(x - 1) \bmod N$, implying that $N \mid (x + 1)$ or $N \mid (x - 1)$ by Proposition 8.3. This can only possibly occur if $x = [\pm 1 \bmod N]$. ■

Let N be an odd prime and fix arbitrary $a \in \mathbb{Z}_N^*$. Let $i \geq 0$ be the minimum value for which $a^{2^i u} = 1 \bmod N$; since $a^{2^r u} = a^{N-1} = 1 \bmod N$ we know that

some such $i \leq r$ exists. If $i = 0$ then $a^u = 1 \bmod N$ and a is not a strong witness. Otherwise,

$$\left(a^{2^{i-1}u}\right)^2 = a^{2^i u} = 1 \bmod N$$

and $a^{2^{i-1}u}$ is a square root of 1. If N is an odd prime, the only square roots of 1 are ± 1 ; by choice of i , however, $a^{2^{i-1}u} \neq 1 \bmod N$. So $a^{2^{i-1}u} = -1 \bmod N$, and a is not a strong witness. We conclude that when N is an odd prime there is no strong witness that N is composite.

A composite integer N is a *prime power* if $N = p^r$ for some prime p and integer $r \geq 1$. We now show that every odd, composite N that is not a prime power has many strong witnesses.

THEOREM 8.40 *Let N be an odd number that is not a prime power. Then at least half the elements of $\mathbb{Z}_N^*$ are strong witnesses that N is composite.*

PROOF Let $\text{Bad} \subseteq \mathbb{Z}_N^*$ denote the set of elements that are not strong witnesses. We define a set Bad' and show that: (1) Bad is a subset of Bad' , and (2) Bad' is a strict subgroup of $\mathbb{Z}_N^*$. This suffices because by combining (2) and Lemma 8.37 we have that $|\text{Bad}'| \leq |\mathbb{Z}_N^*|/2$. Furthermore, by (1) it holds that $\text{Bad} \subseteq \text{Bad}'$, and so $|\text{Bad}| \leq |\text{Bad}'| \leq |\mathbb{Z}_N^*|/2$ as in Theorem 8.38. Thus, at least half the elements of $\mathbb{Z}_N^*$ are strong witnesses. (We stress that we do not claim that Bad is a subgroup of $\mathbb{Z}_N^*$.)

Note first that $-1 \in \text{Bad}$ since $(-1)^u = -1 \bmod N$ (recall u is odd). Let $i \in \{0, \dots, r-1\}$ be the largest integer for which there exists an $a \in \text{Bad}$ with $a^{2^i u} = -1 \bmod N$; alternatively, i is the largest integer for which there exists an $a \in \text{Bad}$ with

$$(a^u, a^{2u}, \dots, a^{2^r u}) = (\underbrace{\star, \dots, \star}_{i+1 \text{ terms}}, -1, 1, \dots, 1).$$

Since $-1 \in \text{Bad}$ and $(-1)^{2^0 u} = -1 \bmod N$, some such i exists.

Fix i as above, and define

$$\text{Bad}' \stackrel{\text{def}}{=} \{a \mid a^{2^i u} = \pm 1 \bmod N\}.$$

We now prove what we claimed above.

CLAIM 8.41 $\text{Bad} \subseteq \text{Bad}'$.

Let $a \in \text{Bad}$. Then either $a^u = 1 \bmod N$ or $a^{2^j u} = -1 \bmod N$ for some $j \in \{0, \dots, r-1\}$. In the first case, $a^{2^i u} = (a^u)^{2^i} = 1 \bmod N$ and so $a \in \text{Bad}'$. In the second case, we have $j \leq i$ by choice of i . If $j = i$ then clearly $a \in \text{Bad}'$.

If $j < i$ then $a^{2^i u} = (a^{2^j u})^{2^{i-j}} = 1 \bmod N$ and $a \in \text{Bad}'$. Since a was arbitrary, this shows $\text{Bad} \subseteq \text{Bad}'$.

CLAIM 8.42 Bad' is a subgroup of $\mathbb{Z}_N^*$.

Clearly $1 \in \text{Bad}'$. Furthermore, if $a, b \in \text{Bad}'$ then

$$(ab)^{2^i u} = a^{2^i u} b^{2^i u} = (\pm 1)(\pm 1) = \pm 1 \bmod N$$

and so $ab \in \text{Bad}'$. By Lemma 8.36, Bad' is a subgroup.

CLAIM 8.43 Bad' is a strict subgroup of $\mathbb{Z}_N^*$.

If N is an odd, composite integer that is not a prime power, then N can be written as $N = N_1 N_2$ with $N_1, N_2 > 1$ odd and $\gcd(N_1, N_2) = 1$. Appealing to the Chinese remainder theorem, let $a \leftrightarrow (a_1, a_2)$ denote the representation of $a \in \mathbb{Z}_N^*$ as an element of $\mathbb{Z}_{N_1}^* \times \mathbb{Z}_{N_2}^*$; that is, $a_1 = [a \bmod N_1]$ and $a_2 = [a \bmod N_2]$. Take $a \in \text{Bad}'$ such that $a^{2^i u} = -1 \bmod N$ (such an a must exist by the way we defined i), and say $a \leftrightarrow (a_1, a_2)$. Since $-1 \leftrightarrow (-1, -1)$ we have

$$(a_1, a_2)^{2^i u} = (a_1^{2^i u}, a_2^{2^i u}) = (-1, -1),$$

and so

$$a_1^{2^i u} = -1 \bmod N_1 \quad \text{and} \quad a_2^{2^i u} = -1 \bmod N_2.$$

Consider the element $b \in \mathbb{Z}_N^*$ with $b \leftrightarrow (a_1, 1)$. Then

$$b^{2^i u} \leftrightarrow (a_1, 1)^{2^i u} = ([a_1^{2^i u} \bmod N_1], 1) = (-1, 1) \not\leftrightarrow \pm 1.$$

That is, $b^{2^i u} \neq \pm 1 \bmod N$ and so we have found an element $b \notin \text{Bad}'$. This proves that Bad' is a *strict* subgroup of $\mathbb{Z}_N^*$ and so, by Lemma 8.37, the size of Bad' (and thus the size of Bad) is at most half the size of $\mathbb{Z}_N^*$. ■

An integer N is a *perfect power* if $N = \hat{N}^e$ for integers $\hat{N}$ and $e \geq 2$ (here it is not required for $\hat{N}$ to be prime, although of course any prime power is also a perfect power). Algorithm 8.44 gives the Miller–Rabin primality test. Exercises 8.12 and 8.13 ask you to show that testing whether N is a perfect power, and testing whether a particular a is a strong witness, can be done in polynomial time. Given these results, the algorithm clearly runs in time polynomial in $\|N\|$ and t . We can now complete the proof of Theorem 8.33:

PROOF If N is an odd prime, there are no strong witnesses and so the Miller–Rabin algorithm always outputs “prime.” If N is even or a prime power, the algorithm always outputs “composite.” The interesting case is when N is an odd, composite integer that is not a prime power. Consider any

ALGORITHM 8.44
The Miller–Rabin primality test

Input: Integer $N > 2$ and parameter 1^t

Output: A decision as to whether N is prime or composite

if N is even, **return** “composite”

if N is a perfect power, **return** “composite”

compute $r \geq 1$ and u odd such that $N - 1 = 2^r u$

for $j = 1$ to t :

$a \leftarrow \{1, \dots, N - 1\}$

if $a^u \not\equiv \pm 1 \pmod N$ and $a^{2^i u} \not\equiv -1 \pmod N$ for $i \in \{1, \dots, r - 1\}$

return “composite”

return “prime”

iteration of the inner loop. Note first that if $a \notin \mathbb{Z}_N^*$ then $a^u \not\equiv \pm 1 \pmod N$ and $a^{2^i u} \not\equiv -1 \pmod N$ for $i \in \{1, \dots, r - 1\}$. The probability of finding either a strong witness or an element not in $\mathbb{Z}_N^*$ is at least $1/2$ (invoking Theorem 8.40). Thus, the probability that the algorithm never outputs “composite” in any of the t iterations is at most 2^{-t} . ■

8.2.3 The Factoring Assumption

Let GenModulus be a polynomial-time algorithm that, on input 1^n , outputs (N, p, q) where $N = pq$, and p and q are n -bit primes except with probability negligible in n . (The natural way to do this is to generate two uniform n -bit primes, as discussed previously, and then multiply them to obtain N .) Then consider the following experiment for a given algorithm $\mathcal{A}$ and parameter n :

The factoring experiment $\text{Factor}_{\mathcal{A}, \text{GenModulus}}(n)$:

1. Run $\text{GenModulus}(1^n)$ to obtain (N, p, q) .
2. $\mathcal{A}$ is given N , and outputs $p', q' > 1$.
3. The output of the experiment is defined to be 1 if $p' \cdot q' = N$, and 0 otherwise.

Note that if the output of the experiment is 1 then $\{p', q'\} = \{p, q\}$, unless p or q are composite (which happens with only negligible probability).

We now formally define the factoring assumption:

DEFINITION 8.45 Factoring is hard relative to GenModulus if for all probabilistic polynomial-time algorithms $\mathcal{A}$ there exists a negligible function negl such that

$$\Pr[\text{Factor}_{\mathcal{A}, \text{GenModulus}}(n) = 1] \leq \text{negl}(n).$$

The *factoring assumption* is the assumption that there exists a GenModulus relative to which factoring is hard.

8.2.4 The RSA Assumption

The factoring problem has been studied for hundreds of years without an efficient algorithm being found. Although the factoring assumption does give a one-way function (see Section 8.4.1), it unfortunately does not *directly* yield practical cryptosystems. (In Section 13.5.2, however, we show how to construct efficient cryptosystems based on a problem whose hardness is *equivalent* to that of factoring.) This has motivated a search for other problems whose difficulty is related to the hardness of factoring. The best known of these is a problem introduced in 1978 by Rivest, Shamir, and Adleman and now called the *RSA problem* in their honor.

Given a modulus N and an integer $e > 2$ relatively prime to $\phi(N)$, Corollary 8.22 shows that exponentiation to the e th power modulo N is a *permutation*. We can therefore define $[y^{1/e} \bmod N]$ (for any $y \in \mathbb{Z}_N^*$) as the unique element of $\mathbb{Z}_N^*$ which yields y when raised to the e th power modulo N ; that is, $x = y^{1/e} \bmod N$ if and only if $x^e = y \bmod N$. The RSA problem, informally, is to compute $[y^{1/e} \bmod N]$ for a modulus N of unknown factorization.

Formally, let **GenRSA** be a probabilistic polynomial-time algorithm that, on input 1^n , outputs a modulus N that is the product of two n -bit primes, as well as integers $e, d > 0$ with $\gcd(e, \phi(N)) = 1$ and $ed = 1 \bmod \phi(N)$. (Such a d exists since e is invertible modulo $\phi(N)$. The purpose of d will become clear later.) The algorithm may fail with probability negligible in n . Consider the following experiment for a given algorithm $\mathcal{A}$ and parameter n :

The RSA experiment $\text{RSA-inv}_{\mathcal{A}, \text{GenRSA}}(n)$:

1. Run **GenRSA**(1^n) to obtain (N, e, d) .
2. Choose a uniform $y \in \mathbb{Z}_N^*$.
3. $\mathcal{A}$ is given N, e, y , and outputs $x \in \mathbb{Z}_N^*$.
4. The output of the experiment is defined to be 1 if $x^e = y \bmod N$, and 0 otherwise.

DEFINITION 8.46 The RSA problem is hard relative to **GenRSA** if for all probabilistic polynomial-time algorithms $\mathcal{A}$ there exists a negligible function negl such that $\Pr[\text{RSA-inv}_{\mathcal{A}, \text{GenRSA}}(n) = 1] \leq \text{negl}(n)$.

The *RSA assumption* is that there exists a **GenRSA** algorithm relative to which the RSA problem is hard. A suitable **GenRSA** algorithm can be constructed from any algorithm **GenModulus** that generates a composite modulus along with its factorization. A high-level outline is provided as Algorithm 8.47, where the only thing left unspecified is how exactly e is chosen. In fact, the RSA problem is believed to be hard for *any* e that is relatively prime to $\phi(N)$. We discuss some typical choices of e below.

ALGORITHM 8.47

GenRSA – high-level outline

Input: Security parameter 1^n
Output: N, e, d as described in the text

 $(N, p, q) \leftarrow \text{GenModulus}(1^n)$
 $\phi(N) := (p-1)(q-1)$
choose $e > 1$ such that $\gcd(e, \phi(N)) = 1$
compute $d := [e^{-1} \bmod \phi(N)]$
return N, e, d
Example 8.48

Say GenModulus outputs $(N, p, q) = (143, 11, 13)$. Then $\phi(N) = 120$. Next, we need to choose an e that is relatively prime to $\phi(N)$; say we take $e = 7$. The next step is to compute d such that $d = [e^{-1} \bmod \phi(N)]$. This can be done as shown in Appendix B.2.2 to obtain $d = 103$. (One can check that $7 \cdot 103 = 721 = 1 \bmod 120$.) Our GenRSA algorithm thus outputs $(143, 7, 103)$.

As an example of the RSA problem relative to these parameters, take $y = 64$ and so the problem is to compute the 7th root of 64 modulo 143 *without* knowledge of d or the factorization of N . $\diamond$

Computing e th roots modulo N becomes easy if $d, \phi(N)$, or the factorization of N is known. (As we show in the next section, any of these can be used to efficiently compute the others.) This follows from Corollary 8.22, which shows that $[y^d \bmod N]$ is the e th root of y modulo N . This asymmetry—namely, that the RSA problem appears to be hard when d or the factorization of N is unknown, but becomes easy when d is known—serves as the basis for applications of the RSA problem to public-key cryptography.

Example 8.49

Continuing the previous example, we can compute the 7th root of 64 modulo 143 using the value $d = 103$; the answer is $25 = 64^d = 64^{103} \bmod 143$. We can verify that this is the correct solution since $25^e = 25^7 = 64 \bmod 143$. $\diamond$

On the choice of e . There does not appear to be any difference in the hardness of the RSA problem for different exponents e and, as such, different methods have been suggested for selecting it. One popular choice is to set $e = 3$, since then computing e th powers modulo N requires only two multiplications (see Appendix B.2.3). If e is to be set equal to 3, then p and q must be chosen with $p, q \not\equiv 1 \bmod 3$ so that $\gcd(e, \phi(N)) = 1$. For similar reasons, another popular choice is $e = 2^{16} + 1 = 65537$, a prime number with low Hamming weight (in Appendix B.2.3, we explain why such exponents are preferable). As compared to choosing $e = 3$, this makes exponentiation

slightly more expensive but reduces the constraints on p and q , and avoids some “low-exponent attacks” (described at the end of Section 11.5.1) that can result from poorly implemented cryptosystems based on the RSA problem.

Note that choosing d small (that is, changing GenRSA to choose small d and then compute $e := [d^{-1} \bmod \phi(N)]$) is a bad idea. If d lies in a very small range then a brute-force search for d can be carried out (and, as noted, once d is known the RSA problem can be solved easily). Even if d is chosen so that $d \approx N^{1/4}$, and so brute-force attacks are ruled out, there are known algorithms that can be used to recover d from N and e in this case.

8.2.5 *Relating the RSA and Factoring Assumptions

Say GenRSA is constructed as in Algorithm 8.47. If N can be factored, then we can compute $\phi(N)$ and use this to compute $d := [e^{-1} \bmod \phi(N)]$ for any given e (using Algorithm B.11). So for the RSA problem to be hard relative to GenRSA, the factoring problem must be hard relative to GenModulus. Put differently, the RSA problem cannot be *more* difficult than factoring; hardness of factoring (relative to GenModulus) can only potentially be a *weaker* assumption than hardness of the RSA problem (relative to GenRSA).

What about the other direction? That is, is hardness of the RSA problem implied by hardness of factoring? This remains an open question. The best we can show is that *computing d from N and e is as hard as factoring*.

THEOREM 8.50 *There is a probabilistic polynomial-time algorithm that, given as input a composite integer N and integers e, d with $ed = 1 \bmod \phi(N)$, outputs a factor of N except with probability negligible in $\|N\|$.*

PROOF The theorem holds for any N , but for simplicity—and because it is the case most relevant to cryptography—we focus here on the case where N is a product of two distinct (odd) primes. We rely on Proposition 8.36 and Lemma 8.37 as well as the following facts (which follow from more general results proved in Sections 13.4.2 and 13.5.2):

- If N is a product of two distinct, odd primes, then 1 has exactly four square roots modulo N . Two of these are the “trivial” square roots ± 1 , and two of these are “nontrivial” square roots.
- Any nontrivial square root of 1 can be used to (efficiently) compute a factor of N . This is by virtue of the fact that $y^2 = 1 \bmod N$ implies

$$0 = y^2 - 1 = (y - 1)(y + 1) \bmod N,$$

and so $N \mid (y - 1)(y + 1)$. However, $N \nmid (y - 1)$ and $N \nmid (y + 1)$ because $y \not\equiv \pm 1 \bmod N$. So it must be the case that $\gcd(y - 1, N)$ is equal to one of the prime factors of N .

Let $k = ed - 1$ and note that $\phi(N) \mid k$. Using Corollary 8.21, we have $x^k = 1 \bmod N$ for all $x \in \mathbb{Z}_N^*$. Let $k = 2^r u$ for u an odd integer; note that

$r \geq 1$ since $\phi(N)$ (and hence k) is even. Our strategy for factoring N will be to repeatedly choose a uniform $x \in \mathbb{Z}_N^*$ and compute the sequence

$$x^u, x^{2u}, \dots, x^{2^r u},$$

all modulo N . Each term in this sequence is the square of the preceding term and, as we have just noted, the final term in the sequence is 1. Take the largest i (if any) for which $y \stackrel{\text{def}}{=} [x^{2^i u} \bmod N] \neq 1$. By our choice of i , we have $y^2 = 1 \bmod N$. If $y \neq -1$ we have found a nontrivial square root of N , and can then factor N as discussed above.

All the above steps can be done in polynomial time, and so the only question is to determine the probability, over choice of x , that y is a nontrivial square root of N . Let $i \in \{0, \dots, r-1\}$ be the largest value of i for which there exists an $x \in \mathbb{Z}_N^*$ such that $x^{2^i u} \neq 1 \bmod N$. (Since u is odd $(-1)^u = -1 \neq 1 \bmod N$, and so the definition is not vacuous.) Then for all $x \in \mathbb{Z}_N^*$, we have $x^{2^{i+1}u} = 1 \bmod N$ and so $[x^{2^i u} \bmod N]$ is a square root of 1. Define

$$\mathbf{Bad} \stackrel{\text{def}}{=} \{x \mid x^{2^i u} = \pm 1 \bmod N\}$$

and observe that if our algorithm chooses $x \notin \mathbf{Bad}$ then it finds a nontrivial square root of 1. We show that $\mathbf{Bad}$ is a *strict* subgroup of $\mathbb{Z}_N^*$; by Lemma 8.37, this implies that $|\mathbf{Bad}| \leq |\mathbb{Z}_N^*|/2$. This means that $x \notin \mathbf{Bad}$ (and the algorithm finds a nontrivial square root of 1) with probability at least $1/2$ in each iteration. Using sufficiently many iterations gives the result of the theorem.

We now prove that $\mathbf{Bad}$ is a strict subgroup of $\mathbb{Z}_N^*$. First note that $\mathbf{Bad}$ is not empty, since $1 \in \mathbf{Bad}$. Furthermore, if $x, x' \in \mathbf{Bad}$ then

$$(xx')^{2^i u} = x^{2^i u} (x')^{2^i u} = (\pm 1) \cdot (\pm 1) = \pm 1 \bmod N,$$

and so $xx' \in \mathbf{Bad}$ and $\mathbf{Bad}$ is a subgroup. To see that $\mathbf{Bad}$ is a *strict* subgroup, let $x \in \mathbb{Z}_N^*$ be such that $x^{2^i u} \neq 1 \bmod N$ (such an x must exist by our definition of i). If $x^{2^i u} \neq -1 \bmod N$, then $x \notin \mathbf{Bad}$ and we are done. Otherwise, let $N = pq$ with p, q prime, and let $x \leftrightarrow (x_p, x_q)$ be the Chinese remaindering representation of x . Since $x^{2^i u} = -1 \bmod N$, we know that

$$(x_p, x_q)^{2^i u} = (x_p^{2^i u}, x_q^{2^i u}) = (-1, -1) \leftrightarrow -1.$$

But then $(x_p, 1)$ (or rather, the element corresponding to it) is not in $\mathbf{Bad}$ since

$$(x_p, 1)^{2^i u} = ([x_p^{2^i u} \bmod p], 1) = (-1, 1) \not\leftrightarrow \pm 1.$$

This completes the proof. ■

Assuming factoring is hard, the above result rules out the possibility of efficiently solving the RSA problem by first computing d from N and e . However, it does not rule out the possibility that there might be some completely

different way of attacking the RSA problem that does not involve (or imply) factoring N . Thus, based on our current knowledge, the RSA assumption is stronger than the factoring assumption—that is, it may be that the RSA problem can be solved in polynomial time even though factoring cannot. Nevertheless, when **GenRSA** is constructed based on **GenModulus** as in Algorithm 8.47, the prevailing conjecture is that the RSA problem is hard relative to **GenRSA** whenever factoring is hard relative to **GenModulus**.

8.3 Cryptographic Assumptions in Cyclic Groups

In this section we introduce a class of cryptographic hardness assumptions in *cyclic groups*. We begin with a general discussion of cyclic groups, followed by abstract definitions of the relevant assumptions. We then look at two concrete and widely used examples of cyclic groups in which these assumptions are believed to hold.

8.3.1 Cyclic Groups and Generators

Let $\mathbb{G}$ be a finite group of order m . For arbitrary $g \in \mathbb{G}$, consider the set

$$\langle g \rangle \stackrel{\text{def}}{=} \{g^0, g^1, \dots\}.$$

(We warn the reader that if $\mathbb{G}$ is an infinite group, $\langle g \rangle$ is defined differently.) By Theorem 8.14, we have $g^m = 1$. Let $i \leq m$ be the smallest positive integer for which $g^i = 1$. Then the above sequence repeats after i terms (i.e., $g^i = g^0$, $g^{i+1} = g^1$, etc.), and so

$$\langle g \rangle = \{g^0, \dots, g^{i-1}\}.$$

We see that $\langle g \rangle$ contains at most i elements. In fact, it contains exactly i elements since if $g^j = g^k$ with $0 \leq j < k < i$ then $g^{k-j} = 1$ and $0 < k-j < i$, contradicting our choice of i as the smallest positive integer for which $g^i = 1$.

It is not hard to verify that $\langle g \rangle$ is a subgroup of $\mathbb{G}$ for any g (see Exercise 8.3); we call $\langle g \rangle$ the *subgroup generated by g* . If the order of the subgroup $\langle g \rangle$ is i , then i is called the *order of g* ; that is:

DEFINITION 8.51 Let $\mathbb{G}$ be a finite group and $g \in \mathbb{G}$. The **order of g** is the smallest positive integer i with $g^i = 1$.

The following is a useful analogue of Corollary 8.15 (the proof is identical):

PROPOSITION 8.52 Let $\mathbb{G}$ be a finite group, and $g \in \mathbb{G}$ an element of order i . Then for any integer x , we have $g^x = g^{[x \bmod i]}$.

We can prove something stronger:

PROPOSITION 8.53 *Let $\mathbb{G}$ be a finite group, and $g \in \mathbb{G}$ an element of order i . Then $g^x = g^y$ if and only if $x = y \bmod i$.*

PROOF If $x = y \bmod i$ then $[x \bmod i] = [y \bmod i]$ and the previous proposition says that

$$g^x = g^{[x \bmod i]} = g^{[y \bmod i]} = g^y.$$

For the more interesting direction, say $g^x = g^y$. Then $1 = g^{x-y} = g^{[x-y \bmod i]}$ (using the previous proposition). Since $[x-y \bmod i] < i$, but i is the smallest positive integer with $g^i = 1$, we must have $[x-y \bmod i] = 0$. ■

The identity element of any group $\mathbb{G}$ is the only element of order 1, and generates the group $\langle 1 \rangle = \{1\}$. At the other extreme, if there is an element $g \in \mathbb{G}$ that has order m (where m is the order of $\mathbb{G}$), then $\langle g \rangle = \mathbb{G}$. In this case, we call $\mathbb{G}$ a *cyclic group* and say that g is a *generator* of $\mathbb{G}$. (A cyclic group may have multiple generators, and so we cannot speak of *the* generator.) If g is a generator of $\mathbb{G}$ then, by definition, every element $h \in \mathbb{G}$ is equal to g^x for some $x \in \{0, \dots, m-1\}$, a point we will return to in the next section.

Different elements of the same group $\mathbb{G}$ may have different orders. We can, however, place some restrictions on what these possible orders might be.

PROPOSITION 8.54 *Let $\mathbb{G}$ be a finite group of order m , and say $g \in \mathbb{G}$ has order i . Then $i \mid m$.*

PROOF By Theorem 8.14 we know that $g^m = 1 = g^0$. Proposition 8.53 implies that $m = 0 \bmod i$. ■

The next corollary illustrates the power of this result:

COROLLARY 8.55 *If $\mathbb{G}$ is a group of prime order p , then $\mathbb{G}$ is cyclic. Furthermore, all elements of $\mathbb{G}$ except the identity are generators of $\mathbb{G}$.*

PROOF By Proposition 8.54, the only possible orders of elements in $\mathbb{G}$ are 1 and p . Only the identity has order 1, and so all other elements have order p and generate $\mathbb{G}$. ■

Groups of prime order form one class of cyclic groups. The additive group $\mathbb{Z}_N$, for $N > 1$, gives another example of a cyclic group (the element 1 is always a generator). The next theorem—a special case of Theorem A.21—

gives an important additional class of cyclic groups; a proof is outside the scope of this book, but can be found in any standard abstract algebra text.

THEOREM 8.56 *If p is prime then $\mathbb{Z}_p^*$ is a cyclic group of order $p - 1$.*

For $p > 3$ prime, $\mathbb{Z}_p^*$ does not have prime order and so the above does not follow from the preceding corollary.

Example 8.57

Consider the (additive) group $\mathbb{Z}_{15}$. As we have noted, $\mathbb{Z}_{15}$ is cyclic and the element 1 is a generator since $15 \cdot 1 = 0 \bmod 15$ and $i \cdot 1 = i \neq 0 \bmod 15$ for any $0 < i < 15$ (recall that in this group the identity is 0).

$\mathbb{Z}_{15}$ has other generators. For example, $\langle 2 \rangle = \{0, 2, 4, \dots, 14, 1, 3, \dots, 13\}$ and so 2 is also a generator.

Not every element generates $\mathbb{Z}_{15}$. For example, the element 3 has order 5 since $5 \cdot 3 = 0 \bmod 15$, and so 3 does not generate $\mathbb{Z}_{15}$. The subgroup $\langle 3 \rangle$ consists of the 5 elements $\{0, 3, 6, 9, 12\}$, and this is indeed a subgroup under addition modulo 15. The element 10 has order 3 since $3 \cdot 10 = 0 \bmod 15$, and the subgroup $\langle 10 \rangle$ consists of the 3 elements $\{0, 5, 10\}$. The orders of the subgroups (i.e., 5 and 3) divide $|\mathbb{Z}_{15}| = 15$ as required by Proposition 8.54. $\diamond$

Example 8.58

Consider the (multiplicative) group $\mathbb{Z}_{15}^*$ of order $(5 - 1)(3 - 1) = 8$. We have $\langle 2 \rangle = \{1, 2, 4, 8\}$, and so the order of 2 is 4. As required by Proposition 8.54, 4 divides 8. $\diamond$

Example 8.59

Consider the (additive) group $\mathbb{Z}_p$ of prime order p . We know this group is cyclic, but Corollary 8.55 tells us more: namely, *every* element except 0 is a generator. Indeed, for any $h \in \{1, \dots, p - 1\}$ and integer $i > 0$ we have $ih = 0 \bmod p$ if and only if $p \mid ih$. But then Proposition 8.3 says that either $p \mid h$ or $p \mid i$. The former cannot occur (since $h < p$), and the smallest positive integer for which the latter can occur is $i = p$. We have thus shown that every nonzero element h has order p (and so generates $\mathbb{Z}_p$), in accordance with Corollary 8.55. $\diamond$

Example 8.60

Consider the (multiplicative) group $\mathbb{Z}_7^*$, which is cyclic by Theorem 8.56. We have $\langle 2 \rangle = \{1, 2, 4\}$, and so 2 is *not* a generator. However,

$$\langle 3 \rangle = \{1, 3, 2, 6, 4, 5\} = \mathbb{Z}_7^*,$$

and so 3 is a generator of $\mathbb{Z}_7^*$. $\diamond$

The following example relies on the material of Section 8.1.5.

Example 8.61

Let $\mathbb{G}$ be a cyclic group of order n , and let g be a generator of $\mathbb{G}$. Then the mapping $f : \mathbb{Z}_n \rightarrow \mathbb{G}$ given by $f(a) = g^a$ is an isomorphism between $\mathbb{Z}_n$ and $\mathbb{G}$. Indeed, for $a, a' \in \mathbb{Z}_n$ we have

$$f(a + a') = g^{[a+a' \bmod n]} = g^{a+a'} = g^a \cdot g^{a'} = f(a) \cdot f(a').$$

Bijectivity of f can be proved using the fact that n is the order of g . ◇

The previous example shows that all cyclic groups of the same order are isomorphic and thus the same from an *algebraic* point of view. We stress that this is not true in a *computational* sense, and in particular an isomorphism $f^{-1} : \mathbb{G} \rightarrow \mathbb{Z}_n$ (which we know must exist) need not be efficiently computable. This point should become clearer from the discussion in the sections below as well as Chapter 9.

8.3.2 The Discrete-Logarithm/Diffie–Hellman Assumptions

We now introduce several computational problems that can be defined for any class of cyclic groups. We will keep the discussion in this section abstract, and consider specific examples of groups in which these problems are believed to be hard in Sections 8.3.3 and 8.3.4.

We let $\mathcal{G}$ denote a generic, polynomial-time, *group-generation algorithm*. This is an algorithm that, on input 1^n , outputs a description of a cyclic group $\mathbb{G}$, its order q (with $\|q\| = n$), and a generator $g \in \mathbb{G}$. The description of a cyclic group specifies how elements of the group are represented as bit-strings; we assume that each group element is represented by a unique bit-string. We require that there are efficient algorithms (namely, algorithms running in time polynomial in n) for computing the group operation in $\mathbb{G}$, as well as for testing whether a given bit-string represents an element of $\mathbb{G}$. Efficient computation of the group operation implies efficient algorithms for exponentiation in $\mathbb{G}$ (see Appendix B.2.3) and for sampling a uniform element $h \in \mathbb{G}$ (simply choose uniform $x \in \mathbb{Z}_q$ and set $h := g^x$).

As discussed at the end of the previous section, although all cyclic groups of a given order are isomorphic, the *representation* of the group determines the computational complexity of mathematical operations in that group.

If $\mathbb{G}$ is a cyclic group of order q with generator g , then $\{g^0, g^1, \dots, g^{q-1}\}$ is all of $\mathbb{G}$. Equivalently, for every $h \in \mathbb{G}$ there is a *unique* $x \in \mathbb{Z}_q$ such that $g^x = h$. When the underlying group $\mathbb{G}$ is understood from the context, we call this x the *discrete logarithm of h with respect to g* and write $x = \log_g h$. (Logarithms in this case are called “discrete” since they take values in a finite range, as opposed to “standard” logarithms from calculus whose values range over the infinite set of real numbers.) Note that if $g^{x'} = h$ for some arbitrary integer x' , then $[x' \bmod q] = \log_g h$.

Discrete logarithms obey many of the same rules as “standard” logarithms. For example, $\log_g 1 = 0$ (where 1 is the identity of $\mathbb{G}$); for any integer r , we have $\log_g h^r = [r \cdot \log_g h \bmod q]$; and $\log_g(h_1 h_2) = [(\log_g h_1 + \log_g h_2) \bmod q]$.

The *discrete-logarithm problem* in a cyclic group $\mathbb{G}$ with generator g is to compute $\log_g h$ for a uniform element $h \in \mathbb{G}$. Consider the following experiment for a group-generation algorithm $\mathcal{G}$, algorithm $\mathcal{A}$, and parameter n :

The discrete-logarithm experiment $\text{DLog}_{\mathcal{A},\mathcal{G}}(n)$:

1. Run $\mathcal{G}(1^n)$ to obtain $(\mathbb{G}, q, g)$, where $\mathbb{G}$ is a cyclic group of order q (with $||q|| = n$), and g is a generator of $\mathbb{G}$.
2. Choose a uniform $h \in \mathbb{G}$.
3. $\mathcal{A}$ is given $\mathbb{G}, q, g, h$, and outputs $x \in \mathbb{Z}_q$.
4. The output of the experiment is defined to be 1 if $g^x = h$, and 0 otherwise.

DEFINITION 8.62 We say that the discrete-logarithm problem is hard relative to $\mathcal{G}$ if for all probabilistic polynomial-time algorithms $\mathcal{A}$ there exists a negligible function negl such that $\Pr[\text{DLog}_{\mathcal{A},\mathcal{G}}(n) = 1] \leq \text{negl}(n)$.

The discrete-logarithm assumption is simply the assumption that there exists a $\mathcal{G}$ for which the discrete-logarithm problem is hard. The following two sections discuss some candidate group-generation algorithms $\mathcal{G}$ for which this is believed to be the case.

The Diffie–Hellman problems. The so-called *Diffie–Hellman* problems are related, but not known to be equivalent, to the problem of computing discrete logarithms. There are two important variants: the *computational* Diffie–Hellman (CDH) problem and the *decisional* Diffie–Hellman (DDH) problem.

Fix a cyclic group $\mathbb{G}$ and a generator $g \in \mathbb{G}$. Given elements $h_1, h_2 \in \mathbb{G}$, define $\text{DH}_g(h_1, h_2) \stackrel{\text{def}}{=} g^{\log_g h_1 \cdot \log_g h_2}$. That is, if $h_1 = g^{x_1}$ and $h_2 = g^{x_2}$ then

$$\text{DH}_g(h_1, h_2) = g^{x_1 \cdot x_2} = h_1^{x_2} = h_2^{x_1}.$$

The *CDH problem* is to compute $\text{DH}_g(h_1, h_2)$ for uniform h_1 and h_2 . Hardness of this problem can be formalized by the natural experiment; we leave the details as an exercise.

If the discrete-logarithm problem relative to some $\mathcal{G}$ is easy, then the CDH problem is, too: given h_1 and h_2 , first compute $x_1 := \log_g h_1$ and then output the answer $h_2^{x_1}$. In contrast, it is not clear whether hardness of the discrete-logarithm problem implies that the CDH problem is hard as well.

The *DDH problem*, roughly speaking, is to distinguish $\text{DH}_g(h_1, h_2)$ from a uniform group element when h_1, h_2 are uniform. That is, given uniform h_1, h_2 and a third group element h' , the problem is to decide whether $h' = \text{DH}_g(h_1, h_2)$ or whether h' was chosen uniformly from $\mathbb{G}$. Formally:

DEFINITION 8.63 We say that the DDH problem is hard relative to $\mathcal{G}$ if for all probabilistic polynomial-time algorithms $\mathcal{A}$ there is a negligible function negl such that

$$\left| \Pr[\mathcal{A}(\mathbb{G}, q, g, g^x, g^y, g^z) = 1] - \Pr[\mathcal{A}(\mathbb{G}, q, g, g^x, g^y, g^{xy}) = 1] \right| \leq \text{negl}(n),$$

where in each case the probabilities are taken over the experiment in which $\mathcal{G}(1^n)$ outputs $(\mathbb{G}, q, g)$, and then uniform $x, y, z \in \mathbb{Z}_q$ are chosen. (Note that when z is uniform in $\mathbb{Z}_q$, then g^z is uniformly distributed in $\mathbb{G}$.)

We have already seen that if the discrete-logarithm problem is easy relative to some $\mathcal{G}$, then the CDH problem is too. Similarly, if the CDH problem is easy relative to $\mathcal{G}$ then so is the DDH problem; you are asked to show this in Exercise 8.15. The converse, however, does not appear to be true, and there are examples of groups in which the discrete-logarithm and CDH problems are believed to be hard even though the DDH problem is easy; see Exercise 13.15.

Using Prime-Order Groups

There are various (classes of) cyclic groups in which the discrete-logarithm and Diffie–Hellman problems are believed to be hard. There is a preference, however, for cyclic groups of *prime order*, for reasons we now explain.

One reason for preferring groups of prime order is because, in a certain sense, the discrete-logarithm problem is hardest in such groups. This is a consequence of the *Pohlig–Hellman algorithm*, described in Chapter 9, which shows that the discrete-logarithm problem in a group of order q becomes easier if q has (small) prime factors. This does not necessarily mean that the discrete-logarithm problem is *easy* in groups of nonprime order; it merely means that the problem becomes *easier*.

Related to the above is the fact that the DDH problem is easy if the group order q has small prime factors. We refer to Exercise 13.15 for one example of this phenomenon.

A second motivation for using prime-order groups is because finding a generator in such groups is trivial. This follows from Corollary 8.55, which says that *every* element of a prime-order group (except the identity) is a generator. In contrast, efficiently finding a generator of an arbitrary cyclic group requires the factorization of the group order to be known (see Appendix B.3).

Proofs of security for some cryptographic constructions require computing multiplicative inverses of certain exponents (we will see an example in Section 8.4.2). When the group order is prime, any nonzero exponent will be invertible, making this computation possible.

A final reason for working with prime-order groups applies in situations when the *decisional* Diffie–Hellman problem should be hard. Fixing a group $\mathbb{G}$ with generator g , the DDH problem boils down to distinguishing between

tuples of the form $(h_1, h_2, \text{DH}_g(h_1, h_2))$ for uniform h_1, h_2 , and tuples of the form (h_1, h_2, y) , for uniform h_1, h_2, y . A necessary condition for the DDH problem to be hard is that $\text{DH}_g(h_1, h_2)$ *by itself* should be indistinguishable from a uniform group element. One can show that $\text{DH}_g(h_1, h_2)$ is “close” to uniform (in a sense we do not define here) when the group order q is prime, something that is not true otherwise.

8.3.3 Working in (Subgroups of) $\mathbb{Z}_p^*$

Groups of the form $\mathbb{Z}_p^*$, for p prime, give one class of cyclic groups in which the discrete-logarithm problem is believed to be hard. Concretely, let $\mathcal{G}$ be an algorithm that, on input 1^n , chooses a uniform n -bit prime p , and outputs p and the group order $q = p - 1$ along with a generator g of $\mathbb{Z}_p^*$. (Section 8.2.1 discusses efficient algorithms for choosing a random prime, and Appendix B.3 shows how to efficiently find a generator of $\mathbb{Z}_p^*$ given the factorization of $p - 1$.) The representation of $\mathbb{Z}_p^*$ here is the trivial one where elements are represented as integers between 1 and $p - 1$. It is conjectured that the discrete-logarithm problem is hard relative to $\mathcal{G}$ of this sort.

The cyclic group $\mathbb{Z}_p^*$ (for $p > 3$ prime), however, does *not* have prime order. (The preference for groups of prime order was discussed in the previous section.) More problematic, the decisional Diffie–Hellman problem is, in general, *not hard* in such groups (see Exercise 13.15), and they are therefore unacceptable for the cryptographic applications based on the DDH assumption that we will explore in later chapters.

These issues can be addressed by using a prime-order *subgroup* of $\mathbb{Z}_p^*$. Let $p = rq + 1$ where both p and q are prime. We prove that $\mathbb{Z}_p^*$ has a subgroup $\mathbb{G}$ of order q given by the set of *r th residues modulo p* , i.e., the set of elements $\{[h^r \bmod p] \mid h \in \mathbb{Z}_p^*\}$ that are equal to the r th power of some $h \in \mathbb{Z}_p^*$.

THEOREM 8.64 *Let $p = rq + 1$ with p, q prime. Then*

$$\mathbb{G} \stackrel{\text{def}}{=} \{[h^r \bmod p] \mid h \in \mathbb{Z}_p^*\}$$

is a subgroup of $\mathbb{Z}_p^$ of order q .*

PROOF The proof that $\mathbb{G}$ is a subgroup is straightforward and is omitted. We prove that $\mathbb{G}$ has order q by showing that the function $f_r : \mathbb{Z}_p^* \rightarrow \mathbb{G}$ defined by $f_r(g) = [g^r \bmod p]$ is an r -to-1 function. (Since $|\mathbb{Z}_p^*| = p - 1$, this shows that $|\mathbb{G}| = (p - 1)/r = q$.) To see this, let g be a generator of $\mathbb{Z}_p^*$ so that $g^0, \dots, g^{p-2}$ are all the elements of $\mathbb{Z}_p^*$. By Proposition 8.53 we have $(g^i)^r = (g^j)^r$ if and only if $ir = jr \bmod (p - 1)$ or, equivalently, $p - 1 \mid (i - j)r$. Since $p - 1 = rq$, this is equivalent to $q \mid (i - j)$. For any fixed $j \in \{0, \dots, p - 2\}$, this means that the set of values $i \in \{0, \dots, p - 2\}$ for which $(g^i)^r = (g^j)^r$ is

exactly the set of r distinct values

$$\{j, j + q, j + 2q, \dots, j + (r - 1)q\},$$

all reduced modulo $p - 1$. (Note that $j + rq = j \bmod (p - 1)$.) This proves that f_r is an r -to-1 function. ■

Besides showing existence of an appropriate subgroup, the theorem also implies that it is easy to generate a uniform element of $\mathbb{G}$ and to test whether a given element of $\mathbb{Z}_p^*$ lies in $\mathbb{G}$. Specifically, choosing a uniform element of $\mathbb{G}$ can be done by choosing uniform $h \in \mathbb{Z}_p^*$ and computing $[h^r \bmod p]$. Since $\mathbb{G}$ has prime order, every element in $\mathbb{G}$ except the identity is a generator of $\mathbb{G}$. Finally, it is possible to determine whether any $h \in \mathbb{Z}_p^*$ is also in $\mathbb{G}$ by checking whether $h^q \stackrel{?}{=} 1 \bmod p$. To see that this works, let $h = g^i$ for g a generator of $\mathbb{Z}_p^*$ and $i \in \{0, \dots, p - 2\}$. Then

$$\begin{aligned} h^q = 1 \bmod p &\iff g^{iq} = 1 \bmod p \\ &\iff iq = 0 \bmod (p - 1) \iff rq \mid iq \iff r \mid i, \end{aligned}$$

using Proposition 8.53. So $h = g^i = g^{cr} = (g^c)^r$ for some c , and $h \in \mathbb{G}$.

Algorithm 8.65 encapsulates the above discussion. In the algorithm, we let n denote the length of q , the order of the group, and let ℓ denote the length of p , the modulus being used. The relationship between these parameters is discussed below.

Choosing ℓ . Let $n = \|q\|$ and $\ell = \|p\|$. Two types of algorithms are known for computing discrete logarithms in order- q subgroups of $\mathbb{Z}_p^*$ (see Section 9.2): those that run in time $\mathcal{O}(\sqrt{q}) = \mathcal{O}(2^{n/2})$ and those that run in time $2^{\mathcal{O}((\log p)^{1/3} \cdot (\log \log p)^{2/3})} = 2^{\mathcal{O}(\ell^{1/3} \cdot (\log \ell)^{2/3})}$. Fixing some desired security parameter n , the parameter ℓ should be chosen so as to balance these times. (If ℓ is any smaller, security is reduced; if ℓ is any larger, operations in $\mathbb{G}$ will be less efficient without any gain in security.) See also Section 9.3.

ALGORITHM 8.65

A group-generation algorithm $\mathcal{G}$

Input: Security parameter 1^n , parameter $\ell = \ell(n)$

Output: Cyclic group $\mathbb{G}$, its (prime) order q , and a generator g

generate a uniform n -bit prime q

generate an ℓ -bit prime p such that $q \mid (p - 1)$

// we omit the details of how this is done

choose a uniform $h \in \mathbb{Z}_p^*$ with $h \neq 1$

set $g := [h^{(p-1)/q} \bmod p]$

return p, q, g // $\mathbb{G}$ is the order- q subgroup of $\mathbb{Z}_p^*$

In practice, standardized values (e.g., recommended by NIST) for p , q , and a generator g are used, and there is no need to generate parameters of one’s own.

Example 8.66

Consider the group $\mathbb{Z}_{11}^*$ of order 10. Let us try to find a generator of this group. Consider trying 2:

Powers of 2:	2^0	2^1	2^2	2^3	2^4	2^5	2^6	2^7	2^8	2^9
Values:	1	2	4	8	5	10	9	7	3	6

(All values above are computed modulo 11.) We got lucky the first time—the number 2 is a generator! Let’s try 3:

Powers of 3:	3^0	3^1	3^2	3^3	3^4	3^5	3^6	3^7	3^8	3^9
Values:	1	3	9	5	4	1	3	9	5	4

We see that 3 is not a generator of the entire group. Rather, it generates a *subgroup* $\mathbb{G} = \{1, 3, 4, 5, 9\}$ of order 5. Now, let’s see what happens with 10:

Powers of 10:	10^0	10^1	10^2	10^3	10^4	10^5	10^6	10^7	10^8	10^9
Values:	1	10	1	10	1	10	1	10	1	10

In this case we generate a subgroup of order 2.

For cryptographic purposes we want to work in a prime-order group. Since $11 = 2 \cdot 5 + 1$ we can apply Theorem 8.64 with $q = 5$ and $r = 2$, or with $q = 2$ and $r = 5$. In the first case, the theorem tells us that the *squares* of all the elements of $\mathbb{Z}_{11}^*$ should give a subgroup of order 5. This can be easily verified:

Element:	1	2	3	4	5	6	7	8	9	10
Square:	1	4	9	5	3	3	5	9	4	1

We have seen above that 3 is a generator of this subgroup. (In fact, since the subgroup is prime, every element of the subgroup besides 1 is a generator of the subgroup.) Taking $q = 2$ and $r = 5$, Theorem 8.64 tells us that taking 5th powers will give a subgroup of order 2. One can check that this gives the order-2 subgroup generated by 10 that we encountered earlier. $\diamond$

Subgroups of finite fields. The discrete-logarithm problem is also believed to be hard in the multiplicative group of a finite field of large characteristic when the polynomial representation is used. (Appendix A.5 provides a brief background on finite fields.) Recall that for any prime p and integer $k \geq 1$ there is a (unique) field $\mathbb{F}_{p^k}$ of order p^k ; the multiplicative group $\mathbb{F}_{p^k}^*$ of that field is a cyclic group of order $p^k - 1$ (cf. Theorem A.21). If q is a large prime factor of $p^k - 1$, then Theorem 8.64 shows that $\mathbb{F}_{p^k}^*$ has a cyclic subgroup of order q . (The only property of $\mathbb{Z}_p^*$ we used in the proof of that theorem was that $\mathbb{Z}_p^*$ is cyclic.) This offers another choice of prime-order groups in which the discrete-logarithm and Diffie–Hellman problems are believed to be hard. Our treatment of $\mathbb{Z}_p^*$ in this section corresponds to the special case $k = 1$.

8.3.4 Elliptic Curves

The groups we have concentrated on thus far have all been based directly on modular arithmetic. Another class of groups important for cryptography is given by groups consisting of *points on elliptic curves*. Such groups are especially interesting from a cryptographic perspective since, in contrast to $\mathbb{Z}_p^*$ or the multiplicative group of a finite field, there are currently no known sub-exponential time algorithms for solving the discrete-logarithm problem in elliptic-curve groups when chosen appropriately. (See Section 9.3 for further discussion.) For cryptosystems based on the discrete-logarithm or Diffie–Hellman assumptions, this means that implementations based on elliptic-curve groups will be more efficient than implementations based on prime-order subgroups of $\mathbb{Z}_p^*$ at any given level of security. In this section we provide only a brief introduction to this area. A deeper understanding of the issues discussed here requires more sophisticated mathematics than we are willing to assume on the part of the reader. Those interested in further exploring this topic are advised to consult the references at the end of this chapter.

Let $p \geq 5$ be a prime.³ Consider an equation E in the variables x and y of the form:

$$y^2 = x^3 + Ax + B \bmod p, \quad (8.1)$$

where $A, B \in \mathbb{Z}_p$ are constants with $4A^3 + 27B^2 \not\equiv 0 \bmod p$. (This ensures that the equation $x^3 + Ax + B = 0 \bmod p$ has no repeated roots.) Let $E(\mathbb{Z}_p)$ denote the set of pairs $(x, y) \in \mathbb{Z}_p \times \mathbb{Z}_p$ satisfying the above equation along with a special value $\mathcal{O}$ whose purpose we will discuss shortly; that is,

$$E(\mathbb{Z}_p) \stackrel{\text{def}}{=} \{(x, y) \mid x, y \in \mathbb{Z}_p \text{ and } y^2 = x^3 + Ax + B \bmod p\} \cup \{\mathcal{O}\}.$$

The elements $E(\mathbb{Z}_p)$ are called the *points* on the *elliptic curve* E defined by Equation (8.1), and $\mathcal{O}$ is called the *point at infinity*.

Example 8.67

An element $y \in \mathbb{Z}_p^*$ is a *quadratic residue modulo* p if there is an $x \in \mathbb{Z}_p^*$ such that $x^2 = y \bmod p$; in that case, we say x is a *square root of* y . For $p > 2$ prime, half the elements in $\mathbb{Z}_p^*$ are quadratic residues, and every quadratic residue has exactly two square roots. (See Section 13.4.1.)

Let $f(x) \stackrel{\text{def}}{=} x^3 + 3x + 3$ and consider the curve $E : y^2 = f(x) \bmod 7$. Each value of x for which $f(x)$ is a quadratic residue modulo 7 yields two points on the curve; values x for which $f(x)$ is a non-quadratic residue are not on

³The theory can be adapted to deal with the case of $p = 2$ or 3 but this introduces additional complications. Elliptic curves can, in fact, be defined over arbitrary finite or infinite *fields* (cf. Section A.5), and our discussion largely carries over to fields of characteristic not equal to 2 or 3. *Binary curves* (i.e., curves over fields of characteristic 2) are particularly important in cryptography, but we will not discuss them here.

the curve; values of x for which $f(x) = 0 \bmod 7$ give one point on the curve. This allows us to determine the points on the curve:

- $f(0) = 3 \bmod 7$, a quadratic non-residue modulo 7.
- $f(1) = 0 \bmod 7$, so we obtain the point $(1, 0) \in E(\mathbb{Z}_7)$.
- $f(2) = 3 \bmod 7$, a quadratic non-residue modulo 7.
- $f(3) = 4 \bmod 7$, a quadratic residue modulo 7 with square roots 2 and 5. This yields the points $(3, 2), (3, 5) \in E(\mathbb{Z}_7)$.
- $f(4) = 2 \bmod 7$, a quadratic residue modulo 7 with square roots 3 and 4. This yields the points $(4, 3), (4, 4) \in E(\mathbb{Z}_7)$.
- $f(5) = 3 \bmod 7$, a quadratic non-residue modulo 7.
- $f(6) = 6 \bmod 7$, a quadratic non-residue modulo 7.

Including the point at infinity, there are 6 points in $E(\mathbb{Z}_7)$. ◇

A useful way to think about $E(\mathbb{Z}_p)$ is to look at the graph of Equation (8.1) over the reals (i.e., the equation $y^2 = x^3 + Ax + B$ without reduction modulo p) as in Figure 8.2. This figure does not correspond exactly to $E(\mathbb{Z}_p)$ because, for example, $E(\mathbb{Z}_p)$ has a finite number of points ($\mathbb{Z}_p$ is, after all, a finite set) while there are an infinite number of solutions to the same equation if we allow x and y to range over all real numbers. Nevertheless, the picture provides useful intuition. In such a figure, one can think of the “point at infinity” $\mathcal{O}$ as sitting at the top of the y -axis and lying on every vertical line.

It can be shown that every line intersecting $E(\mathbb{Z}_p)$ intersects it in exactly 3 points, where (1) a point P is counted twice if the line is tangent to the curve at P , and (2) the point at infinity is also counted when the line is vertical. This fact is used to define a binary operation, called “addition” and denoted by $+$, on points of $E(\mathbb{Z}_p)$ in the following way:

- The point $\mathcal{O}$ is defined to be an (additive) identity; that is, for all $P \in E(\mathbb{Z}_p)$ we define $P + \mathcal{O} = \mathcal{O} + P = P$.
- For two points $P_1, P_2 \neq \mathcal{O}$ on E , we evaluate their sum $P_1 + P_2$ by drawing the line through P_1, P_2 (if $P_1 = P_2$ then draw the line tangent to the curve at P_1) and finding the third point of intersection P_3 of this line with $E(\mathbb{Z}_p)$; the third point of intersection may be $P_3 = \mathcal{O}$ if the line is vertical. If $P_3 = (x, y) \neq \mathcal{O}$ then we define $P_1 + P_2 \stackrel{\text{def}}{=} (x, -y)$. (Graphically, this corresponds to reflecting P_3 in the x -axis.) If $P_3 = \mathcal{O}$ then $P_1 + P_2 \stackrel{\text{def}}{=} \mathcal{O}$.

If $P = (x, y) \neq \mathcal{O}$ is a point of $E(\mathbb{Z}_p)$, then $-P \stackrel{\text{def}}{=} (x, -y)$ (which is clearly also a point of $E(\mathbb{Z}_p)$) is the unique inverse of P . Indeed, the line

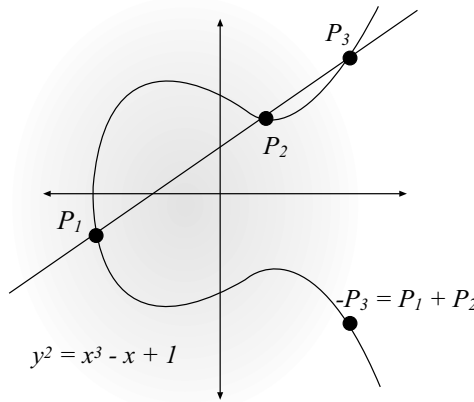


FIGURE 8.2: An elliptic curve over the reals.

through (x, y) and $(x, -y)$ is vertical, and so the addition rule implies that $P + (-P) = \mathcal{O}$. (If $y = 0$ then $P = (x, y) = (x, -y) = -P$ but then the tangent line at P will be vertical and so $P + (-P) = \mathcal{O}$ here as well.) Of course, $-\mathcal{O} = \mathcal{O}$.

It is straightforward, but tedious, to work out the addition law concretely. Let $P_1 = (x_1, y_1)$ and $P_2 = (x_2, y_2)$ be two points in $E(\mathbb{Z}_p)$, with $P_1, P_2 \neq \mathcal{O}$ and E as in Equation (8.1). To keep matters simple, suppose $x_1 \neq x_2$ (dealing with the case $x_1 = x_2$ is still straightforward but even more tedious). The slope of the line through these points is

$$m \stackrel{\text{def}}{=} \left\lfloor \frac{y_2 - y_1}{x_2 - x_1} \bmod p \right\rfloor ;$$

our assumption that $x_1 \neq x_2$ means that the inverse of $(x_2 - x_1)$ modulo p exists. The line passing through P_1 and P_2 has the equation

$$y = m \cdot (x - x_1) + y_1 \bmod p. \quad (8.2)$$

To find the third point of intersection of this line with E , substitute the above into the equation for E to obtain

$$\left(m \cdot (x - x_1) + y_1 \right)^2 = x^3 + Ax + B \bmod p.$$

The values of x that satisfy this equation are x_1 , x_2 , and

$$x_3 \stackrel{\text{def}}{=} [m^2 - x_1 - x_2 \bmod p].$$

The first two solutions correspond to the original points P_1 and P_2 , while the third is the x -coordinate of the third point of intersection P_3 . Plugging x_3 into Equation (8.2) we find that the y -coordinate corresponding to x_3 is

$y_3 = [m \cdot (x_3 - x_1) + y_1 \bmod p]$. To obtain the desired answer $P_1 + P_2$, we flip the sign of y to obtain:

$$(x_1, y_1) + (x_2, y_2) = ([m^2 - x_1 - x_2 \bmod p], [m \cdot (x_1 - x_3) - y_1 \bmod p]).$$

We summarize and extend this in the following proposition.

PROPOSITION 8.68 *Let $p \geq 5$ be prime and let E be the elliptic curve given by $y^2 = x^3 + Ax + B \bmod p$ where $4A^3 + 27B^2 \neq 0 \bmod p$. Let $P_1, P_2 \neq \mathcal{O}$ be points on E , with $P_1 = (x_1, y_1)$ and $P_2 = (x_2, y_2)$.*

1. *If $x_1 \neq x_2$, then $P_1 + P_2 = (x_3, y_3)$ with*

$$x_3 = [m^2 - x_1 - x_2 \bmod p] \quad \text{and} \quad y_3 = [m \cdot (x_1 - x_3) - y_1 \bmod p],$$

$$\text{where } m = \left[\frac{y_2 - y_1}{x_2 - x_1} \bmod p \right].$$

2. *If $x_1 = x_2$ but $y_1 \neq y_2$ then $P_1 = -P_2$ and so $P_1 + P_2 = \mathcal{O}$.*

3. *If $P_1 = P_2$ and $y_1 = 0$ then $P_1 + P_2 = 2P_1 = \mathcal{O}$.*

4. *If $P_1 = P_2$ and $y_1 \neq 0$ then $P_1 + P_2 = 2P_1 = (x_3, y_3)$ with*

$$x_3 = [m^2 - 2x_1 \bmod p] \quad \text{and} \quad y_3 = [m \cdot (x_1 - x_3) - y_1 \bmod p],$$

$$\text{where } m = \left[\frac{3x_1^2 + A}{2y_1} \bmod p \right].$$

Somewhat amazingly, the set of points $E(\mathbb{Z}_p)$ along with the addition rule defined above form an abelian group, called the *elliptic-curve group of E* . Commutativity follows from the way addition is defined, $\mathcal{O}$ acts as the identity, and we have already seen that each point on $E(\mathbb{Z}_p)$ has an inverse in $E(\mathbb{Z}_p)$. The difficult property to verify is associativity, which the disbelieving reader can check through tedious calculation. A more illuminating proof that does not involve explicit calculation relies on algebraic geometry.

Example 8.69

Consider the curve from Example 8.67. We show associativity for three specific points. Let $P_1 = (1, 0)$, $P_2 = Q_2 = (4, 3)$. When computing $P_1 + P_2$ we get $m = [(3 - 0) \cdot (4 - 1)^{-1} \bmod 7] = 1$ and $[1^2 - 1 - 4 \bmod 7] = 3$. Thus,

$$P_3 \stackrel{\text{def}}{=} P_1 + P_2 = (3, [1 \cdot (1 - 3) - 0 \bmod 7]) = (3, 5);$$

note that this is indeed a point on $E(\mathbb{Z}_7)$. If we then compute $P_3 + Q_2$ we get $m = [(3 - 5) \cdot (4 - 3)^{-1} \bmod 7] = 5$ and $[5^2 - 3 - 4 \bmod 7] = 4$. Thus,

$$(P_1 + P_2) + Q_2 = P_3 + Q_2 = (4, [5 \cdot (3 - 4) - 5 \bmod 7]) = (4, 4).$$

If we compute $P_2 + Q_2 = 2P_2$ we obtain $m = [(3 \cdot 4^2 + 3) \cdot (2 \cdot 3)^{-1} \bmod 7] = 5$ and $[5^2 - 2 \cdot 4 \bmod 7] = 3$. Thus,

$$P'_3 \stackrel{\text{def}}{=} P_2 + Q_2 = (3, [5 \cdot (4 - 3) - 3 \bmod 7]) = (3, 2).$$

If we then compute $P_1 + P'_3$ we find $m = [2 \cdot (3 - 1)^{-1} \bmod 7] = 1$ and $[1^2 - 1 - 3 \bmod 7] = 4$. So

$$P_1 + (P_2 + Q_2) = P_1 + P'_3 = (4, [1 \cdot (1 - 4) - 0 \bmod 7]) = (4, 4),$$

and $P_1 + (P_2 + Q_2) = (P_1 + P_2) + Q_2$. ◇

Recall that when a group is written additively, “exponentiation” corresponds to repeated addition. Thus, if we fix some point P in an elliptic-curve group, the discrete-logarithm problem becomes (informally) the problem of computing the integer x from xP , while the decisional Diffie–Hellman problem becomes (informally) the problem of distinguishing tuples of the form (aP, bP, abP) from those of the form (aP, bP, cP) . These problems are believed to be hard in elliptic-curve groups (or subgroups thereof) of large prime order, subject to a few technical conditions we will mention in passing below.

If we want an elliptic-curve group (or subgroup) of large prime order, the first question we must address is: how large are elliptic-curve groups? As noted in Example 8.67, the equation $y^2 = f(x) \bmod p$ has two solutions whenever $f(x)$ is a quadratic residue, and one solution when $f(x) = 0$. Since half the elements in $\mathbb{Z}_p^*$ are quadratic residues, we heuristically expect to find $2 \cdot (p - 1)/2 + 1 = p$ points on the curve. Including the point at infinity, this means there should be about $p + 1$ points in an elliptic-curve group over $\mathbb{Z}_p$. The *Hasse bound* says this heuristic estimate is accurate, in the sense that every elliptic-curve group has “almost” this many points.

THEOREM 8.70 (Hasse bound) *Let p be prime, and let E be an elliptic curve over $\mathbb{Z}_p$. Then $p + 1 - 2\sqrt{p} \leq |E(\mathbb{Z}_p)| \leq p + 1 + 2\sqrt{p}$.*

This bound implies that it is always easy to *find* a point on a given elliptic curve $y^2 = f(x) \bmod p$: simply choose uniform $x \in \mathbb{Z}_p$, check whether $f(x)$ is 0 or a quadratic residue, and—if so—let y be a square root of $f(x)$. (Algorithms for deciding quadratic residuosity and computing square roots modulo a prime are discussed in Chapter 13.) Since points on the elliptic curve are plentiful, we will not have to try very many values of x before finding a point.

The Hasse bound only gives a range for the size of an elliptic-curve group. For a fixed prime p , however, the order of a random elliptic curve over $\mathbb{Z}_p$ (namely, a curve defined by Equation (8.1) in which A, B are chosen uniformly in $\mathbb{Z}_p$ subject to the constraint $4A^3 + 27B^2 \not\equiv 0 \bmod p$) is heuristically found to be “close” to uniformly distributed in the Hasse interval. There also exist efficient algorithms—whose description and analysis are well beyond the scope of this book—for counting the number of points on an elliptic

curve. This suggests an approach to elliptic-curve parameter generation as in Algorithm 8.71.

ALGORITHM 8.71

Elliptic-curve group-generation algorithm $\mathcal{G}$

Input: Security parameter 1^n

Output: Cyclic group $\mathbb{G}$, its (prime) order q , and a generator g

generate a uniform n -bit prime p

until q is an n -bit prime **do**:

choose $A, B \leftarrow \mathbb{Z}_p$ with $4A^3 + 27B^2 \not\equiv 0 \pmod{p}$,
defining elliptic curve E as in Equation (8.1)

let q be the number of points on $E(\mathbb{Z}_p)$

choose $g \in E(\mathbb{Z}_p) \setminus \{\mathcal{O}\}$

return $(A, B, p), q, g$ // $\mathbb{G}$ is the elliptic-curve group of E

Certain classes of curves are considered cryptographically weak and should be avoided. These include elliptic-curve groups over $\mathbb{Z}_p$ whose order is equal to p (*anomalous curves*) or $p+1$ (*supersingular curves*), or whose order divides $p^k - 1$ for “small” k . A full discussion is beyond the scope of this book. In practice, standardized curves (such as those recommended by NIST) are used, and generating a curve of one’s own is not advised.

Efficiency Considerations

We conclude this section with a very brief discussion of some standard efficiency improvements when using elliptic curves.

Point compression. A useful observation is that the number of bits needed to represent a point on an elliptic curve can be reduced almost by half. To see this, note that for any point (x, y) on an elliptic curve $E : y^2 = f(x) \pmod{p}$ there are at most two points on the curve that have x -coordinate x : namely, (x, y) and $(x, -y)$. (It is possible that $y = 0$ in which case these are the same point.) Thus, we can specify any point $P = (x, y)$ by its x -coordinate and a bit b that distinguishes between the (at most) two possibilities for the value of its y -coordinate. One convenient way to do this is to set $b = 0$ if $y < p/2$ and $b = 1$ otherwise. Given x and b we can recover P by computing the two square roots y_1, y_2 of the equation $y^2 = f(x) \pmod{p}$; since $y_1 = -y_2 \pmod{p}$ and so $y_1 = p - y_2$, exactly one of y_1, y_2 will be less than $p/2$.

Projective coordinates. Representing elliptic-curve points as we have been doing until now—in which a point P on an elliptic curve is described by a pair of field elements (x, y) —is called using *affine coordinates*. There are alternate ways to represent points, using *projective coordinates*, that can offer efficiency

improvements. While these alternate representations can be motivated mathematically, we treat them simply as useful computational aids.

Points in projective coordinates are represented using *three* elements of $\mathbb{Z}_p$. An interesting feature is that a point has multiple representations. When using standard projective coordinates, a point $P \neq \mathcal{O}$ with representation (x, y) in affine coordinates is represented by any tuple $(X, Y, Z) \in \mathbb{Z}_p^3$ for which $X/Z = x \bmod p$ and $Y/Z = y \bmod p$. The point $\mathcal{O}$ is represented by any tuple $(0, Y, 0)$ with $Y \neq 0$, and these are the only points (X, Y, Z) with $Z = 0$. We can easily translate between coordinate systems: (x, y) in affine coordinates can be mapped to $(x, y, 1)$ in projective coordinates, and (X, Y, Z) (with $Z \neq 0$) in projective coordinates is mapped to the representation $([X/Z \bmod p], [Y/Z \bmod p])$ in affine coordinates.

The main advantage of using projective coordinates is that we can add points without having to compute inverses modulo p . (Adding points in affine coordinates requires computing inverses; see Proposition 8.68.) We accomplish this by exploiting the fact that points have multiple representations. To see this, let us work out the addition law for two points $P_1 = (X_1, Y_1, Z_1)$ and $P_2 = (X_2, Y_2, Z_2)$ with $P_1, P_2 \neq \mathcal{O}$ (so $Z_1, Z_2 \neq 0$) and $P_1 \neq \pm P_2$ (so $X_1/Z_1 \neq X_2/Z_2 \bmod p$). (If either P_1 or P_2 are equal to $\mathcal{O}$, addition is trivial. The case of $P_1 = \pm P_2$ can be handled as well, but we omit details here.) We can express P_1 and P_2 as $(X_1/Z_1, Y_1/Z_1)$ and $(X_2/Z_2, Y_2/Z_2)$ in affine coordinates, so

$$P_3 \stackrel{\text{def}}{=} P_1 + P_2 = (m^2 - X_1/Z_1 - X_2/Z_2, \\ m \cdot (X_1/Z_1 - m^2 + X_1/Z_1 + X_2/Z_2) - Y_1/Z_1, 1),$$

where

$$m = (Y_2/Z_2 - Y_1/Z_1)(X_2/Z_2 - X_1/Z_1)^{-1} = (Y_2Z_1 - Y_1Z_2)(X_2Z_1 - X_1Z_2)^{-1}$$

and all computations are done modulo p . Note we are using projective coordinates to represent P_3 , setting $Z_3 = 1$ above. But using projective coordinates means we are not limited to $Z_3 = 1$. Multiplying each coordinate by $Z_1Z_2(X_2Z_1 - X_1Z_2)^3 \neq 0 \bmod p$, we find that P_3 can also be represented as

$$P_3 = (vw, u(v^2X_1Z_2 - w) - v^3Y_1Z_2, Z_1Z_2v^3) \quad (8.3)$$

where

$$u = Y_2Z_1 - Y_1Z_2, \quad v = X_2Z_1 - X_1Z_2, \\ w = u^2Z_1Z_2 - v^3 - 2v^2X_1Z_2. \quad (8.4)$$

The point to notice is that the computations in Equations (8.3) and (8.4) can be carried out without having to perform any modular inversions.

Precisely because points have multiple representations in projective coordinates, some subtleties can arise when projective coordinates are used. (We

have explicitly assumed until now that group elements have unique representations as bit-strings.) Specifically, a point expressed in projective coordinates may reveal some information about how that point was obtained, which may depend on some secret information. To address this—as well as for reasons of efficiency—affine coordinates should be used for transmitting and storing points, with projective coordinates used only as an intermediate representation during the course of a computation (with points converted to/from projective coordinates at the beginning/end of the computation).

8.4 *Cryptographic Applications

We have spent a fair bit of time discussing number theory and group theory, and introducing computational hardness assumptions that are widely believed to hold. Applications of these assumptions will occupy us for the rest of the book, but we provide some brief examples here.

8.4.1 One-Way Functions and Permutations

One-way functions are the minimal cryptographic primitive, and they are both necessary and sufficient for private-key encryption and message authentication codes. A more complete discussion of the role of one-way functions in cryptography appears in Chapter 7; here we only provide a definition of one-way functions and demonstrate that their existence follows from the number-theoretic hardness assumptions we have seen in this chapter.

Informally, a function f is *one-way* if it is easy to compute but hard to invert. The following experiment and definition, a restatement of Definition 7.1, formalizes this.

The inverting experiment $\text{Invert}_{\mathcal{A},f}(n)$:

1. Choose uniform $x \in \{0,1\}^n$ and compute $y := f(x)$.
2. $\mathcal{A}$ is given 1^n and y as input, and outputs x' .
3. The output of the experiment is 1 if and only if $f(x') = y$.

DEFINITION 8.72 A function $f : \{0,1\}^* \rightarrow \{0,1\}^*$ is **one-way** if the following two conditions hold:

1. **(Easy to compute:)** There is a polynomial-time algorithm that on input x outputs $f(x)$.
2. **(Hard to invert:)** For all PPT algorithms $\mathcal{A}$ there is a negligible function negl such that $\Pr[\text{Invert}_{\mathcal{A},f}(n) = 1] \leq \text{negl}(n)$.

We now show formally that the factoring assumption implies the existence of a one-way function. Let Gen be a polynomial-time algorithm that, on input 1^n , outputs (N, p, q) where $N = pq$ and p and q are n -bit primes except with probability negligible in n . (We use Gen rather than GenModulus here purely for notational convenience.) Since Gen runs in polynomial time, there is a polynomial upper bound on the number of random bits the algorithm uses. For simplicity, and in order to get the main ideas across, we assume Gen always uses at most n random bits on input 1^n . In Algorithm 8.73 we define a function f_{Gen} that uses its input as the random bits for running Gen . Thus, f_{Gen} is a *deterministic* function as required.

ALGORITHM 8.73

Algorithm computing f_{Gen}

Input: String x of length n

Output: Integer N

compute $(N, p, q) := \text{Gen}(1^n; x)$

// i.e., run $\text{Gen}(1^n)$ using x as the random tape

return N

If the factoring problem is hard relative to Gen then, intuitively, f_{Gen} is a one-way function. Certainly f_{Gen} is easy to compute. As for the hardness of inverting this function, note that the following distributions are identical:

1. The modulus N output by $f_{\text{Gen}}(x)$, when $x \in \{0, 1\}^n$ is chosen uniformly.
2. The modulus N output by (the randomized algorithm) $\text{Gen}(1^n)$.

If moduli N generated according to the second distribution are hard to factor, then the same holds for moduli N generated according to the first distribution. Moreover, given any preimage x' of N with respect to f_{Gen} (i.e., an x' for which $f_{\text{Gen}}(x') = N$; note that we do not require $x' = x$), it is easy to recover a factor of N by running $\text{Gen}(1^n; x')$ to obtain (N, p, q) and outputting the factors p and q . Thus, finding a preimage of N with respect to f_{Gen} is as hard as factoring N . One can easily turn this into a formal proof of the following:

THEOREM 8.74 *If the factoring problem is hard relative to Gen , then f_{Gen} is a one-way function.*

One-Way Permutations

We can also use number-theoretic assumptions to construct a family of one-way *permutations*. We begin with a restatement of Definitions 7.2 and 7.3, specialized to the case of permutations:

DEFINITION 8.75 A triple $\Pi = (\text{Gen}, \text{Samp}, f)$ of probabilistic polynomial-time algorithms is a family of permutations if the following hold:

1. The parameter-generation algorithm Gen , on input 1^n , outputs parameters I with $|I| \geq n$. Each value of I defines a set $\mathcal{D}_I$ that constitutes the domain and range of a permutation (i.e., bijection) $f_I : \mathcal{D}_I \rightarrow \mathcal{D}_I$.
2. The sampling algorithm Samp , on input I , outputs a uniformly distributed element of $\mathcal{D}_I$.
3. The deterministic evaluation algorithm f , on input I and $x \in \mathcal{D}_I$, outputs an element $y \in \mathcal{D}_I$. We write this as $y := f_I(x)$.

Given a family of functions Π , consider the following experiment for any algorithm $\mathcal{A}$ and parameter n :

The inverting experiment $\text{Invert}_{\mathcal{A}, \Pi}(n)$:

1. $\text{Gen}(1^n)$ is run to obtain I , and then $\text{Samp}(I)$ is run to choose a uniform $x \in \mathcal{D}_I$. Finally, $y := f_I(x)$ is computed.
2. $\mathcal{A}$ is given I and y as input, and outputs x' .
3. The output of the experiment is 1 if and only if $f_I(x') = y$.

DEFINITION 8.76 The family of permutations $\Pi = (\text{Gen}, \text{Samp}, f)$ is one-way if for all probabilistic polynomial-time algorithms $\mathcal{A}$ there exists a negligible function negl such that

$$\Pr[\text{Invert}_{\mathcal{A}, \Pi}(n) = 1] \leq \text{negl}(n).$$

Given GenRSA as in Section 8.2.4, Construction 8.77 defines a family of permutations. It is immediate that if the RSA problem is hard relative to GenRSA then this family is one-way. It can similarly be shown that hardness of the discrete-logarithm problem in $\mathbb{Z}_p^*$, with p prime, implies the existence of a one-way family of permutations; see Section 7.1.2.

CONSTRUCTION 8.77

Let GenRSA be as before. Define a family of permutations as follows:

- **Gen**: on input 1^n , run $\text{GenRSA}(1^n)$ to obtain (N, e, d) and output $I = \langle N, e \rangle$. Set $\mathcal{D}_I = \mathbb{Z}_N^*$.
- **Samp**: on input $I = \langle N, e \rangle$, choose a uniform element of $\mathbb{Z}_N^*$.
- **f**: on input $I = \langle N, e \rangle$ and $x \in \mathbb{Z}_N^*$, output $[x^e \bmod N]$.

A family of permutations based on the RSA problem.

8.4.2 Constructing Collision-Resistant Hash Functions

Collision-resistant hash functions were introduced in Section 5.1. Although we have discussed constructions of collision-resistant hash functions used in practice in Section 6.3, we have not yet seen constructions that can be rigorously based on simpler assumptions. We show here a construction based on the discrete-logarithm assumption in prime-order groups. (A construction based on the RSA problem is described in Exercise 8.20.) Although these constructions are less efficient than the hash functions used in practice, they are important since they illustrate the *feasibility* of achieving collision resistance based on standard and well-studied number-theoretic assumptions.

Let $\mathcal{G}$ be a polynomial-time algorithm that, on input 1^n , outputs a (description of a) cyclic group $\mathbb{G}$, its order q (with $\|q\| = n$), and a generator g . Here we also require that q is *prime* except possibly with negligible probability. A fixed-length hash function based on $\mathcal{G}$ is given in Construction 8.78.

CONSTRUCTION 8.78

Let $\mathcal{G}$ be as described in the text. Define a fixed-length hash function (Gen, H) as follows:

- **Gen**: on input 1^n , run $\mathcal{G}(1^n)$ to obtain $(\mathbb{G}, q, g)$ and then select a uniform $h \in \mathbb{G}$. Output $s := \langle \mathbb{G}, q, g, h \rangle$ as the key.
- **H**: given a key $s = \langle \mathbb{G}, q, g, h \rangle$ and input $(x_1, x_2) \in \mathbb{Z}_q \times \mathbb{Z}_q$, output $H^s(x_1, x_2) := g^{x_1} h^{x_2} \in \mathbb{G}$.

A fixed-length hash function.

Note that **Gen** and **H** can be computed in polynomial time. Before continuing with an analysis of the construction, we make some technical remarks:

- For a given $s = \langle \mathbb{G}, q, g, h \rangle$ with $n = \|q\|$, the function H^s is described as taking elements of $\mathbb{Z}_q \times \mathbb{Z}_q$ as input. However, H^s can be viewed as taking bit-strings of length $2 \cdot (n - 1)$ as input if we parse an input $x \in \{0, 1\}^{2(n-1)}$ as two strings x_1, x_2 , each of length $n - 1$, and then view x_1, x_2 as elements of $\mathbb{Z}_q$ in the natural way.
- The output of H^s is similarly specified as being an element of $\mathbb{G}$, but we can view this as a bit-string if we fix some representation of $\mathbb{G}$. To satisfy the requirements of Definition 5.2 (which requires the output length to be fixed as a function of n) we can pad the output as needed.
- Given the above, the construction only compresses its input for groups $\mathbb{G}$ in which elements of $\mathbb{G}$ can be represented using fewer than $2n - 2$ bits. This holds both for the groups output by Algorithm 8.65 (assuming $n \ll \ell$, which is usually the case), as well as for elliptic-curve groups when point compression is used. A generalization of Construction 8.78

can be used to obtain compression from *any* $\mathcal{G}$ for which the discrete-logarithm problem is hard, regardless of the number of bits required to represent group elements; see Exercise 8.21.

THEOREM 8.79 *If the discrete-logarithm problem is hard relative to $\mathcal{G}$, then Construction 8.78 is a fixed-length collision-resistant hash function (subject to the discussion regarding compression, above).*

PROOF Let $\Pi = (\text{Gen}, H)$ be as in Construction 8.78, and let $\mathcal{A}$ be a probabilistic polynomial-time algorithm with

$$\varepsilon(n) \stackrel{\text{def}}{=} \Pr[\text{Hash-coll}_{\mathcal{A}, \Pi}(n) = 1]$$

(cf. Definition 5.2). We show how $\mathcal{A}$ can be used by an algorithm $\mathcal{A}'$ to solve the discrete-logarithm problem with success probability $\varepsilon(n)$:

Algorithm $\mathcal{A}'$:

The algorithm is given $\mathbb{G}, q, g, h$ as input.

1. Let $s := \langle \mathbb{G}, q, g, h \rangle$. Run $\mathcal{A}(s)$ and obtain output x and x' .
2. If $x \neq x'$ and $H^s(x) = H^s(x')$ then:
 - (a) If $h = 1$ return 0.
 - (b) Otherwise ($h \neq 1$), parse x as (x_1, x_2) and parse x' as (x'_1, x'_2) , where $x_1, x_2, x'_1, x'_2 \in \mathbb{Z}_q$, and return the result $[(x_1 - x'_1) \cdot (x'_2 - x_2)^{-1} \bmod q]$.

Clearly, $\mathcal{A}'$ runs in polynomial time. Furthermore, the input s given to $\mathcal{A}$ when run as a subroutine by $\mathcal{A}'$ is distributed exactly as in experiment $\text{Hash-coll}_{\mathcal{A}, \Pi}$ for the same value of the security parameter n . (The input to $\mathcal{A}'$ is generated by running $\mathcal{G}(1^n)$ to obtain $\mathbb{G}, q, g$ and then choosing $h \in \mathbb{G}$ uniformly at random. This is exactly how s is generated by $\text{Gen}(1^n)$.) So, with probability exactly $\varepsilon(n)$ there is a *collision*; i.e., $x \neq x'$ and $H^s(x) = H^s(x')$.

We claim that whenever there is a collision, $\mathcal{A}'$ returns the correct answer $\log_g h$. If $h = 1$ then this is clearly true (since $\log_g h = 0$ in this case). Assuming $h \neq 1$, the existence of a collision means that

$$\begin{aligned} H^s(x_1, x_2) = H^s(x'_1, x'_2) &\Rightarrow g^{x_1} h^{x_2} = g^{x'_1} h^{x'_2} \\ &\Rightarrow g^{x_1 - x'_1} = h^{x'_2 - x_2}. \end{aligned} \tag{8.5}$$

Note that $x'_2 - x_2 \not\equiv 0 \pmod q$; otherwise, we would have $x_1 = x'_1 \pmod q$ but then $x = x'$ and we would not have a collision. Since q is prime, the inverse $\Delta \stackrel{\text{def}}{=} [(x'_2 - x_2)^{-1} \bmod q]$ exists. Raising each side of Equation (8.5) to this power gives:

$$g^{(x_1 - x'_1) \cdot \Delta} = \left(h^{x'_2 - x_2} \right)^\Delta = h^1 = h,$$

and so the output returned by $\mathcal{A}'$ is

$$\log_g h = [(x_1 - x'_1) \cdot \Delta \bmod q] = [(x_1 - x'_1) \cdot (x'_2 - x_2)^{-1} \bmod q].$$

We see that $\mathcal{A}'$ correctly solves the discrete-logarithm problem with probability exactly $\varepsilon(n)$. Since, by assumption, the discrete-logarithm problem is hard relative to $\mathcal{G}$, we conclude that $\varepsilon(n)$ is negligible. ■

Using Exercise 8.21 in combination with the Merkle–Damgård transform (see Section 5.2) we obtain:

THEOREM 8.80 *If the discrete-logarithm problem is hard, then collision-resistant hash functions exist.*

References and Additional Reading

The book by Childs [44] has excellent coverage of the group theory discussed in this chapter (and more), in greater depth but at a similar level of exposition. Shoup [159] gives a more advanced, yet still accessible, treatment of much of this material also, with special focus on algorithmic aspects. (Our statement of Bertrand’s postulate is taken from [159, Theorem 5.8].) Relatively gentle introductions to abstract algebra and group theory that go well beyond what we have space for here are available in the books by Fraleigh [67] and Herstein [89]; the interested reader will have no trouble finding more advanced algebra texts if they are so inclined.

The first efficient primality test was by Solovay and Strassen [164]. The Miller–Rabin test is due to Miller [127] and Rabin [146]. A deterministic primality test was discovered by Agrawal et al. [5]. See Dietzfelbinger [57] for a comprehensive survey of this area.

The RSA problem was publicly introduced by Rivest, Shamir, and Adleman [148], although it was revealed in 1997 that Ellis, Cocks, and Williamson, three members of GCHQ (a British intelligence agency), had explored similar ideas—without fully recognizing their importance—in a classified setting several years earlier. The discrete-logarithm and Diffie–Hellman problems were first considered, at least implicitly, by Diffie and Hellman [58].

Most treatments of elliptic curves require advanced mathematical background on the part of the reader. The book by Silverman and Tate [160] is perhaps an exception. As with many books on the subject written for mathematicians, however, that book has little coverage of elliptic curves over *finite* fields, which is the case most relevant to cryptography. The text by Washington [176], although a bit more advanced, deals heavily (but not exclusively)

with the finite-field case. Implementation issues related to elliptic-curve cryptography are covered by Hankerson et al. [83]. Recommended parameters for elliptic curves, as well as subgroups modulo a prime, are given by NIST [132].

The construction of a collision-resistant hash function based on the discrete-logarithm problem is due to [43], and an earlier construction based on the hardness of factoring is given in [81] (see also Exercise 8.20).

Exercises

8.1 Let $\mathbb{G}$ be an abelian group. Prove that there is a *unique* identity in $\mathbb{G}$, and that every element $g \in \mathbb{G}$ has a *unique* inverse.

8.2 Show that Proposition 8.36 does not necessarily hold when $\mathbb{G}$ is infinite.

Hint: Consider the set $\{1\} \cup \{2, 4, 6, 8, \dots\} \subset \mathbb{R}$.

8.3 Let $\mathbb{G}$ be a finite group, and $g \in \mathbb{G}$. Show that $\langle g \rangle$ is a subgroup of $\mathbb{G}$. Is the set $\{g^0, g^1, \dots\}$ necessarily a subgroup of $\mathbb{G}$ when $\mathbb{G}$ is infinite?

8.4 This question concerns the Euler phi function.

- (a) Let p be prime and $e \geq 1$ an integer. Show that $\phi(p^e) = p^{e-1}(p-1)$.
- (b) Let p, q be relatively prime. Show that $\phi(pq) = \phi(p) \cdot \phi(q)$. (You may use the Chinese remainder theorem.)
- (c) Prove Theorem 8.19.

8.5 Compute the final two (decimal) digits of 3^{1000} (by hand).

Hint: The answer is $[3^{1000} \bmod 100]$.

8.6 Compute $[101^{4,800,000,002} \bmod 35]$ (by hand).

8.7 Compute $[46^{51} \bmod 55]$ (by hand) using the Chinese remainder theorem.

8.8 Prove that if $\mathbb{G}, \mathbb{H}$ are groups, then $\mathbb{G} \times \mathbb{H}$ is a group.

8.9 Let p, N be integers with $p \mid N$. Prove that for any integer X ,

$$[[X \bmod N] \bmod p] = [X \bmod p].$$

Show that, in contrast, $[[X \bmod p] \bmod N]$ need not equal $[X \bmod N]$.

8.10 Corollary 8.21 shows that if $N = pq$ and $ed = 1 \bmod \phi(N)$ then for all $x \in \mathbb{Z}_N^*$ we have $(x^e)^d = x \bmod N$. Show that this holds for all $x \in \{0, \dots, N-1\}$.

Hint: Use the Chinese remainder theorem.

- 8.11 Complete the details of the proof of the Chinese remainder theorem, showing that $\mathbb{Z}_N^*$ is isomorphic to $\mathbb{Z}_p^* \times \mathbb{Z}_q^*$.
- 8.12 This exercise develops an efficient algorithm for testing whether an integer is a perfect power.
- (a) Show that if $N = \hat{N}^e$ for some integers $\hat{N}, e > 1$ then $e \leq \|N\|$.
 - (b) Given N and e with $2 \leq e \leq \|N\| + 1$, show how to determine in $\text{poly}(\|N\|)$ time whether there exists an integer $\hat{N}$ with $\hat{N}^e = N$.
- Hint:** Use binary search.
- (c) Given N , show how to test in $\text{poly}(\|N\|)$ time whether N is a perfect power.
- 8.13 Given N and $a \in \mathbb{Z}_N^*$, show how to test in polynomial time whether a is a strong witness that N is composite.
- 8.14 Fix N, e with $\gcd(e, \phi(N)) = 1$, and assume there is an adversary $\mathcal{A}$ running in time t for which

$$\Pr[\mathcal{A}(x^e \bmod N) = x] = 0.01,$$

where the probability is taken over uniform choice of $x \in \mathbb{Z}_N^*$. Show that it is possible to construct an adversary $\mathcal{A}'$ for which

$$\Pr[\mathcal{A}'(x^e \bmod N) = x] = 0.99$$

for *all* x . The running time t' of $\mathcal{A}'$ should be polynomial in t and $\|N\|$.

Hint: Use the fact that $y^{1/e} \cdot r = (y \cdot r^e)^{1/e} \bmod N$.

- 8.15 Formally define the CDH assumption. Prove that hardness of the CDH problem relative to $\mathcal{G}$ implies hardness of the discrete-logarithm problem relative to $\mathcal{G}$, and that hardness of the DDH problem relative to $\mathcal{G}$ implies hardness of the CDH problem relative to $\mathcal{G}$.
- 8.16 Determine the points on the elliptic curve $E : y^2 = x^3 + 2x + 1$ over $\mathbb{Z}_{11}$. How many points are on this curve?
- 8.17 Consider the elliptic-curve group from Example 8.67. (See also Example 8.69.) Compute $(1, 0) + (4, 3) + (4, 3)$ in this group by first converting to projective coordinates and then using Equations (8.3) and (8.4).
- 8.18 Prove the fourth statement in Proposition 8.68.
- 8.19 Can the following problem be solved in polynomial time? Given a prime p , a value $x \in \mathbb{Z}_{p-1}^*$, and $y := [g^x \bmod p]$ (where g is a uniform value in $\mathbb{Z}_p^*$), find g , i.e., compute $y^{1/x} \bmod p$. If your answer is “yes,” give a polynomial-time algorithm. If your answer is “no,” show a reduction to one of the assumptions introduced in this chapter.

- 8.20 Let **GenRSA** be as in Section 8.2.4. Prove that if the RSA problem is hard relative to **GenRSA** then Construction 8.81 is a fixed-length collision-resistant hash function.

CONSTRUCTION 8.81

Define (Gen, H) as follows:

- **Gen**: on input 1^n , run **GenRSA** (1^n) to obtain N, e, d , and select $y \leftarrow \mathbb{Z}_N^*$. The key is $s := \langle N, e, y \rangle$.
- **H**: if $s = \langle N, e, y \rangle$, then H^s maps inputs in $\{0, 1\}^{3n}$ to outputs in $\mathbb{Z}_N^*$. Let $f_0^s(x) \stackrel{\text{def}}{=} [x^e \bmod N]$ and $f_1^s(x) \stackrel{\text{def}}{=} [y \cdot x^e \bmod N]$. For a $3n$ -bit long string $x = x_1 \cdots x_{3n}$, define

$$H^s(x) \stackrel{\text{def}}{=} f_{x_1}^s \left(f_{x_2}^s \left(\cdots \left(1 \right) \cdots \right) \right).$$

- 8.21 Consider the following generalization of Construction 8.78:

CONSTRUCTION 8.82

Define a fixed-length hash function (Gen, H) as follows:

- (a) **Gen**: on input 1^n , run $\mathcal{G}(1^n)$ to obtain $(\mathbb{G}, q, h_1)$ and then select $h_2, \dots, h_t \leftarrow \mathbb{G}$. Output $s := \langle \mathbb{G}, q, (h_1, \dots, h_t) \rangle$ as the key.
- (b) **H**: given a key $s = \langle \mathbb{G}, q, (h_1, \dots, h_t) \rangle$ and input $(x_1, \dots, x_t)$ with $x_i \in \mathbb{Z}_q$, output $H^s(x_1, \dots, x_t) := \prod_i h_i^{x_i}$.

- (a) Prove that if the discrete-logarithm problem is hard relative to $\mathcal{G}$ and q is prime, then for any $t = \text{poly}(n)$ this construction is a fixed-length collision-resistant hash function.
- (b) Discuss how this construction can be used to obtain *compression* regardless of the number of bits needed to represent elements of $\mathbb{G}$ (as long as it is polynomial in n).

Chapter 9

****Algorithms for Factoring and Computing Discrete Logarithms***

In the last chapter, we introduced several number-theoretic problems—most prominently, *factoring* the product of two large primes and *computing discrete logarithms* in certain groups—that are widely believed to be hard. As defined there, this means there are presumed to be no polynomial-time algorithms for these problems. This *asymptotic* notion of hardness, however, tells us little about how to set the security parameter—sometimes called the *key length*, although the terms are not interchangeable—to achieve some desired, *concrete* level of security in practice. A proper understanding of this issue is extremely important for the real-world deployment of cryptosystems based on these problems. Setting the security parameter too low means a cryptosystem may be vulnerable to attacks more efficient than anticipated; being overly conservative and setting the security parameter too high will give good security, but at the expense of efficiency for the honest users. The relative difficulty of different number-theoretic problems can also play a role in determining which problems to use as the basis for building cryptosystems in the first place.

The fundamental issue, of course, is that a brute-force search may not be the best algorithm for solving a given problem; thus, using key length n does not, in general, give security against attackers running for 2^n time. This is in contrast to the private-key setting where the best attacks on existing block ciphers have roughly the complexity of brute-force search (ignoring pre-computation). As a consequence, the key lengths used in the public-key setting tend to be significantly larger than those used in the private-key setting.

To gain a better appreciation of this point, we explore in this chapter several nonpolynomial-time algorithms for factoring and computing discrete logarithms that are far better than brute-force search. The goal is merely to give a taste of existing algorithms for these problems, as well as to provide some basic guidance for setting parameters in practice. Our focus is on the high-level ideas, and we consciously do not address many important implementation-level details that would be critical to deal with if these algorithms were to be used in practice. We also concentrate exclusively on *classical* algorithms, and refer the reader elsewhere for a discussion of known *quantum* polynomial-time(!) algorithms for factoring and computing discrete logarithms. (This was another conscious decision, both because we did not want to assume a background in quantum mechanics on the part of the reader, and because quantum computers seem unlikely in the near future.)

The reader may also notice that we only describe algorithms for factoring and computing discrete logarithms, and not algorithms for, say, solving the RSA or decisional Diffie–Hellman problems. Our choice is justified by the facts that the best known algorithms for solving RSA require factoring the modulus, and (in the groups discussed in Sections 8.3.3 and 8.3.4) the best known approaches for solving the decisional Diffie–Hellman problem involve computing discrete logarithms.

9.1 Algorithms for Factoring

Throughout, we assume that $N = pq$ is a product of two distinct primes with $p < q$. We will be most interested in the case when p and q each has the same (known) length n , and so $n = \Theta(\log N)$.

We will frequently use the Chinese remainder theorem along with the notation developed in Section 8.1.5. The Chinese remainder theorem states that

$$\mathbb{Z}_N \simeq \mathbb{Z}_p \times \mathbb{Z}_q \quad \text{and} \quad \mathbb{Z}_N^* \simeq \mathbb{Z}_p^* \times \mathbb{Z}_q^*,$$

with isomorphism given by $f(x) \stackrel{\text{def}}{=} ([x \bmod p], [x \bmod q])$. The fact that f is an isomorphism means, in particular, that it gives a bijection between elements $x \in \mathbb{Z}_N$ and pairs $(x_p, x_q) \in \mathbb{Z}_p \times \mathbb{Z}_q$. We write $x \leftrightarrow (x_p, x_q)$, with $x_p = [x \bmod p]$ and $x_q = [x \bmod q]$, to denote this bijection.

Recall from Section 8.2 that *trial division*—a trivial, brute-force factoring method—finds a factor of a given number N in time $\mathcal{O}(N^{1/2} \cdot \text{polylog}(N))$. (This is an exponential-time algorithm, since the size of the input is the length of the binary representation of N , i.e., $\|N\| = \mathcal{O}(\log N)$.¹) We cover three factoring algorithms with better performance:

- *Pollard’s $p-1$ method* is effective if $p-1$ has only “small” prime factors.
- *Pollard’s rho method* applies to arbitrary N . (As such, it is called a *general-purpose* factoring algorithm.) Its running time for N of the form discussed at the beginning of this section is $\mathcal{O}(N^{1/4} \cdot \text{polylog}(N))$. Note this is still *exponential* in n , the length of N .
- The *quadratic sieve algorithm* is a general-purpose factoring algorithm that runs in time *sub-exponential* in the length of N . We give a high-level overview of how this algorithm works, but the details are somewhat complex and are beyond the scope of this book.

¹Thus, a running time of $N^{\mathcal{O}(1)} = 2^{\mathcal{O}(\|N\|)}$ is exponential, a running time of $2^{o(\log N)} = 2^{o(\|N\|)}$ is sub-exponential, and a running time of $\log^{\mathcal{O}(1)} N = \|N\|^{\mathcal{O}(1)}$ is polynomial.

The fastest known general-purpose factoring algorithm is the *general number field sieve*. Heuristically, this algorithm factors its input N in expected time $2^{\mathcal{O}((\log N)^{1/3} \cdot (\log \log N)^{2/3})}$, which is sub-exponential in the length of N .

9.1.1 Pollard's $p - 1$ Algorithm

If $N = pq$ and $p - 1$ has *only* “small” prime factors, Pollard's $p - 1$ algorithm can be used to efficiently factor N . The basic idea is simple. Let B be an integer for which $(p - 1) \mid B$ and $(q - 1) \nmid B$; we defer to below the details of how such a B is found. Say $B = \gamma \cdot (p - 1)$ for some integer γ . Choose a uniform $x \in \mathbb{Z}_N^*$ and compute $y := [x^B - 1 \bmod N]$. (Note that y can be computed using the efficient exponentiation algorithm from Appendix B.2.3.) Since $1 \leftrightarrow (1, 1)$, we have

$$\begin{aligned} y = [x^B - 1 \bmod N] &\leftrightarrow (x_p, x_q)^B - (1, 1) \\ &= (x_p^B - 1 \bmod p, x_q^B - 1 \bmod q) \\ &= ((x_p^{p-1})^\gamma - 1 \bmod p, x_q^B - 1 \bmod q) \\ &= (0, [x_q^B - 1 \bmod q]) \end{aligned}$$

using Theorem 8.14 and the fact that the order of $\mathbb{Z}_p^*$ is $p - 1$. We show below that, with high probability, $x_q^B \neq 1 \bmod q$. Assuming this is the case, we have obtained an integer y for which

$$y = 0 \bmod p \quad \text{but} \quad y \neq 0 \bmod q;$$

that is, $p \mid y$ but $q \nmid y$. This, in turn, implies that $\gcd(y, N) = p$. Thus, a simple gcd computation (which can be done efficiently as described in Appendix B.1.2) yields a prime factor of N .

ALGORITHM 9.1

Pollard's $p - 1$ algorithm for factoring

Input: Integer N

Output: A non-trivial factor of N

```

 $x \leftarrow \mathbb{Z}_N^*$ 
 $y := [x^B - 1 \bmod N]$ 
  //  $B$  is as in the text
 $p := \gcd(y, N)$ 
if  $p \notin \{1, N\}$  return  $p$ 
    
```

Let us first argue that the algorithm works (with high probability). Assume $(p - 1) \mid B$ but $(q - 1) \nmid B$. In that case, as long as $x_q \stackrel{\text{def}}{=} [x \bmod q]$ is a generator of $\mathbb{Z}_q^*$, we have $x_q^B \neq 1 \bmod q$. (This follows from Proposition 8.52.) It remains to analyze the probability that x_q is a generator. Here we rely on

some results proved in Appendix B.3.1. Since q is prime, $\mathbb{Z}_q^*$ is a cyclic group of order $q - 1$ that has exactly $\phi(q - 1)$ generators (cf. Theorem B.16). If x is chosen uniformly from $\mathbb{Z}_N^*$, then x_q is uniformly distributed in $\mathbb{Z}_q^*$. (This is a consequence of the fact that the Chinese remainder theorem gives a bijection between $\mathbb{Z}_N^*$ and $\mathbb{Z}_p^* \times \mathbb{Z}_q^*$.) Thus, the probability that x_q is a generator is $\frac{\phi(q-1)}{q-1} = \Omega(1/\log q) = \Omega(1/n)$ (cf. Theorem B.15). Multiple values of x can be chosen to boost the probability of success.

We are left with the problem of finding B such that $(p-1) \mid B$ but $(q-1) \nmid B$. One possibility is to choose $B = \prod_{i=1}^k p_i^{\lfloor n/\log p_i \rfloor}$ for some k , where p_i denotes the i^{th} prime (i.e., $p_1 = 2, p_2 = 3, p_3 = 5, \dots$) and n is the length of p . (Note that $p_i^{\lfloor n/\log p_i \rfloor}$ is the largest power of p_i that can possibly divide $p - 1$.) If $p - 1$ can be written as $\prod_{i=1}^k p_i^{e_i}$ with $e_i \geq 0$ (that is, if the largest prime factor of $p - 1$ is less than p_k), we will have $(p - 1) \mid B$. In contrast, if $q - 1$ has *any* prime factor larger than p_k , then $(q - 1) \nmid B$.

Choosing a larger value for k increases B and so increases the running time of the algorithm (which performs a modular exponentiation to the power B). A larger value of k also makes it more likely that $(p - 1) \mid B$, but at the same time makes it less likely that $(q - 1) \nmid B$. It is, of course, possible to run the algorithm repeatedly using multiple choices for k .

Pollard's $p - 1$ algorithm is thwarted if both $p - 1$ and $q - 1$ have any large prime factors. (More precisely, the algorithm still works but only for B so large that the algorithm becomes impractical.) If p and q are uniform n -bit primes, then it is unlikely that either $p - 1$ or $q - 1$ will have only small prime factors. Nevertheless, when generating a modulus $N = pq$ for cryptographic applications, p and q are sometimes chosen to be *strong* primes. (Recall that p is a strong prime if $(p - 1)/2$ is also prime.) Selecting p and q in this way is markedly less efficient than simply choosing p and q as *arbitrary* (random) primes. Because better factoring algorithms are available anyway (as we will see below), and due to the observation above, the current consensus is that the added computational cost of generating p and q as strong primes does not yield any appreciable security gains.

9.1.2 Pollard's Rho Algorithm

Pollard's rho algorithm can be used to factor an arbitrary integer $N = pq$; in that sense, it is a *general-purpose* factoring algorithm. Heuristically, the algorithm factors N with constant probability in $\mathcal{O}(N^{1/4} \cdot \text{polylog}(N))$ time; this is still exponential, but is a vast improvement over trial division.

The core idea of the approach is to find distinct values $x, x' \in \mathbb{Z}_N^*$ that are equivalent modulo p (i.e., for which $x = x' \pmod{p}$); call such a pair *good*. Note that for a good pair x, x' it holds that $\gcd(x - x', N) = p$ (since $x \neq x' \pmod{N}$), so computing the gcd gives a non-trivial factor of N .

How can we find a good pair? Say we choose values $x^{(1)}, \dots, x^{(k)}$ uniformly from $\mathbb{Z}_N^*$, where $k = 2^{n/2} = \mathcal{O}(\sqrt{p})$. Viewing these in their Chinese-

ALGORITHM 9.2**Pollard's rho algorithm for factoring****Input:** Integer N , a product of two n -bit primes**Output:** A non-trivial factor of N $x^{(0)} \leftarrow \mathbb{Z}_N^*$, $x' := x := x^{(0)}$ **for** $i = 1$ **to** $2^{n/2}$: $x := F(x)$ $x' := F(F(x'))$ $p := \gcd(x - x', N)$ **if** $p \notin \{1, N\}$ **return** p **and stop**

remaindering representation as $(x_p^{(1)}, x_q^{(1)}), \dots, (x_p^{(k)}, x_q^{(k)})$, we have that each $x_p^{(i)} \stackrel{\text{def}}{=} [x^{(i)} \bmod p]$ is uniform in $\mathbb{Z}_p^*$. (This follows from bijectivity between $\mathbb{Z}_N^*$ and $\mathbb{Z}_p^* \times \mathbb{Z}_q^*$.) Thus, using the birthday bound of Lemma A.16, we see that with high probability there exist distinct i, j with $x_p^{(i)} = x_p^{(j)}$ or, equivalently, $x^{(i)} = x^{(j)} \bmod p$. Moreover, Lemma A.15 shows that $x^{(i)} \neq x^{(j)}$ except with negligible probability. Thus, with high probability we obtain a good pair $x^{(i)}, x^{(j)}$ that can be used to find a non-trivial factor of N , as discussed earlier.

We can generate $k = \mathcal{O}(\sqrt{p})$ uniform elements of $\mathbb{Z}_N^*$ in $\mathcal{O}(\sqrt{p}) = \mathcal{O}(N^{1/4})$ time. *Testing* all pairs of elements in order to identify a good pair, however, would require $\binom{k}{2} = \mathcal{O}(k^2) = \mathcal{O}(p) = \mathcal{O}(N^{1/2})$ time! (Note that since p is unknown we cannot simply compute $x_p^{(1)}, \dots, x_p^{(k)}$ explicitly and then sort the $x_p^{(i)}$ to find a good pair. Instead, for all distinct pairs i, j we must compute $\gcd(x^{(i)} - x^{(j)}, N)$ to see whether this gives a non-trivial factor of N .) Without further optimizations, this will be no better than trial division.

Pollard's idea was to use a technique we have seen in Section 5.4.2 in the context of small-space birthday attacks. Specifically, we compute the sequence $x^{(1)}, x^{(2)}, \dots$ by letting each value be a function of the one before it; i.e., we fix some function $F : \mathbb{Z}_N^* \rightarrow \mathbb{Z}_N^*$, choose a uniform $x^{(0)} \in \mathbb{Z}_N^*$, and then set $x^{(i)} := F(x^{(i-1)})$ for $i = 1, \dots, k$. We require F to have the property that if $x = x' \bmod p$, then $F(x) = F(x') \bmod p$; this ensures that once equivalence modulo p occurs, it persists. (A standard choice is $F(x) = [x^2 + 1 \bmod N]$, but any polynomial F will have this property.) If we model F as a random function (which works heuristically), then with high probability there is a good pair in the first k elements of this sequence. Proceeding roughly as in Algorithm 5.9 from Section 5.4.2, we can detect a good pair (if there is one) using only $\mathcal{O}(k)$ gcd computations; see Algorithm 9.2. In addition to improving the running time, Pollard's idea also drastically reduces the amount of memory needed.

9.1.3 The Quadratic Sieve Algorithm

Pollard's rho algorithm is better than trial division, but still runs in exponential time. The *quadratic sieve* algorithm runs in sub-exponential time. It

was the fastest known factoring algorithm until the early 1990s and remains the factoring algorithm of choice for numbers up to about 300 bits long. We describe the general principles of the algorithm but caution the reader that several important details are omitted.

An element $z \in \mathbb{Z}_N^*$ is a *quadratic residue modulo N* if there is an $x \in \mathbb{Z}_N^*$ such that $x^2 = z \bmod N$; in this case, we say that x is a *square root of z* . The following observations serve as our starting point:

- If N is a product of two distinct, odd primes, then every quadratic residue modulo N has exactly four square roots. (See Section 13.4.2.)
- Given x, y with $x^2 = y^2 \bmod N$ and $x \not\equiv \pm y \bmod N$, it is possible to compute a nontrivial factor of N in polynomial time. This is by virtue of the fact that $x^2 = y^2 \bmod N$ implies

$$0 = x^2 - y^2 = (x - y)(x + y) \bmod N,$$

and so $N \mid (x - y)(x + y)$. However, $N \nmid (x - y)$ and $N \nmid (x + y)$ because $x \not\equiv \pm y \bmod N$. So it must be the case that $\gcd(x - y, N)$ is equal to one of the prime factors of N . (See also Lemma 13.35.)

The quadratic sieve algorithm tries to generate x, y with $x^2 = y^2 \bmod N$ and $x \not\equiv \pm y \bmod N$. A naive way of doing this—which forms the basis of an older factoring algorithm due to Fermat—is to choose an $x \in \mathbb{Z}_N^*$, compute $q := [x^2 \bmod N]$, and then check whether q is a square *over the integers* (i.e., without reduction modulo N). If so, then $q = y^2$ for some integer y and so $x^2 = y^2 \bmod N$. Unfortunately, the probability that $[x^2 \bmod N]$ is a square is so low that this process must be repeated exponentially many times.

A significant improvement is obtained by generating a sequence of values $q_1 := [x_1^2 \bmod N], \dots$ and identifying a subset of those values whose *product* is a square over the integers. In the quadratic sieve algorithm this is accomplished using the following two steps:

Step 1. Fix some bound B . Say an integer is *B -smooth* if all its prime factors are less than or equal to B . In the first phase of the algorithm, we search for integers of the form $q_i = [x_i^2 \bmod N]$ that are B -smooth and factor them. (Although factoring is hard, finding and factoring B -smooth numbers is feasible when B is small enough.) These $\{x_i\}$ are chosen by trying $x = \sqrt{N} + 1, \sqrt{N} + 2, \dots$; this ensures a nontrivial reduction modulo N (since $x > \sqrt{N}$) and has the advantage that $q \stackrel{\text{def}}{=} [x^2 \bmod N] = x^2 - N$ is “small” so that q is more likely to be B -smooth.

Let $\{p_1, \dots, p_k\}$ be the set of prime numbers less than or equal to B . Once we have found and factored the B -smooth $\{q_i\}$ as described above, we have a

set of equations of the form:

$$\begin{aligned} q_1 &= [x_1^2 \bmod N] = \prod_{i=1}^k p_i^{e_{1,i}} \\ &\vdots \\ q_\ell &= [x_\ell^2 \bmod N] = \prod_{i=1}^k p_i^{e_{\ell,i}}. \end{aligned} \tag{9.1}$$

(Note that the above equations are over the integers.)

Step 2. We next want to find some subset of the $\{q_i\}$ whose product is a square. If we multiply some subset S of the $\{q_i\}$, we see that the result

$$z = \prod_{j \in S} q_j = \prod_{i=1}^k p_i^{\sum_{j \in S} e_{j,i}}$$

is a square if and only if the exponent of each prime p_i is even. This suggests that we care about the exponents $\{e_{j,i}\}$ in Equation (9.1) only modulo 2; moreover, we can use linear algebra to find a subset of the $\{q_i\}$ whose “exponent vectors” sum to the 0-vector modulo 2.

In more detail: if we reduce the exponents in Equation (9.1) modulo 2, we obtain the 0/1-matrix Γ given by

$$\begin{pmatrix} \gamma_{1,1} & \gamma_{1,2} & \cdots & \gamma_{1,k} \\ \vdots & \vdots & \ddots & \vdots \\ \gamma_{\ell,1} & \gamma_{\ell,2} & \cdots & \gamma_{\ell,k} \end{pmatrix} \stackrel{\text{def}}{=} \begin{pmatrix} [e_{1,1} \bmod 2] & [e_{1,2} \bmod 2] & \cdots & [e_{1,k} \bmod 2] \\ \vdots & \vdots & \ddots & \vdots \\ [e_{\ell,1} \bmod 2] & [e_{\ell,2} \bmod 2] & \cdots & [e_{\ell,k} \bmod 2] \end{pmatrix}.$$

If $\ell = k + 1$, then Γ has more rows than columns and there must be some nonempty subset S of the rows that sum to the 0-vector modulo 2. Such a subset can be found efficiently using linear algebra. Then:

$$z \stackrel{\text{def}}{=} \prod_{j \in S} q_j = \prod_{i=1}^k p_i^{\sum_{j \in S} e_{j,i}} = \left(\prod_{i=1}^k p_i^{(\sum_{j \in S} e_{j,i})/2} \right)^2,$$

using the fact that all the $\{\sum_{j \in S} e_{j,i}\}$ are even. Since

$$z = \prod_{j \in S} q_j = \prod_{j \in S} x_j^2 = \left(\prod_{j \in S} x_j \right)^2 \bmod N,$$

we have obtained two square roots (modulo N) of z . Although there is no guarantee that these square roots will enable factorization of N (for reasons

discussed at the beginning of this section), heuristically they do with constant probability. By taking $\ell > k + 1$ we can obtain multiple subsets S with the desired property and try to factor N using each possibility.

Example 9.3

Take $N = 377753$. We have $6647 = [620^2 \bmod N]$, and we can factor 6647 (over the integers, without any modular reduction) as

$$[620^2 \bmod N] = 6647 = 17^2 \cdot 23.$$

Similarly,

$$\begin{aligned} [621^2 \bmod N] &= 2^4 \cdot 17 \cdot 29 \\ [645^2 \bmod N] &= 2^7 \cdot 13 \cdot 23 \\ [655^2 \bmod N] &= 2^3 \cdot 13 \cdot 17 \cdot 29. \end{aligned}$$

Letting our subset S include all four of the above equations, we see that

$$\begin{aligned} 620^2 \cdot 621^2 \cdot 645^2 \cdot 655^2 &= 2^{14} \cdot 13^2 \cdot 17^4 \cdot 23^2 \cdot 29^2 \bmod N \\ \Rightarrow [620 \cdot 621 \cdot 645 \cdot 655 \bmod N]^2 &= [2^7 \cdot 13 \cdot 17^2 \cdot 23 \cdot 29 \bmod N]^2 \bmod N \\ &\Rightarrow 127194^2 = 45335^2 \bmod N, \end{aligned}$$

with $127194 \not\equiv \pm 45335 \bmod N$. Computing $\gcd(127194 - 45335, 377753) = 751$ yields a non-trivial factor of N . $\diamond$

Running time. Choosing a larger value of B makes it more likely that a uniform value $q = [x^2 \bmod N]$ is B -smooth; on the other hand, it means we have to work harder to identify and factor B -smooth numbers, and we will have to find more of them (since we require $\ell > k$, where k is the number of primes less than or equal to B). It also means that the matrix Γ will be larger, and so the linear-algebraic step will be slower. Choosing the optimal value of B gives an algorithm that (heuristically, at least) factors N in time $2^{\mathcal{O}(\sqrt{\log N \log \log N})}$. (In fact, the constant term in the exponent can be determined quite precisely.) The important point for our purposes is that this is sub-exponential in the length of N .

9.2 Algorithms for Computing Discrete Logarithms

Let $\mathbb{G}$ be a group of known order q . An instance of the discrete-logarithm problem in $\mathbb{G}$ specifies a *base* $g \in \mathbb{G}$ (which need not be a generator of $\mathbb{G}$) and an element $h \in \langle g \rangle$, the subgroup generated by h ; the goal is to find x

such that $g^x = h$. (See Section 8.3.2.) The solution x is called the *discrete logarithm of h with respect to g* . A trivial brute-force search for x can be done in time $|\langle g \rangle| \leq q$ (by simply trying all possible values), and so we will only be interested in algorithms whose running time is better than this.

Algorithms for solving the discrete-logarithm problem fall into two categories: those that are *generic* and apply to arbitrary groups, and those that are tailored to work for some *specific* class of groups. We begin by discussing three generic algorithms:

- When the group order q is not prime and the factorization of q is known, or easy to determine, the *Pohlig–Hellman algorithm* reduces the problem of finding discrete logarithms in $\mathbb{G}$ to that of finding discrete logarithms in prime-order *subgroups* of $\mathbb{G}$. Roughly speaking, the effect is that the complexity of solving the discrete logarithm in a group of order q is no greater than the complexity of solving the discrete logarithm in a group of order q' , where q' is the largest prime dividing q . This explains the preference for using prime-order groups (cf. Section 8.3.2).
- The *baby-step/giant-step* method, due to Shanks, computes the discrete logarithm in a group of order q in time $\mathcal{O}(\sqrt{q} \cdot \text{polylog}(q))$ and storing $\mathcal{O}(\sqrt{q})$ group elements.
- *Pollard’s rho algorithm* also enables computation of discrete logarithms in time $\mathcal{O}(\sqrt{q} \cdot \text{polylog}(q))$, but using *constant* memory. It can be viewed as exploiting the connection between the discrete-logarithm problem and collision-resistant hashing that we have seen in Section 8.4.2. We describe a different algorithm that more clearly illustrates this connection.

It can be shown that the time complexity of the latter two algorithms is *optimal* as far as generic algorithms are concerned. Thus, to have any hope of doing better we must look at algorithms for specific groups that exploit the *representation* of group elements in those groups, where by “representation” we mean the way group elements are encoded as bit-strings.

This point bears some discussion. From a mathematical point of view, any two cyclic groups of the same order are isomorphic, meaning that the groups are identical up to a “renaming” of the group elements. From a computational/algorithmic point of view, however, this “renaming” can have a significant impact. For example, consider the cyclic group $\mathbb{Z}_q$ of integers $\{0, \dots, q-1\}$ under *addition* modulo q . Computing discrete logarithms in this group is trivial. Say we are given $g, h \in \mathbb{Z}_q$ with g a generator (so $g \neq 0$), and we want to find x such that $x \cdot g = h \bmod q$. Since $g \neq 0$ we have $\gcd(g, q) = 1$ and so g has a multiplicative inverse g^{-1} modulo q . Moreover, g^{-1} can be computed efficiently, as described in Appendix B.2.2. But then $x = h \cdot g^{-1} \bmod q$ is the desired solution. Note that, formally, x here denotes an integer and not a group element—after all, the group operation is addition, not multiplication. Nevertheless, in solving the discrete-logarithm

problem in $\mathbb{Z}_q$ we can make use of the fact that another operation (namely, multiplication) can be defined on the elements of that group.

Turning to groups with cryptographic significance, we focus our attention on (subgroups of) $\mathbb{Z}_p^*$ for p prime. (See Section 8.3.3.) We give a high-level overview of the *index calculus algorithm* for solving the discrete-logarithm problem in such groups in sub-exponential time. Currently, the best known algorithm for this class of groups is the *number field sieve*,² which heuristically runs in time $2^{\mathcal{O}((\log p)^{1/3} \cdot (\log \log p)^{2/3})}$. Sub-exponential algorithms for computing discrete logarithms in multiplicative subgroups of finite fields of large characteristic are also known. In early 2013, significant advances were made in algorithms for computing discrete logarithms in multiplicative subgroups of finite fields of *small* characteristic; it seems prudent to avoid using such groups for cryptographic applications.

Importantly, no sub-exponential algorithms are known for computing discrete logarithms in certain elliptic-curve groups. This means that for a given security level, we can use groups of smaller order when working in elliptic-curve groups as compared to, say, working in $\mathbb{Z}_p^*$, resulting in cryptographic schemes with better asymptotic efficiency.

9.2.1 The Pohlig–Hellman Algorithm

The Pohlig–Hellman algorithm can be used to speed up the computation of discrete logarithms in a group $\mathbb{G}$ when any non-trivial factors of the group order q are known. Recall that the order of an element g , which we denote here by $\text{ord}(g)$, is the smallest positive integer i for which $g^i = 1$. We will need the following lemma:

LEMMA 9.4 *Let $\text{ord}(g) = q$, and say $p \mid q$. Then $\text{ord}(g^p) = q/p$.*

PROOF Since $(g^p)^{q/p} = g^q = 1$, the order of g^p is at most q/p . Let $i > 0$ be such that $(g^p)^i = 1$. Then $g^{pi} = 1$ and, since q is the order of g , we must have $pi \geq q$ or equivalently $i \geq q/p$. The order of g^p is thus exactly q/p . ■

We will also use a generalization of the Chinese remainder theorem: if $q = \prod_{i=1}^k q_i$ and $\gcd(q_i, q_j) = 1$ for all $i \neq j$ then

$$\mathbb{Z}_q \simeq \mathbb{Z}_{q_1} \times \cdots \times \mathbb{Z}_{q_k} \quad \text{and} \quad \mathbb{Z}_q^* \simeq \mathbb{Z}_{q_1}^* \times \cdots \times \mathbb{Z}_{q_k}^*.$$

(This can be proved by induction on k , using the basic Chinese remainder theorem for $k = 2$.) Moreover, by an extension of the algorithm in Section 8.1.5

²It is no accident that the algorithm's name and its running time are similar to those of the general number field sieve for factoring, since the algorithms share many of the same underlying steps.

it is possible to convert efficiently between the representation of an element as an element of $\mathbb{Z}_q$ and its representation as an element of $\mathbb{Z}_{q_1} \times \cdots \times \mathbb{Z}_{q_k}$.

We now describe the Pohlig–Hellman algorithm. We are given a generator g and an element h and wish to find an x such that $g^x = h$. Say a factorization $q = \prod_{i=1}^k q_i$ is known with the $\{q_i\}$ pairwise relatively prime. (This need not be the complete prime factorization of q .) We know that

$$\left(g^{q/q_i}\right)^x = (g^x)^{q/q_i} = h^{q/q_i} \quad \text{for } i = 1, \dots, k. \quad (9.2)$$

Letting $g_i \stackrel{\text{def}}{=} g^{q/q_i}$ and $h_i \stackrel{\text{def}}{=} h^{q/q_i}$, we thus have k instances of a discrete-logarithm problem in k *smaller* groups. Specifically, each problem $g_i^x = h_i$ is in a subgroup of size $\text{ord}(g_i) = q_i$ (by Lemma 9.4). (Note that each such problem only determines $[x \bmod q_i]$; this follows from Proposition 8.53.)

We can solve each of the k resulting instances using any algorithm for solving the discrete-logarithm problem. Solving these instances gives a set of answers $\{x_i\}_{i=1}^k$, with $x_i \in \mathbb{Z}_{q_i}$, for which $g_i^{x_i} = h_i = g_i^x$. Proposition 8.53 implies that $x = x_i \bmod q_i$ for all i . By the generalized Chinese remainder theorem discussed earlier, the constraints

$$\begin{aligned} x &= x_1 \bmod q_1 \\ &\vdots \\ x &= x_k \bmod q_k \end{aligned}$$

uniquely determine x modulo q , and the desired solution x can be efficiently reconstructed from the $\{x_i\}$.

Example 9.5

Consider the problem of computing discrete logarithms in $\mathbb{Z}_{31}^*$, a group of order $q = 30 = 5 \cdot 3 \cdot 2$. Say $g = 3$ and $h = 26 = g^x$ with x unknown. We have:

$$\begin{aligned} (g^{30/5})^x &= h^{30/5} \Rightarrow (3^6)^x = 26^6 \Rightarrow 16^x = 1 \\ (g^{30/3})^x &= h^{30/3} \Rightarrow (3^{10})^x = 26^{10} \Rightarrow 25^x = 5 \\ (g^{30/2})^x &= h^{30/2} \Rightarrow (3^{15})^x = 26^{15} \Rightarrow 30^x = 30. \end{aligned}$$

(All the above equations are modulo 31.) We have $\text{ord}(16) = 5$, $\text{ord}(25) = 3$, and $\text{ord}(30) = 2$. Solving each equation, we obtain

$$x = 0 \bmod 5, \quad x = 2 \bmod 3, \quad \text{and} \quad x = 1 \bmod 2,$$

and so $x = 5 \bmod 30$. Indeed, $3^5 = 26 \bmod 31$. ◇

If q has (known) prime factorization $q = \prod_{i=1}^k p_i^{e_i}$ then, by using the Pohlig–Hellman algorithm, the time to compute discrete logarithms in a group of order q is dominated by the computation of a discrete logarithm in a subgroup of size $\max_i \{p_i^{e_i}\}$. This can be further reduced to computation of a discrete logarithm in a subgroup of size $\max_i \{p_i\}$; see Exercise 9.5.

9.2.2 The Baby-Step/Giant-Step Algorithm

The baby-step/giant-step algorithm computes discrete logarithms in a group of order q using $\mathcal{O}(\sqrt{q})$ group operations. The idea is simple. Given a generator $g \in \mathbb{G}$, we can imagine the powers of g as forming a cycle

$$1 = g^0, g^1, g^2, \dots, g^{q-2}, g^{q-1}, g^q = 1.$$

We know that h must lie somewhere in this cycle. Computing all the points in this cycle would take $\Omega(q)$ time. Instead, we “mark off” the cycle at intervals of size $t \stackrel{\text{def}}{=} \lfloor \sqrt{q} \rfloor$; more precisely, we compute and store the $\lfloor q/t \rfloor + 1 = \mathcal{O}(\sqrt{q})$ elements

$$g^0, g^t, g^{2t}, \dots, g^{\lfloor q/t \rfloor \cdot t}.$$

(These are the “giant steps.”) Note that the gap between any consecutive “marks” is at most t . Furthermore, we know that $h = g^x$ lies in one of these gaps. Thus, if we next take “baby steps” and compute the t elements

$$h \cdot g^1, \dots, h \cdot g^t,$$

each of which corresponds to a “shift” of h , we know that one of these values will be equal to one of the points we have marked off. Say we find $h \cdot g^i = g^{k \cdot t}$. We can then easily compute $\log_g h := [(kt - i) \bmod q]$. Pseudocode for this algorithm follows.

ALGORITHM 9.6

The baby-step/giant-step algorithm

Input: Elements $g, h \in \mathbb{G}$; the order q of $\mathbb{G}$

Output: $\log_g h$

$t := \lfloor \sqrt{q} \rfloor$

for $i = 0$ to $\lfloor q/t \rfloor$:

compute $g_i := g^{i \cdot t}$

sort the pairs (i, g_i) by their second component

for $i = 1$ to t :

compute $h_i := h \cdot g^i$

if $h_i = g_k$ for some k , **return** $[kt - i \bmod q]$

The algorithm requires $\mathcal{O}(\sqrt{q})$ exponentiations/multiplications in $\mathbb{G}$. (In fact, other than the first value $g_1 = g^t$, each value g_i can be computed using a single multiplication as $g_i := g_{i-1} \cdot g_1$. Similarly, each h_i can be computed as $h_i := h_{i-1} \cdot g$.) Sorting the $\mathcal{O}(\sqrt{q})$ pairs $\{(i, g_i)\}$ takes time $\mathcal{O}(\sqrt{q} \cdot \log q)$, and we can then use binary search to check if each h_i is equal to some g_k in time $\mathcal{O}(\log q)$. The overall algorithm thus runs in time $\mathcal{O}(\sqrt{q} \cdot \text{polylog}(q))$.

Example 9.7

We show an application of the algorithm in the cyclic group $\mathbb{Z}_{29}^*$ of order $q = 29 - 1 = 28$. Take $g = 2$ and $h = 17$. We set $t = 5$ and compute:

$$2^0 = 1, \quad 2^5 = 3, \quad 2^{10} = 9, \quad 2^{15} = 27, \quad 2^{20} = 23, \quad 2^{25} = 11.$$

(It should be understood that all operations are in $\mathbb{Z}_{29}^*$.) Then compute:

$$17 \cdot 2^1 = 5, \quad 17 \cdot 2^2 = 10, \quad 17 \cdot 2^3 = 20, \quad 17 \cdot 2^4 = 11,$$

and notice that $17 \cdot 2^4 = 11 = 2^{25}$. We thus have $\log_2 17 = 25 - 4 = 21$. $\diamond$

9.2.3 Discrete Logarithms from Collisions

A drawback of the baby-step/giant-step algorithm is that it uses a large amount of memory, as it requires storage of $\mathcal{O}(\sqrt{q})$ points. We can obtain an algorithm that uses constant memory—and has the same asymptotic running time—by exploiting the connection between the discrete-logarithm problem and collision-resistant hashing shown in Section 8.4.2, and recalling the small-space birthday attack for finding collisions from Section 5.4.2.

We describe the high-level idea. Fix a base $g \in \mathbb{G}$ and some element $h \in \langle g \rangle$. Recall from the results of Section 8.4.2 that if we define the hash function $H_{g,h} : \mathbb{Z}_q \times \mathbb{Z}_q \rightarrow \mathbb{G}$ by $H_{g,h}(x_1, x_2) = g^{x_1} h^{x_2}$, then finding a collision in $H_{g,h}$ implies the ability to compute $\log_g h$. (See the proof of Theorem 8.79.) We have thus reduced the problem of computing $\log_g h$ to that of finding a collision in a hash function, something we know how to do in time $\mathcal{O}(\sqrt{|\mathbb{G}|}) = \mathcal{O}(\sqrt{q})$ using a birthday attack! Moreover, a small-space birthday attack will give a collision in the same time and constant space.

It only remains to address a few technical details. One is that the small-space birthday attack described in Section 5.4.2 assumes that the range of the hash function is a subset of its domain; that is not the case here, and in fact (depending on the representation being used for elements of $\mathbb{G}$) it could even be that $H_{g,h}$ is not compressing. A second issue is that the analysis in Section 5.4.2 treated the hash function as a random function, whereas $H_{g,h}$ has a significant amount of algebraic structure.

Pollard's rho algorithm provides one way to deal with the above issues. We describe a different algorithm that can be viewed as a more direct implementation of the above ideas. (In practice, Pollard's algorithm would be more efficient, although both algorithms use only $\mathcal{O}(\sqrt{q})$ group operations.) Let $F : \mathbb{G} \rightarrow \mathbb{Z}_q \times \mathbb{Z}_q$ denote a cryptographic hash function obtained by, e.g., a suitable modification of SHA-1. Define $H : \mathbb{G} \rightarrow \mathbb{G}$ by $H(k) \stackrel{\text{def}}{=} H_{g,h}(F(k))$. We can use Algorithm 5.9, with natural modifications, to find a collision in H using $\mathcal{O}(\sqrt{|\mathbb{G}|}) = \mathcal{O}(\sqrt{q})$ evaluations of H in expectation (and constant memory). With overwhelming probability, this yields a collision in $H_{g,h}$. You are asked to flesh out the details in Exercise 9.6.

It is interesting to observe here that a security proof based on the hardness of the discrete-logarithm problem—namely, that it implies a collision-resistant hash function—leads to a better algorithm for solving that same problem! A little reflection should convince us that this is not surprising: a proof by reduction demonstrates that an attack on some construction (in this case, finding collisions in the hash function) directly yields an attack on the underlying assumption (here, the hardness of the discrete-logarithm problem), which is exactly the property the above algorithm exploits.

9.2.4 The Index Calculus Algorithm

We conclude with a brief look at the (non-generic) *index calculus algorithm* for computing discrete logarithms in the cyclic group $\mathbb{Z}_p^*$ (for p prime). In contrast to the preceding (generic) algorithms, this approach has sub-exponential running time. The algorithm bears some resemblance to the quadratic sieve algorithm introduced in Section 9.1.3, and we assume readers are familiar with the discussion there. As in that case, we discuss the main ideas of the index calculus method but leave a detailed analysis outside the scope of our treatment. Also, some simplifications are introduced to clarify the presentation.

The index calculus method uses a two-step process. Importantly, the first step requires knowledge only of the modulus p and the base g and so it can be run as a preprocessing step before h —the value whose discrete logarithm we wish to compute—is known. For the same reason, it suffices to run the first step only once in order to solve multiple instances of the discrete-logarithm problem (as long as all those instances share the same p and g).

Step 1. Fix some bound B , and let $\{p_1, \dots, p_k\}$ be the set of prime numbers less than or equal to B . In this step, we find $\ell \geq k$ distinct values $x_1, \dots, x_\ell \in \mathbb{Z}_{p-1}$ for which $g_i \stackrel{\text{def}}{=} [g^{x_i} \bmod p]$ is B -smooth. This is done by simply choosing uniform $\{x_i\}$ until suitable values are found.

Factoring the resulting B -smooth numbers, we have the ℓ equations:

$$\begin{aligned} g^{x_1} &= \prod_{i=1}^k p_i^{e_{1,i}} \bmod p \\ &\vdots \\ g^{x_\ell} &= \prod_{i=1}^k p_i^{e_{\ell,i}} \bmod p. \end{aligned}$$

Taking discrete logarithms, we can transform these into the linear equations

$$x_1 = \sum_{i=1}^k e_{1,i} \cdot \log_g p_i \pmod{p-1} \quad (9.3)$$

$\vdots$

$$x_\ell = \sum_{i=1}^k e_{\ell,i} \cdot \log_g p_i \pmod{p-1}. \quad (9.4)$$

Note that the $\{x_i\}$ and the $\{e_{i,j}\}$ are known, while the $\{\log_g p_i\}$ are unknown.

Step 2. Now we are given an element h and want to compute $\log_g h$. Here, we find a value $x \in \mathbb{Z}_{p-1}$ for which $[g^x \cdot h \bmod p]$ is B -smooth. (Once again, this is done simply by choosing x uniformly.) Say

$$\begin{aligned} g^x \cdot h &= \prod_{i=1}^k p_i^{e_i} \bmod p \\ \Rightarrow x + \log_g h &= \sum_{i=1}^k e_i \cdot \log_g p_i \pmod{p-1}, \end{aligned}$$

where x and the $\{e_i\}$ are known. Combined with Equations (9.3)–(9.4), we have $\ell + 1 \geq k + 1$ linear equations in the $k + 1$ unknowns $\{\log_g p_i\}_{i=1}^k$ and $\log_g h$. Using linear-algebraic³ methods (and assuming the system of equations is not under-defined), we can solve for each of the unknowns and in particular obtain the desired solution $\log_g h$.

Example 9.8

Let $p = 101$, $g = 3$, and $h = 87$. We have $[3^{10} \bmod 101] = 65 = 5 \cdot 13$. Similarly, $[3^{12} \bmod 101] = 80 = 2^4 \cdot 5$ and $[3^{14} \bmod 101] = 13$. We thus have the linear equations

$$\begin{aligned} 10 &= \log_3 5 + \log_3 13 \pmod{100} \\ 12 &= 4 \cdot \log_3 2 + \log_3 5 \pmod{100} \\ 14 &= \log_3 13 \pmod{100}. \end{aligned}$$

We also have $3^5 \cdot 87 = 32 = 2^5 \bmod 101$, or

$$5 + \log_3 87 = 5 \cdot \log_3 2 \pmod{100}. \quad (9.5)$$

Adding the second and third equations and subtracting the first, we derive $4 \cdot \log_3 2 = 16 \bmod 100$. This doesn't determine $\log_3 2$ uniquely (since 4 is

³Technically, things are slightly more complicated since the linear equations are all modulo $p - 1$, which is not prime. Nevertheless, there exist techniques for dealing with this.

not invertible modulo 100), but it does tell us that $\log_3 2 = 4, 29, 54$, or 79 (cf. Exercise 9.3). Trying all possibilities gives $\log_3 2 = 29$. Plugging this into Equation (9.5) gives $\log_3 87 = 40$. $\diamond$

Running time. Choosing a larger value of B makes it more likely that a uniform value in $\mathbb{Z}_p^*$ is B -smooth; however, it means we will have to work harder to identify and factor B -smooth numbers, and we will have to find more of them. Because the system of equations will be larger, solving the system will take longer. Choosing the optimal value of B gives an algorithm that (heuristically, at least) computes discrete logarithms in $\mathbb{Z}_p^*$ in time $2^{\mathcal{O}(\sqrt{\log p} \log \log p)}$. (In fact, the constant term in the exponent can be determined quite precisely.) The important point for our purposes is that this is sub-exponential in the length of p .

9.3 Recommended Key Lengths

Knowledge of the best available algorithms for solving various cryptographic problems is essential for determining an appropriate key length to achieve some desired level of security. The following table summarizes the key lengths currently recommended by the US National Institute of Standards and Technology⁴ (NIST) [13]:

Effective Key Length	RSA	Discrete Logarithm	
	Modulus Length	Order- q Subgroup of $\mathbb{Z}_p^*$	Elliptic-Curve Group Order q
112	2048	p : 2048, q : 224	224
128	3072	p : 3072, q : 256	256
192	7680	p : 7680, q : 384	384
256	15360	p : 15360, q : 512	512

All figures in the table are measured in bits. The “effective key length” is a value n such that the best known algorithm for solving a problem takes time roughly 2^n ; i.e., the computational difficulty of solving a problem is approximately equivalent to that of performing a brute-force search against a symmetric-key scheme with an n -bit key, or the time to find collisions in a hash function with a $2n$ -bit output length. NIST deems a 112-bit effective key length acceptable for security until the year 2030, but recommends 128-bit or higher key lengths for applications where security is required beyond then.

⁴Other groups have made their own recommendations; see <http://keylength.com>.

Given what we have learned in this chapter, it is instructive to look more closely at some of the numbers in the table. The first thing we may notice is that elliptic-curve groups can be used to realize any given level of security with smaller parameters than for RSA or subgroups of $\mathbb{Z}_p^*$. This is simply because no sub-exponential algorithms are known for solving the discrete-logarithm problem in such groups (when chosen appropriately). Achieving n -bit security, however, requires an elliptic-curve group whose order q is $2n$ -bits long. This is a consequence of the generic algorithms we have seen in this chapter, which solve the discrete-logarithm problem (in any group) in time $\mathcal{O}(\sqrt{q}) \approx 2^n$.

Turning to the case of $\mathbb{Z}_p^*$ we see that here, too, a $2n$ -bit value of q is needed for n -bit security. The length of p , however, must be significantly larger, because the number field sieve can be used to compute discrete logarithms in $\mathbb{Z}_p^*$ in time sub-exponential in the length of p . (That is, p and q are chosen such that the running time of the number field sieve, which depends on the length of p , and the running time of a generic algorithm, which depends on the length of q , will be approximately equal.) The practical ramifications of this are that, for any desired security level, elliptic-curve cryptosystems can use significantly smaller parameters, with asymptotically faster group operations for the honest parties, than cryptosystems based on subgroups of $\mathbb{Z}_p^*$.

References and Additional Reading

Pollard's $p-1$ algorithm was published in 1974 [139], and his rho method for factoring was described the following year [140]. The quadratic sieve algorithm is due to Pomerance [142], based on earlier ideas of Dixon [60].

The baby-step/giant-step algorithm is due to Shanks [153]. The Pohlig-Hellman algorithm was published in 1978 [138], as was Pollard's rho algorithm for computing discrete logarithms [141]. The index calculus algorithm as we have described it is by Adleman [4].

The texts by Wagstaff [173], Shoup [159], Crandall and Pomerance [51], Joux [96], and Galbraith [69] provide further information on algorithms for factoring and computing discrete logarithms, including descriptions of the (general) number field sieve. Very recently, an improved algorithm for the discrete-logarithm problem in finite fields of small characteristic was announced [11].

Lower bounds on the running time of generic algorithms for computing discrete logarithms, which asymptotically match the running times of the algorithms described here, were given by Nechaev [133] and Shoup [155].

Lenstra and Verheul [113] provide a comprehensive discussion of how known algorithms for factoring and computing discrete logarithms affect the choice of cryptographic parameters in practice.

Exercises

- 9.1 In order to speed up the key-generation algorithm for RSA, it has been suggested to generate a prime by generating many small random primes, multiplying them together, and adding one (of course, then checking that the result is prime). Ignoring the question of the probability that such a value really is prime, what do you think of this method?
- 9.2 In an execution of Algorithm 9.2, define $x^{(i)} \stackrel{\text{def}}{=} F^{(i)}(x^{(0)})$. Show that if, in a given execution of the algorithm, there exist $i, j \leq 2^{n/2}$ such that $x^{(i)} \neq x^{(j)}$ but $x^{(i)} = x^{(j)} \bmod p$, then that execution of the algorithm outputs p . (The analysis is a little different from the analysis of Algorithm 5.9, since the algorithms—and their goals—are slightly different.)
- 9.3 (a) Show that if $ab = c \bmod N$ and $\gcd(b, N) = d$, then:
- $d \mid c$;
 - $a \cdot (b/d) = (c/d) \bmod (N/d)$; and
 - $\gcd(b/d, N/d) = 1$.
- (b) Describe how to use the above to compute discrete logarithms in $\mathbb{Z}_N$ efficiently even when the base g is not a generator of $\mathbb{Z}_N$.
- 9.4 Here we show how to solve the discrete-logarithm problem in a cyclic group of order $q = p^e$ in time $\mathcal{O}(\text{polylog}(q) \cdot \sqrt{p})$. Given as input a generator g of order $q = p^e$ and a value h , we want to compute $x = \log_g h$.
- (a) Show how to compute $[x \bmod p]$ in time $\mathcal{O}(\text{polylog}(q) \cdot \sqrt{p})$.
Hint: Solve the equation
- $$(g^{p^{e-1}})^{x_0} = h^{p^{e-1}}$$
- and use the same ideas as in the Pohlig–Hellman algorithm.
- (b) Say $x = x_0 + x_1 \cdot p + \cdots + x_{e-1} \cdot p^{e-1}$ with $0 \leq x_i < p$. In the previous step we determined x_0 . Show how to compute in $\text{polylog}(q)$ time a value h_1 such that $(g^p)^{x_1 + x_2 \cdot p + \cdots + x_{e-1} \cdot p^{e-2}} = h_1$.
- (c) Use recursion to obtain the claimed running time for the original problem. (Note that $e = \mathcal{O}(\log q)$.)
- 9.5 Let q have prime factorization $q = \prod_{i=1}^k p_i^{e_i}$. Using the result from the previous problem, show a modification of the Pohlig–Hellman algorithm that solves the discrete-logarithm problem in a group of order q in time $\mathcal{O}\left(\text{polylog}(q) \cdot \sum_{i=1}^k e_i \sqrt{p_i}\right) = \mathcal{O}\left(\text{polylog}(q) \cdot \max_i \{\sqrt{p_i}\}\right)$.
- 9.6 Give pseudocode for the small-space algorithm for computing discrete logarithms described in Section 9.2.3, and give a heuristic analysis of the probability with which it succeeds.

Chapter 10

Key Management and the Public-Key Revolution

10.1 Key Distribution and Key Management

In Chapters 1–7 we have seen how private-key cryptography can be used to ensure secrecy and integrity for two parties communicating over an insecure channel, assuming the two parties are in possession of a shared, secret key. The question we have deferred since Chapter 1, however, is:

How can the parties share a secret key in the first place?

Clearly, the key cannot simply be sent over the public communication channel because an eavesdropping adversary would then be able to observe it *en route*. Some other mechanism must be used instead.

In some situations, the parties may have access to a secure channel that they can use to reliably share a secret key. One common example is when the two parties are co-located at some time, at which point they can share a key. Alternatively, the parties might be able to use a trusted courier service as a secure channel. We stress that the fact that the parties can share a key—and thus must have access to a secure channel at some point—does *not* make private-key cryptography useless: in the first example, the parties have a secure channel at one point in time but not indefinitely; in the second example, utilizing the secure channel might be slower and more costly than communicating over an insecure channel.

The above approaches have been used to share keys in government, diplomatic, and military contexts. For example, the “red phone” connecting Moscow and Washington in the 1960s was encrypted using a one-time pad, with keys shared by couriers who flew from one country to the other carrying briefcases full of print-outs. Such approaches can also be used in corporations, e.g., to set up a shared key between a central database and a new employee on his/her first day of work. (We return to this example in the next section.)

Relying on a secure channel to distribute keys, however, does not work well in many other situations. For example, consider a large, multinational corporation in which *every pair* of employees might need the ability to communicate securely, with their communication protected from other employees as well.

It will be inconvenient, to say the least, for each pair of employees to meet so they can securely share a key; for employees working in different cities, this may even be impossible. Even if the current set of employees could somehow share keys with each other, it would be impractical for them to share keys with new employees who join after this initial sharing is done.

Assuming these N employees are somehow able to securely share keys with each other, another significant drawback is that each employee will have to manage and store $N - 1$ secret keys (one for each other employee). In fact, this may significantly under-count the number of keys stored by each user, because employees may also need keys to communicate securely with remote resources such as databases, servers, printers, and so on. The proliferation of so many secret keys is a significant logistical problem. Moreover, all these keys must be stored *securely*. The more keys there are, the harder it is to protect them, and the higher the chance of some keys being stolen by an attacker. Computer systems are often infected by viruses, worms, and other forms of malicious software which can steal secret keys and send them quietly over the network to an attacker. Thus, storing keys on employees' personal computers is not always a safe solution.

To be clear, potential compromise of secret keys is always a concern, irrespective of the number of keys each party holds. When only a few keys need to be stored, however, there are good solutions available for dealing with this threat. A typical solution today is to store keys on *secure hardware* such as a smartcard. A smartcard can carry out cryptographic computations using the stored secret keys, ensuring that these keys never make their way onto users' personal computers. If designed properly, the smartcard can be much more resilient to attack than a personal computer—for example, it typically cannot be infected by malware—and so offers a good means of protecting users' secret keys. Unfortunately, smartcards are typically quite limited in memory and so cannot store hundreds (or thousands) of keys.

The concerns outlined above can all be addressed—in principle, even if not in practice—in “closed” organizations consisting of a well-defined population of users, all of whom are willing to follow the same policies for distributing and storing keys. They break down, however, in “open systems” where users have transient interactions, cannot arrange a physical meeting, and may not even be aware of each other's existence until the time they want to communicate. This is, in fact, a more common situation than one might at first realize: consider using encryption to send credit-card information to an Internet merchant from whom you have not previously purchased anything, or sending email to someone whom you have never met in person. In such cases, private-key cryptography alone simply does not provide a solution, and we must look further for adequate solutions.

To summarize, there are at least three distinct problems related to the use of private-key cryptography. The first is that of *key distribution*; the second is that of *storing and managing large numbers of secret keys*; the third is the *inapplicability of private-key cryptography to open systems*.

10.2 A Partial Solution: Key-Distribution Centers

One way to address some of the concerns from the previous section is to use a *key-distribution center* (KDC) to establish shared keys. Consider again the case of a large corporation where all pairs of employees must be able to communicate securely. In such a setting, we can leverage the fact that all employees may *trust* some entity—say, the system administrator—at least with respect to the security of work-related information. This trusted entity can then act as a KDC and help all the employees share pairwise keys.

When a new employee joins, the KDC can share a key with that employee (in person, in a secure location) as part of that employee's first day of work. At the same time, the KDC could also distribute shared keys between that employee and all existing employees. That is, when the i th employee joins, the KDC could (in addition to sharing a key between itself and this new employee) generate $i - 1$ keys $k_1, \dots, k_{i-1}$, give these keys to the new employee, and then send key k_j to the j th existing employee by encrypting it using the key that employee already shares with the KDC. Following this, the new employee shares a key with every other employee (as well as with the KDC).

A better approach, which avoids requiring employees to store and manage multiple keys, is to utilize the KDC in an *online* fashion to generate keys “on demand” whenever two employees wish to communicate securely. As before, the KDC will share a (different) key with each employee, something that can be done securely on an employee's first day of work. Say the KDC shares key k_A with employee Alice, and k_B with employee Bob. At some later time, when Alice wishes to communicate securely with Bob, she can simply send the message “**I, Alice, want to talk to Bob**” to the KDC. (If desired, this message can be authenticated using the key shared by Alice and the KDC.) The KDC then chooses a new, random key—called a *session key*—and sends this key k to Alice encrypted using k_A , and to Bob encrypted using k_B . (This protocol is too simplistic to be used in practice; see further discussion below.) Once Alice and Bob both recover this session key, they can use it to communicate securely. When they are done with their conversation, they can (and should) erase the session key because they can always contact the KDC again if they wish to communicate at some later time.

Consider the advantages of this approach:

1. Each employee needs to store only *one* long-term secret key (namely, the one they share with the KDC). Employees still need to manage and store session keys, but these are short-term keys that are erased once a communication session concludes.

The KDC needs to store many long-term keys. However, the KDC can be kept in a secure location and be given the highest possible protection against network attacks.

2. When an employee joins the organization, all that must be done is to set up a key between this employee and the KDC. No other employees need to update the set of keys they hold.

Thus, KDCs can alleviate two of the problems we have seen with regard to private-key cryptography: they can simplify key distribution (since only one new key must be shared when an employee joins, and it is reasonable to assume a secure channel between the KDC and that employee on their first day of work), and can reduce the complexity of key storage (since each employee only needs to store a single key). KDCs go a long way toward making private-key cryptography practical in large organizations where there is a single entity who is trusted by everyone.

There are, however, some drawbacks to relying on KDCs:

1. A successful attack on the KDC will result in a complete break of the system: an attacker can compromise all keys and subsequently eavesdrop on all network traffic. This makes the KDC a high-value target. Note that even if the KDC is well-protected against external attacks, there is always the possibility of insider attacks by employees who have access to the KDC (for example, the IT manager).
2. The KDC is a single point of failure: if the KDC is down, secure communication is temporarily impossible. If employees are constantly contacting the KDC and asking for session keys to be established, the load on the KDC can be very high, thereby increasing the chances that it may fail or be slow to respond.

A simple solution to the second problem is to replicate the KDC. This works (and is done in practice), but also means that there are now more points of attack on the system. Adding more KDCs also makes it more difficult to add new employees, since updates must be securely propagated to every KDC.

Protocols for key distribution using a KDC. There are a number of protocols in the literature for secure key distribution using a KDC. We mention in particular the *Needham–Schroeder protocol*, which forms the core of *Kerberos*, an important and widely used service for performing authentication and supporting secure communication. (Kerberos is used in many universities and corporations, and is the default mechanism for supporting secure networked authentication and communication in Windows and many UNIX systems.) We only highlight one feature of this protocol. When Alice contacts the KDC and asks to communicate with Bob, the KDC does not send the encrypted session key to both Alice and Bob as we have described earlier. Instead, the KDC sends to Alice the session key encrypted under Alice’s key *in addition to* the session key encrypted under Bob’s key. Alice then forwards the second ciphertext to Bob as in Figure 10.1. The second ciphertext is sometimes called a *ticket*, and can be viewed as a credential that allows Alice to talk to Bob (and allows Bob to be assured that he is talking to Alice). Indeed,

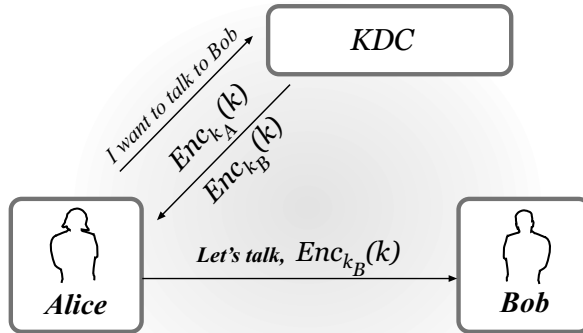


FIGURE 10.1: A general template for key-distribution protocols.

although we have not stressed this point in our discussion, a KDC-based approach can provide a useful means of performing authentication as well. Note also that Alice and Bob need not both be users; Alice might be a user and Bob a resource such as a server, a remote disk, or a printer.

The protocol was designed in this way to reduce the load on the KDC. In the protocol as described, the KDC does not need to initiate a second connection to Bob, and need not worry whether Bob is on-line when Alice initiates the protocol. Moreover, if Alice retains the ticket (and her copy of the session key), then she can re-initiate secure communication with Bob by simply re-sending the ticket to Bob, without the involvement of the KDC at all. (In practice, tickets expire and eventually need to be renewed. But a session could be re-established within some acceptable time period.)

We conclude by noting that in practice the key that Alice shares with the KDC might be a short, easy-to-memorize password. In this case, many additional security problems arise that must be dealt with. We have also been implicitly assuming an attacker who only passively eavesdrops, rather than one who might actively try to interfere with the protocol. We refer the interested reader to the references at the end of this chapter for more information about how such issues can be addressed.

10.3 Key Exchange and the Diffie–Hellman Protocol

KDCs and protocols like Kerberos are commonly used in practice. But these approaches to the key-distribution problem still require, at some point, a *private* and *authenticated* channel that can be used to share keys. (In particular, we assumed the existence of such a channel between the KDC and the employees on their first day.) Thus, they still cannot solve the problem

of key distribution in open systems like the Internet, where there may be no private channel available between two users who wish to communicate.

To achieve private communication without ever communicating over a private channel, a radically different approach is needed. In 1976, Whitfield Diffie and Martin Hellman published a paper with the innocent-looking title “New Directions in Cryptography.” In that work they observed that there is often *asymmetry* in the world; in particular, there are certain actions that can be easily performed but not easily reversed. For example, padlocks can be locked without a key (i.e., easily), but then cannot be reopened. More strikingly, it is easy to shatter a glass vase but extremely difficult to put it back together again. Algorithmically (and more germane for our purposes), it is easy to multiply two large primes but difficult to recover those primes from their product. (This is exactly the factoring problem discussed in previous chapters.) Diffie and Hellman realized that such phenomena could be used to derive interactive protocols for *secure key exchange* that allow two parties to share a secret key, via communication over a public channel, by having the parties perform operations that they can reverse but that an eavesdropper cannot.

The existence of secure key-exchange protocols is quite amazing. It means that you and a friend could agree on a secret by simply shouting across a room (and performing some local computation); the secret would be unknown to anyone else, even if they had listened to everything that was said. Indeed, until 1976 it was generally believed that secure communication could not be achieved without first sharing some secret information using a private channel.

The influence of Diffie and Hellman’s paper was enormous. In addition to introducing a fundamentally new way of looking at cryptography, it was one of the first steps toward moving cryptography out of the private domain and into the public one. We quote the first two paragraphs of their paper:

We stand today on the brink of a revolution in cryptography. The development of cheap digital hardware has freed it from the design limitations of mechanical computing and brought the cost of high grade cryptographic devices down to where they can be used in such commercial applications as remote cash dispensers and computer terminals.

In turn, such applications create a need for new types of cryptographic systems which minimize the necessity of secure key distribution channels. . . . At the same time, theoretical developments in information theory and computer science show promise of providing provably secure cryptosystems, changing this ancient art into a science.

Diffie and Hellman were not exaggerating, and the revolution they spoke of was due in great part to their work.

In this section we present the Diffie–Hellman key-exchange protocol. We prove its security against eavesdropping adversaries or, equivalently, under

the assumption that the parties communicate over a public but *authenticated* channel (so an attacker cannot interfere with their communication). Security against an eavesdropping adversary is a relatively weak guarantee, and in practice key-exchange protocols must satisfy stronger notions of security that are beyond our present scope. (Moreover, we are interested here in the setting where the communicating parties have *no* prior shared information, in which case there is nothing that can be done to prevent an adversary from impersonating one of the parties. We return to this point later.)

The setting and definition of security. We consider a setting with two parties—traditionally called Alice and Bob—who run a probabilistic protocol Π in order to generate a shared, secret key; Π can be viewed as the set of instructions for Alice and Bob in the protocol. Alice and Bob begin by holding the security parameter 1^n ; they then run Π using (independent) random bits. At the end of the protocol, Alice and Bob output keys $k_A, k_B \in \{0, 1\}^n$, respectively. The basic correctness requirement is that $k_A = k_B$. Since we will only deal with protocols that satisfy this requirement, we will speak simply of *the* key $k = k_A = k_B$ generated by an honest execution of Π .

We now turn to defining security. Intuitively, a key-exchange protocol is secure if the key output by Alice and Bob is completely unguessable by an eavesdropping adversary. This is formally defined by requiring that an adversary who has eavesdropped on an execution of the protocol should be unable to distinguish the key k generated by that execution (and now shared by Alice and Bob) from a *uniform* key of length n . This is much stronger than simply requiring that the adversary be unable to *compute* k exactly, and this stronger notion is necessary if the parties will subsequently use k for some cryptographic application (e.g., as a key for a private-key encryption scheme).

Formalizing the above, let Π be a key-exchange protocol, $\mathcal{A}$ an adversary, and n the security parameter. We have the following experiment:

The key-exchange experiment $\text{KE}_{\mathcal{A}, \Pi}^{\text{eav}}(n)$:

1. Two parties holding 1^n execute protocol Π . This results in a transcript trans containing all the messages sent by the parties, and a key k output by each of the parties.
2. A uniform bit $b \in \{0, 1\}$ is chosen. If $b = 0$ set $\hat{k} := k$, and if $b = 1$ then choose $\hat{k} \in \{0, 1\}^n$ uniformly at random.
3. $\mathcal{A}$ is given trans and $\hat{k}$, and outputs a bit b' .
4. The output of the experiment is defined to be 1 if $b' = b$, and 0 otherwise. (In case $\text{KE}_{\mathcal{A}, \Pi}^{\text{eav}}(n) = 1$, we say that $\mathcal{A}$ succeeds.)

$\mathcal{A}$ is given trans to capture the fact that $\mathcal{A}$ eavesdrops on the entire execution of the protocol and thus sees all messages exchanged by the parties. In the real world, $\mathcal{A}$ would not be given any key; in the experiment the adversary is given $\hat{k}$ only as a means of defining what it means for $\mathcal{A}$ to “break” the security of Π .

That is, the adversary succeeds in “breaking” Π if it can correctly determine whether the key $\hat{k}$ is the real key corresponding to the given execution of the protocol, or whether $\hat{k}$ is a uniform key that is independent of the transcript. As expected, we say Π is secure if the adversary succeeds with probability that is at most negligibly greater than $1/2$. That is:

DEFINITION 10.1 *A key-exchange protocol Π is secure in the presence of an eavesdropper if for all probabilistic polynomial-time adversaries $\mathcal{A}$ there is a negligible function negl such that*

$$\Pr [\text{KE}_{\mathcal{A}, \Pi}^{\text{eav}}(n) = 1] \leq \frac{1}{2} + \text{negl}(n).$$

The aim of a key-exchange protocol is almost always to generate a shared key k that will be used by the parties for some further cryptographic purpose, e.g., to encrypt and authenticate their subsequent communication using, say, an authenticated encryption scheme. Intuitively, using a shared key generated by a secure key-exchange protocol should be “as good as” using a key shared over a private channel. It is possible to prove this formally; see Exercise 10.1.

The Diffie–Hellman key-exchange protocol. We now describe the key-exchange protocol that appeared in the original paper by Diffie and Hellman (although they were less formal than we will be here). Let $\mathcal{G}$ be a probabilistic polynomial-time algorithm that, on input 1^n , outputs a description of a cyclic group $\mathbb{G}$, its order q (with $\|\mathbb{G}\| = n$), and a generator $g \in \mathbb{G}$. (See Section 8.3.2.) The Diffie–Hellman key-exchange protocol is described formally as Construction 10.2 and illustrated in Figure 10.2.

CONSTRUCTION 10.2

- **Common input:** The security parameter 1^n
- **The protocol:**
 1. Alice runs $\mathcal{G}(1^n)$ to obtain $(\mathbb{G}, q, g)$.
 2. Alice chooses a uniform $x \in \mathbb{Z}_q$, and computes $h_A := g^x$.
 3. Alice sends $(\mathbb{G}, q, g, h_A)$ to Bob.
 4. Bob receives $(\mathbb{G}, q, g, h_A)$. He chooses a uniform $y \in \mathbb{Z}_q$, and computes $h_B := g^y$. Bob sends h_B to Alice and outputs the key $k_B := h_A^y$.
 5. Alice receives h_B and outputs the key $k_A := h_B^x$.

The Diffie–Hellman key-exchange protocol.

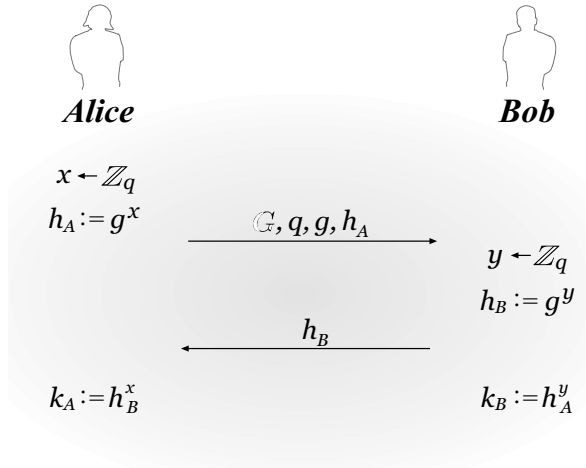


FIGURE 10.2: The Diffie–Hellman key-exchange protocol.

In our description, we have assumed that Alice generates $(\mathbb{G}, q, g)$ and sends these parameters to Bob as part of her first message. In practice, these parameters are *standardized* and are fixed and known to both parties before the protocol begins. In that case Alice need only send h_A , and Bob need not wait to receive Alice’s message before computing and sending h_B .

It is not hard to see that the protocol is correct: Bob computes the key

$$k_B = h_A^y = (g^x)^y = g^{xy}$$

and Alice computes the key

$$k_A = h_B^x = (g^y)^x = g^{xy},$$

and so $k_A = k_B$. (The observant reader will note that the shared key is a group element, not a bit-string. We will return to this point later.)

Diffie and Hellman did not prove security of their protocol; indeed, the appropriate notions (both the definitional framework as well as the idea of formulating precise assumptions) were not yet in place. Let us see what sort of assumption will be needed in order for the protocol to be secure. A first observation, made by Diffie and Hellman, is that a *minimal* requirement for security here is that the discrete-logarithm problem be hard relative to $\mathcal{G}$. If not, then an adversary given the transcript (which, in particular, includes h_A) can compute the secret value of one of the parties (i.e., x) and then easily compute the shared key using that value. So, hardness of the discrete-logarithm problem is necessary for the protocol to be secure. It is not, however, sufficient, as it is possible that there are other ways of computing the key $k_A = k_B$ without explicitly computing x or y . The *computational* Diffie–Hellman assumption—which would only guarantee that the key g^{xy} is

hard to compute in its *entirety* from the transcript—does not suffice either. What is required by Definition 10.1 is that the shared key g^{xy} should be *indistinguishable from uniform* for any adversary given g , g^x , and g^y . This is exactly the *decisional* Diffie–Hellman assumption introduced in Section 8.3.2.

As we will see, a proof of security for the protocol follows almost immediately from the decisional Diffie–Hellman assumption. This should not be surprising, as the Diffie–Hellman assumptions were introduced—well after Diffie and Hellman published their paper—as a way of abstracting the properties underlying the (conjectured) security of the Diffie–Hellman protocol. Given this, it is fair to ask whether anything is gained by defining and proving security here. By this point in the book, hopefully you are convinced the answer is *yes*. Precisely defining secure key exchange forces us to think about exactly what security properties we require; specifying a precise assumption (namely, the decisional Diffie–Hellman assumption) means we can study this assumption independently of any particular application and—once we are convinced of its plausibility—construct other protocols based on it; finally, proving security shows that the assumption does, indeed, suffice for the protocol to meet our desired notion of security.

In our proof of security, we use a modified version of Definition 10.1 in which it is required that the shared key be indistinguishable from a *uniform element* of $\mathbb{G}$ rather than from a *uniform n -bit string*. This discrepancy will need to be addressed before the protocol can be used in practice—after all, group elements are not typically useful as cryptographic keys, and the representation of a uniform group element will not, in general, be a uniform bit-string—and we briefly discuss one standard way to do so following the proof. For now, we let $\widehat{\text{KE}}_{\mathcal{A},\Pi}^{\text{eav}}(n)$ denote a modified experiment where if $b = 1$ the adversary is given $\hat{k}$ chosen uniformly from $\mathbb{G}$ instead of a uniform n -bit string.

THEOREM 10.3 *If the decisional Diffie–Hellman problem is hard relative to $\mathcal{G}$, then the Diffie–Hellman key-exchange protocol Π is secure in the presence of an eavesdropper (with respect to the modified experiment $\widehat{\text{KE}}_{\mathcal{A},\Pi}^{\text{eav}}$).*

PROOF Let $\mathcal{A}$ be a PPT adversary. Since $\Pr[b = 0] = \Pr[b = 1] = 1/2$, we have

$$\begin{aligned} \Pr \left[\widehat{\text{KE}}_{\mathcal{A},\Pi}^{\text{eav}}(n) = 1 \right] \\ = \frac{1}{2} \cdot \Pr \left[\widehat{\text{KE}}_{\mathcal{A},\Pi}^{\text{eav}}(n) = 1 \mid b = 0 \right] + \frac{1}{2} \cdot \Pr \left[\widehat{\text{KE}}_{\mathcal{A},\Pi}^{\text{eav}}(n) = 1 \mid b = 1 \right]. \end{aligned}$$

In experiment $\widehat{\text{KE}}_{\mathcal{A},\Pi}^{\text{eav}}(n)$ the adversary $\mathcal{A}$ receives $(\mathbb{G}, q, g, h_A, h_B, \hat{k})$, where $(\mathbb{G}, q, g, h_A, h_B)$ represents the transcript of the protocol execution, and $\hat{k}$ is either the actual key computed by the parties (if $b = 0$) or a uniform group element (if $b = 1$). Distinguishing between these two cases is exactly

equivalent to solving the decisional Diffie–Hellman problem. That is

$$\begin{aligned}
& \Pr \left[\widehat{\text{KE}}_{\mathcal{A}, \Pi}^{\text{eav}}(n) = 1 \right] \\
&= \frac{1}{2} \cdot \Pr \left[\widehat{\text{KE}}_{\mathcal{A}, \Pi}^{\text{eav}}(n) = 1 \mid b = 0 \right] + \frac{1}{2} \cdot \Pr \left[\widehat{\text{KE}}_{\mathcal{A}, \Pi}^{\text{eav}}(n) = 1 \mid b = 1 \right] \\
&= \frac{1}{2} \cdot \Pr[\mathcal{A}(\mathbb{G}, g, q, g^x, g^y, g^{xy}) = 0] + \frac{1}{2} \cdot \Pr[\mathcal{A}(\mathbb{G}, q, g, g^x, g^y, g^z) = 1] \\
&= \frac{1}{2} \cdot \left(1 - \Pr[\mathcal{A}(\mathbb{G}, g, q, g^x, g^y, g^{xy}) = 1] \right) + \frac{1}{2} \cdot \Pr[\mathcal{A}(\mathbb{G}, q, g, g^x, g^y, g^z) = 1] \\
&= \frac{1}{2} + \frac{1}{2} \cdot \left(\Pr[\mathcal{A}(\mathbb{G}, g, q, g^x, g^y, g^z) = 1] - \Pr[\mathcal{A}(\mathbb{G}, q, g, g^x, g^y, g^{xy}) = 1] \right) \\
&\leq \frac{1}{2} + \frac{1}{2} \cdot \left| \Pr[\mathcal{A}(\mathbb{G}, g, q, g^x, g^y, g^z) = 1] - \Pr[\mathcal{A}(\mathbb{G}, q, g, g^x, g^y, g^{xy}) = 1] \right|,
\end{aligned}$$

where the probabilities are all taken over $(\mathbb{G}, q, g)$ output by $\mathcal{G}(1^n)$, and uniform choice of $x, y, z \in \mathbb{Z}_q$. (Note that since g is a generator, g^z is a uniform element of $\mathbb{G}$ when z is uniformly distributed in $\mathbb{Z}_q$.) If the decisional Diffie–Hellman assumption is hard relative to $\mathcal{G}$, that exactly means that there is a negligible function negl for which

$$\left| \Pr[\mathcal{A}(\mathbb{G}, g, q, g^x, g^y, g^z) = 1] - \Pr[\mathcal{A}(\mathbb{G}, q, g, g^x, g^y, g^{xy}) = 1] \right| \leq \text{negl}(n).$$

We conclude that

$$\Pr \left[\widehat{\text{KE}}_{\mathcal{A}, \Pi}^{\text{eav}}(n) = 1 \right] \leq \frac{1}{2} + \frac{1}{2} \cdot \text{negl}(n),$$

completing the proof. ■

Uniform group elements vs. uniform bit-strings. The previous theorem shows that the key output by Alice and Bob in the Diffie–Hellman protocol is indistinguishable (for a polynomial-time eavesdropper) from a uniform group element. In order to use the key for subsequent cryptographic applications—as well as to meet Definition 10.1—the key output by the parties should instead be indistinguishable from a uniform bit-string of the appropriate length. The Diffie–Hellman protocol can be modified to achieve this by having the parties apply an appropriate *key-derivation function* (cf. Section 5.6.4) to the shared group element g^{xy} they each compute.

Active adversaries. So far we have considered only an eavesdropping adversary. Although eavesdropping attacks are by far the most common (as they are the easiest to carry out), they are by no means the only possible attack. *Active* attacks, in which the adversary sends messages of its own to one or both of the parties, are also a concern, and any protocol used in practice must be resilient to such attacks as well. When considering active attacks, it is useful to distinguish, informally, between *impersonation* attacks where

the adversary impersonates one party while interacting with the other party, and *man-in-the-middle* attacks where both honest parties are executing the protocol and the adversary is intercepting and modifying messages being sent from one party to the other. We will not formally define security against either class of attacks, as such definitions are rather involved and cannot be achieved without the parties sharing *some* information in advance. Nevertheless, it is worth remarking that the Diffie–Hellman protocol is *completely insecure* against man-in-the-middle attacks. In fact, a man-in-the-middle adversary can act in such a way that Alice and Bob terminate the protocol with different keys k_A and k_B that are both known to the adversary, yet neither Alice nor Bob can detect that any attack was carried out. We leave the details of this attack as an exercise.

Diffie–Hellman key exchange in practice. The Diffie–Hellman protocol in its basic form is typically not used in practice due to its insecurity against man-in-the-middle attacks, as discussed above. This does not detract in any way from its importance. The Diffie–Hellman protocol served as the first demonstration that asymmetric techniques (and number-theoretic problems) could be used to alleviate the problems of key distribution in cryptography. Furthermore, the Diffie–Hellman protocol is at the core of standardized key-exchange protocols that are resilient to man-in-the-middle attacks and are in wide use today. One notable example is TLS; see Section 12.8.

10.4 The Public-Key Revolution

In addition to key exchange, Diffie and Hellman also introduced in their ground-breaking work the notion of *public-key* (or *asymmetric*) cryptography. In the public-key setting (in contrast to the private-key setting we have studied in Chapters 1–7), a party who wishes to communicate securely generates a *pair* of keys: a *public key* that is widely disseminated, and a *private key* that it keeps secret. (The fact that there are now two different keys is what makes the scheme asymmetric.) Having generated these keys, a party can use them to ensure secrecy for messages it receives using a *public-key encryption scheme*, or integrity for messages it sends using a *digital signature scheme*. (See Figure 10.3.) We provide a brief taste of these primitives here, and discuss them in extensive detail in Chapters 11 and 12, respectively.

In a public-key encryption scheme, the public key generated by some party serves as an *encryption key*; anyone who knows that public key can use it to encrypt messages and generate corresponding ciphertexts. The private key serves as a *decryption key* and is used by the party who knows it to recover the original message from any ciphertext generated using the matching public key. Furthermore—and it is amazing that something like this exists!—

	Private-Key Setting	Public-Key Setting
Secrecy	Private-key encryption	Public-key encryption
Integrity	Message authentication codes	Digital signature schemes

FIGURE 10.3: Cryptographic primitives in the private-key and the public-key settings.

the secrecy of encrypted messages is preserved *even against an adversary who knows the encryption key* (but not the decryption key). In other words, the (public) encryption key is of no use for an attacker trying to decrypt ciphertexts encrypted using that key. To enable secret communication, then, a receiver can simply send her public key to a potential sender (without having to worry about an eavesdropping adversary who observes her public key), or publicize her public key on her webpage or in some central database. A public-key encryption scheme thus enables private communication without relying on a private channel for key distribution.¹

A digital signature scheme is a public-key analogue of message authentication codes (MACs). Here, the private key serves as an “authentication key” (more typically called a *signing key*) that enables the party who knows this key to generate “authentication tags” (i.e., *signatures*) for messages it sends. The public key acts as a *verification key*, allowing anyone who knows it to verify signatures issued by the sender. As with MACs, a digital signature scheme can be used to prevent undetected tampering of a message. The fact that verification can be done by anyone who knows the public key of the sender, however, turns out to have far-reaching ramifications. Specifically, it makes it possible to take a document that was signed by Alice and present it to a third party (say, a judge) as proof that Alice indeed signed the document. This property is called *non-repudiation* and has extensive applications in electronic commerce. For example, it is possible to digitally sign contracts, send signed electronic purchase orders or promises of payments, and so on. Digital signatures are also used for the secure distribution of public keys as part of a *public-key infrastructure*, as discussed in more detail in Section 12.7.

In their paper, Diffie and Hellman set forth the notion of public-key cryptography but did not give any candidate constructions. A year later, Ron Rivest, Adi Shamir, and Len Adleman proposed the *RSA problem* and presented the first public-key encryption and digital signature schemes based on the hardness of this problem. Variants of their schemes are now among the most widely used cryptographic primitives today. In 1985, Taher El Gamal presented an encryption scheme that is essentially a slight twist on the Diffie–Hellman key-exchange protocol, and is now also widely used. Thus, although

¹For now, however, we do assume an *authenticated* channel that allows the sender to obtain a legitimate copy of the receiver’s public key. In Section 12.7 we show how public-key cryptography can be used to reduce this assumption as well.

Diffie and Hellman did not succeed in constructing a (non-interactive) public-key encryption scheme, they came very close.

We close by summarizing how public-key cryptography addresses the limitations of the private-key setting discussed in Section 10.1:

1. Public-key encryption allows key distribution to be done over public (but authenticated) channels. This can simplify the distribution and updating of key material.
2. Public-key cryptography reduces the need for users to store many secret keys. Consider again the setting of a large corporation where each pair of employees needs the ability to communicate securely. Using public-key cryptography, it suffices for each employee to store just a *single* private key (their own) and the public keys of all other employees. Importantly, these latter keys do *not* need to be stored in secret; they could even be stored in some central (public) repository.
3. Finally, public-key cryptography is (more) suitable for open environments where parties who have never previously interacted want the ability to communicate securely. As one commonplace example, an Internet merchant can post their public key on-line; a user making a purchase can then obtain the merchant's public key, as needed, when they need to encrypt their credit card information.

The invention of public-key encryption was a revolution in cryptography. It is no coincidence that until the late 1970s and early 1980s, encryption and cryptography in general belonged to the domain of intelligence and military organizations, and only with the advent of public-key techniques did the use of cryptography spread to the masses.

References and Additional Reading

We have only briefly discussed the problems of key distribution and key management. For more information, we recommend looking at textbooks on network security. Kaufman et al. [102] provide a good treatment of protocols for secure key distribution, what they aim to achieve, and how they work.

We have not made any attempt to capture the full history of the development of public-key cryptography. Others besides Diffie and Hellman were working on similar ideas in the 1970s. One researcher in particular doing similar and independent work was Ralph Merkle, considered by many to be a co-inventor of public-key cryptography (although he published after Diffie and Hellman). We also mention Michael Rabin, who developed constructions of signature schemes and public-key encryption schemes based on the hardness of factoring about one year after the work of Rivest, Shamir, and Adleman [148].

We highly recommend reading the original paper by Diffie and Hellman [58], and refer the reader to the book by Levy [114] for more on the political and historical aspects of the public-key revolution.

Interestingly, aspects of public-key cryptography were discovered in the intelligence community before being published in the open scientific literature. In the early 1970s, James Ellis, Clifford Cocks, and Malcolm Williamson of the British intelligence agency GCHQ invented the notion of public-key cryptography, a variant of RSA encryption, and a variant of the Diffie–Hellman key-exchange protocol. Their work was not declassified until 1997. Although the underlying mathematics of public-key cryptography may have been discovered before 1976, it is fair to say that the widespread ramifications of this new technology were not appreciated until Diffie and Hellman came along.

Exercises

- 10.1 Let Π be a key-exchange protocol, and (Enc, Dec) be a private-key encryption scheme. Consider the following *interactive* protocol Π' for encrypting a message: first, the sender and receiver run Π to generate a shared key k . Next, the sender computes $c \leftarrow \text{Enc}_k(m)$ and sends c to the other party, who decrypts and recovers m using k .
 - (a) Formulate a definition of indistinguishable encryptions in the presence of an eavesdropper (cf. Definition 3.8) appropriate for this interactive setting.
 - (b) Prove that if Π is secure in the presence of an eavesdropper and (Enc, Dec) has indistinguishable encryptions in the presence of an eavesdropper, then Π' satisfies your definition.
- 10.2 Show that, for either of the groups considered in Sections 8.3.3 or 8.3.4, a uniform group element (expressed using the natural representation) is easily distinguishable from a uniform bit-string of the same length.
- 10.3 Describe a man-in-the-middle attack on the Diffie–Hellman protocol where the adversary shares a key k_A with Alice and a (different) key k_B with Bob, and Alice and Bob cannot detect that anything is wrong.
- 10.4 Consider the following key-exchange protocol:
 - (a) Alice chooses uniform $k, r \in \{0, 1\}^n$, and sends $s := k \oplus r$ to Bob.
 - (b) Bob chooses uniform $t \in \{0, 1\}^n$, and sends $u := s \oplus t$ to Alice.
 - (c) Alice computes $w := u \oplus r$ and sends w to Bob.
 - (d) Alice outputs k and Bob outputs $w \oplus t$.

Show that Alice and Bob output the same key. Analyze the security of the scheme (i.e., either prove its security or show a concrete attack).

Chapter 11

Public-Key Encryption

11.1 Public-Key Encryption – An Overview

The introduction of public-key encryption marked a revolution in cryptography. Until that time, cryptographers had relied exclusively on shared, *secret* keys to achieve private communication. Public-key techniques, in contrast, enable parties to communicate privately without having agreed on *any* secret information in advance. As we have already noted, it is quite amazing and counterintuitive that this is possible: it means that two people on opposite sides of a room who can only communicate by shouting to each other, and have no initial secret, can talk in such a way that no one else in the room learns anything about what they are saying!

In the setting of private-key encryption, two parties agree on a secret key that can be used, by either party, for both encryption and decryption. Public-key encryption is *asymmetric* in both these respects. One party (the *receiver*) generates a *pair* of keys (pk, sk) , called the *public key* and the *private key*, respectively. The public key is used by a *sender* to encrypt a message; the receiver uses the private key to decrypt the resulting ciphertext.

Since the goal is to avoid the need for two parties to meet in advance to agree on any information, how does the sender learn pk ? At an abstract level, there are two ways this can occur. Call the receiver Alice and the sender Bob. In the first approach, when Alice learns that Bob wants to communicate with her, she can at that point generate (pk, sk) (assuming she hasn't done so already) and then send pk to Bob *in the clear*; Bob can then use pk to encrypt his message. We emphasize that the channel between Alice and Bob may be public, but is assumed to be authenticated, meaning that the adversary cannot modify the public key sent by Alice to Bob (and, in particular, cannot replace it with its own key). See Section 12.7 for a discussion of how public keys can be distributed over unauthenticated channels.

An alternative approach is for Alice to generate her keys (pk, sk) in advance, *independently* of any particular sender. (In fact, at the time of key generation Alice need not even be aware that Bob wants to talk to her, or even that Bob exists.) Alice can widely disseminate her public key pk by, say, publishing it on her webpage, putting it on her business cards, or placing it in a public directory. Now, *anyone* who wishes to communicate privately with Alice can

look up her public key and proceed as above. Note that multiple senders can communicate multiple times with Alice using the same public key pk for encrypting all their communication.

Note that pk is inherently public—and can thus be learned easily by an attacker—in either of the above scenarios. In the first case, an adversary eavesdropping on the communication between Alice and Bob obtains pk directly; in the second case, an adversary could just as well look up Alice's public key on its own. We see that the security of public-key encryption cannot rely on secrecy of pk , but must instead rely on secrecy of sk . It is therefore crucial that Alice not reveal her private key to anyone, including the sender Bob.

Comparison to Private-Key Encryption

Perhaps the most obvious difference between private- and public-key encryption is that the former assumes *complete* secrecy of all cryptographic keys, whereas the latter requires secrecy for only the private key sk . Although this may seem like a minor distinction, the ramifications are huge: in the private-key setting the communicating parties must somehow be able to share the secret key without allowing any third party to learn it, whereas in the public-key setting the public key can be sent from one party to the other over a public channel without compromising security. For parties shouting across a room or, more realistically, communicating over a public network like a phone line or the Internet, public-key encryption is the only option.

Another important distinction is that private-key encryption schemes use the same key for both encryption and decryption, whereas public-key encryption schemes use different keys for each operation. That is, public-key encryption is inherently *asymmetric*. This asymmetry in the public-key setting means that the roles of sender and receiver are *not* interchangeable as they are in the private-key setting: a single key-pair allows communication in one direction only. (Bidirectional communication can be achieved in a number of ways; the point is that a single invocation of a public-key encryption scheme forces a distinction between one user who acts as a receiver and other users who act as senders.) In addition, a single instance of a public-key encryption scheme enables multiple senders to communicate privately with a single receiver, in contrast to the private-key case where a secret key shared between two parties enables private communication only between those two parties.

Summarizing and elaborating the preceding discussion, we see that public-key encryption has the following advantages relative to private-key encryption:

- Public-key encryption addresses (to some extent) the *key-distribution problem*, since communicating parties do not need to secretly share a key in advance of their communication. Two parties can communicate secretly even if *all* communication between them is monitored.
- When a single receiver is communicating with N senders (e.g., an on-line merchant processing credit card orders from multiple purchasers), it is

much more convenient for the receiver to store a *single* private key sk rather than to share, store, and manage N different secret keys (i.e., one for each sender). In fact, when using public-key encryption the number and identities of potential senders need not be known at the time of key generation. This allows enormous flexibility in “open systems.”

The fact that public-key encryption schemes allow anyone to act as a sender can be a drawback when a receiver only wants to receive messages from one specific individual. In that case, an authenticated (private-key) encryption scheme would be a better choice than public-key encryption.

The main disadvantage of public-key encryption is that it is roughly *2 to 3 orders of magnitude* slower than private-key encryption.¹ It can be a challenge to implement public-key encryption in severely resource-constrained devices like smartcards or radio-frequency identification (RFID) tags. Even when a desktop computer is performing cryptographic operations, carrying out thousands of such operations per second (as in the case of an on-line merchant processing credit card transactions) may be prohibitive. Thus, when private-key encryption is an option (i.e., if two parties *can* securely share a key in advance), then it typically should be used.

In fact, as we will see in Section 11.3, private-key encryption is used *in the public-key setting* to improve efficiency for the (public-key) encryption of long messages. A thorough understanding of private-key encryption is therefore crucial to appreciate how public-key encryption is implemented in practice.

Secure Distribution of Public Keys

In our entire discussion thus far, we have implicitly assumed that the adversary is *passive*; that is, the adversary only eavesdrops on communication between the sender and receiver but does not *actively* interfere with the communication. If the adversary has the ability to tamper with *all* communication between the honest parties, and these honest parties share no keys in advance, then privacy simply cannot be achieved. For example, if a receiver Alice sends her public key pk to Bob but the adversary replaces it with a key pk' of his own (for which it knows the matching private key sk'), then even though Bob encrypts his message using pk' the adversary will easily be able to recover the message (using sk'). A similar attack works if an adversary is able to change the value of Alice's public key that is stored in some public directory, or if the adversary can tamper with the public key as it is transmitted from the public directory to Bob. If Alice and Bob do not share any information in advance, and are not willing to rely on some mutually trusted third party, there is nothing Alice or Bob can do to prevent active attacks of this sort, or

¹It is difficult to give an exact comparison since the relative efficiency depends on the exact schemes under consideration as well as various implementation details.

even to tell that such an attack is taking place.²

Importantly, our treatment of public-key encryption in this chapter *assumes that senders are able to obtain a legitimate copy of the receiver's public key*. (This will be implicit in the security definitions we provide.) That is, we assume *secure key distribution*. This assumption is made not because active attacks of the type discussed above are of no concern—in fact, they represent a serious threat that must be dealt with in any real-world system that uses public-key encryption. Rather, this assumption is made because there exist other mechanisms for preventing active attacks (see, for example, Section 12.7), and it is therefore convenient (and useful) to decouple the study of secure public-key encryption from the study of secure public-key distribution.

11.2 Definitions

We begin by defining the syntax of public-key encryption. The definition is very similar to Definition 3.7, with the exception that instead of working with just one key, we now have distinct encryption and decryption keys.

DEFINITION 11.1 *A public-key encryption scheme is a triple of probabilistic polynomial-time algorithms $(\text{Gen}, \text{Enc}, \text{Dec})$ such that:*

1. *The key-generation algorithm Gen takes as input the security parameter 1^n and outputs a pair of keys (pk, sk) . We refer to the first of these as the **public key** and the second as the **private key**. We assume for convenience that pk and sk each has length at least n , and that n can be determined from pk, sk .*
2. *The encryption algorithm Enc takes as input a public key pk and a message m from some message space (that may depend on pk). It outputs a ciphertext c , and we write this as $c \leftarrow \text{Enc}_{pk}(m)$. (Looking ahead, Enc will need to be probabilistic to achieve meaningful security.)*
3. *The deterministic decryption algorithm Dec takes as input a private key sk and a ciphertext c , and outputs a message m or a special symbol $\perp$ denoting failure. We write this as $m := \text{Dec}_{sk}(c)$.*

It is required that, except possibly with negligible probability over (pk, sk) output by $\text{Gen}(1^n)$, we have $\text{Dec}_{sk}(\text{Enc}_{pk}(m)) = m$ for any (legal) message m .

The important difference from the private-key setting is that the key-generation algorithm Gen now outputs *two* keys instead of one. The public

²In our “shouting-across-a-room” scenario, Alice and Bob can detect when an adversary interferes with the communication. But this is only because: (1) the adversary cannot prevent Alice’s messages from reaching Bob, and (2) Alice and Bob “share” in advance information (e.g., the sound of their voices) that allows them to “authenticate” their communication.

key pk is used for encryption, while the private key sk is used for decryption. Reiterating our earlier discussion, pk is assumed to be widely distributed so that anyone can encrypt messages for the party who generated this key, but sk must be kept private by the receiver in order for security to possibly hold.

We allow for a negligible probability of decryption error and, indeed, some of the schemes we present will have a negligible error probability (e.g., if a prime needs to be chosen but with negligible probability a composite is obtained instead). Despite this, we will generally ignore the issue from here on.

For practical usage of public-key encryption, we will want the message space to be $\{0, 1\}^n$ or $\{0, 1\}^*$ (and, in particular, to be independent of the public key). Although we will sometimes describe encryption schemes using some message space $\mathcal{M}$ that does not contain all bit-strings of some fixed length (and that may also depend on the public key), we will in such cases also specify how to encode bit-strings as elements of $\mathcal{M}$. This encoding must be both efficiently computable and efficiently reversible, so the receiver can recover the bit-string that was encrypted.

11.2.1 Security against Chosen-Plaintext Attacks

We initiate our treatment of security by introducing the “natural” counterpart of Definition 3.8 in the public-key setting. Since extensive motivation for this definition (as well as the others we will see) has already been given in Chapter 3, the discussion here will be relatively brief and will focus primarily on the differences between the private-key and the public-key settings.

Given a public-key encryption scheme $\Pi = (\text{Gen}, \text{Enc}, \text{Dec})$ and an adversary $\mathcal{A}$, consider the following experiment:

The eavesdropping indistinguishability experiment $\text{PubK}_{\mathcal{A}, \Pi}^{\text{eav}}(n)$:

1. $\text{Gen}(1^n)$ is run to obtain keys (pk, sk) .
2. Adversary $\mathcal{A}$ is given pk , and outputs a pair of equal-length messages m_0, m_1 in the message space.
3. A uniform bit $b \in \{0, 1\}$ is chosen, and then a ciphertext $c \leftarrow \text{Enc}_{pk}(m_b)$ is computed and given to $\mathcal{A}$. We call c the challenge ciphertext.
4. $\mathcal{A}$ outputs a bit b' . The output of the experiment is 1 if $b' = b$, and 0 otherwise. If $b' = b$ we say that $\mathcal{A}$ succeeds.

DEFINITION 11.2 A public-key encryption scheme $\Pi = (\text{Gen}, \text{Enc}, \text{Dec})$ has indistinguishable encryptions in the presence of an eavesdropper if for all probabilistic polynomial-time adversaries $\mathcal{A}$ there is a negligible function negl such that

$$\Pr[\text{PubK}_{\mathcal{A}, \Pi}^{\text{eav}}(n) = 1] \leq \frac{1}{2} + \text{negl}(n).$$

The main difference between the above definition and Definition 3.8 is that here $\mathcal{A}$ is given the public key pk . Furthermore, we allow $\mathcal{A}$ to choose its messages m_0 and m_1 based on this public key. This is essential when defining security of public-key encryption since, as discussed previously, we assume that the adversary knows the public key of the recipient.

The seemingly “minor” modification of giving the adversary $\mathcal{A}$ the public key pk being used to encrypt the message has a tremendous impact: it effectively gives $\mathcal{A}$ access to an encryption oracle *for free*. (The concept of an encryption oracle is explained in Section 3.4.2.) This is true because the adversary, given pk , can encrypt any message m on its own by simply computing $\text{Enc}_{pk}(m)$. (As always, $\mathcal{A}$ is assumed to know the algorithm Enc .) The upshot is that Definition 11.2 is *equivalent* to CPA-security (i.e., security against chosen-plaintext attacks), defined in a manner analogous to Definition 3.22, with the only difference being that the attacker is given the public key in the corresponding experiment. We thus have:

PROPOSITION 11.3 *If a public-key encryption scheme has indistinguishable encryptions in the presence of an eavesdropper, it is CPA-secure.*

This is in contrast to the private-key setting, where there exist schemes that have indistinguishable encryptions in the presence of an eavesdropper but are insecure under a chosen-plaintext attack (see Proposition 3.20). Further differences from the private-key setting that follow almost immediately as consequences of the above are discussed next.

Impossibility of perfectly secret public-key encryption. *Perfectly secret* public-key encryption could be defined analogously to Definition 2.3 by conditioning on the entire view of an eavesdropper (i.e., including the public key). Equivalently, it could be defined by extending Definition 11.2 to require that for *all* adversaries $\mathcal{A}$ (not only efficient ones), we have:

$$\Pr[\text{PubK}_{\mathcal{A},\Pi}^{\text{eav}}(n) = 1] = \frac{1}{2}.$$

In contrast to the private-key setting, however, perfectly secret public-key encryption is *impossible*, regardless of how long the keys are or how small the message space is. In fact, an unbounded adversary given pk and a ciphertext c computed via $c \leftarrow \text{Enc}_{pk}(m)$ can determine m with probability 1. A proof of this is left as Exercise 11.1.

Insecurity of deterministic public-key encryption. As noted in the context of private-key encryption, no deterministic encryption scheme can be CPA-secure. The same is true here:

THEOREM 11.4 *No deterministic public-key encryption scheme is CPA-secure.*

Because Theorem 11.4 is so important, it merits a bit more discussion. The theorem is *not* an “artifact” of our security definition, or an indication that our definition is too strong. Deterministic public-key encryption schemes are vulnerable to *practical* attacks in *realistic* scenarios and should never be used. The reason is that a deterministic scheme not only allows the adversary to determine when the same message is sent twice (as in the private-key setting), but also allows the adversary to recover the message, with probability 1, if the set of possible messages being encrypted is small. For example, consider a professor encrypting students’ grades. Here, an eavesdropper knows that each student’s grade must be one of $\{A, B, C, D, F\}$. If the professor uses a deterministic public-key encryption scheme, an eavesdropper can quickly determine any student’s actual grade by encrypting all possible grades and comparing the result to the given ciphertext.

Although the above theorem seems deceptively simple, for a long time *many real-world systems were designed using deterministic public-key encryption*. When public-key encryption was introduced, it is fair to say that the importance of probabilistic encryption was not yet fully realized. The seminal work of Goldwasser and Micali, in which (something equivalent to) Definition 11.2 was proposed and Theorem 11.4 was stated, marked a turning point in the field of cryptography. The importance of pinning down one’s intuition in a formal definition and looking at things the right way for the first time—even if seemingly simple in retrospect—should not be underestimated.

11.2.2 Multiple Encryptions

As in Chapter 3, it is important to understand the effect of using the same key (in this case, the same public key) for encrypting multiple messages. We could formulate security in such a setting by having an adversary output two *lists* of plaintexts, as in Definition 3.19. For the reasons discussed in Section 3.4.2, however, we choose instead to use a definition in which the attacker is given access to a “left-or-right” oracle $\text{LR}_{pk,b}$ that, on input a pair of equal-length messages m_0, m_1 , computes the ciphertext $c \leftarrow \text{Enc}_{pk}(m_b)$ and returns c . The attacker is allowed to query this oracle as many times as it likes, and the definition therefore models security when multiple (unknown) messages are encrypted using the same public key.

Formally, consider the following experiment defined for a public-key encryption scheme $\Pi = (\text{Gen}, \text{Enc}, \text{Dec})$ and adversary $\mathcal{A}$:

The LR-oracle experiment $\text{PubK}_{\mathcal{A}, \Pi}^{\text{LR-cpa}}(n)$:

1. $\text{Gen}(1^n)$ is run to obtain keys (pk, sk) .
2. A uniform bit $b \in \{0, 1\}$ is chosen.
3. The adversary $\mathcal{A}$ is given input pk and oracle access to $\text{LR}_{pk,b}(\cdot, \cdot)$.
4. The adversary $\mathcal{A}$ outputs a bit b' .
5. The output of the experiment is defined to be 1 if $b' = b$, and 0 otherwise. If $\text{PubK}_{\mathcal{A}, \Pi}^{\text{LR-cpa}}(n) = 1$, we say that $\mathcal{A}$ succeeds.

DEFINITION 11.5 A public-key encryption scheme $\Pi = (\text{Gen}, \text{Enc}, \text{Dec})$ has indistinguishable multiple encryptions if for all probabilistic polynomial-time adversaries $\mathcal{A}$ there exists a negligible function negl such that:

$$\Pr[\text{PubK}_{\mathcal{A}, \Pi}^{\text{LR-cpa}}(n) = 1] \leq \frac{1}{2} + \text{negl}(n).$$

We will show that any CPA-secure scheme *automatically* has indistinguishable multiple encryptions; that is, in the public-key setting, security for encryption of a single message implies security for encryption of multiple messages. This means we can prove security of some scheme with respect to Definition 11.2, which is simpler and easier to work with, and conclude that the scheme satisfies Definition 11.5, a seemingly stronger definition that more accurately models real-world usage of public-key encryption. A proof of the following theorem is given below.

THEOREM 11.6 If public-key encryption scheme Π is CPA-secure, then it also has indistinguishable multiple encryptions.

An analogous result in the private-key setting was stated, but not proved, as Theorem 3.24.

Encrypting arbitrary-length messages. An immediate consequence of Theorem 11.6 is that a CPA-secure public-key encryption scheme for *fixed-length* messages implies a public-key encryption scheme for *arbitrary-length* messages satisfying the same notion of security. We illustrate this in the extreme case when the original scheme encrypts only 1-bit messages. Say $\Pi = (\text{Gen}, \text{Enc}, \text{Dec})$ is an encryption scheme for single-bit messages. We can construct a new scheme $\Pi' = (\text{Gen}, \text{Enc}', \text{Dec}')$ that has message space $\{0, 1\}^*$ by defining Enc' as follows:

$$\text{Enc}'_{pk}(m) = \text{Enc}_{pk}(m_1), \dots, \text{Enc}_{pk}(m_\ell), \quad (11.1)$$

where $m = m_1 \dots m_\ell$. (The decryption algorithm Dec' is constructed in the obvious way.) We have:

CLAIM 11.7 Let Π and Π' be as above. If Π is CPA-secure, then so is Π' .

The claim follows since we can view encryption of the message m using Π' as encryption of t messages $(m_1, \dots, m_t)$ using scheme Π .

A note on terminology. We have introduced three definitions of security for public-key encryption schemes—indistinguishable encryptions in the presence of an eavesdropper, CPA-security, and indistinguishable *multiple* encryptions—that are all equivalent. Following the usual convention in the cryptographic literature, we will simply use the term “CPA-security” to refer to schemes meeting these notions of security.

*Proof of Theorem 11.6

The proof of Theorem 11.6 is rather involved. We therefore provide some intuition before turning to the details. For this intuitive discussion we assume for simplicity that $\mathcal{A}$ makes only *two* calls to the LR oracle in experiment $\text{PubK}_{\mathcal{A},\Pi}^{\text{LR-cpa}}(n)$. (In the full proof, the number of calls is arbitrary.)

Fix an arbitrary PPT adversary $\mathcal{A}$ and a CPA-secure public-key encryption scheme Π , and consider an experiment $\text{PubK}_{\mathcal{A},\Pi}^{\text{LR-cpa}^2}(n)$ where $\mathcal{A}$ can make only *two* queries to the LR oracle. Denote the queries made by $\mathcal{A}$ to the oracle by $(m_{1,0}, m_{1,1})$ and $(m_{2,0}, m_{2,1})$; note that the second pair of messages may depend on the first ciphertext obtained by $\mathcal{A}$ from the oracle. In the experiment, $\mathcal{A}$ receives either a pair of ciphertexts $(\text{Enc}_{pk}(m_{1,0}), \text{Enc}_{pk}(m_{2,0}))$ (if $b = 0$), or a pair of ciphertexts $(\text{Enc}_{pk}(m_{1,1}), \text{Enc}_{pk}(m_{2,1}))$ (if $b = 1$). We write $\mathcal{A}(pk, \text{Enc}_{pk}(m_{1,0}), \text{Enc}_{pk}(m_{2,0}))$ to denote the output of $\mathcal{A}$ in the first case, and analogously for the second.

Let $\vec{C}_0$ denote the distribution of ciphertext pairs in the first case, and $\vec{C}_1$ the distribution of ciphertext pairs in the second case. To show that Definition 11.5 holds (for $\text{PubK}_{\mathcal{A},\Pi}^{\text{LR-cpa}^2}$), we need to prove that $\mathcal{A}$ cannot distinguish between being given a pair of ciphertexts distributed according to $\vec{C}_0$, or a pair of ciphertexts distributed according to $\vec{C}_1$. That is, we need to prove that there is a negligible function negl such that

$$\left| \Pr[\mathcal{A}(pk, \text{Enc}_{pk}(m_{1,0}), \text{Enc}_{pk}(m_{2,0})) = 1] - \Pr[\mathcal{A}(pk, \text{Enc}_{pk}(m_{1,1}), \text{Enc}_{pk}(m_{2,1})) = 1] \right| \leq \text{negl}(n). \quad (11.2)$$

(This is equivalent to Definition 11.5 for the same reason that Definition 3.9 is equivalent to Definition 3.8.) To prove this, we will show that

1. CPA-security of Π implies that $\mathcal{A}$ cannot distinguish between the case when it is given a pair of ciphertexts distributed according to $\vec{C}_0$, or a pair of ciphertexts $(\text{Enc}_{pk}(m_{1,0}), \text{Enc}_{pk}(m_{2,1}))$, which corresponds to encrypting the *first* message in $\mathcal{A}$'s first oracle query and the *second* message in $\mathcal{A}$'s second oracle query. (Although this cannot occur in $\text{PubK}_{\mathcal{A},\Pi}^{\text{LR-cpa}^2}(n)$, we can still ask what $\mathcal{A}$'s behavior would be if given such a ciphertext pair.) Let $\vec{C}_{01}$ denote the distribution of ciphertext pairs in this latter case.
2. Similarly, CPA-security of Π implies that $\mathcal{A}$ cannot distinguish between the case when it is given a pair of ciphertexts distributed according to $\vec{C}_{01}$, or a pair of ciphertexts distributed according to $\vec{C}_1$.

The above says that $\mathcal{A}$ cannot distinguish between distributions $\vec{C}_0$ and $\vec{C}_{01}$, nor between distributions $\vec{C}_{01}$ and $\vec{C}_1$. We conclude (using simple algebra) that $\mathcal{A}$ cannot distinguish between distributions $\vec{C}_0$ and $\vec{C}_1$.

The crux of the proof, then, is showing that $\mathcal{A}$ cannot distinguish between being given a pair of ciphertexts distributed according to $\vec{C}_0$, or a pair of

ciphertexts distributed according to $\tilde{C}_{01}$. (The other case follows similarly.) That is, we want to show that there is a negligible function negl for which

$$\begin{aligned} & \left| \Pr[\mathcal{A}(pk, \text{Enc}_{pk}(m_{1,0}), \text{Enc}_{pk}(m_{2,0})) = 1] \right. \\ & \quad \left. - \Pr[\mathcal{A}(pk, \text{Enc}_{pk}(m_{1,0}), \text{Enc}_{pk}(m_{2,1})) = 1] \right| \leq \text{negl}(n). \end{aligned} \quad (11.3)$$

Note that the only difference between the input of the adversary $\mathcal{A}$ in each case is in the second element. Intuitively, indistinguishability follows from the single-message case since $\mathcal{A}$ can generate $\text{Enc}_{pk}(m_{1,0})$ by itself. Formally, consider the following PPT adversary $\mathcal{A}'$ running in experiment $\text{PubK}_{\mathcal{A}', \Pi}^{\text{eav}}(n)$:

Adversary $\mathcal{A}'$:

1. On input pk , adversary $\mathcal{A}'$ runs $\mathcal{A}(pk)$ as a subroutine.
2. When $\mathcal{A}$ makes its first query $(m_{1,0}, m_{1,1})$ to the LR oracle, $\mathcal{A}'$ computes $c_1 \leftarrow \text{Enc}_{pk}(m_{1,0})$ and returns c_1 to $\mathcal{A}$ as the response from the oracle.
3. When $\mathcal{A}$ makes its second query $(m_{2,0}, m_{2,1})$ to the LR oracle, $\mathcal{A}'$ outputs $(m_{2,0}, m_{2,1})$ and receives back a challenge ciphertext c_2 . This is returned to $\mathcal{A}$ as the response from the LR oracle.
4. $\mathcal{A}'$ outputs the bit b' output by $\mathcal{A}$.

Looking at experiment $\text{PubK}_{\mathcal{A}', \Pi}^{\text{eav}}(n)$, we see that when $b = 0$ then the challenge ciphertext c_2 is computed as $\text{Enc}_{pk}(m_{2,0})$. Thus,

$$\Pr[\mathcal{A}'(\text{Enc}_{pk}(m_{2,0})) = 1] = \Pr[\mathcal{A}(\text{Enc}_{pk}(m_{1,0}), \text{Enc}_{pk}(m_{2,0})) = 1]. \quad (11.4)$$

(We suppress explicit mention of pk to save space.) In contrast, when $b = 1$ in experiment $\text{PubK}_{\mathcal{A}', \Pi}^{\text{eav}}(n)$, then c_2 is computed as $\text{Enc}_{pk}(m_{2,1})$ and so

$$\Pr[\mathcal{A}'(\text{Enc}_{pk}(m_{2,1})) = 1] = \Pr[\mathcal{A}(\text{Enc}_{pk}(m_{1,0}), \text{Enc}_{pk}(m_{2,1})) = 1]. \quad (11.5)$$

CPA-security of Π implies that there is a negligible function negl such that

$$|\Pr[\mathcal{A}'(\text{Enc}_{pk}(m_{2,0})) = 1] - \Pr[\mathcal{A}'(\text{Enc}_{pk}(m_{2,1})) = 1]| \leq \text{negl}(n).$$

This, together with Equations (11.4) and (11.5), yields Equation (11.3).

In almost exactly the same way, we can prove that:

$$\begin{aligned} & \left| \Pr[\mathcal{A}(pk, \text{Enc}_{pk}(m_{1,0}), \text{Enc}_{pk}(m_{2,1})) = 1] \right. \\ & \quad \left. - \Pr[\mathcal{A}(pk, \text{Enc}_{pk}(m_{1,1}), \text{Enc}_{pk}(m_{2,1})) = 1] \right| \leq \text{negl}(n). \end{aligned} \quad (11.6)$$

Equation (11.2) follows by combining Equations (11.3) and (11.6).

The main complication that arises in the general case is that the number of queries to the LR oracle is no longer fixed but may instead be an arbitrary

polynomial of n . In the formal proof this is handled using a *hybrid argument*. (Hybrid arguments were used also in Chapter 7.)

PROOF (of Theorem 11.6) Let Π be a CPA-secure public-key encryption scheme and $\mathcal{A}$ an arbitrary PPT adversary in experiment $\text{PubK}_{\mathcal{A}, \Pi}^{\text{LR-cpa}}(n)$. Let $t = t(n)$ be a polynomial upper bound on the number of queries made by $\mathcal{A}$ to the LR oracle, and assume without loss of generality that $\mathcal{A}$ always queries the oracle *exactly* this many times. For a given public key pk and $0 \leq i \leq t$, let LR_{pk}^i denote the oracle that on input (m_0, m_1) returns $\text{Enc}_{pk}(m_0)$ for the first i queries it receives, and returns $\text{Enc}_{pk}(m_1)$ for the next $t - i$ queries it receives. (That is, for the first i queries the first message in the input pair is encrypted, and in response to the remaining queries the second message in the input pair is encrypted.) We stress that each encryption is computed using uniform, independent randomness. Using this notation, we have

$$\Pr \left[\text{PubK}_{\mathcal{A}, \Pi}^{\text{LR-cpa}}(n) = 1 \right] = \frac{1}{2} \cdot \Pr[\mathcal{A}^{\text{LR}_{pk}^t}(pk) = 0] + \frac{1}{2} \cdot \Pr[\mathcal{A}^{\text{LR}_{pk}^0}(pk) = 1]$$

because, from the point of view of $\mathcal{A}$ (who makes exactly t queries), oracle LR_{pk}^t is equivalent to $\text{LR}_{pk,0}$, and oracle LR_{pk}^0 is equivalent to $\text{LR}_{pk,1}$. To prove that Π satisfies Definition 11.5, we will show that for any PPT $\mathcal{A}$ there is a negligible function negl' such that

$$\left| \Pr[\mathcal{A}^{\text{LR}_{pk}^t}(pk) = 1] - \Pr[\mathcal{A}^{\text{LR}_{pk}^0}(pk) = 1] \right| \leq \text{negl}'(n). \quad (11.7)$$

(As before, this is equivalent to Definition 11.5 for the same reason that Definition 3.9 is equivalent to Definition 3.8.)

Consider the following PPT adversary $\mathcal{A}'$ that eavesdrops on the encryption of a *single* message:

Adversary $\mathcal{A}'$:

1. $\mathcal{A}'$, given pk , chooses a uniform index $i \leftarrow \{1, \dots, t\}$.
2. $\mathcal{A}'$ runs $\mathcal{A}(pk)$, answering its j th oracle query $(m_{j,0}, m_{j,1})$ as follows:
 - (a) For $j < i$, adversary $\mathcal{A}'$ computes $c_j \leftarrow \text{Enc}_{pk}(m_{j,0})$ and returns c_j to $\mathcal{A}$ as the response from its oracle.
 - (b) For $j = i$, adversary $\mathcal{A}'$ outputs $(m_{j,0}, m_{j,1})$ and receives back a challenge ciphertext c_j . This is returned to $\mathcal{A}$ as the response from its oracle.
 - (c) For $j > i$, adversary $\mathcal{A}'$ computes $c_j \leftarrow \text{Enc}_{pk}(m_{j,1})$ and returns c_j to $\mathcal{A}$ as the response from its oracle.
3. $\mathcal{A}'$ outputs the bit b' that is output by $\mathcal{A}$.

Consider experiment $\text{PubK}_{\mathcal{A}', \Pi}^{\text{eav}}(n)$. Fixing some choice of $i = i^*$, note that if c_{i^*} is an encryption of $m_{i^*,0}$ then the interaction of $\mathcal{A}$ with its oracle is identical to an interaction with oracle $\text{LR}_{pk}^{i^*}$. Thus,

$$\begin{aligned} \Pr[\mathcal{A}' \text{ outputs } 1 \mid b = 0] &= \sum_{i^*=1}^t \Pr[i = i^*] \cdot \Pr[\mathcal{A}' \text{ outputs } 1 \mid b = 0 \wedge i = i^*] \\ &= \sum_{i^*=1}^t \frac{1}{t} \cdot \Pr[\mathcal{A}^{\text{LR}_{pk}^{i^*}}(pk) = 1]. \end{aligned}$$

On the other hand, if c_{i^*} is an encryption of $m_{i^*,1}$ then the interaction of $\mathcal{A}$ with its oracle is identical to an interaction with oracle $\text{LR}_{pk}^{i^*-1}$, and so

$$\begin{aligned} \Pr[\mathcal{A}' \text{ outputs } 1 \mid b = 1] &= \sum_{i^*=1}^t \Pr[i = i^*] \cdot \Pr[\mathcal{A}' \text{ outputs } 1 \mid b = 1 \wedge i = i^*] \\ &= \sum_{i^*=1}^t \frac{1}{t} \cdot \Pr[\mathcal{A}^{\text{LR}_{pk}^{i^*-1}}(pk) = 1] \\ &= \sum_{i^*=0}^{t-1} \frac{1}{t} \cdot \Pr[\mathcal{A}^{\text{LR}_{pk}^{i^*}}(pk) = 1], \end{aligned}$$

where the third equality is obtained just by shifting the indices of summation.

Since $\mathcal{A}'$ runs in polynomial time, the assumption that Π is CPA-secure means that there exists a negligible function negl such that

$$|\Pr[\mathcal{A}' \text{ outputs } 1 \mid b = 0] - \Pr[\mathcal{A}' \text{ outputs } 1 \mid b = 1]| \leq \text{negl}(n).$$

But this means that

$$\begin{aligned} \text{negl}(n) &\geq \left| \sum_{i^*=1}^t \frac{1}{t} \cdot \Pr[\mathcal{A}^{\text{LR}_{pk}^{i^*}}(pk) = 1] - \sum_{i^*=0}^{t-1} \frac{1}{t} \cdot \Pr[\mathcal{A}^{\text{LR}_{pk}^{i^*}}(pk) = 1] \right| \\ &= \frac{1}{t} \cdot \left| \Pr[\mathcal{A}^{\text{LR}_{pk}^t}(pk) = 1] - \Pr[\mathcal{A}^{\text{LR}_{pk}^0}(pk) = 1] \right|, \end{aligned}$$

since all but one of the terms in each summation cancel. We conclude that

$$\left| \Pr[\mathcal{A}^{\text{LR}_{pk}^t}(pk) = 1] - \Pr[\mathcal{A}^{\text{LR}_{pk}^0}(pk) = 1] \right| \leq t(n) \cdot \text{negl}(n).$$

Because t is polynomial, the function $t \cdot \text{negl}(n)$ is negligible. Since $\mathcal{A}$ was an arbitrary PPT adversary, this shows that Equation (11.7) holds and so completes the proof that Π has indistinguishable multiple encryptions. $\blacksquare$

11.2.3 Security against Chosen-Ciphertext Attacks

Chosen-ciphertext attacks, in which an adversary is able to obtain the decryption of arbitrary ciphertexts of its choice (with one technical restriction described below), are a concern in the public-key setting just as they are in the private-key setting. In fact, they are arguably *more* of a concern in the public-key setting since there a receiver expects to receive ciphertexts from multiple senders who are possibly unknown in advance, whereas a receiver in the private-key setting intends to communicate only with a single, known sender using any particular secret key.

Assume an eavesdropper $\mathcal{A}$ observes a ciphertext c sent by a sender $\mathcal{S}$ to a receiver $\mathcal{R}$. Broadly speaking, in the public-key setting there are two classes of chosen-ciphertext attacks:

- $\mathcal{A}$ might send a modified ciphertext c' to $\mathcal{R}$ *on behalf of* $\mathcal{S}$. (For example, in the context of encrypted e-mail, $\mathcal{A}$ might construct an encrypted e-mail c' and forge the “From” field so that it appears the e-mail originated from $\mathcal{S}$.) In this case, although it is unlikely that $\mathcal{A}$ would be able to obtain the entire decryption m' of c' , it might be possible for $\mathcal{A}$ to infer some information about m' based on the subsequent behavior of $\mathcal{R}$. Based on this information, $\mathcal{A}$ might be able to learn something about the original message m .
- $\mathcal{A}$ might send a modified ciphertext c' to $\mathcal{R}$ *in its own name*. In this case, $\mathcal{A}$ might obtain the entire decryption m' of c' if $\mathcal{R}$ responds directly to $\mathcal{A}$. Even if $\mathcal{A}$ learns nothing about m' , this modified message may have a known *relation* to the original message m that can be exploited by $\mathcal{A}$; see the third scenario below for an example.

The second class of attacks is specific to the context of public-key encryption, and has no analogue in the private-key setting.

It is not hard to identify a number of realistic scenarios illustrating the above types of attacks:

Scenario 1. Say a user $\mathcal{S}$ logs in to her bank account by sending to her bank an encryption of her password pw concatenated with a timestamp. Assume further that there are two types of error messages the bank sends: it returns “password incorrect” if the encrypted password does not match the stored password of $\mathcal{S}$, and “timestamp incorrect” if the password is correct but the timestamp is not.

If an adversary obtains a ciphertext c sent by $\mathcal{S}$ to the bank, the adversary can now mount a chosen-ciphertext attack by sending ciphertexts c' to the bank on behalf of $\mathcal{S}$ and observing the error messages that result. (This is similar to the padding-oracle attack that we saw in Section 3.7.2.) In some cases, this information may be enough to allow the adversary to determine the user’s entire password.

Scenario 2. Say $\mathcal{S}$ sends an encrypted e-mail c to $\mathcal{R}$, and this e-mail is observed by $\mathcal{A}$. If $\mathcal{A}$ sends, in its own name, an encrypted e-mail c' to $\mathcal{R}$, then $\mathcal{R}$ might reply to this e-mail *and quote the decrypted text m' corresponding to c'* . In this case, $\mathcal{R}$ is essentially acting as a decryption oracle for $\mathcal{A}$ and might potentially decrypt any ciphertext that $\mathcal{A}$ sends it.

Scenario 3. An issue that is closely related to that of chosen-ciphertext security is potential *malleability* of ciphertexts. Since a formal definition is quite involved, we do not provide one here but instead only give the intuitive idea. A scheme is *malleable* if it has the following property: given an encryption c of some unknown message m , it is possible to come up with a ciphertext c' that is an encryption of a message m' *that is related in some known way to m* . For example, perhaps given an encryption of m , it is possible to construct an encryption of $2m$. (We will see natural examples of CPA-secure schemes with this and similar properties later; see Section 13.2.3.)

Now imagine that $\mathcal{R}$ is running an auction, where two parties $\mathcal{S}$ and $\mathcal{A}$ submit their bids by encrypting them using the public key of $\mathcal{R}$. If a malleable encryption scheme is used, it may be possible for an adversary $\mathcal{A}$ to always place the highest bid (without bidding the maximum) by carrying out the following attack: wait until $\mathcal{S}$ sends a ciphertext c corresponding to its bid m (that is unknown to $\mathcal{A}$); then send a ciphertext c' corresponding to the bid $m' = 2m$. Note that m (and m' , for that matter) remain unknown to $\mathcal{A}$ until $\mathcal{R}$ announces the results, and so the possibility of such an attack does not contradict the fact that the encryption scheme is CPA-secure. CCA-secure schemes, on the other hand, can be shown to be non-malleable, meaning they are not vulnerable to such attacks.

The definition. Security against chosen-ciphertext attacks is defined by suitable modification of the analogous definition from the private-key setting (Definition 3.33). Given a public-key encryption scheme Π and an adversary $\mathcal{A}$, consider the following experiment:

The CCA indistinguishability experiment $\text{PubK}_{\mathcal{A}, \Pi}^{\text{cca}}(n)$:

1. $\text{Gen}(1^n)$ is run to obtain keys (pk, sk) .
2. The adversary $\mathcal{A}$ is given pk and access to a decryption oracle $\text{Dec}_{sk}(\cdot)$. It outputs a pair of messages m_0, m_1 of the same length. (These messages must be in the message space associated with pk .)
3. A uniform bit $b \in \{0, 1\}$ is chosen, and then a ciphertext $c \leftarrow \text{Enc}_{pk}(m_b)$ is computed and given to $\mathcal{A}$.
4. $\mathcal{A}$ continues to interact with the decryption oracle, but may not request a decryption of c itself. Finally, $\mathcal{A}$ outputs a bit b' .
5. The output of the experiment is defined to be 1 if $b' = b$, and 0 otherwise.

DEFINITION 11.8 A public-key encryption scheme $\Pi = (\text{Gen}, \text{Enc}, \text{Dec})$ has indistinguishable encryptions under a chosen-ciphertext attack (or is CCA-secure) if for all probabilistic polynomial-time adversaries $\mathcal{A}$ there exists a negligible function negl such that

$$\Pr[\text{PubK}_{\mathcal{A}, \Pi}^{\text{cca}}(n) = 1] \leq \frac{1}{2} + \text{negl}(n).$$

The natural analogue of Theorem 11.6 holds for CCA-security as well. That is, if a scheme has indistinguishable encryptions under a chosen-ciphertext attack then it has indistinguishable *multiple* encryptions under a chosen-ciphertext attack, where this is defined appropriately. Interestingly, however, the analogue of Claim 11.7 does *not* hold for CCA-security.

As in Definition 3.33, we must prevent the attacker from submitting the challenge ciphertext c to the decryption oracle for the definition to be achievable. But this restriction does not make the definition meaningless and, in particular, for each of the three motivating scenarios given earlier one can argue that setting $c' = c$ is of no benefit to the attacker:

- In the first scenario involving password-based login, the attacker learns nothing about $\mathcal{S}$'s password by replaying c to the bank since in this case it already knows that no error message will be generated.
- In the second scenario involving encrypted email, sending $c' = c$ to the receiver would likely make the receiver suspicious and so it would refuse to respond at all.
- In the final scenario involving an auction, $\mathcal{R}$ could easily detect cheating if the adversary's encrypted bid is identical to the other party's encrypted bid. Anyway, in that case all the attacker achieves by replaying c is that it submits the *same* bid as the honest party.

11.3 Hybrid Encryption and the KEM/DEM Paradigm

Claim 11.7 shows that any CPA-secure public-key encryption scheme for ℓ' -bit messages can be used to obtain a CPA-secure public-key encryption scheme for messages of arbitrary length. Encrypting an ℓ -bit message using this approach requires $\gamma \stackrel{\text{def}}{=} \lceil \ell/\ell' \rceil$ invocations of the original encryption scheme, meaning that both the computation and the ciphertext length are increased by a multiplicative factor of γ relative to the underlying scheme.

It is possible to do better by using private-key encryption *in tandem with* public-key encryption. This improves efficiency because private-key encryption is significantly faster than public-key encryption, and improves bandwidth because private-key schemes have lower ciphertext expansion. The resulting combination is called *hybrid encryption* and is used extensively in

practice. The basic idea is to use public-key encryption to obtain a shared key k , and then encrypt the message m using a private-key encryption scheme and key k . The receiver uses its long-term (asymmetric) private key to derive k , and then uses private-key decryption (with key k) to recover the original message. We stress that although private-key encryption is used as a component, this is a full-fledged public-key encryption scheme by virtue of the fact that the sender and receiver do not share any secret key *in advance*.

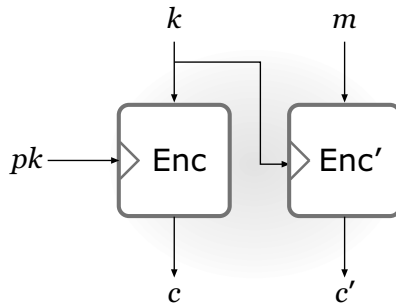


FIGURE 11.1: Hybrid encryption. Enc denotes a public-key encryption scheme, while Enc' is a private-key encryption scheme.

In a direct implementation of this idea (see Figure 11.1), the sender would share k by (1) choosing a uniform value k and then (2) encrypting k using a public-key encryption scheme. A more direct approach is to use a public-key primitive called a *key-encapsulation mechanism* (KEM) to accomplish both of these “in one shot.” This is advantageous both from a conceptual point of view and in terms of efficiency, as we will see later.

A KEM has three algorithms similar in spirit to those of a public-key encryption scheme. As before, the key-generation algorithm Gen is used to generate a pair of public and private keys. In place of encryption, we now have an *encapsulation* algorithm Encaps that takes only a public key as input (and no message), and outputs a ciphertext c along with a key k . A corresponding *decapsulation* algorithm Decaps is run by the receiver to recover k from the ciphertext c using the private key. Formally:

DEFINITION 11.9 A key-encapsulation mechanism (KEM) is a tuple of probabilistic polynomial-time algorithms $(\text{Gen}, \text{Encaps}, \text{Decaps})$ such that:

1. The key-generation algorithm Gen takes as input the security parameter 1^n and outputs a public-/private-key pair (pk, sk) . We assume pk and sk each has length at least n , and that n can be determined from pk .
2. The encapsulation algorithm Encaps takes as input a public key pk and the security parameter 1^n . It outputs a ciphertext c and a key $k \in \{0, 1\}^{\ell(n)}$ where ℓ is the key length. We write this as $(c, k) \leftarrow \text{Encaps}_{pk}(1^n)$.

3. The deterministic decapsulation algorithm **Decaps** takes as input a private key sk and a ciphertext c , and outputs a key k or a special symbol $\perp$ denoting failure. We write this as $k := \text{Decaps}_{sk}(c)$.

It is required that with all but negligible probability over (sk, pk) output by $\text{Gen}(1^n)$, if $\text{Encaps}_{pk}(1^n)$ outputs (c, k) then $\text{Decaps}_{sk}(c)$ outputs k .

In the definition we assume for simplicity that **Encaps** always outputs (a ciphertext c and) a key of some fixed length $\ell(n)$. One could also consider a more general definition in which **Encaps** takes 1^ℓ as an additional input and outputs a key of length ℓ .

Any public-key encryption scheme trivially gives a KEM by choosing a random key k and encrypting it. As we will see, however, dedicated constructions of KEMs can be more efficient.

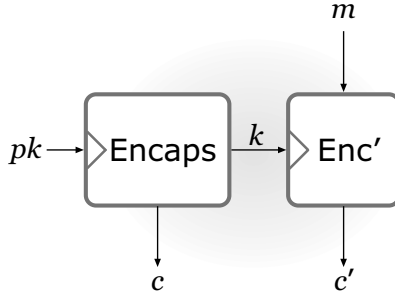


FIGURE 11.2: Hybrid encryption using the KEM/DEM approach.

Using a KEM (with key length n), we can implement hybrid encryption as in Figure 11.2. The sender runs $\text{Encaps}_{pk}(1^n)$ to obtain c along with a key k ; it then uses a private-key encryption scheme to encrypt its message m , using k as the key. In this context, the private-key encryption scheme is called a *data-encapsulation mechanism* (DEM) for obvious reasons. The ciphertext sent to the receiver includes both c and the ciphertext c' from the private-key scheme. Construction 11.10 gives a formal specification.

What is the efficiency of the resulting hybrid encryption scheme Π^{hy} ? For some fixed value of n , let α denote the cost of encapsulating an n -bit key using **Encaps**, and let β denote the cost (per bit of plaintext) of encryption using **Enc'**. Assume $|m| > n$, which is the interesting case. Then the cost, per bit of plaintext, of encrypting a message m using Π^{hy} is

$$\frac{\alpha + \beta \cdot |m|}{|m|} = \frac{\alpha}{|m|} + \beta, \quad (11.8)$$

which approaches β for sufficiently long m . In the limit of very long messages, then, the cost per bit incurred by the *public-key* encryption scheme Π^{hy} is the

CONSTRUCTION 11.10

Let $\Pi = (\text{Gen}, \text{Encaps}, \text{Decaps})$ be a KEM with key length n , and let $\Pi' = (\text{Gen}', \text{Enc}', \text{Dec}')$ be a private-key encryption scheme. Construct a public-key encryption scheme $\Pi^{\text{hy}} = (\text{Gen}^{\text{hy}}, \text{Enc}^{\text{hy}}, \text{Dec}^{\text{hy}})$ as follows:

- Gen^{hy} : on input 1^n run $\text{Gen}(1^n)$ and use the public and private keys (pk, sk) that are output.
- Enc^{hy} : on input a public key pk and a message $m \in \{0, 1\}^*$ do:
 1. Compute $(c, k) \leftarrow \text{Encaps}_{pk}(1^n)$.
 2. Compute $c' \leftarrow \text{Enc}'_k(m)$.
 3. Output the ciphertext $\langle c, c' \rangle$.
- Dec^{hy} : on input a private key sk and a ciphertext $\langle c, c' \rangle$ do:
 1. Compute $k := \text{Decaps}_{sk}(c)$.
 2. Output the message $m := \text{Dec}'_k(c')$.

Hybrid encryption using the KEM/DEM paradigm.

same as the cost per bit of the *private-key* scheme Π' . Hybrid encryption thus allows us to achieve the *functionality* of public-key encryption at the *efficiency* of private-key encryption, at least for sufficiently long messages.

A similar calculation can be used to measure the effect of hybrid encryption on the ciphertext length. For some fixed value of n , let L denote the length of the ciphertext output by Encaps , and say the private-key encryption of a message m using Enc' results in a ciphertext of length $n + |m|$ (this can be achieved using one of the modes of encryption discussed in Section 3.6; actually, even ciphertext length $|m|$ is possible since, as we will see, Π' need not be CPA-secure). Then the total length of a ciphertext in scheme Π^{hy} is

$$L + n + |m|. \quad (11.9)$$

In contrast, when using block-by-block encryption as in Equation (11.1), and assuming that public-key encryption of an n -bit message using Enc results in a ciphertext of length L , encryption of a message m would result in a ciphertext of length $L \cdot \lceil |m|/n \rceil$. The ciphertext length reported in Equation (11.9) is a significant improvement for sufficiently long m .

We can use some rough estimates to get a sense for what the above results mean in practice. (We stress that these numbers are only meant to give the reader a feel for the improvement; realistic values would depend on a variety of factors.) A typical value for the length of the key k might be $n = 128$. Furthermore, a “native” public-key encryption scheme might yield 256-bit ciphertexts when encrypting 128-bit messages; assume a KEM has ciphertexts of the same length when encapsulating a 128-bit key. Letting α , as before, denote the computational cost of public-key encryption/encapsulation of a 128-bit key, we see that block-by-block encryption as in Equation (11.1) would

encrypt a 1 MB ($= 10^6$ -bit) message with computational cost $\alpha \cdot \lceil 10^6/128 \rceil \approx 7800 \cdot \alpha$ and the ciphertext would be 2 MB long. Compare this to the efficiency of hybrid encryption. Letting β , as before, denote the per-bit computational cost of private-key encryption, a reasonable approximation is $\beta \approx \alpha/10^5$. Using Equation (11.8), we see that the overall computational cost for hybrid encryption for a 1 Mb message is

$$\alpha + 10^6 \cdot \frac{\alpha}{10^5} = 11 \cdot \alpha,$$

and the ciphertext would be only slightly longer than 1 MB. Thus, hybrid encryption improves the computational efficiency in this case by a factor of 700, and the ciphertext length by a factor of 2.

It remains to analyze the security of Π^{hy} . This, of course, depends on the security of its underlying components Π and Π' . In the following sections we define notions of CPA-security and CCA-security for KEMs, and show:

- If Π is a CPA-secure KEM and the private-key scheme Π' has indistinguishable encryptions in the presence of an eavesdropper, then Π^{hy} is a CPA-secure public-key encryption scheme. Notice that it suffices for Π' to satisfy a weaker definition of security—which, recall, does *not* imply CPA-security in the private-key setting—in order for the hybrid scheme Π^{hy} to be CPA-secure. Intuitively, the reason is that a *fresh*, uniform key k is chosen each time a new message is encrypted. Since each key k is used only once, indistinguishability of a single encryption of Π' suffices for security of the hybrid scheme Π^{hy} . This means that basic private-key encryption using a pseudorandom generator (or stream cipher), as in Construction 3.17, suffices.
- If Π is a CCA-secure KEM and Π' is a CCA-secure private-key encryption scheme, then Π^{hy} is a CCA-secure public-key encryption scheme.

11.3.1 CPA-Security

For simplicity, we assume in this and the next section a KEM with key length n . We define a notion of CPA-security for KEMs by analogy with Definition 11.2. As there, the adversary here eavesdrops on a single ciphertext c . Definition 11.2 requires that the attacker is unable to distinguish whether c is an encryption of some message m_0 or some other message m_1 . With a KEM there is no message, and we require instead that the encapsulated key k is indistinguishable from a uniform key that is independent of the ciphertext c .

Let $\Pi = (\text{Gen}, \text{Encaps}, \text{Decaps})$ be a KEM and $\mathcal{A}$ an arbitrary adversary.

The CPA indistinguishability experiment $\text{KEM}_{\mathcal{A}, \Pi}^{\text{cpa}}(n)$:

1. $\text{Gen}(1^n)$ is run to obtain keys (pk, sk) . Then $\text{Encaps}_{pk}(1^n)$ is run to generate (c, k) with $k \in \{0, 1\}^n$.
2. A uniform bit $b \in \{0, 1\}$ is chosen. If $b = 0$ set $\hat{k} := k$. If $b = 1$ then choose a uniform $\hat{k} \in \{0, 1\}^n$.
3. Give $(pk, c, \hat{k})$ to $\mathcal{A}$, who outputs a bit b' . The output of the experiment is defined to be 1 if $b' = b$, and 0 otherwise.

In the experiment, $\mathcal{A}$ is given the ciphertext c and either the actual key k corresponding to c , or an independent, uniform key. The KEM is CPA-secure if no efficient adversary can distinguish between these possibilities.

DEFINITION 11.11 A key-encapsulation mechanism Π is CPA-secure if for all probabilistic polynomial-time adversaries $\mathcal{A}$ there exists a negligible function negl such that

$$\Pr[\text{KEM}_{\mathcal{A}, \Pi}^{\text{cpa}}(n) = 1] \leq \frac{1}{2} + \text{negl}(n).$$

In the remainder of this section we prove the following theorem:

THEOREM 11.12 If Π is a CPA-secure KEM and Π' is a private-key encryption scheme that has indistinguishable encryptions in the presence of an eavesdropper, then Π^{hy} as in Construction 11.10 is a CPA-secure public-key encryption scheme.

Before proving the theorem formally, we give some intuition. Let the notation “ $X \stackrel{c}{=} Y$ ” denote the event that no polynomial-time adversary can distinguish between two distributions X and Y . (This concept is treated more formally in Section 7.8, although we do not rely on that section here.) For example, let $\text{Encaps}_{pk}^{(1)}(1^n)$ (resp., $\text{Encaps}_{pk}^{(2)}(1^n)$) denote the ciphertext (resp., key) output by Encaps . The fact that Π is CPA-secure means that

$$(pk, \text{Encaps}_{pk}^{(1)}(1^n), \text{Encaps}_{pk}^{(2)}(1^n)) \stackrel{c}{=} (pk, \text{Encaps}_{pk}^{(1)}(1^n), k'),$$

where pk is generated by $\text{Gen}(1^n)$ and k' is chosen independently and uniformly from $\{0, 1\}^n$. Similarly, the fact that Π' has indistinguishable encryptions in the presence of an eavesdropper means that for any m_0, m_1 output by $\mathcal{A}$ we have $\text{Enc}'_k(m_0) \stackrel{c}{=} \text{Enc}'_k(m_1)$ if k is chosen uniformly at random.

In order to prove CPA-security of Π^{hy} we need to show that

$$(pk, \text{Encaps}_{pk}^{(1)}(1^n), \text{Enc}'_k(m_0)) \stackrel{c}{=} (pk, \text{Encaps}_{pk}^{(1)}(1^n), \text{Enc}'_k(m_1)) \quad (11.10)$$

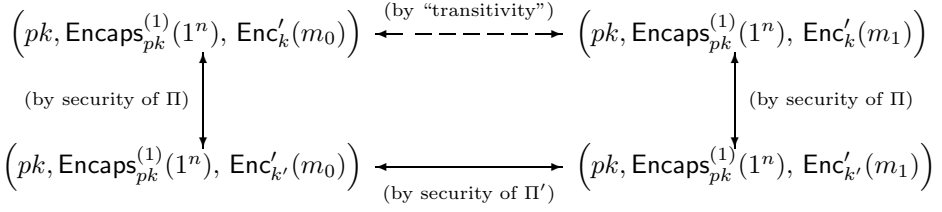


FIGURE 11.3: High-level structure of the proof of Theorem 11.12 (the arrows represent indistinguishability).

for m_0, m_1 output by a PPT adversary $\mathcal{A}$. (Equation (11.10) suffices to show that Π^{hy} has indistinguishable encryptions in the presence of an eavesdropper, and by Proposition 11.3 this implies that Π^{hy} is CPA-secure.)

The proof proceeds in three steps. (See Figure 11.3.) First we prove that

$$\left(pk, \text{Encaps}_{pk}^{(1)}(1^n), \text{Enc}'_{k'}(m_0)\right) \stackrel{c}{\equiv} \left(pk, \text{Encaps}_{pk}^{(1)}(1^n), \text{Enc}'_{k'}(m_0)\right), \quad (11.11)$$

where on the left k is output by $\text{Encaps}_{pk}^{(2)}(1^n)$, and on the right k' is an independent, uniform key. This follows via a fairly straightforward reduction, since CPA-security of Π means exactly that $\text{Encaps}_{pk}^{(2)}(1^n)$ cannot be distinguished from a uniform key k' even given pk and $\text{Encaps}_{pk}^{(1)}(1^n)$.

Next, we prove that

$$\left(pk, \text{Encaps}_{pk}^{(1)}(1^n), \text{Enc}'_{k'}(m_0)\right) \stackrel{c}{\equiv} \left(pk, \text{Encaps}_{pk}^{(1)}(1^n), \text{Enc}'_{k'}(m_1)\right). \quad (11.12)$$

Here the difference is between encrypting m_0 or m_1 using Π' and a uniform, independent key k' . Equation (11.12) thus follows using the fact that Π' has indistinguishable encryptions in the presence of an eavesdropper.

Exactly as in the case of Equation (11.11), we can also show that

$$\left(pk, \text{Encaps}_{pk}^{(1)}(1^n), \text{Enc}'_{k'}(m_1)\right) \stackrel{c}{\equiv} \left(pk, \text{Encaps}_{pk}^{(1)}(1^n), \text{Enc}'_{k'}(m_1)\right), \quad (11.13)$$

by relying again on the CPA-security of Π . Equations (11.11)–(11.13) imply, by transitivity, the desired result of Equation (11.10). (Transitivity will be implicit in the proof we give below.)

We now present the full proof.

PROOF (of Theorem 11.12) We prove that Π^{hy} has indistinguishable encryptions in the presence of an eavesdropper; by Proposition 11.3, this implies it is CPA-secure.

Fix an arbitrary PPT adversary $\mathcal{A}^{\text{hy}}$, and consider experiment $\text{PubK}_{\mathcal{A}^{\text{hy}}, \Pi^{\text{hy}}}^{\text{eav}}(n)$. Our goal is to prove that there is a negligible function negl such that

$$\Pr[\text{PubK}_{\mathcal{A}^{\text{hy}}, \Pi^{\text{hy}}}^{\text{eav}}(n) = 1] \leq \frac{1}{2} + \text{negl}(n).$$

By definition of the experiment, we have

$$\begin{aligned} \Pr[\text{PubK}_{\mathcal{A}^{\text{hy}}, \Pi^{\text{hy}}}^{\text{eav}}(n) = 1] &= \frac{1}{2} \cdot \Pr[\mathcal{A}^{\text{hy}}(pk, \text{Encaps}_{pk}^{(1)}(1^n), \text{Enc}'_k(m_0)) = 0] \\ &\quad + \frac{1}{2} \cdot \Pr[\mathcal{A}^{\text{hy}}(pk, \text{Encaps}_{pk}^{(1)}(1^n), \text{Enc}'_k(m_1)) = 1], \end{aligned} \tag{11.14}$$

where in each case k equals $\text{Encaps}_{pk}^{(2)}(1^n)$. Consider the following PPT adversary $\mathcal{A}_1$ attacking Π .

Adversary $\mathcal{A}_1$:

1. $\mathcal{A}_1$ is given $(pk, c, \hat{k})$.
2. $\mathcal{A}_1$ runs $\mathcal{A}^{\text{hy}}(pk)$ to obtain two messages m_0, m_1 . Then $\mathcal{A}_1$ computes $c' \leftarrow \text{Enc}'_{\hat{k}}(m_0)$, gives ciphertext $\langle c, c' \rangle$ to $\mathcal{A}^{\text{hy}}$, and outputs the bit b' that $\mathcal{A}^{\text{hy}}$ outputs.

Consider the behavior of $\mathcal{A}_1$ when attacking Π in experiment $\text{KEM}_{\mathcal{A}_1, \Pi}^{\text{cpa}}(n)$. When $b = 0$ in that experiment, then $\mathcal{A}_1$ is given $(pk, c, \hat{k})$ where c and $\hat{k}$ were both output by $\text{Encaps}_{pk}(1^n)$. This means that $\mathcal{A}^{\text{hy}}$ is given a ciphertext of the form $\langle c, c' \rangle = \langle c, \text{Enc}'_k(m_0) \rangle$, where k is the key encapsulated by c . So,

$$\Pr[\mathcal{A}_1 \text{ outputs } 0 \mid b = 0] = \Pr[\mathcal{A}^{\text{hy}}(pk, \text{Encaps}_{pk}^{(1)}(1^n), \text{Enc}'_k(m_0)) = 0].$$

On the other hand, when $b = 1$ in experiment $\text{KEM}_{\mathcal{A}_1, \Pi}^{\text{cpa}}(n)$ then $\mathcal{A}_1$ is given $(pk, c, \hat{k})$ with $\hat{k}$ uniform and independent of c . If we denote such a key by k' , this means $\mathcal{A}^{\text{hy}}$ is given a ciphertext of the form $\langle c, \text{Enc}'_{k'}(m_0) \rangle$, and

$$\Pr[\mathcal{A}_1 \text{ outputs } 1 \mid b = 1] = \Pr[\mathcal{A}^{\text{hy}}(pk, \text{Encaps}_{pk}^{(1)}(1^n), \text{Enc}'_{k'}(m_0)) = 1].$$

Since Π is a CPA-secure KEM, there is a negligible function negl_1 such that

$$\begin{aligned} \frac{1}{2} + \text{negl}_1(n) &\geq \Pr[\text{KEM}_{\mathcal{A}_1, \Pi}^{\text{cpa}}(n) = 1] \\ &= \frac{1}{2} \cdot \Pr[\mathcal{A}_1 \text{ outputs } 0 \mid b = 0] + \frac{1}{2} \cdot \Pr[\mathcal{A}_1 \text{ outputs } 1 \mid b = 1] \\ &= \frac{1}{2} \cdot \Pr[\mathcal{A}^{\text{hy}}(pk, \text{Encaps}_{pk}^{(1)}(1^n), \text{Enc}'_k(m_0)) = 0] \\ &\quad + \frac{1}{2} \cdot \Pr[\mathcal{A}^{\text{hy}}(pk, \text{Encaps}_{pk}^{(1)}(1^n), \text{Enc}'_{k'}(m_0)) = 1] \end{aligned} \tag{11.15}$$

where k is equal to $\text{Encaps}_{pk}^{(2)}(1^n)$ and k' is a uniform and independent key.

Next, consider the following PPT adversary $\mathcal{A}'$ that eavesdrops on a message encrypted using the private-key scheme Π' .

Adversary $\mathcal{A}'$:

1. $\mathcal{A}'(1^n)$ runs $\text{Gen}(1^n)$ on its own to generate keys (pk, sk) . It also computes $c \leftarrow \text{Encaps}_{pk}^{(1)}(1^n)$.
2. $\mathcal{A}'$ runs $\mathcal{A}^{\text{hy}}(pk)$ to obtain two messages m_0, m_1 . These are output by $\mathcal{A}'$, and it is given in return a ciphertext c' .
3. $\mathcal{A}'$ gives the ciphertext $\langle c, c' \rangle$ to $\mathcal{A}^{\text{hy}}$, and outputs the bit b' that $\mathcal{A}^{\text{hy}}$ outputs.

When $b = 0$ in experiment $\text{PrivK}_{\mathcal{A}', \Pi'}^{\text{eav}}(n)$, adversary $\mathcal{A}'$ is given a ciphertext c' which is an encryption of m_0 using a key k' that is uniform and independent of anything else. So $\mathcal{A}^{\text{hy}}$ is given a ciphertext of the form $\langle c, \text{Enc}_{k'}'(m_0) \rangle$ where k' is uniform and independent of c , and

$$\Pr[\mathcal{A}' \text{ outputs } 0 \mid b = 0] = \Pr[\mathcal{A}^{\text{hy}}(pk, \text{Encaps}_{pk}^{(1)}(1^n), \text{Enc}_{k'}'(m_0)) = 0].$$

On the other hand, when $b = 1$ in experiment $\text{PrivK}_{\mathcal{A}', \Pi'}^{\text{eav}}(n)$, then $\mathcal{A}'$ is given an encryption of m_1 using a uniform, independent key k' . This means $\mathcal{A}^{\text{hy}}$ is given a ciphertext of the form $\langle c, \text{Enc}_{k'}'(m_1) \rangle$ and so

$$\Pr[\mathcal{A}' \text{ outputs } 1 \mid b = 1] = \Pr[\mathcal{A}^{\text{hy}}(pk, \text{Encaps}_{pk}^{(1)}(1^n), \text{Enc}_{k'}'(m_1)) = 1].$$

Since Π' has indistinguishable encryptions in the presence of an eavesdropper, there is a negligible function negl' such that

$$\begin{aligned} \frac{1}{2} + \text{negl}'(n) &\geq \Pr[\text{PrivK}_{\mathcal{A}', \Pi'}^{\text{eav}}(n) = 1] \\ &= \frac{1}{2} \cdot \Pr[\mathcal{A}' \text{ outputs } 0 \mid b = 0] + \frac{1}{2} \cdot \Pr[\mathcal{A}' \text{ outputs } 1 \mid b = 1] \\ &= \frac{1}{2} \cdot \Pr[\mathcal{A}^{\text{hy}}(pk, \text{Encaps}_{pk}^{(1)}(1^n), \text{Enc}_{k'}'(m_0)) = 0] \\ &\quad + \frac{1}{2} \cdot \Pr[\mathcal{A}^{\text{hy}}(pk, \text{Encaps}_{pk}^{(1)}(1^n), \text{Enc}_{k'}'(m_1)) = 1]. \end{aligned} \tag{11.16}$$

Proceeding exactly as we did to prove Equation (11.15), we can show that there is a negligible function negl_2 such that

$$\begin{aligned} \frac{1}{2} + \text{negl}_2(n) &\geq \Pr[\text{KEM}_{\mathcal{A}_2, \Pi}^{\text{cpa}}(n) = 1] \\ &= \frac{1}{2} \cdot \Pr[\mathcal{A}_2 \text{ outputs } 0 \mid b = 0] + \frac{1}{2} \cdot \Pr[\mathcal{A}_2 \text{ outputs } 1 \mid b = 1] \\ &= \frac{1}{2} \cdot \Pr[\mathcal{A}^{\text{hy}}(pk, \text{Encaps}_{pk}^{(1)}(1^n), \text{Enc}_k'(m_1)) = 1] \\ &\quad + \frac{1}{2} \cdot \Pr[\mathcal{A}^{\text{hy}}(pk, \text{Encaps}_{pk}^{(1)}(1^n), \text{Enc}_{k'}'(m_1)) = 0]. \end{aligned} \tag{11.17}$$

Summing Equations (11.15)–(11.17) and using the fact that the sum of three negligible functions is negligible, we see there exists a negligible function negl such that

$$\begin{aligned} \frac{3}{2} + \text{negl}(n) \geq & \frac{1}{2} \cdot \left(\Pr[\mathcal{A}^{\text{hy}}(pk, c, \text{Enc}'_k(m_0)) = 0] + \Pr[\mathcal{A}^{\text{hy}}(pk, c, \text{Enc}'_{k'}(m_0)) = 1] \right. \\ & + \Pr[\mathcal{A}^{\text{hy}}(pk, c, \text{Enc}'_{k'}(m_0)) = 0] + \Pr[\mathcal{A}^{\text{hy}}(pk, c, \text{Enc}'_{k'}(m_1)) = 1] \\ & \left. + \Pr[\mathcal{A}^{\text{hy}}(pk, c, \text{Enc}'_k(m_1)) = 1] + \Pr[\mathcal{A}^{\text{hy}}(pk, c, \text{Enc}'_{k'}(m_1)) = 0] \right), \end{aligned}$$

where $c = \text{Encaps}_{pk}^{(1)}(1^n)$ in all the above. Note that

$$\Pr[\mathcal{A}^{\text{hy}}(pk, c, \text{Enc}'_{k'}(m_0)) = 1] + \Pr[\mathcal{A}^{\text{hy}}(pk, c, \text{Enc}'_{k'}(m_0)) = 0] = 1,$$

since the probabilities of complementary events always sum to 1. Similarly,

$$\Pr[\mathcal{A}^{\text{hy}}(pk, c, \text{Enc}'_{k'}(m_1)) = 1] + \Pr[\mathcal{A}^{\text{hy}}(pk, c, \text{Enc}'_{k'}(m_1)) = 0] = 1.$$

Therefore,

$$\begin{aligned} & \frac{1}{2} + \text{negl}(n) \\ & \geq \frac{1}{2} \cdot \left(\Pr[\mathcal{A}^{\text{hy}}(pk, c, \text{Enc}'_k(m_0)) = 0] + \Pr[\mathcal{A}^{\text{hy}}(pk, c, \text{Enc}'_k(m_1)) = 1] \right) \\ & = \Pr[\text{PubK}_{\mathcal{A}^{\text{hy}}, \Pi^{\text{hy}}}^{\text{eav}}(n) = 1] \end{aligned}$$

(using Equation (11.14) for the last equality), proving the theorem. ■

11.3.2 CCA-Security

If the private-key encryption scheme Π' is not itself secure against chosen-ciphertext attacks, then (regardless of the KEM used) neither is the resulting hybrid encryption scheme Π^{hy} . As a simple, illustrative example, say we take Construction 3.17 as our private-key encryption scheme. Then, leaving the KEM unspecified, encryption of a message m by Π^{hy} is done by computing $(c, k) \leftarrow \text{Encaps}_{pk}(1^n)$ and then outputting the ciphertext

$$\langle c, G(k) \oplus m \rangle,$$

where G is a pseudorandom generator. Given a ciphertext $\langle c, c' \rangle$, an attacker can simply flip the last bit of c' to obtain a modified ciphertext that is a valid encryption of m with its last bit flipped.

The natural way to fix this is to use a CCA-secure private-key encryption scheme. But this is clearly not enough if the KEM is susceptible to chosen-ciphertext attacks. Since we have not yet defined this notion, we do so now.

As in Definition 11.11, we require that an adversary given a ciphertext c cannot distinguish the key k encapsulated by that ciphertext from a uniform and independent key k' . Now, however, we additionally allow the attacker to request *decapsulation* of ciphertexts of its choice (as long as they are different from the challenge ciphertext).

Formally, let $\Pi = (\text{Gen}, \text{Encaps}, \text{Decaps})$ be a KEM with key length n and $\mathcal{A}$ an adversary, and consider the following experiment:

The CCA indistinguishability experiment $\text{KEM}_{\mathcal{A}, \Pi}^{\text{cca}}(n)$:

1. $\text{Gen}(1^n)$ is run to obtain keys (pk, sk) . Then $\text{Encaps}_{pk}(1^n)$ is run to generate (c, k) with $k \in \{0, 1\}^n$.
2. A uniform bit $b \in \{0, 1\}$ is chosen. If $b = 0$ set $\hat{k} := k$. If $b = 1$ then choose a uniform $\hat{k} \in \{0, 1\}^n$.
3. $\mathcal{A}$ is given $(pk, c, \hat{k})$ and access to an oracle $\text{Decaps}_{sk}(\cdot)$, but may not request decapsulation of c itself.
4. $\mathcal{A}$ outputs a bit b' . The output of the experiment is defined to be 1 if $b' = b$, and 0 otherwise.

DEFINITION 11.13 A key-encapsulation mechanism Π is CCA-secure if for all probabilistic polynomial-time adversaries $\mathcal{A}$ there is a negligible function negl such that

$$\Pr[\text{KEM}_{\mathcal{A}, \Pi}^{\text{cca}}(n) = 1] \leq \frac{1}{2} + \text{negl}(n).$$

Fortunately, we can show that using a CCA-secure KEM in combination with a CCA-secure private-key encryption scheme results in a public-key encryption scheme secure against chosen-ciphertext attacks.

THEOREM 11.14 If Π is a CCA-secure KEM and Π' is a CCA-secure private-key encryption scheme, then Π^{hy} as in Construction 11.10 is a CCA-secure public-key encryption scheme.

A proof is obtained by suitable modification of the proof of Theorem 11.12.

11.4 CDH/DDH-Based Encryption

So far we have discussed public-key encryption abstractly, but have not yet seen any concrete examples of public-key encryption schemes (or KEMs). Here we explore some constructions based on the *Diffie–Hellman problems*. (The Diffie–Hellman problems are introduced in Section 8.3.2.)

11.4.1 El Gamal Encryption

In 1985, Taher El Gamal observed that the Diffie–Hellman key-exchange protocol (cf. Section 10.3) could be adapted to give a public-key encryption scheme. Recall that in the Diffie–Hellman protocol, Alice sends a message to Bob and then Bob responds with a message to Alice; based on these messages, Alice and Bob can derive a shared value k which is indistinguishable (to an eavesdropper) from a uniform element of some group $\mathbb{G}$. We could imagine Bob using that shared value to encrypt a message $m \in \mathbb{G}$ by simply sending $k \cdot m$ to Alice; Alice can clearly recover m using her knowledge of k , and we will argue below that an eavesdropper learns nothing about m .

In the *El Gamal encryption scheme* we simply change our perspective on the above interaction. We view Alice’s initial message as her public key, and Bob’s reply (both his initial response and $k \cdot m$) as a ciphertext. CPA-security based on the decisional Diffie–Hellman (DDH) assumption follows fairly easily from security of the Diffie–Hellman key-exchange protocol (Theorem 10.3).

In our formal treatment, we begin by stating and proving a simple lemma that underlies the El Gamal encryption scheme. Let $\mathbb{G}$ be a finite group, and let $m \in \mathbb{G}$ be an arbitrary element. The lemma states that multiplying m by a uniform group element k yields a uniformly distributed group element k' . Importantly, the distribution of k' is independent of m ; this means that k' contains no information about m .

LEMMA 11.15 *Let $\mathbb{G}$ be a finite group, and let $m \in \mathbb{G}$ be arbitrary. Then choosing uniform $k \in \mathbb{G}$ and setting $k' := k \cdot m$ gives the same distribution for k' as choosing uniform $k' \in \mathbb{G}$. Put differently, for any $\hat{g} \in \mathbb{G}$ we have*

$$\Pr[k \cdot m = \hat{g}] = 1/|\mathbb{G}|,$$

where the probability is taken over uniform choice of $k \in \mathbb{G}$.

PROOF Let $\hat{g} \in \mathbb{G}$ be arbitrary. Then

$$\Pr[k \cdot m = \hat{g}] = \Pr[k = \hat{g} \cdot m^{-1}].$$

Since k is uniform, the probability that k is equal to the fixed element $\hat{g} \cdot m^{-1}$ is exactly $1/|\mathbb{G}|$. ■

The above lemma suggests a way to construct a perfectly secret *private-key* encryption scheme with message space $\mathbb{G}$. The sender and receiver share as their secret key a uniform element $k \in \mathbb{G}$. To encrypt the message $m \in \mathbb{G}$, the sender computes the ciphertext $k' := k \cdot m$. The receiver can recover the message from the ciphertext k' by computing $m := k'/k$. Perfect secrecy follows immediately from the lemma above. In fact, we have already seen this scheme in a different guise—the one-time pad encryption scheme is an

instantiation of this approach, with the underlying group being the set of strings of some fixed length under the operation of bit-wise XOR.

We can adapt the above ideas to the public-key setting by providing the parties with a way to generate a shared, “random-looking” value k by interacting over a public channel. This should sound familiar since it is exactly what the Diffie–Hellman protocol provides. We proceed with the details.

As in Section 8.3.2, let $\mathcal{G}$ be a polynomial-time algorithm that takes as input 1^n and (except possibly with negligible probability) outputs a description of a cyclic group $\mathbb{G}$, its order q (with $\|q\| = n$), and a generator g . The El Gamal encryption scheme is described in Construction 11.16.

CONSTRUCTION 11.16

Let $\mathcal{G}$ be as in the text. Define a public-key encryption scheme as follows:

- **Gen:** on input 1^n run $\mathcal{G}(1^n)$ to obtain $(\mathbb{G}, q, g)$. Then choose a uniform $x \in \mathbb{Z}_q$ and compute $h := g^x$. The public key is $\langle \mathbb{G}, q, g, h \rangle$ and the private key is $\langle \mathbb{G}, q, g, x \rangle$. The message space is $\mathbb{G}$.
- **Enc:** on input a public key $pk = \langle \mathbb{G}, q, g, h \rangle$ and a message $m \in \mathbb{G}$, choose a uniform $y \in \mathbb{Z}_q$ and output the ciphertext

$$\langle g^y, h^y \cdot m \rangle.$$

- **Dec:** on input a private key $sk = \langle \mathbb{G}, q, g, x \rangle$ and a ciphertext $\langle c_1, c_2 \rangle$, output

$$\hat{m} := c_2 / c_1^x.$$

The El Gamal encryption scheme.

To see that decryption succeeds, let $\langle c_1, c_2 \rangle = \langle g^y, h^y \cdot m \rangle$ with $h = g^x$. Then

$$\hat{m} = \frac{c_2}{c_1^x} = \frac{h^y \cdot m}{(g^y)^x} = \frac{(g^x)^y \cdot m}{g^{xy}} = \frac{g^{xy} \cdot m}{g^{xy}} = m.$$

Example 11.17

Let $q = 83$ and $p = 2q + 1 = 167$, and let $\mathbb{G}$ denote the group of *quadratic residues* (i.e., squares) modulo p . (Since p and q are prime, $\mathbb{G}$ is a subgroup of $\mathbb{Z}_p^*$ with order q . See Section 8.3.3.) Since the order of $\mathbb{G}$ is prime, any element of $\mathbb{G}$ except 1 is a generator; take $g = 2^2 = 4 \bmod 167$. Say the receiver chooses secret key $37 \in \mathbb{Z}_{83}$ and so the public key is

$$pk = \langle p, q, g, h \rangle = \langle 167, 83, 4, [4^{37} \bmod 167] \rangle = \langle 167, 83, 4, 76 \rangle,$$

where we use p to represent $\mathbb{G}$ (it is assumed that the receiver knows that the group is the set of quadratic residues modulo p).

Say a sender encrypts the message $m = 65 \in \mathbb{G}$ (note $65 = 30^2 \bmod 167$ and so 65 is an element in the subgroup). If $y = 71$, the ciphertext is

$$\langle [4^{71} \bmod 167], [76^{71} \cdot 65 \bmod 167] \rangle = \langle 132, 44 \rangle.$$

To decrypt, the receiver first computes $124 = [132^{37} \bmod 167]$; then, since $66 = [124^{-1} \bmod 167]$, the receiver recovers $m = 65 = [44 \cdot 66 \bmod 167]$. $\diamond$

We now prove security of the scheme. (The reader may want to compare the proof of the following to the proofs of Theorems 3.18 and 10.3.)

THEOREM 11.18 *If the DDH problem is hard relative to $\mathcal{G}$, then the El Gamal encryption scheme is CPA-secure.*

PROOF Let Π denote the El Gamal encryption scheme. We prove that Π has indistinguishable encryptions in the presence of an eavesdropper; by Proposition 11.3, this implies it is CPA-secure.

Let $\mathcal{A}$ be a probabilistic polynomial-time adversary. We want to show that there is a negligible function negl such that

$$\Pr[\text{PubK}_{\mathcal{A}, \Pi}^{\text{eav}}(n) = 1] \leq \frac{1}{2} + \text{negl}(n).$$

Consider the modified “encryption scheme” $\tilde{\Pi}$ where Gen is the same as in Π , but encryption of a message m with respect to the public key $\langle \mathbb{G}, q, g, h \rangle$ is done by choosing uniform $y, z \in \mathbb{Z}_q$ and outputting the ciphertext

$$\langle g^y, g^z \cdot m \rangle.$$

Although $\tilde{\Pi}$ is not actually an encryption scheme (as there is no way for the receiver to decrypt), the experiment $\text{PubK}_{\mathcal{A}, \tilde{\Pi}}^{\text{eav}}(n)$ is still well-defined since that experiment depends only on the key-generation and encryption algorithms.

Lemma 11.15 and the discussion that immediately follows it imply that the second component of the ciphertext in scheme $\tilde{\Pi}$ is a uniformly distributed group element and, in particular, is independent of the message m being encrypted. (Remember that g^z is a uniform element of $\mathbb{G}$ when z is chosen uniformly from $\mathbb{Z}_q$.) The first component of the ciphertext is trivially independent of m . Taken together, this means that the entire ciphertext contains no information about m . It follows that

$$\Pr[\text{PubK}_{\mathcal{A}, \tilde{\Pi}}^{\text{eav}}(n) = 1] = \frac{1}{2}.$$

Now consider the following PPT algorithm D that attempts to solve the DDH problem relative to $\mathcal{G}$. Recall that D receives $(\mathbb{G}, q, g, h_1, h_2, h_3)$ where $h_1 = g^x$, $h_2 = g^y$, and h_3 is either g^{xy} or g^z (for uniform x, y, z); the goal of D is to determine which is the case.

Algorithm D :

The algorithm is given $(\mathbb{G}, q, g, h_1, h_2, h_3)$ as input.

- Set $pk = \langle \mathbb{G}, q, g, h_1 \rangle$ and run $\mathcal{A}(pk)$ to obtain two messages $m_0, m_1 \in \mathbb{G}$.
- Choose a uniform bit b , and set $c_1 := h_2$ and $c_2 := h_3 \cdot m_b$.
- Give the ciphertext $\langle c_1, c_2 \rangle$ to $\mathcal{A}$ and obtain an output bit b' .
If $b' = b$, output 1; otherwise, output 0.

Let us analyze the behavior of D . There are two cases to consider:

Case 1: Say the input to D is generated by running $\mathcal{G}(1^n)$ to obtain $(\mathbb{G}, q, g)$, then choosing uniform $x, y, z \in \mathbb{Z}_q$, and finally setting $h_1 := g^x$, $h_2 := g^y$, and $h_3 := g^z$. Then D runs $\mathcal{A}$ on a public key constructed as

$$pk = \langle \mathbb{G}, q, g, g^x \rangle$$

and a ciphertext constructed as

$$\langle c_1, c_2 \rangle = \langle g^y, g^z \cdot m_b \rangle.$$

We see that in this case the view of $\mathcal{A}$ when run as a subroutine by D is distributed identically to $\mathcal{A}$'s view in experiment $\text{PubK}_{\mathcal{A}, \tilde{\Pi}}^{\text{eav}}(n)$. Since D outputs 1 exactly when the output b' of $\mathcal{A}$ is equal to b , we have that

$$\Pr[D(\mathbb{G}, q, g, g^x, g^y, g^z) = 1] = \Pr[\text{PubK}_{\mathcal{A}, \tilde{\Pi}}^{\text{eav}}(n) = 1] = \frac{1}{2}.$$

Case 2: Say the input to D is generated by running $\mathcal{G}(1^n)$ to obtain $(\mathbb{G}, q, g)$, then choosing uniform $x, y \in \mathbb{Z}_q$, and finally setting $h_1 := g^x$, $h_2 := g^y$, and $h_3 := g^{xy}$. Then D runs $\mathcal{A}$ on a public key constructed as

$$pk = \langle \mathbb{G}, q, g, g^x \rangle$$

and a ciphertext constructed as

$$\langle c_1, c_2 \rangle = \langle g^y, g^{xy} \cdot m_b \rangle = \langle g^y, (g^x)^y \cdot m_b \rangle.$$

We see that in this case the view of $\mathcal{A}$ when run as a subroutine by D is distributed identically to $\mathcal{A}$'s view in experiment $\text{PubK}_{\mathcal{A}, \Pi}^{\text{eav}}(n)$. Since D outputs 1 exactly when the output b' of $\mathcal{A}$ is equal to b , we have that

$$\Pr[D(\mathbb{G}, q, g, g^x, g^y, g^{xy}) = 1] = \Pr[\text{PubK}_{\mathcal{A}, \Pi}^{\text{eav}}(n) = 1].$$

Under the assumption that the DDH problem is hard relative to $\mathcal{G}$, there is a negligible function negl such that

$$\begin{aligned} \text{negl}(n) &\geq \left| \Pr[D(\mathbb{G}, q, g, g^x, g^y, g^z) = 1] - \Pr[D(\mathbb{G}, q, g, g^x, g^y, g^{xy}) = 1] \right| \\ &= \left| \frac{1}{2} - \Pr[\text{PubK}_{\mathcal{A}, \Pi}^{\text{eav}}(n) = 1] \right|. \end{aligned}$$

This implies $\Pr[\text{PubK}_{\mathcal{A}, \Pi}^{\text{eav}}(n) = 1] \leq \frac{1}{2} + \text{negl}(n)$, completing the proof. ■

El Gamal Implementation Issues

We briefly discuss some practical issues related to El Gamal encryption.

Sharing public parameters. Our description of the El Gamal encryption scheme in Construction 11.16 requires the receiver to run $\mathcal{G}$ to generate $\mathbb{G}, q, g$. In practice, it is common for these parameters to be generated and fixed “once-and-for-all,” and then shared by multiple receivers. (Of course, each receiver must choose their own secret value x and publish their own public key $h = g^x$.) For example, NIST has published a set of recommended parameters suitable for use in the El Gamal encryption scheme. Sharing parameters in this way does not impact security (assuming the parameters were generated correctly and honestly in the first place). Looking ahead, we remark that this is in contrast to the case of RSA, where parameters cannot safely be shared (see Section 11.5.1).

Choice of group. As discussed in Section 8.3.2, the group order q is generally chosen to be prime. As far as specific groups are concerned, elliptic curves are one increasingly popular choice; an alternative is to let $\mathbb{G}$ be a prime-order subgroup of $\mathbb{Z}_p^*$, for p prime. We refer to Section 9.3 for a tabulation of recommended key lengths for achieving different levels of security.

The message space. An inconvenient aspect of the El Gamal encryption scheme is that the message space is a group $\mathbb{G}$ rather than bit-strings of some specified length. For some choices of the group, it is possible to address this by defining a reversible encoding of bit-strings as group elements. In such cases, the sender can first encode their message $m \in \{0, 1\}^\ell$ as a group element $\hat{m} \in \mathbb{G}$ and then apply El Gamal encryption to $\hat{m}$. The receiver can decrypt as in Construction 11.16 to obtain the encoded message $\hat{m}$, and then reverse the encoding to recover the original message m .

A simpler approach is to use (a variant of) El Gamal encryption as part of a hybrid encryption scheme. For example, the sender could choose a uniform group element $m \in \mathbb{G}$, encrypt this using the El Gamal encryption scheme, and then encrypt their actual message using a private-key encryption scheme and key $H(m)$, where $H : \mathbb{G} \rightarrow \{0, 1\}^n$ is an appropriate key-derivation function (see the following section). In this case, it would be more efficient to use the DDH-based KEM that we describe next.

11.4.2 DDH-Based Key Encapsulation

At the end of the previous section we noted that El Gamal encryption can be used as part of a hybrid encryption scheme by simply encrypting a uniform group element m and using a hash of that element as a key. But this is wasteful! The proof of security for El Gamal encryption shows that c_1^x (where c_1 is the first component of the ciphertext, and x is the private key of the receiver) is already indistinguishable from a uniform group element, so the sender/receiver may as well use that. Construction 11.19 illustrates the KEM

that follows this approach. Note that the resulting encapsulation consists of just a single group element. In contrast, if we were to use El Gamal encryption of a uniform group element, the ciphertext would contain two group elements.

CONSTRUCTION 11.19

Let $\mathcal{G}$ be as in the previous section. Define a KEM as follows:

- **Gen**: on input 1^n run $\mathcal{G}(1^n)$ to obtain $(\mathbb{G}, q, g)$. choose a uniform $x \in \mathbb{Z}_q$ and set $h := g^x$. Also specify a function $H : \mathbb{G} \rightarrow \{0, 1\}^{\ell(n)}$ for some function ℓ (see text). The public key is $\langle \mathbb{G}, q, g, h, H \rangle$ and the private key is $\langle \mathbb{G}, q, g, x \rangle$.
- **Encaps**: on input a public key $pk = \langle \mathbb{G}, q, g, h, H \rangle$ choose a uniform $y \in \mathbb{Z}_q$ and output the ciphertext g^y and the key $H(h^y)$.
- **Decaps**: on input a private key $sk = \langle \mathbb{G}, q, g, x \rangle$ and a ciphertext $c \in \mathbb{G}$, output the key $H(c^x)$.

An “El Gamal-like” KEM.

As described, the construction leaves the key-derivation function H unspecified, and there are several options for it. (See Section 5.6.4 for more on key derivation in general.) One possibility is to choose a function $H : \mathbb{G} \rightarrow \{0, 1\}^{\ell}$ that is (close to) *regular*, meaning that for each possible key $k \in \{0, 1\}^{\ell}$ the number of group elements that map to k is approximately the same. (Formally, we need a negligible function negl such that for each $k \in \{0, 1\}^{\ell}$

$$2^{\ell} \cdot |\Pr[H(g) = k] - 2^{-\ell}| \leq \text{negl}(n),$$

where the probability is taken over uniform choice of $g \in \mathbb{G}$. This ensures that the distribution on the key k is statistically close to uniform.) Both the complexity of H , as well as the achievable key length ℓ , will depend on the specific group $\mathbb{G}$ being used.

A second possibility is to let H be a *keyed* function, where the (uniform) key for H is included as part of the receiver’s public key. This works if H is a strong extractor, as mentioned briefly in Section 5.6.4. Appropriate choice of ℓ here (to ensure that the resulting key is statistically close to uniform) will depend on the size of $\mathbb{G}$.

In either of the above cases, a proof of CPA-security based on the decisional Diffie–Hellman (DDH) assumption follows easily by adapting the proof of security for the Diffie–Hellman key-exchange protocol (Theorem 10.3).

THEOREM 11.20 *If the DDH problem is hard relative to $\mathcal{G}$, and H is chosen as described, then Construction 11.19 is a CPA-secure KEM.*

If one is willing to model H as a *random oracle*, then Construction 11.19 can be proven CPA-secure based on the (weaker) *computational* Diffie–Hellman (CDH) assumption. We discuss this in the following section.

11.4.3 *A CDH-Based KEM in the Random-Oracle Model

In this section, we show that if one is willing to model H as a *random oracle*, then Construction 11.19 can be proven CPA-secure based on the CDH assumption. (Readers may want to review Section 5.5 to remind themselves of the random-oracle model.) Intuitively, the CDH assumption implies that an attacker observing $h = g^x$ (from the public key) and the ciphertext $c = g^y$ cannot compute $\text{DH}_g(h, c) = h^y$. In particular, then, an attacker cannot query h^y to the random oracle. But this means that the encapsulated key $H(h^y)$ is completely random from the attacker's point of view. This intuition is turned into a formal proof below.

As indicated by the intuition above, the proof inherently relies on modeling H as a random oracle.³ Specifically, the proof relies on the facts that (1) the only way to learn $H(h^y)$ is to explicitly query h^y to H , which would mean that the attacker has solved a CDH instance (this is called “extractability” in Section 5.5.1), and (2) if an attacker does *not* query h^y to H , then the value $H(h^y)$ is uniform from the attacker's point of view. These properties only hold—indeed, they only make sense—if H is modeled as a random oracle.

THEOREM 11.21 *If the CDH problem is hard relative to $\mathcal{G}$, and H is modeled as a random oracle, then Construction 11.19 is CPA-secure.*

PROOF Let Π denote Construction 11.19, and let $\mathcal{A}$ be a PPT adversary. We want to show that there is a negligible function negl such that

$$\Pr[\text{KEM}_{\mathcal{A}, \Pi}^{\text{cpa}}(n) = 1] \leq \frac{1}{2} + \text{negl}(n).$$

The above probability is also taken over uniform choice of the function H , to which $\mathcal{A}$ is given oracle access.

Consider an execution of experiment $\text{KEM}_{\mathcal{A}, \Pi}^{\text{cpa}}(n)$ in which the public key is $\langle \mathbb{G}, q, g, h \rangle$ and the ciphertext is $c = g^y$, and let Query be the event that $\mathcal{A}$ queries $\text{DH}_g(h, c) = h^y$ to H . We have

$$\begin{aligned} \Pr[\text{KEM}_{\mathcal{A}, \Pi}^{\text{cpa}}(n) = 1] &= \Pr[\text{KEM}_{\mathcal{A}, \Pi}^{\text{cpa}}(n) = 1 \wedge \overline{\text{Query}}] \\ &\quad + \Pr[\text{KEM}_{\mathcal{A}, \Pi}^{\text{cpa}}(n) = 1 \wedge \text{Query}] \\ &\leq \Pr[\text{KEM}_{\mathcal{A}, \Pi}^{\text{cpa}}(n) = 1 \wedge \overline{\text{Query}}] + \Pr[\text{Query}]. \end{aligned} \quad (11.18)$$

If $\Pr[\overline{\text{Query}}] = 0$ then $\Pr[\text{KEM}_{\mathcal{A}, \Pi}^{\text{cpa}}(n) = 1 \wedge \overline{\text{Query}}] = 0$. Otherwise,

$$\begin{aligned} \Pr[\text{KEM}_{\mathcal{A}, \Pi}^{\text{cpa}}(n) = 1 \wedge \overline{\text{Query}}] &= \Pr[\text{KEM}_{\mathcal{A}, \Pi}^{\text{cpa}}(n) = 1 \mid \overline{\text{Query}}] \cdot \Pr[\overline{\text{Query}}] \\ &\leq \Pr[\text{KEM}_{\mathcal{A}, \Pi}^{\text{cpa}}(n) = 1 \mid \overline{\text{Query}}]. \end{aligned}$$

³This is true as long as we wish to rely only on the CDH assumption. As noted earlier, a proof without random oracles is possible if we rely on the stronger DDH assumption.

In experiment $\text{KEM}_{\mathcal{A},\Pi}^{\text{cpa}}(n)$, the adversary $\mathcal{A}$ is given the public key and the ciphertext, plus either the encapsulated key $k \stackrel{\text{def}}{=} H(h^y)$ or a uniform key. If **Query** does not occur, then k is uniformly distributed from the perspective of the adversary, and so there is no way $\mathcal{A}$ can distinguish between these two possibilities. This means that

$$\Pr[\text{KEM}_{\mathcal{A},\Pi}^{\text{cpa}}(n) = 1 \mid \overline{\text{Query}}] = \frac{1}{2}.$$

Returning to Equation (11.18), we thus have

$$\Pr[\text{KEM}_{\mathcal{A},\Pi}^{\text{cpa}}(n) = 1] \leq \frac{1}{2} + \Pr[\text{Query}].$$

We next show that $\Pr[\text{Query}]$ is negligible, completing the proof.

Let $t = t(n)$ be a (polynomial) upper bound on the number of queries that $\mathcal{A}$ makes to the random oracle H . Define the following PPT algorithm $\mathcal{A}'$ for the CDH problem relative to $\mathcal{G}$:

Algorithm $\mathcal{A}'$:

The algorithm is given $\mathbb{G}, q, g, h, c$ as input.

- Set $pk = \langle \mathbb{G}, q, g, h \rangle$ and choose a uniform $k \in \{0, 1\}^\ell$.
- Run $\mathcal{A}(pk, c, k)$. When $\mathcal{A}$ makes a query to H , answer it by choosing a fresh, uniform ℓ -bit string.
- At the end of $\mathcal{A}$'s execution, let $y_1, \dots, y_t$ be the list of queries that $\mathcal{A}$ has made to H . Choose a uniform index $i \in \{1, \dots, t\}$ and output y_i .

We are interested in the probability with which $\mathcal{A}'$ solves the CDH problem, i.e., $\Pr[\mathcal{A}'(\mathbb{G}, q, g, h, c) = \text{DH}_g(h, c)]$, where the probability is taken over $\mathbb{G}, q, g$ output by $\mathcal{G}(1^n)$, uniform $h, c \in \mathbb{G}$, and the randomness of $\mathcal{A}'$. To analyze this probability, note first that event **Query** is still well-defined in the execution of $\mathcal{A}'$, even though $\mathcal{A}'$ cannot detect whether it occurs. Moreover, the probability of event **Query** when $\mathcal{A}$ is run as a sub-routine by $\mathcal{A}'$ is identical to the probability of event **Query** in experiment $\text{KEM}_{\mathcal{A},\Pi}^{\text{cpa}}(n)$. This follows because the view of $\mathcal{A}$ is identical in both cases until event **Query** occurs: in each case, $\mathbb{G}, q, g$ are output by $\mathcal{G}(1^n)$; in each case, h and c are uniform elements of $\mathbb{G}$ and k is a uniform, ℓ -bit string; and in each case, queries to H other than $H(\text{DH}_g(h, c))$ are answered with a uniform ℓ -bit string. (In $\text{KEM}_{\mathcal{A},\Pi}^{\text{cpa}}(n)$, the query $H(\text{DH}_g(h, c))$ is answered with the actual encapsulated key, which is equal to k with probability $1/2$, whereas when $\mathcal{A}$ is run as a subroutine by $\mathcal{A}'$ the query $H(\text{DH}_g(h, c))$ is answered with a uniform ℓ -bit string that is independent of k . But when this query is made, event **Query** occurs.)

Finally, observe that when **Query** occurs then $\text{DH}_g(h, c) \in \{y_1, \dots, y_t\}$ by definition, and so $\mathcal{A}'$ outputs the correct result $\text{DH}_g(h, c)$ with probability at

least $1/t$. We therefore conclude that

$$\Pr[\mathcal{A}'(\mathbb{G}, q, g, h, c) = \text{DH}_g(h, c)] \geq \Pr[\text{Query}]/t,$$

or $\Pr[\text{Query}] \leq t \cdot \Pr[\mathcal{A}'(\mathbb{G}, q, g, h, c) = \text{DH}_g(h, c)]$. Since the CDH problem is hard for $\mathcal{G}$, this latter probability is negligible; since t is polynomial, this implies that $\Pr[\text{Query}]$ is negligible as well. This completes the proof. $\blacksquare$

In the next section we will see that Construction 11.19 can be shown to be CCA-secure under a stronger variant of the CDH assumption (if we continue to model H as a random oracle).

11.4.4 Chosen-Ciphertext Security and DHIES/ECIES

The El Gamal encryption scheme is vulnerable to chosen-ciphertext attacks. This follows from the fact that it is *malleable*. Recall that an encryption scheme is malleable, informally, if given a ciphertext c that is an encryption of some unknown message m , it is possible to generate a modified ciphertext c' that is an encryption of a message m' having some known relation to m . In the case of El Gamal encryption, consider an adversary $\mathcal{A}$ who intercepts a ciphertext $c = \langle c_1, c_2 \rangle$ encrypted using the public key $pk = \langle \mathbb{G}, q, g, h \rangle$, and who then constructs the modified ciphertext $c' = \langle c_1, c'_2 \rangle$ where $c'_s = c_2 \cdot \alpha$ for some $\alpha \in \mathbb{G}$. If c is an encryption of a message $m \in \mathbb{G}$ (which may be unknown to $\mathcal{A}$), we have $c_1 = g^y$ and $c_2 = h^y \cdot m$ for some $y \in \mathbb{Z}_q$. But then

$$c_1 = g^y \quad \text{and} \quad c'_2 = h^y \cdot (\alpha \cdot m),$$

and so c' is a valid encryption of the message $\alpha \cdot m$. In other words, $\mathcal{A}$ can transform an encryption of the (unknown) message m into an encryption of the (unknown) message $\alpha \cdot m$. As discussed in Scenario 3 in Section 11.2.3, this sort of attack can have serious consequences.

The KEM discussed in the previous section might also be malleable depending on the specific key-derivation function H being used. If H is modeled as a random oracle, however, then such attacks no longer seem possible. In fact, one can prove in this case that Construction 11.19 is CCA-secure based on the so-called *gap-CDH assumption*. Recall that the CDH assumption says that given group elements g^x and g^y (for some generator g), it is infeasible to compute g^{xy} . The gap-CDH assumption says that this remains infeasible even given access to an oracle $\mathcal{O}$ such that $\mathcal{O}(U, V)$ returns 1 exactly when $V = U^y$. Stated differently, the CDH problem remains hard even given an oracle that solves the DDH problem. (We do not give a formal definition since we will not use this assumption in the rest of the book.) This assumption is believed to hold for the classes of groups we have discussed in this book. A proof of the following is very similar to the proof of Theorem 11.38.

THEOREM 11.22 *If the gap-CDH problem is hard relative to $\mathcal{G}$, and H is modeled as a random oracle, then Construction 11.19 is a CCA-secure KEM.*

It is interesting to observe that the same construction (namely, Construction 11.19) can be analyzed under different assumptions and in different models, yielding different results. Assuming only that the DDH problem is hard (and for H chosen appropriately), the scheme is CPA-secure. If we model H as a random oracle (which imposes more stringent requirements on H), then under the weaker CDH assumption we obtain CPA-security, and under the stronger gap-CDH assumption we obtain CCA-security.

CONSTRUCTION 11.23

Let $\mathcal{G}$ be as in the text. Let $\Pi_E = (\text{Enc}', \text{Dec}')$ be a private-key encryption scheme, and let $\Pi_M = (\text{Mac}, \text{Vrfy})$ be a message authentication code. Define a public-key encryption scheme as follows:

- **Gen:** On input 1^n run $\mathcal{G}(1^n)$ to obtain $(\mathbb{G}, q, g)$. choose a uniform $x \in \mathbb{Z}_q$, set $h := g^x$, and specify a function $H : \mathbb{G} \rightarrow \{0, 1\}^{2n}$. The public key is $\langle \mathbb{G}, q, g, h, H \rangle$ and the private key is $\langle \mathbb{G}, q, g, x, H \rangle$.
- **Enc:** On input a public key $pk = \langle \mathbb{G}, q, g, h, H \rangle$, choose a uniform $y \in \mathbb{Z}_q$ and set $k_E \| k_M := H(h^y)$. Compute $c' \leftarrow \text{Enc}'_{k_E}(m)$, and output the ciphertext $\langle g^y, c', \text{Mac}_{k_M}(c') \rangle$.
- **Dec:** On input a private key $sk = \langle \mathbb{G}, q, g, x, H \rangle$ and a ciphertext $\langle c, c', t \rangle$, output $\perp$ if $c \notin \mathbb{G}$. Else, compute $k_E \| k_M := H(c^x)$. If $\text{Vrfy}_{k_M}(c', t) \neq 1$ then output $\perp$; otherwise, output $\text{Dec}'_{k_E}(c')$.

DHIES/ECIES.

CCA-secure encryption with Construction 11.19. Combining the KEM in Construction 11.19 with any CCA-secure private-key encryption scheme yields a CCA-secure public-key encryption scheme. (See Theorem 11.14.) Instantiating this approach using Construction 4.18 for the private-key component matches what is done in DHIES/ECIES, variants of which are included in the ISO/IEC 18033-2 standard for public-key encryption. (See Construction 11.23.) Encryption of a message m in these schemes takes the form

$$\langle g^y, \text{Enc}'_{k_E}(m), \text{Mac}_{k_M}(c') \rangle,$$

where Enc' denotes a CPA-secure private-key encryption scheme and c' denotes $\text{Enc}'_{k_E}(m)$. DHIES, the *Diffie–Hellman Integrated Encryption Scheme*, can be used generically to refer to any scheme of this form, or to refer specifically to the case when the group $\mathbb{G}$ is a cyclic subgroup of a finite field. ECIES, the *Elliptic Curve Integrated Encryption Scheme*, refers to the case when $\mathbb{G}$ is an elliptic-curve group. We remark that, in Construction 11.23, it is critical to check during decryption that c , the first component of the ciphertext, is in $\mathbb{G}$. Otherwise, an attacker might request decryption of a malformed

ciphertext $\langle c, c', t \rangle$ in which $c \notin \mathbb{G}$; decrypting such a ciphertext (i.e., without returning $\perp$) might leak information about the private key.

By Theorem 4.19, encrypting a message and then applying a (strong) message authentication code yields a CCA-secure private-key encryption scheme. Combining this with Theorem 11.14, we conclude:

COROLLARY 11.24 *Let Π_E be a CPA-secure private-key encryption scheme, and let Π_M be a strongly secure message authentication code. If the gap-CDH problem is hard relative to $\mathcal{G}$, and H is modeled as a random oracle, then Construction 11.23 is a CCA-secure public-key encryption scheme.*

11.5 RSA Encryption

In this section we turn our attention to encryption schemes based on the *RSA assumption* defined in Section 8.2.4. We remark that although RSA-based encryption is in widespread use today, there is also currently a gradual shift away from using RSA—and toward using CDH/DDH-based cryptosystems relying on elliptic-curve groups—because of the longer key lengths required for RSA-based schemes. We refer to Section 9.3 for further discussion.

11.5.1 Plain RSA

We begin by describing a simple encryption scheme based on the RSA problem. Although the scheme is insecure, it provides a useful starting point for the secure schemes that follow.

Let **GenRSA** be a PPT algorithm that, on input 1^n , outputs a modulus N that is the product of two n -bit primes, along with integers e, d satisfying $ed = 1 \bmod \phi(N)$. (As usual, the algorithm may fail with negligible probability but we ignore that here.) Recall from Section 8.2.4 that such an algorithm can be easily constructed from any algorithm **GenModulus** that outputs a composite modulus N along with its factorization; see Algorithm 11.25.

ALGORITHM 11.25 **RSA key generation GenRSA**

Input: Security parameter 1^n
Output: N, e, d as described in the text
 $(N, p, q) \leftarrow \text{GenModulus}(1^n)$
 $\phi(N) := (p-1)(q-1)$
choose $e > 1$ such that $\gcd(e, \phi(N)) = 1$
compute $d := [e^{-1} \bmod \phi(N)]$
return N, e, d

Let N, e, d be as above, and let $c = m^e \bmod N$. RSA encryption relies on the fact that someone who knows d can recover m from c by computing $[c^d \bmod N]$; this works because

$$c^d = (m^e)^d = m^{ed} = m \bmod N,$$

as discussed in Section 8.2.4. On the other hand, *without* knowledge of d (even if N and e are known) the RSA assumption (cf. Definition 8.46) implies that it is difficult to recover m from c , at least if m is chosen uniformly from $\mathbb{Z}_N^*$. This naturally suggests the public-key encryption scheme shown as Construction 11.26: The receiver runs **GenRSA** to obtain N, e, d ; it publishes N and e as its public key, and keeps d in its private key. To encrypt a message⁴ $m \in \mathbb{Z}_N^*$, a sender computes the ciphertext $c := [m^e \bmod N]$. As we have just noted, the receiver—who knows d —can decrypt c and recover m .

CONSTRUCTION 11.26

Let **GenRSA** be as in the text. Define a public-key encryption scheme as follows:

- **Gen**: on input 1^n run **GenRSA**(1^n) to obtain N, e , and d . The public key is $\langle N, e \rangle$ and the private key is $\langle N, d \rangle$.
- **Enc**: on input a public key $pk = \langle N, e \rangle$ and a message $m \in \mathbb{Z}_N^*$, compute the ciphertext

$$c := [m^e \bmod N].$$

- **Dec**: on input a private key $sk = \langle N, d \rangle$ and a ciphertext $c \in \mathbb{Z}_N^*$, compute the message

$$m := [c^d \bmod N].$$

The plain RSA encryption scheme.

The following gives a worked example of the above (see also Example 8.49).

Example 11.27

Say **GenRSA** outputs $(N, e, d) = (391, 3, 235)$. (Note that $391 = 17 \cdot 23$ and so $\phi(391) = 16 \cdot 22 = 352$. Moreover, $3 \cdot 235 = 1 \bmod 352$.) So the public key is $(391, 3)$ and the private key is $(391, 235)$.

To encrypt the message $m = 158 \in \mathbb{Z}_{391}^*$ using the public key $(391, 3)$, we simply compute $c := [158^3 \bmod 391] = 295$; this is the ciphertext. To decrypt, the receiver computes $[295^{235} \bmod 391] = 158$. $\diamond$

Is the plain RSA encryption scheme secure? The factoring assumption implies that it is computationally infeasible for an attacker who is given the

⁴We assume that $m \in \mathbb{Z}_N^*$. If factoring N is hard, it is computationally difficult to find an $m \in \{1, \dots, N-1\}$ with $m \notin \mathbb{Z}_N^*$ (since then $\gcd(m, N)$ is a nontrivial factor of N).

public key to derive the corresponding private key; see Section 8.2.5. This is necessary—but not sufficient—for a public-key encryption scheme to be secure. The RSA assumption implies that if the message m is *chosen uniformly from* $\mathbb{Z}_N^*$ then an eavesdropper given N , e , and c (namely, the public key and the ciphertext) *cannot recover* m . But these are weak guarantees, and fall far short of the level of security we want! In particular, they leave open the possibility that an attacker can recover the message when it is *not* chosen uniformly from $\mathbb{Z}_N^*$ (indeed, when m is chosen from a small range it is easy to see that an attacker can compute m from the public key and ciphertext). In addition, it does not rule out the possibility that an attacker can learn *partial information* about the message, even when it is uniform (in fact, this is known to be possible). Moreover, plain RSA encryption is *deterministic* and so must be insecure, as we have already discussed in Section 11.2.1.

More Attacks on Plain RSA

We have already noted that plain RSA encryption is not CPA-secure. Nevertheless, there may be a temptation to use plain RSA for encrypting “random messages” and/or in situations where leaking a few bits of information about the message is acceptable. We warn against this in general, and provide here just a few examples of what can go wrong.

(Some of the attacks that follow assume $e = 3$. In some cases the attacks can be extended, at least partially, to larger e ; in any case, as noted in Section 8.2.4, setting $e = 3$ is often done in practice. The attacks should be taken as demonstrating that Construction 11.26 is inadequate, not as indicating that setting $e = 3$ is necessarily a bad choice.)

A quadratic improvement in recovering m . Since plain RSA encryption is deterministic, we know that if $m < B$ then an attacker can determine m from the ciphertext $c = [m^e \bmod N]$ in time $\mathcal{O}(B)$ using the brute-force attack discussed in Section 11.2.1. One might hope, however, that plain RSA encryption can be used if B is large, i.e., if the message is chosen from a reasonably large set of values. One possible scenario where this might occur is in the context of hybrid encryption (see Section 11.3), where the “message” is a random n -bit key and so $B = 2^n$. Unfortunately, there is a clever attack that recovers m , with high probability, in time roughly $\mathcal{O}(\sqrt{B})$. This can make a significant difference in practice: a 2^{80} -time attack (say) is infeasible, but an attack running in time 2^{40} is relatively easy to carry out.

A description of the attack is given as Algorithm 11.28. In our description, we assume $B = 2^n$ and let $\alpha \in (\frac{1}{2}, 1)$ denote some fixed constant (see below).

The time complexity of the algorithm is dominated by the time required to sort the $2^{\alpha n}$ pairs (r, x_r) ; this can be done in time $\mathcal{O}(n \cdot 2^{\alpha n})$. Binary search is used in the second-to-last line to check whether there exists an r with $x_r = [s^e \bmod N]$.

We now sketch why the attack recovers m with high probability. Let $c =$

ALGORITHM 11.28**An attack on plain RSA encryption****Input:** Public key $\langle N, e \rangle$; ciphertext c **Output:** $m < 2^n$ such that $m^e = c \bmod N$ **set** $T := 2^{\alpha n}$ **for** $r = 1$ to T : $x_r := [c/r^e \bmod N]$ **sort** the pairs $\{(r, x_r)\}_{r=1}^T$ by their second component**for** $s = 1$ to T :**if** $x_r \stackrel{?}{=} [s^e \bmod N]$ for some r **return** $[r \cdot s \bmod N]$

$m^e \bmod N$. For appropriate choice of $\alpha > \frac{1}{2}$, it can be shown that if m is a uniform n -bit integer then with high probability there exist r, s with $1 < r \leq s \leq 2^{\alpha n}$ for which $m = r \cdot s$. (For example, if $n = 64$ and so m is a random 64-bit string, then with probability 0.35 there exist r, s of length at most 34 bits such that $m = r \cdot s$. See the references at the end of the chapter for details.) Assuming this to be the case, the above algorithm finds m since

$$c = m^e = (r \cdot s)^e = r^e \cdot s^e \bmod N,$$

and so $x_r = c/r^e = s^e \bmod N$ with $r, s < T$.

Encrypting short messages using small e . The previous attack shows how to recover a message m known to be smaller than some bound B in time roughly $\mathcal{O}(\sqrt{B})$. Here we show how to do the same thing in time $\text{poly}(\|N\|)$ if $B \leq N^{1/e}$ (where this means the e th root of N as a real number).

The attack relies on the observation that when $m < N^{1/e}$, raising m to the e th power modulo N involves no modular reduction; i.e., $[m^e \bmod N]$ is equal to the integer m^e . This means that given the ciphertext $c = [m^e \bmod N]$, an attacker can determine m by computing $m := c^{1/e}$ over the integers (i.e., not modulo N); this can be done easily in time $\text{poly}(\|c\|) = \text{poly}(\|N\|)$ since finding e th roots is easy over the integers and hard only when working mod N .

For small e this represents a serious weakness of plain RSA encryption. For example, if we take $e = 3$ and assume $\|N\| \approx 1024$ bits, then the attack works even when m is a uniform 300-bit integer; this once again rules out security of plain RSA even when used as part of a hybrid encryption scheme.

Encrypting a partially known message. This attack can be viewed as a generalization of the previous one. It assumes a sender who encrypts a message, part of which is known (something that should not result in an attack when using a secure encryption scheme). Here we rely on a powerful result of Coppersmith that we state without proof:

THEOREM 11.29 *Let $p(x)$ be a polynomial of degree e . Then in time $\text{poly}(\|N\|, e)$ one can find all m such that $p(m) = 0 \bmod N$ and $|m| \leq N^{1/e}$.*

Due to the dependence of the running time on e , the attack is only practical for small e . In what follows we assume $e = 3$ for concreteness.

Assume a sender encrypts a message $m = m_1 \| m_2$ to a receiver with public key $\langle N, 3 \rangle$, where the first portion m_1 of the message is known but the second portion m_2 is not. For concreteness, say m_2 is k bits long, so $m = B \cdot m_1 + m_2$ where we let $B = 2^k$. Given the resulting ciphertext $c = [(m_1 \| m_2)^3 \bmod N]$, an eavesdropper can define $p(x) \stackrel{\text{def}}{=} (2^k \cdot m_1 + x)^3 - c$, a cubic polynomial. This polynomial has m_2 as a root (modulo N), and $|m_2| < B$. Theorem 11.29 thus implies that the attacker can compute m_2 efficiently as long as $B \leq N^{1/3}$. A similar attack works when m_2 is known but m_1 is not.

Encrypting related messages.⁵ This attack assumes a sender who encrypts two related messages to the same receiver (something that should not result in an attack when using a secure encryption scheme). Assume the sender encrypts both m and $m + \delta$ to a receiver with public key $\langle N, e \rangle$, where the offset δ is known but m is not. Given the two ciphertexts $c_1 = [m^e \bmod N]$ and $c_2 = [(m + \delta)^e \bmod N]$, an eavesdropper can define the two polynomials $f_1(x) \stackrel{\text{def}}{=} x^e - c_1$ and $f_2(x) \stackrel{\text{def}}{=} (x + \delta)^e - c_2$ (modulo N), each of degree e . Note that $x = m$ is a root (modulo N) of both polynomials, and so the linear term $(x - m)$ is a factor of both. Thus, if the greatest common divisor of $f_1(x)$ and $f_2(x)$ (as polynomials over $\mathbb{Z}_N^*$) is linear, it will reveal m . The greatest common divisor can be computed in time $\text{poly}(\|N\|, e)$ using an algorithm similar to the one in Appendix B.1.2; thus, this attack is feasible for small e .

Sending the same message to multiple receivers.⁶ Our final attack assumes a sender who encrypts the same message to multiple receivers (something that, once again, should not result in an attack when using a secure encryption scheme). Let $e = 3$, and say the same message m is encrypted to three different parties holding public keys $pk_1 = \langle N_1, 3 \rangle$, $pk_2 = \langle N_2, 3 \rangle$, and $pk_3 = \langle N_3, 3 \rangle$, respectively. Assume $\gcd(N_i, N_j) = 1$ for distinct i, j ; if not, then at least one of the moduli can be factored immediately and the message m can be easily recovered. An eavesdropper sees

$$c_1 = [m^3 \bmod N_1], \quad c_2 = [m^3 \bmod N_2], \quad \text{and} \quad c_3 = [m^3 \bmod N_3].$$

Let $N^* = N_1 N_2 N_3$. An extended version of the Chinese remainder theorem says that there exists a unique non-negative integer $\hat{c} < N^*$ such that

$$\begin{aligned} \hat{c} &= c_1 \bmod N_1 \\ \hat{c} &= c_2 \bmod N_2 \\ \hat{c} &= c_3 \bmod N_3. \end{aligned}$$

⁵This attack relies on some algebra slightly beyond what we have covered in this book.

⁶This attack relies on the Chinese remainder theorem presented in Section 8.1.5.

Moreover, using techniques similar to those shown in Section 8.1.5 it is possible to compute $\hat{c}$ efficiently given the public keys and the above ciphertexts. Note finally that m^3 satisfies the above equations, and $m^3 < N^*$ since $m < \min\{N_1, N_2, N_3\}$. As in the previous attack, this means that $\hat{c} = m^3$ *over the integers* (i.e., with no modular reduction taking place), and so the message m can be recovered by computing the integer cube root of $\hat{c}$.

11.5.2 Padded RSA and PKCS #1 v1.5

Although plain RSA is insecure, it does suggest one general approach to public-key encryption based on the RSA problem: to encrypt a message m using public key $\langle N, e \rangle$, first map m to an element $\hat{m} \in \mathbb{Z}_N^*$; then compute the ciphertext $c = [\hat{m}^e \bmod N]$. To decrypt a ciphertext c , the receiver computes $\hat{m} = [c^d \bmod N]$ and then recovers the original message m . For the receiver to be able to recover the message, the mapping from messages to elements of $\mathbb{Z}_N^*$ must be (efficiently) *reversible*. For a scheme following this approach to have a hope of being CPA-secure, the mapping must be *randomized* so encryption is not deterministic. This is, of course, a necessary condition but not a sufficient one, and security of the encryption scheme depends critically on the specific mapping that is used.

One simple implementation of the above idea is to *randomly pad* the message before encrypting. That is, to map a message m (viewed as a bit-string) to an element of $\mathbb{Z}_N^*$, the sender chooses a uniform bit-string $r \in \{0, 1\}^\ell$ (for some appropriate ℓ) and sets $\hat{m} := r \| m$; the resulting value can naturally be interpreted as an integer in $\mathbb{Z}_N^*$, and this mapping is clearly reversible. See Construction 11.30. (The bounds on $\ell(n)$ and the length of m ensure that the integer $\hat{m}$ is smaller than N .)

The construction is parameterized by a value ℓ that determines the length of the random padding used. Security of the scheme depends on ℓ . There is an obvious brute-force attack on the scheme that runs in time 2^ℓ , so if ℓ is too short (in particular, if $\ell(n) = \mathcal{O}(\log n)$), the scheme is insecure. At the other extreme, we show in the following section (essentially) that when the padding is as large as possible, and m is just a single bit, then it is possible to prove security based on the RSA assumption. In intermediate cases, the situation is less clear: for certain ranges of ℓ we cannot prove security based on the RSA assumption but no polynomial-time attacks are known either. We defer further discussion until after our treatment of PKCS #1 v1.5 next.

RSA PKCS #1 v1.5. The *RSA Laboratories Public-Key Cryptography Standard (PKCS) #1 version 1.5*, issued in 1993, utilizes a variant of padded RSA encryption. For a public key $pk = \langle N, e \rangle$ of the usual form, let k denote the length of N in bytes; i.e., k is the integer satisfying $2^{8(k-1)} \leq N < 2^{8k}$. Messages m to be encrypted are assumed to be a multiple of 8 bits long and can have length anywhere from one to $k - 11$ bytes. Encryption of a message

CONSTRUCTION 11.30

Let **GenRSA** be as before, and let ℓ be a function with $\ell(n) \leq 2n - 4$ for all n . Define a public-key encryption scheme as follows:

- **Gen**: on input 1^n , run **GenRSA**(1^n) to obtain (N, e, d) . Output the public key $pk = \langle N, e \rangle$, and the private key $sk = \langle N, d \rangle$.
- **Enc**: on input a public key $pk = \langle N, e \rangle$ and a message $m \in \{0, 1\}^{\|N\| - \ell(n) - 2}$, choose a uniform string $r \in \{0, 1\}^{\ell(n)}$ and interpret $\hat{m} := r \| m$ as an element of $\mathbb{Z}_N^*$. Output the ciphertext

$$c := [\hat{m}^e \bmod N].$$

- **Dec**: on input a private key $sk = \langle N, d \rangle$ and a ciphertext $c \in \mathbb{Z}_N^*$, compute

$$\hat{m} := [c^d \bmod N],$$

and output the $\|N\| - \ell(n) - 2$ least-significant bits of $\hat{m}$.

The padded RSA encryption scheme.

m that is D -bytes long is computed as

$$[(0x00 \| 0x02 \| r \| 0x00 \| m)^e \bmod N],$$

where r is a randomly generated, $(k - D - 3)$ -byte string with none of its bytes equal to $0x00$. (This latter condition enables the message to be unambiguously recovered upon decryption.) Note that the maximum allowed length of m ensures that the length of r is at least 8 bytes.

Unfortunately, PKCS #1 v1.5 as specified is not CPA-secure because it allows using random padding that is too short. This is best illustrated by showing that an attacker can determine the initial portion of a message known to have many trailing 0s. For simplicity, say $m = b \| \underbrace{0 \cdots 0}_L$ where $b \in \{0, 1\}$ is

unknown and m is as long as possible (so $L = 8 \cdot (k - 11) - 1$). Encryption of m gives a ciphertext c with

$$c = (0x00 \| 0x02 \| r \| 0x00 \| b \| 0 \cdots 0)^e \bmod N.$$

An attacker can compute $c' = c / (2^L)^e \bmod N$; note that

$$c' = \left(\frac{0x00 \| 0x02 \| r \| 0x00 \| b \| 0 \cdots 0}{10 \cdots 0} \right)^e = (0x00 \| 0x02 \| r \| 0x00 \| b)^e \bmod N.$$

The integer $0x02 \| r \| 0x00 \| b$ is 75 bits long (note that $0x02 = 0000\,0010$, and all the high-order 0-bits don't count), and so an attacker can now apply the “short-message attack,” or the attack based on encrypting a partially known message, from the previous section. To avoid these attacks we need to take r of length at least $\|N\|/e$. Even if e is large, however, the “quadratic-improvement attack” from the previous section shows that r can be recovered, with high probability, in time roughly $2^{\|r\|/2}$.

If we force r to be roughly half the length of N , and correspondingly reduce the maximum message length, then it is reasonable to conjecture that the encryption scheme in PKCS #1 v1.5 is CPA-secure. (We stress, however, that no proof of security based on the RSA assumption is known.) Nevertheless, because of a serious chosen-ciphertext attack on the scheme, described briefly in Section 11.5.5, newer versions of the PKCS #1 standard have been introduced and should be used instead.

11.5.3 *CPA-Secure Encryption without Random Oracles

In this section we show an encryption scheme that can be proven to be CPA-secure based on the RSA assumption. We begin by describing a specific hard-core predicate (see Section 7.1.3) for the RSA problem and then show how to use that hard-core predicate to encrypt a single bit. We then extend this scheme to give a KEM.

The schemes described in this section are mainly of theoretical interest and are not used in practice. This is because they are less efficient than alternative RSA-based constructions that can be proven secure in the random-oracle model (cf. Section 5.5). We will see examples of such encryption schemes in the sections that follow.

A hard-core predicate for the RSA problem. Loosely speaking, the RSA assumption says that given N, e , and $[x^e \bmod N]$ (for x chosen uniformly from $\mathbb{Z}_N^*$), it is infeasible to recover x . By itself, this says nothing about the computational difficulty of computing some specific information about x . Can we isolate some particular bit of information about x that is hard to compute from N, e and $[x^e \bmod N]$? The notion of a *hard-core predicate* captures exactly this requirement. (Hard-core predicates were introduced in Section 7.1.3. The fact that the RSA assumption gives a family of one-way permutations is discussed in Section 8.4.1. Our treatment here is self-contained, however.) It turns out that the least-significant bit of x , denoted $\text{lsb}(x)$, is a hard-core predicate for the RSA problem.

Define the following experiment for a given algorithm GenRSA (with the usual behavior) and algorithm $\mathcal{A}$:

The RSA hard-core predicate experiment $\text{RSA-lsb}_{\mathcal{A}, \text{GenRSA}}(1^n)$:

1. Run $\text{GenRSA}(1^n)$ to obtain (N, e, d) .
2. choose a uniform $x \in \mathbb{Z}_N^*$ and compute $y := [x^e \bmod N]$.
3. $\mathcal{A}$ is given N, e, y , and outputs a bit b .
4. The output of the experiment is 1 if and only if $\text{lsb}(x) = b$.

Observe that $\text{lsb}(x)$ is a uniform bit when $x \in \mathbb{Z}_N^*$ is uniform. $\mathcal{A}$ can guess $\text{lsb}(x)$ with probability $1/2$ by simply outputting a uniform bit b . The following theorem states that if the RSA problem is hard, then no efficient

algorithm $\mathcal{A}$ can do significantly better than this; i.e., the least-significant bit is a hard-core predicate of the RSA permutation.

THEOREM 11.31 *If the RSA problem is hard relative to GenRSA then for all probabilistic polynomial-time algorithms $\mathcal{A}$ there is a negligible function negl such that $\Pr[\text{RSA-lsb}_{\mathcal{A}, \text{GenRSA}}(n) = 1] \leq \frac{1}{2} + \text{negl}(n)$.*

A full proof of this theorem is beyond the scope of this book. However, we provide some intuition for the theorem by sketching a proof of a weaker result: that the RSA assumption implies $\Pr[\text{RSA-lsb}_{\mathcal{A}, \text{GenRSA}}(n) = 1] < 1$ for all probabilistic polynomial-time $\mathcal{A}$. To prove this we show that an efficient algorithm that *always* correctly computes $\text{lsb}(r)$ from N, e , and $[r^e \bmod N]$ can be used to efficiently recover x (in its entirety) from N, e , and $[x^e \bmod N]$.

Fix N and e , and let $\mathcal{A}$ be an algorithm such that $\mathcal{A}([r^e \bmod N]) = \text{lsb}(r)$. Given N, e , and $y = [x^e \bmod N]$, we will recover the bits of x one-by-one, from least to most significant. To determine $\text{lsb}(x)$ we simply run $\mathcal{A}(y)$. There are now two cases:

Case 1: $\text{lsb}(x) = 0$. Note that $y/2^e = (x/2)^e \bmod N$, and because x is even (i.e., $\text{lsb}(x) = 0$), 2 divides the integer x . So $x/2$ is just the right-wise bit-shift of x , and $\text{lsb}(x/2)$ is equal to $2\text{sb}(x)$, the 2nd-least-significant bit of x . So we can obtain $2\text{sb}(x)$ by computing $y' := [y/2^e \bmod N]$ and then running $\mathcal{A}(y')$.

Case 2: $\text{lsb}(x) = 1$. Here $[x/2 \bmod N] = (x + N)/2$. So $\text{lsb}([x/2 \bmod N])$ is equal to $2\text{sb}(x + N)$; the latter is equal to $1 \oplus 2\text{sb}(N) \oplus 2\text{sb}(x)$ (we have a carry bit in the second position because both x and N are odd). So if we compute $y' := [y/2^e \bmod N]$, then $2\text{sb}(x) = \mathcal{A}(y') \oplus 1 \oplus 2\text{sb}(N)$.

Continuing in this way, we can recover all the bits of x .

Encrypting one bit. We can use the hard-core predicate identified above to encrypt a single bit. The idea is straightforward: to encrypt the message $m \in \{0, 1\}$, the sender chooses uniform $r \in \mathbb{Z}_N^*$ subject to the constraint that $\text{lsb}(r) = m$; the ciphertext is $c := [r^e \bmod N]$. See Construction 11.32.

THEOREM 11.33 *If the RSA problem is hard relative to GenRSA then Construction 11.32 is CPA-secure.*

PROOF Let Π denote Construction 11.32. We prove that Π has indistinguishable encryptions in the presence of an eavesdropper; by Proposition 11.3, this implies it is CPA-secure.

Let $\mathcal{A}$ be a probabilistic polynomial-time adversary. Without loss of gener-

CONSTRUCTION 11.32

Let GenRSA be as usual, and define a public-key encryption scheme as follows:

- **Gen**: on input 1^n , run $\text{GenRSA}(1^n)$ to obtain (N, e, d) . Output the public key $pk = \langle N, e \rangle$, and the private key $sk = \langle N, d \rangle$.
- **Enc**: on input a public key $pk = \langle N, e \rangle$ and a message $m \in \{0, 1\}$, choose a uniform $r \in \mathbb{Z}_N^*$ subject to the constraint that $\text{lsb}(r) = m$. Output the ciphertext $c := [r^e \bmod N]$.
- **Dec**: on input a private key $sk = \langle N, d \rangle$ and a ciphertext c , compute $r := [c^d \bmod N]$ and output $\text{lsb}(r)$.

Single-bit encryption using a hard-core predicate for RSA.

ality, we may assume $m_0 = 0$ and $m_1 = 1$ in experiment $\text{PubK}_{\mathcal{A}, \Pi}^{\text{eav}}(n)$. So

$$\begin{aligned} \Pr[\text{PubK}_{\mathcal{A}, \Pi}^{\text{eav}}(n) = 1] &= \frac{1}{2} \cdot \Pr[\mathcal{A}(N, e, c) = 0 \mid c \text{ is an encryption of } 0] \\ &\quad + \frac{1}{2} \cdot \Pr[\mathcal{A}(N, e, c) = 1 \mid c \text{ is an encryption of } 1]. \end{aligned}$$

Consider running $\mathcal{A}$ in experiment RSA-lsb . By definition,

$$\Pr[\text{RSA-lsb}_{\mathcal{A}, \text{GenRSA}}(n) = 1] = \Pr[\mathcal{A}(N, e, [r^e \bmod N]) = \text{lsb}(r)],$$

where r is uniform in $\mathbb{Z}_N^*$. Since $\Pr[\text{lsb}(r) = 1] = 1/2$, we have

$$\begin{aligned} \Pr[\text{RSA-lsb}_{\mathcal{A}, \text{GenRSA}}(n) = 1] &= \frac{1}{2} \cdot \Pr[\mathcal{A}(N, e, [r^e \bmod N]) = 0 \mid \text{lsb}(r) = 0] \\ &\quad + \frac{1}{2} \cdot \Pr[\mathcal{A}(N, e, [r^e \bmod N]) = 1 \mid \text{lsb}(r) = 1]. \end{aligned}$$

Noting that encrypting $m \in \{0, 1\}$ corresponds exactly to choosing uniform r subject to the constraint that $\text{lsb}(r) = m$, we see that

$$\Pr[\text{PubK}_{\mathcal{A}, \Pi}^{\text{eav}}(n) = 1] = \Pr[\text{RSA-lsb}_{\mathcal{A}, \text{GenRSA}}(n) = 1].$$

Theorem 11.31 thus implies that there is a negligible function negl such that

$$\Pr[\text{PubK}_{\mathcal{A}, \Pi}^{\text{eav}}(n) = 1] \leq \frac{1}{2} + \text{negl}(n),$$

as desired. ■

Constructing a KEM. We now show how to extend Construction 11.32 so as to obtain a KEM with key length n . A naive way of doing this would be to simply choose a uniform, n -bit key k and then encrypt the bits of k one-by-one

using n invocations of Construction 11.32. This would result in a rather long ciphertext consisting of n elements of $\mathbb{Z}_N$.

A better approach is for the sender to apply the RSA permutation (namely, raising to the e th power modulo N) repeatedly, starting from an initial, uniform value c_1 . That is, the sender will successively compute c_1^e , followed by $(c_1^e)^e = c_1^{e^2}$, and so on, up to $c_1^{e^n}$ (all modulo N). The final value $[c_1^{e^n} \bmod N]$ will be the ciphertext, and the sequence of bits $\text{lsb}(c_1), \text{lsb}(c_1^e), \dots, \text{lsb}(c_1^{e^{n-1}})$ is the key. To decrypt a ciphertext c , the receiver simply reverses this process, successively computing $c^d, (c^d)^d = c^{d^2}$ up to c^{d^n} (again, all modulo N) to recover the initial value $c_1 = c^{d^n}$ used by the sender. Having recovered c_1 , the receiver can then recompute $c_1^e, \dots, c_1^{e^n}$ and obtain the key.

It is possible to implement decryption more efficiently using the fact that the receiver knows the order of the group $\mathbb{Z}_N^*$. At key-generation time, the receiver can pre-compute $d' := [d^n \bmod \phi(N)]$ and store d' as part of its private key. Then, rather than computing the n successive modular exponentiations $c^d, c^{d^2}, \dots$ to obtain c_1 , the receiver can instead directly compute $c_1 := [c^{d'} \bmod N]$. This works, of course, since

$$c^{d^n} \bmod N = c^{[d^n \bmod \phi(N)]} = c^{d'} \bmod N.$$

The above is formally described as Construction 11.34.

CONSTRUCTION 11.34

Let **GenRSA** be as usual, and define a KEM as follows:

- **Gen**: on input 1^n , run **GenRSA**(1^n) to obtain (N, e, d) . Then compute $d' := [d^n \bmod \phi(N)]$ (note that $\phi(N)$ can be computed from (N, e, d) or obtained during the course of running **GenRSA**). Output $pk = \langle N, e \rangle$ and $sk = \langle N, d' \rangle$.
- **Encaps**: on input $pk = \langle N, e \rangle$ and 1^n , choose a uniform $c_1 \in \mathbb{Z}_N^*$. Then for $i = 1, \dots, n$ do:

1. Compute $k_i := \text{lsb}(c_i)$.
2. Compute $c_{i+1} := [c_i^e \bmod N]$.

Output the ciphertext c_{n+1} and the key $k = k_1 \cdots k_n$.

- **Decaps**: on input $sk = \langle N, d' \rangle$ and a ciphertext c , compute $c_1 := [c^{d'} \bmod N]$. Then for $i = 1, \dots, n$ do:

1. Compute $k_i := \text{lsb}(c_i)$.
2. Compute $c_{i+1} := [c_i^e \bmod N]$.

Output the key $k = k_1 \cdots k_n$.

A KEM using a hard-core predicate for RSA.

The construction is reminiscent of the approach used to construct a pseudorandom generator from a one-way permutation toward the end of Section 7.4.2. If we let f denote the RSA permutation relative to some public key $\langle N, e \rangle$ (i.e., $f(x) \stackrel{\text{def}}{=} [x^e \bmod N]$), then CPA-security of Construction 11.34 is equivalent to pseudorandomness of $\text{lsb}(f^{n-1}(c_1)), \dots, \text{lsb}(c_1)$ even conditioned on the value $c = f^n(c_1)$. This, in turn, can be proven using Theorem 11.31 and the techniques from Section 7.4.2. (The only difference is that in Section 7.4.2 the value $f^n(c_1)$ was itself a uniform n -bit string, whereas here it is a uniform element of $\mathbb{Z}_N^*$. Pseudorandomness of the successive hard-core predicates is independent of the domain of f .) Summarizing:

THEOREM 11.35 *If the RSA problem is hard relative to GenRSA then Construction 11.34 is a CPA-secure KEM.*

Efficiency. Construction 11.34 is reasonably efficient. To be concrete, assume that $n = 128$, the RSA modulus N is 2048 bits long, and the public exponent e is 3 so that exponentiation to the power e modulo N can be computed using two modular multiplications. (See Appendix B.2.3.) Encryption then requires $2n = 256$ modular multiplications. Decryption can be done with one full modular exponentiation (at the cost of approximately $1.5 \cdot 2048 = 3072$ modular multiplications) plus an additional 256 modular multiplications. The cost of decryption is thus only about 8% less efficient than for the plain RSA encryption scheme. In contrast, encryption is significantly more expensive than in plain RSA, but in many applications decryption time is much more crucial (since it may be implemented by a server that is performing thousands of decryptions simultaneously).

11.5.4 OAEP and RSA PKCS #1 v2.0

We have so far not considered RSA-based encryption schemes that are CCA-secure. We first show that all the RSA-based encryption schemes we have seen so far are vulnerable to chosen-ciphertext attacks.

Plain RSA encryption. Plain RSA is not even CPA-secure. But it does ensure that if $m \in \mathbb{Z}_N^*$ is uniform then an attacker who eavesdrops on the encryption $c = [m^e \bmod N]$ of m with respect to the public key $\langle N, e \rangle$ cannot recover m . Even this weak guarantee no longer holds in a setting where chosen-ciphertext attacks are possible. As in the case of El Gamal encryption, this is a consequence of the fact that plain RSA is *malleable*: given the encryption $c = [m^e \bmod N]$ of an unknown message m , it is easy to generate a ciphertext c' that is an encryption of $[2m \bmod N]$ by setting

$$\begin{aligned} c' &:= [2^e \cdot c \bmod N] \\ &= 2^e \cdot m^e = (2m)^e \bmod N. \end{aligned}$$

RSA PKCS #1 v1.5. Padded RSA encryption, which is conjectured to be CPA-secure for the right setting of the parameters, is vulnerable to essentially the same attack as plain RSA encryption is. But there is also a more interesting chosen-ciphertext attack on PKCS #1 v1.5 encryption that, in contrast to the attack presented above, does *not* require full access to a decryption oracle; it only requires access to a “partial” decryption oracle that indicates whether or not decryption of some ciphertext returns an error. This makes the attack much more practical, as the attack can be carried out whenever an attacker can distinguish the behavior of the receiver after a decryption success from its behavior after a decryption failure, as in the case of the padding-oracle attack shown in Section 3.7.2.

Recall that the public-key encryption scheme defined in the PKCS #1 v1.5 standard uses a variant of padded RSA encryption where the padding is done in a specific way. In particular, the two high-order bytes of the padded message are always $0x00\|0x02$. When decrypting, the receiver is supposed to check that the two high-order bytes match these values, and return an error if this is not the case. In 1998, Bleichenbacher developed a chosen-ciphertext attack that exploits the fact that this check is done. Roughly, given a ciphertext c that corresponds to an honest encryption of some unknown message m with respect to a public key $\langle N, e \rangle$, the attack repeatedly chooses uniform $s \in \mathbb{Z}_N^*$ and submits the ciphertext $c' := [s^e \cdot c \bmod N]$ to the receiver. Say $c = [\hat{m}^e \bmod N]$ where

$$\hat{m} = 0x00\|0x02\|r\|0x00\|m,$$

as specified by PKCS #1 v1.5. Then decryption of c' will give the intermediate result $\hat{m}' = [s \cdot \hat{m} \bmod N]$, and the receiver will return an error unless the top two bytes of $\hat{m}'$ are exactly $0x00\|0x02$. (Other checks are done as well, but we ignore those for simplicity.) Any time decryption succeeds, the attacker learns that the top two bytes of $s \cdot \hat{m} \bmod N$ are $0x00\|0x02$, where s is known. A number of equations of this type suffice for the attacker to learn $\hat{m}$ and recover all of the original message m .

The CPA-secure KEM. In Section 11.5.3 we showed a construction of a KEM that can be proven CPA-secure based on the RSA assumption. That construction is also insecure against a chosen-ciphertext attack; we leave the details as an exercise.

RSA-OAEP

In this section we explore a construction of RSA-based CCA-secure encryption using *optimal asymmetric encryption padding* (OAEP). The resulting RSA-OAEP scheme follows the idea (used also in Section 11.5.2) of taking a message m , transforming it to an element $\hat{m} \in \mathbb{Z}_N^*$, and then letting $c = [\hat{m}^e \bmod N]$ be the ciphertext. The transformation here, however, is

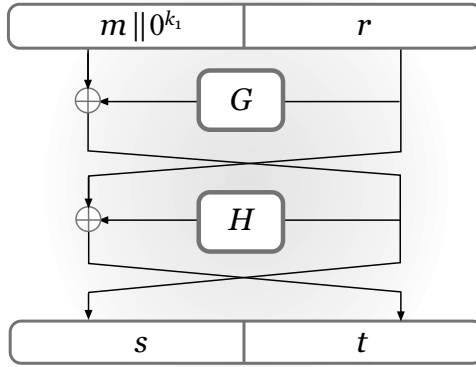


FIGURE 11.4: The OAEP mechanism.

more complex than before. A version of RSA-OAEP has been standardized as part of RSA PKCS #1 since version 2.0.

Let $\ell(n), k_0(n), k_1(n)$ be integer-valued functions with $k_0(n), k_1(n) = \Theta(n)$ and such that $\ell(n) + k_0(n) + k_1(n)$ is less than the minimum bit-length of moduli output by $\text{GenRSA}(1^n)$. Fix n , and let $\ell = \ell(n)$, $k_0 = k_0(n)$, and $k_1 = k_1(n)$. Let $G : \{0, 1\}^{k_0} \rightarrow \{0, 1\}^{\ell+k_1}$ and $H : \{0, 1\}^{\ell+k_1} \rightarrow \{0, 1\}^{k_0}$ be two hash functions that will be modeled as independent random oracles. (Although using more than one random oracle was not discussed in Section 5.5.1, this is interpreted in the natural way.) The transformation defined by OAEP is a two-round Feistel network with G and H as round functions; see Figure 11.4. In detail, padding of a message $m \in \{0, 1\}^\ell$ is done as follows: first set $m' := m \parallel 0^{k_1}$ and choose a uniform $r \in \{0, 1\}^{k_0}$. Then compute

$$s := m' \oplus G(r) \in \{0, 1\}^{\ell+k_1}, \quad t := r \oplus H(s) \in \{0, 1\}^{k_0},$$

and set $\hat{m} := s \parallel t$.

To encrypt a message m with respect to the public key $\langle N, e \rangle$, the sender generates $\hat{m}$ as above and outputs the ciphertext $c := \hat{m}^e \bmod N$. (Note that $\hat{m}$, interpreted as an integer, is less than N because of the constraints on ℓ, k_0, k_1 .) To decrypt, the receiver computes $\hat{m} := [c^d \bmod N]$ and lets $s \parallel t := \hat{m}$ with s and t of the appropriate lengths. It then inverts the Feistel network by computing $r := H(s) \oplus t$ and $m' := G(r) \oplus s$. Importantly, the receiver then verifies that the trailing k_1 bits of m' are all 0; if not, the ciphertext is rejected and an error message is returned. Otherwise, the k_1 least-significant 0s of m' are discarded, and the remaining ℓ bits of m' are output as the message. This process is described in Construction 11.36.

The proof of CCA-security for RSA-OAEP is rather complicated, and we do not give it here. Instead, we merely provide some intuition. First consider

CONSTRUCTION 11.36

Let GenRSA be as in the previous sections, and ℓ, k_0, k_1 be as described in the text. Let $G : \{0, 1\}^{k_0} \rightarrow \{0, 1\}^{\ell+k_1}$ and $H : \{0, 1\}^{\ell+k_1} \rightarrow \{0, 1\}^{k_0}$ be functions. Construct a public-key encryption scheme as follows:

- **Gen**: on input 1^n , run $\text{GenRSA}(1^n)$ to obtain (N, e, d) . The public key is $\langle N, e \rangle$ and the private key is $\langle N, d \rangle$.
- **Enc**: on input a public key $\langle N, e \rangle$ and a message $m \in \{0, 1\}^\ell$, set $m' := m \| 0^{k_1}$ and choose a uniform $r \in \{0, 1\}^{k_0}$. Then compute

$$s := m' \oplus G(r), \quad t := r \oplus H(s)$$

and set $\hat{m} := s \| t$. Output the ciphertext $c := [\hat{m}^e \bmod N]$.

- **Dec**: on input a private key $\langle N, d \rangle$ and a ciphertext $c \in \mathbb{Z}_N^*$, compute $\hat{m} := [c^d \bmod N]$. If $\|\hat{m}\| > \ell + k_0 + k_1$, output $\perp$. Otherwise, parse $\hat{m}$ as $s \| t$ with $s \in \{0, 1\}^{\ell+k_1}$ and $t \in \{0, 1\}^{k_0}$. Compute $r := H(s) \oplus t$ and $m' := G(r) \oplus s$. If the least-significant k_1 bits of m' are not all 0, output $\perp$. Otherwise, output the ℓ most-significant bits of $\hat{m}$.

The RSA-OAEP encryption scheme.

CPA-security. During encryption the sender computes

$$m' := m \| 0^{k_1}, \quad s := m' \oplus G(r), \quad t := r \oplus H(s)$$

for uniform r ; the ciphertext is $[(s \| t)^e \bmod N]$. If the attacker never queries r to G then, since we model G as a random function, the value $G(r)$ is uniform from the attacker's point of view and so m is masked with a uniform string just as in the one-time pad encryption scheme. Thus, if the attacker never queries r to G then no information about the message is leaked.

Can the attacker query r to G ? Note that the value of r is itself masked by $H(s)$. So the attacker has no information about r unless it first queries s to H . If the attacker does not query s to H then the attacker may get lucky and guess r anyway, but if we set the length of r (i.e., k_0) sufficiently long then the probability of this is negligible.

So, the only way for the attacker to learn anything about m is to first query s to H . This would require the attacker to compute s from the (uniform) ciphertext $[(s \| t)^e \bmod N]$. Note that computing s from $[(s \| t)^e \bmod N]$ is not the RSA problem, which instead involves computing *both* s and t . Nevertheless, for the right settings of the parameters we can use Theorem 11.29 to show that recovering s enables recovery of t in polynomial time, and so recovering s is computationally infeasible if the RSA problem is hard.

Arguing CCA-security involves additional complications, but the basic idea is to show that every decryption-oracle query c made by the attacker falls into one of two categories: either the attacker obtained c by legally encrypting some message m (in which case the attacker learns nothing from the decryption query), or else decryption of c returns an error. This is a consequence of

the fact that the receiver checks that the k_1 low-order bits of $\hat{m}$ are 0 during decryption; if the attacker did not construct some ciphertext c by legally encrypting some message, the probability that this condition holds is negligible. The formal proof is complicated by the fact that the attacker's decryption-oracle queries must be answered correctly without knowledge of the private key, which means there must be an efficient way to determine whether to return an error or not and, if not, what message to return. This is accomplished by looking at the adversary's queries to the random oracles G, H .

Manger's chosen-ciphertext attack on PKCS #1 v2.0. In 2001, James Manger showed a chosen-ciphertext attack against certain implementations of the RSA encryption scheme specified in PKCS #1 v2.0—even though what was specified was a variant of RSA-OAEP! Since Construction 11.36 is CCA-secure (assuming the RSA problem is hard), how is this possible?

Examining the decryption algorithm in Construction 11.36, note that there are two ways an error can occur: either $\hat{m} \in \mathbb{Z}_N^*$ is too large, or $m' \in \{0, 1\}^{\ell+k_1}$ does not have enough trailing 0s. In Construction 11.36, the receiver is supposed to return the *same* error (denoted $\perp$) in either case. In some implementations, however, the receiver would output *different* errors depending on which step failed. This single bit of additional information enables an attacker to mount a chosen-ciphertext attack that recovers a message m in its entirety from an encryption of that message, using only about $\|N\|$ queries to an oracle that leaks the error message upon decryption. This shows the importance of implementing cryptographic schemes exactly as specified, since the resulting proof and analysis may no longer apply if aspects of the scheme are changed.

Even if the same error is returned in both cases, an attacker can determine where the error occurs if the *time* to return the error is different. (This is a great example of how an attacker is not limited to examining the inputs/outputs of an algorithm, but can use *side-channel information* to attack a scheme.) Implementations must be careful to ensure that the time to return an error is identical regardless of where the error occurs.

11.5.5 *A CCA-Secure KEM in the Random-Oracle Model

We show here a construction of an RSA-based KEM that is CCA-secure in the random-oracle model. (Recall from Theorem 11.14 that any such construction can be used in conjunction with any CCA-secure *private*-key encryption scheme to give a CCA-secure *public*-key encryption scheme.) As compared to the RSA-OAEP scheme from the previous section, the main advantage is the simplicity of both the construction and its proof of security. Its main disadvantage is that it results in longer ciphertexts when encrypting short messages since it requires the KEM/DEM paradigm whereas RSA-OAEP does not. For encrypting long messages, however, RSA-OAEP would also be used as part of a hybrid encryption scheme, and would result in an encryption scheme having similar efficiency to what could be obtained using the KEM shown here.

The KEM we describe is included as part of the ISO/IEC 18033-2 standard for public-key encryption. In the scheme, the public key includes $\langle N, e \rangle$ as usual, and a function $H : \mathbb{Z}_N^* \rightarrow \{0, 1\}^n$ is specified that will be modeled as a random oracle in the analysis. (This function can be based on some underlying cryptographic hash function, as discussed in Section 5.5. We omit the details.) To encapsulate a key, the sender chooses uniform $r \in \mathbb{Z}_N^*$ and then computes the ciphertext $c := [r^e \bmod N]$ and the key $k := H(r)$. To decrypt a ciphertext c , the receiver simply recovers r in the usual way and then re-derives the same key $k := H(r)$. See Construction 11.37.

CONSTRUCTION 11.37

Let GenRSA be as usual, and construct a KEM as follows:

- **Gen:** on input 1^n , run GenRSA(1^n) to compute (N, e, d) . The public key is $\langle N, e \rangle$, and the private key is $\langle N, d \rangle$.
As part of key generation, a function $H : \mathbb{Z}_N^* \rightarrow \{0, 1\}^n$ is specified, but we leave this implicit.
- **Encaps:** on input public key $\langle N, e \rangle$ and 1^n , choose a uniform $r \in \mathbb{Z}_N^*$. Output the ciphertext $c := [r^e \bmod N]$ and the key $k := H(r)$.
- **Decaps:** on input private key $\langle N, d \rangle$ and a ciphertext $c \in \mathbb{Z}_N^*$, compute $r := [c^d \bmod N]$ and output the key $k := H(r)$.

A CCA-secure KEM (in the random-oracle model).

CPA-security of the scheme is immediate. Indeed, the ciphertext c is equal to $[r^e \bmod N]$ for uniform $r \in \mathbb{Z}_N^*$, and so the RSA assumption implies that an eavesdropper who observes c will be unable to compute r . This means, in turn, that the eavesdropper will not query r to H , and thus the value of the key $k \stackrel{\text{def}}{=} H(r)$ remains uniform from the attacker's point of view.

In fact, the above extends to show CCA-security as well. This is because answering a decapsulation-oracle query for any ciphertext $\tilde{c} \neq c$ only involves evaluating H at some input $[\tilde{c}^d \bmod N] = \tilde{r} \neq r$. Thus, the attacker's decapsulation-oracle queries do not reveal any additional information about the key $H(r)$ encapsulated by the challenge ciphertext. (A formal proof is slightly more involved since we must show how it is possible to *simulate* the answers to decapsulation-oracle queries without knowledge of the private key. Nevertheless, this turns out not to be very difficult.)

THEOREM 11.38 *If the RSA problem is hard relative to GenRSA and H is modeled as a random oracle, then Construction 11.37 is CCA-secure.*

PROOF Let Π denote Construction 11.37, and let $\mathcal{A}$ be a probabilistic polynomial-time adversary. For convenience, and because this is the first proof where we use the full power of the random-oracle model, we explicitly describe the steps of experiment $\text{KEM}_{\mathcal{A},\Pi}^{\text{cca}}(n)$:

1. $\text{GenRSA}(1^n)$ is run to obtain (N, e, d) . In addition, a random function $H : \mathbb{Z}_N^* \rightarrow \{0, 1\}^n$ is chosen.
2. Uniform $r \in \mathbb{Z}_N^*$ is chosen, and the ciphertext $c := [r^e \bmod N]$ and key $k := H(r)$ are computed.
3. A uniform bit $b \in \{0, 1\}$ is chosen. If $b = 0$ set $\hat{k} := k$. If $b = 1$ then choose a uniform $\hat{k} \in \{0, 1\}^n$.
4. $\mathcal{A}$ is given $pk = \langle N, e \rangle$, c , and $\hat{k}$, and may query $H(\cdot)$ (on any input) and the decapsulation oracle $\text{Decaps}_{\langle N, d \rangle}(\cdot)$ on any ciphertext $\hat{c} \neq c$.
5. $\mathcal{A}$ outputs a bit b' . The output of the experiment is defined to be 1 if $b' = b$, and 0 otherwise.

In an execution of experiment $\text{KEM}_{\mathcal{A},\Pi}^{\text{cca}}(n)$, let **Query** be the event that, at any point during its execution, $\mathcal{A}$ queries r to the random oracle H . We let **Success** denote the event that $b' = b$ (i.e., the experiment outputs 1). Then

$$\begin{aligned} \Pr[\text{Success}] &= \Pr[\text{Success} \wedge \overline{\text{Query}}] + \Pr[\text{Success} \wedge \text{Query}] \\ &\leq \Pr[\text{Success} \wedge \overline{\text{Query}}] + \Pr[\text{Query}], \end{aligned}$$

where all probabilities are taken over the randomness used in experiment $\text{KEM}_{\mathcal{A},\Pi}^{\text{cca}}(n)$. We show that $\Pr[\text{Success} \wedge \overline{\text{Query}}] \leq \frac{1}{2}$ and that $\Pr[\text{Query}]$ is negligible. The theorem follows.

We first argue that $\Pr[\text{Success} \wedge \overline{\text{Query}}] \leq \frac{1}{2}$. If $\Pr[\overline{\text{Query}}] = 0$ this is immediate. Otherwise, $\Pr[\text{Success} \wedge \overline{\text{Query}}] \leq \Pr[\text{Success} | \overline{\text{Query}}]$. Now, conditioned on $\overline{\text{Query}}$, the value of the correct key $k = H(r)$ is uniform because H is a random function. Consider $\mathcal{A}$'s information about k in experiment $\text{KEM}_{\mathcal{A},\Pi}^{\text{cca}}(n)$. The public key pk and ciphertext c , by themselves, do not contain any information about k . (They do uniquely determine r , but since H is chosen independently of anything else, this gives no information about $H(r)$.) Queries that $\mathcal{A}$ makes to H also do not reveal any information about r , unless $\mathcal{A}$ queries r to H (in which case **Query** occurs); this, again, relies on the fact that H is a random function. Finally, queries that $\mathcal{A}$ makes to its decapsulation oracle only reveal $H(\tilde{r})$ for $\tilde{r} \neq r$. This follows from the fact that $\text{Decaps}_{\langle N, d \rangle}(\tilde{c}) = H(\tilde{r})$ where $\tilde{r} = [\tilde{c}^d \bmod N]$, but $\tilde{c} \neq c$ implies $\tilde{r} \neq r$. Once again, this and the fact that H is a random function mean that no information about $H(r)$ is revealed unless **Query** occurs.

The above shows that, as long as **Query** does not occur, the value of the correct key k is uniform even given $\mathcal{A}$'s view of the public key, ciphertext, and

the answers to all its oracle queries. In that case, then, there is no way $\mathcal{A}$ can distinguish (any better than random guessing) whether $\hat{k}$ is the correct key or a uniform, independent key. Therefore, $\Pr[\text{Success}|\overline{\text{Query}}] = \frac{1}{2}$.

We highlight that nowhere in the above argument did we rely on the fact that $\mathcal{A}$ is computationally bounded, and in fact $\Pr[\text{Success} \wedge \overline{\text{Query}}] \leq \frac{1}{2}$ even if no computational restrictions are placed on $\mathcal{A}$. This indicates part of the power of the random-oracle model.

To complete the proof of the theorem, we show

CLAIM 11.39 *If the RSA problem is hard relative to GenRSA and H is modeled as a random oracle, then $\Pr[\text{Query}]$ is negligible.*

To prove this, we construct an algorithm $\mathcal{A}'$ that uses $\mathcal{A}$ as a subroutine. $\mathcal{A}'$ is given an instance N, e, c of the RSA problem, and its goal is to compute r for which $r^e = c \bmod N$. To do so, it will run $\mathcal{A}$, answering its queries to H and Decaps. Handling queries to H is simple, since $\mathcal{A}'$ can just return a random value. Queries to Decaps are trickier, however, since $\mathcal{A}'$ does not know the private key associated with the effective public key $\langle N, e \rangle$.

On further thought, however, decapsulation queries are also easy to answer since $\mathcal{A}'$ can just return a random value here as well. That is, although the query Decaps($\tilde{c}$) is supposed to be computed by first computing $\tilde{r}$ such that $\tilde{r}^e = \tilde{c} \bmod N$ and then evaluating $H(\tilde{r})$, the result is just a uniform value. Thus, $\mathcal{A}'$ can simply return a random value without performing the intermediate computation. The only “catch” is that $\mathcal{A}'$ must ensure consistency between its answers to H -queries and Decaps-queries; namely, it must ensure that for any $\tilde{r}, \tilde{c}$ with $\tilde{r}^e = \tilde{c} \bmod N$ it holds that $H(\tilde{r}) = \text{Decaps}(\tilde{c})$. This is handled using simple bookkeeping and lists L_H and L_{Decaps} that keep track of the answers $\mathcal{A}'$ has given in response to the respective oracle queries. We now give the details.

Algorithm $\mathcal{A}'$:

The algorithm is given (N, e, c) as input.

1. Initialize empty lists L_H, L_{Decaps} . choose a uniform $k \in \{0, 1\}^n$ and store (c, k) in L_{Decaps} .
2. Choose a uniform bit $b \in \{0, 1\}$. If $b = 0$ set $\hat{k} := k$. If $b = 1$ then choose a uniform $\hat{k} \in \{0, 1\}^n$. Run $\mathcal{A}$ on $\langle N, e \rangle, c$, and $\hat{k}$.

When $\mathcal{A}$ makes a query $H(\tilde{r})$, answer it as follows:

- If there is an entry in L_H of the form $(\tilde{r}, k)$ for some k , return k .
- Otherwise, let $\tilde{c} := [\tilde{r}^e \bmod N]$. If there is an entry in L_{Decaps} of the form $(\tilde{c}, k)$ for some k , return k and store $(\tilde{r}, k)$ in L_H .

- Otherwise, choose a uniform $k \in \{0, 1\}^n$, return k , and store $(\tilde{r}, k)$ in L_H .

When $\mathcal{A}$ makes a query $\text{Decaps}(\tilde{c})$, answer it as follows:

- If there is an entry in L_{Decaps} of the form $(\tilde{c}, k)$ for some k , return k .
 - Otherwise, for each entry $(\tilde{r}, k) \in L_H$, check if $\tilde{r}^e = \tilde{c} \bmod N$ and, if so, output k .
 - Otherwise, choose a uniform $k \in \{0, 1\}^n$, return k , and store $(\tilde{c}, k)$ in L_{Decaps} .
3. At the end of $\mathcal{A}$'s execution, if there is an entry (r, k) in L_H for which $r^e = c \bmod N$ then return r .

Clearly $\mathcal{A}'$ runs in polynomial time, and the view of $\mathcal{A}$ when run as a subroutine by $\mathcal{A}'$ in experiment $\text{RSA-inv}_{\mathcal{A}', \text{GenRSA}}(n)$ is *identical* to the view of $\mathcal{A}$ in experiment $\text{KEM}_{\mathcal{A}, \Pi}^{\text{cca}}(n)$: the inputs given to $\mathcal{A}$ clearly have the right distribution, the answers to $\mathcal{A}$'s oracle queries are consistent, and the responses to all H -queries are uniform and independent. Finally, $\mathcal{A}'$ outputs the correct solution exactly when **Query** occurs. Hardness of the RSA problem relative to GenRSA thus implies that $\Pr[\text{Query}]$ is negligible, as required. ■

It is worth remarking on the various properties of the random-oracle model (see Section 5.5.1) that are used in the above proof. First, we rely on the fact that the value $H(r)$ is uniform unless r is queried to H —even if H is queried on multiple other values $\tilde{r} \neq r$. We also, implicitly, use extractability to argue that the attacker cannot query r to H ; otherwise, we could use this attacker to solve the RSA problem. Finally, the proof relies on programmability in order to simulate the adversary's decapsulation-oracle queries.

11.5.6 RSA Implementation Issues and Pitfalls

We close this section with a brief discussion of some issues related to the implementation of RSA-based schemes, and some pitfalls to be aware of.

Using Chinese remaindering. In implementations of RSA-based encryption, the receiver can use the Chinese remainder theorem (Section 8.1.5) to speed up computation of e th roots modulo N during decryption. Specifically, let $N = pq$ and say the receiver wishes to compute the e th root of some value y using $d = [e^{-1} \bmod \phi(N)]$. The receiver can use the correspondence $[y^d \bmod N] \leftrightarrow ([y^d \bmod p], [y^d \bmod q])$ to compute the partial results

$$x_p := [y^d \bmod p] = [y^{[d \bmod (p-1)]} \bmod p] \quad (11.19)$$

and

$$x_q := [y^d \bmod q] = [y^{[d \bmod (q-1)]} \bmod q], \quad (11.20)$$

and then combine these to obtain $x \leftrightarrow (x_p, x_q)$, as discussed in Section 8.1.5. Note that $[d \bmod (p-1)]$ and $[d \bmod (q-1)]$ could be pre-computed since they are independent of y .

Why is this better? Assume exponentiation modulo an ℓ -bit integer takes $\gamma \cdot \ell^3$ operations for some constant γ . If p, q are each n bits long, then naively computing $[y^d \bmod N]$ takes $\gamma \cdot (2n)^3 = 8\gamma \cdot n^3$ steps (because $\|N\| = 2n$). Using Chinese remaindering reduces this to roughly $2 \cdot (\gamma \cdot n^3)$ steps (because $\|p\| = \|q\| = n$), or roughly 1/4 of the time.

Example 11.40

We revisit Example 8.49. Recall that $N = 143 = 11 \cdot 13$ and $d = 103$, and $y = 64$ there. To calculate $[64^{103} \bmod 143]$ we compute

$$\begin{aligned} ([64 \bmod 11], [64 \bmod 13])^{103} &= ([-2]^{103} \bmod 11, [(-1)^{103} \bmod 13]) \\ &= ([(-2)^{103 \bmod 10}] \bmod 11, -1) \\ &= ([-8 \bmod 11], -1) = (3, -1). \end{aligned}$$

We can compute $1_p = 78 \leftrightarrow (1, 0)$ and $1_q = 66 \leftrightarrow (0, 1)$, as discussed in Section 8.1.5. (Note these values can be pre-computed, as they are independent of y .) Then $(3, -1) \leftrightarrow 3 \cdot 1_p - 1_q = 3 \cdot 78 - 66 = 168 = 25 \bmod 143$, in agreement with the answer previously obtained. $\diamond$

A fault attack when using Chinese remaindering. When using Chinese remaindering as just described, one should be aware of a potential attack that can be carried out if *faults* occur (or can be induced to occur by an attacker, e.g., by hardware tampering) during the course of the computation.

Consider what happens if $[y^d \bmod N]$ is computed twice: the first time with no error (giving the correct result x), but the second time with an error during computation of Equation (11.20) but not Equation (11.19) (the same attack applies in the opposite case). The second computation yields an incorrect result x' for which $x' = x \bmod p$ but $x' \neq x \bmod q$. This means that $p \mid (x' - x)$ but $q \nmid (x' - x)$. But then $\gcd(x' - x, N) = p$, yielding the factorization of N .

One possible countermeasure is to verify correctness of the result before using it, by checking that $x^e = y \bmod N$. (Since $\|e\| \ll \|d\|$, using Chinese remaindering still gives better efficiency.) This is recommended in hardware implementations.

Dependent public keys I. When multiple receivers wish to utilize the same encryption scheme, they should use *independent* public keys. This and the following attack demonstrate what can go wrong when this is not done.

Imagine a company wants to use the same modulus N for each of its employees. Since it is not desirable for messages encrypted to one employee to be read by any other employee, the company issues different (e_i, d_i) pairs to

each employee. That is, the public key of the i th employee is $pk_i = \langle N, e_i \rangle$ and their private key is $sk_i = \langle N, d_i \rangle$, where $e_i \cdot d_i = 1 \bmod \phi(N)$ for all i .

This approach is insecure and allows any employee to read messages encrypted to all other employees. The reason is that, as noted in Section 8.2.4, given N and e_i, d_i with $e_i \cdot d_i = 1 \bmod \phi(N)$, the factorization of N can be efficiently computed. Given the factorization of N , of course, it is possible to compute $d_j := e_j^{-1} \bmod \phi(N)$ for any j .

Dependent public keys II. The attack just shown allows any employee to decrypt messages sent to any other employee. This still leaves the possibility that sharing the modulus N is fine as long as all employees trust each other (or, alternatively, as long as confidentiality need only be preserved against outsiders but not against other members of the company). Here we show a scenario indicating that sharing a modulus is still a bad idea, at least when plain RSA encryption is used.

Say the same message m is encrypted and sent to two different (known) employees with public keys (N, e_1) and (N, e_2) where $e_1 \neq e_2$. Assume further that $\gcd(e_1, e_2) = 1$. Then an eavesdropper sees the two ciphertexts

$$c_1 = m^{e_1} \bmod N \quad \text{and} \quad c_2 = m^{e_2} \bmod N.$$

Since $\gcd(e_1, e_2) = 1$, there exist integers X, Y such that $Xe_1 + Ye_2 = 1$ by Proposition 8.2. Moreover, given the public exponents e_1 and e_2 it is possible to efficiently compute X and Y using the extended Euclidean algorithm (see Appendix B.1.2). We claim that $m = [c_1^X \cdot c_2^Y \bmod N]$, which can easily be calculated. This is true because

$$c_1^X \cdot c_2^Y = m^{Xe_1} m^{Ye_2} = m^{Xe_1 + Ye_2} = m^1 = m \bmod N.$$

A similar attack applies when using padded RSA or RSA-OAEP if the sender uses the same transformed message $\hat{m}$ when encrypting to two users.

Randomness quality in RSA key generation. Throughout this book, we always assume that honest parties have access to sufficient, high-quality randomness. When this assumption is violated then security may fail to hold. In particular, if an ℓ -bit string is chosen from some set $S \subset \{0, 1\}^\ell$ rather than uniformly from $\{0, 1\}^\ell$, then an attacker can perform a brute-force search (in time $\mathcal{O}(|S|)$) to attack the system.

In some cases the situation may be even worse. Consider in particular the case of RSA key generation, where one sample of random bits r_p is used to choose the first prime p , and a second sample r_q is used to generate the second prime q . Assume further that many public/private keys are generated using the same source of poor-quality randomness, in which r_p, r_q are chosen uniformly from some set S of size 2^s . After generating roughly $2^{s/2}$ public keys (see Appendix A.4), we expect to obtain two different moduli N, N' that were generated using identical randomness $r_p = r'_p$. These two moduli share a prime factor which can be easily found by computing $\gcd(N, N')$. An attacker

can thus scrape the Internet for a large set of RSA public keys, compute their pairwise gcd's, and hope to factor some subset of them. Although computing pairwise gcd's of $2^{s/2}$ moduli would naively take time $\mathcal{O}(2^s)$, it turns out that this can be significantly improved using a “divide-and-conquer” approach that is beyond the scope of this book. The upshot is that an attacker can perform a brute-force search in time much less than 2^s . Moreover, the attack works even if the set S is unknown to the attacker!

The above scenario was verified experimentally by two research teams working independently, who carried out exactly the above attack on public keys scraped over the Internet and were able to successfully factor a significant fraction of the keys they found.

References and Additional Reading

The idea of public-key encryption was first proposed in the open literature by Diffie and Hellman [58]. Rivest, Shamir, and Adleman [148] introduced the RSA assumption and proposed a public-key encryption scheme based on this assumption. As pointed out in the previous chapter, other pioneers of public-key cryptography include Merkle and Rabin (in academic publications) and Ellis, Cocks, and Williamson (in classified publications).

Definition 11.2 is rooted in the seminal work of Goldwasser and Micali [80], who were also the first to recognize the necessity of *probabilistic* encryption for satisfying this definition. As noted in Chapter 4, chosen-ciphertext attacks were first formally defined by Naor and Yung [129] and Rackoff and Simon [147]. The expository article by Shoup [156] discusses the importance of security against chosen-ciphertext attacks. Bellare et al. give a unified, modern treatment of various security notions for public-key encryption [16].

A proof of CPA-security for hybrid encryption was first given by Blum and Goldwasser [36]. The case of CCA-security was treated in [56].

Somewhat amazingly, the El Gamal encryption scheme [70] was not suggested until 1984, even though it can be viewed as a direct transformation of the Diffie–Hellman key-exchange protocol (see Exercise 11.4). DHIES was introduced in [2]. The ISO/IEC 18033-2 standard for public-key encryption can be found at <http://www.shoup.net/iso>.

Plain RSA encryption corresponds to the original scheme introduced by Rivest, Shamir, and Adleman [148]. The attacks on plain RSA encryption described in Section 11.5.1 are due to [161, 55, 84, 47, 40]; see [120, Chapter 8] and [38] for additional attacks and further information. Proofs of Copper-smith's theorem can be found in the original work [46] or several subsequent expositions (e.g., [119]).

The *PKCS #1 RSA Cryptography Standards* (both previous and current

versions) are available at <http://www.emc.com/emc-plus/rsa-labs>. The chosen-plaintext attack on PKCS #1 v1.5 described here is due to [49]. A description of Bleichenbacher's chosen-ciphertext attack on PKCS #1 v1.5 can be found in the original paper [34]. See [12] for subsequent improvements.

Proofs of Theorem 11.31, and generalizations, can be found in [8, 86, 66, 7]. See Section 13.1.2 for a general treatment of schemes of this form. Construction 11.37 appears to have been introduced and first analyzed by Shoup [157]. OAEP was introduced by Bellare and Rogaway [22]. The original proof of OAEP was later found to be flawed; the interested reader is referred to [39, 158, 68]. For details of Manger's chosen-ciphertext attack on implementations of PKCS #1 v2.0, see [117].

The pairwise-gcd attack described in Section 11.5.6 was carried out by Lenstra et al. [112] and Heninger et al. [88].

When using any encryption scheme in practice, the question arises as to what key length to use. This issue should not be taken lightly, and we refer the reader to Section 9.3 and references therein for an in-depth treatment.

The first efficient CCA-secure public-key encryption scheme not relying on the random-oracle model was shown by Cramer and Shoup [50] based on the DDH assumption. Subsequently, Hoffheinz and Kiltz have shown an efficient CCA-secure scheme without random oracles based on the RSA assumption [92].

Exercises

- 11.1 Assume a public-key encryption scheme for single-bit messages with no decryption error. Show that, given pk and a ciphertext c computed via $c \leftarrow \text{Enc}_{pk}(m)$, it is possible for an unbounded adversary to determine m with probability 1.
- 11.2 Show that for any CPA-secure public-key encryption scheme for single-bit messages, the length of the ciphertext must be superlogarithmic in the security parameter.

Hint: If not, the range of possible ciphertexts has polynomial size.
- 11.3 Say a public-key encryption scheme $(\text{Gen}, \text{Enc}, \text{Dec})$ for n -bit messages is *one-way* if any PPT adversary $\mathcal{A}$ has negligible probability of success in the following experiment:

- $\text{Gen}(1^n)$ is run to obtain keys (pk, sk) .
- A message $m \in \{0, 1\}^n$ is chosen uniformly at random, and a ciphertext $c \leftarrow \text{Enc}_{pk}(m)$ is computed.
- $\mathcal{A}$ is given pk and c , and outputs a message m' . We say $\mathcal{A}$ succeeds if $m' = m$.

- (a) Give a construction of a CPA-secure KEM in the random-oracle model based on any one-way public-key encryption scheme.
 - (b) Can a *deterministic* public-key encryption scheme be one-way? If not, prove impossibility; if so, give a construction based on any of the assumptions introduced in this book.
- 11.4 Show that any two-round key-exchange protocol (that is, where each party sends a single message) satisfying Definition 10.1 can be converted into a CPA-secure public-key encryption scheme.
- 11.5 Show that Claim 11.7 does not hold in the setting of CCA-security.
- 11.6 Consider the following public-key encryption scheme. The public key is $(\mathbb{G}, q, g, h)$ and the private key is x , generated exactly as in the El Gamal encryption scheme. In order to encrypt a bit b , the sender does the following:
- (a) If $b = 0$ then choose a uniform $y \in \mathbb{Z}_q$ and compute $c_1 := g^y$ and $c_2 := h^y$. The ciphertext is $\langle c_1, c_2 \rangle$.
 - (b) If $b = 1$ then choose independent uniform $y, z \in \mathbb{Z}_q$, compute $c_1 := g^y$ and $c_2 := g^z$, and set the ciphertext equal to $\langle c_1, c_2 \rangle$.

Show that it is possible to decrypt efficiently given knowledge of x . Prove that this encryption scheme is CPA-secure if the decisional Diffie–Hellman problem is hard relative to $\mathcal{G}$.

- 11.7 Consider the following variant of El Gamal encryption. Let $p = 2q + 1$, let $\mathbb{G}$ be the group of squares modulo p (so $\mathbb{G}$ is a subgroup of $\mathbb{Z}_p^*$ of order q), and let g be a generator of $\mathbb{G}$. The private key is $(\mathbb{G}, g, q, x)$ and the public key is $(\mathbb{G}, g, q, h)$, where $h = g^x$ and $x \in \mathbb{Z}_q$ is chosen uniformly. To encrypt a message $m \in \mathbb{Z}_q$, choose a uniform $r \in \mathbb{Z}_q$, compute $c_1 := g^r \bmod p$ and $c_2 := h^r + m \bmod p$, and let the ciphertext be $\langle c_1, c_2 \rangle$. Is this scheme CPA-secure? Prove your answer.
- 11.8 Consider the following protocol for two parties A and B to flip a fair coin (more complicated versions of this might be used for Internet gambling): (1) a trusted party T publishes her public key pk ; (2) then A chooses a uniform bit b_A , encrypts it using pk , and announces the ciphertext c_A to B and T ; (3) next, B acts symmetrically and announces a ciphertext $c_B \neq c_A$; (4) T decrypts both c_A and c_B , and the parties XOR the results to obtain the value of the coin.
- (a) Argue that even if A is dishonest (but B is honest), the final value of the coin is uniformly distributed.
 - (b) Assume the parties use El Gamal encryption (where the bit b is encoded as the group element g^b before being encrypted—note that efficient decrypt is still possible). Show how a dishonest B can bias the coin to any value he likes.

- (c) Suggest what type of encryption scheme would be appropriate to use here. Can you define an appropriate notion of security and prove that your suggestion achieves this definition?
- 11.9 Prove formally that the El Gamal encryption scheme is not CCA-secure.
- 11.10 In Section 11.4.4 we showed that El Gamal encryption is malleable, and specifically that given a ciphertext $\langle c_1, c_2 \rangle$ that is the encryption of some unknown message m , it is possible to produce a ciphertext $\langle c_1, c'_2 \rangle$ that is the encryption of $\alpha \cdot m$ (for known α). A receiver who receives both these ciphertexts might be suspicious since both ciphertexts share the first component. Show that it is possible to generate $\langle c'_1, c'_2 \rangle$ that is the encryption of $\alpha \cdot m$, with $c'_1 \neq c_1$ and $c'_2 \neq c_2$.
- 11.11 Prove Theorem 11.22.
- 11.12 One of the attacks on plain RSA discussed in Section 11.5.1 involves a sender who encrypts two related messages using the same public key. Formulate an appropriate definition of security ruling out such attacks, and show that any CPA-secure public-key encryption scheme satisfies your definition.
- 11.13 One of the attacks on plain RSA discussed in Section 11.5.1 involves a sender who encrypts the same message to three different receivers. Formulate an appropriate definition of security ruling out such attacks, and show that any CPA-secure public-key encryption scheme satisfies your definition.
- 11.14 Consider the following modified version of padded RSA encryption: Assume messages to be encrypted have length exactly $\|N\|/2$. To encrypt, first compute $\hat{m} := 0x00\|r\|0x00\|m$ where r is a uniform string of length $\|N\|/2 - 16$. Then compute the ciphertext $c := [\hat{m}^e \bmod N]$. When decrypting a ciphertext c , the receiver computes $\hat{m} := [c^d \bmod N]$ and returns an error if $\hat{m}$ does not consist of $0x00$ followed by $\|N\|/2 - 16$ arbitrary bits followed by $0x00$. Show that this scheme is not CCA-secure. Why is it easier to construct a chosen-ciphertext attack on this scheme than on PKCS #1 v1.5?
- 11.15 Consider the RSA-based encryption scheme in which a user encrypts a message $m \in \{0, 1\}^\ell$ with respect to the public key $\langle N, e \rangle$ by computing $\hat{m} := H(m)\|m$ and outputting the ciphertext $[\hat{m}^e \bmod N]$. (Here, let $H : \{0, 1\}^\ell \rightarrow \{0, 1\}^n$ and assume $\ell + n < \|N\|$.) The receiver recovers $\hat{m}$ in the usual way and verifies that it has the correct form before outputting the ℓ least-significant bits as m . Prove or disprove that this scheme is CCA-secure if H is modeled as a random oracle.
- 11.16 Show a chosen-ciphertext attack on Construction 11.34.

- 11.17 Let $\Pi = (\text{Gen}, \text{Enc}, \text{Dec})$ be a CPA-secure public-key encryption scheme, and let $\Pi' = (\text{Gen}', \text{Enc}', \text{Dec}')$ be a CCA-secure private-key encryption scheme. Consider the following construction:

CONSTRUCTION 11.41

Let $H : \{0, 1\}^n \rightarrow \{0, 1\}^n$ be a function. Construct a public-key encryption scheme as follows:

- **Gen***: on input 1^n , run $\text{Gen}(1^n)$ to obtain (pk, sk) . Output these as the public and private keys, respectively.
- **Enc***: on input a public key pk and a message $m \in \{0, 1\}^n$, choose a uniform $r \in \{0, 1\}^n$ and output the ciphertext

$$\langle \text{Enc}_{pk}(r), \text{Enc}'_{H(r)}(m) \rangle.$$

- **Dec***: on input a private key sk and a ciphertext $\langle c_1, c_2 \rangle$, compute $r := \text{Dec}_{sk}(c_1)$ and set $k := H(r)$. Then output $\text{Dec}'_k(c_2)$.

Does the above construction have indistinguishable encryptions under a chosen-ciphertext attack, if H is modeled as a random oracle? If yes, provide a proof. If not, where does the approach used to prove Theorem 11.38 break down?

- 11.18 Consider the following variant of Construction 11.32:

CONSTRUCTION 11.42

Let GenRSA be as usual, and define a public-key encryption scheme as follows:

- **Gen**: on input 1^n , run $\text{GenRSA}(1^n)$ to obtain (N, e, d) . Output the public key $pk = \langle N, e \rangle$, and the private key $sk = \langle N, d \rangle$.
- **Enc**: on input a public key $pk = \langle N, e \rangle$ and a message $m \in \{0, 1\}$, choose a uniform $r \in \mathbb{Z}_N^*$. Output the ciphertext $\langle [r^e \bmod N], \text{lsb}(r) \oplus m \rangle$.
- **Dec**: on input a private key $sk = \langle N, d \rangle$ and a ciphertext $\langle c, b \rangle$, compute $r := [c^d \bmod N]$ and output $\text{lsb}(r) \oplus b$.

Prove that this scheme is CPA-secure. Discuss its advantages and disadvantages relative to Construction 11.32.

- 11.19 Say three users have RSA public keys $\langle N_1, 3 \rangle$, $\langle N_2, 3 \rangle$, and $\langle N_3, 3 \rangle$ (i.e., they all use $e = 3$), with $N_1 < N_2 < N_3$. Consider the following method for sending the same message $m \in \{0, 1\}^\ell$ to each of these parties: choose a uniform $r \leftarrow \mathbb{Z}_{N_1}^*$, and send to everyone the same ciphertext

$$\langle [r^3 \bmod N_1], [r^3 \bmod N_2], [r^3 \bmod N_3], H(r) \oplus m \rangle,$$

where $H : \mathbb{Z}_{N_1}^* \rightarrow \{0, 1\}^\ell$. Assume $\ell \gg n$.

- (a) Show that this is not CPA-secure, and an adversary can recover m from the ciphertext even when H is modeled as a random oracle.

Hint: See Section 11.5.1.

- (b) Show a simple way to fix this and get a CPA-secure method that transmits a ciphertext of length $3\ell + \mathcal{O}(n)$.
- (c) Show a better approach that is still CPA-secure but with a ciphertext of length $\ell + \mathcal{O}(n)$.

11.20 Fix an RSA public key $\langle N, e \rangle$ and assume we have an algorithm $\mathcal{A}$ that always correctly computes $\text{lsb}(x)$ given $[x^e \bmod N]$. Write full pseudocode for an algorithm $\mathcal{A}'$ that computes x from $[x^e \bmod N]$.

11.21 Fix an RSA public key $\langle N, e \rangle$ and define

$$\text{half}(x) = \begin{cases} 0 & \text{if } 0 < x < N/2 \\ 1 & \text{if } N/2 < x < N \end{cases}$$

Prove that half is a hard-core predicate for the RSA problem.

Hint: Reduce to hardness of computing lsb .

Chapter 12

Digital Signature Schemes

12.1 Digital Signatures – An Overview

In the previous chapter we explored how public-key encryption can be used to achieve *secrecy* in the public-key setting. *Integrity* (or *authenticity*) in the public-key setting is provided using *digital signature schemes*. These can be viewed as the public-key analogue of message authentication codes, although, as we will see, there are several important differences between these primitives.

Signature schemes allow a *signer* S who has established a public key pk to “sign” a message using the associated private key sk in such a way that anyone who knows pk (and knows that this public key was established by S) can *verify* that the message originated from S and was not modified in transit. (Note that, in contrast to public-key encryption, in the context of digital signatures the owner of the public key acts as the *sender*.) As a prototypical application, consider a software company that wants to disseminate software updates in an authenticated manner; that is, when the company releases an update it should be possible for any of its clients to verify that the update is authentic, and a malicious third party should never be able to fool a client into accepting an update that was not actually released by the company. To do this, the company can generate a public key pk along with a private key sk , and then distribute pk in some reliable manner to its clients while keeping sk secret. (As in the case of public-key encryption, we assume that this initial distribution of the public key is carried out correctly so that all clients have a correct copy of pk . In the current example, pk could be bundled with the original software purchased by a client.) When releasing a software update m , the company computes a digital signature σ on m using its private key sk , and sends (m, σ) to every client. Each client can verify the authenticity of m by checking that σ is a correct signature on m with respect to the public key pk .

A malicious party might try to issue a fraudulent update by sending (m', σ') to a client, where m' represents an update that was never released by the company. This m' might be a modified version of some previous update, or it might be completely new and unrelated to any prior updates. If the signature scheme is “secure” (in a sense we will define more carefully soon), however, then when the client attempts to verify σ' it will find that this is an *invalid* signature on m' with respect to pk , and will therefore *reject* the signature. The

client will reject even if m' is modified only slightly from a genuine update m .

The above is not just a theoretical application of digital signatures, but one that is used extensively today for distributing software updates.

Comparison to Message Authentication Codes

Both message authentication codes and digital signature schemes are used to ensure the integrity of transmitted messages. Although the discussion in Chapter 10 comparing the public-key and private-key settings focused mainly on encryption, that discussion applies also to message integrity. Using digital signatures rather than message authentication codes simplifies key distribution and management, especially when a sender needs to communicate with multiple receivers as in the software-update example above. By using a digital signature scheme the sender avoids having to establish a distinct secret key with each potential receiver, and avoids having to compute a separate MAC tag with respect to each such key. Instead, the sender need only compute a single signature that can be verified by all recipients.

A *qualitative* advantage that digital signatures have as compared to message authentication codes is that signatures are *publicly verifiable*. This means that if a receiver verifies that a signature on a given message is legitimate, then all other parties who receive this signed message will also verify it as legitimate. This feature is not achieved by message authentication codes if the signer shares a separate key with each receiver: in such a setting a malicious sender might compute a correct MAC tag with respect to the key it shares with receiver A but an incorrect MAC tag with respect to the key it shares with a different user B . In this case, A knows that he received an authentic message from the sender but has no guarantee that B will agree.

Public verifiability implies that signatures are *transferable*: a signature σ on a message m by a signer S can be shown to a third party, who can then verify herself that σ is a legitimate signature on m with respect to S 's public key (here, we assume this third party also knows S 's public key). By making a copy of the signature, this third party can then show the signature to another party and convince *them* that S authenticated m , and so on. Public verifiability and transferability are essential for the application of digital signatures to certificates and public-key infrastructures, as we will discuss in Section 12.7.

Digital signature schemes also provide the very important property of *non-repudiation*. This means that once S signs a message he cannot later deny having done so (assuming the public key of S is widely publicized and distributed). This aspect of digital signatures is crucial for legal applications where a recipient may need to prove to a third party (say, a judge) that a signer did indeed “certify” a particular message (e.g., a contract): assuming S 's public key is known to the judge, or is otherwise publicly available, a valid signature on a message serves as convincing evidence that S indeed signed this message. Message authentication codes simply cannot provide non-repudiation. To see this, say users S and R share a key k_{SR} , and S

sends a message m to R along with a (valid) MAC tag t computed using this key. Since the judge does *not* know k_{SR} (indeed, this key is kept secret by S and R), there is no way for the judge to determine whether t is valid or not. If R were to reveal the key k_{SR} to the judge, there would be no way for the judge to know whether this is the “actual” key that S and R shared, or whether it is some “fake” key manufactured by R . Finally, even if we assume the judge can somehow obtain the actual key k_{SR} shared by the parties, there is no way for the judge to distinguish whether S generated t or whether R did—this follows from the fact that message authentication codes are a *symmetric*-key primitive; anything S can do, R can do also.

As in the case of private-key vs. public-key encryption, message authentication codes have the advantage of being shorter and roughly 2–3 orders of magnitude more efficient to generate/verify than digital signatures. Thus, in situations where public verifiability, transferability, and/or non-repudiation are not needed, and the sender communicates primarily with a single recipient (with whom it is able to share a secret key), message authentication codes should be used.

Relation to Public-Key Encryption

Digital signatures are often mistakenly viewed as the “inverse” of public-key encryption, with the roles of the sender and receiver interchanged. Historically,¹ in fact, it has been suggested that digital signatures can be obtained by “reversing” public-key encryption, i.e., signing a message m by decrypting it (using the private key) to obtain σ , and verifying a signature σ by encrypting it (using the corresponding public key) and checking whether the result is m . The suggestion to construct signature schemes in this way is *completely unfounded*: in most cases, it is simply inapplicable, and in cases where it is applicable it results in signature schemes that are not secure.

12.2 Definitions

Digital signatures are the public-key counterpart of message authentication codes, and their syntax and security guarantees are analogous. The algorithm that the sender applies to a message is here denoted **Sign** (rather than **Mac**), and the output of this algorithm is now called a *signature* (rather than a tag).

¹This view no doubt arose because, as we will see in Section 12.4.1, plain RSA signatures are the reverse of plain RSA encryption. However, neither plain RSA signatures nor plain RSA encryption meet even minimal notions of security.

The algorithm that the receiver applies to a message and a signature in order to verify validity is still denoted Vrfy .

DEFINITION 12.1 A (digital) signature scheme consists of three probabilistic polynomial-time algorithms $(\text{Gen}, \text{Sign}, \text{Vrfy})$ such that:

1. The key-generation algorithm Gen takes as input a security parameter 1^n and outputs a pair of keys (pk, sk) . These are called the public key and the private key, respectively. We assume that pk and sk each has length at least n , and that n can be determined from pk or sk .
2. The signing algorithm Sign takes as input a private key sk and a message m from some message space (that may depend on pk). It outputs a signature σ , and we write this as $\sigma \leftarrow \text{Sign}_{sk}(m)$.
3. The deterministic verification algorithm Vrfy takes as input a public key pk , a message m , and a signature σ . It outputs a bit b , with $b = 1$ meaning valid and $b = 0$ meaning invalid. We write this as $b := \text{Vrfy}_{pk}(m, \sigma)$.

It is required that except with negligible probability over (pk, sk) output by $\text{Gen}(1^n)$, it holds that $\text{Vrfy}_{pk}(m, \text{Sign}_{sk}(m)) = 1$ for every (legal) message m .

If there is a function ℓ such that for every (pk, sk) output by $\text{Gen}(1^n)$ the message space is $\{0, 1\}^{\ell(n)}$, then we say that $(\text{Gen}, \text{Sign}, \text{Vrfy})$ is a signature scheme for messages of length $\ell(n)$.

We call σ a *valid signature* on a message m (with respect to some public key pk understood from the context) if $\text{Vrfy}_{pk}(m, \sigma) = 1$.

A signature scheme is used in the following way. One party S , who acts as the *sender*, runs $\text{Gen}(1^n)$ to obtain keys (pk, sk) . The public key pk is then publicized as belonging to S ; e.g., S can put the public key on its webpage or place it in some public directory. As in the case of public-key encryption, we assume that any other party is able to obtain a legitimate copy of S 's public key (see discussion below). When S wants to authenticate a message m , it computes the signature $\sigma \leftarrow \text{Sign}_{sk}(m)$ and sends (m, σ) . Upon receipt of (m, σ) , a receiver who knows pk can verify the authenticity of m by checking whether $\text{Vrfy}_{pk}(m, \sigma) \stackrel{?}{=} 1$. This establishes both that S sent m , and also that m was not modified in transit. As in the case of message authentication codes, however, it does not say anything about *when* m was sent, and replay attacks are still possible (see Section 4.2).

The assumption that parties are able to obtain a legitimate copy of S 's public key implies that S is able to transmit at least one message (namely, pk itself) in a reliable and authenticated manner. If we are able to transmit messages reliably, however, then why do we need a signature scheme at all? The answer is that reliable distribution of pk is a difficult and expensive task, but using a signature scheme means that this need only be carried out

once, after which an unlimited number of messages can subsequently be sent reliably. Furthermore, as we will discuss in Section 12.7, signature schemes themselves are used to ensure the reliable distribution of *other* public keys. They thus serve as a central tool for setting up a “public-key infrastructure” to address the key-distribution problem.

Security of signature schemes. For a fixed public key pk generated by a signer S , a *forgery* is a message m along with a valid signature σ , where m was not previously signed by S . Security of a signature scheme means that an adversary should be unable to output a forgery even if it obtains signatures on many other messages of its choice. This is the direct analogue of the definition of security for message authentication codes, and we refer the reader to Section 4.2 for motivation and further discussion.

We now present the formal definition of security, which is essentially the same as Definition 4.2. Let $\Pi = (\text{Gen}, \text{Sign}, \text{Vrfy})$ be a signature scheme, and consider the following experiment for an adversary $\mathcal{A}$ and parameter n :

The signature experiment $\text{Sig-forge}_{\mathcal{A}, \Pi}(n)$:

1. $\text{Gen}(1^n)$ is run to obtain keys (pk, sk) .
2. Adversary $\mathcal{A}$ is given pk and access to an oracle $\text{Sign}_{sk}(\cdot)$. The adversary then outputs (m, σ) . Let $\mathcal{Q}$ denote the set of all queries that $\mathcal{A}$ asked its oracle.
3. $\mathcal{A}$ succeeds if and only if (1) $\text{Vrfy}_{pk}(m, \sigma) = 1$ and (2) $m \notin \mathcal{Q}$. In this case the output of the experiment is defined to be 1.

DEFINITION 12.2 A signature scheme $\Pi = (\text{Gen}, \text{Sign}, \text{Vrfy})$ is *existentially unforgeable under an adaptive chosen-message attack*, or *just secure*, if for all probabilistic polynomial-time adversaries $\mathcal{A}$, there is a negligible function negl such that:

$$\Pr[\text{Sig-forge}_{\mathcal{A}, \Pi}(n) = 1] \leq \text{negl}(n).$$

12.3 The Hash-and-Sign Paradigm

As in the case of public-key vs. private-key encryption, “native” signature schemes are orders of magnitude less efficient than message authentication codes. Fortunately, as with hybrid encryption (see Section 11.3), it is possible to obtain the functionality of digital signatures at the asymptotic cost of a private-key operation, at least for sufficiently long messages. This can be done using the *hash-and-sign* approach, discussed next.

The intuition behind the hash-and-sign approach is straightforward. Say we have a signature scheme for messages of length ℓ , and wish to sign a (longer) message $m \in \{0,1\}^*$. Rather than sign m itself, we can instead use a hash function H to *hash* the message down to a fixed-length *digest* of length ℓ , and then sign the resulting digest. This approach is exactly analogous to the hash-and-MAC approach discussed in Section 5.3.1.

CONSTRUCTION 12.3

Let $\Pi = (\text{Gen}, \text{Sign}, \text{Vrfy})$ be a signature scheme for messages of length $\ell(n)$, and let $\Pi_H = (\text{Gen}_H, H)$ be a hash function with output length $\ell(n)$. Construct signature scheme $\Pi' = (\text{Gen}', \text{Sign}', \text{Vrfy}')$ as follows:

- **Gen'**: on input 1^n , run $\text{Gen}(1^n)$ to obtain (pk, sk) and run $\text{Gen}_H(1^n)$ to obtain s ; the public key is $\langle pk, s \rangle$ and the private key is $\langle sk, s \rangle$.
- **Sign'**: on input a private key $\langle sk, s \rangle$ and a message $m \in \{0,1\}^*$, output $\sigma \leftarrow \text{Sign}_{sk}(H^s(m))$.
- **Vrfy'**: on input a public key $\langle pk, s \rangle$, a message $m \in \{0,1\}^*$, and a signature σ , output 1 if and only if $\text{Vrfy}_{pk}(H^s(m), \sigma) \stackrel{?}{=} 1$.

The hash-and-sign paradigm.

THEOREM 12.4 *If Π is a secure signature scheme for messages of length ℓ and Π_H is collision resistant, then Construction 12.3 is a secure signature scheme (for arbitrary-length messages).*

The proof of this theorem is almost identical to that of Theorem 5.6.

12.4 RSA Signatures

We begin our consideration of concrete signature schemes with a discussion of schemes based on the RSA assumption.

12.4.1 Plain RSA

We first describe a simple, RSA-based signature scheme. Although the scheme is insecure, it serves as a useful starting point.

As usual, let GenRSA be a PPT algorithm that, on input 1^n , outputs a

modulus N that is the product of two n -bit primes (except with negligible probability), along with integers e, d satisfying $ed = 1 \bmod \phi(N)$. Key generation in plain RSA involves simply running **GenRSA**, and outputting $\langle N, e \rangle$ as the public key and $\langle N, d \rangle$ as the private key. To sign a message $m \in \mathbb{Z}_N^*$, the signer computes $\sigma := [m^d \bmod N]$. Verification of a signature σ on a message m with respect to the public key $\langle N, e \rangle$ is carried out by checking whether $m \stackrel{?}{=} \sigma^e \bmod N$. See Construction 12.5.

CONSTRUCTION 12.5

Let **GenRSA** be as in the text. Define a signature scheme as follows:

- **Gen**: on input 1^n run **GenRSA**(1^n) to obtain (N, e, d) . The public key is $\langle N, e \rangle$ and the private key is $\langle N, d \rangle$.
- **Sign**: on input a private key $sk = \langle N, d \rangle$ and a message $m \in \mathbb{Z}_N^*$, compute the signature

$$\sigma := [m^d \bmod N].$$

- **Vrfy**: on input a public key $pk = \langle N, e \rangle$, a message $m \in \mathbb{Z}_N^*$, and a signature $\sigma \in \mathbb{Z}_N^*$, output 1 if and only if

$$m \stackrel{?}{=} [\sigma^e \bmod N].$$

The plain RSA signature scheme.

It is easy to see that verification of a legitimately generated signature is always successful since

$$\sigma^e = (m^d)^e = m^{[ed \bmod \phi(N)]} = m^1 = m \bmod N.$$

One might expect this scheme to be secure since, for an adversary knowing only the public key $\langle N, e \rangle$, computing a valid signature on a message m seems to require solving the RSA problem (since the signature is exactly the e th root of m). Unfortunately, this reasoning is incorrect. For one thing, the RSA assumption only implies hardness of computing a signature (that is, computing an e th root) of a *uniform* message m ; it says nothing about hardness of computing a signature on a nonuniform m or on some message m of the attacker's choice. Moreover, the RSA assumption says nothing about what an attacker might be able to do once it learns signatures on *other* messages. The following examples demonstrate that both of these observations lead to attacks on the plain RSA signature scheme.

A no-message attack. The first attack we describe generates a forgery *using the public key alone*, without obtaining any signatures from the legitimate signer. The attack works as follows: given a public key $pk = \langle N, e \rangle$, choose a uniform $\sigma \in \mathbb{Z}_N^*$ and compute $m := [\sigma^e \bmod N]$. Then output the forgery

(m, σ) . It is immediate that σ is a valid signature on m , and this is a forgery since no signatures at all were issued by the owner of the public key. We conclude that the plain RSA signature scheme does not satisfy Definition 12.2.

One might argue that this does not constitute a “realistic” attack since the adversary has “no control” over the message m for which it forges a valid signature. This is irrelevant as far as Definition 12.2 is concerned, and we have already discussed (in Chapter 4) why it is dangerous to assume any semantics for messages that are going to be authenticated using any cryptographic scheme. Moreover, the adversary does have *some* control over m : for example, by choosing multiple, uniform values of σ it can (with high probability) obtain an m with a few bits set in some desired way. By choosing σ in some specific manner, it may also be possible to influence the resulting message for which a forgery is output.

Forging a signature on an arbitrary message. A more damaging attack on the plain RSA signature scheme requires the adversary to obtain *two* signatures from the signer, but allows the adversary to output a forged signature on any message of its choice. Say the adversary wants to forge a signature on the message $m \in \mathbb{Z}_N^*$ with respect to the public key $pk = \langle N, e \rangle$. The adversary chooses arbitrary $m_1, m_2 \in \mathbb{Z}_N^*$ distinct from m such that $m = m_1 \cdot m_2 \bmod N$. It then obtains signatures σ_1, σ_2 on m_1, m_2 , respectively. Finally, it outputs $\sigma := [\sigma_1 \cdot \sigma_2 \bmod N]$ as a valid signature on m . This works because

$$\sigma^e = (\sigma_1 \cdot \sigma_2)^e = (m_1^d \cdot m_2^d)^e = m_1^{ed} \cdot m_2^{ed} = m_1 \cdot m_2 = m \bmod N,$$

using the fact that σ_1, σ_2 are valid signatures on m_1, m_2 .

Being able to forge a signature on an arbitrary message is devastating. Nevertheless, one might argue that this attack is unrealistic since an adversary will not be able to convince a signer to sign the exact messages m_1 and m_2 . Once again, this is irrelevant as far as Definition 12.2 is concerned. Furthermore, it is dangerous to make assumptions about what messages the signer may or may not be willing to sign. For example, a client may use a signature scheme to authenticate to a server by signing a random challenge sent by the server. Here, a malicious server would be able to obtain a signature on any message(s) of its choice. More generally, it may be possible for the adversary to choose m_1 and m_2 as “legitimate” messages that the signer will agree to sign. Finally, note that the attack can be generalized: if an adversary obtains valid signatures on q arbitrary messages $M = \{m_1, \dots, m_q\}$, then the adversary can output a valid signature on any of $2^q - q$ other messages obtained by taking products of subsets of M (of size different from 1).

12.4.2 RSA-FDH and PKCS #1 v2.1

One can attempt to prevent the attacks from the previous section by applying some transformation to messages before signing them. That is, the signer will now specify as part of its public key a (deterministic) function H

with certain cryptographic properties (described below) mapping messages to $\mathbb{Z}_N^*$; the signature on a message m will be $\sigma := [H(m)^d \bmod N]$, and verification of the signature σ on the message m will be done by checking whether $\sigma^e \stackrel{?}{=} H(m) \bmod N$. See Construction 12.6.

CONSTRUCTION 12.6

Let **GenRSA** be as in the previous sections, and construct a signature scheme as follows:

- **Gen**: on input 1^n , run **GenRSA**(1^n) to compute (N, e, d) . The public key is $\langle N, e \rangle$ and the private key is $\langle N, d \rangle$.
As part of key generation, a function $H : \{0, 1\}^* \rightarrow \mathbb{Z}_N^*$ is specified, but we leave this implicit.

- **Sign**: on input a private key $\langle N, d \rangle$ and a message $m \in \{0, 1\}^*$, compute

$$\sigma := [H(m)^d \bmod N].$$

- **Vrfy**: on input a public key $\langle N, e \rangle$, a message m , and a signature σ , output 1 if and only if $\sigma^e \stackrel{?}{=} H(m) \bmod N$.

The RSA-FDH signature scheme.

What properties does H need in order for this construction to be secure? At a minimum, to prevent the no-message attack it should be infeasible for an attacker to start with σ , compute $\hat{m} := [\sigma^e \bmod N]$, and then find a message m such that $H(m) = \hat{m}$. This, in particular, means that H should be hard to invert in some sense. To prevent the second attack, we need an H that does not admit “multiplicative relations,” that is, for which it is hard to find three messages m, m_1, m_2 with $H(m) = H(m_1) \cdot H(m_2) \bmod N$. Finally, it must be hard to find collisions in H : if $H(m_1) = H(m_2)$, then m_1 and m_2 have the same signature and forgery becomes trivial.

There is no known way to choose H so that Construction 12.6 can be proven secure. However, it *is* possible to prove security if H is modeled as a random oracle that maps its inputs uniformly onto $\mathbb{Z}_N^*$; the scheme in this case is called the *RSA full-domain hash* (RSA-FDH) signature scheme. One can check that a random function of this sort satisfies the requirements discussed in the previous paragraph: a random function (with large range) is hard to invert, does not have any easy-to-find multiplicative relations, and is collision resistant. Of course, this informal reasoning does not rule out all possible attacks, but the proof of security below does.

Before turning to the formal proof, we provide some intuition. Our goal is to prove that if the RSA problem is hard relative to **GenRSA**, then RSA-FDH is secure when H is modeled as a random oracle. We consider first security against a no-message attack, i.e., when the adversary $\mathcal{A}$ cannot request any

signatures. Here the adversary is limited to making queries to the random oracle, and we assume without loss of generality that $\mathcal{A}$ always makes exactly q (distinct) queries to H and that if the adversary outputs a forgery (m, σ) then it had previously queried m to H .

Say there is an efficient adversary $\mathcal{A}$ that carries out a no-message attack and makes exactly q queries to H . We construct an efficient algorithm $\mathcal{A}'$ solving the RSA problem relative to **GenRSA**. Given input (N, e, y) , algorithm $\mathcal{A}'$ runs $\mathcal{A}$ on the public key $pk = \langle N, e \rangle$. Let $m_1, \dots, m_q$ denote the q (distinct) queries that $\mathcal{A}$ makes to H . Our algorithm $\mathcal{A}'$ answers these random-oracle queries of $\mathcal{A}$ with uniform elements of $\mathbb{Z}_N^*$ except for one query—say, the i th query, chosen uniformly from the oracle queries of $\mathcal{A}$ —that is answered with y itself. Note that, from the point of view of $\mathcal{A}$, all its random-oracle queries are answered with uniform elements of $\mathbb{Z}_N^*$ (recall that y is uniform as well, although it is not chosen by $\mathcal{A}'$), and so $\mathcal{A}$ has no information about i . Moreover, the view of $\mathcal{A}$ when run as a subroutine by $\mathcal{A}'$ is identically distributed to the view of $\mathcal{A}$ when attacking the original signature scheme.

If $\mathcal{A}$ outputs a forgery (m, σ) then, because $m \in \{m_1, \dots, m_q\}$, with probability $1/q$ we will have $m = m_i$. In that case,

$$\sigma^e = H(m) = H(m_i) = y \bmod N$$

and $\mathcal{A}'$ can output σ as the solution to its given RSA instance (N, e, y) . We conclude that if $\mathcal{A}$ outputs a forgery with probability ε , then $\mathcal{A}'$ solves the RSA problem with probability ε/q . Since q is polynomial, we conclude that ε must be negligible if the RSA problem is hard relative to **GenRSA**.

Handling the case when the adversary is allowed to request signatures on messages of its choice is more difficult. The complication arises since our algorithm $\mathcal{A}'$ above does not know the decryption exponent d , yet now has to compute valid signatures on messages queried by $\mathcal{A}$ to its signing oracle. This seems impossible (and possibly even contradictory!) until we realize that $\mathcal{A}'$ *can* correctly compute a signature on a message m as long as it sets $H(m)$ to equal to $[\sigma^e \bmod N]$ for a *known* value σ . (Here we are using the fact that the random oracle is “programmable.”) If σ is uniform then $[\sigma^e \bmod N]$ is uniform as well, and so the random oracle is still emulated “properly” by $\mathcal{A}'$.

The above intuition is formalized in the proof of the following:

THEOREM 12.7 *If the RSA problem is hard relative to **GenRSA** and H is modeled as a random oracle, then Construction 12.6 is secure.*

PROOF Let $\Pi = (\text{Gen}, \text{Sign}, \text{Vrfy})$ denote Construction 12.6, and let $\mathcal{A}$ be a probabilistic polynomial-time adversary. We assume without loss of generality that if $\mathcal{A}$ requests a signature on a message m , or outputs a forgery (m, σ) , then it previously queried m to H . Let $q(n)$ be a polynomial upper bound on the number of queries $\mathcal{A}$ makes to H on security parameter n ; we assume without loss of generality that $\mathcal{A}$ makes exactly $q(n)$ distinct queries to H .

For convenience, we list the steps of experiment $\text{Sig-forg}_{\mathcal{A},\Pi}(n)$:

1. $\text{GenRSA}(1^n)$ is run to obtain (N, e, d) . A random function $H : \{0, 1\}^* \rightarrow \mathbb{Z}_N^*$ is chosen.
2. The adversary $\mathcal{A}$ is given $pk = \langle N, e \rangle$, and may query H as well as a signing oracle $\text{Sign}_{\langle N, d \rangle}(\cdot)$ that, on input a message m , returns $\sigma := [H(m)^d \bmod N]$.
3. $\mathcal{A}$ outputs (m, σ) , where it had not previously requested a signature on m . The output of the experiment is 1 if and only if $\sigma^e = H(m) \bmod N$.

We define a modified experiment $\text{Sig-forg}'_{\mathcal{A},\Pi}(n)$ in which a guess is made at the outset as to which message (from among the q messages that $\mathcal{A}$ queries to H) will correspond to the eventual forgery (if any) output by $\mathcal{A}$:

1. Choose uniform $j \in \{1, \dots, q\}$.
2. $\text{GenRSA}(1^n)$ is run to obtain (N, e, d) . A random function $H : \{0, 1\}^* \rightarrow \mathbb{Z}_N^*$ is chosen.
3. The adversary $\mathcal{A}$ is given $pk = \langle N, e \rangle$, and may query H as well as a signing oracle $\text{Sign}_{\langle N, d \rangle}(\cdot)$ that, on input a message m , returns $\sigma := [H(m)^d \bmod N]$.
4. $\mathcal{A}$ outputs (m, σ) , where it had not previously requested a signature on m . Let i be such that $m = m_i$.² The output of the experiment is 1 if and only if $\sigma^e = H(m) \bmod N$ and $j = i$.

Since j is uniform and independent of everything else, the probability that $j = i$ (even conditioned on the event that $\mathcal{A}$ outputs a forgery) is exactly $1/q$. Therefore $\Pr[\text{Sig-forg}'_{\mathcal{A},\Pi}(n) = 1] = \frac{1}{q(n)} \cdot \Pr[\text{Sig-forg}_{\mathcal{A},\Pi}(n) = 1]$.

Now consider the modified experiment $\text{Sig-forg}''_{\mathcal{A},\Pi}(n)$ in which the experiment is aborted if $\mathcal{A}$ ever requests a signature on the message m_j (where m_j denotes the j th message queried to H , and j is the uniform value chosen at the outset). This does not change the probability that the output of the experiment is 1, since if $\mathcal{A}$ ever requests a signature on m_j then it cannot possible output a forgery on m_j . In words,

$$\begin{aligned} \Pr[\text{Sig-forg}''_{\mathcal{A},\Pi}(n) = 1] &= \Pr[\text{Sig-forg}'_{\mathcal{A},\Pi}(n) = 1] \\ &= \frac{\Pr[\text{Sig-forg}_{\mathcal{A},\Pi}(n) = 1]}{q(n)}. \end{aligned} \quad (12.1)$$

²Here m_i denotes the i th query made to H . Recall, by assumption, that if $\mathcal{A}$ requests a signature on a message m , then it must have previously queried m to H .

Finally, consider the following algorithm $\mathcal{A}'$ solving the RSA problem:

Algorithm $\mathcal{A}'$:

The algorithm is given (N, e, y) as input.

1. Choose uniform $j \in \{1, \dots, q\}$.
2. Run $\mathcal{A}$ on input the public key $pk = \langle N, e \rangle$. Store triples $(\cdot, \cdot, \cdot)$ in a table, initially empty. An entry (m_i, σ_i, y_i) indicates that $\mathcal{A}'$ has set $H(m_i) = y_i$, and $\sigma_i^e = y_i \bmod N$.
3. When $\mathcal{A}$ makes its i th random-oracle query $H(m_i)$, answer it as follows:
 - If $i = j$, return y as the answer to the query.
 - Else choose uniform $\sigma_i \in \mathbb{Z}_N^*$, compute $y_i := [\sigma_i^e \bmod N]$, return y_i as the answer to the query, and store (m_i, σ_i, y_i) in the table.

When $\mathcal{A}$ requests a signature on message m , let i be such that $m = m_i$ and answer the query as follows³

- If $i = j$ then $\mathcal{A}'$ aborts.
 - If $i \neq j$ then there is an entry (m_i, σ_i, y_i) in the table. Return σ_i as the answer to the query.
4. At the end of $\mathcal{A}$'s execution, it outputs (m, σ) . If $m = m_j$ and $\sigma^e = y \bmod N$, then output σ .

Clearly, $\mathcal{A}'$ runs in probabilistic polynomial time. Say the input (N, e, y) to $\mathcal{A}'$ is generated by running $\text{GenRSA}(1^n)$ to obtain (N, e, d) , and then choosing uniform $y \in \mathbb{Z}_N^*$. The crucial observation is that the view of $\mathcal{A}$ when run as a subroutine by $\mathcal{A}'$ is identical to the view of $\mathcal{A}$ in experiment $\text{Sig-forge}_{\mathcal{A}, \Pi}'(n)$. In particular, all Sign-oracle queries are answered correctly, and each of the random-oracle queries of $\mathcal{A}$ when run as a subroutine by $\mathcal{A}'$ is answered with a uniform element of $\mathbb{Z}_N^*$:

- The query $H(m_j)$ is answered with y , a uniform element of $\mathbb{Z}_N^*$.
- Queries $H(m_i)$ with $i \neq j$ are answered with $y_i = [\sigma_i^e \bmod N]$, where σ_i is uniform in $\mathbb{Z}_N^*$. Since exponentiation to the e th power is a one-to-one function, y_i is uniformly distributed as well.

Finally, observe that whenever experiment $\text{Sig-forge}_{\mathcal{A}, \Pi}''(n)$ would output 1, then $\mathcal{A}'$ outputs a correct solution to its given RSA instance. This follows since $\text{Sig-forge}_{\mathcal{A}, \Pi}''(n) = 1$ implies that $j = i$ and $\sigma^e = H(m_i) \bmod N$. Now, when $j = i$, algorithm $\mathcal{A}'$ does not abort and in addition $H(m_i) = y$. Thus, $\sigma^e =$

³Here m_i denotes the i th query made to H . Recall, by assumption, that if $\mathcal{A}$ requests a signature on a message m , then it must have previously queried m to H .

$H(m_i) = y \bmod N$, and so σ is the desired inverse. Using Equation (12.1), this means that

$$\begin{aligned} \Pr[\text{RSA-inv}_{\mathcal{A}', \text{GenRSA}}(n) = 1] &= \Pr[\text{Sig-forge}_{\mathcal{A}, \Pi}''(n) = 1] \\ &= \frac{\Pr[\text{Sig-forge}_{\mathcal{A}, \Pi}(n) = 1]}{q(n)}. \end{aligned} \quad (12.2)$$

If the RSA problem is hard relative to GenRSA , there is a negligible function negl such that $\Pr[\text{RSA-inv}_{\mathcal{A}', \text{GenRSA}}(n) = 1] \leq \text{negl}(n)$. Since $q(n)$ is polynomial, we conclude from Equation (12.2) that $\Pr[\text{Sig-forge}_{\mathcal{A}, \Pi}(n) = 1]$ is negligible as well. This completes the proof. ■

RSA PKCS #1 v2.1. The RSA PKCS #1 v2.1 standard includes a signature scheme that can be viewed as a variant of RSA-FDH. A bit more precisely, the standard defines a scheme in which the signature on a message depends on a *salt* (i.e., a random value) chosen by the signer at the time of signature generation. If this salt is fixed to NULL by the signer—something that is allowed by the standard—the resulting scheme is very similar to RSA-FDH.

It is critical for the range of H to be (close to) *all* of $\mathbb{Z}_N^*$; in particular it does not suffice to simply let H be an “off-the-shelf” cryptographic hash function such as SHA-1. (The output length of SHA-1 is much smaller than the length of RSA moduli used in practice.) Indeed, *practical attacks* on Construction 12.6 are known if the output length of H is too small (e.g., if the output length is 160 bits as would be the case if SHA-1 were used directly as H). In the PKCS #1 v2.1 signature scheme, H is constructed via repeated application of an underlying cryptographic hash function.

12.5 Signatures from the Discrete-Logarithm Problem

Signature schemes can be based on the discrete-logarithm assumption as well, although the assumption does not lend itself as readily to signatures as the RSA assumption does. In Section 12.5.1 we describe a signature scheme introduced by Claus Schnorr that can be proven secure in the random-oracle model. In Section 12.5.2 we describe the DSA and ECDSA signature schemes which are used widely.

12.5.1 The Schnorr Signature Scheme

The underlying intuition for the Schnorr signature scheme is best explained by taking a slight detour to discuss (public-key) *identification schemes*. We then describe the *Fiat-Shamir transform* that can be used to convert identification schemes to signature schemes in the random-oracle model. Finally,

we present the Schnorr identification scheme—and corresponding signature scheme—based on the discrete-logarithm problem.

Identification Schemes

An identification scheme is an interactive protocol that allows one party to prove its identity (i.e., to *authenticate* itself) to another. This is a very natural notion, and it is common nowadays to authenticate oneself when logging in to a website. We call the party identifying herself (e.g., the user) the “prover,” and the party verifying the identity (e.g., the web server) the “verifier.” Here, we are interested in the public-key setting where the prover and verifier do not share any secret information (such as a password) in advance; instead, the verifier only knows the public key of the prover. Successful execution of the identification protocol convinces the verifier that it is communicating with the intended prover rather than an imposter.

We will only consider three-round identification protocols of a specific form, where the prover is specified by two algorithms $\mathcal{P}_1, \mathcal{P}_2$ and the verifier’s side of the protocol is specified by an algorithm $\mathcal{V}$. The prover runs $\mathcal{P}_1(sk)$ using its private key sk to obtain an initial message I along with some state st , and initiates the protocol by sending I to the verifier. In response, the verifier sends a challenge r chosen uniformly from some set Ω_{pk} defined by the prover’s public key pk . Next, the prover runs $\mathcal{P}_2(sk, st, r)$ to compute a response s that it sends back to the verifier. Finally, the verifier computes $\mathcal{V}(pk, r, s)$ and accepts if and only if this results in the initial message I ; see Figure 12.1. Of course, for correctness we require that if the legitimate prover executes the protocol correctly then the verifier should always accept.

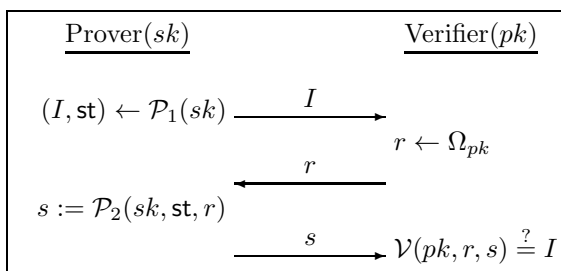


FIGURE 12.1: A three-round identification scheme.

For technical reasons, we assume identification schemes that are “non-degenerate,” which intuitively means that there are many possible initial messages I , and none has a high probability of being sent. Formally, a scheme is *non-degenerate* if for every private key sk and any fixed initial message I , the

probability that $\mathcal{P}_1(sk)$ outputs I is negligible. (Any identification scheme can be trivially modified to be non-degenerate by sending a uniform n -bit string along with the initial message.)

The basic security requirement of an identification scheme is that an adversary who does not know the prover's secret key should be unable to fool the verifier into accepting. This should hold even if the attacker is able to passively eavesdrop on multiple (honest) executions of the protocol between the prover and verifier. We formalize such eavesdropping via an oracle Trans_{sk} that, when called without any input, runs an honest execution of the protocol and returns to the adversary the entire transcript (I, r, s) of the interaction.

Let $\Pi = (\text{Gen}, \mathcal{P}_1, \mathcal{P}_2, \mathcal{V})$ be an identification scheme, and consider the following experiment for an adversary $\mathcal{A}$ and parameter n :

The identification experiment $\text{Ident}_{\mathcal{A}, \Pi}(n)$:

1. $\text{Gen}(1^n)$ is run to obtain keys (pk, sk) .
2. Adversary $\mathcal{A}$ is given pk and access to an oracle Trans_{sk} that it can query as often as it likes.
3. At any point during the experiment, $\mathcal{A}$ outputs a message I . A uniform challenge $r \in \Omega_{pk}$ is chosen and given to $\mathcal{A}$, who responds with some s . ($\mathcal{A}$ may continue to query Trans_{sk} even after receiving r .)
4. The experiment outputs 1 if and only if $\mathcal{V}(pk, r, s) \stackrel{?}{=} I$.

DEFINITION 12.8 An identification scheme $\Pi = (\text{Gen}, \mathcal{P}_1, \mathcal{P}_2, \mathcal{V})$ is secure against a passive attack, or just secure, if for all probabilistic polynomial-time adversaries $\mathcal{A}$, there exists a negligible function negl such that:

$$\Pr[\text{Ident}_{\mathcal{A}, \Pi}(n) = 1] \leq \text{negl}(n).$$

It is also possible to consider stronger notions of security, for example, where the adversary can also carry out *active* attacks on the protocol by impersonating a verifier and possibly sending maliciously chosen values r . We will not need this for our application to signature schemes.

From Identification Schemes to Signatures

The Fiat–Shamir transform (Construction 12.9) provides a way to convert any (interactive) identification scheme into a (non-interactive) signature scheme. The basic idea is for the signer to act as a prover, running the identification protocol *by itself*. That is, to sign a message m , the signer first computes I , and next generates the challenge r by applying some function H to I and m . It then derives the correct response s . The signature on m is (r, s) , which can be verified by (1) recomputing $I := \mathcal{V}(pk, r, s)$ and then (2) checking that $H(I, m) \stackrel{?}{=} r$.

CONSTRUCTION 12.9

Let $(\text{Gen}_{\text{id}}, \mathcal{P}_1, \mathcal{P}_2, \mathcal{V})$ be an identification scheme, and construct a signature scheme as follows:

- **Gen**: on input 1^n , simply run $\text{Gen}_{\text{id}}(1^n)$ to obtain keys pk, sk .
The public key pk specifies a set of challenges Ω_{pk} . As part of key generation, a function $H : \{0, 1\}^* \rightarrow \Omega_{pk}$ is specified, but we leave this implicit.
- **Sign**: on input a private key sk and a message $m \in \{0, 1\}^*$, do:
 1. Compute $(I, \text{st}) \leftarrow \mathcal{P}_1(sk)$.
 2. Compute $r := H(I, m)$.
 3. Compute $s := \mathcal{P}_2(sk, \text{st}, r)$.
 Output the signature (r, s) .
- **Vrfy**: on input a public key pk , a message m , and a signature (r, s) , compute $I := \mathcal{V}(pk, r, s)$ and output 1 if and only if $H(I, m) \stackrel{?}{=} r$.

The Fiat–Shamir transform.

A signature (r, s) is “bound” to a specific message m because r is a function of both I and m ; changing m thus results in a completely different r . If H is modeled as a random oracle mapping inputs uniformly onto Ω_{pk} , then the challenge r is uniform; intuitively, it will be just as difficult for an adversary (who does not know sk) to find a valid signature (r, s) on a message m as it would be to impersonate the prover in an honest execution of the protocol. This intuition is formalized in the proof of the following theorem.

THEOREM 12.10 *Let Π be an identification scheme, and let Π' be the signature scheme that results by applying the Fiat–Shamir transform to it. If Π is secure and H is modeled as a random oracle, then Π' is secure.*

PROOF Let $\mathcal{A}'$ be a probabilistic polynomial-time adversary attacking the signature scheme Π' , with $q = q(n)$ an upper bound on the number of queries that $\mathcal{A}'$ makes to H . We make a number of simplifying assumptions without loss of generality. First, we assume that $\mathcal{A}'$ makes any given query to H only once. We also assume that after being given a signature (r, s) on a message m with $\mathcal{V}(pk, r, s) = I$, the adversary $\mathcal{A}'$ never queries $H(I, m)$ (since it knows the answer will be r). Finally, we assume that if $\mathcal{A}'$ outputs a forged signature (r, s) on a message m with $\mathcal{V}(pk, r, s) = I$, then $\mathcal{A}'$ had previously queried $H(I, m)$.

We construct an efficient adversary $\mathcal{A}$ that uses $\mathcal{A}'$ as a subroutine and attacks the identification scheme Π :

Algorithm $\mathcal{A}$:

The algorithm is given pk and access to an oracle Trans_{sk} .

1. Choose uniform $j \in \{1, \dots, q\}$.
2. Run $\mathcal{A}'(pk)$. Answer its queries as follows:
 When $\mathcal{A}'$ makes its i th random-oracle query $H(I_i, m_i)$, answer it as follows:
 - If $i = j$, output I_j and receive in return a challenge r . Return r to $\mathcal{A}'$ as the answer to its query.
 - If $i \neq j$, choose a uniform $r \in \Omega_{pk}$ and return r as the answer to the query.

When $\mathcal{A}'$ requests a signature on m , answer it as follows:

- (a) Query Trans_{sk} to obtain a transcript (I, r, s) of an honest execution of the protocol.
- (b) Return the signature (r, s) .
3. If $\mathcal{A}'$ outputs a forged signature (r, s) on a message m , compute $I := \mathcal{V}(pk, r, s)$ and check whether $(I, m) \stackrel{?}{=} (I_j, m_j)$. If so, then output s . Otherwise, abort.

The view of $\mathcal{A}'$ when run as a subroutine by $\mathcal{A}$ in experiment $\text{Ident}_{\mathcal{A}, \Pi}(n)$ is *almost* identical to the view of $\mathcal{A}'$ in experiment $\text{Sig-forg}_{\mathcal{A}', \Pi'}(n)$. Indeed, all the H -queries that $\mathcal{A}'$ makes are answered with a uniform value from Ω_{pk} , and all the signing queries that $\mathcal{A}'$ makes are answered with valid signatures having the correct distribution. The only difference between the views is that when $\mathcal{A}'$ is run as a subroutine by $\mathcal{A}$ it is possible for there to be an inconsistency in the answers $\mathcal{A}'$ receives from its queries to H : specifically, this happens if $\mathcal{A}$ ever answers a signing query for a message m using a transcript (I, r, s) for which $H(I, m)$ is already defined (that is, $\mathcal{A}'$ had previously queried (I, m) to H) and $H(I, m) \neq r$. However, if Π is non-degenerate then this only ever happens with negligible probability. Thus, the probability that $\mathcal{A}'$ outputs a forgery when run as a subroutine by $\mathcal{A}$ is $\text{Sig-forg}_{\mathcal{A}', \Pi'}(n) - \text{negl}(n)$ for some negligible function negl .

Consider an execution of experiment $\text{Ident}_{\mathcal{A}, \Pi}(n)$ in which $\mathcal{A}'$ outputs a forged signature (r, s) on a message m , and let $I := \mathcal{V}(pk, r, s)$. Since j is uniform and independent of everything else, the probability that $(I, m) = (I_j, m_j)$ (even conditioned on the event that $\mathcal{A}'$ outputs a forgery) is exactly $1/q$. (Recall we assume that if $\mathcal{A}'$ outputs a forged signature (r, s) on a message m with $\mathcal{V}(pk, r, s) = I$, then $\mathcal{A}'$ had previously queried $H(I, m)$.) When both events happen, $\mathcal{A}$ successfully impersonates the prover. Indeed, $\mathcal{A}$ sends I_j as its initial message, receives in response a challenge r , and responds with s . But $H(I_j, m_j) = r$ and (since the forged signature is valid) $\mathcal{V}(pk, r, s) = I$. Putting everything together, we see that

$$\Pr[\text{Ident}_{\mathcal{A}, \Pi}(n) = 1] \geq \frac{1}{q(n)} \cdot (\Pr[\text{Sig-forg}_{\mathcal{A}', \Pi'}(n) = 1] - \text{negl}(n))$$

or

$$\Pr[\text{Sig-forg}_{\mathcal{A}', \Pi'}(n) = 1] \leq q(n) \cdot \Pr[\text{Ident}_{\mathcal{A}, \Pi}(n) = 1] + \text{negl}(n).$$

If Π is secure then $\Pr[\text{Ident}_{\mathcal{A}, \Pi}(n) = 1]$ is negligible; since $q(n)$ is polynomial this implies that $\Pr[\text{Sig-forg}_{\mathcal{A}', \Pi'}(n) = 1]$ is also negligible. Because $\mathcal{A}'$ was arbitrary, this means Π' is secure. $\blacksquare$

The Schnorr Identification Scheme

The Schnorr identification scheme is based on hardness of the discrete-logarithm problem. Let $\mathcal{G}$ be a polynomial-time algorithm that takes as input 1^n and (except possibly with negligible probability) outputs a description of a cyclic group $\mathbb{G}$, its order q (with $\|q\| = n$), and a generator g . To generate its keys, the prover runs $\mathcal{G}(1^n)$ to obtain $(\mathbb{G}, q, g)$, chooses a uniform $x \in \mathbb{Z}_q$, and sets $y := g^x$; the public key is $\langle \mathbb{G}, q, g, y \rangle$ and the private key is x . To execute the protocol (see Figure 12.2), the prover begins by choosing a uniform $k \in \mathbb{Z}_q^*$ and setting $I := g^k$; it sends I as the initial message. The verifier chooses and sends a uniform challenge $r \in \mathbb{Z}_q$; in response, the prover computes $s := [rx + k \bmod q]$. The verifier accepts if and only if $g^s \cdot y^{-r} \stackrel{?}{=} I$. Correctness holds because

$$g^s \cdot y^{-r} = g^{rx+k} \cdot (g^x)^{-r} = g^k = I.$$

Note that I is uniform in $\mathbb{G}$, and so the scheme is non-degenerate.

Before giving the proof, we provide some high-level intuition. A first important observation is that passive eavesdropping is of no help to the attacker. The reason is that the attacker can *simulate* transcripts of honest executions on its own, based only on the public key and *without* knowledge of the private key. To do this, the attacker just reverses the order of the steps: it first chooses uniform and independent $r, s \in \mathbb{Z}_q$ and then sets $I := g^s \cdot y^{-r}$. In an honest transcript (I, r, s) , the initial message I is a uniform element of $\mathbb{G}$, the challenge is an independent, uniform element of $\mathbb{Z}_q$, and s is then uniquely determined as $s = \log_g(I \cdot y^r)$. Simulated transcripts constructed by an attacker

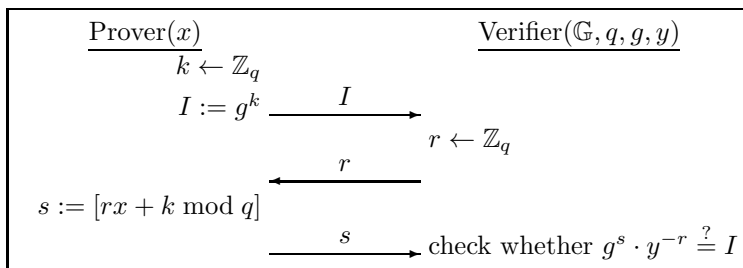


FIGURE 12.2: An execution of the Schnorr identification scheme.

have the same distribution: $r \in \mathbb{Z}_q$ is uniform and, because s is uniform in $\mathbb{Z}_q$ and independent of r , we see that I is uniform in $\mathbb{G}$ and independent of r . Finally, s is uniquely determined as satisfying the same constraint as before. Due to this, we may effectively assume that when attacking the identification scheme, an attacker does not eavesdrop on honest executions at all.

So, we have reduced to an attacker who gets a public key y , sends an initial message I , is given in response a uniform challenge r , and then must send a response s for which $g^s \cdot y^{-r} = I$. Informally, if an attacker is able to do this with high probability then it must, in particular, be able to compute correct responses s_1, s_2 to at least two different challenges $r_1, r_2 \in \mathbb{Z}_q$. Note

$$g^{s_1} \cdot y^{-r_1} = I = g^{s_2} \cdot y^{-r_2},$$

and so $g^{s_1-s_2} = y^{r_1-r_2}$. But this implies that the attacker (who, recall, is able to generate s_1 in response to r_1 , and s_2 in response to r_2) can implicitly compute the discrete logarithm

$$\log_g y = [(s_1 - s_2) \cdot (r_1 - r_2)^{-1} \bmod q],$$

contradicting the assumed hardness of the discrete-logarithm problem.

THEOREM 12.11 *If the discrete-logarithm problem is hard relative to $\mathcal{G}$, then the Schnorr identification scheme is secure.*

PROOF Let Π denote the Schnorr identification scheme, and let $\mathcal{A}$ be a PPT adversary attacking the scheme. We construct the following PPT algorithm $\mathcal{A}'$ solving the discrete-logarithm problem relative to $\mathcal{G}$:

Algorithm $\mathcal{A}'$:

The algorithm is given $\mathbb{G}, q, g, y$ as input.

1. Run $\mathcal{A}(pk)$, answering all its queries to Trans_{sk} as described in the intuition given previously.
2. When $\mathcal{A}$ outputs I , choose a uniform $r_1 \in \mathbb{Z}_q$ as the challenge. Give r_1 to $\mathcal{A}$, who responds with s_1 .
3. Run $\mathcal{A}(pk)$ a second time (from the beginning), using the same randomness as before except for uniform and independent $r_2 \in \mathbb{Z}_q$. Eventually, $\mathcal{A}$ responds with s_2 .
4. If $g^{s_1} \cdot y^{-r_1} = I$ and $g^{s_2} \cdot y^{-r_2} = I$ and $r_1 \neq r_2$ then output $[(s_1 - s_2) \cdot (r_1 - r_2)^{-1} \bmod q]$. Else, output nothing.

Considering a single run of $\mathcal{A}$ as a subroutine of $\mathcal{A}'$, let ω denote the randomness used in that execution except for the challenge itself. So, ω comprises any randomness used by $\mathcal{G}$, the choice of (unknown) private key x , any randomness used by $\mathcal{A}$ itself, and the randomness used by $\mathcal{A}'$ when answering queries

to Trans_{sk} . Define $V(\omega, r)$ to be equal to 1 if and only if $\mathcal{A}$ correctly responds to challenge r when randomness ω is used in the rest of the execution. For any fixed ω , define $\delta_\omega \stackrel{\text{def}}{=} \Pr_r[V(\omega, r) = 1]$; having fixed ω , this is the probability over choice of the challenge r that $\mathcal{A}$ responds correctly.

Define $\delta(n) \stackrel{\text{def}}{=} \Pr[\text{Ident}_{\mathcal{A}, \Pi}(n) = 1]$. Since the simulation of the Trans_{sk} oracle is perfect, we have

$$\delta(n) = \Pr_{\omega, r}[V(\omega, r) = 1] = \sum_{\omega} \Pr[\omega] \cdot \delta_\omega.$$

Moreover, the intuition preceding the proof shows that $\mathcal{A}'$ correctly computes the discrete logarithm of y whenever $\mathcal{A}$ succeeds twice and $r_1 \neq r_2$. Thus:

$$\begin{aligned} \Pr[\text{DLog}_{\mathcal{A}', \mathcal{G}}(n) = 1] &= \Pr_{\omega, r_1, r_2}[V(\omega, r_1) \wedge V(\omega, r_2) \wedge r_1 \neq r_2] \\ &\geq \Pr_{\omega, r_1, r_2}[V(\omega, r_1) \wedge V(\omega, r_2)] - \Pr_{\omega, r_1, r_2}[r_1 = r_2] \\ &= \sum_{\omega} \Pr[\omega] \cdot (\delta_\omega)^2 - 1/q \\ &\geq \left(\sum_{\omega} \Pr[\omega] \cdot \delta_\omega\right)^2 - 1/q \\ &= \delta(n)^2 - 1/q, \end{aligned}$$

using Jensen's inequality⁴ in the second-to-last step. If the discrete-logarithm problem is hard relative to $\mathcal{G}$ then $\Pr[\text{DLog}_{\mathcal{A}', \mathcal{G}}(n) = 1]$ is negligible. Since $1/q$ is negligible (because $\|q\| = n$), this implies that $\delta(n)$ is also negligible, and so Π is a secure identification scheme. $\blacksquare$

The Schnorr signature scheme is obtained by applying the Fiat–Shamir transform to the Schnorr identification scheme. See Construction 12.12.

CONSTRUCTION 12.12

Let $\mathcal{G}$ be as described in the text.

- **Gen**: run $\mathcal{G}(1^n)$ to obtain $(\mathbb{G}, q, g)$. Choose a uniform $x \in \mathbb{Z}_q$ and set $y := g^x$. The private key is x and the public key is $(\mathbb{G}, q, g, y)$. As part of key generation, a function $H : \{0, 1\}^* \rightarrow \mathbb{Z}_q$ is specified, but we leave this implicit.
- **Sign**: on input a private key x and a message $m \in \{0, 1\}^*$, choose uniform $k \in \mathbb{Z}_q$ and set $I := g^k$. Then compute $r := H(I, m)$, followed by $s := [rx + k \bmod q]$. Output the signature (r, s) .
- **Vrfy**: on input a public key $(\mathbb{G}, q, g, y)$, a message m , and a signature (r, s) , compute $I := g^s \cdot y^{-r}$ and output 1 if $H(I, m) \stackrel{?}{=} r$.

The Schnorr signature scheme.

⁴Jensen's inequality says that $\sum_i a_i \cdot b_i^2 \geq (\sum_i a_i)^{-1} \cdot (\sum_i a_i \cdot b_i)^2$ for positive $\{a_i\}$.

12.5.2 DSA and ECDSA

The *Digital Signature Algorithm* (DSA) and *Elliptic Curve Digital Signature Algorithm* (ECDSA) are based on the discrete-logarithm problem in different classes of groups. They have been around in some form since 1991, and are both included in the current *Digital Signature Standard* (DSS) issued by NIST.

Both schemes follow a common template and can be viewed as being constructed from an underlying identification scheme (see the previous section). Let $\mathbb{G}$ be a cyclic group of prime order q with generator g . Consider the following identification scheme in which the prover's private key is x and public key is $(\mathbb{G}, q, g, y)$ with $y = g^x$:

1. The prover chooses uniform $k \in \mathbb{Z}_q^*$ and sends $I := g^k$.
2. The verifier chooses and sends uniform $\alpha, r \in \mathbb{Z}_q$ as the challenge.
3. The prover sends $s := [k^{-1} \cdot (\alpha + xr) \bmod q]$ as the response.
4. The verifier accepts if $s \neq 0$ and $g^{\alpha s^{-1}} \cdot y^{r s^{-1}} \stackrel{?}{=} I$.

Note $s \neq 0$ unless $\alpha = -xr \bmod q$, which occurs with negligible probability. Assuming $s \neq 0$, the inverse $s^{-1} \bmod q$ exists and

$$g^{\alpha s^{-1}} \cdot y^{r s^{-1}} = g^{\alpha s^{-1}} \cdot g^{x r s^{-1}} = g^{(\alpha + xr) \cdot s^{-1}} = g^{(\alpha + xr) \cdot k \cdot (\alpha + xr)^{-1}} = I.$$

We thus see that correctness holds with all but negligible probability.

One can show that this identification scheme is secure if the discrete-logarithm problem is hard relative to $\mathcal{G}$. We merely sketch the argument, assuming familiarity with the results of the previous section. First of all, transcripts of honest executions can be simulated: to do so, simply choose uniform $\alpha, r \in \mathbb{Z}_q$ and $s \in \mathbb{Z}_q^*$, and then set $I := g^{\alpha s^{-1}} \cdot y^{r s^{-1}}$. (This no longer gives a *perfect* simulation, but it is close enough.) Moreover, if an attacker outputs an initial message I for which it can give correct responses $s_1, s_2 \in \mathbb{Z}_q^*$ to distinct challenges $(\alpha, r_1), (\alpha, r_2)$ then

$$g^{\alpha s_1^{-1}} \cdot y^{r_1 s_1^{-1}} = I = g^{\alpha s_2^{-1}} \cdot y^{r_2 s_2^{-1}},$$

and so $g^{\alpha(s_1^{-1} - s_2^{-1})} = h^{r_1 s_1^{-1} - r_2 s_2^{-1}}$ and $\log_g h$ can be computed as in the previous section. The same holds if the attacker gives correct responses to distinct challenges $(\alpha_1, r), (\alpha_2, r)$.

The DSA/ECDSA signature schemes are constructed by “collapsing” the above identification scheme into a non-interactive algorithm run by the signer. In contrast to the Fiat–Shamir transform, however, the transformation here is carried out as follows (see Construction 12.13):

- Set $\alpha := H(m)$, where m is the message being signed and H is a cryptographic hash function.

- Set $r := F(I)$ for a (specified) function $F : \mathbb{G} \rightarrow \mathbb{Z}_q$. Here, F is a “simple” function that is *not* intended to act like a random oracle.

The function F depends on the group $\mathbb{G}$, which in turn depends on the scheme. In DSA, $\mathbb{G}$ is taken to be an order- q subgroup of $\mathbb{Z}_p^*$, for p prime (cf. Section 8.3.3), and $F(I) \stackrel{\text{def}}{=} [I \bmod q]$. In ECDSA, $\mathbb{G}$ is an order- q subgroup of an elliptic-curve group $E(\mathbb{Z}_p)$, for p prime.⁵ Recall from Section 8.3.4 that any element of such a group can be represented as a pair $(x, y) \in \mathbb{Z}_p \times \mathbb{Z}_p$. The function F in this case is defined as $F((x, y)) \stackrel{\text{def}}{=} [x \bmod q]$.

CONSTRUCTION 12.13

Let $\mathcal{G}$ be as in the text.

- **Gen:** on input 1^n , run $\mathcal{G}(1^n)$ to obtain $(\mathbb{G}, q, g)$. Choose uniform $x \in \mathbb{Z}_q$ and set $y := g^x$. The public key is $\langle \mathbb{G}, q, g, y \rangle$ and the private key is x .

As part of key generation, two functions $H : \{0, 1\}^* \rightarrow \mathbb{Z}_q$ and $F : \mathbb{G} \rightarrow \mathbb{Z}_q$ are specified, but we leave this implicit.

- **Sign:** on input the private key x and a message $m \in \{0, 1\}^*$, choose uniform $k \in \mathbb{Z}_q^*$ and set $r := F(g^k)$. Then compute $s := [k^{-1} \cdot (H(m) + xr) \bmod q]$. (If $r = 0$ or $s = 0$ then start again with a fresh choice of k .) Output the signature (r, s) .
- **Vrfy:** on input a public key $\langle \mathbb{G}, q, g, y \rangle$, a message $m \in \{0, 1\}^*$, and a signature (r, s) with $r, s \neq 0 \bmod q$, output 1 if and only if

$$r \stackrel{?}{=} F\left(g^{H(m) \cdot s^{-1}} y^{r \cdot s^{-1}}\right).$$

DSA and ECDSA—abstractly.

Assuming hardness of the discrete-logarithm problem, DSA and ECDSA can be proven secure if H and F are modeled as random oracles. As we have discussed above, however, while the random-oracle model may be reasonable for H , it is *not* an appropriate model for F . No proofs of security are known for the specific choices of F in the standard. Nevertheless, DSA and ECDSA have been used and studied for decades without any attacks being found.

Proper generation of k . The DSA/ECDSA schemes specify that the signer should choose a uniform $k \in \mathbb{Z}_q^*$ when computing a signature. Failure to choose k properly (e.g., due to poor random-number generation) can lead to catastrophic results. For starters, if an attacker can predict the value of k used to compute a signature (r, s) on a message m , then they can compute

⁵ECDSA also allows elliptic curves over other fields, but we have only covered the case of prime fields in Section 8.3.4.

the signer's private key. This is true because $s = k^{-1} \cdot (H(m) + xr) \bmod q$, and if k is known then the only unknown is the private key x .

Even if k is unpredictable, the attacker can compute the signer's private key if the *same* k is ever used to generate two different signatures. The attacker can easily tell when this happens because then r repeats as well. Say (r, s_1) and (r, s_2) are signatures on messages m_1 and m_2 , respectively. Then

$$\begin{aligned}s_1 &= k^{-1} \cdot (H(m_1) + xr) \bmod q \\ s_2 &= k^{-1} \cdot (H(m_2) + xr) \bmod q.\end{aligned}$$

Subtracting gives $s_1 - s_2 = k^{-1} (H(m_1) - H(m_2)) \bmod q$, from which k can be computed; given k , the attacker can determine the private key x as in the previous paragraph. This very attack was used by hackers to extract the master private key from the Sony PlayStation (PS3) in 2010.

12.6 *Signatures from Hash Functions

Interestingly—and perhaps somewhat surprisingly—signature schemes can be constructed based on cryptographic hash functions, *without* relying on number-theoretic assumptions. (This is in contrast to public-key encryption, which seems to require hard problems with some algebraic structure.) In this section we explore such constructions. The schemes we will see are also interesting in that they do not rely on random oracles, as opposed to all the constructions we have described previously in this chapter.

12.6.1 Lamport's Signature Scheme

We initiate our study of signature schemes based on hash functions by considering the relatively weak notion of *one-time-secure signature schemes*. Informally, such schemes are “secure” as long as a given private key is used to sign only a *single* message. Schemes satisfying this notion of security may be appropriate for some applications, and also serve as useful “building blocks” for achieving stronger notions of security, as we will see in the following section.

Let $\Pi = (\text{Gen}, \text{Sign}, \text{Vrfy})$ be a signature scheme, and consider the following experiment for an adversary $\mathcal{A}$ and parameter n :

The one-time signature experiment $\text{Sig-forge}_{\mathcal{A}, \Pi}^{1\text{-time}}(n)$:

1. $\text{Gen}(1^n)$ is run to obtain keys (pk, sk) .
2. Adversary $\mathcal{A}$ is given pk and asks a single query m' to its oracle $\text{Sign}_{sk}(\cdot)$. $\mathcal{A}$ then outputs (m, σ) with $m \neq m'$.
3. The output of the experiment is defined to be 1 if and only if $\text{Vrfy}_{pk}(m, \sigma) = 1$.

DEFINITION 12.14 *Signature scheme $\Pi = (\text{Gen}, \text{Sign}, \text{Vrfy})$ is existentially unforgeable under a single-message attack, or is a one-time-secure signature scheme, if for all probabilistic polynomial-time adversaries $\mathcal{A}$, there exists a negligible function negl such that:*

$$\Pr \left[\text{Sig-forge}_{\mathcal{A}, \Pi}^{1\text{-time}}(n) = 1 \right] \leq \text{negl}(n).$$

Leslie Lamport showed a construction of a one-time-secure signature scheme in 1979. We illustrate the idea for the case of signing 3-bit messages. Let H be a cryptographic hash function. A private key consists of six uniform values $x_{1,0}, x_{1,1}, x_{2,0}, x_{2,1}, x_{3,0}, x_{3,1} \in \{0,1\}^n$, and the corresponding public key contains the results obtained by applying H to each of these elements. These keys can be visualized as two-dimensional arrays:

$$pk = \begin{pmatrix} y_{1,0} & y_{2,0} & y_{3,0} \\ y_{1,1} & y_{2,1} & y_{3,1} \end{pmatrix} \quad sk = \begin{pmatrix} x_{1,0} & x_{2,0} & x_{3,0} \\ x_{1,1} & x_{2,1} & x_{3,1} \end{pmatrix}.$$

To sign a message $m = m_1m_2m_3$ (where $m_i \in \{0,1\}$), the signer releases the appropriate preimage x_{i,m_i} for each bit of the message; the signature σ consists of the three values $(x_{1,m_1}, x_{2,m_2}, x_{3,m_3})$. Verification is carried out in the natural way: presented with the candidate signature (x_1, x_2, x_3) on the message $m = m_1m_2m_3$, accept if and only if $H(x_i) \stackrel{?}{=} y_{i,m_i}$ for $1 \leq i \leq 3$. This is shown graphically in Figure 12.3, and the general case—for messages of any length ℓ —is described formally in Construction 12.15.

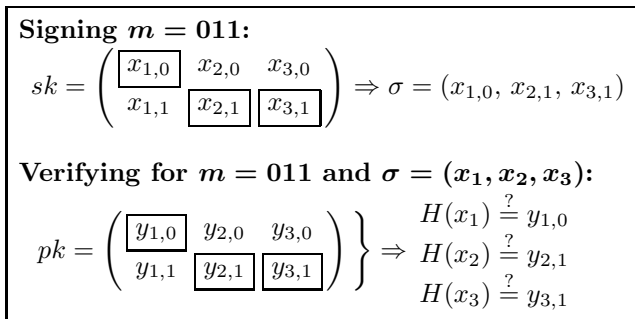


FIGURE 12.3: The Lamport scheme used to sign the message $m = 011$.

After observing a signature on a message, an attacker who wishes to forge a signature on any *other* message must find a preimage of one of the three “unused” elements in the public key. If H is *one-way* (see Definition 8.72), then finding any such preimage is computationally difficult.

THEOREM 12.16 *Let ℓ be any polynomial. If H is a one-way function, then Construction 12.15 is a one-time-secure signature scheme.*

CONSTRUCTION 12.15

Let $H : \{0, 1\}^* \rightarrow \{0, 1\}^*$ be a function. Construct a signature scheme for messages of length $\ell = \ell(n)$ as follows:

- **Gen**: on input 1^n , proceed as follows for $i \in \{1, \dots, \ell\}$:

1. Choose uniform $x_{i,0}, x_{i,1} \in \{0, 1\}^n$.
2. Compute $y_{i,0} := H(x_{i,0})$ and $y_{i,1} := H(x_{i,1})$.

The public key pk and the private key sk are

$$pk = \begin{pmatrix} y_{1,0} & y_{2,0} & \cdots & y_{\ell,0} \\ y_{1,1} & y_{2,1} & \cdots & y_{\ell,1} \end{pmatrix} \quad sk = \begin{pmatrix} x_{1,0} & x_{2,0} & \cdots & x_{\ell,0} \\ x_{1,1} & x_{2,1} & \cdots & x_{\ell,1} \end{pmatrix}.$$

- **Sign**: on input a private key sk as above and a message $m \in \{0, 1\}^\ell$ with $m = m_1 \cdots m_\ell$, output the signature $(x_{1,m_1}, \dots, x_{\ell,m_\ell})$.
- **Vrfy**: on input a public key pk as above, a message $m \in \{0, 1\}^\ell$ with $m = m_1 \cdots m_\ell$, and a signature $\sigma = (x_1, \dots, x_\ell)$, output 1 if and only if $H(x_i) = y_{i,m_i}$ for all $1 \leq i \leq \ell$.

The Lamport signature scheme.

PROOF Let $\ell = \ell(n)$ throughout. As noted a moment ago, the key observation is this: say an attacker $\mathcal{A}$ requests a signature on a message m' , and consider any other message $m \neq m'$. There must be at least one position $i^* \in \{1, \dots, \ell\}$ on which m and m' differ. Say $m_{i^*} = b \neq m'_{i^*}$. Then forging a signature on m requires, at least, finding a preimage (under H) of element y_{i^*,b^*} of the public key. Since H is one-way, this should be infeasible. We now formalize this intuition.

Let Π denote the Lamport scheme, and let $\mathcal{A}$ be a probabilistic polynomial-time adversary. In a particular execution of $\text{Sig-forg}_{\mathcal{A},\Pi}^{1-\text{time}}(n)$, let m' denote the message whose signature is requested by $\mathcal{A}$ (we assume without loss of generality that $\mathcal{A}$ always requests a signature on a message), and let (m, σ) be the final output of $\mathcal{A}$. We say that $\mathcal{A}$ *outputs a forgery at (i, b)* if $\text{Vrfy}_{pk}(m, \sigma) = 1$ and furthermore $m_i \neq m'_i$ (i.e., messages m and m' differ on their i th position) and $m_i = b \neq m'_{i^*}$. Note that whenever $\mathcal{A}$ outputs a forgery, it outputs a forgery at *some* (i, b) .

Consider the following PPT algorithm $\mathcal{I}$ attempting to invert H :

Algorithm $\mathcal{I}$:

The algorithm is given 1^n and y as input.

1. Choose uniform $i^* \in \{1, \dots, \ell\}$ and $b^* \in \{0, 1\}$. Set $y_{i^*,b^*} := y$.
2. For all $i \in \{1, \dots, \ell\}$ and $b \in \{0, 1\}$ with $(i, b) \neq (i^*, b^*)$:
 - Choose uniform $x_{i,b} \in \{0, 1\}^n$ and set $y_{i,b} := H(x_{i,b})$.
3. Run $\mathcal{A}$ on input $pk := \begin{pmatrix} y_{1,0} & y_{2,0} & \cdots & y_{\ell,0} \\ y_{1,1} & y_{2,1} & \cdots & y_{\ell,1} \end{pmatrix}$.

4. When $\mathcal{A}$ requests a signature on the message m' :
 - If $m'_{i^*} = b^*$, then $\mathcal{I}$ aborts the execution.
 - Otherwise, $\mathcal{I}$ returns the signature $\sigma = (x_{1,m'_1}, \dots, x_{\ell,m'_\ell})$.
5. When $\mathcal{A}$ outputs (m, σ) with $\sigma = (x_1, \dots, x_\ell)$:
 - If $\mathcal{A}$ outputs a forgery at (i^*, b^*) , then output x_{i^*} .

Whenever $\mathcal{A}$ outputs a forgery at (i^*, b^*) , algorithm $\mathcal{I}$ succeeds in inverting its given input y . We are interested in the probability that this occurs when the input to $\mathcal{I}$ is generated by choosing uniform $x \in \{0,1\}^n$ and setting $y := H(x)$ (cf. Definition 8.72). Imagine a “mental experiment” in which $\mathcal{I}$ is given x at the outset, sets $x_{i^*,b^*} := x$, and then always returns a signature to $\mathcal{A}$ in step 4 (i.e., even if $m'_{i^*} = b^*$). The view of $\mathcal{A}$ when run as a subroutine by $\mathcal{I}$ in this mental experiment is distributed identically to the view of $\mathcal{A}$ in experiment $\text{Sig-forg}_{\mathcal{A},\Pi}^{1\text{-time}}(n)$. Because (i^*, b^*) was chosen uniformly at the beginning of the experiment, and the view of $\mathcal{A}$ is independent of this choice, the probability that $\mathcal{A}$ outputs a forgery at (i^*, b^*) , conditioned on the fact that $\mathcal{A}$ outputs a forgery at all, is at least $1/2\ell$. (This is because a signature forgery implies a forgery for at least one point (i, b) . Since there are $2\ell(n)$ points, the probability of the forgery being at (i^*, b^*) is at least $1/2\ell$.) We conclude that, in this mental experiment, the probability that $\mathcal{A}$ outputs a forgery at (i^*, b^*) is at least $\frac{1}{2\ell} \cdot \Pr[\text{Sig-forg}_{\mathcal{A},\Pi}^{1\text{-time}}(n) = 1]$.

Returning to the real experiment involving $\mathcal{I}$ as initially described, the key observation is that *the probability that $\mathcal{A}$ outputs a forgery at (i^*, b^*) is unchanged*. This is because the mental experiment and the real experiment coincide if $\mathcal{A}$ outputs a forgery at (i^*, b^*) . That is, the experiments only differ if $m'_{i^*} = b^*$, but if this happens then it is impossible (by definition) for $\mathcal{A}$ to subsequently output a forgery at (i^*, b^*) . So the probability that $\mathcal{A}$ outputs a forgery at (i^*, b^*) is still at least $\frac{1}{2\ell} \cdot \Pr[\text{Sig-forg}_{\mathcal{A},\Pi}^{1\text{-time}}(n) = 1]$. In other words,

$$\Pr[\text{Invert}_{\mathcal{I},H}(n) = 1] \geq \frac{1}{2\ell} \cdot \Pr[\text{Sig-forg}_{\mathcal{A},\Pi}^{1\text{-time}}(n) = 1].$$

Because H is a one-way function, there is a negligible function negl such that

$$\text{negl}(n) \geq \Pr[\text{Invert}_{\mathcal{I},H}(n) = 1].$$

Since ℓ is polynomial this implies that $\Pr[\text{Sig-forg}_{\mathcal{A},\Pi}^{1\text{-time}}(n) = 1]$ is negligible, completing the proof. ■

COROLLARY 12.17 *If one-way functions exist, then for any polynomial ℓ there is a one-time-secure signature scheme for messages of length ℓ .*

12.6.2 Chain-Based Signatures

Being able to sign only a single message with a given private key is obviously a significant drawback. We show here an approach based on collision-resistant hash functions that allows a signer to sign *arbitrarily many* messages, at the expense of maintaining *state* that must be updated after each signature is generated. In Section 12.6.3 we discuss a more efficient variant of this approach (that still requires state), and then describe how this modified construction can be made *stateless*. The result shows that signature schemes satisfying Definition 12.2 can be constructed based on collision-resistant hash functions.

We first define signature schemes that allow the signer to maintain *state* that is updated after every signature is produced.

DEFINITION 12.18 *A stateful signature scheme is a tuple of probabilistic polynomial-time algorithms $(\text{Gen}, \text{Sign}, \text{Vrfy})$ satisfying the following:*

1. *The key-generation algorithm Gen takes as input a security parameter 1^n and outputs (pk, sk, s_0) . These are called the public key, private key, and initial state, respectively. We assume pk and sk each has length at least n , and that n can be determined from pk, sk .*
2. *The signing algorithm Sign takes as input a private key sk , a value s_{i-1} , and a message $m \in \{0, 1\}^*$. It outputs a signature σ and a value s_i .*
3. *The deterministic verification algorithm Vrfy takes as input a public key pk , a message m , and a signature σ . It outputs a bit b .*

We require that for every n , every (pk, sk, s_0) output by $\text{Gen}(1^n)$, and any messages $m_1, \dots, m_t \in \{0, 1\}^$, if we iteratively compute $(\sigma_i, s_i) \leftarrow \text{Sign}_{sk, s_{i-1}}(m_i)$ for $i = 1, \dots, t$, then for every $i \in \{1, \dots, t\}$, it holds that $\text{Vrfy}_{pk}(m_i, \sigma_i) = 1$.*

We emphasize that the verifier does not need to know the signer's state in order to verify a signature; in fact, in some schemes the state must be kept secret by the signer in order for security to hold. Signature schemes that do not maintain state (as in Definition 12.1) are called *stateless* to distinguish them from stateful schemes. Clearly, stateless schemes are preferable (although stateful schemes can still potentially be useful). We introduce stateful signatures as a stepping stone to an eventual stateless construction.

Security for stateful signatures schemes is exactly analogous to Definition 12.2, with the only subtleties being that the signing oracle returns only the signature (and *not* the state), and that the signing oracle updates the state each time it is invoked.

For any polynomial $t = t(n)$, we can easily construct a stateful “ t -time-secure” signature scheme. (The definition of security here would be the obvious generalization of Definition 12.14.) We can do this by simply letting the public key (resp., private key) consist of t independently generated public keys (resp., private keys) for any one-time-secure signature scheme; i.e.,

set $pk := \langle pk_1, \dots, pk_t \rangle$ and $sk := \langle sk_1, \dots, sk_t \rangle$ where each (pk_i, sk_i) is an independently generated key-pair for some one-time-secure signature scheme. The state is a counter i initially set to 1. To sign a message m using the private key sk and current state $i \leq t$, compute $\sigma \leftarrow \text{Sign}_{sk_i}(m)$ (that is, generate a signature on m using the private key sk_i) and output (σ, i) ; the state is updated to $i := i + 1$. Since the state starts at 1, this means the i th message is signed using sk_i . Verification of a signature (σ, i) on a message m is done by checking whether σ is a valid signature on m with respect to pk_i . This scheme is secure if used to sign t messages since each private key of the underlying one-time-secure scheme is used to sign only a *single* message.

As described, signatures have constant length (i.e., independent of t), but the public key has length *linear* in t . It is possible to trade off the length of the public key and signature by having the signer compute a Merkle tree $h := \mathcal{MT}_t(pk_1, \dots, pk_t)$ (see Section 5.6.2) over the t underlying public keys from the one-time-secure scheme. That is, the public key will now be $\langle t, h \rangle$, and the signature on the i th message will include (σ, i) , as before, along with the i th value pk_i and a proof π_i that this is the correct value corresponding to h . (Verification is done in the natural way.) The public key now has constant size, and the signature length grows only logarithmically with t .

Since t can be an arbitrary polynomial, why don't the previous schemes give us the solution we are looking for? The main drawback is that they require the upper bound t on the number of messages that can be signed *to be fixed in advance*, at the time of key generation. This is a potentially severe limitation since once the upper bound is reached a new public key would have to be generated and distributed. We would prefer instead to have a single, fixed public key that can be used to sign an *unbounded* number of messages.

Let $\Pi = (\text{Gen}, \text{Sign}, \text{Vrfy})$ be a one-time-secure signature scheme. In the scheme we have just described (ignoring the Merkle-tree optimization), the signer runs t invocations of Gen to obtain public keys $pk_1, \dots, pk_t$, and includes each of these in its actual public key pk . The signer is then restricted to signing at most t messages. We can do better by using a “chain-based” scheme in which the signer generates additional public keys *on-the-fly*, as needed.

In the chain-based scheme, the public key consists of just a single public key pk_1 generated using Gen , and the private key is just the associated private key sk_1 . To sign the first message m_1 , the signer first generates a new key-pair (pk_2, sk_2) using Gen , and then signs both m_1 and pk_2 using sk_1 to obtain $\sigma_1 \leftarrow \text{Sign}_{sk_1}(m_1 \| pk_2)$. The signature that is output includes both pk_2 and σ_1 , and the signer adds $(m_1, pk_2, sk_2, \sigma_1)$ to its current state. In general, when it comes time to sign the i th message the signer will have stored $\{(m_j, pk_{j+1}, sk_{j+1}, \sigma_j)\}_{j=1}^{i-1}$ as part of its state. To sign the i th message m_i , the signer first generates a new key-pair (pk_{i+1}, sk_{i+1}) using Gen , and then signs m_i and pk_{i+1} using sk_i to obtain a signature $\sigma_i \leftarrow \text{Sign}_{sk_i}(m_i \| pk_{i+1})$. The actual signature that is output includes pk_{i+1} , σ_i , and also the values $\{m_j, pk_{j+1}, \sigma_j\}_{j=1}^{i-1}$. The signer then adds $(m_i, pk_{i+1}, sk_{i+1}, \sigma_i)$ to its state. See Figure 12.4 for a graphical depiction of this process.

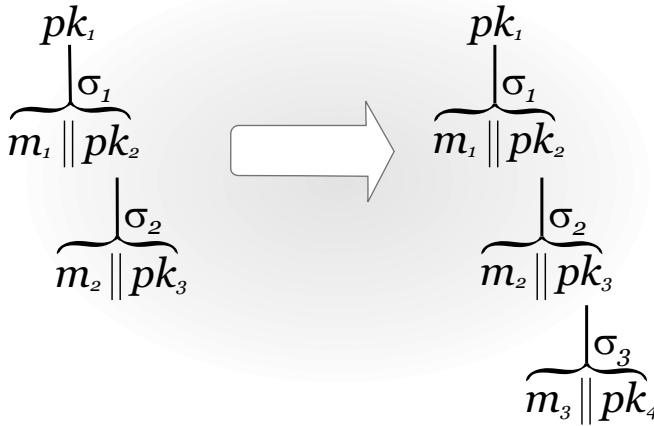


FIGURE 12.4: Chain-based signatures: the situation before and after signing the third message m_3 .

To verify a signature $(pk_{i+1}, \sigma_i, \{m_j, pk_{j+1}, \sigma_j\}_{j=1}^{i-1})$ on a message $m = m_i$ with respect to public key pk_1 , the receiver verifies each link between a public key pk_j and the next public key pk_{j+1} in the chain, as well as the link between the last public key pk_i and m . That is, verification outputs 1 if and only if $\text{Vrfy}_{pk_j}(m_j || pk_{j+1}, \sigma_j) \stackrel{?}{=} 1$ for all $j \in \{1, \dots, i\}$. (Refer to Figure 12.4.)

It is not hard to be convinced—at least on an intuitive level—that this signature scheme is existentially unforgeable under an adaptive chosen-message attack (regardless of how many messages are signed). Informally, this is once again due to the fact that each key-pair (pk_i, sk_i) is used to sign only a single “message,” where in this case the “message” is actually a message/public-key pair $m_i || pk_{i+1}$. Since we will prove security of a more efficient scheme in the next section, we do not prove security for the chain-based scheme here.

In the chain-based scheme, each public key pk_i is used to sign both a message and another public key. Thus, it is essential that the underlying one-time-secure signature scheme Π is capable of signing messages longer than the public key. The Lamport scheme presented in Section 12.6.1 does *not* have this property. However, if we apply the hash-and-sign paradigm from Section 12.3 to the Lamport scheme, we *do* obtain a one-time-secure signature scheme that can sign messages of arbitrary length. (Although Theorem 12.4 was stated only with regard to signature schemes satisfying Definition 12.2, it is not hard to see that an identical proof works for one-time-secure signature schemes.) Because this result is crucial for the next section, we state it formally. (Note that the existence of collision-resistant hash functions implies the existence of one-way functions; see Exercise 7.4.)

LEMMA 12.19 *If collision-resistant hash functions exist, then there exists a one-time-secure signature scheme (for messages of arbitrary length).*

The chain-based signature scheme is a stateful signature scheme that is existentially unforgeable under an adaptive chosen-message attack. It has a number of disadvantages, though. For one, there is no immediate way to eliminate the state (recall that our ultimate goal is a stateless scheme satisfying Definition 12.2). It is also not very efficient, in that the signature length, size of the state, and verification time are all linear in the number of messages that have been signed. Finally, each signature reveals all previously signed messages, and this may be undesirable in some contexts.

12.6.3 Tree-Based Signatures

The signer in the chain-based scheme of the previous section can be viewed as maintaining a *tree* of degree 1, rooted at the public key pk_1 , and with depth equal to the number of messages signed so far (cf. Figure 12.4). A natural way to improve efficiency is to use a *binary* tree in which each node has degree 2. As before, a signature will correspond to a “signed” path in the tree from a leaf to the root; as long as the tree has polynomial depth (even if it has exponential size!), verification can be done in polynomial time.

Concretely, to sign messages of length n we will work with a binary tree of depth n having 2^n leaves. As before, the signer will add nodes to the tree “on-the-fly,” as needed. In contrast to the chain-based scheme, however, only leaves (and not internal nodes) will be used for signing messages. Each leaf of the tree will correspond to one of the possible messages of length n .

In more detail, we imagine a binary tree of depth n where the root is labeled by ε (i.e., the empty string), and a node that is labeled with the binary string w (of length less than n) has left-child labeled $w0$ and right-child labeled $w1$. This tree is never constructed in its entirety (note that it has exponential size), but is instead built up by the signer as needed.

For every node w , we associate a pair of keys pk_w, sk_w for a one-time-secure signature scheme Π . The public key of the root, pk_ε , is the actual public key of the signer. To sign a message $m \in \{0, 1\}^n$, the signer does the following:

1. It first generates keys (as needed) for all nodes on the path from the root to the leaf labeled m . (Some of these public keys may have been generated in the process of signing previous messages, and in that case are not generated again.)
2. Next, it “certifies” the path from the root to the leaf labeled m by computing a signature on $pk_{w0} \| pk_{w1}$, using private key sk_w , for each string w that is a proper prefix of m .
3. Finally, it “certifies” m itself by computing a signature on m using the private key sk_m .

The final signature on m consists of the signature on m with respect to pk_m , as well as all the information needed to verify the path from the leaf labeled

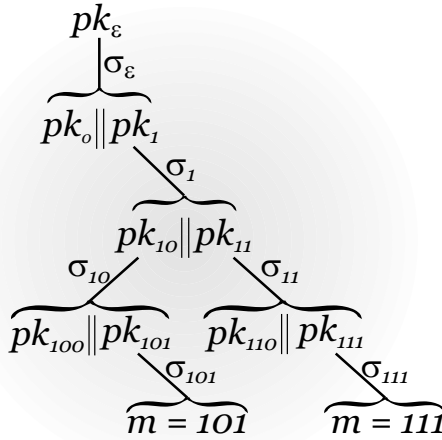


FIGURE 12.5: Tree-based signatures (conceptually).

m to the root; see Figure 12.5. Additionally, the signer updates its state by storing all the keys generated as part of the above signing process. A formal description of this scheme is given as Construction 12.20.

Notice that each of the underlying keys in this scheme is used to sign only a *single* “message.” Each key associated with an internal node signs a pair of public keys, and a key at a leaf is used to sign only a single message. Since each key is used to sign a *pair* of other keys, we again need the one-time-secure signature scheme Π to be capable of signing messages longer than the public key. Lemma 12.19 shows that such schemes can be constructed based on collision-resistant hash functions.

Before proving security of this tree-based approach, note that it improves on the chain-based scheme in a number of respects. It still allows for signing an unbounded number of messages. (Although there are only 2^n leaves, the message space contains only 2^n messages. In any case, 2^n is eventually larger than any polynomial function of n .) In terms of efficiency, the signature length and verification time are now proportional to the message length n but are *independent* of the number of messages signed. The scheme is still stateful, but we will see how this can be avoided after we prove the following result.

THEOREM 12.21 *Let Π be a one-time-secure signature scheme. Then Construction 12.20 is a secure signature scheme.*

PROOF Let Π^* denote Construction 12.20. Let $\mathcal{A}^*$ be a probabilistic polynomial time adversary, let $\ell^* = \ell^*(n)$ be a (polynomial) upper bound on the number of signing queries made by $\mathcal{A}^*$, and set $\ell = \ell(n) \stackrel{\text{def}}{=} 2n\ell^*(n) + 1$.

CONSTRUCTION 12.20

Let $\Pi = (\text{Gen}, \text{Sign}, \text{Vrfy})$ be a signature scheme. For a binary string m , let $m|_i \stackrel{\text{def}}{=} m_1 \cdots m_i$ denote the i -bit prefix of m (with $m|_0 \stackrel{\text{def}}{=} \varepsilon$, the empty string). Construct the scheme $\Pi^* = (\text{Gen}^*, \text{Sign}^*, \text{Vrfy}^*)$ as follows:

- **Gen***: on input 1^n , compute $(pk_\varepsilon, sk_\varepsilon) \leftarrow \text{Gen}(1^n)$ and output the public key pk_ε . The private key and initial state are sk_ε .
- **Sign***: on input a message $m \in \{0, 1\}^n$, carry out the following.
 1. For $i = 0$ to $n - 1$:
 - If $pk_{m|_i 0}, pk_{m|_i 1}$, and $\sigma_{m|_i}$ are not in the state, compute $(pk_{m|_i 0}, sk_{m|_i 0}) \leftarrow \text{Gen}(1^n)$, $(pk_{m|_i 1}, sk_{m|_i 1}) \leftarrow \text{Gen}(1^n)$, and $\sigma_{m|_i} \leftarrow \text{Sign}_{sk_{m|_i}}(pk_{m|_i 0} \parallel pk_{m|_i 1})$. In addition, add all of these values to the state.
 2. If σ_m is not yet included in the state, compute $\sigma_m \leftarrow \text{Sign}_{sk_m}(m)$ and store it as part of the state.
 3. Output the signature $(\{\sigma_{m|_i}, pk_{m|_i 0}, pk_{m|_i 1}\}_{i=0}^{n-1}, \sigma_m)$.
- **Vrfy***: on input a public key pk_ε , message m , and signature $(\{\sigma_{m|_i}, pk_{m|_i 0}, pk_{m|_i 1}\}_{i=0}^{n-1}, \sigma_m)$, output 1 if and only if:
 1. $\text{Vrfy}_{pk_{m|_i}}(pk_{m|_i 0} \parallel pk_{m|_i 1}, \sigma_{m|_i}) \stackrel{?}{=} 1$ for all $i \in \{0, \dots, n - 1\}$.
 2. $\text{Vrfy}_{pk_m}(m, \sigma_m) \stackrel{?}{=} 1$.

A “tree-based” signature scheme.

Note that ℓ upper bounds the number of public keys from Π that are needed to generate ℓ^* signatures using Π^* . This is because each signature in Π^* requires at most $2n$ new keys from Π (in the worst case), and one additional key from Π is used as the actual public key pk_ε .

Consider the following PPT adversary $\mathcal{A}$ attacking the one-time-secure signature scheme Π :

Adversary $\mathcal{A}$:

$\mathcal{A}$ is given as input a public key pk (the security parameter n is implicit).

- Choose a uniform index $i^* \in \{1, \dots, \ell\}$. Construct a list $pk^1, \dots, pk^\ell$ of keys as follows:
 - Set $pk^{i^*} := pk$.
 - For $i \neq i^*$, compute $(pk^i, sk^i) \leftarrow \text{Gen}(1^n)$.
- Run $\mathcal{A}^*$ on input public key $pk_\varepsilon = pk^1$. When $\mathcal{A}^*$ requests a signature on a message m do:
 1. For $i = 0$ to $n - 1$:

- If the values $pk_{m|i0}$, $pk_{m|i1}$, and $\sigma_{m|i}$ have not yet been defined, then set $pk_{m|i0}$ and $pk_{m|i1}$ equal to the next two unused public keys pk^j and pk^{j+1} , and compute a signature $\sigma_{m|i}$ on $pk_{m|i0} \parallel pk_{m|i1}$ with respect to $pk_{m|i}$.⁶
- 2. If σ_m is not yet defined, compute a signature σ_m on m with respect to pk_m (see footnote 6).
- 3. Give $\left(\{ \sigma_{m|i}, pk_{m|i0}, pk_{m|i1} \}_{i=0}^{n-1}, \sigma_m \right)$ to $\mathcal{A}^*$.
- Say $\mathcal{A}^*$ outputs a message m (for which it had not previously requested a signature) and a signature $\left(\{ \sigma'_{m|i}, pk'_{m|i0}, pk'_{m|i1} \}_{i=0}^{n-1}, \sigma'_m \right)$. If this is a valid signature on m , then:

Case 1: Say there exists a $j \in \{0, \dots, n-1\}$ for which $pk'_{m|j0} \neq pk_{m|j0}$ or $pk'_{m|j1} \neq pk_{m|j1}$; this includes the case when $pk_{m|j0}$ or $pk_{m|j1}$ were never defined by $\mathcal{A}$. Take the minimal such j , and let i be such that $pk^i = pk_{m|j} = pk'_{m|j}$ (such an i exists by the minimality of j). If $i = i^*$, output $(pk'_{m|j0} \parallel pk'_{m|j1}, \sigma'_{m|j})$.

Case 2: If case 1 does not hold, then $pk'_m = pk_m$. Let i be such that $pk^i = pk_m$. If $i = i^*$, output (m, σ'_m) .

In experiment $\text{Sig-forge}_{\mathcal{A}, \Pi}^{1\text{-time}}(n)$, the view of $\mathcal{A}^*$ being run as a subroutine by $\mathcal{A}$ is distributed identically to the view of $\mathcal{A}^*$ in experiment $\text{Sig-forge}_{\mathcal{A}^*, \Pi^*}(n)$.⁷ Thus, the probability that $\mathcal{A}^*$ outputs a forgery is exactly $\delta(n)$ when it is run as a subroutine by $\mathcal{A}$ in this experiment. Given that $\mathcal{A}^*$ outputs a forgery, consider each of the two possible cases described above:

Case 1: Since i^* is uniform and independent of the view of $\mathcal{A}^*$, the probability that $i = i^*$ is exactly $1/\ell$. If $i = i^*$, then $\mathcal{A}$ requested a signature on the message $pk_{m|j0} \parallel pk_{m|j1}$ with respect to the public key $pk = pk^{i^*} = pk_{m|j}$ that it was given (and requested no other signatures). Moreover,

$$pk'_{m|j0} \parallel pk'_{m|j1} \neq pk_{m|j0} \parallel pk_{m|j1}$$

and yet $\sigma'_{m|j}$ is a valid signature on $pk'_{m|j0} \parallel pk'_{m|j1}$ with respect to pk . Thus, $\mathcal{A}$ outputs a forgery in this case.

Case 2: Again, since i^* was chosen uniformly at random and is independent of the view of $\mathcal{A}^*$, the probability that $i = i^*$ is exactly $1/\ell$. If $i = i^*$, then $\mathcal{A}$

⁶If $i \neq i^*$ then $\mathcal{A}$ can compute a signature with respect to pk^i by itself. $\mathcal{A}$ can also obtain a (single) signature with respect to pk^{i^*} by making the appropriate query to its signing oracle. This is what is meant here.

⁷As we have mentioned, $\mathcal{A}$ never “runs out” of public keys. A signing query of $\mathcal{A}^*$ uses $2n$ public keys; thus, even if new public keys were required to answer *every* signing query of $\mathcal{A}^*$ (which will in general not be the case), only $2n\ell^*(n)$ public keys would be needed by $\mathcal{A}$ in addition to the “root” public key pk_ε .

did not request any signatures with respect to the public key $pk = pk^i = pk_m$ and yet σ'_m is a valid signature on m with respect to pk .

We see that, conditioned on $\mathcal{A}^*$ outputting a forgery, $\mathcal{A}$ outputs a forgery with probability exactly $1/\ell$. This means that

$$\Pr[\text{Sig-forge}_{\mathcal{A}, \Pi}^{1\text{-time}}(n) = 1] = \Pr[\text{Sig-forge}_{\mathcal{A}^*, \Pi^*}(n) = 1]/\ell(n).$$

Because Π is a one-time-secure signature scheme, there is a negligible function negl for which

$$\Pr[\text{Sig-forge}_{\mathcal{A}, \Pi}^{1\text{-time}}(n) = 1] \leq \text{negl}(n).$$

Since ℓ is polynomial, this means $\Pr[\text{Sig-forge}_{\mathcal{A}^*, \Pi^*}(n) = 1]$ is negligible. ■

A Stateless Solution

As described, the signer generates state on-the-fly as needed. However, we can imagine having the signer generate the necessary information for all the nodes in the entire tree *in advance*, at the time of key generation. (That is, at the time of key generation the signer could generate the keys $\{(pk_w, sk_w)\}$ and the signatures $\{\sigma_w\}$ for all binary strings w of length at most n .) If key generation were done in this way, the signer would not have to update its state at all; these values could all be stored as part of a (huge) private key, and we would obtain a stateless scheme. The problem with this approach, of course, is that generating all these values would require *exponential* time, and storing them all would require exponential memory.

An alternative is to store some *randomness* that can be used to generate the values $\{(pk_w, sk_w)\}$ and $\{\sigma_w\}$, as needed, rather than storing the values themselves. That is, the signer could store a random string r_w for each w , and whenever the values pk_w, sk_w are needed the signer can compute $(pk_w, sk_w) := \text{Gen}(1^n; r_w)$, where this denotes the generation of a length- n key using random coins r_w . Similarly, if the signing procedure is probabilistic, the signer can store r'_w and then set $\sigma_w := \text{Sign}_{sk_w}(pk_{w0} || pk_{w1}; r'_w)$ (assuming here that $|w| < n$). Generating and storing sufficiently many random strings, however, still requires exponential time and memory.

A simple modification of this alternative gives a polynomial-time solution. Instead of storing random r_w and r'_w as suggested above, the signer can store two keys k, k' for a pseudorandom function F . When needed, the values pk_w, sk_w can now be generated by the following two-step process:

1. Compute $r_w := F_k(w)$.⁸
2. Compute $(pk_w, sk_w) := \text{Gen}(1^n; r_w)$ (as before).

⁸We assume that the output length of F is sufficiently long, and that w is padded to some fixed-length string in a one-to-one fashion. We ignore these technicalities here.

In addition, the key k' is used to generate the value r'_w that is used to compute the signature σ_w . This gives a *stateless* scheme in which key generation (as well as signing and verifying) can be done in polynomial time. Intuitively, this is secure because storing a random function is equivalent to storing all the r_w and r'_w values that are needed, and storing a pseudorandom function is “just as good.” We leave it as an exercise to give a formal proof that this modified scheme remains secure.

Since the existence of collision-resistant hash functions implies the existence of one-way functions (cf. Exercise 7.4), and the latter implies the existence of pseudorandom functions (see Chapter 7), we have:

THEOREM 12.22 *If collision-resistant hash functions exist, then there exists a (stateless) secure signature scheme.*

We remark that it is possible to construct signature schemes satisfying Definition 12.2 from the (minimal) assumption that one-way functions exist; a proof of this result is beyond the scope of this book.

12.7 *Certificates and Public-Key Infrastructures

In this section we briefly discuss one of the primary applications of digital signatures: the secure distribution of public keys. This brings us full circle in our discussion of public-key cryptography. In this and the previous chapter we have seen how to *use* public-key cryptography once public keys are securely distributed. Now we show how public-key cryptography itself can be used to securely distribute public keys. This may sound circular, but it is not. What we will show is that once a *single* public key, belonging to a trusted party, is distributed in a secure fashion, that key can be used to “bootstrap” the secure distribution of arbitrarily many other public keys. Thus, at least in principle, the problem of secure key distribution need only be solved *once*.

The key notion here is a *digital certificate*, which is simply a signature binding an entity to some public key. To be concrete, say a party Charlie has generated a key-pair (pk_C, sk_C) for a secure digital signature scheme (in this section, we will only be concerned with signature schemes satisfying Definition 12.2). Assume further that another party Bob has also generated a key-pair (pk_B, sk_B) (in the present discussion, these may be keys for either a signature scheme or a public-key encryption scheme), and that Charlie *knows* that pk_B is Bob’s public key. Then Charlie can compute the signature

$$\text{cert}_{C \rightarrow B} \stackrel{\text{def}}{=} \text{Sign}_{sk_C}(\text{'Bob's key is } pk_B')$$

and give this signature to Bob. We call $\text{cert}_{C \rightarrow B}$ a *certificate* for Bob’s key

issued by Charlie. In practice a certificate should unambiguously identify the party holding a particular public key and so a more uniquely descriptive term than “Bob” would be used, for example, Bob’s full name and email address, or the URL of Bob’s website.

Now say Bob wants to communicate with some other party Alice who already knows pk_C . Bob can send $(pk_B, \text{cert}_{C \rightarrow B})$ to Alice, who can then verify that $\text{cert}_{C \rightarrow B}$ is indeed a valid signature on the message ‘Bob’s key is pk_B ’ with respect to pk_C . Assuming verification succeeds, Alice now knows that Charlie has signed the indicated message. If Alice trusts Charlie, she can accept pk_B as Bob’s legitimate public key.

All communication between Bob and Alice can occur over an *insecure* and *unauthenticated* channel. If an active adversary interferes with the transmission of $(pk_B, \text{cert}_{C \rightarrow B})$ from Bob to Alice, that adversary will be unable to generate a valid certificate linking Bob to any *other* public key pk'_B unless Charlie had previously signed some other certificate linking Bob with pk'_B (in which case this is anyway not much of an attack). This all assumes that Charlie is not dishonest and that his private key has not been compromised.

We have omitted many details in the above description. Most prominently, we have not discussed how Alice learns pk_C in the first place; how Charlie can be sure that pk_B is Bob’s public key; and how Alice decides whether to trust Charlie. Fully specifying such details (and others) defines a *public-key infrastructure* (PKI) that enables the widespread distribution of public keys. A variety of different PKI models have been suggested, and we mention a few of the more popular ones now. Our treatment here will be kept at a relatively high level, and the reader interested in further details is advised to consult the references at the end of this chapter.

A single certificate authority. The simplest PKI assumes a single *certificate authority* (CA) who is completely trusted by everybody and who issues certificates for everyone’s public key. A certificate authority would not typically be a person, but would more likely be a company whose business it is to certify public keys, a government agency, or perhaps a department within an organization (although in this latter case the CA would likely only be used by people within the organization). Anyone who wants to rely on the services of the CA would have to obtain a legitimate copy of the CA’s public key pk_{CA} . Clearly, this step must be carried out in a secure fashion since if some party obtains an incorrect version of pk_{CA} then that party may not be able to obtain an authentic copy of anyone else’s public key. This means that pk_{CA} must be distributed over an *authenticated* channel. The easiest way of doing this is via physical means: for example, if the CA is within an organization then any employee can obtain an authentic copy of pk_{CA} directly from the CA on their first day of work. If the CA is a company, then other users would have to go to this company at some point and, say, pick up a USB stick that contains the CA’s public key. This inconvenient step need only be carried out once.

A common way for a CA to distribute its public key in practice is to “bun-

dle” this public key with some other software. For example, this occurs today in many popular web browsers: a CA’s public key is provided together with the browser, and the browser is programmed to automatically verify certificates as they arrive. (Actually, modern web browsers have public keys of *multiple* CAs hard-wired into their code, and so more accurately fall into the “multiple CA” model discussed below.)

The mechanism by which a CA issues a certificate to some party Bob must also be very carefully controlled, although the details may vary from CA to CA. As one example, Bob may have to show up in person with a copy of his public key pk_B along with identification proving that his name (or his email address) is what he claims. Only then would the CA issue the certificate.

In the model where there is a single CA, parties completely trust this CA to issue certificates only when appropriate; this is why it is crucial that a detailed verification process be used before a certificate is issued. As a consequence, if Alice receives a certificate $\text{cert}_{CA \rightarrow B}$ certifying that pk_B is Bob’s public key, Alice will accept this assertion as valid, and use pk_B as Bob’s public key.

Multiple certificate authorities. While the model in which there is only one CA is simple and appealing, it is not very practical. For one thing, outside of a single organization it is unlikely for *everyone* to trust the same CA. This need not imply that anyone thinks the CA is corrupt; it could simply be the case that someone finds the CA’s verification process to be insufficient (say, the CA asks for only one form of identification when generating a certificate but Alice would prefer that two be used instead). Moreover, the CA is a single point of failure for the entire system. If the CA is corrupt, or can be bribed, or even if the CA is merely lax with the way it protects its private key, the legitimacy of issued certificates may be called into question. It is also inconvenient for all parties who want certificates to have to contact this CA.

One approach to alleviating these issues is to rely on multiple CAs. A party Bob who wants to obtain a certificate on his public key can choose which CA(s) it wants to issue a certificate, and a party Alice who is presented with a certificate, or even multiple certificates issued by different CAs, can choose which CA’s certificates she trusts. There is no harm in having Bob obtain a certificate from more than one CA (apart from some inconvenience and expense for Bob), but Alice must be more careful since the security of her communication is ultimately only as good as the least-secure CA that she trusts. That is, say Alice trusts two CAs CA_1 and CA_2 , and CA_2 is corrupted by an adversary. Then, although this adversary will not be able to forge certificates issued by CA_1 , it will be able to issue fake certificates in the name of CA_2 for any identity/public key of its choice. This is a real problem in current systems. As mentioned earlier, operating systems/web browsers typically come pre-configured with many CAs’ public keys, and the default setting is for all these CAs to be treated as equally trustworthy. Essentially any company willing to pay, however, can be included as a CA. So the list of pre-configured CAs includes some reputable, well-established companies

along with other, newer companies whose trustworthiness cannot be easily established. It is left to the user to manually configure their settings so as to only accept certificates from CAs the user trusts.

Delegation and certificate chains. Another approach which alleviates some of the burden on a single CA (but does not address the security concerns of having a single point of failure) is to use *certificate chains*. We present the idea for certificate chains of length 2, although it is easy to see that everything we say generalizes to chains of arbitrary length.

Say Charlie, acting as a CA, issues a certificate for Bob as in our original discussion. Assume further that Bob's key pk_B is a public key for a signature scheme. Bob, in turn, can issue his own certificates for other parties. For example, Bob may issue a certificate for Alice of the form

$$\text{cert}_{B \rightarrow A} \stackrel{\text{def}}{=} \text{Sign}_{sk_B}(\text{'Alice's key is } pk_A \text{'}).$$

Now, if Alice wants to communicate with some fourth party Dave who knows Charlie's public key (but not Bob's), then Alice can send

$$pk_A, \text{cert}_{B \rightarrow A}, pk_B, \text{cert}_{C \rightarrow B},$$

to Dave. What can Dave deduce from this? Well, he can first verify that Charlie, whom he trusts and whose public key is already in his possession, has signed a certificate $\text{cert}_{C \rightarrow B}$ indicating that pk_B indeed belongs to someone named Bob. Dave can also verify that this person named Bob has signed a certificate $\text{cert}_{B \rightarrow A}$ indicating that pk_A indeed belongs to Alice. If Dave trusts Charlie to issue certificates only to trustworthy people, then Dave may accept pk_A as being the authentic key of Alice.

We highlight that in this example stronger semantics are associated with a certificate $\text{cert}_{C \rightarrow B}$. In our prior discussion, a certificate of this form was only an assertion that Bob holds public key pk_B . Now, a certificate asserts that Bob holds public key pk_B and *Bob is trusted to issue other certificates*. When Charlie signs a certificate for Bob having these stronger semantics, Charlie is, in effect, *delegating* his ability to issue certificates to Bob. Bob can now act as a proxy for Charlie, issuing certificates on Charlie's behalf.

Coming back to a CA-based PKI, we can imagine one "root" CA and n "second-level" CAs $CA_1, \dots, CA_n$. The root CA can issue certificates for each of the second-level CAs, who can then in turn issue certificates for other principles holding public keys. This eases the burden on the root CA, and also makes it more convenient for parties to obtain certificates (since they may now contact the second-level CA who is closest to them, for example). On the other hand, managing these second-level CAs may be difficult, and their presence means that there are now more points of attack in the system.

The "web of trust" model. The last example of a PKI we will discuss is a fully distributed model, with no central points of trust, called the "web of

trust.” A variant of this model is used by the PGP (“Pretty Good Privacy”) email-encryption software for distribution of public keys.

In the “web of trust” model, anyone can issue certificates to anyone else and each user has to make their own decision about how much trust to place in certificates issued by other users. As an example of how this might work, say a user Alice is already in possession of public keys pk_1, pk_2, pk_3 for some users C_1, C_2, C_3 . (We discuss below how these public keys might initially be obtained by Alice.) Another user Bob who wants to communicate with Alice might have certificates $\text{cert}_{C_1 \rightarrow B}$, $\text{cert}_{C_3 \rightarrow B}$, and $\text{cert}_{C_4 \rightarrow B}$, and will send these certificates (along with his public key pk_B) to Alice. Alice cannot verify $\text{cert}_{C_4 \rightarrow B}$ (since she doesn’t have C_4 ’s public key), but she can verify the other two certificates. Now she has to decide how much trust she places in C_1 and C_3 . She may decide to accept pk_B if she unequivocally trusts C_1 , or if she trusts both C_1 and C_3 to a lesser extent. (She may, for example, consider it likely that either C_1 or C_3 is corrupt, but consider it unlikely for them *both* to be corrupt.)

In this model, as described, users are expected to collect both public keys of other parties, as well as certificates on their own public key. In the context of PGP, this used to be done at “key-signing parties” where PGP users got together (say, at a conference), gave each other authentic copies of their public keys, and issued certificates for each other. In general the users at a key-signing party may not know each other, but they can check a driver’s license, say, before accepting or issuing a certificate for someone’s public key.

Public keys and certificates can also be stored in a central database, and this is done for PGP (see <http://pgp.mit.edu>). When Alice wants to send an encrypted message to Bob, she can search for Bob’s public key in this database; along with Bob’s public key, the database will return a list of all certificates it holds that have been issued for Bob’s public key. It is also possible that multiple public keys for Bob will be found in the database, and each of these public keys may be certified by certificates issued by a different set of parties. Once again, Alice then needs to decide how much trust to place in any of these public keys before using them.

The web of trust model is attractive because it does not require trust in any central authority. On the other hand, while it may work well for the average user encrypting their email, it does not seem appropriate for settings where security is more critical, or for the distribution of organizational public keys (e.g., for e-commerce on the web). If a user wants to communicate with his bank, for example, it is unlikely that he would trust people he met at a conference to certify his bank’s public key, and also unlikely that a bank representative will go to a key-signing party to get the bank’s key certified.

Invalidating Certificates

One important issue we have not yet touched upon at all is the fact that certificates should generally not be valid indefinitely. An employee may leave

a company, in which case he or she is no longer allowed to receive encrypted communication from others within the company; a user's private key might also be stolen, at which point the user (assuming they know about the theft) will want to generate a new key-pair and have the old public key removed from circulation. In either of these scenarios, we need a way to render previously issued certificates invalid.

Approaches for handling these issues are varied and complex, and we will only mention two relatively simple ideas that, in some sense, represent opposite extremes. (Improving these methods is an active area of real-world network-security research.)

Expiration. One method for preventing certificates from being used indefinitely is to include an *expiry date* as part of the certificate. A certificate issued by a CA Charlie for Bob's public key might now have the form

$$\text{cert}_{C \rightarrow B} \stackrel{\text{def}}{=} \text{Sign}_{sk_C}(\text{'Bob's key is } pk_B, \text{ date}),$$

where **date** is some date in the future at which point the certificate becomes invalid. (For example, one year from the day the certificate is issued.) When another user verifies this certificate, they need to know not only pk_B but also the expiry date, and they now need to check not only that the signature is valid, but also that the expiry date has not passed. A user who holds a certificate must contact the CA to get a new certificate issued whenever their current one expires; at this point, the CA verifies the identity/credentials of the user again before issuing another certificate.

Using expiry dates provides a very coarse-grained solution to the problems mentioned earlier. If an employee leaves a company the day after getting a certificate, and the certificate expires one year after its issuance date, then this employee can use his or her public key illegitimately for an entire year until the expiry date passes. For this reason, this approach is typically used in conjunction with other methods such as the one we describe next.

Revocation. When an employee leaves an organization, or a user's private key is stolen, we would like the certificates that have been issued for their public keys to become invalid immediately, or at least as soon as possible. This can be achieved by having the CA explicitly *revoke* the certificate. For simplicity we assume a single CA, but everything we say applies more generally if the user had certificates issued by multiple CAs.

There are many different ways revocation can be handled. One possibility (the only one we will discuss) is for the CA to include a serial number in every certificate it issues; that is, a certificate will now have the form

$$\text{cert}_{C \rightarrow B} \stackrel{\text{def}}{=} \text{Sign}_{sk_C}(\text{'Bob's key is } pk_B, \text{ ###}),$$

where “###” represents the serial number of this certificate. Each certificate should have a unique serial number, and the CA will store the information $(\text{Bob}, pk_B, \text{###})$ for each certificate it generates.

If a user Bob's private key corresponding to a public key pk_B is stolen, Bob can alert the CA to this fact. (The CA must verify Bob's identity here, to prevent another user from falsely revoking a certificate issued to Bob.) The CA will then search its database to find the serial number associated with the certificate issued for Bob and pk_B . At the end of each day, say, the CA will then generate a *certificate revocation list* (CRL) with the serial numbers of all revoked certificates, and sign the CRL and the current date. The signed CRL is then widely distributed or otherwise made available to potential verifiers. Verification of a certificate now requires checking that the signature in the certificate is valid, checking that the serial number does not appear on the most current revocation list, and verifying the CA's signature on the revocation list itself.

In this approach the way we have described it, there is a gap of at most one day before a certificate becomes invalid. This offers more flexibility than an approach based only on expiry dates.

12.8 Putting It All Together – SSL/TLS

As a culmination of what we have covered in the book so far, we discuss the *Transport Layer Security* (TLS) protocol that is used extensively to secure communication over the web; TLS is the protocol used by your browser any time you connect to a website using **https** rather than **http**. We should be clear that our goal here is to focus on the underlying cryptography used in the core TLS protocol, and not various other aspects that, although interesting from a network-security standpoint, are irrelevant for our concerns. As usual, we have slightly simplified and abstracted parts of the protocol in order to convey the main point, and our description should not be relied upon for an implementation. Finally, we do not formally define or claim security for the protocol; indeed, a formal analysis of TLS is the subject of active research.

TLS is a standardized protocol based on a precursor called SSL (or *Secure Socket Layer*) that was developed by Netscape in the mid-1990s; the last version available was SSL 3.0. TLS version 1.0 was released in 1999, updated to version 1.1 in 2006, and updated again to version 1.2 (the current version) in 2008. As of the time of this writing, approximately 50% of websites still use TLS 1.0 rather than a more recent version; all major web browsers support TLS 1.2, although in some cases earlier versions of TLS are used by default. For the most part, our description below is at a sufficiently high level that the differences between the versions are unimportant for our purposes. We caution, however, that there are several known attacks on earlier versions.

TLS allows a client (e.g., a web browser) and a server (e.g., a website) to agree on a set of shared keys and then to use those keys to encrypt and

authenticate their subsequent communication. It consists of two parts: a *handshake protocol* that performs authenticated key exchange to establish the shared keys, and a *record-layer protocol* that uses those shared keys to encrypt/authenticate the parties' communication. Although TLS allows for clients to authenticate to servers, it is primarily used only for authentication of servers to clients because only servers typically have certificates. (After a TLS session is established, user-to-server authentication—if desired—can be done at the application layer of the network stack by, e.g., sending a password.)

The handshake protocol. We now describe the basic flow of the handshake protocol. At the outset of the protocol, the client C holds a set of CA's public keys $\{pk_1, \dots, pk_n\}$, and the server S has a key-pair (pk_S, sk_S) for a KEM⁹ along with a certificate $\text{cert}_{i \rightarrow S}$ issued by one of the CAs whose public key C knows. To connect to S , the parties run the following steps.

1. C begins by sending a message to S that includes information about versions of the protocol supported by the client, the *ciphersuites* supported by the client (e.g., which hash functions or block ciphers the client allows), and a uniform value (a “nonce”) N_C .
2. S responds by selecting the latest version of the protocol it supports as well as an appropriate ciphersuite. In addition, it sends its public key pk_S , its certificate $\text{cert}_{i \rightarrow S}$, and its own uniform value N_S .
3. C checks whether one of the CA's public keys it holds, say pk_i , matches the CA who issued S 's certificate. If so, C verifies the certificate (and also checks that it has not expired or been revoked) and, if successful, learns that pk_S is S 's public key. It then runs $(c, \text{pmk}) \leftarrow \text{Encaps}_{pk_S}(1^n)$ (see Section 11.3) to obtain a ciphertext c and what is called a *pre-master key pmk*. It sends c to the server.

pmk is used to derive a *master key* mk using a key-derivation function (cf. Section 5.6.4) applied to pmk , N_C , and N_S . The client then applies a pseudorandom generator to mk to derive four keys k_C, k'_C, k_S, k'_S .

Finally, C computes $\tau_C \leftarrow \text{Mac}_{\text{mk}}(\text{transcript})$, where **transcript** denotes all messages exchanged between C and S thus far. The client then sends τ_C to S . (In fact, τ_C is itself encrypted and authenticated as done for communication in the record-layer, described below.)

4. S computes $\text{pmk} := \text{Decaps}_{sk_S}(c)$, from which it can derive mk and k_C, k'_C, k_S, k'_S as the client did. If $\text{Vrfy}_{\text{mk}}(\text{transcript}, \tau_C) \neq 1$, then S aborts. Otherwise, it sets $\tau_S \leftarrow \text{Mac}_{\text{mk}}(\text{transcript}')$, where **transcript'** denotes all messages exchanged between C and S thus far (i.e., including

⁹It is also possible for S to have a public key for a signature scheme and to use an *ephemeral* KEM public key in the protocol below. We do not discuss this more complicated option.

the last message received from C). S then sends τ_S to C . (Again, τ_S is actually encrypted and authenticated as record-layer traffic is.)

5. If $\text{Vrfy}_{\text{mk}}(\text{transcript}', \tau_S) \neq 1$, the client aborts.

At the end of a successful execution of the handshake protocol, C and S share a set of four symmetric keys k_C, k'_C, k_S, k'_S .

TLS 1.2 supports two KEMs: a CDH/DDH-based KEM as in Construction 11.19, or the RSA-based encryption scheme PKCS #1 v1.5. To prevent Bleichenbacher-style attacks on the scheme in the latter case, the server should not report a decryption error if **Decaps** fails in step 4. Instead, if decryption fails it should choose a uniform **pmk** and then continue running the protocol using that value. (In that case, of course, the value τ_C will not verify and S will abort; the point is that the attacker cannot distinguish whether this was due to a decryption failure or not.)

The intuition for the security of the handshake protocol is that since C verifies the certificate, it knows that only the legitimate server S can learn **pmk** and hence **mk**. Thus, if the protocol terminates successfully, C knows that it shares keys k_C, k'_C, k_S, k'_S with the legitimate S , and that no adversary learned anything about those keys. This is supposed to hold even in the presence of an active attacker, although a formal proof of this is quite involved. We remark, however, that the message authentication code on the transcript is used to prevent a man-in-the-middle attacker from changing the protocol version and ciphersuites sent at the beginning of the handshake protocol. This prevents an attacker from causing S or C to use old, weaker versions of the protocol, or weak block ciphers and short keys.

The record-layer protocol. Once keys have been agreed upon by C and S , the parties use those keys to encrypt and authenticate all their subsequent communication. C uses k_C (resp., k'_C) to encrypt (resp., authenticate) all messages it sends to S ; similarly, S uses k_S and k'_S to encrypt and authenticate all the messages it sends. Sequence numbers are used to prevent replay attacks, as discussed in Section 4.5.3. TLS 1.2 uses an authenticate-then-encrypt approach, which, as we have seen in Section 4.5.2, can be problematic.

12.9 *Signcryption

To close this chapter, we briefly and informally discuss the issue of joint secrecy and integrity in the public-key setting. While this parallels our treatment from Section 4.5, the fact that we are now in the public-key setting introduces several additional complications.

For simplicity, we consider a network in which all relevant parties have public/private key-pairs for *both* encrypting and signing. We let (ek, dk) denote

a (public) encryption key and (private) decryption key, and use (vk, sk) for a (public) verification key and (private) signing key. We assume all parties know everyone else's public keys.

Informally, our goal is to design a mechanism that allows a sender S to send a message m to a receiver R while ensuring that (1) no other party in the network can learn any information about m (i.e., secrecy) and (2) R is assured that the message came from S (i.e., integrity). We will want to consider both of these security properties even against active (e.g., chosen-ciphertext) attacks by other parties in the network.

Following our discussion in Section 4.5, a natural idea is to use an “encrypt-then-authenticate” approach in which S sends $\langle S, c, \text{Sign}_{sk_S}(c) \rangle$ to R , where c is an encryption of m using R 's encryption key ek_R . (We explicitly include the sender's identity here for convenience.) However, there is a clever chosen-ciphertext attack here regardless of the encryption scheme used. Having observed a transmission as above, another (adversarial) party A can strip off S 's signature and replace it with its own, sending $\langle A, c, \text{Sign}_{sk_A}(c) \rangle$ to R . In this case, R would not detect anything wrong, and would mistakenly think that A has sent it the message m . If R replies to A , or otherwise behaves toward A in a way that depends on the contents of the message, then A can potentially learn the unknown message m .

(Another problem with this scheme, although somewhat independent of our discussion here, is that it no longer provides *non-repudiation*. That is, R cannot easily prove to a third party that S has signed the message m , at least not without divulging its own decryption key dk_R .)

One could instead try an “authenticate-then-encrypt” approach. Here, S would first compute a signature $\sigma \leftarrow \text{Sign}_{sk_S}(m)$ and then send

$$\langle S, \text{Enc}_{ek_R}(m \parallel \sigma) \rangle.$$

(Note that this solves the non-repudiation issue mentioned above.) If the encryption scheme is only CPA-secure then problems just like those mentioned in Section 4.5 apply, so let us assume a CCA-secure encryption scheme is used instead. Even then, there is an attack that can be carried by a malicious R . Upon receiving $\langle S, \text{Enc}_{ek_R}(m \parallel \sigma) \rangle$ from S , a malicious R can decrypt to obtain $m \parallel \sigma$, and then re-encrypt and send $\langle S, \text{Enc}_{ek_{R'}}(m \parallel \sigma) \rangle$ to another receiver R' . This (honest) receiver R' will then think that S sent it the message m . This can have serious consequences, e.g., if m is the message “I owe you \$100.”

These attacks can be prevented if parties are more careful about how they handle identifiers. When encrypting, a sender should include its own identity along with the message; when signing, a party should sign the identity of the intended recipient along with what is being signed. For example, the second approach would be modified so that S first computes $\sigma \leftarrow \text{Sign}_{sk_S}(m \parallel R)$, and then sends $\langle S, \text{Enc}_{ek_R}(S \parallel m \parallel \sigma) \rangle$ to R . When decrypting, the receiver should check that the resulting decrypted value includes the (purported) sender's identity; when verifying, the receiver should check that what was

signed incorporates its own identity. When including identities in this way, both authenticate-then-encrypt and encrypt-then-authenticate are secure if a CCA-secure encryption scheme and a strongly secure signature scheme (whose definition parallels Definition 4.3 for MACs) are used.

References and Additional Reading

Notable early work on signatures includes that of Diffie and Hellman [58], Rabin [144, 145], Rivest, Shamir, and Adleman [148], and Goldwasser, Micali, and Yao [82]. For an extensive treatment of signature schemes beyond what could be covered here, see [98].

Goldwasser, Micali, and Rivest [81] defined the notion of existential unforgeability under an adaptive chosen-message attack, and also gave the first construction of a stateful signature scheme satisfying this definition. Goldreich [74] suggested an approach to make the Goldwasser–Micali–Rivest scheme stateless, and we have essentially adopted Goldreich’s ideas in Section 12.6.3.

Plain RSA signatures date to the original RSA paper [148]. RSA-FDH was proposed by Bellare and Rogaway in their original paper on the random-oracle model [21], although the idea (without proof) of using a cryptographic hash function to prevent algebraic attacks can be traced back to Rabin [145]. A later improvement [23] was standardized as part of PKCS #1 v2.1, available at <http://www.emc.com/emc-plus/rsa-labs>.

The Fiat–Shamir transform [65] and the Schnorr signature scheme [152] both date to the late-1980s, although our proof of Theorem 12.10 is due to [1] and our proof of Theorem 12.11 is inspired by [20]. The DSA and ECDSA standards are described in [132].

Lamport’s signature scheme was published in 1979 [111], although it was already described in [58]. A tree-based construction similar in spirit to Construction 12.20 was suggested by Merkle [121, 122], although a tree-based approach was also used in [81]. Naor and Yung [128] showed that one-way permutations suffice for constructing one-time-secure signatures that can sign messages of arbitrary length, and this was improved by Rompel [151], who showed that one-way functions are sufficient. (See also [99].) As we have seen in Section 12.6.3, one-time-secure signatures of this sort can be used to construct secure signature schemes, implying that one-way functions suffice for the existence of (stateless) secure signatures.

The notion of certificates was first described by Kohnfelder [107] in his undergraduate thesis. Public-key infrastructures are discussed in greater detail in [102, Chapter 15]. See also [3, 62]. Further details of the TLS protocol can be found in, e.g., [102, 165]. A formal treatment of combined secrecy and integrity in the public-key setting is given by An et al. [10].

Exercises

- 12.1 Show that Construction 4.7 for constructing a variable-length MAC from any fixed-length MAC can also be used (with appropriate modifications) to construct a signature scheme for arbitrary-length messages from any signature scheme for messages of fixed length $\ell(n) \geq n$.
- 12.2 Prove that the existence of a one-time-secure signature scheme for 1-bit messages implies the existence of one-way functions.
- 12.3 In Section 12.4.1 we showed an attack on the plain RSA signature scheme in which an attacker forges a signature on an arbitrary message using two signing queries. Show how an attacker can forge a signature on an arbitrary message using a *single* signing query.
- 12.4 Assume the RSA problem is hard. Show that the plain RSA signature scheme satisfies the following weak definition of security: an attacker is given the public key $\langle N, e \rangle$ and a uniform message $m \in \mathbb{Z}_N^*$. The adversary succeeds if it can output a valid signature on m without making any signing queries.
- 12.5 Another approach (besides hashing) that has been tried to construct secure RSA-based signatures is to *encode* the message before applying the RSA permutation. Here the signer fixes a public encoding function $\text{enc} : \{0, 1\}^\ell \rightarrow \mathbb{Z}_N^*$ as part of its public key, and the signature on a message m is $\sigma := [\text{enc}(m)^d \bmod N]$.
- How is verification performed in encoded RSA?
 - Discuss why appropriate choice of encoding function for $\ell \ll \|N\|$ prevents the “no-message attack” described in Section 12.4.1.
 - Show that encoded RSA is insecure if $\text{enc}(m) = 0\mathbf{x}00\|m\|0^{\kappa/10}$ (where $\kappa \stackrel{\text{def}}{=} \|N\|$, $\ell = |m| \stackrel{\text{def}}{=} 4\kappa/5$, and m is not the all-0 message). Assume $e = 3$.
 - Show that encoded RSA is insecure for $\text{enc}(m) = 0\|m\|0\|m$ (where $\ell = |m| \stackrel{\text{def}}{=} (\|N\| - 1)/2$ and m is not the all-0 message). Assume $e = 3$.
 - Solve parts (c) and (d) for arbitrary e .
- 12.6 Consider a variant of the Fiat–Shamir transform in which the signature is (I, s) rather than (r, s) and verification is changed in the natural way. Show that if the underlying identification scheme is secure, then the resulting signature scheme is secure here as well.

12.7 Consider a variant of DSA in which the message space is $\mathbb{Z}_q$ and H is omitted. (So the second component of the signature is now $s := [k^{-1} \cdot (m + xr) \bmod q]$.) Show that this variant is not secure.

12.8 Let f be a one-way permutation. Consider the following signature scheme for messages in the set $\{1, \dots, n\}$:

- To generate keys, choose uniform $x \in \{0, 1\}^n$ and set $y := f^{(n)}(x)$ (where $f^{(i)}(\cdot)$ refers to i -fold iteration of f , and $f^{(0)}(x) \stackrel{\text{def}}{=} x$). The public key is y and the private key is x .
 - To sign message $i \in \{1, \dots, n\}$, output $f^{(n-i)}(x)$.
 - To verify signature σ on message i with respect to public key y , check whether $y \stackrel{?}{=} f^{(i)}(\sigma)$.
- (a) Show that the above is not a one-time-secure signature scheme. Given a signature on a message i , for what messages j can an adversary output a forgery?
- (b) Prove that no PPT adversary given a signature on i can output a forgery on any message $j > i$ except with negligible probability.
- (c) Suggest how to modify the scheme so as to obtain a one-time-secure signature scheme.

Hint: Include two values y, y' in the public key.

12.9 A *strong* one-time-secure signature scheme satisfies the following (informally): given a signature σ' on a message m' , it is infeasible to output $(m, \sigma) \neq (m', \sigma')$ for which σ is a valid signature on m (note that $m = m'$ is allowed).

- (a) Give a formal definition of strong one-time-secure signatures.
- (b) Assuming the existence of one-way functions, show a one-way function for which Lamport's scheme is *not* a strong one-time-secure signature scheme.
- (c) Construct a strong one-time-secure signature scheme based on any assumption used in this book.

Hint: Use a particular one-way function in Lamport's scheme.

12.10 Consider the Lamport signature scheme. Describe an adversary who obtains signatures on *two* messages of its choice and can then forge signatures on any message it likes.

12.11 The Lamport scheme uses 2ℓ values in the public key to sign messages of length ℓ . Consider the variant in which the private key contains 2ℓ values $x_1, \dots, x_{2\ell}$ and the public key contains the values $y_1, \dots, y_{2\ell}$ with $y_i := f(x_i)$. A message $m \in \{0, 1\}^{\ell'}$ is mapped in a one-to-one fashion to a subset $S_m \subset \{1, \dots, 2\ell\}$ of size ℓ . To sign m , the signer reveals $\{x_i\}_{i \in S_m}$. Prove that this gives a one-time-secure signature scheme. What is the maximum message length ℓ' that this scheme supports?

- 12.12 At the end of Section 12.6.3, we show how a pseudorandom function can be used to make Construction 12.20 stateless. Does a similar approach work for the chain-based scheme described in Section 12.6.2? If so, sketch a construction and proof. If not, explain why and modify the scheme to obtain a stateless variant.
- 12.13 Prove Theorem 12.22.
- 12.14 Assume revocation of certificates is handled in the following way: when a user Bob claims that the private key corresponding to his public key pk_B has been stolen, the user sends to the CA a statement of this fact *signed with respect to pk_B* . Upon receiving such a signed message, the CA revokes the appropriate certificate.
- Explain why it is not necessary for the CA to check Bob's identity in this case. In particular, explain why it is of no concern that an adversary who has stolen Bob's private key can forge signatures with respect to pk_B .

Chapter 13

**Advanced Topics in Public-Key Encryption*

In Chapter 11 we saw several examples of public-key encryption schemes used in practice. Here, we explore some schemes that are currently more of theoretical interest—although in some cases it is possible that these schemes (or variants thereof) will be used more widely in the future.

We begin with a treatment of *trapdoor permutations*, a generalization of one-way permutations, and show how to use them to construct public-key encryption schemes. Trapdoor permutations neatly encapsulate the key characteristics of the RSA permutation that make it so useful. As such, they often provide a useful abstraction for designing new cryptosystems.

Next, we present three schemes based on problems related to factoring:

- The *Paillier encryption scheme* is an example of an encryption scheme that is *homomorphic*. This property turns out to be useful for constructing more-complex cryptographic protocols, something we touch on briefly in Section 13.3.
- The *Goldwasser–Micali encryption scheme* is of historical interest as the first scheme to be proven CPA-secure. It is also homomorphic, and uses some interesting number theory that can be applied in other contexts.
- Finally, we discuss the *Rabin trapdoor permutation*, which can be used to construct a public-key encryption scheme. Although superficially similar to the RSA trapdoor permutation, the Rabin trapdoor permutation is distinguished by the fact that its security is based *directly* on the hardness of factoring. (Recall from Section 8.2.5 that hardness of the RSA problem appears to be a stronger assumption.)

13.1 Encryption from Trapdoor Permutations

In Section 11.5.3 we saw how to construct a CPA-secure public-key encryption scheme based on the RSA assumption. By distilling those properties of

RSA that are used in the construction, and defining an abstract notion that encapsulates those properties, we obtain a general *template* for constructing secure encryption schemes based on any primitive satisfying the same set of properties. *Trapdoor permutations* turn out to be the “right” abstraction here.

In the following section we define (families of) trapdoor permutations and observe that the RSA family of one-way permutations (Construction 8.77) satisfies the additional requirements needed to be a family of *trapdoor* permutations. In Section 13.1.2 we generalize the construction from Section 11.5.3 and show that public-key encryption can be constructed from any trapdoor permutation. These results will be used again in Section 13.5, where we show a second example of a trapdoor permutation, this time based directly on the factoring assumption.

In this section we rely on the material from Section 8.4.1 or, alternately, Chapter 7.

13.1.1 Trapdoor Permutations

Recall the definitions of families of functions and families of one-way permutations from Section 8.4.1. In that section, we showed that the RSA assumption naturally gives rise to a family of one-way permutations. The astute reader may have noticed that the construction we gave (Construction 8.77) has a special property that was not remarked upon there: namely, the parameter-generation algorithm Gen outputs some additional information along with I that *enables efficient inversion of f_I* . We refer to such additional information as a *trapdoor*, and call families of one-way permutations with this additional property *families of trapdoor permutations*. A formal definition follows.

DEFINITION 13.1 *A tuple of polynomial-time algorithms $(\text{Gen}, \text{Samp}, f, \text{Inv})$ is a family of trapdoor permutations (or a trapdoor permutation) if:*

- *The probabilistic parameter-generation algorithm Gen , on input 1^n , outputs (I, td) with $|I| \geq n$. Each value of I defines a set $\mathcal{D}_I$ that constitutes the domain and range of a permutation (i.e., bijection) $f_I : \mathcal{D}_I \rightarrow \mathcal{D}_I$.*
- *Let Gen_1 denote the algorithm that results by running Gen and outputting only I . Then $(\text{Gen}_1, \text{Samp}, f)$ is a family of one-way permutations.*
- *Let (I, td) be an output of $\text{Gen}(1^n)$. The deterministic inverting algorithm Inv , on input td and $y \in \mathcal{D}_I$, outputs $x \in \mathcal{D}_I$. We denote this by $x := \text{Inv}_{\text{td}}(y)$. It is required that with all but negligible probability over (I, td) output by $\text{Gen}(1^n)$ and uniform choice of $x \in \mathcal{D}_I$, we have*

$$\text{Inv}_{\text{td}}(f_I(x)) = x.$$

As shorthand, we drop explicit mention of Samp and simply refer to trapdoor permutation $(\text{Gen}, f, \text{Inv})$. For (I, td) output by Gen we write $x \leftarrow \mathcal{D}_I$ to

denote uniform selection of $x \in \mathcal{D}_I$ (with the understanding that this is done by algorithm **Samp**).

The second condition above implies that f_I cannot be efficiently inverted without **td**, but the final condition means that f_I can be efficiently inverted with **td**. It is immediate that Construction 8.77 can be modified to give a family of trapdoor permutations if the RSA problem is hard relative to **GenRSA**, and so we refer to that construction as the *RSA trapdoor permutation*.

13.1.2 Public-Key Encryption from Trapdoor Permutations

We now sketch how a public-key encryption scheme can be constructed from an arbitrary family of trapdoor permutations. The construction is simply a generalization of what was already done for the specific RSA trapdoor permutation in Section 11.5.3.

We begin by (re-)introducing the notion of a hard-core predicate. This is the natural adaptation of Definition 7.4 to our context, and also generalizes our previous discussion of one specific hard-core predicate for the RSA trapdoor permutation in Section 11.5.3.

DEFINITION 13.2 *Let $\Pi = (\text{Gen}, f, \text{Inv})$ be a family of trapdoor permutations, and let hc be a deterministic polynomial-time algorithm that, on input I and $x \in \mathcal{D}_I$, outputs a single bit $\text{hc}_I(x)$. We say that hc is a hard-core predicate of Π if for every probabilistic polynomial-time algorithm $\mathcal{A}$ there is a negligible function negl such that*

$$\Pr[\mathcal{A}(I, f_I(x)) = \text{hc}_I(x)] \leq \frac{1}{2} + \text{negl}(n),$$

where the probability is taken over the experiment in which $\text{Gen}(1^n)$ is run to generate (I, td) and then x is chosen uniformly from $\mathcal{D}_I$.

The asymmetry provided by trapdoor permutations implies that anyone who knows the trapdoor **td** associated with I can recover x from $f_I(x)$ and thus compute $\text{hc}_I(x)$ from $f_I(x)$. But given only I , it is infeasible to compute $\text{hc}_I(x)$ from $f_I(x)$ for a uniform x .

The following can be proved by a suitable modification of Theorem 7.5:

THEOREM 13.3 *Given a family of trapdoor permutations Π , there is a family of trapdoor permutations $\widehat{\Pi}$ with a hard-core predicate hc for $\widehat{\Pi}$.*

Given a family of trapdoor permutations $\widehat{\Pi} = (\widehat{\text{Gen}}, f, \text{Inv})$ with hard-core predicate hc , we can construct a single-bit encryption scheme via the following approach (see Construction 13.4 below, and compare to Construction 11.32): To generate keys, run $\widehat{\text{Gen}}(1^n)$ to obtain (I, td) ; the public key is I and the

private key is td . Given a public key I , encryption of a message $m \in \{0, 1\}$ works by choosing uniform $r \in \mathcal{D}_I$ subject to the constraint that $\text{hc}_I(r) = m$, and then setting the ciphertext equal to $f_I(r)$. In order to decrypt, the receiver uses td to recover r from $f_I(r)$ and then outputs the message $m := \text{hc}_I(r)$.

CONSTRUCTION 13.4

Let $\widehat{\Pi} = (\widehat{\text{Gen}}, f, \text{Inv})$ be a family of trapdoor permutations with hard-core predicate hc . Define a public-key encryption scheme as follows:

- **Gen**: on input 1^n , run $\widehat{\text{Gen}}(1^n)$ to obtain (I, td) . Output the public key I and the private key td .
- **Enc**: on input a public key I and a message $m \in \{0, 1\}$, choose a uniform $r \in \mathcal{D}_I$ subject to the constraint that $\text{hc}_I(r) = m$. Output the ciphertext $c := f_I(r)$.
- **Dec**: on input a private key td and a ciphertext c , compute the value $r := \text{Inv}_I(c)$ and output the message $\text{hc}_I(r)$.

Public-key encryption from any family of trapdoor permutations.

A proof of security follows along the lines of the proof of Theorem 11.33.

THEOREM 13.5 *If $\widehat{\Pi}$ is a family of trapdoor permutations with hard-core predicate hc , then Construction 13.4 is CPA-secure.*

PROOF Let Π denote Construction 13.4. We prove that Π has indistinguishable encryptions in the presence of an eavesdropper; by Proposition 11.3, this implies it is CPA-secure.

We first observe that hc must be *unbiased* in the following sense. Let

$$\delta_0(n) \stackrel{\text{def}}{=} \Pr_{(I, \text{td}) \leftarrow \widehat{\text{Gen}}(1^n); x \leftarrow \mathcal{D}_I} [\text{hc}_I(x) = 0]$$

and

$$\delta_1(n) \stackrel{\text{def}}{=} \Pr_{(I, \text{td}) \leftarrow \widehat{\text{Gen}}(1^n); x \leftarrow \mathcal{D}_I} [\text{hc}_I(x) = 1].$$

Then there is a negligible function negl such that

$$\delta_0(n), \delta_1(n) \geq \frac{1}{2} - \text{negl}(n);$$

if not, then an attacker who simply outputs the more frequently occurring bit would violate Definition 13.2.

Now let $\mathcal{A}$ be a probabilistic polynomial-time adversary. Without loss of generality, we may assume $m_0 = 0$ and $m_1 = 1$ in experiment $\text{PubK}_{\mathcal{A}, \Pi}^{\text{eav}}(n)$.

We then have

$$\begin{aligned} \Pr[\text{PubK}_{\mathcal{A},\Pi}^{\text{eav}}(n) = 1] &= \frac{1}{2} \cdot \Pr[\mathcal{A}(pk, c) = 0 \mid c \text{ is an encryption of } 0] \\ &\quad + \frac{1}{2} \cdot \Pr[\mathcal{A}(pk, c) = 1 \mid c \text{ is an encryption of } 1]. \end{aligned}$$

But then

$$\begin{aligned} &\Pr[\mathcal{A}(I, f_I(x)) = \text{hc}_I(x)] \\ &= \delta_0(n) \cdot \Pr[\mathcal{A}(I, f_I(x)) = 0 \mid \text{hc}_I(x) = 0] \\ &\quad + \delta_1(n) \cdot \Pr[\mathcal{A}(I, f_I(x)) = 1 \mid \text{hc}_I(x) = 1] \\ &\geq \left(\frac{1}{2} - \text{negl}(n)\right) \cdot \Pr[\mathcal{A}(I, f_I(x)) = 0 \mid \text{hc}_I(x) = 0] \\ &\quad + \left(\frac{1}{2} - \text{negl}(n)\right) \cdot \Pr[\mathcal{A}(I, f_I(x)) = 1 \mid \text{hc}_I(1) = 1] \\ &\geq \frac{1}{2} \cdot \Pr[\mathcal{A}(I, f_I(x)) = 0 \mid \text{hc}_I(x) = 0] \\ &\quad + \frac{1}{2} \cdot \Pr[\mathcal{A}(I, f_I(x)) = 1 \mid \text{hc}_I(1) = 1] - 2 \cdot \text{negl}(n) \\ &= \Pr[\text{PubK}_{\mathcal{A},\Pi}^{\text{eav}}(n) = 1] - 2 \cdot \text{negl}(n). \end{aligned}$$

Since hc is a hard-core predicate for $\widehat{\Pi}$, there is a negligible function negl' such that $\text{negl}'(n) \geq \Pr[\mathcal{A}(I, f_I(x)) = \text{hc}_I(x)]$; this means that

$$\Pr[\text{PubK}_{\mathcal{A},\Pi}^{\text{eav}}(n) = 1] \leq \text{negl}'(n) + 2 \cdot \text{negl}(n),$$

completing the proof. ■

Encrypting longer messages. Using Claim 11.7, we know that we can extend Construction 13.4 to encrypt ℓ -bit messages using ciphertexts ℓ times as long. Better efficiency can be obtained by constructing a KEM, following along the lines of Construction 11.34. We leave the details as an exercise.

13.2 The Paillier Encryption Scheme

In this section we describe the *Paillier encryption scheme*, a public-key encryption scheme whose security is based on an assumption related (but not known to be equivalent) to the hardness of factoring. This encryption scheme is particularly interesting because it possesses some nice *homomorphic* properties, as we will discuss further in Section 13.2.3.

The Paillier encryption scheme utilizes the group $\mathbb{Z}_{N^2}^*$, the multiplicative group of elements in the range $\{1, \dots, N^2\}$ that are relatively prime to N , for N a product of two distinct primes. To understand the scheme it is helpful to first understand the structure of $\mathbb{Z}_{N^2}^*$. A useful characterization of this group is given by the following proposition, which says, among other things, that $\mathbb{Z}_{N^2}^*$ is isomorphic to $\mathbb{Z}_N \times \mathbb{Z}_N^*$ (cf. Definition 8.23) for N of the form we will be interested in. We prove the proposition in the next section. (The reader willing to accept the proposition on faith can skip to Section 13.2.2.)

PROPOSITION 13.6 *Let $N = pq$, where p, q are distinct odd primes of equal length. Then:*

1. $\gcd(N, \phi(N)) = 1$.

2. For any integer $a \geq 0$, we have $(1 + N)^a = (1 + aN) \bmod N^2$.

As a consequence, the order of $(1 + N)$ in $\mathbb{Z}_{N^2}^*$ is N . That is, $(1 + N)^N = 1 \bmod N^2$ and $(1 + N)^a \neq 1 \bmod N^2$ for any $1 \leq a < N$.

3. $\mathbb{Z}_N \times \mathbb{Z}_N^*$ is isomorphic to $\mathbb{Z}_{N^2}^*$, with isomorphism $f: \mathbb{Z}_N \times \mathbb{Z}_N^* \rightarrow \mathbb{Z}_{N^2}^*$ given by

$$f(a, b) = [(1 + N)^a \cdot b^N \bmod N^2].$$

In light of the last part of the above proposition, we introduce some convenient notation. With N understood, and $x \in \mathbb{Z}_{N^2}^*$, $a \in \mathbb{Z}_N$, $b \in \mathbb{Z}_N^*$, we write $x \leftrightarrow (a, b)$ if $f(a, b) = x$ where f is the isomorphism from the proposition above. One way to think about this notation is that it means “ x in $\mathbb{Z}_{N^2}^*$ corresponds to (a, b) in $\mathbb{Z}_N \times \mathbb{Z}_N^*$.” We have used the same notation in this book with regard to the isomorphism $\mathbb{Z}_N^* \simeq \mathbb{Z}_p^* \times \mathbb{Z}_q^*$ given by the Chinese remainder theorem; we keep the notation because in both cases it refers to an isomorphism of groups. Nevertheless, there should be no confusion since the group $\mathbb{Z}_{N^2}^*$ and the above proposition are only used in this section. We remark that here the isomorphism—but not its inverse—is efficiently computable even without the factorization of N .

13.2.1 The Structure of $\mathbb{Z}_{N^2}^*$

This section is devoted to a proof of Proposition 13.6. Throughout, we let N, p, q be as in the proposition.

CLAIM 13.7 $\gcd(N, \phi(N)) = 1$.

PROOF Recall that $\phi(N) = (p - 1)(q - 1)$. Assume $p > q$ without loss of generality. Since p is prime and $p > p - 1 > q - 1$, clearly $\gcd(p, \phi(N)) = 1$. Similarly, $\gcd(q, q - 1) = 1$. Now, if $\gcd(q, p - 1) \neq 1$ then $\gcd(q, p - 1) = q$

since q is prime. But then $(p-1)/q \geq 2$, contradicting the assumption that p and q have the same length. ■

CLAIM 13.8 *For $a \geq 0$ an integer, we have $(1+N)^a = 1 + aN \bmod N^2$. Thus, the order of $(1+N)$ in $\mathbb{Z}_{N^2}^*$ is N .*

PROOF Using the binomial expansion theorem (Theorem A.1):

$$(1+N)^a = \sum_{i=0}^a \binom{a}{i} N^i.$$

Reducing the right-hand side modulo N^2 , all terms with $i \geq 2$ become 0 and so $(1+N)^a = 1 + aN \bmod N^2$. The smallest nonzero a such that $(1+N)^a = 1 \bmod N^2$ is therefore $a = N$. ■

CLAIM 13.9 *The group $\mathbb{Z}_N \times \mathbb{Z}_N^*$ is isomorphic to the group $\mathbb{Z}_{N^2}^*$, with isomorphism $f: \mathbb{Z}_N \times \mathbb{Z}_N^* \rightarrow \mathbb{Z}_{N^2}^*$ given by $f(a, b) = [(1+N)^a \cdot b^N \bmod N^2]$.*

PROOF Note that $(1+N)^a \cdot b^N$ does not have a factor in common with N^2 since $\gcd((1+N), N^2) = 1$ and $\gcd(b, N^2) = 1$ (because $b \in \mathbb{Z}_N^*$). So $[(1+N)^a \cdot b^N \bmod N^2]$ lies in $\mathbb{Z}_{N^2}^*$. We now prove that f is an isomorphism.

We first show that f is a bijection. Since

$$\begin{aligned} |\mathbb{Z}_{N^2}^*| &= \phi(N^2) = p \cdot (p-1) \cdot q \cdot (q-1) = pq \cdot (p-1)(q-1) \\ &= |\mathbb{Z}_N| \cdot |\mathbb{Z}_N^*| = |\mathbb{Z}_N \times \mathbb{Z}_N^*| \end{aligned}$$

(see Theorem 8.19 for the second equality), it suffices to show that f is one-to-one. Say $a_1, a_2 \in \mathbb{Z}_N$ and $b_1, b_2 \in \mathbb{Z}_N^*$ are such that $f(a_1, b_1) = f(a_2, b_2)$. Then:

$$(1+N)^{a_1-a_2} \cdot (b_1/b_2)^N = 1 \bmod N^2. \quad (13.1)$$

(Note that $b_2 \in \mathbb{Z}_N^*$ and thus $b_2 \in \mathbb{Z}_{N^2}^*$, and so b_2 has a multiplicative inverse modulo N^2 .) Raising both sides to the power $\phi(N)$ and using the fact that the order of $\mathbb{Z}_{N^2}^*$ is $\phi(N^2) = N \cdot \phi(N)$ we obtain

$$\begin{aligned} (1+N)^{(a_1-a_2) \cdot \phi(N)} \cdot (b_1/b_2)^{N \cdot \phi(N)} &= 1 \bmod N^2 \\ \Rightarrow (1+N)^{(a_1-a_2) \cdot \phi(N)} &= 1 \bmod N^2. \end{aligned}$$

By Claim 13.8, $(1+N)$ has order N modulo N^2 . Applying Proposition 8.53, we see that $(a_1 - a_2) \cdot \phi(N) = 0 \bmod N$ and so N divides $(a_1 - a_2) \cdot \phi(N)$. Since $\gcd(N, \phi(N)) = 1$ by Claim 13.7, it follows that $N \mid (a_1 - a_2)$. Since $a_1, a_2 \in \mathbb{Z}_N$, this can only occur if $a_1 = a_2$.

Returning to Equation (13.1) and setting $a_1 = a_2$, we thus have $b_1^N = b_2^N \bmod N^2$. This implies $b_1^N = b_2^N \bmod N$. Since N is relatively prime to $\phi(N)$, the order of $\mathbb{Z}_N^*$, exponentiation to the power N is a bijection in $\mathbb{Z}_N^*$ (cf. Corollary 8.17). This means that $b_1 = b_2 \bmod N$; since $b_1, b_2 \in \mathbb{Z}_N^*$, we have $b_1 = b_2$. We conclude that f is one-to-one, and hence a bijection.

To show that f is an isomorphism, we show that $f(a_1, b_1) \cdot f(a_2, b_2) = f(a_1 + a_2, b_1 \cdot b_2)$. (Note that multiplication on the left-hand side of the equality takes place modulo N^2 , while addition/multiplication on the right-hand side takes place modulo N .) We have:

$$\begin{aligned} f(a_1, b_1) \cdot f(a_2, b_2) &= ((1 + N)^{a_1} \cdot b_1^N) \cdot ((1 + N)^{a_2} \cdot b_2^N) \bmod N^2 \\ &= (1 + N)^{a_1 + a_2} \cdot (b_1 b_2)^N \bmod N^2. \end{aligned}$$

Since $(1 + N)$ has order N modulo N^2 (by Claim 13.8), we can apply Proposition 8.52 and obtain

$$\begin{aligned} f(a_1, b_1) \cdot f(a_2, b_2) &= (1 + N)^{a_1 + a_2} \cdot (b_1 b_2)^N \bmod N^2 \\ &= (1 + N)^{[a_1 + a_2 \bmod N]} \cdot (b_1 b_2)^N \bmod N^2. \end{aligned} \quad (13.2)$$

We are not yet done, since $b_1 b_2$ in Equation (13.2) represents multiplication modulo N^2 whereas we would like it to be modulo N . Let $b_1 b_2 = r + \gamma N$, where γ, r are integers with $1 \leq r < N$ (r cannot be 0 since $b_1, b_2 \in \mathbb{Z}_N^*$ and so their product cannot be divisible by N). Note that $r = b_1 b_2 \bmod N$. We also have

$$\begin{aligned} (b_1 b_2)^N &= (r + \gamma N)^N \bmod N^2 \\ &= \sum_{k=0}^N \binom{N}{k} r^{N-k} (\gamma N)^k \bmod N^2 \\ &= r^N + N \cdot r^{N-1} \cdot (\gamma N) = r^N = ([b_1 b_2 \bmod N])^N \bmod N^2, \end{aligned}$$

using the binomial expansion theorem as in Claim 13.8. Plugging this in to Equation (13.2) we get the desired result:

$$\begin{aligned} f(a_1, b_1) \cdot f(a_2, b_2) &= (1 + N)^{[a_1 + a_2 \bmod N]} \cdot (b_1 b_2 \bmod N)^N \bmod N^2 \\ &= f(a_1 + a_2, b_1 b_2), \end{aligned}$$

proving that f is an isomorphism from $\mathbb{Z}_N \times \mathbb{Z}_N^*$ to $\mathbb{Z}_{N^2}^*$. ■

13.2.2 The Paillier Encryption Scheme

Let $N = pq$ be a product of two distinct primes of equal length. Proposition 13.6 says that $\mathbb{Z}_N \times \mathbb{Z}_N^*$ is isomorphic to $\mathbb{Z}_{N^2}^*$, with isomorphism given by $f(a, b) = [(1 + N)^a \cdot b^N \bmod N^2]$. A consequence is that a uniform element

$y \in \mathbb{Z}_{N^2}^*$ corresponds to a uniform element $(a, b) \in \mathbb{Z}_N \times \mathbb{Z}_N^*$ or, in other words, an element (a, b) with uniform $a \in \mathbb{Z}_N$ and uniform $b \in \mathbb{Z}_N^*$.

Call $y \in \mathbb{Z}_{N^2}^*$ an N th residue modulo N^2 if y is an N th power, that is, if there exists an $x \in \mathbb{Z}_{N^2}^*$ with $y = x^N \bmod N^2$. We denote the set of N th residues modulo N^2 by $\text{Res}(N^2)$. Let us characterize the N th residues in $\mathbb{Z}_{N^2}^*$. Taking any $x \in \mathbb{Z}_{N^2}^*$ with $x \leftrightarrow (a, b)$ and raising it to the N th power gives:

$$[x^N \bmod N^2] \leftrightarrow (a, b)^N = (N \cdot a \bmod N, b^N \bmod N) = (0, b^N \bmod N).$$

(Recall that the group operation in $\mathbb{Z}_N \times \mathbb{Z}_N^*$ is addition modulo N in the first component and multiplication modulo N in the second component.) Moreover, we claim that any element y with $y \leftrightarrow (0, b)$ is an N th residue. To see this, recall that $\gcd(N, \phi(N)) = 1$ and so $d \stackrel{\text{def}}{=} [N^{-1} \bmod \phi(N)]$ exists. So

$$(a, [b^d \bmod N])^N = (Na \bmod N, [b^{dN} \bmod N]) = (0, b) \leftrightarrow y$$

for any $a \in \mathbb{Z}_N$. We have thus shown that $\text{Res}(N^2)$ corresponds to the set

$$\{(0, b) \mid b \in \mathbb{Z}_N^*\}.$$

The above also demonstrates that the number of N th roots of any $y \in \text{Res}(N^2)$ is exactly N , and so computing N th powers is an N -to-1 function. As such, if $r \in \mathbb{Z}_{N^2}^*$ is uniform then $[r^N \bmod N^2]$ is a uniform element of $\text{Res}(N^2)$.

The *decisional composite residuosity problem*, roughly speaking, is to distinguish a uniform element of $\mathbb{Z}_{N^2}^*$ from a uniform element of $\text{Res}(N^2)$. Formally, let **GenModulus** be a polynomial-time algorithm that, on input 1^n , outputs (N, p, q) where $N = pq$, and p and q are n -bit primes (except with probability negligible in n). Then:

DEFINITION 13.10 *The decisional composite residuosity problem is hard relative to **GenModulus** if for all probabilistic polynomial-time algorithms D there is a negligible function negl such that*

$$\left| \Pr[D(N, [r^N \bmod N^2]) = 1] - \Pr[D(N, r) = 1] \right| \leq \text{negl}(n),$$

where in each case the probabilities are taken over the experiment in which **GenModulus**(1^n) outputs (N, p, q) , and then a uniform $r \in \mathbb{Z}_{N^2}^*$ is chosen. (Recall that $[r^N \bmod N^2]$ is a uniform element of $\text{Res}(N^2)$.)

The *decisional composite residuosity (DCR) assumption* is the assumption that there is a **GenModulus** relative to which the decisional composite residuosity problem is hard.

As we have discussed, elements of $\mathbb{Z}_{N^2}^*$ have the form (r', r) with r' and r arbitrary (in the appropriate groups), whereas N th residues have the form $(0, r)$ with $r \in \mathbb{Z}_N^*$ arbitrary. The DCR assumption is that it is hard to

distinguish uniform elements of the first type from uniform elements of the second type. This suggests the following abstract way to encrypt a message $m \in \mathbb{Z}_N$ with respect to a public key N : choose a uniform N th residue $(0, r)$ and set the ciphertext equal to

$$c \leftrightarrow (m, 1) \cdot (0, r) = (m + 0, 1 \cdot r) = (m, r).$$

Without worrying for now how this can be carried out efficiently by the sender, or how the receiver can decrypt, let us simply convince ourselves (on an intuitive level) that this is secure. Since a uniform N th residue $(0, r)$ cannot be distinguished from a uniform element (r', r) , the ciphertext as constructed above is indistinguishable (from the point of an eavesdropper who does not know the factorization of N) from the ciphertext

$$c' \leftrightarrow (m, 1) \cdot (r', r) = ([m + r' \bmod N], r)$$

for uniform $r' \in \mathbb{Z}_N$ and $r \in \mathbb{Z}_N^*$. Lemma 11.15 shows that $[m + r' \bmod N]$ is uniformly distributed in $\mathbb{Z}_N$ and so, in particular, this ciphertext c' is independent of the message m . CPA-security follows. A formal proof that proceeds exactly along these lines is given further below.

Before turning to the formal description and proof of security, we show how encryption and decryption can be performed efficiently.

Encryption. We have described encryption above as though it is taking place in $\mathbb{Z}_N \times \mathbb{Z}_N^*$. In fact it takes place in the isomorphic group $\mathbb{Z}_{N^2}^*$. That is, the sender generates a ciphertext $c \in \mathbb{Z}_{N^2}^*$ by choosing a uniform¹ $r \in \mathbb{Z}_N^*$ and then computing

$$c := [(1 + N)^m \cdot r^N \bmod N^2].$$

Observe that

$$c = ((1 + N)^m \cdot 1^N) \cdot ((1 + N)^0 \cdot r^N) \bmod N^2 \leftrightarrow (m, 1) \cdot (0, r),$$

and so $c \leftrightarrow (m, r)$ as desired.

Decryption. We now describe how decryption can be performed efficiently given the factorization of N . For c constructed as above, we claim that m is recovered by the following steps:

- Set $\hat{c} := [c^{\phi(N)} \bmod N^2]$.
- Set $\hat{m} := (\hat{c} - 1)/N$. (Note that this is carried out over the integers.)
- Set $m := [\hat{m} \cdot \phi(N)^{-1} \bmod N]$.

¹We remark that it does not make any difference whether the sender chooses uniform $r \in \mathbb{Z}_N^*$ or uniform $r \in \mathbb{Z}_{N^2}^*$, since in either case the distribution of $[r^N \bmod N^2]$ is the same (as can be verified by looking at what happens in the isomorphic group $\mathbb{Z}_N \times \mathbb{Z}_N^*$).

To see why this works, let $c \leftrightarrow (m, r)$ for an arbitrary $r \in \mathbb{Z}_N^*$. Then

$$\begin{aligned}\hat{c} &\stackrel{\text{def}}{=} [c^{\phi(N)} \bmod N^2] \\ &\leftrightarrow (m, r)^{\phi(N)} \\ &= ([m \cdot \phi(N) \bmod N], [r^{\phi(N)} \bmod N]) \\ &= ([m \cdot \phi(N) \bmod N], 1).\end{aligned}$$

By Proposition 13.6(3), this means that $\hat{c} = (1 + N)^{[m \cdot \phi(N) \bmod N]} \bmod N^2$. Using Proposition 13.6(2), we know that

$$\hat{c} = (1 + N)^{[m \cdot \phi(N) \bmod N]} = (1 + [m \cdot \phi(N) \bmod N] \cdot N) \bmod N^2.$$

Since $1 + [m \cdot \phi(N) \bmod N] \cdot N$ is always less than N^2 we can drop the $\bmod N^2$ at the end and view the above as an equality over the integers. Thus, $\hat{m} \stackrel{\text{def}}{=} (\hat{c} - 1)/N = [m \cdot \phi(N) \bmod N]$ and, finally,

$$m = [\hat{m} \cdot \phi(N)^{-1} \bmod N],$$

as required. (Note that $\phi(N)$ is invertible modulo N since $\gcd(N, \phi(N)) = 1$.)

We give a complete description of the Paillier encryption scheme, followed by an example of the above calculations.

CONSTRUCTION 13.11

Let **GenModulus** be a polynomial-time algorithm that, on input 1^n , outputs (N, p, q) where $N = pq$ and p and q are n -bit primes (except with probability negligible in n). Define the following encryption scheme:

- **Gen**: on input 1^n run **GenModulus**(1^n) to obtain (N, p, q) . The public key is N , and the private key is $\langle N, \phi(N) \rangle$.
- **Enc**: on input a public key N and a message $m \in \mathbb{Z}_N$, choose a uniform $r \leftarrow \mathbb{Z}_N^*$ and output the ciphertext

$$c := [(1 + N)^m \cdot r^N \bmod N^2].$$

- **Dec**: on input a private key $\langle N, \phi(N) \rangle$ and a ciphertext c , compute

$$m := \left\lceil \frac{[c^{\phi(N)} \bmod N^2] - 1}{N} \cdot \phi(N)^{-1} \bmod N \right\rceil.$$

The Paillier encryption scheme.

Example 13.12

Let $N = 11 \cdot 17 = 187$ (and so $N^2 = 34969$), and consider encrypting the message $m = 175$ and then decrypting the corresponding ciphertext. Choosing

$r = 83 \in \mathbb{Z}_{187}^*$, we compute the ciphertext

$$c := [(1 + 187)^{175} \cdot 83^{187} \bmod 34969] = 23911$$

corresponding to $(175, 83)$. To decrypt, note that $\phi(N) = 160$. So we first compute $\hat{c} := [23911^{160} \bmod 34969] = 25620$. Subtracting 1 and dividing by 187 gives $\hat{m} := (25620 - 1)/187 = 137$; since $90 = [160^{-1} \bmod 187]$, the message is recovered as $m := [137 \cdot 90 \bmod 187] = 175$. $\diamond$

THEOREM 13.13 *If the decisional composite residuosity problem is hard relative to GenModulus , then the Paillier encryption scheme is CPA-secure.*

PROOF Let Π denote the Paillier encryption scheme. We prove that Π has indistinguishable encryptions in the presence of an eavesdropper; by Theorem 11.6 this implies that it is CPA-secure.

Let $\mathcal{A}$ be an arbitrary probabilistic polynomial-time adversary. Consider the following PPT algorithm D that attempts to solve the decisional composite residuosity problem relative to GenModulus :

Algorithm D :

The algorithm is given N, y as input.

- Set $pk = N$ and run $\mathcal{A}(pk)$ to obtain two messages m_0, m_1 .
- Choose a uniform bit b and set $c := [(1 + N)^{m_b} \cdot y \bmod N^2]$.
- Give the ciphertext c to $\mathcal{A}$ and obtain an output bit b' . If $b' = b$, output 1; otherwise, output 0.

Let us analyze the behavior of D . There are two cases to consider:

Case 1: Say the input to D was generated by running $\text{GenModulus}(1^n)$ to obtain (N, p, q) , choosing uniform $r \in \mathbb{Z}_{N^2}^*$, and setting $y := [r^N \bmod N^2]$. (That is, y is a uniform element of $\text{Res}(N^2)$.) In this case,

$$c = [(1 + N)^{m_b} \cdot r^N \bmod N^2]$$

for uniform $r \in \mathbb{Z}_{N^2}^*$. Recalling that the distribution on $[r^N \bmod N^2]$ is the same whether r is chosen uniformly from $\mathbb{Z}_N^*$ or from $\mathbb{Z}_{N^2}^*$, we see that in this case the view of $\mathcal{A}$ when run as a subroutine by D is distributed identically to $\mathcal{A}$'s view in experiment $\text{PubK}_{\mathcal{A}, \Pi}^{\text{eav}}(n)$. Since D outputs 1 exactly when the output b' of $\mathcal{A}$ is equal to b , we have

$$\Pr [D(N, [r^N \bmod N^2]) = 1] = \Pr [\text{PubK}_{\mathcal{A}, \Pi}^{\text{eav}}(n) = 1],$$

where the first probability is taken over the experiment as in Definition 13.10.

Case 2: Say the input to D was generated by running $\text{GenModulus}(1^n)$ to obtain (N, p, q) and choosing uniform $y \in \mathbb{Z}_{N^2}^*$. We claim that the view of $\mathcal{A}$

in this case is *independent* of the bit b . This follows because y is a uniform element of the group $\mathbb{Z}_{N^2}^*$, and so the ciphertext c is uniformly distributed in $\mathbb{Z}_{N^2}^*$ (see Lemma 11.15), independent of m . Thus, the probability that $b' = b$ in this case is exactly $\frac{1}{2}$. That is,

$$\Pr[D(N, r) = 1] = \frac{1}{2},$$

where the probability is taken over the experiment as in Definition 13.10.

Combining the above, we see that

$$\begin{aligned} & \left| \Pr[D(N, [r^N \bmod N^2]) = 1] - \Pr[D(N, r) = 1] \right| \\ &= \left| \Pr[\text{PubK}_{\mathcal{A}, \Pi}^{\text{eav}}(n) = 1] - \frac{1}{2} \right|. \end{aligned}$$

By the assumption that the decisional composite residuosity problem is hard relative to **GenModulus**, there is a negligible function negl such that

$$\left| \Pr[\text{PubK}_{\mathcal{A}, \Pi}^{\text{eav}}(n) = 1] - \frac{1}{2} \right| \leq \text{negl}(n).$$

Thus $\Pr[\text{PubK}_{\mathcal{A}, \Pi}^{\text{eav}}(n) = 1] \leq \frac{1}{2} + \text{negl}(n)$, completing the proof. ■

13.2.3 Homomorphic Encryption

The Paillier encryption scheme is useful in a number of settings because it is *homomorphic*. Roughly, a homomorphic encryption scheme enables (certain) computations to be performed on encrypted data, yielding a ciphertext containing the encrypted result. In the case of Paillier encryption, the computation that can be performed is (modular) *addition*. Specifically, fix a public key $pk = N$. Then the Paillier scheme has the property that multiplying an encryption of m_1 and an encryption of m_2 (with multiplication done modulo N^2) results in an encryption of $[m_1 + m_2 \bmod N]$; this is because

$$\begin{aligned} & ((1 + N)^{m_1} \cdot r_1^N) \cdot ((1 + N)^{m_2} \cdot r_2^N) \\ &= (1 + N)^{[m_1 + m_2 \bmod N]} \cdot (r_1 r_2)^N \bmod N^2. \end{aligned}$$

Although the ability to add encrypted values may not seem very useful, it suffices for several interesting applications including voting, discussed below.

We present a general definition, of which Paillier encryption is a special case.

DEFINITION 13.14 *A public-key encryption scheme $(\text{Gen}, \text{Enc}, \text{Dec})$ is homomorphic if for all n and all (pk, sk) output by $\text{Gen}(1^n)$, it is possible to define groups $\mathbb{M}, \mathbb{C}$ (depending on pk only) such that:*

- *The message space is $\mathbb{M}$, and all ciphertexts output by Enc_{pk} are elements of $\mathbb{C}$. For notational convenience, we write $\mathbb{M}$ as an additive group and $\mathbb{C}$ as a multiplicative group.*

- For any $m_1, m_2 \in \mathbb{M}$, any c_1 output by $\text{Enc}_{pk}(m_1)$, and any c_2 output by $\text{Enc}_{pk}(m_2)$, it holds that

$$\text{Dec}_{sk}(c_1 \cdot c_2) = m_1 + m_2.$$

Moreover, the distribution on ciphertexts obtained by encrypting m_1 , encrypting m_2 , and then multiplying the results is identical to the distribution on ciphertexts obtained by encrypting $m_1 + m_2$.

The last part of the definition ensures that if ciphertexts $c_1 \leftarrow \text{Enc}_{pk}(m_1)$ and $c_2 \leftarrow \text{Enc}_{pk}(m_2)$ are generated and the result $c_3 := c_1 \cdot c_2$ is computed, then the resulting ciphertext c_3 contains no more information about m_1 or m_2 than the sum m_3 .

The Paillier encryption scheme with $pk = N$ is homomorphic with $\mathbb{M} = \mathbb{Z}_N$ and $\mathbb{C} = \mathbb{Z}_{N^2}^*$. This is not the first example of a homomorphic encryption scheme we have seen; El Gamal encryption is also homomorphic. Specifically, for public key $pk = \langle \mathbb{G}, g, q, h \rangle$ we can take $\mathbb{M} = \mathbb{G}$ and $\mathbb{C} = \mathbb{G} \times \mathbb{G}$; then

$$\langle g^{y_1}, h^{y_1} \cdot m_1 \rangle \cdot \langle g^{y_2}, h^{y_2} \cdot m_2 \rangle = \langle g^{y_1+y_2}, h^{y_1+y_2} \cdot m_1 m_2 \rangle,$$

where multiplication of ciphertexts is component-wise. The Goldwasser-Micali encryption scheme we will see later is also homomorphic (see Exercise 13.10).

A nice feature of Paillier encryption is that it is homomorphic over a large additive group (namely, $\mathbb{Z}_N$). To see an application of this, consider the following distributed voting scheme, where ℓ voters can vote “no” or “yes” and the goal is to tabulate the number of “yes” votes:

1. A voting authority generates a public key N for the Paillier encryption scheme and publicizes N .
2. Let 0 stand for a “no,” and let 1 stand for a “yes.” Each voter casts their vote by encrypting it. That is, voter i casts her vote v_i by computing $c_i := [(1 + N)^{v_i} \cdot (r_i)^N \bmod N^2]$ for a uniform $r_i \in \mathbb{Z}_N^*$.
3. Each voter broadcasts their vote c_i . These votes are then publicly *aggregated* by computing

$$c_{\text{total}} := \left[\prod_{i=1}^{\ell} c_i \bmod N^2 \right].$$

4. The authority is given c_{total} . (We assume the authority has not been able to observe what goes on until now.) By decrypting it, the authority obtains the vote total

$$v_{\text{total}} \stackrel{\text{def}}{=} \sum_{i=1}^{\ell} v_i \bmod N.$$

If ℓ is small (so that $v_{\text{total}} \ll N$), there is no wrap-around modulo N and $v_{\text{total}} = \sum_{i=1}^{\ell} v_i$.

Key features of the above are that *no voter learns anyone else's vote*, and calculation of the total is *publicly verifiable* if the authority is trusted to correctly compute v_{total} from c_{total} . Also, the authority obtains the correct total *without learning any individual votes*. (Here, we assume the authority cannot see voters' ciphertexts. In Section 13.3.3 we show a protocol in which votes are kept hidden from authorities even if they see all the communication.) We assume all voters act honestly (and only try to learn others' votes based on information they observe); an entire research area of cryptography is dedicated to addressing potential threats from participants who might be malicious and not follow the protocol.

13.3 Secret Sharing and Threshold Encryption

Motivated by the discussion of distributed voting in the previous section, we briefly consider *secure (interactive) protocols*. Such protocols can be significantly more complicated than the basic cryptographic primitives (e.g., encryption and signature schemes) we have focused on until now, both because they can involve multiple parties exchanging several rounds of messages, as well as because they are intended to realize more-complex security requirements.

The goal of this section is mainly to give the reader a taste of this fascinating area, and no attempt is made at being comprehensive or complete. Although the protocols presented here can be proven secure (with respect to appropriate definitions), we omit formal definitions, details, and proofs and instead rely on informal discussion.

13.3.1 Secret Sharing

Consider the following problem. A *dealer* holds a *secret* s —say, a nuclear-launch code—that it wishes to share among some set of N users $P_1, \dots, P_N$ by giving each user a *share*. Any t users should be able to pool their shares and reconstruct the secret, but no coalition of fewer than t users should get *any* information about s from their collective shares (beyond whatever information they had about s already). We refer to such a sharing mechanism as a (t, N) -*threshold secret-sharing scheme*. Such a scheme ensures that s is not revealed without sufficient authorization, while also guaranteeing availability of s when needed (since any t users can reconstruct it). Beyond their direct application, secret-sharing schemes are also a building block of many cryptographic protocols.

Consider a simple solution for the case $t = N$, assuming $s \in \{0, 1\}^\ell$. The dealer chooses uniform $s_1, \dots, s_{N-1} \in \{0, 1\}^\ell$ and sets $s_N := s \oplus \left(\bigoplus_{i=1}^{N-1} s_i \right)$; the share of user P_i is s_i . Since $\bigoplus_{i=1}^N s_i = s$ by construction, clearly all the users together can recover s . However, the shares of any coalition of

$N - 1$ users are (jointly) uniform and independent of s , and thus reveal no information about s . This is clear when the coalition is $P_1, \dots, P_{N-1}$. In the general case, when the coalition includes everyone except for P_j ($j \neq N$), this is true because $s_1, \dots, s_{j-1}, s_{j+1}, \dots, s_{N-1}$ are uniform and independent of s by construction, and

$$s_N = s \oplus \left(\bigoplus_{i < N, i \neq j} s_i \right) \oplus s_j;$$

thus, even conditioned on some fixed values for s and the shares of the other members of the coalition, the share s_N of user P_N is uniform because s_j is uniform and independent of s .

We can extend this to obtain a solution for $t < N$. The basic idea is to replicate the above scheme for each subset $T \subset N$ of size t . That is, for each such subset $T = \{P_{i_1}, \dots, P_{i_t}\}$, we choose uniform shares $s_{T,i_1}, \dots, s_{T,i_t}$ subject to the constraint that $\bigoplus_{j=1}^t s_{T,i_j} = s$, and give s_{T,i_j} to user P_{i_j} . It is not hard to see that this satisfies the requirements.

Unfortunately, this extension of the original scheme is not *efficient*. Each user now stores a share $s_{T,i}$ for *every* subset T of which she is a member. For each user there are $\binom{N-1}{t-1}$ such subsets, which is exponential in N if $t \approx N/2$.

Shamir's scheme. Fortunately, it is possible to do significantly better using a secret-sharing scheme introduced by Adi Shamir (of RSA fame). This scheme is based on polynomials² over a finite field $\mathbb{F}$, where $\mathbb{F}$ is chosen so that $s \in \mathbb{F}$ and $|\mathbb{F}| > N$. (See Appendix A.5 for a brief discussion of finite fields.) Before describing the scheme, we briefly review some background related to polynomials over a field $\mathbb{F}$.

A value $x \in \mathbb{F}$ is a *root* of a polynomial p if $p(x) = 0$. We use the well-known fact that any nonzero, degree- t polynomial over a field has at most t roots. This implies:

COROLLARY 13.15 *Any two distinct degree- t polynomials p and q agree on at most t points.*

PROOF If not, then the nonzero, degree- t polynomial $p - q$ would have more than t roots. ■

Shamir's scheme relies on the fact that for any t pairs of elements $(x_1, y_1), \dots, (x_t, y_t)$ from $\mathbb{F}$ (with the $\{x_i\}$ distinct), there is a *unique* polynomial p of degree $(t-1)$ such that $p(x_i) = y_i$ for $1 \leq i \leq t$. We can prove this quite easily. The fact that there exists such a p uses standard polynomial interpolation.

²A degree- t polynomial p over $\mathbb{F}$ is given by $p(X) = \sum_{i=0}^t a_i X^i$, where $a_i \in \mathbb{F}$ and X is a formal variable. (Note that we allow $a_t = 0$ and so we really mean a polynomial of degree at most t .) Any such polynomial naturally defines a function mapping $\mathbb{F}$ to itself, given by evaluating the polynomial on its input.

In detail: for $i = 1, \dots, t$, define the degree- $(t-1)$ polynomial

$$\delta_i(X) \stackrel{\text{def}}{=} \frac{\prod_{j=1, j \neq i}^t (X - x_j)}{\prod_{j=1, j \neq i}^t (x_i - x_j)}.$$

Note that $\delta_i(x_j) = 0$ for any $j \neq i$, and $\delta_i(x_i) = 1$. So $p(X) \stackrel{\text{def}}{=} \sum_{i=1}^t \delta_i(X) \cdot y_i$ is a polynomial of degree $(t-1)$ with $p(x_i) = y_i$ for $1 \leq i \leq t$. (We remark that this, in fact, demonstrates that the desired polynomial p can be found *efficiently*.) Uniqueness follows from Corollary 13.15.

We now describe Shamir's (t, N) -threshold secret-sharing scheme. Let $\mathbb{F}$ be a finite field that contains the domain of possible secrets, and with $|\mathbb{F}| > N$. Let $x_1, \dots, x_N \in \mathbb{F}$ be distinct, nonzero elements that are fixed and publicly known. (Such elements exist since $|\mathbb{F}| > N$.) The scheme works as follows:

Sharing: Given a secret $s \in \mathbb{F}$, the dealer chooses uniform $a_1, \dots, a_{t-1} \in \mathbb{F}$ and defines the polynomial $p(X) \stackrel{\text{def}}{=} s + \sum_{i=1}^{t-1} a_i X^i$. This is a uniform degree- $(t-1)$ polynomial with constant term s . The share of user P_i is $s_i := p(x_i) \in \mathbb{F}$.

Reconstruction: Say t users $P_{i_1}, \dots, P_{i_t}$ pool their shares $s_{i_1}, \dots, s_{i_t}$. Using polynomial interpolation, they compute the unique degree- $(t-1)$ polynomial p' for which $p'(x_{i_j}) = s_{i_j}$ for $1 \leq j \leq t$. The secret is $p'(0)$.

It is clear that reconstruction works since $p' = p$ and $p(0) = s$.

It remains to show that any $t-1$ users learn nothing about the secret s from their shares. By symmetry, it suffices to consider the shares of users $P_1, \dots, P_{t-1}$. We claim that for *any* secret s , the shares $s_1, \dots, s_{t-1}$ are (jointly) uniform. Since the dealer chooses $a_1, \dots, a_{t-1}$ uniformly, this follows if we show that there is a one-to-one correspondence between the polynomial p chosen by the dealer and the shares $s_1, \dots, s_{t-1}$. But this is a direct consequence of Corollary 13.15.

13.3.2 Verifiable Secret Sharing

So far we have considered *passive* attacks in which $t-1$ users may try to use their shares to learn information about the secret. But we may also be concerned about *active*, malicious behavior. Here there are two separate concerns: First, a corrupted *dealer* may give inconsistent shares to the users, i.e., such that different secrets are recovered depending on which t users pool their shares. Second, in the reconstruction phase a malicious user may present a *different* share from the one given to them by the dealer, and thus affect the recovered secret. (While this could be addressed by having the dealer sign the shares, this does not work when the dealer itself may be dishonest.) *Verifiable secret-sharing (VSS) schemes* prevent both these attacks.

More formally, we allow any $t-1$ users to be corrupted and to collude with each other and, possibly, the dealer. We require (1a) at the end of the

sharing phase, a secret s is defined such that any collection that includes t uncorrupted users (whether or not this collection also includes some corrupted users) will successfully recover s in the reconstruction phase; moreover, (1b) if the dealer is honest, then s corresponds to the dealer's secret. In addition, (2) when the dealer is honest then, as before, the $t - 1$ corrupted users learn nothing about the secret from their shares and any public information the dealer publishes. Since we want there to be t uncorrupted users even if $t - 1$ users are corrupted, we require $N \geq t + (t - 1) > 2(t - 1)$; in other words, we assume a majority of the users remain uncorrupted.

We describe a VSS scheme due to Feldman that relies on an algorithm $\mathcal{G}$ relative to which the discrete-logarithm problem is hard. For simplicity, we describe it in the random-oracle model and let H denote a function to be modeled as a random oracle. We also assume that some trusted parameters $(\mathbb{G}, q, g)$, generated using $\mathcal{G}(1^n)$, are published in advance, where q is prime and so $\mathbb{Z}_q$ is a field. Finally, we assume that all users have access to a broadcast channel, such that a message broadcast by any user is heard by everyone.

The sharing phase now involves the N users running an interactive protocol with the dealer that proceeds as follows:

1. To share a secret s , the dealer chooses uniform $a_0 \in \mathbb{Z}_q$ and then shares a_0 as in Shamir's scheme. That is, the dealer chooses uniform $a_1, \dots, a_{t-1} \in \mathbb{Z}_q$ and defines the polynomial $p(X) \stackrel{\text{def}}{=} \sum_{i=0}^{t-1} a_i X^i$. The dealer sends the share $s_i := p(i) = \sum_{j=0}^{t-1} a_j \cdot i^j$ to user P_i .³

In addition, the dealer publicly broadcasts the values $A_0 := g^{a_0}, \dots, A_{t-1} := g^{a_{t-1}}$, and the "masked secret" $c := H(a_0) \oplus s$.

2. Each user P_i verifies that its share s_i satisfies

$$g^{s_i} \stackrel{?}{=} \prod_{j=0}^{t-1} (A_j)^{i^j}. \quad (13.3)$$

If not, P_i publicly broadcasts a complaint.

Note that if the dealer is honest, we have

$$\prod_{j=0}^{t-1} (A_j)^{i^j} = \prod_{j=0}^{t-1} (g^{a_j})^{i^j} = g^{\sum_{j=0}^{t-1} a_j \cdot i^j} = g^{p(i)} = g^{s_i},$$

and so no honest user will complain. Since there are at most $t - 1$ corrupted users, there are at most $t - 1$ complaints if the dealer is honest.

3. If more than $t - 1$ users complain, the dealer is disqualified and the protocol is aborted. Otherwise, the dealer responds to a complaint from P_i by broadcasting s_i . If this share does not satisfy Equation (13.3) (or if the dealer refuses to respond to a complaint at all), the dealer is disqualified and the protocol is aborted. Otherwise, P_i uses the broadcast value (rather than the value it received in the first round) as its share.

³Note that we are now setting $x_i = i$, which is fine since we are using the field $\mathbb{Z}_q$.

In the reconstruction phase, say a group of users (that includes at least t uncorrupted users) pool their shares. A share s_i provided by a user P_i is discarded if it does not satisfy Equation (13.3). Among the remaining shares, any t of them are used to recover a_0 exactly as in Shamir's scheme. The original secret is then computed as $s := c \oplus H(a_0)$.

We now argue that this protocol meets the desired security requirements. We first show that, assuming the dealer is not disqualified, the value recovered in the reconstruction phase is uniquely determined by the public information; specifically, the recovered value is $c \oplus H(\log_g A_0)$. (Combined with the fact that an honest dealer is never disqualified, this proves that conditions (1a) and (1b) hold.) Define $a_i := \log_g A_i$ for $0 \leq i \leq t-1$; the $\{a_i\}$ cannot be computed efficiently if the discrete-logarithm problem is hard, but they are still well-defined. Define the polynomial $p(X) \stackrel{\text{def}}{=} \sum_{i=0}^{t-1} a_i X^i$. Any share s_i , contributed by party P_i , that is not discarded during the reconstruction phase must satisfy Equation (13.3), and hence satisfies $s_i = p(i)$. It follows that, regardless of which shares are used, the parties will reconstruct polynomial p , compute $a_0 = p(0)$, and then recover $s = c \oplus H(a_0)$.

It is also possible to show that condition (2) holds for computationally bounded adversaries if the discrete-logarithm problem is hard for $\mathcal{G}$. (In contrast to Shamir's secret-sharing scheme, secrecy here is no longer unconditional. Unconditionally secure VSS schemes are possible, but are beyond the scope of our treatment.) Intuitively, this is because the secret s is masked by the random value $H(a_0)$, and the information given to any $t-1$ users in the sharing phase—namely, their shares and the public values $\{A_i\}$ —reveals only g^{a_0} , from which it is hard to compute a_0 . This intuition can be made rigorous, but we do not do so here.

13.3.3 Threshold Encryption and Electronic Voting

In Section 13.2.3 we introduced the notion of homomorphic encryption schemes and gave the Paillier encryption scheme as an example. Here we show a different homomorphic encryption scheme that is a variant of El Gamal encryption. Specifically, given a public key $pk = \langle \mathbb{G}, q, g, h \rangle$ as in regular El Gamal encryption, we now encrypt a message $m \in \mathbb{Z}_q$ by setting $M := g^m$, choosing a uniform $y \in \mathbb{Z}_q$, and sending the ciphertext $c := \langle g^y, h^y \cdot M \rangle$. To decrypt, the receiver recovers M as in standard El Gamal decryption and then computes $m := \log_g M$. Although this is not efficient if m comes from a large domain, if m is from a small domain—as it will be in our application—then the receiver can compute $\log_g M$ efficiently using exhaustive search. The advantage of this variant scheme is that it is homomorphic with respect to addition in $\mathbb{Z}_q$. That is,

$$\langle g^{y_1}, h^{y_1} \cdot g^{m_1} \rangle \cdot \langle g^{y_2}, h^{y_2} \cdot g^{m_2} \rangle = \langle g^{y_1+y_2}, h^{y_1+y_2} \cdot g^{m_1+m_2} \rangle.$$

Recall that the basic approach to electronic voting using homomorphic en-

ryption has each voter i encrypt her vote $v_i \in \{0, 1\}$ to obtain a ciphertext c_i . Once everyone has voted, the ciphertexts are multiplied to obtain an encryption of the sum $v_{total} \stackrel{\text{def}}{=} \sum_i v_i \bmod q = \sum_i c_i$. (The value q is, in practice, large enough so that no wrap-around modulo q occurs.) Since $0 \leq v_{total} \leq \ell$, where ℓ is the total number of voters, an authority with the private key can efficiently decrypt the final ciphertext and recover v_{total} .

A drawback of this approach is that the authority is trusted, both to (correctly) decrypt the final ciphertext as well as not to decrypt any of the individual voters' ciphertexts. (In Section 13.2.3 we assumed the authority could not see the individual voters' ciphertexts.) We might instead prefer to *distribute* trust among a set of N authorities, such that any set of t authorities is able to jointly decrypt an agreed-upon ciphertext (this ensures availability even if some authorities are down or unwilling to help decrypt), but no collection of $t - 1$ authorities is able to decrypt any ciphertext on their own (this ensures privacy as long as fewer than t authorities are corrupted).

At first glance, it may seem that secret sharing solves the problem. If we share the private key among the N authorities, then no set of $t - 1$ authorities learns the private key and so they cannot decrypt. On the other hand, any t authorities can pool their shares, recover the private key, and then decrypt any desired ciphertext.

A little thought shows that this does not quite work. If the authorities reconstruct the private key in order to decrypt some ciphertext, then as part of this process *all the authorities learn the private key!* Thus, afterward, any authority could decrypt any ciphertext of its choice, on its own.

We need instead a modified approach in which the “secret” (namely, the private key) is never reconstructed in the clear, yet is implicitly reconstructed only enough to enable decryption of one, agreed-upon ciphertext. We can achieve this for the specific case of El Gamal encryption in the following way. Fix a public key $pk = \langle \mathbb{G}, q, g, h \rangle$, and let $x \in \mathbb{Z}_q$ be the private key, i.e., $g^x = h$. Each authority is given a share $x_i \in \mathbb{Z}_q$ exactly as in Shamir's secret-sharing scheme. That is, a uniform degree- $(t - 1)$ polynomial p with $p(0) = x$ is chosen, and the i th authority is given $x_i := p(i)$. (We assume a trusted dealer who knows x and securely deletes it once it is shared. It is possible to eliminate the dealer entirely, but this is beyond our present scope.)

Now, say some t authorities $i_1, \dots, i_t$ wish to jointly decrypt a ciphertext $\langle c_1, c_2 \rangle$. To do so, authority i_j first publishes the value $w_j := c_1^{x_{i_j}}$. Recall from the previous section that there exist publicly computable polynomials $\{\delta_j(X)\}$ (that depend on the identities of these t authorities) such that $p(X) \stackrel{\text{def}}{=} \sum_{j=1}^t \delta_j(X) \cdot x_{i_j}$. Setting $\delta_j \stackrel{\text{def}}{=} \delta_j(0)$, we see that there exist publicly computable values $\delta_1, \dots, \delta_t \in \mathbb{Z}_q$ for which $x = p(0) = \sum_{j=1}^t \delta_j \cdot x_{i_j}$. Any authority can then compute

$$M' := \frac{c_2}{\prod_{j=1}^t w_j^{\delta_j}}.$$

(They can then each compute $\log_g M$, if desired.) To see that this correctly recovers the message, say $c_1 = g^y$ and $c_2 = h^y \cdot M$. Then

$$\prod_{j=1}^t w_j^{\delta_j} = \prod_{j=1}^t c_1^{x_{i_j} \delta_j} = c_1^{\sum_{j=1}^t x_{i_j} \delta_j} = c_1^{p(0)} = c_1^x,$$

and so

$$M' \stackrel{\text{def}}{=} \frac{c_2}{\prod_{j=1}^t w_j^{\delta_j}} = \frac{h^y \cdot M}{c_1^x} = \frac{(g^x)^y \cdot M}{(g^y)^x} = M.$$

Note that any set of $t - 1$ corrupted authorities learns nothing about the private key x from their shares. Moreover, it is possible to show that they learn nothing from the decryption process beyond the recovered value M .

Malicious (active) adversaries. Our treatment above assumes that the authorities decrypting some ciphertext all behave correctly. (If they do not, it would be easy for any of them to cause an incorrect result by publishing an arbitrary value w_j .) We also assume that voters behave honestly, and encrypt a vote of either 0 or 1. (Note that a voter could unfairly sway the election by encrypting a large value or a negative value.) Potential malicious behavior of this sort can be prevented using techniques beyond the scope of this book.

13.4 The Goldwasser–Micali Encryption Scheme

Before we present the Goldwasser–Micali encryption scheme, we need to develop a better understanding of *quadratic residues*. We first explore the easier case of quadratic residues modulo a prime p , and then look at the slightly more complicated case of quadratic residues modulo a composite N .

Throughout this section, p and q denote odd primes, and $N = pq$ denotes a product of two distinct, odd primes.

13.4.1 Quadratic Residues Modulo a Prime

In a group $\mathbb{G}$, an element $y \in \mathbb{G}$ is a *quadratic residue* if there exists an $x \in \mathbb{G}$ with $x^2 = y$. In this case, we call x a *square root* of y . An element that is not a quadratic residue is called a *quadratic non-residue*. In an abelian group, the set of quadratic residues forms a subgroup.

In the specific case of $\mathbb{Z}_p^*$, we have that y is a quadratic residue if there exists an x with $x^2 = y \bmod p$. We begin with an easy observation.

PROPOSITION 13.16 *Let $p > 2$ be prime. Every quadratic residue in $\mathbb{Z}_p^*$ has exactly two square roots.*

PROOF Let $y \in \mathbb{Z}_p^*$ be a quadratic residue. Then there exists an $x \in \mathbb{Z}_p^*$ such that $x^2 = y \bmod p$. Clearly, $(-x)^2 = x^2 = y \bmod p$. Furthermore, $-x \neq x \bmod p$: if $-x = x \bmod p$ then $2x = 0 \bmod p$, which implies $p \mid 2x$. Since p is prime, this would mean that either $p \mid 2$ (which is impossible since $p > 2$) or $p \mid x$ (which is impossible since $0 < x < p$). So, $[x \bmod p]$ and $[-x \bmod p]$ are distinct elements of $\mathbb{Z}_p^*$, and y has at least two square roots.

Let $x' \in \mathbb{Z}_p^*$ be a square root of y . Then $x^2 = y = (x')^2 \bmod p$, implying that $x^2 - (x')^2 = 0 \bmod p$. Factoring the left-hand side we obtain

$$(x - x')(x + x') = 0 \bmod p,$$

so that (by Proposition 8.3) either $p \mid (x - x')$ or $p \mid (x + x')$. In the first case, $x' = x \bmod p$ and in the second case $x' = -x \bmod p$, showing that y indeed has only $[\pm x \bmod p]$ as square roots. ■

Let $\text{sq}_p : \mathbb{Z}_p^* \rightarrow \mathbb{Z}_p^*$ be the function $\text{sq}_p(x) \stackrel{\text{def}}{=} [x^2 \bmod p]$. The above shows that sq_p is a two-to-one function when $p > 2$ is prime. This immediately implies that *exactly half the elements of $\mathbb{Z}_p^*$ are quadratic residues*. We denote the set of quadratic residues modulo p by $\mathcal{QR}_p$, and the set of quadratic non-residues by $\mathcal{QNR}_p$. We have just seen that for $p > 2$ prime

$$|\mathcal{QR}_p| = |\mathcal{QNR}_p| = \frac{|\mathbb{Z}_p^*|}{2} = \frac{p-1}{2}.$$

Define $\mathcal{J}_p(x)$, the *Jacobi symbol of x modulo p* , as follows.⁴ Let $p > 2$ be prime, and $x \in \mathbb{Z}_p^*$. Then

$$\mathcal{J}_p(x) \stackrel{\text{def}}{=} \begin{cases} +1 & \text{if } x \text{ is a quadratic residue modulo } p \\ -1 & \text{if } x \text{ is not a quadratic residue modulo } p. \end{cases}$$

The notation can be extended in the natural way for any x relatively prime to p by setting $\mathcal{J}_p(x) \stackrel{\text{def}}{=} \mathcal{J}_p([x \bmod p])$.

Can we characterize the quadratic residues in $\mathbb{Z}_p^*$? We begin with the fact that $\mathbb{Z}_p^*$ is a cyclic group of order $p-1$ (see Theorem 8.56). Let g be a generator of $\mathbb{Z}_p^*$. This means that

$$\mathbb{Z}_p^* = \{g^0, g^1, g^2, \dots, g^{\frac{p-1}{2}-1}, g^{\frac{p-1}{2}}, g^{\frac{p-1}{2}+1}, \dots, g^{p-2}\}$$

(recall that p is odd, so $p-1$ is even). Squaring each element in this list and reducing modulo $p-1$ in the exponent (cf. Corollary 8.15) yields a list of all the quadratic residues in $\mathbb{Z}_p^*$:

$$\mathcal{QR}_p = \{g^0, g^2, g^4, \dots, g^{p-3}, g^0, g^2, \dots, g^{p-3}\}.$$

⁴For p prime, $\mathcal{J}_p(x)$ is also sometimes called the *Legendre symbol* of x and denoted by $\mathcal{L}_p(x)$; we have chosen our notation to be consistent with notation introduced later.

Each quadratic residue appears twice in this list. Therefore, the quadratic residues in $\mathbb{Z}_p^*$ are exactly those elements that can be written as g^i with $i \in \{0, \dots, p-2\}$ an *even* integer.

The above characterization leads to a simple way to compute the Jacobi symbol and thus tell whether an element $x \in \mathbb{Z}_p^*$ is a quadratic residue or not.

PROPOSITION 13.17 *Let $p > 2$ be a prime. Then $\mathcal{J}_p(x) = x^{\frac{p-1}{2}} \bmod p$.*

PROOF Let g be an arbitrary generator of $\mathbb{Z}_p^*$. If x is a quadratic residue modulo p , our earlier discussion shows that $x = g^i$ for some even integer i . Writing $i = 2j$ with j an integer we then have

$$x^{\frac{p-1}{2}} = (g^{2j})^{\frac{p-1}{2}} = g^{(p-1)j} = (g^{p-1})^j = 1^j = 1 \bmod p,$$

and so $x^{\frac{p-1}{2}} = +1 = \mathcal{J}_p(x) \bmod p$ as claimed.

On the other hand, if x is not a quadratic residue then $x = g^i$ for some odd integer i . Writing $i = 2j + 1$ with j an integer we have

$$x^{\frac{p-1}{2}} = (g^{2j+1})^{\frac{p-1}{2}} = (g^{2j})^{\frac{p-1}{2}} \cdot g^{\frac{p-1}{2}} = 1 \cdot g^{\frac{p-1}{2}} = g^{\frac{p-1}{2}} \bmod p.$$

Now,

$$\left(g^{\frac{p-1}{2}}\right)^2 = g^{p-1} = 1 \bmod p,$$

and so $g^{\frac{p-1}{2}} = \pm 1 \bmod p$ since $[\pm 1 \bmod p]$ are the two square roots of 1 (cf. Proposition 13.16). Since g is a generator, it has order $p-1$ and so $g^{\frac{p-1}{2}} \neq 1 \bmod p$. It follows that $x^{\frac{p-1}{2}} = -1 = \mathcal{J}_p(x) \bmod p$. ■

Proposition 13.17 directly gives a polynomial-time algorithm (cf. Algorithm 13.18) for testing whether an element $x \in \mathbb{Z}_p^*$ is a quadratic residue.

ALGORITHM 13.18

Deciding quadratic residuosity modulo a prime

Input: A prime p ; an element $x \in \mathbb{Z}_p^*$

Output: $\mathcal{J}_p(x)$ (or, equivalently, whether x is a quadratic residue or quadratic non-residue)

$b := \left[x^{\frac{p-1}{2}} \bmod p\right]$

if $b = 1$ **return** “quadratic residue”

else return “quadratic non-residue”

We conclude this section by noting a nice multiplicative property of quadratic residues and non-residues modulo p .

PROPOSITION 13.19 *Let $p > 2$ be a prime, and $x, y \in \mathbb{Z}_p^*$. Then*

$$\mathcal{J}_p(xy) = \mathcal{J}_p(x) \cdot \mathcal{J}_p(y).$$

PROOF Using the previous proposition,

$$\mathcal{J}_p(xy) = (xy)^{\frac{p-1}{2}} = x^{\frac{p-1}{2}} \cdot y^{\frac{p-1}{2}} = \mathcal{J}_p(x) \cdot \mathcal{J}_p(y) \bmod p.$$

Since $\mathcal{J}_p(xy), \mathcal{J}_p(x), \mathcal{J}_p(y) = \pm 1$, equality holds over the integers as well. ■

COROLLARY 13.20 *Let $p > 2$ be prime, and say $x, x' \in \mathcal{QR}_p$ and $y, y' \in \mathcal{QNR}_p$. Then:*

1. $[xx' \bmod p] \in \mathcal{QR}_p$.
2. $[yy' \bmod p] \in \mathcal{QR}_p$.
3. $[xy \bmod p] \in \mathcal{QNR}_p$.

13.4.2 Quadratic Residues Modulo a Composite

We now turn our attention to quadratic residues in the group $\mathbb{Z}_N^*$, where $N = pq$. Characterizing the quadratic residues modulo N is easy if we use the results of the previous section in conjunction with the Chinese remainder theorem. Recall that the Chinese remainder theorem says that $\mathbb{Z}_N^* \simeq \mathbb{Z}_p^* \times \mathbb{Z}_q^*$, and we let $y \leftrightarrow (y_p, y_q)$ denote the correspondence guaranteed by the theorem (i.e., $y_p = [y \bmod p]$ and $y_q = [y \bmod q]$). The key observation is:

PROPOSITION 13.21 *Let $N = pq$ with p, q distinct primes, and $y \in \mathbb{Z}_N^*$ with $y \leftrightarrow (y_p, y_q)$. Then y is a quadratic residue modulo N if and only if y_p is a quadratic residue modulo p and y_q is a quadratic residue modulo q .*

PROOF If y is a quadratic residue modulo N then, by definition, there exists an $x \in \mathbb{Z}_N^*$ such that $x^2 = y \bmod N$. Let $x \leftrightarrow (x_p, x_q)$. Then

$$(y_p, y_q) \leftrightarrow y = x^2 \leftrightarrow (x_p, x_q)^2 = ([x_p^2 \bmod p], [x_q^2 \bmod q]),$$

where $(x_p, x_q)^2$ is simply the square of the element (x_p, x_q) in the group $\mathbb{Z}_p^* \times \mathbb{Z}_q^*$. We have thus shown that

$$y_p = x_p^2 \bmod p \quad \text{and} \quad y_q = x_q^2 \bmod q \tag{13.4}$$

and y_p, y_q are quadratic residues (with respect to the appropriate moduli).

Conversely, if $y \leftrightarrow (y_p, y_q)$ and y_p, y_q are quadratic residues modulo p and q , respectively, then there exist $x_p \in \mathbb{Z}_p^*$ and $x_q \in \mathbb{Z}_q^*$ such that Equation (13.4) holds. Let $x \in \mathbb{Z}_N^*$ be such that $x \leftrightarrow (x_p, x_q)$. Reversing the above steps shows that x is a square root of y modulo N . ■

The above proposition characterizes the quadratic residues modulo N . A careful examination of the proof yields another important observation: each quadratic residue $y \in \mathbb{Z}_N^*$ has exactly *four* square roots. To see this, let $y \leftrightarrow (y_p, y_q)$ be a quadratic residue modulo N and let x_p, x_q be square roots of y_p and y_q modulo p and q , respectively. Then the four square roots of y are given by the elements in $\mathbb{Z}_N^*$ corresponding to

$$(x_p, x_q), \quad (-x_p, x_q), \quad (x_p, -x_q), \quad (-x_p, -x_q). \quad (13.5)$$

Each of these is a square root of y since

$$\begin{aligned} (\pm x_p, \pm x_q)^2 &= \left([(\pm x_p)^2 \bmod p], [(\pm x_q)^2 \bmod q] \right) \\ &= ([x_p^2 \bmod p], [x_q^2 \bmod q]) = (y_p, y_q) \leftrightarrow y \end{aligned}$$

(where again the notation $(\cdot, \cdot)^2$ refers to squaring in the group $\mathbb{Z}_p \times \mathbb{Z}_q$). The Chinese remainder theorem guarantees that the four elements in Equation (13.5) correspond to *distinct* elements of $\mathbb{Z}_N^*$, since x_p and $-x_p$ are unique modulo p (and similarly for x_q and $-x_q$ modulo q).

Example 13.22

Consider $\mathbb{Z}_{15}^*$ (the correspondence given by the Chinese remainder theorem is tabulated in Example 8.25). Element 4 is a quadratic residue modulo 15 with square root 2. Since $2 \leftrightarrow (2, 2)$, the other square roots of 4 are given by

- $(2, [-2 \bmod 3]) = (2, 1) \leftrightarrow 7$;
- $([-2 \bmod 5], 2) = (3, 2) \leftrightarrow 8$; and
- $([-2 \bmod 5], [-2 \bmod 3]) = (3, 1) \leftrightarrow 13$.

One can verify that $7^2 = 8^2 = 13^2 = 4 \bmod 15$. ◇

Let $\mathcal{QR}_N$ denote the set of quadratic residues modulo N . Since squaring modulo N is a four-to-one function, we see that exactly $1/4$ of the elements of $\mathbb{Z}_N^*$ are quadratic residues. Alternately, we could note that since $y \in \mathbb{Z}_N^*$ is a quadratic residue if and only if y_p, y_q are quadratic residues, there is a one-to-one correspondence between $\mathcal{QR}_N$ and $\mathcal{QR}_p \times \mathcal{QR}_q$. Thus, the fraction of quadratic residues modulo N is

$$\frac{|\mathcal{QR}_N|}{|\mathbb{Z}_N^*|} = \frac{|\mathcal{QR}_p| \cdot |\mathcal{QR}_q|}{|\mathbb{Z}_p^*|} = \frac{\frac{p-1}{2} \cdot \frac{q-1}{2}}{(p-1)(q-1)} = \frac{1}{4},$$

in agreement with the above.

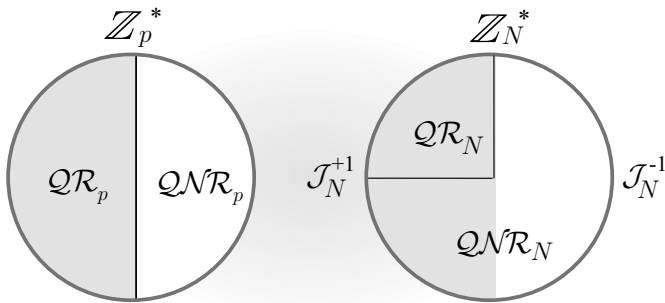


FIGURE 13.1: The structure of $\mathbb{Z}_p^*$ and $\mathbb{Z}_N^*$.

In the previous section, we defined the Jacobi symbol $\mathcal{J}_p(x)$ for $p > 2$ prime. We extend the definition to the case of N a product of distinct, odd primes p and q as follows. For any x relatively prime to $N = pq$,

$$\begin{aligned}\mathcal{J}_N(x) &\stackrel{\text{def}}{=} \mathcal{J}_p(x) \cdot \mathcal{J}_q(x) \\ &= \mathcal{J}_p([x \bmod p]) \cdot \mathcal{J}_q([x \bmod q]).\end{aligned}$$

We define $\mathcal{J}_N^{+1}$ as the set of elements in $\mathbb{Z}_N^*$ having Jacobi symbol $+1$, and define $\mathcal{J}_N^{-1}$ analogously.

We know from Proposition 13.21 that if x is a quadratic residue modulo N , then $[x \bmod p]$ and $[x \bmod q]$ are quadratic residues modulo p and q , respectively; that is, $\mathcal{J}_p(x) = \mathcal{J}_q(x) = +1$. So $\mathcal{J}_N(x) = +1$ and we see that:

If x is a quadratic residue modulo N , then $\mathcal{J}_N(x) = +1$.

However, $\mathcal{J}_N(x) = +1$ can also occur when $\mathcal{J}_p(x) = \mathcal{J}_q(x) = -1$, that is, when *both* $[x \bmod p]$ and $[x \bmod q]$ are *not* quadratic residues modulo p and q (and so x is not a quadratic residue modulo N). This turns out to be useful for the Goldwasser–Micali encryption scheme, and we therefore introduce the notation $\mathcal{QNR}_N^{+1}$ for the set of elements of this type. That is,

$$\mathcal{QNR}_N^{+1} \stackrel{\text{def}}{=} \left\{ x \in \mathbb{Z}_N^* \mid \begin{array}{l} x \text{ is not a quadratic residue modulo } N, \\ \text{but } \mathcal{J}_N(x) = +1 \end{array} \right\}.$$

It is now easy to prove the following (see Figure 13.1):

PROPOSITION 13.23 *Let $N = pq$ with p, q distinct, odd primes. Then:*

1. *Exactly half the elements of $\mathbb{Z}_N^*$ are in $\mathcal{J}_N^{+1}$.*
2. *$\mathcal{QR}_N$ is contained in $\mathcal{J}_N^{+1}$.*
3. *Exactly half the elements of $\mathcal{J}_N^{+1}$ are in $\mathcal{QR}_N$ (the other half are in $\mathcal{QNR}_N^{+1}$).*

PROOF We know that $\mathcal{J}_N(x) = +1$ if either $\mathcal{J}_p(x) = \mathcal{J}_q(x) = +1$ or $\mathcal{J}_p(x) = \mathcal{J}_q(x) = -1$. We also know (from the previous section) that exactly half the elements of $\mathbb{Z}_p^*$ have Jacobi symbol $+1$, and half have Jacobi symbol -1 (and similarly for $\mathbb{Z}_q^*$). Defining $\mathcal{J}_p^{+1}$, $\mathcal{J}_p^{-1}$, $\mathcal{J}_q^{+1}$, and $\mathcal{J}_q^{-1}$ in the natural way, we thus have

$$\begin{aligned} |\mathcal{J}_N^{+1}| &= |\mathcal{J}_p^{+1} \times \mathcal{J}_q^{+1}| + |\mathcal{J}_p^{-1} \times \mathcal{J}_q^{-1}| \\ &= |\mathcal{J}_p^{+1}| \cdot |\mathcal{J}_q^{+1}| + |\mathcal{J}_p^{-1}| \cdot |\mathcal{J}_q^{-1}| \\ &= \frac{(p-1)}{2} \frac{(q-1)}{2} + \frac{(p-1)}{2} \frac{(q-1)}{2} = \frac{\phi(N)}{2}. \end{aligned}$$

So $|\mathcal{J}_N^{+1}| = |\mathbb{Z}_N^*|/2$, proving that half the elements of $\mathbb{Z}_N^*$ are in $\mathcal{J}_N^{+1}$.

We have noted earlier that all quadratic residues modulo N have Jacobi symbol $+1$, showing that $\mathcal{QR}_N \subseteq \mathcal{J}_N^{+1}$.

Since $x \in \mathcal{QR}_N$ if and only if $\mathcal{J}_p(x) = \mathcal{J}_q(x) = +1$, we have

$$|\mathcal{QR}_N| = |\mathcal{J}_p^{+1} \times \mathcal{J}_q^{+1}| = \frac{(p-1)}{2} \frac{(q-1)}{2} = \frac{\phi(N)}{4},$$

and so $|\mathcal{QR}_N| = |\mathcal{J}_N^{+1}|/2$. Since $\mathcal{QR}_N$ is a subset of $\mathcal{J}_N^{+1}$, this proves that half the elements of $\mathcal{J}_N^{+1}$ are in $\mathcal{QR}_N$. ■

The next two results are analogues of Proposition 13.19 and Corollary 13.20.

PROPOSITION 13.24 *Let $N = pq$ be a product of distinct, odd primes, and $x, y \in \mathbb{Z}_N^*$. Then $\mathcal{J}_N(xy) = \mathcal{J}_N(x) \cdot \mathcal{J}_N(y)$.*

PROOF Using the definition of $\mathcal{J}_N(\cdot)$ and Proposition 13.19:

$$\begin{aligned} \mathcal{J}_N(xy) &= \mathcal{J}_p(xy) \cdot \mathcal{J}_q(xy) = \mathcal{J}_p(x) \cdot \mathcal{J}_p(y) \cdot \mathcal{J}_q(x) \cdot \mathcal{J}_q(y) \\ &= \mathcal{J}_p(x) \cdot \mathcal{J}_q(x) \cdot \mathcal{J}_p(y) \cdot \mathcal{J}_q(y) = \mathcal{J}_N(x) \cdot \mathcal{J}_N(y). \end{aligned}$$
■

COROLLARY 13.25 *Let $N = pq$ be a product of distinct, odd primes, and say $x, x' \in \mathcal{QR}_N$ and $y, y' \in \mathcal{QNR}_N^{+1}$. Then:*

1. $[xx' \bmod N] \in \mathcal{QR}_N$.
2. $[yy' \bmod N] \in \mathcal{QR}_N$.
3. $[xy \bmod N] \in \mathcal{QNR}_N^{+1}$.

PROOF We prove the final claim; proofs of the others are similar. Since $x \in \mathcal{QR}_N$, we have $\mathcal{J}_p(x) = \mathcal{J}_q(x) = +1$. Since $y \in \mathcal{QNR}_N^{+1}$, we have $\mathcal{J}_p(y) = \mathcal{J}_q(y) = -1$. Using Proposition 13.19,

$$\mathcal{J}_p(xy) = \mathcal{J}_p(x) \cdot \mathcal{J}_p(y) = -1 \quad \text{and} \quad \mathcal{J}_q(xy) = \mathcal{J}_q(x) \cdot \mathcal{J}_q(y) = -1,$$

and so $\mathcal{J}_N(xy) = +1$. But xy is not a quadratic residue modulo N , since $\mathcal{J}_p(xy) = -1$ and so $[xy \bmod p]$ is not a quadratic residue modulo p . We conclude that $xy \in \mathcal{QNR}_N^{+1}$. ■

In contrast to Corollary 13.20, it is *not* true that $y, y' \in \mathcal{QNR}_N$ implies $yy' \in \mathcal{QR}_N$. (Instead, as indicated in the corollary, this is only guaranteed if $y, y' \in \mathcal{QNR}_N^{+1}$.) For example, we could have $\mathcal{J}_p(y) = +1$, $\mathcal{J}_q(y) = -1$ and $\mathcal{J}_p(y') = -1$, $\mathcal{J}_q(y') = +1$, so $\mathcal{J}_p(yy') = \mathcal{J}_q(yy') = -1$ and yy' is not a quadratic residue even though $\mathcal{J}_N(yy') = +1$.

13.4.3 The Quadratic Residuosity Assumption

In Section 13.4.1, we showed an efficient algorithm for deciding whether an input x is a quadratic residue modulo a prime p . Can we adapt the algorithm to work modulo a composite number N ? Proposition 13.21 gives an easy solution to this problem *provided the factorization of N is known*. See Algorithm 13.26.

ALGORITHM 13.26

Deciding quadratic residuosity modulo a composite of known factorization

Input: Composite $N = pq$; the factors p and q ; element $x \in \mathbb{Z}_N^*$

Output: A decision as to whether $x \in \mathcal{QR}_N$

compute $\mathcal{J}_p(x)$ and $\mathcal{J}_q(x)$

if $\mathcal{J}_p(x) = \mathcal{J}_q(x) = +1$ **return** “quadratic residue”

else return “quadratic non-residue”

(As always, we assume the factors of N are distinct odd primes.) A simple modification of the above algorithm allows for computing $\mathcal{J}_N(x)$ when the factorization of N is known.

When the factorization of N is *unknown*, however, there is no known polynomial-time algorithm for deciding whether a given x is a quadratic residue modulo N or not. Somewhat surprisingly, a polynomial-time algorithm *is* known for computing $\mathcal{J}_N(x)$ without the factorization of N . (Although the algorithm itself is not that complicated, its proof of correctness is beyond the scope of this book and we therefore do not present the algorithm at all. The interested reader can refer to the references listed at the

end of this chapter.) This leads to a partial test of quadratic residuosity: if, for a given input x , it holds that $\mathcal{J}_N(x) = -1$, then x cannot possibly be a quadratic residue. (See Proposition 13.23.) This test says nothing when $\mathcal{J}_N(x) = +1$, and there is *no* known polynomial-time algorithm for deciding quadratic residuosity in that case (that does better than random guessing).

We now formalize the assumption that this problem is hard. Let **GenModulus** be a polynomial-time algorithm that, on input 1^n , outputs (N, p, q) where $N = pq$, and p and q are n -bit primes except with probability negligible in n .

DEFINITION 13.27 *We say deciding quadratic residuosity is hard relative to **GenModulus** if for all probabilistic polynomial-time algorithms D there exists a negligible function negl such that*

$$\left| \Pr[D(N, \text{qr}) = 1] - \Pr[D(N, \text{qnr}) = 1] \right| \leq \text{negl}(n),$$

where in each case the probabilities are taken over the experiment in which **GenModulus**(1^n) is run to give (N, p, q) , qr is chosen uniformly from $\mathcal{QR}_N$, and qnr is chosen uniformly from $\mathcal{QNR}_N^{+1}$.

It is crucial in the above that qnr is chosen from $\mathcal{QNR}_N^{+1}$ rather than $\mathcal{QNR}_N$; if qnr were chosen from $\mathcal{QNR}_N$ then with probability $2/3$ it would be the case that $\mathcal{J}_N(x) = -1$ and so distinguishing qnr from a uniform quadratic residue would be easy. (Recall that $\mathcal{J}_N(x)$ can be computed efficiently even without the factorization of N .)

The quadratic residuosity assumption is simply the assumption that there exists a **GenModulus** relative to which deciding quadratic residuosity is hard. It is easy to see that if deciding quadratic residuosity is hard relative to **GenModulus**, then factoring must be hard relative to **GenModulus** as well.

13.4.4 The Goldwasser–Micali Encryption Scheme

The preceding section immediately suggests a public-key encryption scheme for single-bit messages based on the quadratic residuosity assumption:

- The public key is a modulus N , and the private key is its factorization.
- To encrypt a ‘0,’ send a uniform quadratic residue; to encrypt a ‘1,’ send a uniform quadratic non-residue with Jacobi symbol $+1$.
- The receiver can decrypt a ciphertext c with its private key by using the factorization of N to decide whether c is a quadratic residue or not.

CPA-security of this scheme follows almost trivially from the hardness of the quadratic residuosity problem as formalized in Definition 13.27.

One thing missing from the above description is a specification of how the sender, who does not know the factorization of N , can choose a uniform

element of $\mathcal{QR}_N$ (to encrypt a 0) or a uniform element of $\mathcal{QNR}_N^{+1}$ (to encrypt a 1). The first of these is easy, while the second requires some ingenuity.

Choosing a uniform quadratic residue. Choosing a uniform element $y \in \mathcal{QR}_N$ is easy: simply pick a uniform $x \in \mathbb{Z}_N^*$ (see Appendix B.2.5) and set $y := x^2 \bmod N$. Clearly $y \in \mathcal{QR}_N$. The fact that y is uniformly distributed in $\mathcal{QR}_N$ follows from the facts that squaring modulo N is a 4-to-1 function (see Section 13.4.2) and that x is chosen uniformly from $\mathbb{Z}_N^*$. In more detail, fix any $\hat{y} \in \mathcal{QR}_N$ and let us compute the probability that $y = \hat{y}$ after the above procedure. Denote the four square roots of $\hat{y}$ by $\pm\hat{x}, \pm\hat{x}'$. Then:

$$\begin{aligned} \Pr[y = \hat{y}] &= \Pr[x \text{ is a square root of } \hat{y}] \\ &= \Pr[x \in \{\pm\hat{x}, \pm\hat{x}'\}] \\ &= \frac{4}{|\mathbb{Z}_N^*|} = \frac{1}{|\mathcal{QR}_N|}. \end{aligned}$$

Since the above holds for every $\hat{y} \in \mathcal{QR}_N$, we see that y is distributed uniformly in $\mathcal{QR}_N$.

Choosing a uniform element of $\mathcal{QNR}_N^{+1}$. In general, it is not known how to choose a uniform element of $\mathcal{QNR}_N^{+1}$ if the factorization of N is unknown. What saves us in the present context is that *the receiver can help* by including certain information in the public key. Specifically, we modify the scheme so that the receiver additionally chooses a uniform $z \in \mathcal{QNR}_N^{+1}$ and includes z as part of its public key. (This is easy for the receiver to do since it knows the factorization of N ; see Exercise 13.6.) The sender can choose a uniform element $y \in \mathcal{QNR}_N^{+1}$ by choosing a uniform $x \in \mathbb{Z}_N^*$ (as above) and setting $y := [z \cdot x^2 \bmod N]$. It follows from Corollary 13.25 that $y \in \mathcal{QNR}_N^{+1}$. We leave it as an exercise to show that y is uniformly distributed in $\mathcal{QNR}_N^{+1}$; we do not use this fact directly in the proof of security given below.

We give a complete description of the Goldwasser–Micali encryption scheme, implementing the above ideas, in Construction 13.28.

THEOREM 13.29 *If the quadratic residuosity problem is hard relative to GenModulus, then the Goldwasser–Micali encryption scheme is CPA-secure.*

PROOF Let Π denote the Goldwasser–Micali encryption scheme. We prove that Π has indistinguishable encryptions in the presence of an eavesdropper; by Theorem 11.6 this implies that it is CPA-secure.

Let $\mathcal{A}$ be an arbitrary probabilistic polynomial-time adversary. Consider the following PPT adversary D that attempts to solve the quadratic residuosity problem relative to GenModulus:

Algorithm D :

The algorithm is given N and z as input, and its goal is to determine if $z \in \mathcal{QR}_N$ or $z \in \mathcal{QNR}_N^{+1}$.

CONSTRUCTION 13.28

Let **GenModulus** be as usual. Construct a public-key encryption scheme as follows:

- **Gen**: on input 1^n , run **GenModulus**(1^n) to obtain (N, p, q) , and choose a uniform $z \in \mathcal{QNR}_N^{+1}$. The public key is $pk = \langle N, z \rangle$ and the private key is $sk = \langle p, q \rangle$.
- **Enc**: on input a public key $pk = \langle N, z \rangle$ and a message $m \in \{0, 1\}$, choose a uniform $x \in \mathbb{Z}_N^*$ and output the ciphertext

$$c := [z^m \cdot x^2 \bmod N].$$

- **Dec**: on input a private key sk and a ciphertext c , determine whether c is a quadratic residue modulo N using, e.g., Algorithm 13.26. If yes, output 0; otherwise, output 1.

The Goldwasser–Micali encryption scheme.

- Set $pk = \langle N, z \rangle$ and run $\mathcal{A}(pk)$ to obtain two single-bit messages m_0, m_1 .
- Choose a uniform bit b and a uniform $x \in \mathbb{Z}_N^*$, and then set $c := [z^{m_b} \cdot x^2 \bmod N]$.
- Give the ciphertext c to $\mathcal{A}$, who in turn outputs a bit b' . If $b' = b$, output 1; otherwise, output 0.

Let us analyze the behavior of D . There are two cases to consider:

Case 1: Say the input to D was generated by running **GenModulus**(1^n) to obtain (N, p, q) and then choosing a uniform $z \in \mathcal{QNR}_N^{+1}$. Then D runs $\mathcal{A}$ on a public key constructed exactly as in Π , and we see that in this case the view of $\mathcal{A}$ when run as a subroutine by D is distributed identically to $\mathcal{A}$'s view in experiment $\text{PubK}_{\mathcal{A}, \Pi}^{\text{eav}}(n)$. Since D outputs 1 exactly when the output b' of $\mathcal{A}$ is equal to b , we have

$$\Pr[D(N, \text{qnr}) = 1] = \Pr[\text{PubK}_{\mathcal{A}, \Pi}^{\text{eav}}(n) = 1],$$

where qnr represents a uniform element of $\mathcal{QNR}_N^{+1}$ as in Definition 13.27.

Case 2: Say the input to D was generated by running **GenModulus**(1^n) to obtain (N, p, q) and then choosing a uniform $z \in \mathcal{QR}_N$. We claim that the view of $\mathcal{A}$ in this case is *independent* of the bit b . To see this, note that the ciphertext c given to $\mathcal{A}$ is a uniform quadratic residue regardless of whether a 0 or a 1 is encrypted:

- When a 0 is encrypted, $c = [x^2 \bmod N]$ for a uniform $x \in \mathbb{Z}_N^*$, and so c is a uniform quadratic residue.
- When a 1 is encrypted, $c = [z \cdot x^2 \bmod N]$ for a uniform $x \in \mathbb{Z}_N^*$. Let $\hat{x} \stackrel{\text{def}}{=} [x^2 \bmod N]$, and note that $\hat{x}$ is a uniformly distributed element

of the group $\mathcal{QR}_N$. Since $z \in \mathcal{QR}_N$, we can apply Lemma 11.15 to conclude that c is uniformly distributed in $\mathcal{QR}_N$ as well.

Since $\mathcal{A}$'s view is independent of b , the probability that $b' = b$ in this case is exactly $\frac{1}{2}$. That is,

$$\Pr[D(N, \mathbf{qr}) = 1] = \frac{1}{2},$$

where $\mathbf{qr}$ represents a uniform element of $\mathcal{QR}_N$ as in Definition 13.27.

Thus,

$$\left| \Pr[D(N, \mathbf{qr}) = 1] - \Pr[D(N, \mathbf{qnr}) = 1] \right| = \left| \Pr[\text{PubK}_{\mathcal{A}, \Pi}^{\text{eav}}(n) = 1] - \frac{1}{2} \right|.$$

By the assumption that the quadratic residuosity problem is hard relative to GenModulus , there is a negligible function negl such that

$$|\varepsilon(n) - \frac{1}{2}| \leq \text{negl}(n);$$

thus, $\varepsilon(n) \leq \frac{1}{2} + \text{negl}(n)$. This completes the proof. ■

13.5 The Rabin Encryption Scheme

As mentioned at the beginning of this chapter, the Rabin encryption scheme is attractive because its security is *equivalent* to the assumption that factoring is hard. An analogous result is *not* known for RSA-based encryption, and the RSA problem may potentially be easier than factoring. (The same is true of the Goldwasser–Micali encryption scheme, and it is possible that deciding quadratic residuosity modulo N is easier than factoring N .)

Interestingly, the Rabin encryption scheme is (superficially, at least) very similar to the RSA encryption scheme yet has the advantage of being based on a potentially weaker assumption. The fact that RSA is more widely used than the former seems to be due more to historical factors than technical ones; we discuss this further at the end of this section.

We begin with some preliminaries about computing modular square roots. We then introduce a trapdoor permutation that can be based directly on the assumption that factoring is hard. The Rabin encryption scheme (or, at least, one instantiation of it) is then obtained by applying the results from Section 13.1. Throughout this section, we continue to let p and q denote odd primes, and let $N = pq$ denote a product of two distinct, odd primes.

13.5.1 Computing Modular Square Roots

The Rabin encryption scheme requires the receiver to compute modular square roots, and so in this section we explore the algorithmic complexity of

this problem. We first show an efficient algorithm for computing square roots modulo a prime p , and then extend this algorithm to enable computation of square roots modulo a composite N of *known* factorization. The reader willing to accept the existence of these algorithms on faith can skip to the following section, where we show that computing square roots modulo a composite N with *unknown* factorization is equivalent to factoring N .

Let p be an odd prime. Computing square roots modulo p is relatively simple when $p = 3 \bmod 4$, but much more involved when $p = 1 \bmod 4$. (The easier case is all we need for the Rabin encryption scheme as presented in Section 13.5.3; we include the second case for completeness.) In both cases, we show how to compute one of the square roots of a quadratic residue $a \in \mathbb{Z}_p^*$. Note that if x is one of the square roots of a , then $[-x \bmod p]$ is the other.

We tackle the easier case first. Say $p = 3 \bmod 4$, meaning we can write $p = 4i + 3$ for some integer i . Since $a \in \mathbb{Z}_p^*$ is a quadratic residue, we have $\mathcal{J}_p(a) = 1 = a^{\frac{p-1}{2}} \bmod p$ (see Proposition 13.17). Multiplying both sides by a we obtain

$$a = a^{\frac{p-1}{2}+1} = a^{2i+2} = (a^{i+1})^2 \bmod p,$$

and so $a^{i+1} = a^{\frac{p+1}{4}} \bmod p$ is a square root of a . That is, we obtain a square root of a modulo p by simply computing $x := [a^{\frac{p+1}{4}} \bmod p]$.

It is crucial above that $(p+1)/2$ is *even* because this ensures that $(p+1)/4$ is an *integer* (this is necessary in order for $a^{\frac{p+1}{4}} \bmod p$ to be well-defined; recall that the exponent must be an integer). This approach does not succeed when $p = 1 \bmod 4$, in which case $p+1$ is an integer that is *not* divisible by 4.

When $p = 1 \bmod 4$ we proceed slightly differently. Motivated by the above approach, we might hope to find an odd integer r for which it holds that $a^r = 1 \bmod p$. Then, as above, $a^{r+1} = a \bmod p$ and $a^{\frac{r+1}{2}} \bmod p$ would be a square root of a with $(r+1)/2$ an integer. Although we will not be able to do this, we *can* do something just as good: we will find an odd integer r along with an element $b \in \mathbb{Z}_p^*$ and an *even* integer r' such that

$$a^r \cdot b^{r'} = 1 \bmod p.$$

Then $a^{r+1} \cdot b^{r'} = a \bmod p$ and $a^{\frac{r+1}{2}} \cdot b^{\frac{r'}{2}} \bmod p$ is a square root of a (with the exponents $(r+1)/2$ and $r'/2$ being integers).

We now describe the general approach to finding r, b , and r' with the stated properties. Let $\frac{p-1}{2} = 2^\ell \cdot m$ where ℓ, m are integers with $\ell \geq 1$ and m odd.⁵ Since a is a quadratic residue, we know that

$$a^{2^\ell m} = a^{\frac{p-1}{2}} = 1 \bmod p. \quad (13.6)$$

This means that $a^{2^{\ell-1}m} = a^{2^{\ell-1}m} \bmod p$ is a square root of 1. The square roots of 1 modulo p are $\pm 1 \bmod p$, so $a^{2^{\ell-1}m} = \pm 1 \bmod p$. If $a^{2^{\ell-1}m} = 1 \bmod p$, we

⁵The integers ℓ and m can be computed easily by taking out factors of 2 from $(p-1)/2$.

are in the same situation as in Equation (13.6) except that the exponent of a is now divisible by a smaller power of 2. This is progress in the right direction: if we can get to the point where the exponent of a is not divisible by *any* power of 2 (as would be the case here if $\ell = 1$), then the exponent of a is *odd* and we can compute a square root as discussed earlier. We give an example, and discuss in a moment how to deal with the case when $a^{2^{\ell-1}m} = -1 \bmod p$.

Example 13.30

Take $p = 29$ and $a = 7$. Since 7 is a quadratic residue modulo 29, we have $7^{14} \bmod 29 = 1$ and we know that $7^7 \bmod 29$ is a square root of 1. In fact,

$$7^7 = 1 \bmod 29,$$

and the exponent 7 is odd. So $7^{(7+1)/2} = 7^4 = 23 \bmod 29$ is a square root of 7 modulo 29. $\diamond$

To summarize the algorithm so far: we begin with $a^{2^\ell m} = 1 \bmod p$ and we pull out factors of 2 from the exponent until one of two things happen: either $a^m = 1 \bmod p$, or $a^{2^{\ell'} m} = -1 \bmod p$ for some $\ell' < \ell$. In the first case, since m is odd we can immediately compute a square root of a as in Example 13.30. In the second case, we will “restore” the +1 on the right-hand side of the equation by multiplying each side of the equation by $-1 \bmod p$. However, as motivated at the beginning of this discussion, we want to achieve this by multiplying the left-hand side of the equation by some element b raised to an *even* power. If we have available a quadratic *non-residue* $b \in \mathbb{Z}_p^*$, this is easy: since $b^{2^\ell m} = b^{\frac{p-1}{2}} = -1 \bmod p$, we have

$$a^{2^{\ell'} m} \cdot b^{2^\ell m} = (-1)(-1) = +1 \bmod p.$$

With this we can proceed as before, taking a square root of the left-hand side to reduce the largest power of 2 dividing the exponent of a , and multiplying by $b^{2^\ell m}$ (as needed) so the right-hand side is always +1. Observe that the exponent of b is always divisible by a larger power of 2 than the exponent of a (and so we can indeed take square roots by dividing by 2 in both exponents). We continue performing these steps until the exponent of a is odd, and can then compute a square root of a as described earlier. Pseudocode for this algorithm, which gives another way of viewing what is going on, is given below in Algorithm 13.31. It can be verified that the algorithm runs in polynomial time given a quadratic non-residue b since the number of iterations of the inner loop is $\ell = \mathcal{O}(\log p)$.

One point we have not yet addressed is how to find b in the first place. In fact, no *deterministic* polynomial-time algorithm for finding a quadratic non-residue modulo p is known. Fortunately, it is easy to find a quadratic non-residue probabilistically: simply choose uniform elements of $\mathbb{Z}_p^*$ until a

ALGORITHM 13.31

Computing square roots modulo a prime

Input: Prime p ; quadratic residue $a \in \mathbb{Z}_p^*$

Output: A square root of a

case $p = 3 \bmod 4$:

return $[a^{\frac{p+1}{4}} \bmod p]$

case $p = 1 \bmod 4$:

let b be a quadratic non-residue modulo p

compute ℓ and m odd with $2^\ell \cdot m = \frac{p-1}{2}$

$r := 2^\ell \cdot m, r' := 0$

for $i = \ell$ to 1 {

 // maintain the invariant $a^r \cdot b^{r'} = 1 \bmod p$

$r := r/2, r' := r'/2$

if $a^r \cdot b^{r'} = -1 \bmod p$

$r' := r' + 2^\ell \cdot m$

 }

 // now $r = m, r'$ is even, and $a^r \cdot b^{r'} = 1 \bmod p$

return $[a^{\frac{r+1}{2}} \cdot b^{\frac{r'}{2}} \bmod p]$

quadratic non-residue is found. This works because exactly half the elements of $\mathbb{Z}_p^*$ are quadratic non-residues, and because a polynomial-time algorithm for deciding quadratic residuosity modulo a prime is known (see Section 13.4.1 for proofs of both these statements). This means that the algorithm we have shown is actually randomized when $p = 1 \bmod 4$; a deterministic polynomial-time algorithm for computing square roots in this case is not known.

Example 13.32

Here we consider the “worst case,” when taking a square root always gives -1 . Let $a \in \mathbb{Z}_p^*$ be the element whose square root we are trying to compute; let $b \in \mathbb{Z}_p^*$ be a quadratic non-residue; and let $\frac{p-1}{2} = 2^3 \cdot m$ where m is odd.

In the first step, we have $a^{2^3 m} = 1 \bmod p$. Since $a^{2^3 m} = \left(a^{2^2 m}\right)^2$ and the square roots of 1 are ± 1 , this means that $a^{2^2 m} = \pm 1 \bmod p$; assuming the worst case, $a^{2^2 m} = -1 \bmod p$. So, we multiply by $b^{\frac{p-1}{2}} = b^{2^3 m} = -1 \bmod p$ to obtain

$$a^{2^2 m} \cdot b^{2^3 m} = 1 \bmod p.$$

In the second step, we observe that $a^{2^m} \cdot b^{2^2 m}$ is a square root of 1; again assuming the worst case, we thus have $a^{2^m} \cdot b^{2^2 m} = -1 \bmod p$. Multiplying by $b^{2^3 m}$ to “correct” this gives

$$a^{2^m} \cdot b^{2^2 m} \cdot b^{2^3 m} = 1 \bmod p.$$

In the third step, taking square roots and assuming the worst case (as

above) we obtain $a^m \cdot b^{2m} \cdot b^{2^2m} = -1 \pmod p$; multiplying by the “correction factor” b^{2^3m} we get

$$a^m \cdot b^{2m} \cdot b^{2^2m} \cdot b^{2^3m} = 1 \pmod p.$$

We are now where we want to be. To conclude the algorithm, multiply both sides by a to obtain

$$a^{m+1} \cdot b^{2m+2^2m+2^3m} = a \pmod p.$$

Since m is odd, $(m+1)/2$ is an integer and $a^{\frac{m+1}{2}} \cdot b^{m+2m+2^2m} \pmod p$ is a square root of a . $\diamond$

Example 13.33

Here we work out a concrete example. Let $p = 17$, $a = 2$, and $b = 3$. Note that here $(p-1)/2 = 2^3$ and $m = 1$.

We begin with $2^{2^3} = 1 \pmod{17}$. So 2^{2^2} should be equal to $\pm 1 \pmod{17}$; by calculation, $2^{2^2} = -1 \pmod{17}$. Multiplying by 3^{2^3} gives $2^{2^2} \cdot 3^{2^3} = 1 \pmod{17}$.

Continuing, we know that $2^2 \cdot 3^{2^2}$ is a square root of 1 and so must be equal to $\pm 1 \pmod{17}$; calculation gives $2^2 \cdot 3^{2^2} = 1 \pmod{17}$. So no correction term is needed here.

Halving the exponents again we find that $2 \cdot 3^2 = 1 \pmod{17}$. We are now almost done: multiplying both sides by 2 gives $2^2 \cdot 3^2 = 2 \pmod{17}$, and so $2 \cdot 3 = 6 \pmod{17}$ is a square root of 2. $\diamond$

Computing Square Roots Modulo N

It is not hard to see that the algorithm we have shown for computing square roots modulo a prime extends easily to the case of computing square roots modulo a composite $N = pq$ of known factorization. Specifically, let $a \in \mathbb{Z}_N^*$ be a quadratic residue with $a \leftrightarrow (a_p, a_q)$ via the Chinese remainder theorem. Computing the square roots x_p, x_q of a_p, a_q modulo p and q , respectively, gives a square root (x_p, x_q) of a (see Section 13.4.2). Given x_p and x_q , the representation x corresponding to (x_p, x_q) can be recovered as discussed in Section 8.1.5. That is, to compute a square root of a modulo an integer $N = pq$ of known factorization:

- Compute $a_p := [a \pmod p]$ and $a_q := [a \pmod q]$.
- Using Algorithm 13.31, compute a square root x_p of a_p modulo p and a square root x_q of a_q modulo q .
- Convert from the representation $(x_p, x_q) \in \mathbb{Z}_p^* \times \mathbb{Z}_q^*$ to $x \in \mathbb{Z}_N^*$ with $x \leftrightarrow (x_p, x_q)$. Output x , which is a square root of a modulo N .

It is easy to modify the algorithm so that it returns all four square roots of a .

13.5.2 A Trapdoor Permutation Based on Factoring

We have seen that computing square roots modulo N can be carried out in polynomial time if the factorization of N is known. We show here that, in contrast, computing square roots modulo a composite N of *unknown* factorization is as hard as factoring N .

More formally, let **GenModulus** be a polynomial-time algorithm that, on input 1^n , outputs (N, p, q) where $N = pq$ and p and q are n -bit primes except with probability negligible in n . Consider the following experiment for a given algorithm $\mathcal{A}$ and parameter n :

The square-root computation experiment $\text{SQR}_{\mathcal{A}, \text{GenModulus}}(n)$:

1. Run **GenModulus** (1^n) to obtain output N, p, q .
2. Choose a uniform $y \in \mathcal{QR}_N$.
3. $\mathcal{A}$ is given (N, y) , and outputs $x \in \mathbb{Z}_N^*$.
4. The output of the experiment is defined to be 1 if $x^2 = y \bmod N$, and 0 otherwise.

DEFINITION 13.34 We say that computing square roots is hard relative to **GenModulus** if for all probabilistic polynomial-time algorithms $\mathcal{A}$ there exists a negligible function negl such that

$$\Pr[\text{SQR}_{\mathcal{A}, \text{GenModulus}}(n) = 1] \leq \text{negl}(n).$$

It is easy to see that if computing square roots is hard relative to **GenModulus** then factoring must be hard relative to **GenModulus** too: if moduli N output by **GenModulus** could be factored easily, then it would be easy to compute square roots modulo N by first factoring N and then applying the algorithm discussed in the previous section. Our goal now is to show the converse: that if factoring is *hard* relative to **GenModulus** then so is the problem of computing square roots. We emphasize again that an analogous result is not known for the RSA problem or the problem of deciding quadratic residuosity.

The key is the following lemma, which says that two “unrelated” square roots of any element in $\mathbb{Z}_N^*$ can be used to factor N .

LEMMA 13.35 Let $N = pq$ with p, q distinct, odd primes. Given $x, \hat{x}$ such that $x^2 = y = \hat{x}^2 \bmod N$ but $x \not\equiv \pm \hat{x} \bmod N$, it is possible to factor N in time polynomial in $\|N\|$.

PROOF We claim that either $\gcd(N, x + \hat{x})$ or $\gcd(N, x - \hat{x})$ is equal to one of the prime factors of N .⁶ Since gcd computations can be carried out in polynomial time (see Appendix B.1.2), this proves the lemma.

⁶In fact, both of these are equal to one of the prime factors of N .

If $x^2 = \hat{x}^2 \bmod N$ then

$$0 = x^2 - \hat{x}^2 = (x - \hat{x}) \cdot (x + \hat{x}) \bmod N,$$

and so $N \mid (x - \hat{x})(x + \hat{x})$. Then $p \mid (x - \hat{x})(x + \hat{x})$ and so p divides one of these terms. Say $p \mid (x + \hat{x})$ (the proof proceeds similarly if $p \mid (x - \hat{x})$). If $q \mid (x + \hat{x})$ then $N \mid (x + \hat{x})$, but this cannot be the case since $x \not\equiv -\hat{x} \bmod N$. So $q \nmid (x + \hat{x})$ and $\gcd(N, x + \hat{x}) = p$. ■

An alternative way of proving the above is to look at what happens in the Chinese remaindering representation. Say $x \leftrightarrow (x_p, x_q)$. Then, because x and $\hat{x}$ are square roots of the same value y , we know that $\hat{x}$ corresponds to either $(-x_p, x_q)$ or $(x_p, -x_q)$. (It cannot correspond to (x_p, x_q) or $(-x_p, -x_q)$ since the first corresponds to x while the second corresponds to $[-x \bmod N]$, and both possibilities are ruled out by the assumption of the lemma.) Say $\hat{x} \leftrightarrow (-x_p, x_q)$. Then

$$[x + \hat{x} \bmod N] \leftrightarrow (x_p, x_q) + (-x_p, x_q) = (0, [2x_q \bmod q]),$$

and we see that $x + \hat{x} \equiv 0 \bmod p$ while $x + \hat{x} \not\equiv 0 \bmod q$. It follows that $\gcd(N, x + \hat{x}) = p$, a factor of N .

We can now prove the main result of this section.

THEOREM 13.36 *If factoring is hard relative to **GenModulus**, then computing square roots is hard relative to **GenModulus**.*

PROOF Let $\mathcal{A}$ be a probabilistic polynomial-time algorithm computing square roots (as in Definition 13.34). Consider the following probabilistic polynomial-time algorithm $\mathcal{A}_{\text{fact}}$ for factoring moduli output by **GenModulus**:

Algorithm $\mathcal{A}_{\text{fact}}$:

The algorithm is given a modulus N as input.

- Choose a uniform $x \in \mathbb{Z}_N^*$ and compute $y := [x^2 \bmod N]$.
- Run $\mathcal{A}(N, y)$ to obtain output $\hat{x}$.
- If $\hat{x}^2 = y \bmod N$ and $\hat{x} \not\equiv \pm x \bmod N$, then factor N using Lemma 13.35.

By Lemma 13.35, we know that $\mathcal{A}_{\text{fact}}$ succeeds in factoring N exactly when $\hat{x} \not\equiv \pm x \bmod N$ and $\hat{x}^2 = y \bmod N$. That is,

$$\begin{aligned} & \Pr[\text{Factor}_{\mathcal{A}_{\text{fact}}, \text{GenModulus}}(n) = 1] \\ &= \Pr[\hat{x} \not\equiv \pm x \bmod N \wedge \hat{x}^2 = y \bmod N] \\ &= \Pr[\hat{x} \not\equiv \pm x \bmod N \mid \hat{x}^2 = y \bmod N] \cdot \Pr[\hat{x}^2 = y \bmod N], \quad (13.7) \end{aligned}$$

where the above probabilities all refer to experiment $\text{Factor}_{\mathcal{A}_{\text{fact}}, \text{GenModulus}}(n)$ (see Section 8.2.3 for a description of this experiment). In the experiment, the modulus N given as input to $\mathcal{A}_{\text{fact}}$ is generated by $\text{GenModulus}(1^n)$, and y is a uniform quadratic residue modulo N since x was chosen uniformly from $\mathbb{Z}_N^*$ (see Section 13.4.4). So the view of $\mathcal{A}$ when run as a subroutine by $\mathcal{A}_{\text{fact}}$ is distributed exactly as $\mathcal{A}$'s view in experiment $\text{SQR}_{\mathcal{A}, \text{GenModulus}}(n)$. Therefore,

$$\Pr[\hat{x}^2 = y \bmod N] = \Pr[\text{SQR}_{\mathcal{A}, \text{GenModulus}}(n) = 1]. \quad (13.8)$$

Conditioned on the value of the quadratic residue y used in experiment $\text{Factor}_{\mathcal{A}_{\text{fact}}, \text{GenModulus}}(n)$, the value x is equally likely to be any of the four possible square roots of y . This means that from the point of view of algorithm $\mathcal{A}$ (being run as a subroutine by $\mathcal{A}_{\text{fact}}$), x is equally likely to be each of the four square roots of y . This in turn means that, conditioned on $\mathcal{A}$ outputting *some* square root $\hat{x}$ of y , the probability that $\hat{x} = \pm x \bmod N$ is exactly $1/2$. (We stress that we do not make any assumption about how $\hat{x}$ is distributed among the square roots of y , and in particular are not assuming here that $\mathcal{A}$ outputs a uniform square root of y . Rather we are using the fact that x is uniformly distributed among the square roots of y .) That is,

$$\Pr[\hat{x} \neq \pm x \bmod N \mid \hat{x}^2 = y \bmod N] = \frac{1}{2}. \quad (13.9)$$

Combining Equations (13.7)–(13.9), we see that

$$\Pr[\text{Factor}_{\mathcal{A}_{\text{fact}}, \text{GenModulus}}(n) = 1] = \frac{1}{2} \cdot \Pr[\text{SQR}_{\mathcal{A}, \text{GenModulus}}(n) = 1].$$

Since factoring is hard relative to GenModulus , there is a negligible function negl such that

$$\Pr[\text{Factor}_{\mathcal{A}_{\text{fact}}, \text{GenModulus}}(n) = 1] \leq \text{negl}(n),$$

which implies $\Pr[\text{SQR}_{\mathcal{A}, \text{GenModulus}}(n) = 1] \leq 2 \cdot \text{negl}(n)$. Since $\mathcal{A}$ was arbitrary, this completes the proof. ■

The previous theorem leads directly to a family of one-way functions (see Definition 8.76) based on any GenModulus relative to which factoring is hard:

- Algorithm **Gen**, on input 1^n , runs $\text{GenModulus}(1^n)$ to obtain (N, p, q) and outputs $I = N$. The domain $\mathcal{D}_I$ is $\mathbb{Z}_N^*$ and the range $\mathcal{R}_I$ is $\mathcal{QR}_N$.
- Algorithm **Samp**, on input N , chooses a uniform element $x \in \mathbb{Z}_N^*$.
- Algorithm f , on input N and $x \in \mathbb{Z}_N^*$, outputs $[x^2 \bmod N]$.

The preceding theorem shows that this family is one-way if factoring is hard relative to GenModulus .

We can turn this into a family of one-way *permutations* by using moduli N of a special form and letting $\mathcal{D}_I$ be a subset of $\mathbb{Z}_N^*$. (See Exercise 13.19 for another way to make this a permutation.) Call $N = pq$ a *Blum integer* if p and q are distinct primes with $p = q = 3 \bmod 4$. The key to building a permutation is the following proposition.

PROPOSITION 13.37 *Let N be a Blum integer. Then every quadratic residue modulo N has exactly one square root that is also a quadratic residue.*

PROOF Say $N = pq$ with $p = q = 3 \bmod 4$. Using Proposition 13.17, we see that -1 is not a quadratic residue modulo p or q . This is because for $p = 3 \bmod 4$ it holds that $p = 4i + 3$ for some i and so

$$(-1)^{\frac{p-1}{2}} = (-1)^{2i+1} = -1 \bmod p$$

(because $2i + 1$ is odd). Now let $y \leftrightarrow (y_p, y_q)$ be an arbitrary quadratic residue modulo N with the four square roots

$$(x_p, x_q), \quad (-x_p, x_q), \quad (x_p, -x_q), \quad (-x_p, -x_q).$$

We claim that exactly one of these is a quadratic residue modulo N . To see this, assume $\mathcal{J}_p(x_p) = +1$ and $\mathcal{J}_q(x_q) = -1$ (the proof proceeds similarly in any other case). Using Proposition 13.19, we have

$$\mathcal{J}_q(-x_q) = \mathcal{J}_q(-1) \cdot \mathcal{J}_q(x_q) = +1,$$

and so $(x_p, -x_q)$ corresponds to a quadratic residue modulo N (using Proposition 13.21). Similarly, $\mathcal{J}_p(-x_p) = -1$ and so none of the other square roots of y are quadratic residues modulo N . ■

Expressed differently, the above proposition says that when N is a Blum integer, the function $f_N : \mathcal{QR}_N \rightarrow \mathcal{QR}_N$ given by $f_N(x) = [x^2 \bmod N]$ is a permutation over $\mathcal{QR}_N$. Modifying the sampling algorithm **Samp** above to choose a uniform $x \in \mathcal{QR}_N$ (which, as we have already seen, can be done easily by choosing uniform $r \in \mathbb{Z}_N^*$ and setting $x := [r^2 \bmod N]$) gives a family of one-way *permutations*. Finally, because square roots modulo N can be computed in polynomial time given the factorization of N , a straightforward modification yields a family of *trapdoor* permutations based on any **GenModulus** relative to which factoring is hard. This is sometimes called the *Rabin* family of trapdoor permutations. In summary:

THEOREM 13.38 *Let **GenModulus** be an algorithm that, on input 1^n , outputs (N, p, q) where $N = pq$ and p and q are distinct primes (except possibly with negligible probability) with $p = q = 3 \bmod 4$. If factoring is hard relative to **GenModulus**, then there exists a family of trapdoor permutations.*

13.5.3 The Rabin Encryption Scheme

We can apply the results of Section 13.1.2 to the Rabin trapdoor permutation to obtain a public-key encryption scheme whose security is based on factoring. To do this, we first need to identify a hard-core predicate for this trapdoor permutation. Although we could appeal to Theorem 13.3, which states that a suitable hard-core predicate always exists, it turns out that the least significant bit lsb is a hard-core predicate for the Rabin trapdoor permutation just as it is for the case of RSA (see Section 11.5.3). Using this as our hard-core predicate, we obtain the scheme of Construction 13.39.

CONSTRUCTION 13.39

Let GenModulus be a polynomial-time algorithm that, on input 1^n , outputs (N, p, q) where $N = pq$ and p and q are n -bit primes (except with probability negligible in n) with $p = q = 3 \pmod{4}$. Construct a public-key encryption scheme as follows:

- **Gen:** on input 1^n run $\text{GenModulus}(1^n)$ to obtain (N, p, q) . The public key is N , and the private key is $\langle p, q \rangle$.
- **Enc:** on input a public-key N and message $m \in \{0, 1\}$, choose a uniform $x \in \mathcal{QR}_N$ subject to the constraint that $\text{lsb}(x) = m$. Output the ciphertext $c := [x^2 \bmod N]$.
- **Dec:** on input a private key $\langle p, q \rangle$ and a ciphertext c , compute the unique $x \in \mathcal{QR}_N$ such that $x^2 = c \bmod N$, and output $\text{lsb}(x)$.

The Rabin encryption scheme.

Theorems 13.5 and 13.38 imply the following result.

THEOREM 13.40 *If factoring is hard relative to GenModulus , then Construction 13.39 is CPA-secure.*

Rabin Encryption vs. RSA Encryption

It is worthwhile to remark on the similarities and differences between the Rabin and RSA cryptosystems. (The discussion here applies to any cryptographic construction—not necessarily a public-key encryption scheme—based on the Rabin or RSA trapdoor permutations.)

The RSA and Rabin trapdoor permutations appear quite similar, with squaring in the case of Rabin corresponding to taking $e = 2$ in the case of RSA. (Of course, 2 is *not* relatively prime to $\phi(N)$ and so Rabin is not a special case of RSA.) In terms of the security offered by each construction, hardness of computing modular square roots is equivalent to hardness of factoring, whereas hardness of solving the RSA problem is not known to

be implied by the hardness of factoring. The Rabin trapdoor permutation is thus based on a potentially *weaker* assumption: it is theoretically possible that someone might develop an efficient algorithm for solving the RSA problem, yet computing square roots will remain hard. Or, someone may develop an algorithm that solves the RSA problem faster than known factoring algorithms. Lemma 13.35 ensures, however, that computing square roots modulo N can never be much faster than the best available algorithm for factoring N .

In terms of their efficiency, the RSA and Rabin permutations are essentially the same. Actually, if a large exponent e is used in the case of RSA then computing e th powers (as in RSA) is slightly slower than squaring (as in Rabin). On the other hand, a bit more care is required when working with the Rabin permutation since it is only a permutation over a *subset* of $\mathbb{Z}_N^*$, in contrast to RSA, which gives a permutation over all of $\mathbb{Z}_N^*$.

A “plain Rabin” encryption scheme, constructed in a manner analogous to plain RSA encryption, is vulnerable to a chosen-ciphertext attack that enables an adversary to learn the entire private key (see Exercise 13.17). Although plain RSA is not CCA-secure either, known chosen-ciphertext attacks on plain RSA are less damaging since they recover the message but not the private key. Perhaps the existence of such an attack on “plain Rabin” influenced cryptographers, early on, to reject the use of Rabin encryption entirely.

In summary, the RSA permutation is much more widely used in practice than the Rabin permutation, but in light of the above this appears to be due more to historical accident than to any compelling technical justification.

References and Additional Reading

The existence of public-key encryption based on arbitrary trapdoor permutations was shown by Yao [179], and the efficiency improvement discussed at the end of Section 13.1.2 is due to Blum and Goldwasser [36].

Childs [44] and Shoup [159] provide further coverage of the (computational) number theory used in this chapter. A good description of the algorithm for computing the Jacobi symbol modulo a composite of unknown factorization, along with a proof of correctness, is given in [57].

The Paillier encryption scheme was introduced in [136]. Shoup [159, Section 7.5] gives a characterization of $\mathbb{Z}_{N^e}^*$ for arbitrary integers N, e (and not just $N = pq$, $e = 2$ as done here).

The problem of deciding quadratic residuosity modulo a composite of unknown factorization goes back to Gauss [71] and is related to other (conjectured) hard number-theoretic problems. The Goldwasser–Micali encryption scheme [80], introduced in 1982, was the first public-key encryption scheme with a proof of security.

Rabin [145] showed that computing square roots modulo a composite is equivalent to factoring. The results of Section 13.5.2 are due to Blum [35]. Hard-core predicates for the Rabin trapdoor permutation are discussed in [8, 86, 7] and references therein.

Exercises

- 13.1 Construct and prove CPA-security for a KEM based on any trapdoor permutation by suitably generalizing Construction 11.34.
- 13.2 Show that the isomorphism of Proposition 13.6 can be efficiently inverted when the factorization of N is known.
- 13.3 Let $\Phi(N^2)$ denote the set $\{(a, 1) \mid a \in \mathbb{Z}_N\} \subset \mathbb{Z}_{N^2}^*$. Show that it is *not* hard to decide whether a given element $y \in \mathbb{Z}_{N^2}^*$ is in $\Phi(N^2)$.
- 13.4 Let $\mathbb{G}$ be an abelian group. Show that the set of quadratic residues in $\mathbb{G}$ forms a subgroup.
- 13.5 This question concerns the quadratic residues in the additive group $\mathbb{Z}_N$. (An element $y \in \mathbb{Z}_N$ is a quadratic residue if and only if there exists an $x \in \mathbb{Z}_N$ with $2x = y \bmod N$.)
 - (a) Let p be an odd prime. How many elements of $\mathbb{Z}_p$ are quadratic residues?
 - (b) Let $N = pq$ be a product of two odd primes p and q . How many elements of $\mathbb{Z}_N$ are quadratic residues?
 - (c) Let N be an even integer. How many elements of $\mathbb{Z}_N$ are quadratic residues?
- 13.6 Let $N = pq$ with p, q distinct, odd primes. Show a PPT algorithm for choosing a uniform element of $\mathcal{QR}_N^{+1}$ *when the factorization of N is known*. (Your algorithm can have failure probability negligible in $\|N\|$.)
- 13.7 Let $N = pq$ with p, q distinct, odd primes. Prove that if $x \in \mathcal{QR}_N$ then $[x^{-1} \bmod N] \in \mathcal{QR}_N$, and if $x \in \mathcal{QR}_N^{+1}$ then $[x^{-1} \bmod N] \in \mathcal{QR}_N^{+1}$.
- 13.8 Let $N = pq$ with p, q distinct, odd primes, and fix $z \in \mathcal{QR}_N^{+1}$. Show that choosing uniform $x \in \mathcal{QR}_N$ and setting $y := [z \cdot x \bmod N]$ gives a y that is uniformly distributed in $\mathcal{QR}_N^{+1}$. That is, for any $\hat{y} \in \mathcal{QR}_N^{+1}$

$$\Pr[z \cdot x = \hat{y} \bmod N] = 1/|\mathcal{QR}_N^{+1}|,$$

where the probability is taken over uniform choice of $x \in \mathcal{QR}_N$.

Hint: Use the previous exercise.

- 13.9 Let N be the product of 5 distinct, odd primes. If $y \in \mathbb{Z}_N^*$ is a quadratic residue, how many solutions are there to the equation $x^2 = y \bmod N$?
- 13.10 Show that the Goldwasser–Micali encryption scheme is homomorphic if the message space $\{0, 1\}$ is viewed as the group $\mathbb{Z}_2$.
- 13.11 Consider the following variation of the Goldwasser–Micali encryption scheme: **GenModulus**(1^n) is run to obtain (N, p, q) where $N = pq$ and $p = q = 3 \bmod 4$. (I.e., N is a Blum integer.) The public key is N and the private key is $\langle p, q \rangle$. To encrypt $m \in \{0, 1\}$, the sender chooses uniform $x \in \mathbb{Z}_N$ and computes the ciphertext $c := [(-1)^m \cdot x^2 \bmod N]$.
- Prove that for N of the stated form, $[-1 \bmod N] \in \mathcal{QR}_N^{+1}$.
 - Prove that the scheme described has indistinguishable encryptions under a chosen-plaintext attack if deciding quadratic residuosity is hard relative to **GenModulus**.
- 13.12 Assume deciding quadratic residuosity is hard for **GenModulus**. Show that this implies the hardness of distinguishing a uniform element of $\mathcal{QR}_N$ from a uniform element of $\mathcal{J}_N^{+1}$.
- 13.13 Show that plain RSA encryption of a message m leaks $\mathcal{J}_N(m)$.
- 13.14 Consider the following variation of the Goldwasser–Micali encryption scheme: **GenModulus**(1^n) is run to obtain (N, p, q) . The public key is N and the private key is $\langle p, q \rangle$. To encrypt a 0, the sender chooses n uniform elements $c_1, \dots, c_n \in \mathcal{QR}_N$. To encrypt a 1, the sender chooses n uniform elements $c_1, \dots, c_n \in \mathcal{J}_N^{+1}$. In each case, the resulting ciphertext is $c^* = \langle c_1, \dots, c_n \rangle$.
- Show how the sender can generate a uniform element of $\mathcal{J}_N^{+1}$ in polynomial time, where failing with negligible probability.
 - Suggest a way for the receiver to decrypt efficiently, although with negligible error probability.
 - Prove that if deciding quadratic residuosity is hard relative to **GenModulus**, this scheme is CPA-secure.
- Hint:** Use the previous exercise.
- 13.15 Let $\mathcal{G}$ be a polynomial-time algorithm that, on input 1^n , outputs a prime p with $\|p\| = n$ and a generator g of $\mathbb{Z}_p^*$. Prove that the DDH problem is *not* hard relative to $\mathcal{G}$.
- Hint:** Use the fact that quadratic residuosity can be decided efficiently modulo a prime.
- 13.16 The discrete logarithm problem is believed to be hard for $\mathcal{G}$ as in the previous exercise. This means that the function (family) $f_{p,g}$ where $f_{p,g}(x) \stackrel{\text{def}}{=} [g^x \bmod p]$ is one-way. Let $\text{lsb}(x)$ denote the least-significant bit of x . Show that lsb is *not* a hard-core predicate for $f_{p,g}$.

- 13.17 Consider the plain Rabin encryption scheme in which a message $m \in \mathcal{QR}_N$ is encrypted relative to a public key N (where N is a Blum integer) by computing the ciphertext $c := [m^2 \bmod N]$. Show a chosen-ciphertext attack on this scheme that recovers the entire private key.
- 13.18 The plain Rabin signature scheme is like the plain RSA signature scheme, except using the Rabin trapdoor permutation. Show an attack on plain Rabin signatures by which the attacker learns the signer's private key.
- 13.19 Let N be a Blum integer.

- (a) Define the set $S \stackrel{\text{def}}{=} \{x \in \mathbb{Z}_N^* \mid x < N/2 \text{ and } \mathcal{J}_N(x) = +1\}$. Define the function $f_N : S \rightarrow \mathbb{Z}_N^*$ by:

$$f_N(x) = \begin{cases} [x^2 \bmod N] & \text{if } [x^2 \bmod N] < N/2 \\ [-x^2 \bmod N] & \text{if } [x^2 \bmod N] > N/2 \end{cases}$$

Show that f_N is a permutation over S .

- (b) Define a family of trapdoor permutations based on factoring using f_N as defined above.

- 13.20 Let N be a Blum integer. Define the function $\text{half}_N : \mathbb{Z}_N^* \rightarrow \{0, 1\}$ as

$$\text{half}_N(x) = \begin{cases} -1 & \text{if } x < N/2 \\ +1 & \text{if } x > N/2 \end{cases}$$

Show that the function $f : \mathbb{Z}_N^* \rightarrow \mathcal{QR}_N \times \{-1, +1\}^2$ defined as

$$f(x) = ([x^2 \bmod N], \mathcal{J}_N(x), \text{half}_N(x))$$

is one-to-one.

Index of Common Notation

General notation:

- $:=$ refers to deterministic assignment
- If S is a set, then $x \leftarrow S$ denotes that x is chosen uniformly from S
- If A is a randomized algorithm, then $y \leftarrow A(x)$ denotes running A on input x with a uniform random tape and assigning the output to y . We write $y := A(x; r)$ to denote running A on input x using random tape r and assigning the output to y
- $\wedge$ denotes Boolean conjunction (the AND operator)
- $\vee$ denotes Boolean disjunction (the OR operator)
- $\oplus$ denotes the exclusive-or (XOR) operator; this operator can be applied to single bits or entire strings (in the latter case, the XOR is bitwise)
- $\{0, 1\}^n$ is the set of all bit-strings of length n
- $\{0, 1\}^{\leq n}$ is the set of all bit-strings of length at most n
- $\{0, 1\}^*$ is the set of all finite bit-strings
- 0^n (resp., 1^n) denotes the string comprised of n zeroes (resp., n ones)
- $\|x\|$ denotes the length of the binary representation of the (positive) integer x , written with leading bit 1. Note that $\log x < \|x\| \leq \log x + 1$
- $|x|$ denotes the length of the binary string x (which may have leading 0s), or the absolute value of the real number x
- $\mathcal{O}(\cdot), \Theta(\cdot), \Omega(\cdot), \omega(\cdot)$ see Appendix A.2
- $0x$ denotes that digits are being represented in hexadecimal
- $x\|y$ denotes unambiguous concatenation of the strings x and y (“unambiguous” means that x and y can be recovered from $x\|y$)
- $\Pr[X]$ denotes the probability of event X
- $\log x$ denotes the base-2 logarithm of x

Crypto-specific notation:

- n is the security parameter
- PPT stands for “probabilistic polynomial time”
- $\mathcal{A}^{\mathcal{O}(\cdot)}$ denotes the algorithm $\mathcal{A}$ with oracle access to $\mathcal{O}$
- k typically denotes a secret key (as in private-key encryption and MACs)
- (pk, sk) denotes a public/private key-pair (for public-key encryption and digital signatures)
- negl denotes a negligible function; see Definition 3.4
- $\text{poly}(n)$ denotes an arbitrary polynomial
- $\text{polylog}(n)$ denotes $\text{poly}(\log(n))$
- Func_n denotes the set of functions mapping n -bit strings to n -bit strings
- Perm_n denotes the set of bijections on n -bit strings
- IV denotes an initialization vector (used for modes of operation and collision-resistant hash functions)

Algorithms and procedures:

- G denotes a pseudorandom generator
- F denotes a keyed function that is typically a pseudorandom function or permutation
- $(\text{Gen}, \text{Enc}, \text{Dec})$ denote the key-generation, encryption, and decryption procedures, respectively, for both private- and public-key encryption. For the case of private-key encryption, when Gen is unspecified then $\text{Gen}(1^n)$ outputs a uniform $k \in \{0, 1\}^n$
- $(\text{Gen}, \text{Mac}, \text{Vrfy})$ denote the key-generation, tag-generation, and verification procedures, respectively, for a message authentication code. When Gen is unspecified then $\text{Gen}(1^n)$ outputs a uniform $k \in \{0, 1\}^n$
- $(\text{Gen}, \text{Sign}, \text{Vrfy})$ denote the key-generation, signature-generation, and verification procedures, respectively, for a digital signature scheme
- GenPrime denotes a PPT algorithm that, on input 1^n , outputs an n -bit prime except with probability negligible in n
- GenModulus denotes a PPT algorithm that, on input 1^n , outputs (N, p, q) where $N = pq$ and (except with negligible probability) p and q are n -bit primes

- **GenRSA** denotes a PPT algorithm that, on input 1^n , outputs (except with negligible probability) a modulus N , an integer $e > 0$ with $\gcd(e, \phi(N)) = 1$, and an integer d satisfying $ed = 1 \bmod \phi(N)$
- $\mathcal{G}$ denotes a PPT algorithm that, on input 1^n , outputs (except with negligible probability) a description of a cyclic group $\mathbb{G}$, the group order q (with $\|q\| = n$), and a generator $g \in \mathbb{G}$.

Number theory:

- $\mathbb{Z}$ denotes the set of integers
- $a \mid b$ means a divides b
- $a \nmid b$ means that a does not divide b
- $\gcd(a, b)$ denotes the greatest common divisor of a and b
- $[a \bmod b]$ denotes the remainder of a when divided by b . Note that $0 \leq [a \bmod b] < b$.
- $x_1 = x_2 = \dots = x_n \bmod N$ means that $x_1, \dots, x_n$ are all congruent modulo N

Note: $x = y \bmod N$ means that x and y are congruent modulo N , whereas $x = [y \bmod N]$ means that x is equal to the remainder of y when divided by N .

- $\mathbb{Z}_N$ denotes the additive group of integers modulo N as well as the set $\{0, \dots, N-1\}$
- $\mathbb{Z}_N^*$ denotes the multiplicative group of invertible integers modulo N (i.e., those that are relatively prime to N)
- $\phi(N)$ denotes the size of $\mathbb{Z}_N^*$
- $\mathbb{G}$ and $\mathbb{H}$ denote groups
- $\mathbb{G}_1 \simeq \mathbb{G}_2$ means that groups $\mathbb{G}_1$ and $\mathbb{G}_2$ are isomorphic. If this isomorphism is given by f and $f(x_1) = x_2$ then we write $x_1 \leftrightarrow x_2$
- g is typically a generator of a group
- $\log_g h$ denotes the discrete logarithm of h to the base g
- $\langle g \rangle$ denotes the group generated by g
- p and q usually denote primes
- N typically denotes the product of two distinct primes p and q of equal length

- $\mathcal{QR}_p$ is the set of quadratic residues modulo p
- $\mathcal{QNR}_p$ is the set of quadratic non-residues modulo p
- $\mathcal{J}_p(x)$ is the Jacobi symbol of x modulo p
- $\mathcal{J}_N^{+1}$ is the set of elements with Jacobi symbol $+1$ modulo N
- $\mathcal{J}_N^{-1}$ is the set of elements with Jacobi symbol -1 modulo N
- $\mathcal{QNR}_N^{+1}$ is the set of quadratic non-residues modulo N having Jacobi symbol $+1$

Appendix A

Mathematical Background

A.1 Identities and Inequalities

We list some standard identities and inequalities that are used at various points throughout the text.

THEOREM A.1 (Binomial expansion theorem) *Let x, y be real numbers, and let n be a positive integer. Then*

$$(x + y)^n = \sum_{i=0}^n \binom{n}{i} x^i y^{n-i}.$$

PROPOSITION A.2 *For all $x \geq 1$ it holds that $(1 - 1/x)^x \leq e^{-1}$.*

PROPOSITION A.3 *For all x it holds that $1 - x \leq e^{-x}$.*

PROPOSITION A.4 *For all x with $0 \leq x \leq 1$ it holds that*

$$e^{-x} \leq 1 - \left(1 - \frac{1}{e}\right) \cdot x \leq 1 - \frac{x}{2}.$$

A.2 Asymptotic Notation

We use standard notation for expressing asymptotic behavior of functions.

DEFINITION A.5 *Let $f(n), g(n)$ be functions from non-negative integers to non-negative reals. Then:*

- $f(n) = \mathcal{O}(g(n))$ means that there exist positive integers c and n' such that for all $n > n'$ it holds that $f(n) \leq c \cdot g(n)$.

- $f(n) = \Omega(g(n))$ means that there exist positive integers c and n' such that for all $n > n'$ it holds that $f(n) \geq c \cdot g(n)$.
- $f(n) = \Theta(g(n))$ means that there exist positive integers c_1, c_2 , and n' such that for all $n > n'$ it holds that $c_1 \cdot g(n) \leq f(n) \leq c_2 \cdot g(n)$.
- $f(n) = o(g(n))$ means that $\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = 0$.
- $f(n) = \omega(g(n))$ means that $\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = \infty$.

Example A.6

Let $f(n) = n^4 + 3n + 500$. Then:

- $f(n) = \mathcal{O}(n^4)$.
- $f(n) = \mathcal{O}(n^5)$. In fact, $f(n) = o(n^5)$.
- $f(n) = \Omega(n^3 \log n)$. In fact, $f(n) = \omega(n^3 \log n)$.
- $f(n) = \Theta(n^4)$.

◇

A.3 Basic Probability

We assume the reader is familiar with basic probability theory, on the level of what is covered in a typical undergraduate course on discrete mathematics. Here we simply remind the reader of some notation and basic facts.

If E is an event, then $\bar{E}$ denotes the complement of that event; i.e., $\bar{E}$ is the event that E does *not* occur. By definition, $\Pr[E] = 1 - \Pr[\bar{E}]$. If E_1 and E_2 are events, then $E_1 \wedge E_2$ denotes their conjunction; i.e., $E_1 \wedge E_2$ is the event that *both* E_1 and E_2 occur. By definition, $\Pr[E_1 \wedge E_2] \leq \Pr[E_1]$. Events E_1 and E_2 are said to be *independent* if $\Pr[E_1 \wedge E_2] = \Pr[E_1] \cdot \Pr[E_2]$.

If E_1 and E_2 are events, then $E_1 \vee E_2$ denotes the disjunction of E_1 and E_2 ; that is, $E_1 \vee E_2$ is the event that *either* E_1 *or* E_2 occurs. It follows from the definition that $\Pr[E_1 \vee E_2] \geq \Pr[E_1]$. The *union bound* is often a very useful upper bound of this quantity.

PROPOSITION A.7 (Union Bound)

$$\Pr[E_1 \vee E_2] \leq \Pr[E_1] + \Pr[E_2].$$

Repeated application of the union bound for any events $E_1, \dots, E_k$ gives

$$\Pr \left[\bigvee_{i=1}^k E_i \right] \leq \sum_{i=1}^k \Pr[E_i].$$

The *conditional probability of E_1 given E_2* , denoted $\Pr[E_1 \mid E_2]$, is defined as

$$\Pr[E_1 \mid E_2] \stackrel{\text{def}}{=} \frac{\Pr[E_1 \wedge E_2]}{\Pr[E_2]}$$

as long as $\Pr[E_2] \neq 0$. (If $\Pr[E_2] = 0$ then $\Pr[E_1 \mid E_2]$ is undefined.) This represents the probability that event E_1 occurs, given that event E_2 has occurred. It follows immediately from the definition that

$$\Pr[E_1 \wedge E_2] = \Pr[E_1 \mid E_2] \cdot \Pr[E_2];$$

equality holds even if $\Pr[E_2] = 0$ as long as we interpret multiplication by zero on the right-hand side in the obvious way.

We can now easily derive Bayes' theorem.

THEOREM A.8 (Bayes' Theorem) *If $\Pr[E_2] \neq 0$ then*

$$\Pr[E_1 \mid E_2] = \frac{\Pr[E_2 \mid E_1] \cdot \Pr[E_1]}{\Pr[E_2]}.$$

PROOF This follows because

$$\Pr[E_1 \mid E_2] = \frac{\Pr[E_1 \wedge E_2]}{\Pr[E_2]} = \frac{\Pr[E_2 \wedge E_1]}{\Pr[E_2]} = \frac{\Pr[E_2 \mid E_1] \cdot \Pr[E_1]}{\Pr[E_2]}.$$

■

Let $E_1, \dots, E_n$ be events such that $\Pr[E_1 \vee \dots \vee E_n] = 1$ and $\Pr[E_i \wedge E_j] = 0$ for all $i \neq j$. That is, the $\{E_i\}$ *partition* the space of all possible events, so that with probability 1 exactly one of the events E_i occurs. Then for any F

$$\Pr[F] = \sum_{i=1}^n \Pr[F \wedge E_i].$$

A special case is when $n = 2$ and $E_2 = \bar{E}_1$, giving

$$\begin{aligned} \Pr[F] &= \Pr[F \wedge E_1] + \Pr[F \wedge \bar{E}_1] \\ &= \Pr[F \mid E_1] \cdot \Pr[E_1] + \Pr[F \mid \bar{E}_1] \cdot \Pr[\bar{E}_1]. \end{aligned}$$

Taking $F = E_1 \vee E_2$, we get a tighter version of the union bound:

$$\begin{aligned} \Pr[E_1 \vee E_2] &= \Pr[E_1 \vee E_2 \mid E_1] \cdot \Pr[E_1] + \Pr[E_1 \vee E_2 \mid \bar{E}_1] \cdot \Pr[\bar{E}_1] \\ &\leq \Pr[E_1] + \Pr[E_2 \mid \bar{E}_1]. \end{aligned}$$

Extending this to events $E_1, \dots, E_n$ we obtain

PROPOSITION A.9

$$\Pr \left[\bigvee_{i=1}^k E_i \right] \leq \Pr[E_1] + \sum_{i=2}^k \Pr[E_i \mid \bar{E}_1 \wedge \dots \wedge \bar{E}_{i-1}].$$

*** Useful Probability Bounds**

We review some terminology and state probability bounds that are standard, but may not be encountered in a basic discrete mathematics course. The material here is used only in Section 7.3.

A (discrete, real-valued) random variable X is a variable whose value is assigned probabilistically from some finite set S of real numbers. X is non-negative if it does not take negative values; it is a 0/1-random variable if $S = \{0, 1\}$. The 0/1-random variables $X_1, \dots, X_k$ are *independent* if for all $b_1, \dots, b_k$ it holds that $\Pr[X_1 = b_1 \wedge \dots \wedge X_k = b_k] = \prod_{i=1}^k \Pr[X_i = b_i]$.

We let $\text{Exp}[X]$ denote the expectation of a random variable X ; if X takes values in a set S then $\text{Exp}[X] \stackrel{\text{def}}{=} \sum_{s \in S} s \cdot \Pr[X = s]$. One of the most important facts is that expectation is *linear*; for random variables $X_1, \dots, X_k$ (with arbitrary dependencies) we have $\text{Exp}[\sum_i X_i] = \sum_i \text{Exp}[X_i]$. If X_1, X_2 are independent, then $\text{Exp}[X_i \cdot X_j] = \text{Exp}[X_i] \cdot \text{Exp}[X_j]$.

Markov's inequality is useful when little is known about X .

PROPOSITION A.10 (Markov's inequality) *Let X be a non-negative random variable and $v > 0$. Then $\Pr[X \geq v] \leq \text{Exp}[X]/v$.*

PROOF Say X takes values in a set S . We have

$$\begin{aligned} \text{Exp}[X] &= \sum_{s \in S} s \cdot \Pr[X = s] \\ &\geq \sum_{x \in S, x < v} \Pr[X = s] \cdot 0 + \sum_{x \in S, x \geq v} v \cdot \Pr[X = s] \\ &= v \cdot \Pr[X \geq v]. \end{aligned}$$

■

The variance of X , denoted $\text{Var}[X]$, measures how much X deviates from its expectation. We have $\text{Var}[X] \stackrel{\text{def}}{=} \text{Exp}[(X - \text{Exp}[X])^2] = \text{Exp}[X^2] - \text{Exp}[X]^2$, and one can easily show that $\text{Var}[aX + b] = a^2 \text{Var}[X]$. For a 0/1-random variable X_i , we have $\text{Var}[X_i] \leq 1/4$ because in this case $\text{Exp}[X_i] = \text{Exp}[X_i^2]$ and so $\text{Var}[X_i] = \text{Exp}[X_i](1 - \text{Exp}[X_i])$, which is maximized when $\text{Exp}[X_i] = \frac{1}{2}$.

PROPOSITION A.11 (Chebyshev's inequality) *Let X be a random variable and $\delta > 0$. Then:*

$$\Pr[|X - \text{Exp}[X]| \geq \delta] \leq \frac{\text{Var}[X]}{\delta^2}.$$

PROOF Define the non-negative random variable $Y \stackrel{\text{def}}{=} (X - \text{Exp}[X])^2$ and then apply Markov's inequality. So,

$$\begin{aligned} \Pr[|X - \text{Exp}[X]| \geq \delta] &= \Pr[(X - \text{Exp}[X])^2 \geq \delta^2] \\ &\leq \frac{\text{Exp}[(X - \text{Exp}[X])^2]}{\delta^2} = \frac{\text{Var}[X]}{\delta^2}. \end{aligned}$$

■

The 0/1-random variables $X_1, \dots, X_m$ are *pairwise independent* if for every $i \neq j$ and every $b_i, b_j \in \{0, 1\}$ it holds that

$$\Pr[X_i = b_i \wedge X_j = b_j] = \Pr[X_i = b_i] \cdot \Pr[X_j = b_j].$$

If $X_1, \dots, X_m$ are pairwise independent then $\text{Var}[\sum_{i=1}^m X_i] = \sum_{i=1}^m \text{Var}[X_i]$. (This follows since $\text{Exp}[X_i \cdot X_j] = \text{Exp}[X_i] \cdot \text{Exp}[X_j]$ when $i \neq j$, using pairwise independence.) An important corollary of Chebyshev's inequality follows.

COROLLARY A.12 *Let $X_1, \dots, X_m$ be pairwise-independent random variables with the same expectation μ and variance σ^2 . Then for every $\delta > 0$,*

$$\Pr\left[\left|\frac{\sum_{i=1}^m X_i}{m} - \mu\right| \geq \delta\right] \leq \frac{\sigma^2}{\delta^2 m}.$$

PROOF By linearity of expectation, $\text{Exp}[\sum_{i=1}^m X_i/m] = \mu$. Applying Chebyshev's inequality to the random variable $\sum_{i=1}^m X_i/m$, we have

$$\Pr\left[\left|\frac{\sum_{i=1}^m X_i}{m} - \mu\right| \geq \delta\right] \leq \frac{\text{Var}\left[\frac{1}{m} \cdot \sum_{i=1}^m X_i\right]}{\delta^2}.$$

Using pairwise independence, it follows that

$$\text{Var}\left[\frac{1}{m} \cdot \sum_{i=1}^m X_i\right] = \frac{1}{m^2} \sum_{i=1}^m \text{Var}[X_i] = \frac{1}{m^2} \sum_{i=1}^m \sigma^2 = \frac{\sigma^2}{m}.$$

The inequality is obtained by combining the above two equations. ■

Say 0/1-random variables $X_1, \dots, X_m$ each provides an estimate of some fixed (unknown) bit b . That is, $\Pr[X_i = b] \geq 1/2 + \varepsilon$ for all i , where $\varepsilon > 0$.

We can estimate b by looking at the value of X_1 ; this estimate will be correct with probability $\Pr[X_1 = b]$. A better estimate can be obtained by looking at the values of $X_1, \dots, X_m$ and taking the value that occurs the majority of the time. We analyze how well this does when $X_1, \dots, X_m$ are pairwise independent.

PROPOSITION A.13 Fix $\varepsilon > 0$ and $b \in \{0, 1\}$, and let $\{X_i\}$ be pairwise-independent, 0/1-random variables for which $\Pr[X_i = b] \geq \frac{1}{2} + \varepsilon$ for all i . Consider the process in which m values $X_1, \dots, X_m$ are recorded and X is set to the value that occurs a strict majority of the time. Then

$$\Pr[X \neq b] \leq \frac{1}{4 \cdot \varepsilon^2 \cdot m}.$$

PROOF Assume $b = 1$; by symmetry, this is without loss of generality. Then $\mathbb{E}[X_i] = \frac{1}{2} + \varepsilon$. Let X denote the strict majority of the $\{X_i\}$ as in the proposition, and note that $X \neq 1$ if and only if $\sum_{i=1}^m X_i \leq m/2$. So

$$\begin{aligned} \Pr[X \neq 1] &= \Pr\left[\sum_{i=1}^m X_i \leq m/2\right] \\ &= \Pr\left[\frac{\sum_{i=1}^m X_i}{m} - \frac{1}{2} \leq 0\right] \\ &= \Pr\left[\frac{\sum_{i=1}^m X_i}{m} - \left(\frac{1}{2} + \varepsilon\right) \leq -\varepsilon\right] \\ &\leq \Pr\left[\left|\frac{\sum_{i=1}^m X_i}{m} - \left(\frac{1}{2} + \varepsilon\right)\right| \geq \varepsilon\right]. \end{aligned}$$

Since $\text{Var}[X_i] \leq 1/4$ for all i , applying the previous corollary shows that $\Pr[X \neq 1] \leq \frac{1}{4\varepsilon^2 m}$ as claimed. ■

A better bound is obtained if the $\{X_i\}$ are independent:

PROPOSITION A.14 (Chernoff bound) Fix $\varepsilon > 0$ and $b \in \{0, 1\}$, and let $\{X_i\}$ be independent 0/1-random variables with $\Pr[X_i = b] = \frac{1}{2} + \varepsilon$ for all i . The probability that their majority value is not b is at most $e^{-\varepsilon^2 m/2}$.

A.4 The “Birthday” Problem

If we choose q elements $y_1, \dots, y_q$ uniformly from a set of size N , what is the probability that there exist distinct i, j with $y_i = y_j$? We refer to the stated

event as a *collision*, and denote the probability of this event by $\text{coll}(q, N)$. This problem is related to the so-called *birthday problem*, which asks what size group of people we need such that with probability $1/2$ some pair of people in the group share a birthday. To see the relationship, let y_i denote the birthday of the i th person in the group. If there are q people in the group then we have q values $y_1, \dots, y_q$ chosen uniformly from $\{1, \dots, 365\}$, making the simplifying assumption that birthdays are uniformly and independently distributed among the 365 days of a non-leap year. Furthermore, matching birthdays correspond to a collision, i.e., distinct i, j with $y_i = y_j$. So the desired solution to the birthday problem is given by the minimal (integer) value of q for which $\text{coll}(q, 365) \geq 1/2$. (The answer may surprise you—taking $q = 23$ people suffices!)

In this section, we prove lower and upper bounds on $\text{coll}(q, N)$. Taken together and summarized at a high level, they show that if $q < \sqrt{N}$ then the probability of a collision is $\Theta(q^2/N)$; alternately, for $q = \Theta(\sqrt{N})$ the probability of a collision is constant.

An upper bound for the collision probability is easy to obtain.

LEMMA A.15 *Fix a positive integer N , and say q elements $y_1, \dots, y_q$ are chosen uniformly and independently at random from a set of size N . Then the probability that there exist distinct i, j with $y_i = y_j$ is at most $\frac{q^2}{2N}$. That is,*

$$\text{coll}(q, N) \leq \frac{q^2}{2N}.$$

PROOF The proof is a simple application of the union bound (Proposition A.7). Recall that a *collision* means that there exist distinct i, j with $y_i = y_j$. Let Coll denote the event of a collision, and let $\text{Coll}_{i,j}$ denote the event that $y_i = y_j$. It is immediate that $\Pr[\text{Coll}_{i,j}] = 1/N$ for any distinct i, j . Furthermore, $\text{Coll} = \bigvee_{i \neq j} \text{Coll}_{i,j}$ and so repeated application of the union bound implies that

$$\begin{aligned} \Pr[\text{Coll}] &= \Pr\left[\bigvee_{i \neq j} \text{Coll}_{i,j}\right] \\ &\leq \sum_{i \neq j} \Pr[\text{Coll}_{i,j}] = \binom{q}{2} \cdot \frac{1}{N} \leq \frac{q^2}{2N}. \end{aligned}$$



LEMMA A.16 Fix a positive integer N , and say $q \leq \sqrt{2N}$ elements $y_1, \dots, y_q$ are chosen uniformly and independently at random from a set of size N . Then the probability that there exist distinct i, j with $y_i = y_j$ is at least $\frac{q(q-1)}{4N}$. In fact,

$$\text{coll}(q, N) \geq 1 - e^{-q(q-1)/2N} \geq \frac{q(q-1)}{4N}.$$

PROOF Recall that a *collision* means that there exist distinct i, j with $y_i = y_j$. Let Coll denote this event. Let NoColl_i be the event that there is *no* collision among $y_1, \dots, y_i$; that is, $y_j \neq y_k$ for all $j < k \leq i$. Then $\text{NoColl}_q = \overline{\text{Coll}}$ is the event that there is no collision at all.

If NoColl_q occurs then NoColl_i must also have occurred for all $i \leq q$. Thus,

$$\Pr[\text{NoColl}_q] = \Pr[\text{NoColl}_1] \cdot \Pr[\text{NoColl}_2 \mid \text{NoColl}_1] \cdots \Pr[\text{NoColl}_q \mid \text{NoColl}_{q-1}].$$

Now, $\Pr[\text{NoColl}_1] = 1$ since y_1 cannot collide with itself. Furthermore, if event NoColl_i occurs then $\{y_1, \dots, y_i\}$ contains i distinct values; so, the probability that y_{i+1} collides with one of these values is $\frac{i}{N}$ and hence the probability that y_{i+1} does *not* collide with any of these values is $1 - \frac{i}{N}$. This means

$$\Pr[\text{NoColl}_{i+1} \mid \text{NoColl}_i] = 1 - \frac{i}{N},$$

and so

$$\Pr[\text{NoColl}_q] = \prod_{i=1}^{q-1} \left(1 - \frac{i}{N}\right).$$

Since $i/N < 1$ for all i , we have $1 - \frac{i}{N} \leq e^{-i/N}$ (by Inequality A.3) and so

$$\Pr[\text{NoColl}_q] \leq \prod_{i=1}^{q-1} e^{-i/N} = e^{-\sum_{i=1}^{q-1} (i/N)} = e^{-q(q-1)/2N}.$$

We conclude that

$$\Pr[\text{Coll}] = 1 - \Pr[\text{NoColl}_q] \geq 1 - e^{-q(q-1)/2N} \geq \frac{q(q-1)}{4N},$$

using Inequality A.4 in the last step (note that $q(q-1)/2N < 1$). ■

A.5 *Finite Fields

We use finite fields only sparingly in the book, but we include a definition and some basic facts for completeness. Further details can be found in any textbook on abstract algebra.

DEFINITION A.17 A (finite) field is a (finite) set $\mathbb{F}$ along with two binary operations $+, \cdot$ for which the following hold:

- $\mathbb{F}$ is an abelian group with respect to the operation $+$. We let 0 denote the identity element of this group.
 - $\mathbb{F} \setminus \{0\}$ is an abelian group with respect to the operation $\cdot$. We let 1 denote the identity element of this group.
- As usual, we often write ab in place of $a \cdot b$.
- **(Distributivity:)** For all $a, b, c \in \mathbb{F}$, we have $a \cdot (b + c) = ab + ac$.

The *additive inverse* of $a \in \mathbb{F}$, denoted by $-a$, is the unique element satisfying $a + (-a) = 0$; we write $b - a$ in place of $b + (-a)$. The *multiplicative inverse* of $a \in \mathbb{F} \setminus \{0\}$, denoted by a^{-1} , is the unique element satisfying $aa^{-1} = 1$; we often write b/a in place of ba^{-1} .

Example A.18

It follows from the results of Section 8.1.4 that for any prime p the set $\{0, \dots, p-1\}$ is a finite field with respect to addition and multiplication modulo p . We denote this field by $\mathbb{F}_p$. $\diamond$

Finite fields have a rich theory. For our purposes, we need only a few basic facts. The *order* of $\mathbb{F}$ is the number of elements in $\mathbb{F}$ (assuming it is finite). Recall also that q is a prime power if $q = p^r$ for some prime p and integer $r \geq 1$.

THEOREM A.19 If $\mathbb{F}$ is a finite field, then the order of $\mathbb{F}$ is a prime power. Conversely, for every prime power q there is a finite field of order q , which is moreover the unique such field (up to relabeling of the elements).

For $q = p^r$ with p prime, we let $\mathbb{F}_q$ denote the (unique) field of order q . We call p the *characteristic* of $\mathbb{F}_q$. The preceding theorem tells us that the characteristic of any finite field is prime.

As in the case of groups, if n is a positive integer and $a \in \mathbb{F}$ then

$$n \cdot a \stackrel{\text{def}}{=} \underbrace{a + \dots + a}_{n \text{ times}} \quad \text{and} \quad a^n \stackrel{\text{def}}{=} \underbrace{a \cdot \dots \cdot a}_{n \text{ times}}.$$

The notation is extended for $n \leq 0$ in the natural way.

THEOREM A.20 Let $\mathbb{F}_q$ be a finite field of characteristic p . Then for all $a \in \mathbb{F}_q$ we have $p \cdot a = 0$.

Let $q = p^r$ with p prime. For $r = 1$, we have seen in Example A.18 that $\mathbb{F}_q = \mathbb{F}_p$ can be taken to be the set $\{0, \dots, p-1\}$ under addition and multiplication

modulo p . We caution, however, that for $r > 1$ the set $\{0, \dots, q-1\}$ is *not* a field under addition and multiplication modulo q . For example, if we take $q = 3^2 = 9$ then the element 3 does not have a multiplicative inverse modulo 9.

Finite fields of characteristic p can be represented using polynomials over $\mathbb{F}_p$. We give an example to demonstrate the flavor of the construction, without discussing why the construction works or describing the general case. We construct the field $\mathbb{F}_4$ by working with polynomials over $\mathbb{F}_2$. Fix the polynomial $r(x) = x^2 + x + 1$, and note that $r(x)$ has no roots over $\mathbb{F}_2$ since $r(0) = r(1) = 1$ (recall that we are working in $\mathbb{F}_2$, which means that all operations are carried out modulo 2). In the same way that we can introduce the imaginary number i to be a root of $x^2 + 1$ over the reals, we can introduce a value ω to be a root of $r(x)$ over $\mathbb{F}_2$; that is, $\omega^2 = -\omega - 1$. We then define $\mathbb{F}_4$ to be the set of all degree-1 polynomials in ω over $\mathbb{F}_2$; that is, $\mathbb{F}_4 = \{0, 1, \omega, \omega + 1\}$. Addition in $\mathbb{F}_4$ will just be regular polynomial addition, remembering that operations on the coefficients are done in $\mathbb{F}_2$ (that is, modulo 2). Multiplication in $\mathbb{F}_4$ will be polynomial multiplication (again, with operations on the coefficients carried out modulo 2) followed by the substitution $\omega^2 = -\omega - 1$; this also ensures that the result lies in $\mathbb{F}_4$. So, for example,

$$\omega + (\omega + 1) = 2\omega + 1 = 1$$

and

$$(\omega + 1) \cdot (\omega + 1) = \omega^2 + 2\omega + 1 = (-\omega - 1) + 1 = -\omega = \omega.$$

Although not obvious, one can check that this is a field; the only difficult condition to verify is that every nonzero element has a multiplicative inverse.

We need only one other result.

THEOREM A.21 *Let $\mathbb{F}_q$ be a finite field of order q . Then the abelian group $\mathbb{F}_q \setminus \{0\}$ with respect to $\cdot$ is a **cyclic** group of order $q - 1$.*

Appendix B

Basic Algorithmic Number Theory

For the cryptographic constructions given in this book to be efficient (i.e., to run in time polynomial in the lengths of their inputs), it is necessary for these constructions to utilize efficient (that is, polynomial-time) algorithms for performing basic number-theoretic operations. Although in some cases there exist “trivial” algorithms that would work, it is still worthwhile to carefully consider their efficiency since for cryptographic applications it is not uncommon to use integers that are *thousands* of bits long. In other cases obtaining any polynomial-time algorithm requires a bit of cleverness, and an analysis of their performance may rely on non-trivial group-theoretic results.

In Appendix B.1 we describe basic algorithms for integer arithmetic. Here we cover the familiar algorithms for addition, subtraction, etc., as well as the Euclidean algorithm for computing greatest common divisors. We also discuss the extended Euclidean algorithm, assuming there that the reader has covered the material in Section 8.1.1.

In Appendix B.2 we show various algorithms for modular arithmetic. In addition to a brief discussion of basic modular operations (i.e., modular reduction, addition, multiplication, and inversion), we also describe *Montgomery multiplication*, which can greatly simplify (and speed up) implementations of modular arithmetic. We then discuss algorithms for problems that are less common outside the field of cryptography: exponentiation modulo N (as well as in arbitrary groups) and choosing a uniform element of $\mathbb{Z}_N$ or $\mathbb{Z}_N^*$ (or in an arbitrary group). This section assumes familiarity with the basic group theory covered in Section 8.1.

The material above is used implicitly throughout the second half of the book, although it is not absolutely necessary to read this material in order to follow the book. (In particular, the reader willing to accept the results of this Appendix without proof can simply read the summary of those results in the theorems below.) Appendix B.3, which discusses finding generators in cyclic groups (when the factorization of the group order is known) and assumes the results of Section 8.3.1, contains material that is hardly used at all; it is included for completeness and reference.

Since our goal is only to establish that certain problems can be solved in polynomial time, we have opted for *simplicity* rather than *efficiency* in our selection of algorithms and their descriptions (as long as the algorithms run in polynomial time). For this reason, we generally will not be interested in

the exact running times of the algorithms we present beyond establishing that they indeed run in polynomial time. The reader who is seriously interested in implementing these algorithms is forewarned to look at other sources for more efficient alternatives as well as various techniques for speeding up the necessary computations.

The results in this Appendix are summarized by the theorems that follow. Throughout, we assume that any integer a provided as input is written using exactly $\|a\|$ bits; i.e., the high-order bit is 1. In Appendix B.1 we show:

THEOREM B.1 (Integer operations) *Given integers a and b , it is possible to perform the following operations in time polynomial in $\|a\|$ and $\|b\|$:*

1. *Computing the sum $a + b$ and the difference $a - b$;*
2. *Computing the product ab ;*
3. *Computing positive integers q and $r < b$ such that $a = qb + r$ (i.e., computing division with remainder);*
4. *Computing the greatest common divisor of a and b , $\gcd(a, b)$;*
5. *Computing integers X, Y with $Xa + Yb = \gcd(a, b)$.*

The following results are proved in Appendix B.2:

THEOREM B.2 (Modular operations) *Given integers $N > 1$, a , and b , it is possible to perform the following operations in time polynomial in $\|a\|$, $\|b\|$, and $\|N\|$:*

1. *Computing the modular reduction $[a \bmod N]$;*
2. *Computing the sum $[(a + b) \bmod N]$, the difference $[(a - b) \bmod N]$, and the product $[ab \bmod N]$;*
3. *Determining whether a is invertible modulo N ;*
4. *Computing the multiplicative inverse $[a^{-1} \bmod N]$, assuming a is invertible modulo N ;*
5. *Computing the exponentiation $[a^b \bmod N]$.*

The following generalizes Theorem B.2(5) to arbitrary groups:

THEOREM B.3 (Group exponentiation) *Let $\mathbb{G}$ be a group, written multiplicatively. Let g be an element of the group and let b be a non-negative integer. Then g^b can be computed using $\text{poly}(\|b\|)$ group operations.*

THEOREM B.4 (Choosing uniform elements) *There exists a randomized algorithm with the following properties: on input N ,*

- *The algorithm runs in time polynomial in $\|N\|$;*
- *The algorithm outputs fail with probability negligible in $\|N\|$; and*
- *Conditioned on not outputting fail, the algorithm outputs a uniformly distributed element of $\mathbb{Z}_N$.*

An algorithm with analogous properties exists for $\mathbb{Z}_N^$ as well.*

Since the probability that either algorithm referenced in the above theorem outputs fail is negligible, we ignore this possibility (and instead leave it implicit). In Appendix B.2 we also discuss generalizations of the above to the case of selecting a uniform element from any finite group (subject to certain requirements on the representation of group elements).

A proof of the following is in Appendix B.3:

THEOREM B.5 (Testing and finding generators) *Let $\mathbb{G}$ be a cyclic group of order q , and assume that the group operation and selection of a uniform group element can be carried out in unit time.*

1. *There is an algorithm that on input q , the prime factorization of q , and an element $g \in \mathbb{G}$, runs in $\text{poly}(\|q\|)$ time and decides whether g is a generator of $\mathbb{G}$.*
2. *There is a randomized algorithm that on input q and the prime factorization of q , runs in $\text{poly}(\|q\|)$ time and outputs a generator of $\mathbb{G}$ except with probability negligible in $\|q\|$. Conditioned on the output being a generator, it is uniformly distributed among the generators of $\mathbb{G}$.*

B.1 Integer Arithmetic

B.1.1 Basic Operations

We begin our exploration of algorithmic number theory with a discussion of integer addition/subtraction, multiplication, and division with remainder. A little thought shows that all these operations can be carried out in time polynomial in the input length using the standard “grade-school” algorithms for these problems. For example, addition of two positive integers a and b with $a > b$ can be done in time linear in $\|a\|$ by stepping one-by-one through the bits of a and b , starting with the low-order bits, and computing the corresponding output bit and a “carry bit” at each step. (Details are omitted.) Multiplication of two n -bit integers a and b , to take another example, can

be done by first generating a list of n integers of length at most $2n$ (each of which is equal to $a \cdot 2^{i-1} \cdot b_i$, where b_i is the i th bit of b) and then adding these n integers together to obtain the final result. (Division with remainder is trickier to implement efficiently, but can also be done.)

Although these grade-school algorithms suffice to demonstrate that the aforementioned problems can be solved in polynomial time, it is interesting to note that these algorithms are in some cases not the best ones available. As an example, the simple algorithm for multiplication given above multiplies two n -bit numbers in time $\mathcal{O}(n^2)$, but there exists a better algorithm running in time $\mathcal{O}(n^{\log_2 3})$ (and even that is not the best possible). While the difference is insignificant for numbers of the size we encounter daily, it becomes noticeable when the numbers are large. In cryptographic applications it is not uncommon to use integers that are thousands of bits long (i.e., $n > 1000$), and a judicious choice of which algorithms to use then becomes critical.

B.1.2 The Euclidean and Extended Euclidean Algorithms

Recall from Section 8.1 that $\gcd(a, b)$, the *greatest common divisor* of two integers a and b , is the largest integer d that divides both a and b . We state an easy proposition regarding the greatest common divisor, and then show how this leads to an efficient algorithm for computing gcd's.

PROPOSITION B.6 *Let $a, b > 1$ with $b \nmid a$. Then*

$$\gcd(a, b) = \gcd(b, [a \bmod b]).$$

PROOF If $b > a$ the stated claim is immediate. So assume $a > b$. Write $a = qb + r$ for q, r positive integers and $r < b$ (cf. Proposition 8.1); note that $r > 0$ because $b \nmid a$. Since $r = [a \bmod b]$, we prove the proposition by showing that $\gcd(a, b) = \gcd(b, r)$.

Let $d = \gcd(a, b)$. Then d divides both a and b , and so d also divides $r = a - qb$. By definition of the greatest common divisor, we thus have $\gcd(b, r) \geq d = \gcd(a, b)$.

Let $d' = \gcd(b, r)$. Then d' divides both b and r , and so d' also divides $a = qb + r$. By definition of the greatest common divisor, we thus have $\gcd(a, b) \geq d' = \gcd(b, r)$.

Since $d \geq d'$ and $d' \geq d$, we conclude that $d = d'$. ■

The above suggests the recursive *Euclidean algorithm* (Algorithm B.7) for computing the greatest common divisor $\gcd(a, b)$ of two integers a and b . Correctness of the algorithm follows readily from Proposition B.6. As for its running time, we show below that on input (a, b) the algorithm makes fewer than $2 \cdot \|b\|$ recursive calls. Since checking whether b divides a and computing

ALGORITHM B.7**The Euclidean algorithm GCD****Input:** Integers a, b with $a \geq b > 0$ **Output:** The greatest common divisor of a and b **if** b divides a **return** b **else return** $\text{GCD}(b, [a \bmod b])$

$[a \bmod b]$ can both be done in time polynomial in $\|a\|$ and $\|b\|$, this implies that the entire algorithm runs in polynomial time.

PROPOSITION B.8 Consider an execution of $\text{GCD}(a_0, b_0)$, and let a_i, b_i (for $i = 1, \dots, \ell$) denote the arguments to the i th recursive call of GCD . Then $b_{i+2} \leq b_i/2$ for $0 \leq i \leq \ell - 2$.

PROOF First note that for any $a > b$ we have $[a \bmod b] < a/2$. To see this, consider the two cases: If $b \leq a/2$ then $[a \bmod b] < b \leq a/2$ is immediate. On the other hand, if $b > a/2$ then $[a \bmod b] = a - b < a/2$.

Now fix arbitrary i with $0 \leq i \leq \ell - 2$. Then $b_{i+2} = [a_{i+1} \bmod b_{i+1}] < a_{i+1}/2 = b_i/2$. ■

COROLLARY B.9 In an execution of algorithm $\text{GCD}(a, b)$, there are at most $2\|b\| - 2$ recursive calls to GCD .

PROOF Let a_i, b_i (for $i = 1, \dots, \ell$) denote the arguments to the i th recursive call of GCD . The $\{b_i\}$ are always greater than zero, and the algorithm makes no further recursive calls if it ever happens that $b_i = 1$ (since then $b_i \mid a_i$). The previous proposition indicates that the $\{b_i\}$ decrease by a multiplicative factor of (at least) 2 in every two iterations. It follows that the number of recursive calls to GCD is at most $2 \cdot (\|b\| - 1)$. ■

The Extended Euclidean Algorithm

By Proposition 8.2, we know that for positive integers a, b there exist integers X, Y with $Xa + Yb = \gcd(a, b)$. A simple modification of the Euclidean algorithm, called the *extended Euclidean algorithm*, can be used to find X, Y in addition to computing $\gcd(a, b)$; see Algorithm B.10. You are asked to show correctness of the extended Euclidean algorithm in Exercise B.1, and to prove that the algorithm runs in polynomial time in Exercise B.2.

ALGORITHM B.10**The extended Euclidean algorithm eGCD****Input:** Integers a, b with $a \geq b > 0$ **Output:** (d, X, Y) with $d = \gcd(a, b)$ and $Xa + Yb = d$ **if** b divides a **return** $(b, 0, 1)$ **else** Compute integers q, r with $a = qb + r$ and $0 < r < b$ $(d, X, Y) := \text{eGCD}(b, r)$ // note that $Xb + Yr = d$ **return** $(d, Y, X - Yq)$

B.2 Modular Arithmetic

We now turn our attention to basic arithmetic operations modulo $N > 1$. We will use $\mathbb{Z}_N$ to refer both to the set $\{0, \dots, N-1\}$ as well as to the group that results by considering addition modulo N among the elements of this set.

B.2.1 Basic Operations

Efficient algorithms for the basic arithmetic operations over the integers immediately imply efficient algorithms for the corresponding arithmetic operations modulo N . For example, computing the modular reduction $[a \bmod N]$ can be done in time polynomial in $\|a\|$ and $\|N\|$ by computing division-with-remainder over the integers. Next consider modular operations on two elements $a, b \in \mathbb{Z}_N$ where $\|N\| = n$. (Note that a, b have length at most n . Actually, it is convenient to simply assume that all elements of $\mathbb{Z}_N$ have length *exactly* n , padding to the left with 0s if necessary.) Addition of a and b modulo N can be done by first computing $a+b$, an integer of length at most $n+1$, and then reducing this intermediate result modulo N . Similarly, multiplication modulo N can be performed by first computing the integer ab of length at most $2n$ and then reducing the result modulo N . Since addition, multiplication, and division-with-remainder can all be done in polynomial time, these give polynomial-time algorithms for addition and multiplication modulo N .

B.2.2 Computing Modular Inverses

Our discussion thus far has shown how to add, subtract, and multiply modulo N . One operation we are missing is “division” or, equivalently, computing multiplicative inverses modulo N . Recall from Section 8.1.2 that the multiplicative inverse (modulo N) of an element $a \in \mathbb{Z}_N$ is an element $a^{-1} \in \mathbb{Z}_N$ such that $a \cdot a^{-1} = 1 \bmod N$. Proposition 8.7 shows that a has an inverse if and only if $\gcd(a, N) = 1$, i.e., if and only if $a \in \mathbb{Z}_N^*$. Thus, using the

Euclidean algorithm we can easily determine whether a given element a has a multiplicative inverse modulo N .

Given N and $a \in \mathbb{Z}_N$ with $\gcd(a, N) = 1$, Proposition 8.2 tells us that there exist integers X, Y with $Xa + YN = 1$. This means that $[X \bmod N]$ is the multiplicative inverse of a . Integers X and Y satisfying $Xa + YN = 1$ can be found efficiently using the extended Euclidean algorithm **eGCD** shown in Section B.1.2. This leads to the following polynomial-time algorithm for computing multiplicative inverses:

ALGORITHM B.11

Computing modular inverses

Input: Modulus N ; element a

Output: $[a^{-1} \bmod N]$ (if it exists)

$(d, X, Y) := \mathbf{eGCD}(a, N)$ // note that $Xa + YN = \gcd(a, N)$

if $d \neq 1$ **return** “ a is not invertible modulo N ”

else return $[X \bmod N]$

B.2.3 Modular Exponentiation

A more challenging task is that of *exponentiation* modulo N , that is, computing $[a^b \bmod N]$ for base $a \in \mathbb{Z}_N$ and integer exponent $b > 0$. (When $b = 0$ the problem is easy. When $b < 0$ and $a \in \mathbb{Z}_N^*$ then $a^b = (a^{-1})^{-b} \bmod N$ and the problem is reduced to the case of exponentiation with a positive exponent given that we can compute inverses, as discussed in the previous section.) Notice that the basic approach used in the case of addition and multiplication (i.e., computing the integer a^b and then reducing this intermediate result modulo N) does *not* work here: the integer a^b has length $\|a^b\| = \Theta(\log a^b) = \Theta(b \cdot \|a\|)$, and so even storing the intermediate result a^b would require time exponential in $\|b\| = \Theta(\log b)$.

We can address this problem by reducing modulo N at all intermediate steps of the computation, rather than only reducing modulo N at the end. This has the effect of keeping the intermediate results “small” throughout the computation. Even with this important initial observation, it is still non-trivial to design a polynomial-time algorithm for modular exponentiation. Consider the naïve approach of Algorithm B.12, which simply performs b multiplications by a . This still runs in time that is *exponential* in $\|b\|$.

This naïve algorithm can be viewed as relying on the following recurrence:

$$[a^b \bmod N] = [a \cdot a^{b-1} \bmod N] = [a \cdot a \cdot a^{b-2} \bmod N] = \dots$$

Any algorithm based on this relationship will require $\Theta(b)$ time. We can do

ALGORITHM B.12**A naïve algorithm for modular exponentiation****Input:** Modulus N ; base $a \in \mathbb{Z}_N$; integer exponent $b > 0$ **Output:** $[a^b \bmod N]$ $x := 1$ **for** $i = 1$ to b : $x := [x \cdot a \bmod N]$ **return** x

better by relying on the following recurrence:

$$[a^b \bmod N] = \begin{cases} \left[\left(a^{\frac{b}{2}} \right)^2 \bmod N \right] & \text{when } b \text{ is even} \\ \left[a \cdot \left(a^{\frac{b-1}{2}} \right)^2 \bmod N \right] & \text{when } b \text{ is odd.} \end{cases}$$

Doing so leads to an algorithm—called, for obvious reasons, “square-and-multiply” (or “repeated squaring”)—that requires only $\mathcal{O}(\log b) = \mathcal{O}(\|b\|)$ modular squarings/multiplications; see Algorithm B.13. In this algorithm, the length of b decreases by 1 in each iteration; it follows that the number of iterations is $\|b\|$, and so the overall algorithm runs in time polynomial in $\|a\|$, $\|b\|$, and $\|N\|$. More precisely, the number of modular squarings is exactly $\|b\|$, and the number of additional modular multiplications is exactly the Hamming weight of b (i.e., the number of 1s in the binary representation of b). This explains the preference, discussed in Section 8.2.4, for choosing the public RSA exponent e to have small length/Hamming weight.

ALGORITHM B.13**Algorithm ModExp for efficient modular exponentiation****Input:** Modulus N ; base $a \in \mathbb{Z}_N$; integer exponent $b > 0$ **Output:** $[a^b \bmod N]$ $x := a$ $t := 1$ // maintain the invariant that the answer is $[t \cdot x^b \bmod N]$ **while** $b > 0$ do:**if** b is odd $t := [t \cdot x \bmod N], \quad b := b - 1$ $x := [x^2 \bmod N], \quad b := b/2$ **return** t

Fix a and N and consider the modular exponentiation function given by $f_{a,N}(b) = [a^b \bmod N]$. We have just seen that computing $f_{a,N}$ is easy. In contrast, computing the *inverse* of this function—that is, computing b given a , N , and $[a^b \bmod N]$ —is believed to be hard for appropriate choice of a

and N . Inverting this function requires solving the *discrete-logarithm problem*, something we discuss in detail in Section 8.3.2.

Using precomputation. If the base a is known in advance, and there is a bound on the length of the exponent b , then one can use precomputation and a small amount of memory to speed up computation of $[a^b \bmod N]$. Say $\|b\| \leq n$. Then we precompute and store the n values

$$x_0 := a, \quad x_1 := [a^2 \bmod N], \quad \dots, \quad x_{n-1} := [a^{2^{n-1}} \bmod N].$$

Given exponent b with binary representation $b_{n-1} \cdots b_0$ (written from most to least significant bit), we then have

$$a^b = a^{\sum_{i=0}^{n-1} 2^i \cdot b_i} = \prod_{i=0}^{n-1} x_i^{b_i} \bmod N.$$

Since $b_i \in \{0, 1\}$, the number of multiplications needed to compute the result is exactly one less than the Hamming weight of b .

Exponentiation in Arbitrary Groups

The efficient modular exponentiation algorithm given above carries over in a straightforward way to enable efficient exponentiation in any group, as long as the underlying group operation can be performed efficiently. Specifically, if $\mathbb{G}$ is a group and g is an element of $\mathbb{G}$, then g^b can be computed using at most $2 \cdot \|b\|$ applications of the underlying group operation. Precomputation could also be used, exactly as described above.

If the order q of $\mathbb{G}$ is known, then $a^b = a^{[b \bmod q]}$ (cf. Proposition 8.52) and this can be used to speed up the computation by reducing b modulo q first.

Considering the (additive) group $\mathbb{Z}_N$, the group exponentiation algorithm just described gives a method for computing the “exponentiation”

$$[b \cdot g \bmod N] \stackrel{\text{def}}{=} [\underbrace{g + \cdots + g}_{b \text{ times}} \bmod N]$$

that differs from the method discussed earlier that relies on standard integer multiplication followed by a modular reduction. In comparing the two approaches to solving the same problem, note that the original algorithm uses specific information about $\mathbb{Z}_N$; in particular, it (essentially) treats the “exponent” b as an element of $\mathbb{Z}_N$ (possibly by reducing b modulo N first). In contrast, the “square-and-multiply” algorithm just presented treats $\mathbb{Z}_N$ only as an abstract group. (Of course, the group operation of addition modulo N relies on the specifics of $\mathbb{Z}_N$.) The point of this discussion is merely to illustrate that some group algorithms are *generic* (i.e., they apply equally well to all groups) while some group algorithms rely on specific properties of a *particular* group or class of groups. We saw some examples of this phenomenon in Chapter 9.

B.2.4 *Montgomery Multiplication

Although division over the integers (and hence modular reduction) can be done in polynomial time, algorithms for integer division are slow in comparison to, say, algorithms for integer multiplication. Montgomery multiplication provides a way to perform modular multiplication *without* carrying out any expensive modular reductions. Since pre- and postprocessing is required, the method is advantageous only when several modular multiplications will be done in sequence as, e.g., when computing a modular exponentiation.

Fix an odd modulus N with respect to which modular operations are to be done. Let $R > N$ be a power of two, say $R = 2^w$, and note that $\gcd(R, N) = 1$. The key property we will exploit is that division by R is fast: the quotient of x upon division by R is obtained by simply shifting x to the right w positions, and $[x \bmod R]$ is just the w least-significant bits of x .

Define the *Montgomery representation* of $x \in \mathbb{Z}_N^*$ by $\bar{x} \stackrel{\text{def}}{=} [xR \bmod N]$. Montgomery multiplication of $\bar{x}, \bar{y} \in \mathbb{Z}_N^*$ is defined as

$$\text{Mont}(\bar{x}, \bar{y}) \stackrel{\text{def}}{=} [\bar{x}\bar{y}R^{-1} \bmod N].$$

(We show below how this can be computed without any expensive modular reductions.) Note that

$$\text{Mont}(\bar{x}, \bar{y}) = \bar{x}\bar{y}R^{-1} = (xR)(yR)R^{-1} = (xy)R = \overline{xy} \bmod N.$$

This means we can multiply several values in $\mathbb{Z}_N$ by (1) converting to the Montgomery representation, (2) carrying out all multiplications using Montgomery multiplication to obtain the final result, and then (3) converting the result from Montgomery representation back to the standard representation.

Let $\alpha \stackrel{\text{def}}{=} [-N^{-1} \bmod R]$, a value which can be precomputed. (Computation of α , and conversion to/from Montgomery representation, can also be done without any expensive modular reductions; details are beyond our scope.) To compute $c \stackrel{\text{def}}{=} \text{Mont}(x, y)$ without any expensive modular reductions do:

1. Let $z := x \cdot y$ (over the integers).
2. Set $c' := (z + [z\alpha \bmod R] \cdot N) / R$.
3. If $c' < N$ then set $c := c'$; else set $c := c' - N$.

To see that this works, we first need to verify that step 2 is well-defined, namely, that the numerator is divisible by R . This follows because

$$z + [z\alpha \bmod R] \cdot N = z + z\alpha N = z - zN^{-1}N = 0 \bmod R.$$

Next, note that $c' = z/R \bmod N$ after step 2; moreover, since $z < N^2 < RN$ we have $0 < c' < (z + RN)/R < 2RN/R = 2N$. But then $[c' \bmod N] = c'$ if $c' < N$, and $[c' \bmod N] = c' - N$ if $c' > N$. We conclude that

$$c = [c' \bmod N] = [z/R \bmod N] = [xyR^{-1} \bmod N],$$

as desired.

B.2.5 Choosing a Uniform Group Element

For cryptographic applications, it is often necessary to choose a uniform element of a group $\mathbb{G}$. We first treat the problem in an abstract setting, and then focus specifically on the cases of $\mathbb{Z}_N$ and $\mathbb{Z}_N^*$.

Note that if $\mathbb{G}$ is a cyclic group of order q , and a generator $g \in \mathbb{G}$ is known, then choosing a uniform element $h \in \mathbb{G}$ reduces to choosing a uniform integer $x \in \mathbb{Z}_q$ and setting $h := g^x$. In what follows we make no assumptions on $\mathbb{G}$.

Elements of a group $\mathbb{G}$ must be specified using some *representation* of these elements as bit-strings, where we assume without any real loss of generality that all elements are represented using strings of the same length. (It is also crucial that there is a *unique* string representing each group element.) For example, if $\|N\| = n$ then elements of $\mathbb{Z}_N$ can all be represented as strings of length n , where the integer $a \in \mathbb{Z}_N$ is padded to the left with 0s if $\|a\| < n$.

We do not focus much on the issue of representation, since for all the groups considered in this text the representation can simply be taken to be the “natural” one (as in the case of $\mathbb{Z}_N$, above). Note, however, that different representations of the same group can affect the complexity of performing various computations, and so choosing the “right” representation for a given group is often important in practice. Since our goal is only to show *polynomial-time* algorithms for each of the operations we need (and not to show the most efficient algorithms known), the exact representation used is less important for our purposes. Moreover, most of the “higher-level” algorithms we present use the group operation in a “black-box” manner, so that as long as the group operation can be performed in polynomial time (in some parameter), the resulting algorithm will run in polynomial time as well.

Given a group $\mathbb{G}$ where elements are represented by strings of length ℓ , a uniform group element can be selected by choosing uniform ℓ -bit strings until the first string that corresponds to a group element is found. (Note this assumes that testing group membership can be done efficiently.) To obtain an algorithm with bounded running time, we introduce a parameter t bounding the maximum number of times this process is repeated; if all t iterations fail to find an element of $\mathbb{G}$, then the algorithm outputs **fail**. (An alternative is to output an arbitrary element of $\mathbb{G}$.) That is:

ALGORITHM B.14

Choosing a uniform group element

Input: A (description of a) group $\mathbb{G}$; length-parameter ℓ ;
parameter t

Output: A uniform element of $\mathbb{G}$

for $i = 1$ to t :

 Choose uniform $x \in \{0, 1\}^\ell$

if $x \in \mathbb{G}$ **return** x

return “fail”

It is clear that whenever the above algorithm does *not* output **fail**, it outputs a uniformly distributed element of $\mathbb{G}$. This is simply because each element of $\mathbb{G}$ is equally likely to be chosen in any iteration. Formally, if we let **Fail** be the event that the algorithm outputs **fail**, then for any element $g \in \mathbb{G}$ we have

$$\Pr \left[\text{output of the algorithm equals } g \mid \overline{\text{Fail}} \right] = \frac{1}{|\mathbb{G}|}.$$

What is the probability that the algorithm outputs **fail**? In any iteration the probability that $x \in \mathbb{G}$ is exactly $|\mathbb{G}|/2^\ell$, and so the probability that x does *not* lie in $\mathbb{G}$ in any of the t iterations is

$$\left(1 - \frac{|\mathbb{G}|}{2^\ell}\right)^t. \quad (\text{B.1})$$

There is a trade-off between the running time of Algorithm B.14 and the probability that the algorithm outputs **fail**: increasing t decreases the probability of failure but increases the worst-case running time. For cryptographic applications we need an algorithm where the worst-case running time is polynomial in the security parameter n , while the failure probability is negligible in n . Let $K \stackrel{\text{def}}{=} 2^\ell/|\mathbb{G}|$. If we set $t := K \cdot n$ then the probability that the algorithm outputs **fail** is:

$$\left(1 - \frac{1}{K}\right)^{K \cdot n} = \left(\left(1 - \frac{1}{K}\right)^K\right)^n \leq (e^{-1})^n = e^{-n},$$

using Proposition A.2. Thus, if $K = \text{poly}(n)$ (we assume some group-generation algorithm that depends on the security parameter n , and so both $|\mathbb{G}|$ and ℓ are functions of n), we obtain an algorithm with the desired properties.

The case of $\mathbb{Z}_N$. Consider the group $\mathbb{Z}_N$, with $n = \|N\|$. Checking whether an n -bit string x (interpreted as a positive integer of length at most n) is an element of $\mathbb{Z}_N$ simply requires checking whether $x < N$. Furthermore,

$$\frac{2^n}{|\mathbb{Z}_N|} = \frac{2^n}{N} \leq \frac{2^n}{2^{n-1}} = 2,$$

and so we can sample a uniform element of $\mathbb{Z}_N$ in $\text{poly}(n)$ time and with failure probability negligible in n .

The case of $\mathbb{Z}_N^*$. Consider next the group $\mathbb{Z}_N^*$, with $n = \|N\|$ as before. Determining whether an n -bit string x is an element of $\mathbb{Z}_N^*$ is also easy (see the exercises). Moreover,

$$\frac{2^n}{|\mathbb{Z}_N^*|} = \frac{2^n}{\phi(N)} = \frac{2^n}{N} \cdot \frac{N}{\phi(N)} \leq 2 \cdot \frac{N}{\phi(N)}.$$

A $\text{poly}(n)$ upper-bound is a consequence of the following theorem.

THEOREM B.15 For $N \geq 3$ of length n , we have $\frac{N}{\phi(N)} < 2n$.

(Stronger bounds are known, but the above suffices for our purpose.) The theorem can be proved using Bertrand's Postulate (Theorem 8.32), but we content ourselves with a proof in two special cases: when N is prime and when N is a product of two equal-length (distinct) primes.

The analysis is easy when N is an odd prime. Here $\phi(N) = N - 1$ and so

$$\frac{N}{\phi(N)} \leq \frac{2^n}{\phi(N)} = \frac{2^n}{N-1} \leq \frac{2^n}{2^{n-1}} = 2$$

(using the fact that N is odd for the second inequality). Consider next the case of $N = pq$ for p and q distinct, odd primes. Then

$$\frac{N}{\phi(N)} = \frac{pq}{(p-1)(q-1)} = \frac{p}{p-1} \cdot \frac{q}{q-1} < \left(\frac{3}{2}\right) \cdot \left(\frac{5}{4}\right) < 2.$$

We conclude that when N is prime or the product of two distinct, odd primes, there is an algorithm for generating a uniform element of $\mathbb{Z}_N^*$ that runs in time polynomial in $n = \|N\|$ and outputs fail with probability negligible in n .

Throughout this book, when we speak of sampling a uniform element of $\mathbb{Z}_N$ or $\mathbb{Z}_N^*$ we simply ignore the negligible probability of outputting fail with the understanding that this has no significant effect on the analysis.

B.3 *Finding a Generator of a Cyclic Group

In this section we address the problem of finding a generator of an arbitrary cyclic group $\mathbb{G}$ of order q . Here, q does not necessarily denote a prime number; indeed, finding a generator when q is prime is trivial by Corollary 8.55.

We actually show how to sample a *uniform* generator, proceeding in a manner very similar to that of Section B.2.5. Here, we repeatedly sample uniform elements of $\mathbb{G}$ until we find an element that is a generator. As in Section B.2.5, an analysis of this method requires understanding two things:

- How to efficiently test whether a given element is a generator; and
- the fraction of group elements that are generators.

In order to understand these issues, we first develop a bit of additional group-theoretic background.

B.3.1 Group-Theoretic Background

We tackle the second issue first. Recall that the order of an element h is the smallest positive integer i for which $h^i = 1$. Let g be a generator of a

group $\mathbb{G}$ of order $q > 1$; this means the order of g is q . Consider an element $h \in \mathbb{G}$ that is not the identity (the identity cannot be a generator of $\mathbb{G}$), and let us ask whether h might also be a generator of $\mathbb{G}$. Since g generates $\mathbb{G}$, we can write $h = g^x$ for some $x \in \{1, \dots, q-1\}$ (note $x \neq 0$ since h is not the identity). Consider two cases:

Case 1: $\gcd(x, q) = r > 1$. Write $x = \alpha \cdot r$ and $q = \beta \cdot r$ with α, β non-zero integers less than q . Then:

$$h^\beta = (g^x)^\beta = g^{\alpha r \beta} = (g^q)^\alpha = 1.$$

So the order of h is at most $\beta < q$, and h cannot be a generator of $\mathbb{G}$.

Case 2: $\gcd(x, q) = 1$. Let $i \leq q$ be the order of h . Then

$$g^0 = 1 = h^i = (g^x)^i = g^{xi},$$

implying $xi = 0 \bmod q$ by Proposition 8.53. This means that $q \mid xi$. Since $\gcd(x, q) = 1$, however, Proposition 8.3 shows that $q \mid i$ and so $i = q$. We conclude that h is a generator of $\mathbb{G}$.

Summarizing the above, we see that for $x \in \{1, \dots, q-1\}$ the element $h = g^x$ is a generator of $\mathbb{G}$ exactly when $\gcd(x, q) = 1$. We have thus proved the following:

THEOREM B.16 *Let $\mathbb{G}$ be a cyclic group of order $q > 1$ with generator g . There are $\phi(q)$ generators of $\mathbb{G}$, and these are exactly given by $\{g^x \mid x \in \mathbb{Z}_q^*\}$.*

In particular, if $\mathbb{G}$ is a group of prime order q , then it has $\phi(q) = q - 1$ generators—exactly in agreement with Corollary 8.55.

We turn next to the first issue, that of deciding whether a given element h is a generator of $\mathbb{G}$. Of course, one way to check whether h generates $\mathbb{G}$ is to enumerate $\{h^0, h^1, \dots, h^{q-1}\}$ and see whether this list includes every element of $\mathbb{G}$. This requires time linear in q (i.e., exponential in $\|q\|$) and is therefore unacceptable for our purposes. Another approach, if we already know a generator g , is to compute the discrete logarithm $x = \log_g h$ and then apply the previous theorem; in general, however, we may not have such a g , and anyway computing the discrete logarithm may itself be a hard problem.

If we know the factorization of q , we can do better.

PROPOSITION B.17 *Let $\mathbb{G}$ be a group of order q , and let $q = \prod_{i=1}^k p_i^{e_i}$ be the prime factorization of q , where the $\{p_i\}$ are distinct primes and $e_i \geq 1$. Set $q_i = q/p_i$. Then $h \in \mathbb{G}$ is a generator of $\mathbb{G}$ if and only if*

$$h^{q_i} \neq 1 \quad \text{for } i = 1, \dots, k.$$

PROOF One direction is easy. Say $h^{q_i} = 1$ for some i . Then the order of h is at most $q_i < q$, and so h cannot be a generator.

Conversely, say h is not a generator but instead has order $q' < q$. By Proposition 8.54, we know $q' \mid q$. This implies that q' can be written as $q' = \prod_{i=1}^k p_i^{e'_i}$, where $e'_i \geq 0$ and for at least one index j we have $e'_j < e_j$. But then q' divides $q_j = p_j^{e_j-1} \cdot \prod_{i \neq j} p_i^{e_i}$, and so (using Proposition 8.53) $h^{q_j} = h^{[q_j \bmod q']} = h^0 = 1$. ■

The proposition does not require $\mathbb{G}$ to be cyclic; if $\mathbb{G}$ is not cyclic then every element $h \in \mathbb{G}$ will satisfy $h^{q_i} = 1$ for some i and there are no generators.

B.3.2 Efficient Algorithms

Armed with the results of the previous section, we show how to efficiently *test* whether a given element is a generator, as well as how to efficiently *find* a generator in an arbitrary group.

Testing if an element is a generator. Proposition B.17 immediately suggests an efficient algorithm for deciding whether a given element h is a generator or not.

ALGORITHM B.18

Testing whether an element is a generator

Input: Group order q ; prime factors $\{p_i\}_{i=1}^k$ of q ; element $h \in \mathbb{G}$

Output: A decision as to whether h is a generator of $\mathbb{G}$

for $i = 1$ to k :

if $h^{q/p_i} = 1$ **return** “ h is not a generator”

return “ h is a generator”

Correctness of the algorithm is evident from Proposition B.17. We now show that the algorithm terminates in time polynomial in $\|q\|$. Since, in each iteration, h^{q/p_i} can be computed in polynomial time, we need only show that the number of iterations k is polynomial. This is the case since an integer q can have no more than $\log_2 q = \mathcal{O}(\|q\|)$ prime factors; this is because

$$q = \prod_{i=1}^k p_i^{e_i} \geq \prod_{i=1}^k p_i \geq \prod_{i=1}^k 2 = 2^k$$

and so $k \leq \log_2 q$.

Algorithm B.18 requires the prime factors of the group order q to be provided as input. Interestingly, there is no known efficient algorithm for testing whether an element of an arbitrary group is a generator when the factors of the group order are *not* known.

The fraction of elements that are generators. As shown in Theorem B.16, the fraction of elements of a group $\mathbb{G}$ of order q that are generators is $\phi(q)/q$. Theorem B.15 says that $\phi(q)/q = \Omega(1/\|q\|)$. The fraction of elements that are generators is thus sufficiently high to ensure that sampling a polynomial number of elements from the group will yield a generator with all but negligible probability. (The analysis is the same as in Section B.2.5.)

Concrete examples in $\mathbb{Z}_p^*$. Putting everything together, we see there is an efficient probabilistic algorithm for finding a generator of a group $\mathbb{G}$ *as long as the factorization of the group order is known*. When selecting a group for cryptographic applications, it is therefore important that the group is chosen in such a way that this holds. This explains again the preference, discussed extensively in Section 8.3.2, for working in an appropriate prime-order subgroup of $\mathbb{Z}_p^*$. Another possibility is to use $\mathbb{G} = \mathbb{Z}_p^*$ for p a strong prime (i.e., $p = 2q + 1$ with q also prime), in which case the prime factorization of the group order $p - 1$ is known. One final possibility is to generate a prime p in such a way that the factorization of $p - 1$ is known. Further details are beyond the scope of this book.

References and Additional Reading

The book by Shoup [159] is highly recommended for those seeking to explore the topics of this chapter in further detail. In particular, bounds on $\phi(N)/N$ (and an asymptotic version of Theorem B.15) can be found in [159, Chapter 5]. Hankerson et al. [83] also provide extensive detail on the implementation of number-theoretic algorithms for cryptography.

Exercises

- B.1 Prove correctness of the extended Euclidean algorithm.
- B.2 Prove that the extended Euclidean algorithm runs in time polynomial in the lengths of its inputs.

Hint: First prove a proposition analogous to Proposition B.8.
- B.3 Show how to determine that an n -bit string is in $\mathbb{Z}_N^*$ in polynomial time.

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